

Protein synthesis and maturation

A. Protein synthesis in the rER ●

With all proteins, protein biosynthesis (**Translation**; for details, see p. 250) starts on free ribosomes in the cytoplasm (1). Proteins that are exported out of the cell or into lysosomes, and membrane proteins of the ER and the plasma membrane, carry a **signal peptide for the ER** at their N-terminus. This is a section of 15–60 amino acids in which one or two strongly basic residues (Lys, Arg) near the N-terminus are followed by a strongly hydrophobic sequence of 10–15 residues (see p. 228).

As soon as the signal peptide (red) appears on the surface of the ribosome (2), an RNA-containing *signal recognition particle* (**SRP**, green) binds to the sequence and initially interrupts translation (3). The SRP then binds to an **SRP receptor** in the rER membrane, and in this way attaches the ribosome to the ER (4). After this, the SRP dissociates from the signal peptide and from the SRP receptor and is available again for step 3. This endergonic process is driven by GTP hydrolysis (5). Translation now resumes. The remainder of the protein, still unfolded, is gradually introduced into a channel (the **translocon**) in the lumen of the rER (6), where a *signal peptidase* located in the inner ER membrane cleaves the signal peptide while translation is still taking place (7). This converts the **preprotein** into a **proprotein**, from which the mature protein finally arises after additional post-translational modifications (8) in the ER and in the Golgi apparatus.

If the growing polypeptide contains a **stop-transfer signal** (see p. 228), then this hydrophobic section of the chain remains stuck in the membrane outside the translocon, and an *integral membrane protein* arises. In the course of translation, an additional signal sequence can re-start the transfer of the chain through the translocon. Several repetitions of this process produce integral membrane proteins with several transmembrane helices (see p. 214).

B. Protein glycosylation ○

Most extracellular proteins contain covalently bound oligosaccharide residues. For example,

all plasma proteins with the exception of albumin are glycosylated. Together with glycolipids, numerous **glycoproteins** on the cell surface form the **glycocalyx**. Inside the ER, the carbohydrate parts of the glycoproteins are cotranslationally transferred to the growing chain, and are then converted into their final form while passing through the ER and Golgi apparatus.

N-bound oligosaccharides (see p. 44) are always bound to the acid-amide group of asparagine residues. If a **glycosylation sequence** (–Asn–X–Ser(Thr)–, where X can be any amino acid) appears in the growing peptide chain, then a *transglycosylase* in the ER membrane [1] transfers a previously produced **core oligosaccharide** consisting of 14 hexose residues *en-bloc* from the carrier molecule **dolichol diphosphate** to the peptide.

Dolichol is a long-chain isoprenoid (see p. 52) consisting of 10–20 isoprene units, which is embedded in the ER membrane. A hydroxyl group at the end of the molecule is bound to diphosphate, on which the nuclear oligosaccharide is built up in an extended reaction sequence (not shown here in detail). The core structure consists of two residues of N-acetylglucosamine (GlcNAc), a branched group of nine mannose residues (Man) and three terminal glucose residues (Glc).

As the proprotein passes through the ER, *glycosidases* [2] remove the glucose residues completely and the mannoses partially (“trimming”), thereby producing the **mannose-rich type** of oligosaccharide residues. Subsequently, various *glycosyltransferases* [3] transfer additional monosaccharides (e.g., GlcNAc, galactose, fucose, and N-acetylneuraminic acid; see p. 38) to the mannose-rich intermediate and thereby produce the **complex type** of oligosaccharide. The structure of the final oligosaccharide depends on the type and activity of the glycosyltransferases present in the ER of the cell concerned, and is therefore genetically determined (although indirectly).

Protein maturation

After translation, proteins destined for the secretory pathway (see p. 228) first have to fold into their native conformation within the rER (see p. 230). During this process they are supported by various *auxiliary proteins*.

A. Protein folding in the rER ●

To prevent incorrect folding of the growing protein during protein biosynthesis, **chaperones** (see B) in the lumen of the rER bind to the peptide chain and stabilize it until translation has been completed. Binding protein (**BiP**) is an important chaperone in the ER.

Many secretory proteins—e.g., pancreatic ribonuclease (RNase; see p. 74)—contain several disulfide bonds that are only formed oxidatively from SH groups after translation. The eight cysteine residues of the RNase can in principle form 105 different pairings, but only the combination of the four disulfide bonds shown on p. 75 provides active enzyme. Incorrect pairings can block further folding or lead to unstable or insoluble conformations. The enzyme *protein disulfide isomerase* [1] accelerates the equilibration between paired and unpaired cysteine residues, so that incorrect pairs can be quickly split before the protein finds its final conformation.

Most peptide bonds in proteins take on the *trans* conformation (see p. 66). Only bonds with proline residues (–X–Pro–) can be present in both *cis* and *trans* forms.

In the protein's native conformation, every X–Pro bond has to have the correct conformation (*cis* or *trans*). As the uncatalyzed transition between the two forms is very slow, there is a *proline cis–trans isomerase* [2] in the ER that accelerates the conversion.

B. Chaperones and chaperonins ●

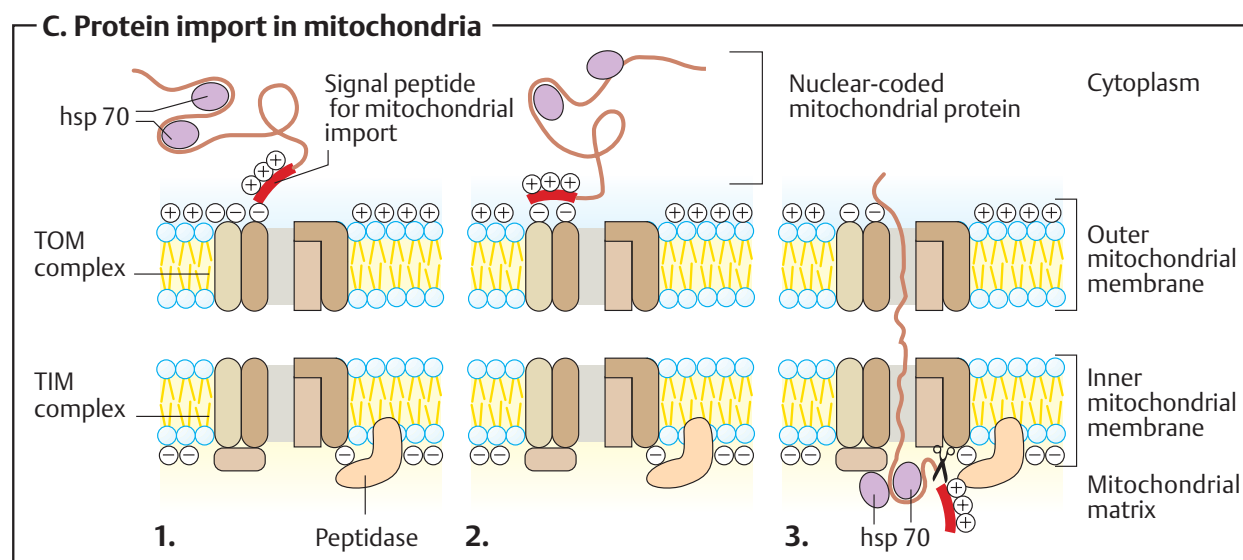
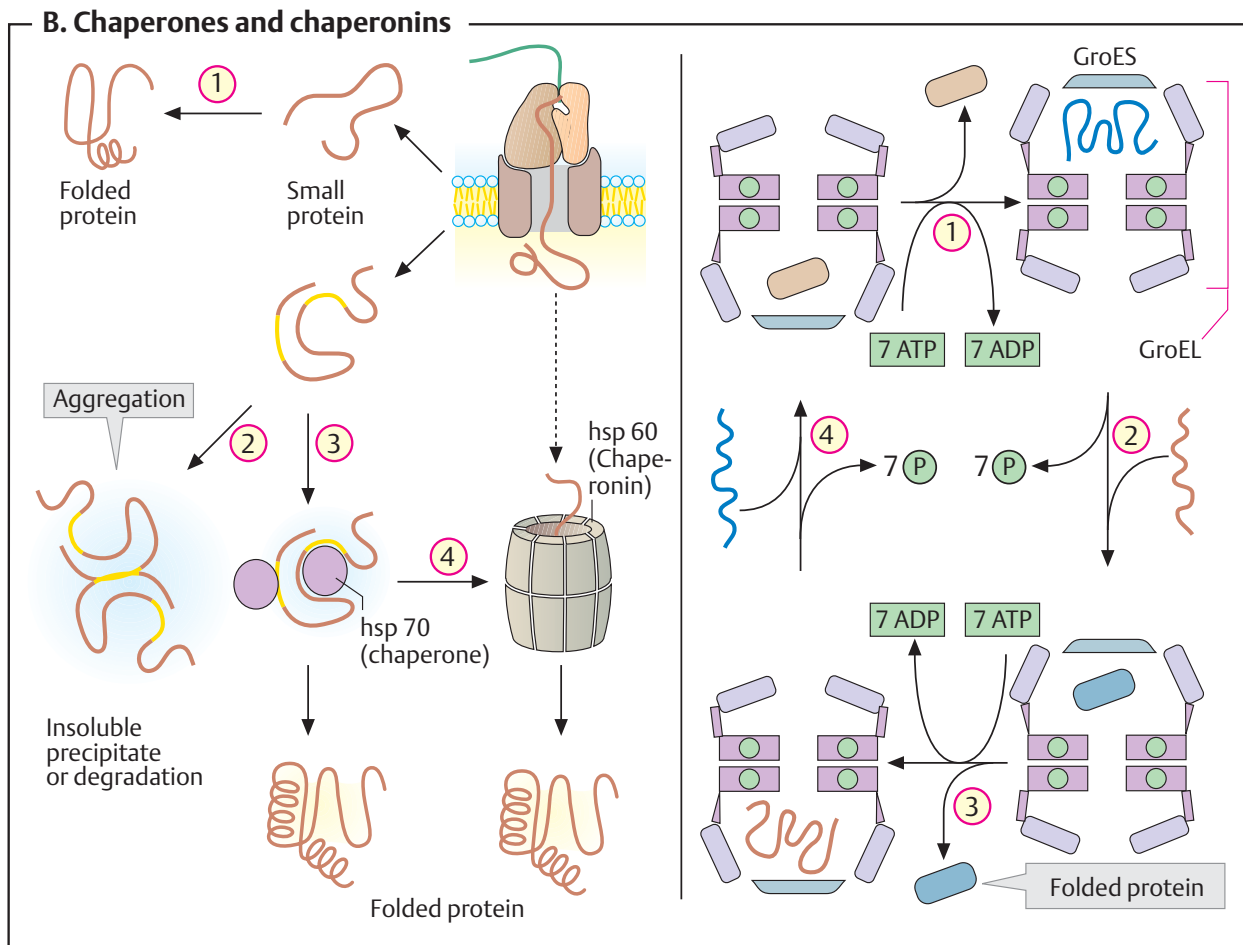
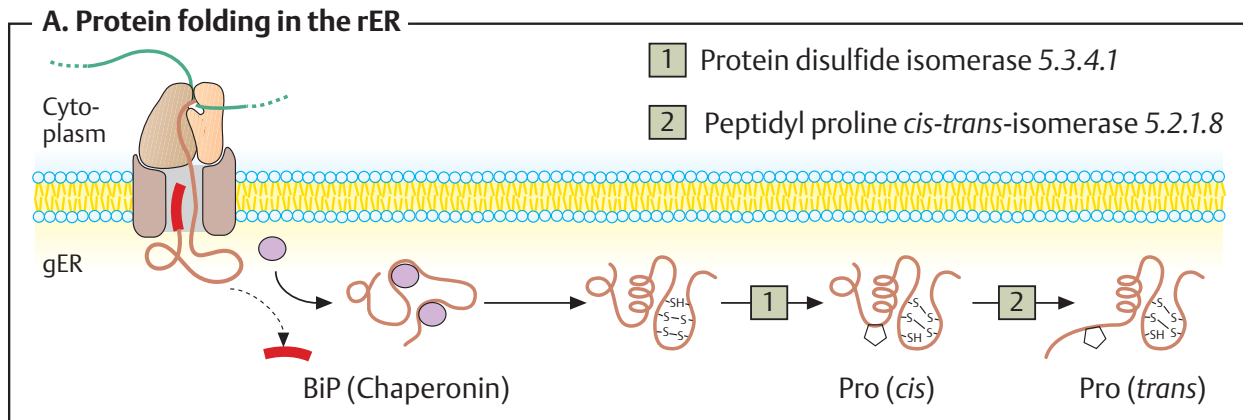
Most proteins fold spontaneously into their native conformation, even in the test tube. In the cell, where there are very high concentrations of proteins (around 350 g L^{-1}), this is more difficult. In the unfolded state, the apolar regions of the peptide chain (yellow) tend to aggregate—due to the hydrophobic effect (see p. 28)—with other proteins or with each other to form insoluble products (2). In addition, unfolded proteins are suscep-

tible to proteinases. To protect partly folded proteins, there are auxiliary proteins called **chaperones** because they guard immature proteins against damaging contacts. Chaperones are formed increasingly during temperature stress and are therefore also known as **heat-shock proteins** (hsp). Several classes of hsp are distinguished. Chaperones of the hsp70 type (Dna K in bacteria) are common, as are type hsp60 **chaperonins** (GroEL/ES in bacteria). Class hsp90 chaperones have special tasks (see p. 378).

While small proteins can often reach their native conformation without any help (1), larger molecules require hsp70 proteins for protection against aggregation which bind as monomers and can dissociate again, dependent on ATP (3). By contrast, type hsp60 chaperonins form large, barrel-shaped complexes with 14 subunits in which proteins can fold independently while shielded from their environment (4). The function of hsp60 has been investigated in detail in the bacterial chaperonin **GroEL** (right). The barrel has two chambers, which are closed with a lid (**GroES**) during folding of the guest protein. Driven by ATP hydrolysis, the chambers open and close alternately—i.e., the release of the fully folded protein from one chamber is coupled to the uptake of an unfolded peptide in the second chamber.

C. Protein import in mitochondria ●

Class hsp70 chaperones are also needed for translocation of nuclear-coded proteins from the cytoplasm into the mitochondria (see p. 228). As two membranes have to be crossed to reach the matrix, there are two translocator complexes: **TOM** (“transport outer membrane”) and **TIM** (“transport inner membrane”). For transport, proteins are unfolded in the cytoplasm and protected by hsp70. TOM recognizes the positively charged signal sequence at the protein's N terminus (see p. 228) and with the help of the membrane potential threads the chains through the central pores of the two complexes. Inside TIM, further hsp70 molecules bind and pull the chain completely into the matrix. As with import into the ER, the signal peptide is proteolytically removed by a signal peptidase during translocation.



Lysosomes

A. Structure and contents ①

Animal lysosomes are organelles with a diameter of 0.2–2.0 μm with various shapes that are surrounded by a single membrane. There are usually several hundred lysosomes per cell. ATP-driven V-type proton pumps are active in their membranes (see p. 220). As these accumulate H^+ in the lysosomes, the content of lysosomes with pH values of 4.5–5 is much more acidic than the cytoplasm (pH 7–7.3).

The lysosomes are the cell's "stomach," serving to break down various cell components. For this purpose, they contain some 40 different types of **hydrolases**, which are capable of breaking down every type of macromolecule. The marker enzyme of lysosomes is *acid phosphatase*. The pH optimum of lysosomal enzymes is adjusted to the acid pH value and is also in the range of pH 5. At neutral pH, as in the cytoplasm, lysosomal enzymes only have low levels of activity. This appears to be a mechanism for protecting the cells from digesting themselves in case lysosomal enzymes enter the cytoplasm at any time. In plants and fungi, the **cell vacuoles** (see p. 43) have the function of lysosomes.

B. Functions ①

Lysosomes serve for enzymatic degradation of macromolecules and cell organelles, which are supplied in various ways. The example shows the degradation of an overaged mitochondrion by *autophagy*. To accomplish this, the lysosome encloses the organelle (1). During this process, the **primary lysosome** converts into a **secondary lysosome**, in which the hydrolytic degradation takes place (2). Finally, **residual bodies** contain the indigestible residues of the lysosomal degradation process. Lysosomes are also responsible for the degradation of macromolecules and particles taken up by cells via *endocytosis* and *phagocytosis*—e.g., lipoproteins, proteohormones, and bacteria (*heterophagy*). In the process, lysosomes fuse with the **endosomes** (3) in which the endocytosed substances are supplied.

C. Synthesis and transport of lysosomal proteins ①

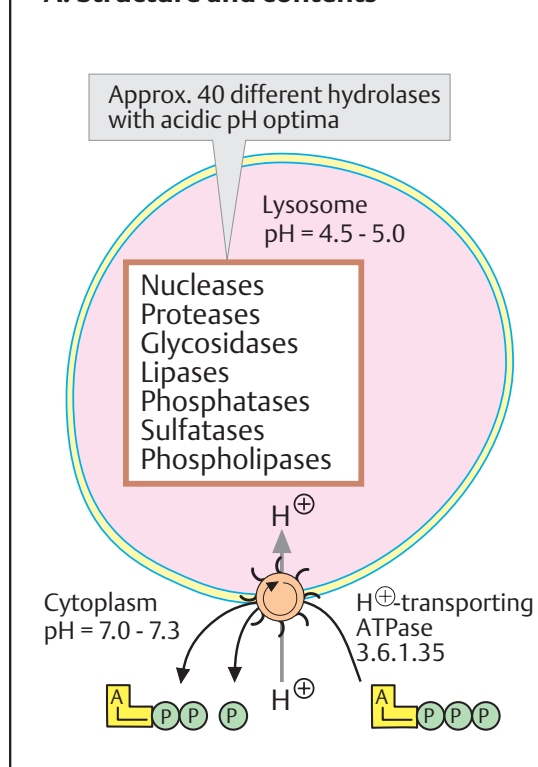
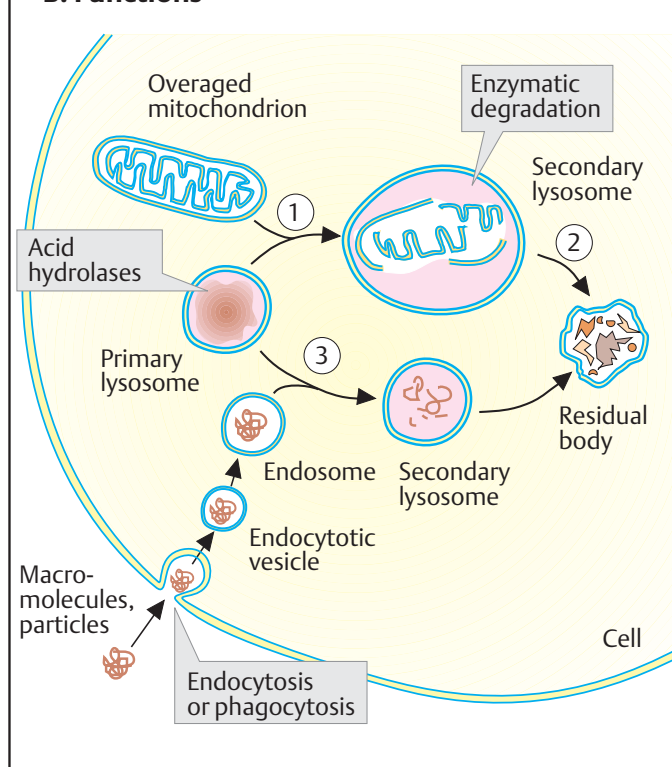
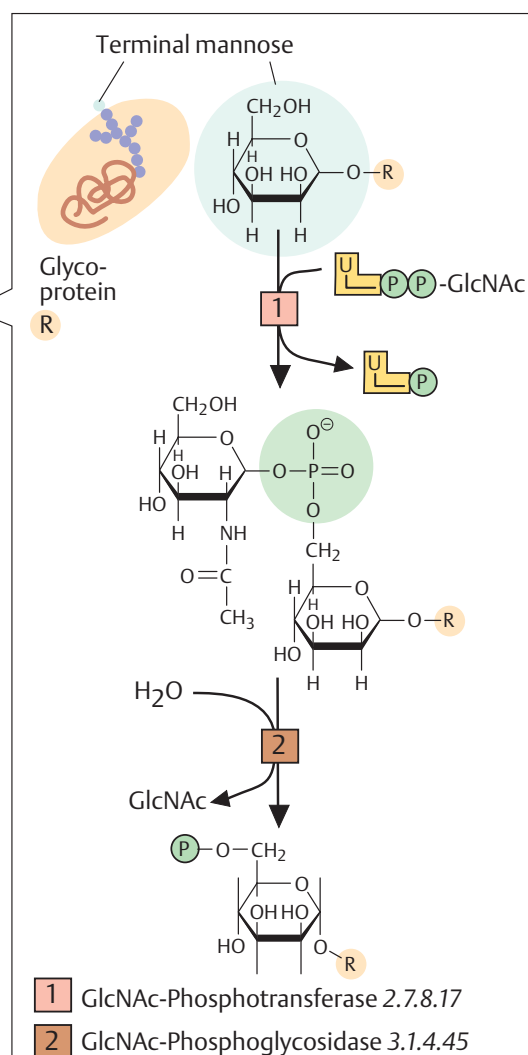
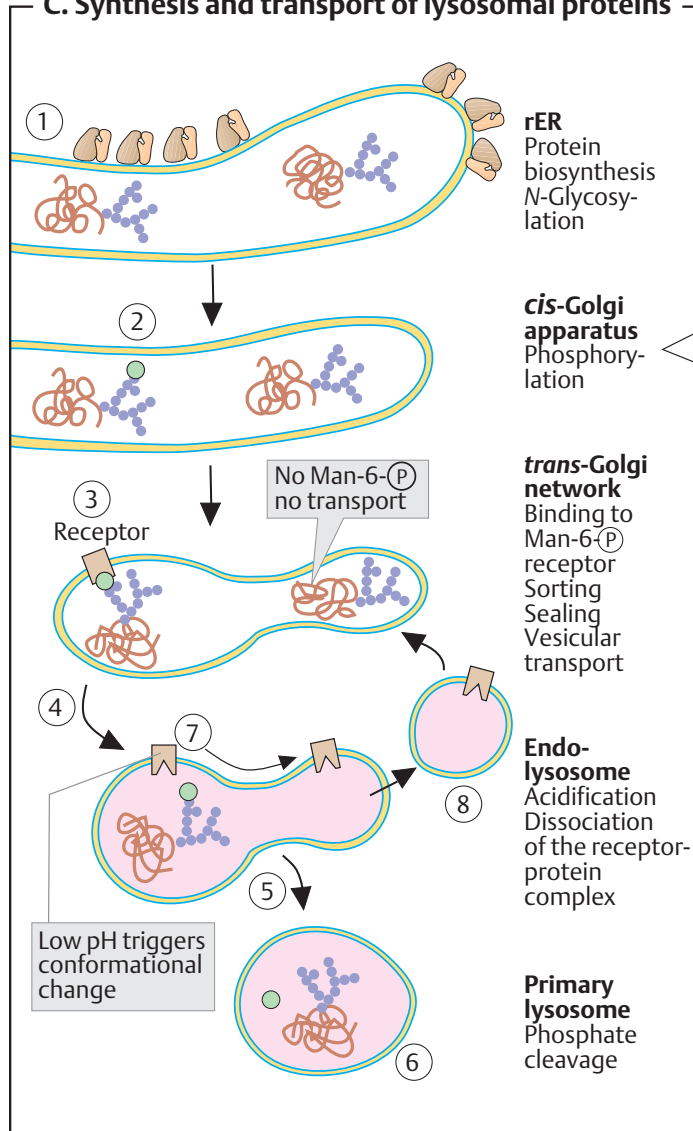
Primary lysosomes arise in the region of the Golgi apparatus. **Lysosomal proteins** are synthesized in the rER and are glycosylated there as usual (1; see p. 228). The next steps are specific for lysosomal proteins (right part of the illustration). In a two-step reaction, terminal mannose residues (Man) are phosphorylated at the C-6 position of the mannose. First, *N*-acetylglucosamine 1-phosphate is transferred to the OH group at C-6 in a terminal mannose residue, and *N*-acetylglucosamine is then cleaved again. Lysosomal proteins therefore carry a terminal **mannose 6-phosphate** (Man6-P; 2).

The membranes of the Golgi apparatus contain receptor molecules that bind Man 6-P. They recognize lysosomal proproteins by this residue and bind them (3). With the help of *clathrin*, the receptors are concentrated locally. This allows the appropriate membrane sections to be pinched off and transported to the endolysosomes with the help of transport vesicles (4), from which primary lysosomes arise through maturation (5). Finally, the phosphate groups are removed from Man 6-P (6).

The *Man 6-P receptors* are reused. The fall in the pH value in the endolysosomes releases the receptors from the bound proteins (7) which are then transported back to the Golgi apparatus with the help of transport vesicles.

Further information

Many hereditary diseases are due to genetic defects in lysosomal enzymes. The metabolism of glycogen ($\rightarrow$ *glycogenoses*), lipids ($\rightarrow$ *lipidoses*), and proteoglycans ($\rightarrow$ *mucopolysaccharidoses*) is particularly affected. As the lysosomal enzymes are indispensable for the intracellular breakdown of macromolecules, unmetabolized macromolecules or degradation products accumulate in the lysosomes in these diseases and lead to irreversible cell damage over time. In the longer term, enlargement takes place, and in severe cases there may be failure of the organ affected—e.g., the liver.

A. Structure and contents**B. Functions****C. Synthesis and transport of lysosomal proteins**

Molecular genetics: overview

Nucleic acids (DNA and various RNAs) are of central importance in the storage, transmission, and expression of genetic information. The decisive factor involved is their ability to enter into specific base pairings with each other (see p. 84). The individual processes involved, which are summed up in an overview here, are discussed in more detail on the following pages.

A. Expression and transmission of genetic information ●

Storage. The genetic information of all cells is stored in the base sequence of their DNA (RNA only occurs as a genetic material in viruses; see p. 404). Functional sections of DNA that code for inheritable structures or functions are referred to as **genes**. The 30 000–40 000 human genes represent only a few percent of the **genome**, which consists of approximately $5 \cdot 10^9$ base pairs (bp). Most genes code for proteins—i.e., they contain the information for the sequence of amino acid residues of a protein (its sequence). Every amino acid residue is represented in DNA by a code word (a **codon**) consisting of a sequence of three base pairs (a triplet). At the level of DNA, codons are defined as sequences of the sense strand read in the 5'→3' direction (see p. 84). A DNA codon for the amino acid phenylalanine, for example, is thus *TTC* (2).

Replication. During cell division, all of the genetic information has to be passed on to the daughter cells. To achieve this, the whole of the DNA is copied during the S phase of the cell cycle (see p. 394). In this process, each strand serves as a matrix for the synthesis of a complementary second strand (1; see p. 240).

Transcription. For *expression* of a gene—i.e., synthesis of the coded protein—the DNA sequence information has to be converted into a protein sequence. As DNA itself is not involved in protein synthesis, the information is transferred from the nucleus to the site of synthesis in the cytoplasm. To achieve this, the *template strand* in the relevant part of the gene is transcribed into an **RNA** (hnRNA). The sequence of this RNA is thus complementary to that of the *template strand* (3), but—with the exception of the exchange of thy-

mine for uracil—it is identical to that of the sense strand. In this way, the DNA triplet *TTC* gives rise in hnRNA to the RNA codon *UUC*.

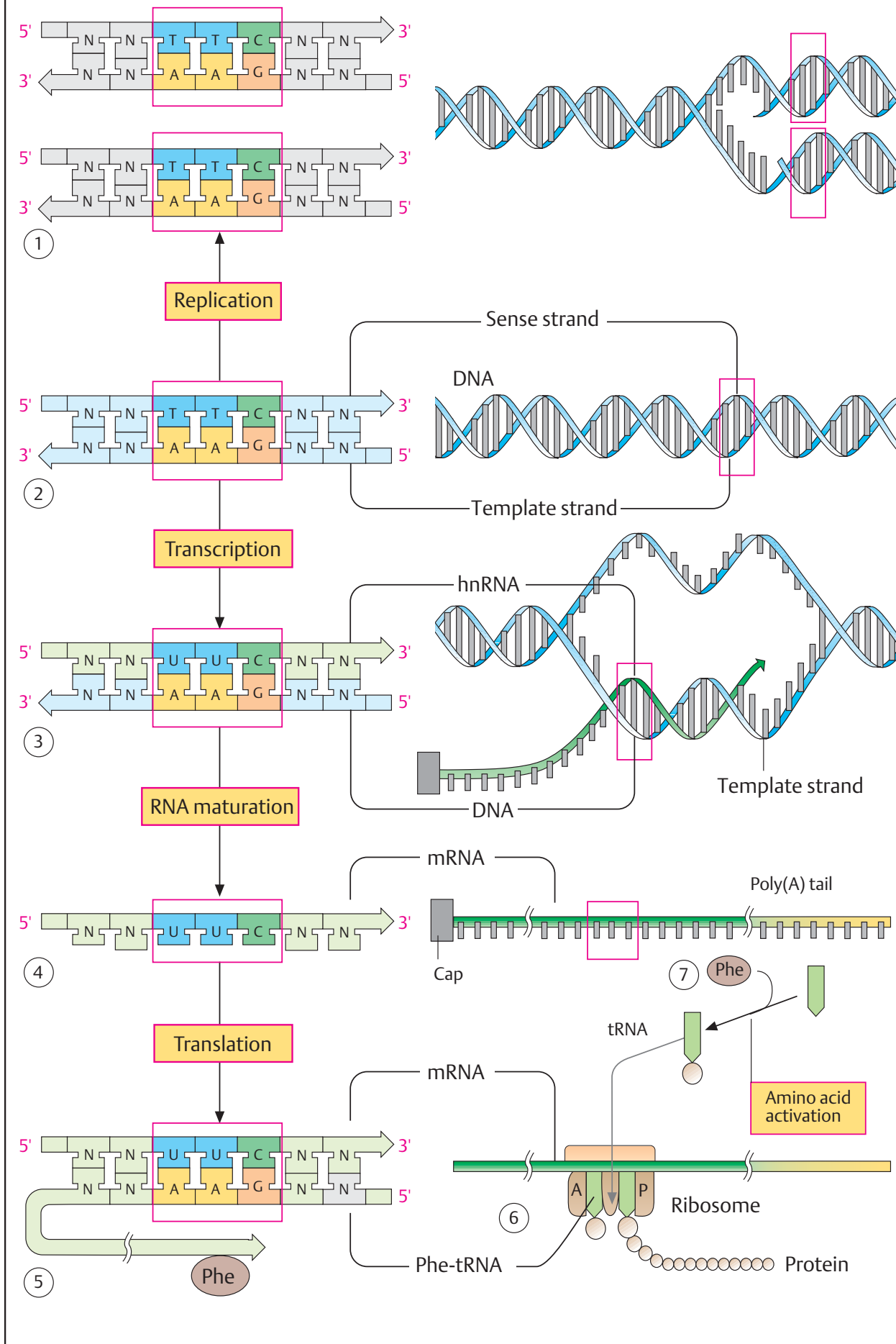
RNA maturation. In eukaryotes, the hnRNA initially formed is modified several times before it can leave the nucleus as **messenger RNA** (mRNA, 4). During RNA maturation, superfluous (“intervening”) sequences (introns) are removed from the molecule, and both ends of the transcript are protected by the addition of further nucleotides (see p. 246).

Translation. Mature mRNA enters the cytoplasm, where it binds to **ribosomes**, which convert the RNA information into a peptide sequence. The ribosomes (see p. 250) consist of more than 100 proteins and several RNA molecules (rRNA; see p. 82). rRNA plays a role as a ribosomal structural element and is also involved in the binding of mRNA to the ribosome and the formation of the peptide bond.

The actual information transfer is based on the interaction between the mRNA codons and another type of RNA, **transfer RNA** (tRNA; see p. 82). tRNAs, of which there are numerous types, always provide the correct amino acid to the ribosome according to the sequence information in the mRNA. tRNAs are loaded with an amino acid residue at the 3' end. Approximately in the middle, they present the triplet that is complementary to each mRNA codon, known as the *anticodon* (*GAA* in the example shown). If the codon *UUC* appears on the mRNA, the anticodon binds a molecule of Phe-t-RNA^{Phe} to the mRNA (5) and thus brings the phenylalanine residue at the other end of the molecule into a position in which it can take over the growing polypeptide chain from the neighboring tRNA (6).

Amino acid activation. Before binding to the ribosomes, tRNAs are loaded with the correct amino acids by specific ligases (7; see p. 248). It is the amino acid tRNA ligases that carry out the transfer (translation) of the genetic information from the nucleic acid level to the protein level.

A. Expression and transmission of genetic information



Genome

A. Chromatin ●

In the nuclei of eukaryotes (see p. 196), DNA is closely associated with proteins and RNA. These nucleoprotein complexes, with a DNA proportion of approximately one-third, are known as **chromatin**. It is only during cell division (see p. 394) that chromatin condenses into **chromosomes** that are visible under light microscopy. During interphase, most of the chromatin is loose, and in these conditions a morphological distinction can be made between tightly packed **heterochromatin** and the less dense **euchromatin**. Euchromatin is the site of active transcription.

The proteins contained in chromatin are classified as either histone or non-histone proteins. **Histones (B)** are small, strongly basic proteins that are directly associated with DNA. They contribute to the structural organization of chromatin, and their basic amino acids also neutralize the negatively charged phosphate groups, allowing the dense packing of DNA in the nucleus. This makes it possible for the 46 DNA molecules of the diploid human genome, with their 5×10^9 base pairs (bp) and a total length of about 2 m, to be accommodated in a nucleus with a diameter of only 10 μm . Histones also play a central role in regulating transcription (see p. 244).

Two histone molecules each of types **H2A** (blue), **H2B** (green), **H3** (yellow), and **H4** (red) form an octameric complex, around which 146 bp of DNA are wound in 1.8 turns. These particles, with a diameter of 7 nm, are referred to as **nucleosomes**. Another histone (**H1**) binds to DNA segments that are not directly in contact with the histone octamers ("linker" DNA). It covers about 20 bp and supports the formation of spirally wound superstructures with diameters of 30 nm, known as **solenoids**. When chromatin condenses into chromosomes, the solenoids form **loops** about 200 nm long, which already contain about 80 000 bp. The loops are bound to a protein framework (the **nuclear scaffolding**), which in turn organizes some 20 loops to form **minibands**. A large number of stacked minibands finally produces a chromosome. In the chromosome, the DNA is so densely packed that the smallest human chromosome already contains more than 50 million bp.

The **non-histone proteins** are very heterogeneous. This group includes *structural proteins* of the nucleus, as well as many *enzymes* and *transcription factors* (see p. 118), which selectively bind to specific segments of DNA and regulate gene expression and other processes.

B. Histones ○

The histones are remarkable in several ways. With their high proportions of lysine and arginine (blue shading), they are strongly basic, as mentioned above. In addition, their amino acid sequence has hardly changed at all in the course of evolution. This becomes clear when one compares the histone sequences in mammals, plants, and fungi (yeasts are single-celled fungi; see p. 148). For example, the H4 histones in humans and wheat differ only in a single amino acid residue, and there are only a few changes between humans and yeast. In addition, all of these changes are "conservative"—i.e., the size and polarity barely differ. It can be concluded from this that the histones were already "optimized" when the last common predecessor of animals, plants, and fungi was alive on Earth (more than 700 million years ago). Although countless mutations in histone genes have taken place since, almost all of these evidently led to the extinction of the organisms concerned.

The histones in the octamer carry N-terminal mobile "tails" consisting of some 20 amino acid residues that project out of the nucleosomes and are important in the regulation of chromatin structure and in controlling gene expression (see **A2**; only two of the eight tails are shown in full length). For example, the condensation of chromatin into chromosomes is associated with *phosphorylation* (P) of the histones, while the transcription of genes is initiated by *acetylation* (A) of lysine residues in the N-terminal region (see p. 244).

Replication

For genetic information to be passed on during cell division, a complete copy of the genome has to be produced before each mitosis. This process is known as **DNA replication**.

A. Mechanism of DNA polymerases ①

Replication is catalyzed by *DNA-dependent DNA polymerases*. These enzymes require a single strand of DNA, known as the **template**. Beginning at a short starting sequence of RNA (the **primer**), they synthesize a second complementary strand on the basis of this template, and thus create a complete DNA double helix again. The substrates of the DNA polymerases are the four deoxynucleoside triphosphates **dATP**, **dGTP**, **dCTP**, and **dTTP**. In each step, base pairing first binds the nucleotide that is complementary to the current base in the template strand. The α -phosphate residue of the newly bound nucleoside triphosphate is then subjected to nucleophilic attack by the 3'-OH group of the nucleotide incorporated immediately previously. This is followed by the elimination of diphosphate and the formation of a new phosphoric acid diester bond. These steps are repeated again for each nucleotide. The mechanism described means that the matrix can only be read in the 3'→5' direction. In other words, the newly synthesized strand always grows in the **5'→3' direction**. The same mechanism is also used in transcription by *DNA-dependent RNA polymerases* (see p. 242). Most DNA and RNA polymerases consist of more than 10 subunits, the role of which is still unclear to some extent.

B. Replication in *E. coli* ①

Although replication in prokaryotes is now well understood, many details in eukaryotes are still unclear. However, it is certain that the process is in principle similar. A simplified scheme of replication in the bacterium *Escherichia coli* is shown here.

In bacteria, replication starts at a specific point in the circular DNA—the **origin of replication**—and proceeds in both directions. This results in two diverging **replication forks**, in which the two strands are replicated simultaneously. Numerous proteins are involved in

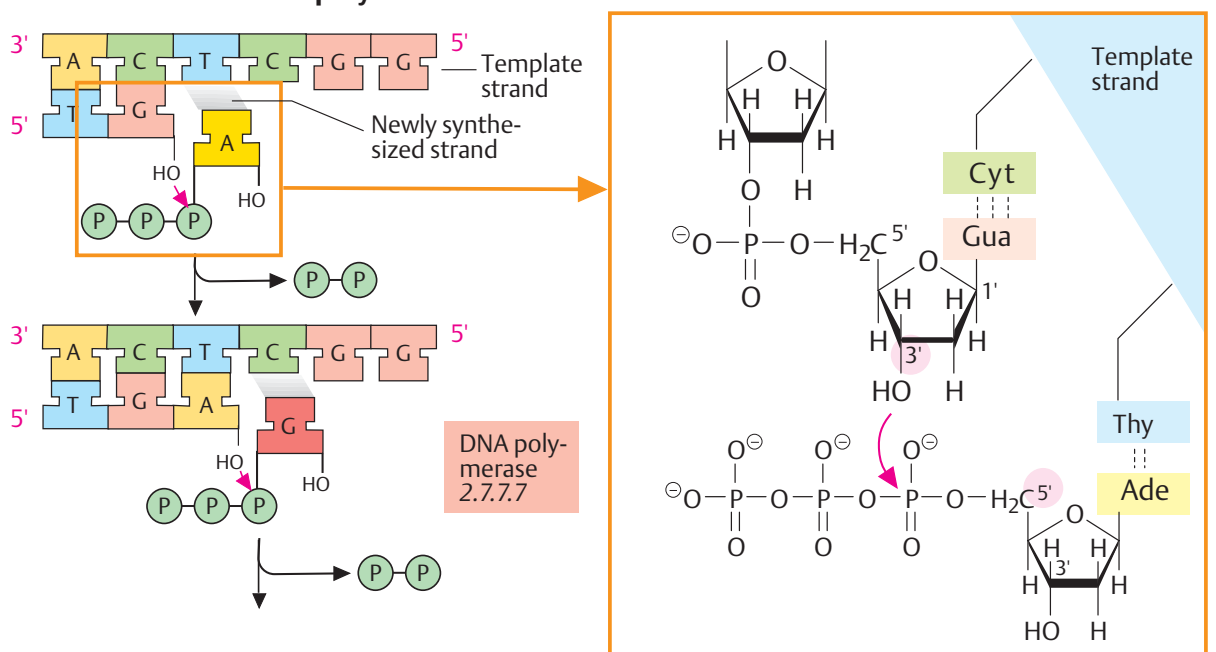
the processes taking place in this type of fork, and only the most important are shown here. The two strands of the initial DNA (**1**) are shown in blue and violet, while the newly formed strands are pink and orange.

Each fork (**2**) contains two molecules of **DNA polymerase III** and a number of helper proteins. The latter include **DNA topoisomerases** and **single-strand-binding proteins**. Topoisomerases are enzymes that unwind the superhelical DNA double strand (*gyrase*, *topoisomerase II*) and then separate it into the two individual strands (*helicase*, *topoisomerase I*). Since the template strand is always read from 3' to 5' (see above), only one of the strands (known as the **leading** strand; violet/pink) can undergo *continuous* replication. For the **lagging strand** (light blue), the reading direction is the *opposite* of the direction of movement of the fork. In this matrix, the new strand is first synthesized in individual pieces, which are known as **Okazaki fragments** after their discoverer (green/orange).

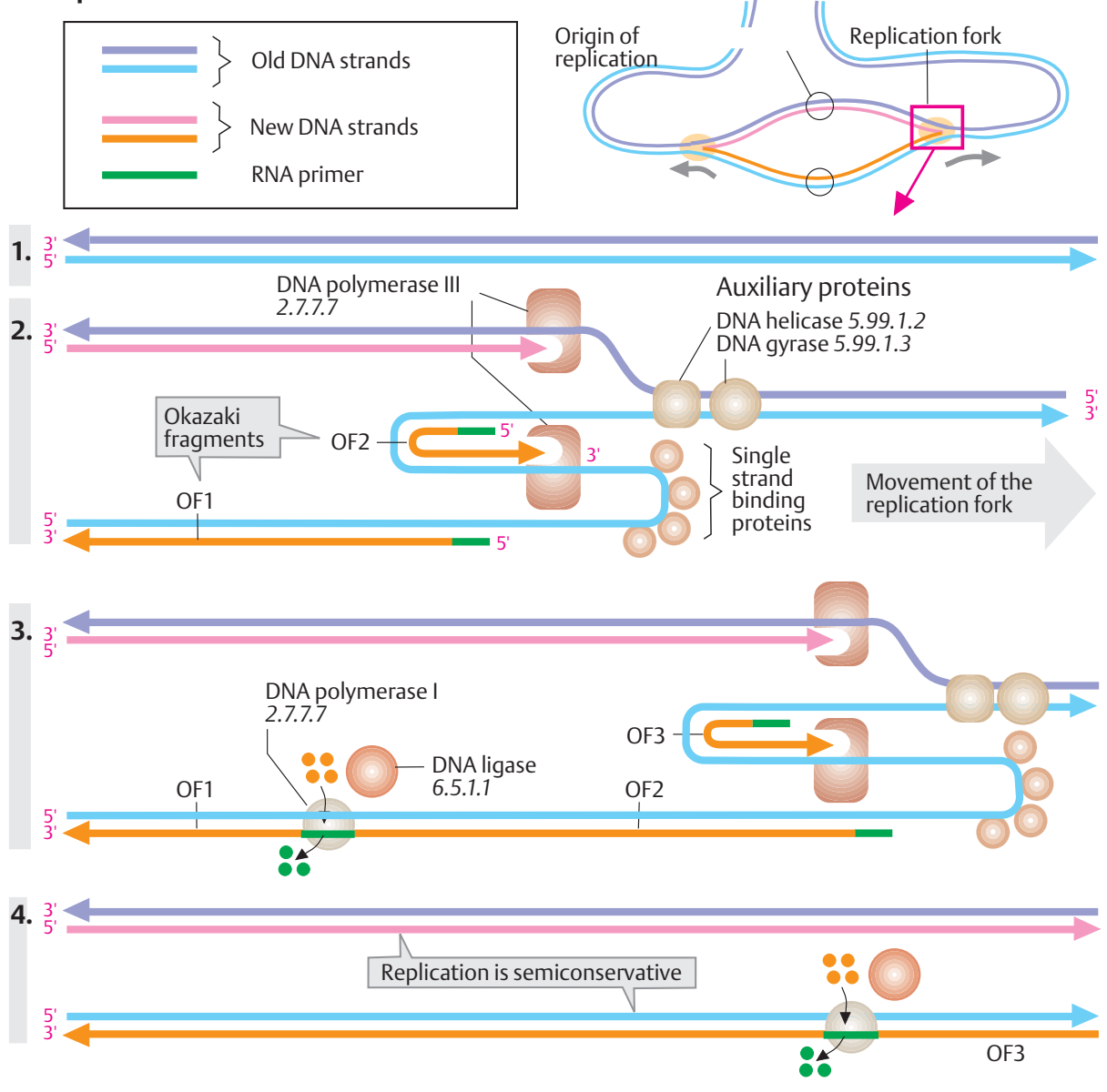
Each fragment starts with a short RNA primer (green), which is necessary for the functioning of the DNA polymerase and is synthesized by a special *RNA polymerase* ("primase," not shown). The primer is then extended by *DNA polymerase III* (orange). After 1000–2000 nucleotides have been included, synthesis of the fragment is interrupted and a new one is begun, starting with another RNA primer that has been synthesized in the interim. The individual Okazaki fragments are initially not bound to one another and still have RNA at the 5' end (**3**). At some distance from the fork, *DNA polymerase I* therefore starts to remove the RNA primer and replace it with DNA components. Finally, the gaps still remaining are closed by a *DNA ligase*. In DNA double helices formed in this way, only *one* of the strands has been newly synthesized—i.e., replication is *semiconservative*.

In bacteria, some 1000 nucleotides are replicated per second. In eukaryotes, replication takes place more slowly (about 50 nucleotides s^{-1}) and the genome is larger. Thousands of replication forks are therefore active simultaneously in eukaryotes.

A. Mechanism of DNA polymerases



B. Replication in *E. coli*



Transcription

For the genetic information stored in DNA to become effective, it has to be rewritten (transcribed) into RNA. DNA only serves as a template and is not altered in any way by the **transcription** process. Transcribable segments of DNA that code for a defined product are called **genes**. It is estimated that the mammalian genome contains 30 000–40 000 genes, which together account for less than 5% of the DNA.

A. Transcription and maturation of RNA: overview ❶

Transcription is catalyzed by *DNA-dependent RNA polymerases*. These act in a similar way to DNA polymerases (see p. 240), except that they incorporate *ribonucleotides* instead of deoxyribonucleotides into the newly synthesized strand; also, they do not require a primer. Eukaryotic cells contain at least three different types of RNA polymerase. *RNA polymerase I* synthesizes an RNA with a sedimentation coefficient (see p. 200) of 45 S, which serves as precursor for three ribosomal RNAs. The products of *RNA polymerase II* are hnRNAs, from which mRNAs later develop, as well as precursors for snRNAs. Finally, *RNA polymerase III* transcribes genes that code for tRNAs, 5S rRNA, and certain snRNAs. These precursors give rise to functional RNA molecules by a process called **RNA maturation** (see p. 246). Polymerases II and III are inhibited by α -amanitin, a toxin in the *Amanita phalloides* mushroom.

B. Organization of the PEP-CK gene ❷

The way in which a typical eukaryotic gene is organized is illustrated here using a gene that codes for a key enzyme in gluconeogenesis (see p. 154)—the *phosphoenolpyruvate carboxykinase* (PEP-CK).

In the rat, the PEP-CK gene is nearly 7 kbp (kilobase pairs) long. Only 1863 bp, distributed over 10 coding segments (**exons**, dark blue) carry the information for the protein's 621 amino acids. The remainder is allotted to the promoter (pink) and intervening sequences (**introns**, light blue). The gene's promoter region (approximately 1 kbp) serves for regulation (see p. 188). Transcription starts at the

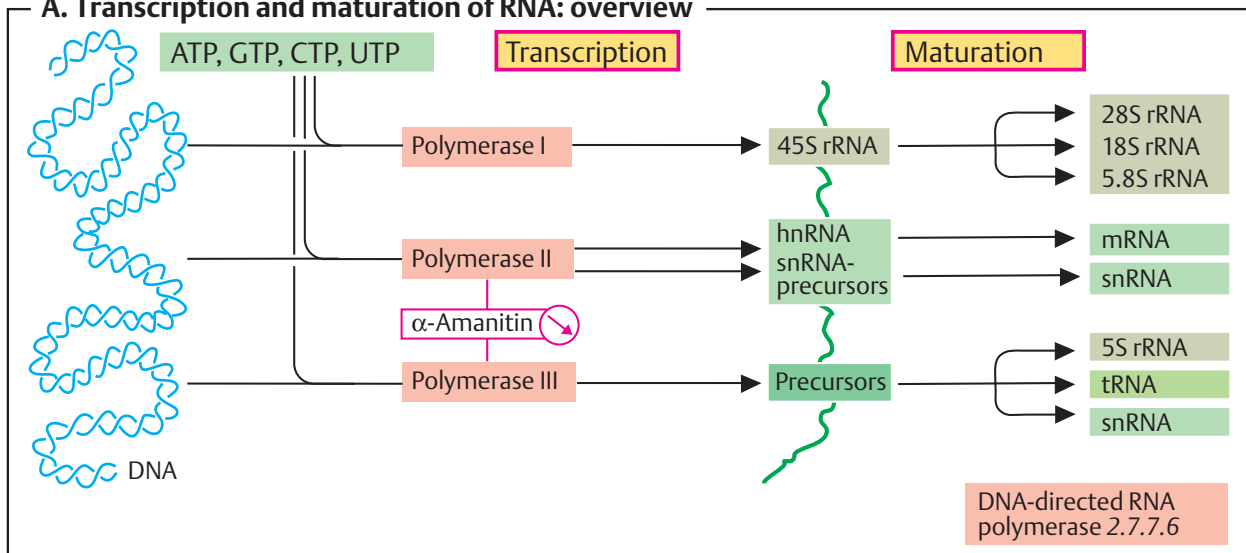
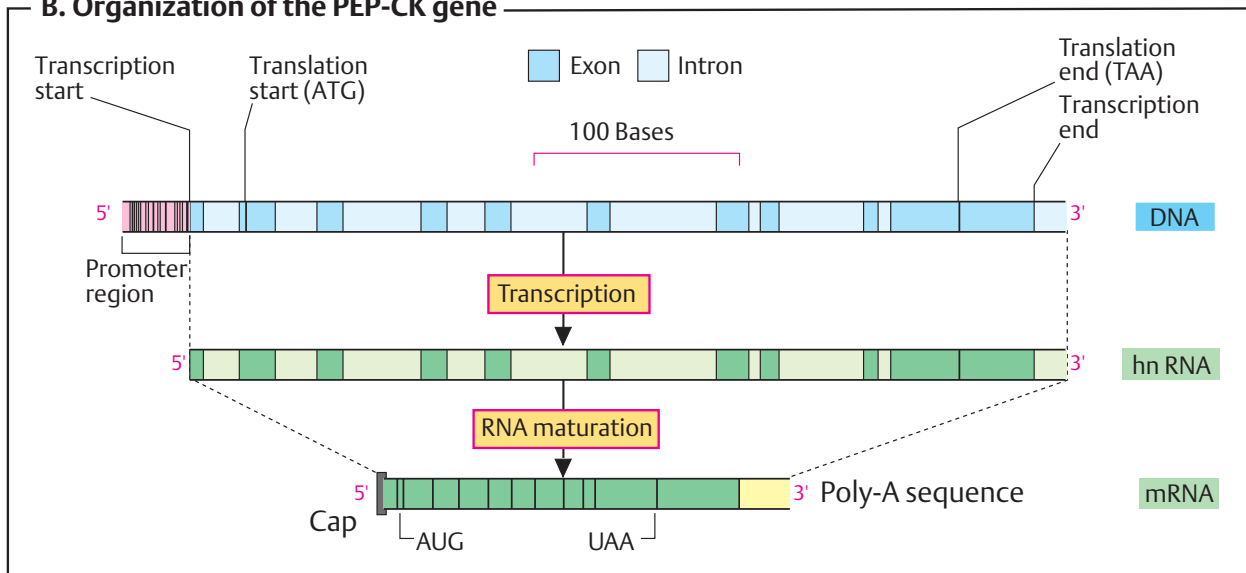
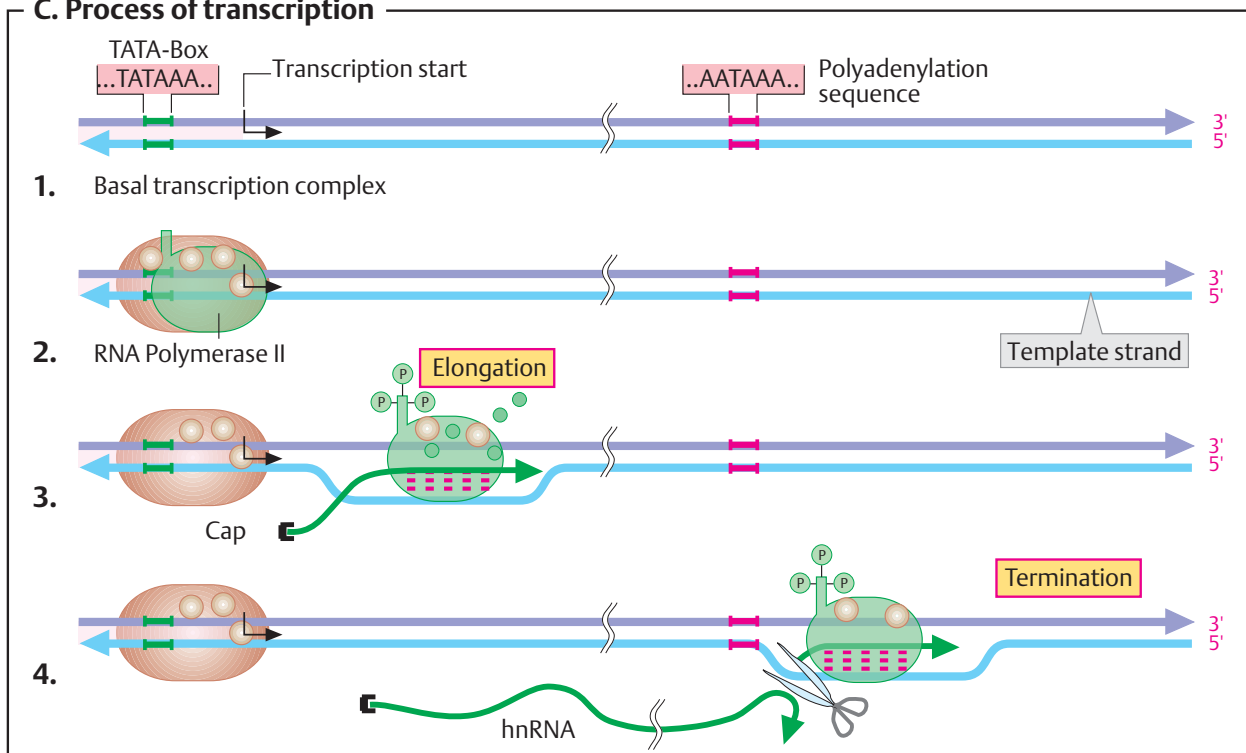
3' end of the promoter ("transcription start") and continues until the polyadenylation sequence (see below) is reached. The primary transcript (**hnRNA**) still has a length of about 6.2 kbp. During RNA maturation, the non-coding sequences corresponding to the introns are removed, and the two ends of the hnRNA are modified. The translatable mRNA still has half the length of the hnRNA and is modified at both ends (see p. 246).

In many eukaryotic genes, the proportion of introns is even higher. For example, the gene for *dihydrofolate reductase* (see p. 402) is over 30 kbp long. The information is distributed over six exons, which together have a length of only about 6 kbp.

C. Transcription process ❸

As mentioned above, *RNA polymerase II* (green) binds to the 3' end of the promoter region. A sequence that is important for this binding is known as the **TATA box**—a short A- and T-rich sequence that varies slightly from gene to gene. A typical base sequence ("*consensus sequence*") is ...TATAAA... Numerous proteins known as *basal transcription factors* are necessary for the interaction of the polymerase with this region. Additional factors can promote or inhibit the process (transcriptional control; see p. 244). Together with the polymerase, they form the **basal transcription complex**.

At the end of **initiation** (2), the polymerase is repeatedly phosphorylated, frees itself from the basal complex, and starts moving along the DNA in the 3' direction. The enzyme separates a short stretch of the DNA double helix into two single strands. The complementary nucleoside triphosphates are bound by base pairing in the *template strand* and are linked step by step to the hnRNA as it grows in the 5'→3' direction (3). Shortly after the beginning of **elongation**, the 5' end of the transcript is protected by a "cap" (see p. 246). Once the polyadenylation sequence has been reached (typical sequence: ...AATAA...), the transcript is released (4). Shortly after this, the RNA polymerase stops transcribing and dissociates from the DNA.

A. Transcription and maturation of RNA: overview**B. Organization of the PEP-CK gene****C. Process of transcription**

Transcriptional control

Although all cells contain the complete genome, they only use a fraction of the information in it. The genes known as “housekeeping genes,” which code for structural molecules and enzymes of intermediate metabolism, are the only ones that undergo constant transcription. The majority of genes are only active in certain cell types, in specific metabolic conditions, or during differentiation. Which genes are transcribed and which are not is regulated by **transcriptional control** (see also p. 118). This involves *control elements* (*cis*-active elements) in the gene’s promoter region and gene-specific regulatory proteins (transcription factors, *trans*-active factors), which bind to the control elements and thereby activate or inhibit transcription.

A. Initiation of transcription ○

In the higher organisms, DNA is blocked by histones (see p. 238) and is therefore not capable of being transcribed without special positive regulation. In eukaryotes, it is therefore histones that play the role of *repressors* (see p. 118). For transcription to be set in motion at all, the chromatin first has to be restructured.

In the resting state, the lysine residues in the N-terminal “tail” of the histones (see p. 238) are not acetylated. In this state, which can be produced by *histone deacetylases* [1], the nucleosomes are stable. It is only the interaction of activator and regulator proteins with their control elements that allows the binding of coactivator complexes that have *histone acetylase* activity [2]. They acetylate the histone tails and thereby loosen the nucleosome structure sufficiently for the basal transcription complex to form.

This consists of *DNA-dependent RNA polymerase II* and basal transcription factors (TFIIX, X = A – H). First, the basal factor TFIID binds to the promoter. TFIID, a large complex of numerous proteins, contains *TATA box-binding protein* (TBP) and so-called TAFs (TBP-associated factors). The polymerase is attached to this core with the help of TFIIB. Before transcription starts, additional TFs have to bind, including TFIIF, which has *helicase* activity and separates the two strands of DNA during elongation. In all, some 35 differ-

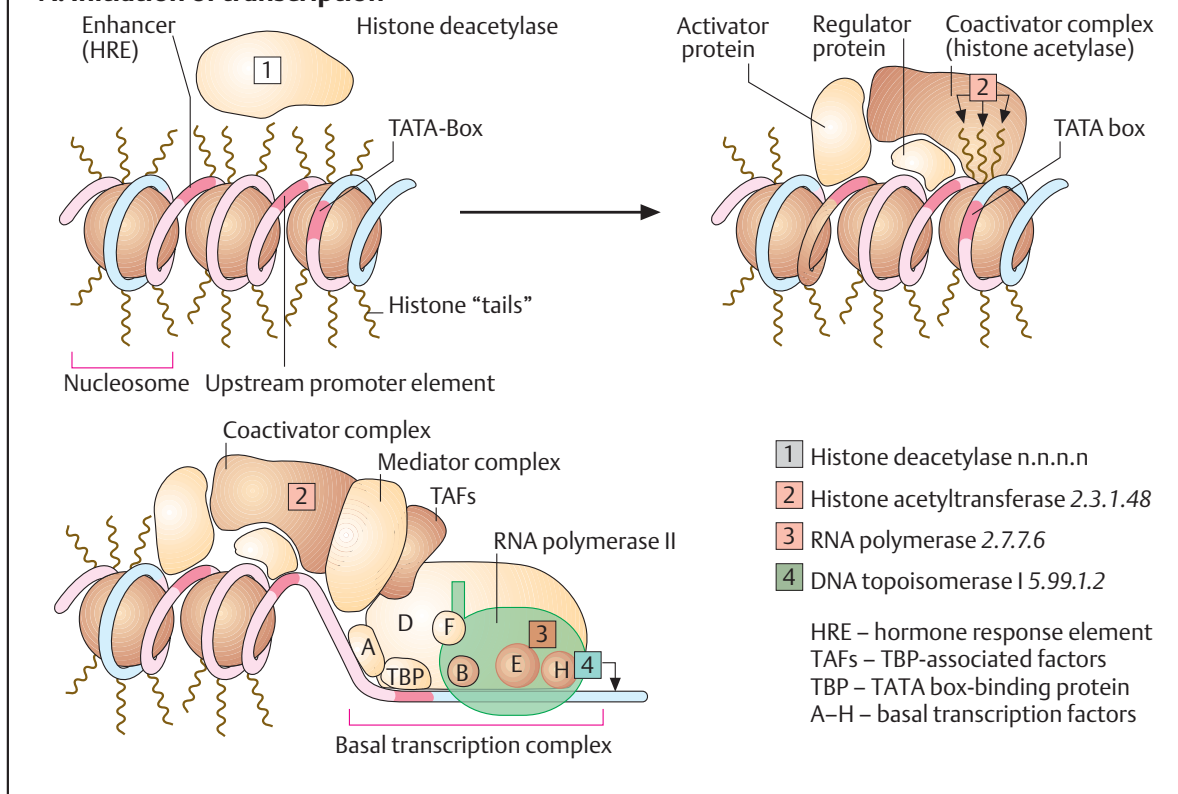
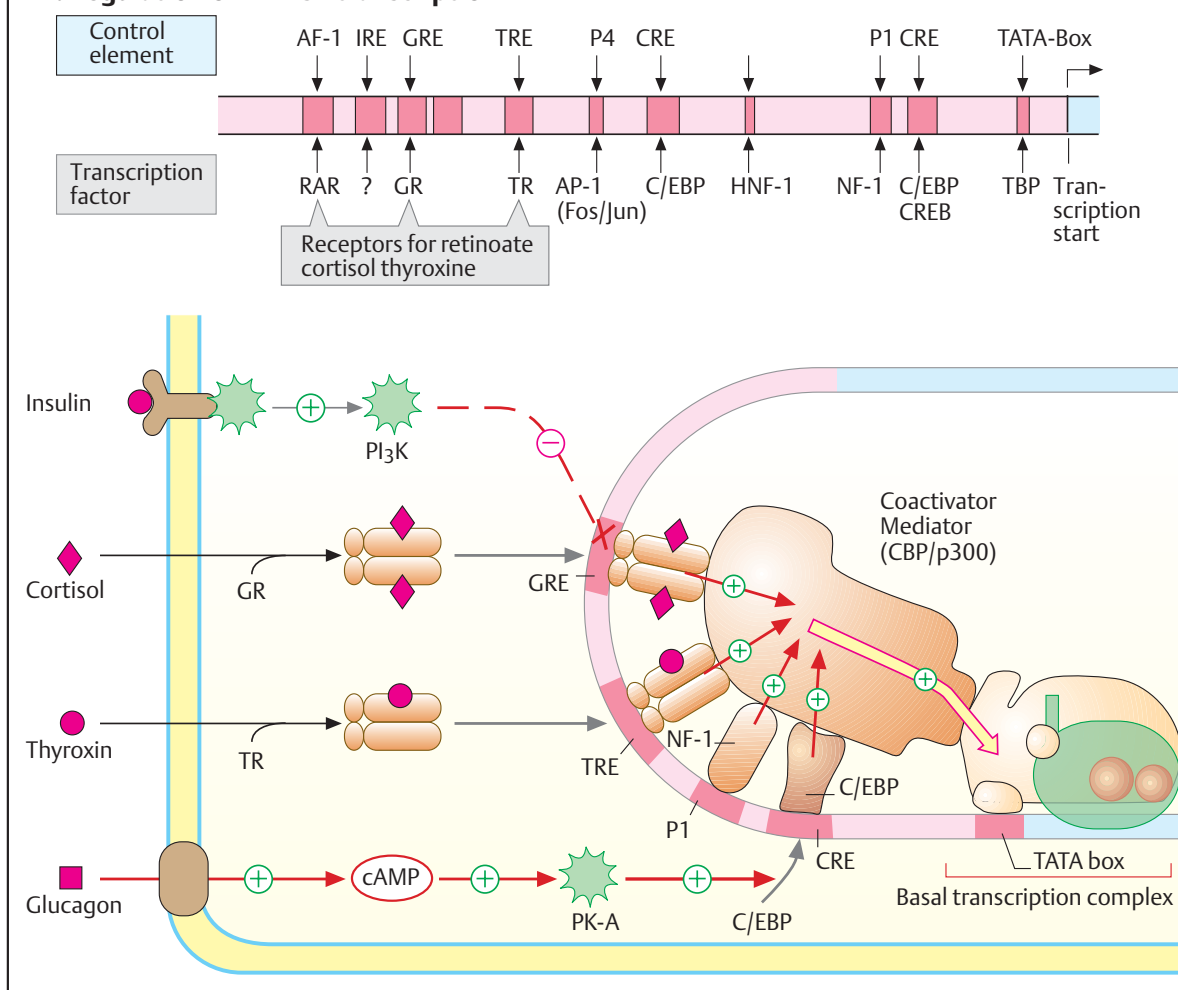
ent proteins are involved in the basal complex. This alone, however, is still not sufficient for transcription to start. In addition, positive signals have to be emitted by more distant *trans*-active factors, integrated by the coactivator/mediator complex, and passed on to the basal complex (see B).

The actual signal for starting elongation consists of the multiple phosphorylation of a domain in the C-terminal region of the polymerase. In phosphorylated form, it releases itself from the basal complex along with a few TFs and starts to synthesize hnRNA.

B. Regulation of PEP-CK transcription ○

Phosphoenolpyruvate carboxykinase (PEP-CK), a key enzyme in gluconeogenesis, is regulated by several hormones, all of which affect the transcription of the PEP-CK gene. Cortisol, glucagon, and thyroxine induce PEP-CK, while insulin inhibits its induction (see p. 158).

More than ten **control elements** (dark red), distributed over approximately 1 kbp, have so far been identified in the promoter of the PEP-CK gene (top). These include response elements for the glucocorticoid receptor (GRE; see p. 378), for the thyroxine receptor (TRE), and for the steroid-like retinoic acid (AF-1). Additional control elements (CRE, cAMP-responsive element) bind the transcription factor C/EBP, which is activated by cAMP-dependent protein kinase A through phosphorylation. This is the way in which glucagon, which raises the cAMP level (see p. 158), works. Control element P1 binds the hormone-independent factor NF-1 (nuclear factor-1). All proteins that bind to the control elements mentioned above are in contact with a **coactivator/mediator complex** (CBP/p300), which integrates their input like a computer and transmits the result in the form of stronger or weaker signals to the basal transcription complex. Inhibition of PEP-CK transcription by insulin is mediated by an insulin-responsive element (IRE) in the vicinity of the GRE. Binding of an as yet unknown factor takes place here, inhibiting the binding of the glucocorticoid receptor to the GRE.

A. Initiation of transcription**B. Regulation of PEP-CK transcription**

RNA maturation

Before the hnRNA produced by RNA polymerase II (see p. 242) can leave the nucleus in order to serve as a template for protein synthesis in the cytoplasm, it has to undergo several modifications first. Even during transcription, the two ends of the transcript have additional nucleotides added (**A**). The sections that correspond to the intervening gene sequences in the DNA (introns) are then cut out (splicing; see **B**). Other transcripts—e.g., the 45 S precursor of rRNA formed by polymerase I (see p. 242)—are broken down into smaller fragments by nucleases before export into the cytoplasm.

A. 5' and 3' modification of mRNA ①

Shortly after transcription begins in eukaryotes, the end of the growing RNA is blocked in several reaction steps by a structure known as a “**cap**.” In hnRNAs, this consists of a GTP residue that is methylated at N-7 of the guanine ring. The β -phosphate residue of the cap is linked to the free 5'-OH group of the terminal ribose via an ester bond. After the “polyadenylation signal” has been reached (typical sequence: ...AAUAAA...; see p. 242), a **polyadenylate “tail”** consisting of up to 200 AMP nucleotides is also added at the free 3' end of the transcript. This reaction is catalyzed by a special *polyadenylate polymerase*. It is only at this point that the mRNA leaves the nucleus as a complex with RNA-binding proteins.

Both the cap and the poly-A tail play a vital part in initiating eukaryotic translation (see p. 250). They help position the ribosome correctly on the mRNA near to the starting codon. The protection which the additional nucleotides provide against premature enzymatic degradation appears to be of lesser importance.

B. Splicing of hnRNA ①

Immediately after transcription, the hnRNA introns are removed and the exons are linked to form a continuous coding sequence. This process, known as **splicing**, is supported by complicated RNA-protein complexes in the nucleus, the so-called **spliceosomes**. The components of these macromolecular machines

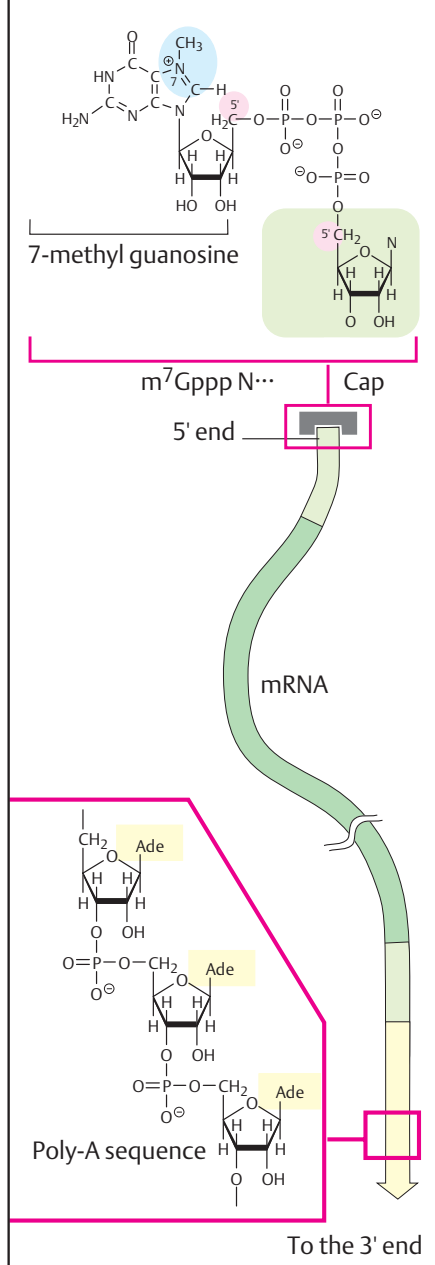
are called **snRNPs** (*small nuclear ribonucleoprotein particles*, pronounced “snurps”). SnRNPs occur in five different forms (U1, U2, U4, U5, and U6). They consist of numerous *proteins* and one molecule of *snRNA* each (see p. 82).

To ensure that the RNA message is not destroyed, splicing has to take place in a very precise fashion. The start and end of the hnRNA introns are recognized by a characteristic sequence (...AGGT... at the 5' end or ...[C,U]AGG... at the 3' end). Another important structure is the so-called *branching point* inside the intron. Its sequence is less conserved than the terminal splicing sites, but it always contains one adenosine residue (A). During splicing, the 2'-OH group of this residue—supported by the spliceosome (see **C**)—attacks the phosphoric acid diester bond at the 5' end of the intron and cleaves it (**b**). Simultaneously, an unusual 2'→5' bond is formed inside the intron, which thereby takes on a *lasso shape* (**c**; see formula). In the second step of the splicing process, the free 3'-OH group at the end of the 5' terminal exon attacks the A-G bond at the 3' end of the intron. As a result, the two exons are linked and the intron is released, still in a lasso shape (**d**).

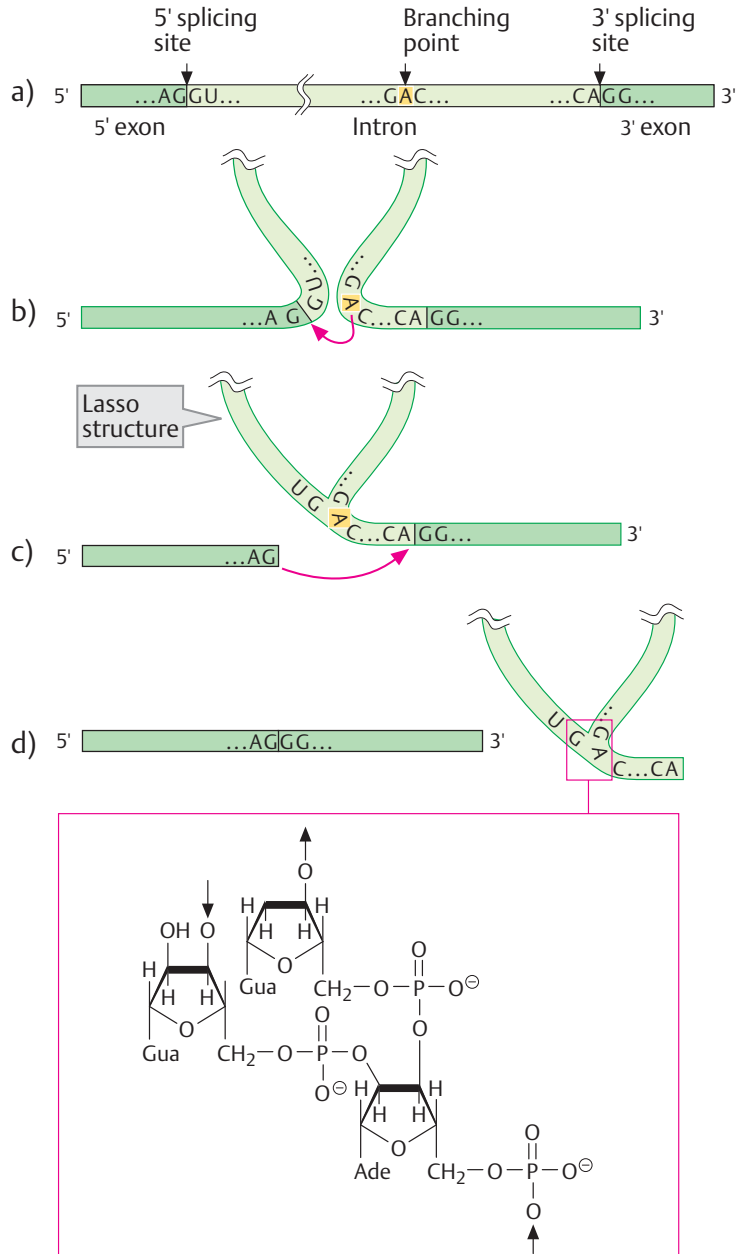
C. Spliceosome ①

As described above, it is residues of the hnRNA that carry out bond cleavage and bond formation during the splicing process. It is therefore not a protein enzyme that acts as a catalyst here, but rather an RNA. Catalytic RNAs of this type are called *ribozymes* (see also p. 88). The task of the spliceosomes is to fix and orientate the reacting groups by establishing base pairings between snRNAs and segments of the hnRNA. The probable situation before the adenosine attack at the branching point on the 5' splicing site (see **B**, Fig. **b**) is shown schematically on the right side of the illustration. In this phase, the U1 snRNA fixes the 5' splicing site, U2 fixes the branching site, and U5 fixes the ends of the two exons.

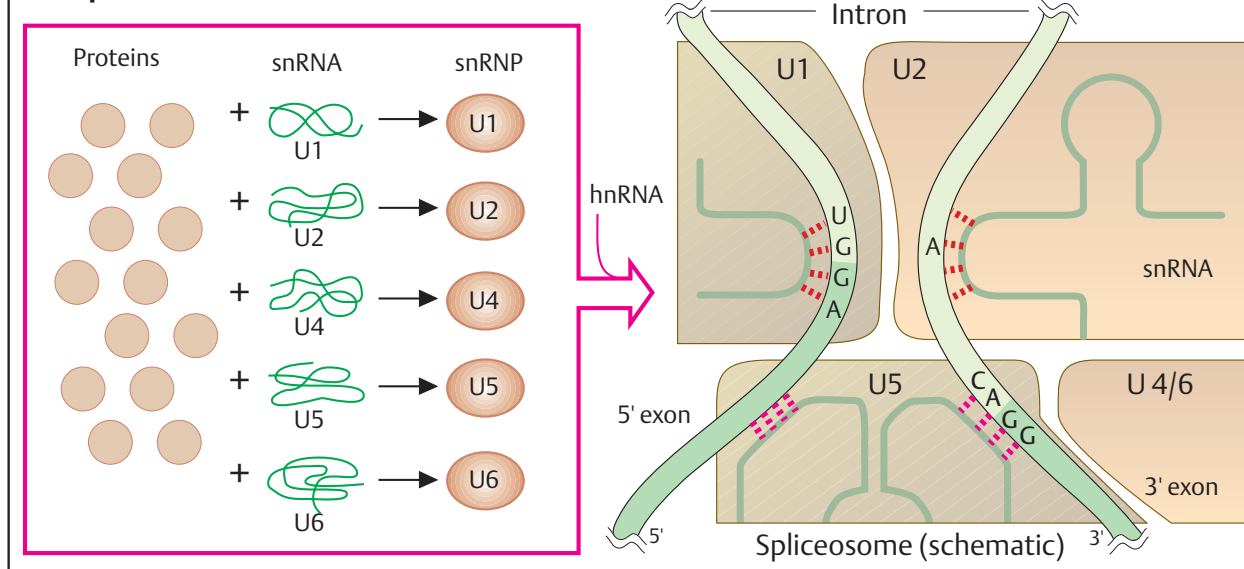
A. 5' and 3' modification of mRNA



B. Splicing of hnRNA: mechanism



C. Spliceosome



Amino acid activation

A. The genetic code ●

Most of the genetic information stored in the genome codes for the amino acid sequences of proteins. For these proteins to be expressed, a text in “nucleic acid language” therefore has to be translated into “protein language.” This is the origin of the use of the term **translation** to describe protein biosynthesis. The dictionary used for the translation is the **genetic code**.

As there are 20 proteinogenic amino acids (see p. 60), the nucleic acid language has to contain at least as many words (**codons**). However, there are only four letters in the nucleic acid alphabet (A, G, C, and U or T). To obtain 20 different words from these, each word has to be at least three letters long (with two letters, there would only be $4^2 = 16$ possibilities). And in fact the codons do consist of three sequential bases (**triplets**).

Figure 1 shows the standard code in “DNA language” (i. e., as a sequence of triplets in the *sense strand* of DNA, read in the 5'→3' direction; see p. 84), represented as a circular diagram. The scheme is read from the inside to the outside. For example, the triplet CAT codes for the amino acid histidine. With the exception of the exchange of U for T, the DNA codons are identical to those of mRNA.

As the genetic code provides $4^3 = 64$ codons for the 20 amino acids, there are several synonymous codons for most amino acids—the code is **degenerate**. Three triplets do not code for amino acids, but instead signal the end of translation (**stop codons**). Another special codon, the **start codon**, marks the start of translation. The code shown here is almost universally applicable; only the mitochondria (see p. 210) and a few microorganisms deviate from it slightly.

As an example of the way in which the code is read, Fig. 2 shows small sections from the normal and a mutated form of the β -globin gene (see p. 280), as well as the corresponding mRNA and protein sequences. The **point mutation** shown, which is relatively frequent, leads to replacement of a glutamate residue in position 6 of the β -chain by valine (GAG → GTG). As a consequence, the mutated hemoglobin tends to aggregate in the deoxygenated form. This leads to sickle-shaped dis-

tortions of the erythrocytes and disturbances of O₂ transport (sickle-cell anemia).

B. Amino acid activation ●

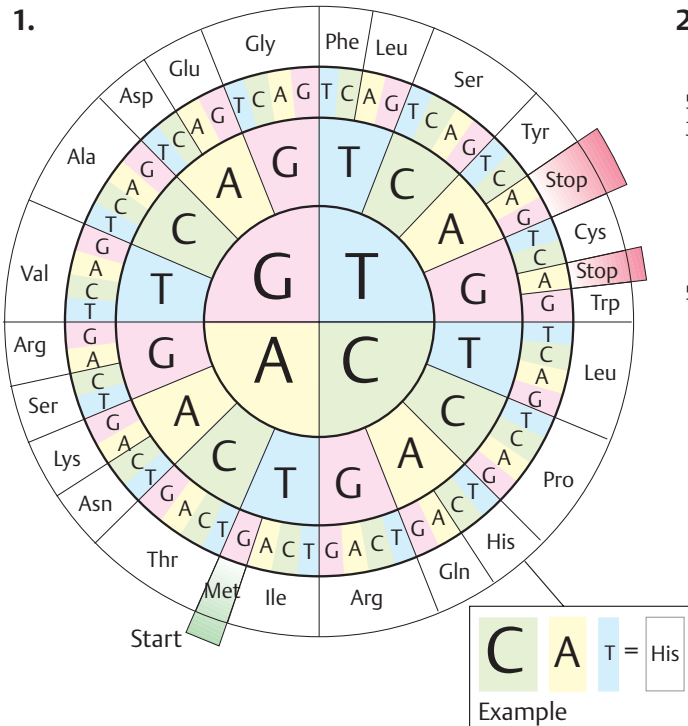
Some 20 different *amino acid tRNA ligases* in the cytoplasm each bind one type of tRNA (see p. 82) with the corresponding amino acid. This reaction, known as **amino acid activation**, is endergonic and is therefore coupled to ATP cleavage in two steps.

First, the amino acid is bound by the enzyme and reacts there with ATP to form diphosphate and an “energy-rich” mixed acid anhydride (**aminoacyl adenylate**). In the second step, the 3'-OH group (in other ligases it is the 2'-OH group) of the terminal ribose residue of the tRNA takes over the amino acid residue from the aminoacyl adenylate. In **aminoacyl tRNAs**, the carboxyl group of the amino acid is therefore esterified with the ribose residue of the terminal adenosine of the sequence ...CCA-3'.

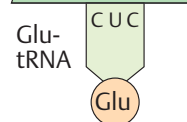
The accuracy of translation primarily depends on the specificity of the amino acid tRNA ligases, as incorrectly incorporated amino acid residues are not recognized by the ribosome later. A “*proofreading mechanism*” in the active center of the ligase therefore ensures that incorrectly incorporated amino acid residues are immediately removed again. On average, an error only occurs once every 1300 amino acid residues. This is a surprisingly low rate considering how similar some amino acids are—e. g., leucine and isoleucine.

C. Asp-tRNA ligase (dimer) ○

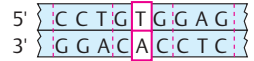
The illustration shows the ligase responsible for the activation of aspartate. Each subunit of the dimeric enzyme (protein parts shown in orange) binds one molecule of tRNA^{Asp} (blue). The active centers can be located by the bound ATP (green). They are associated with the 3' end of the tRNA. Another domain in the protein (upper left) is responsible for “recognition” of the tRNA anticodon.

A. The genetic code**1.****2. Normal globin gene (β -chain)**

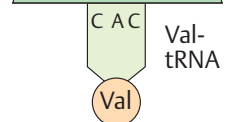
Transcription



Translation

**Sickle-cell globin gene (β -chain)**

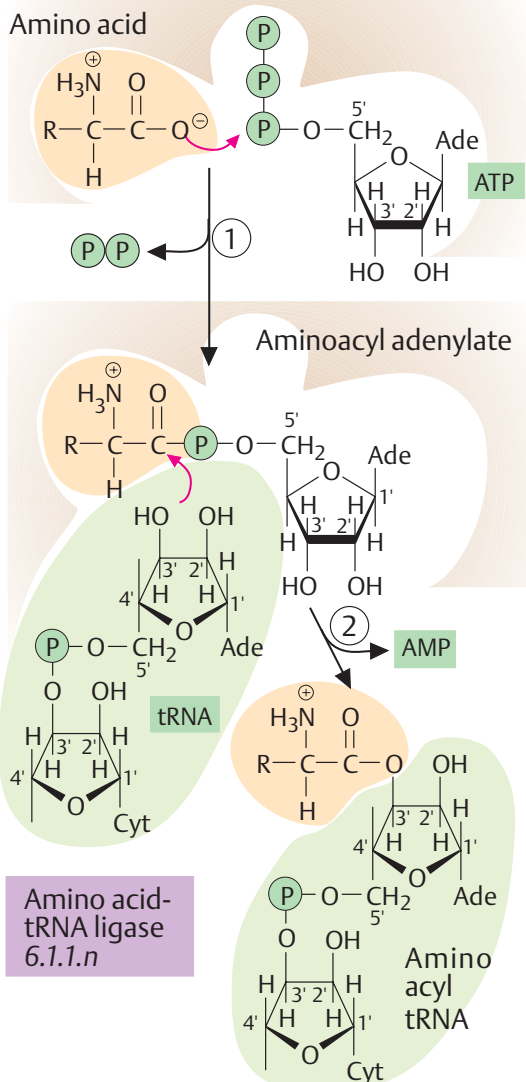
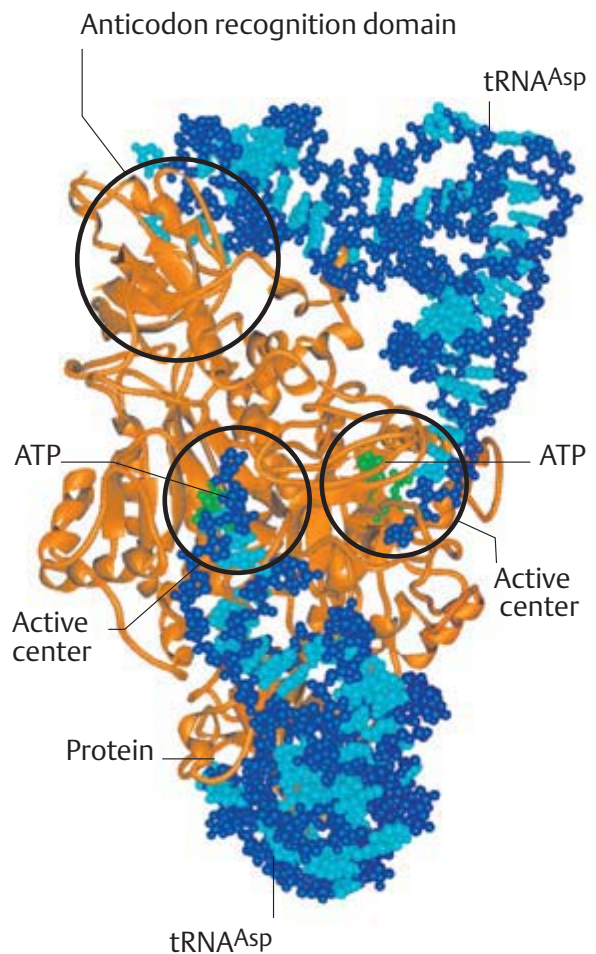
Transcription



Translation



Mutation

B. Amino acid activation**C. Asp-tRNA-Ligase (Dimer)**

Translation I: initiation

Like amino acid activation (see p. 248), protein biosynthesis (**translation**) takes place in the cytoplasm. It is catalyzed by complex nucleoprotein particles, the **ribosomes**, and mainly requires GTP to cover its energy requirements.

A. Structure of the eukaryotic ribosome ●

Ribosomes consist of two subunits of different size, made up of **ribosomal RNA (rRNA)** and nearly 80 **proteins** (the number of proteins applies to rat liver ribosomes). It is customary to give the sedimentation coefficients (see p. 200) of ribosomes and their components instead of their masses. For example, the eukaryotic ribosome has a sedimentation coefficient of 80 Svedberg units (80 S), while the sedimentation coefficients of its subunits are 40 S and 60 S (S values are not additive).

The smaller 40 S subunit consists of one molecule of 18 S rRNA and 33 protein molecules. The larger 60 S subunit contains three types of rRNA with sedimentation coefficients of 5 S, 5.8 S, and 28 S and 47 proteins. In the presence of mRNA, the subunits assemble to form the complete ribosome, with a mass about 650 times larger than that of a hemoglobin molecule.

The arrangement of the individual components of a ribosome has now been determined for prokaryotic ribosomes. It is known that filamentous mRNA passes through a cleft between the two subunits near the characteristic “horn” on the small subunit. tRNAs also bind near this site. The illustration shows the size of a tRNA molecule for comparison.

Prokaryotic ribosomes have a similar structure, but are somewhat smaller than those of eukaryotes (sedimentation coefficient 70 S for the complete ribosome, 30 S and 50 S for the subunits). Mitochondrial and chloroplast ribosomes are comparable to prokaryotic ones.

B. Polysomes ●

In cells that are carrying out intensive protein synthesis, ribosomes are often found in a linear arrangement like a string of pearls; these are known as **polysomes**. This arrangement arises because several ribosomes are translat-

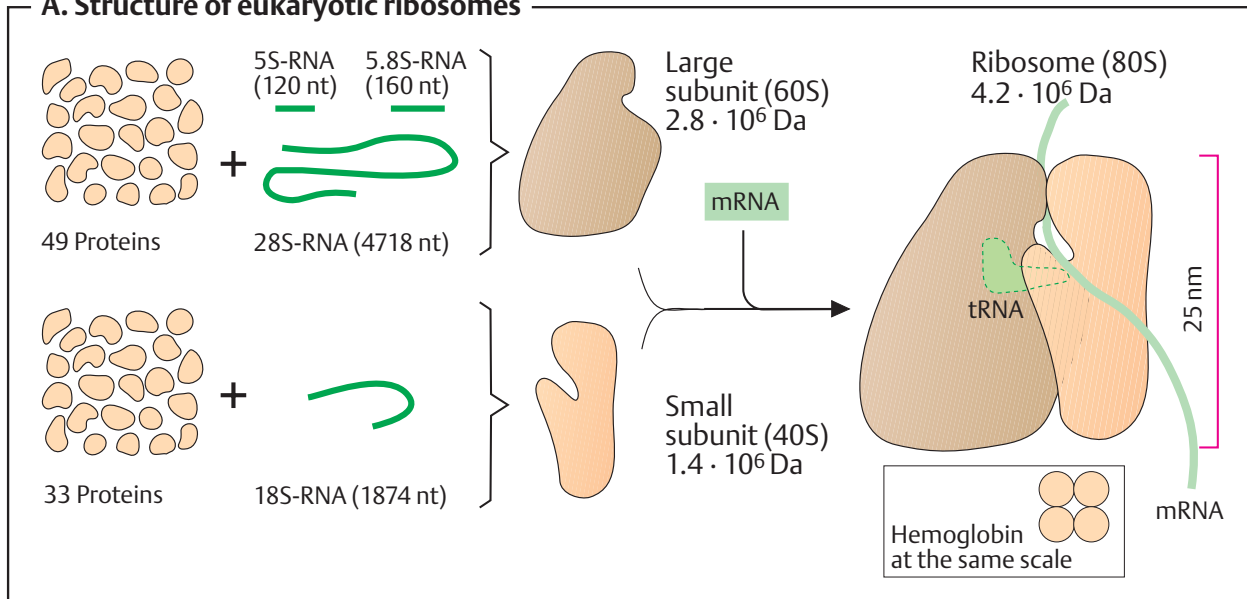
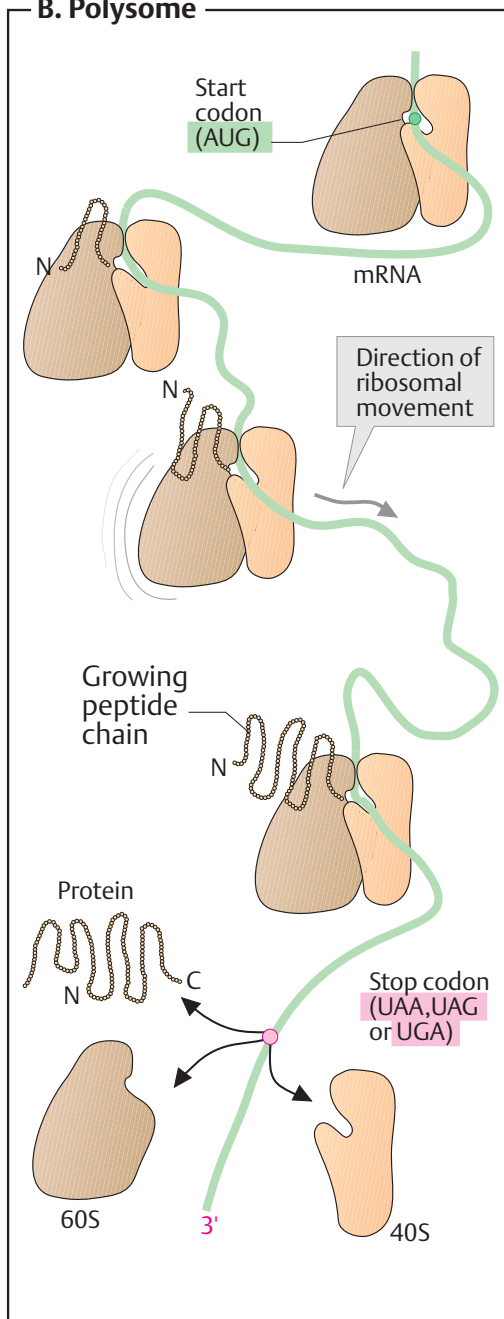
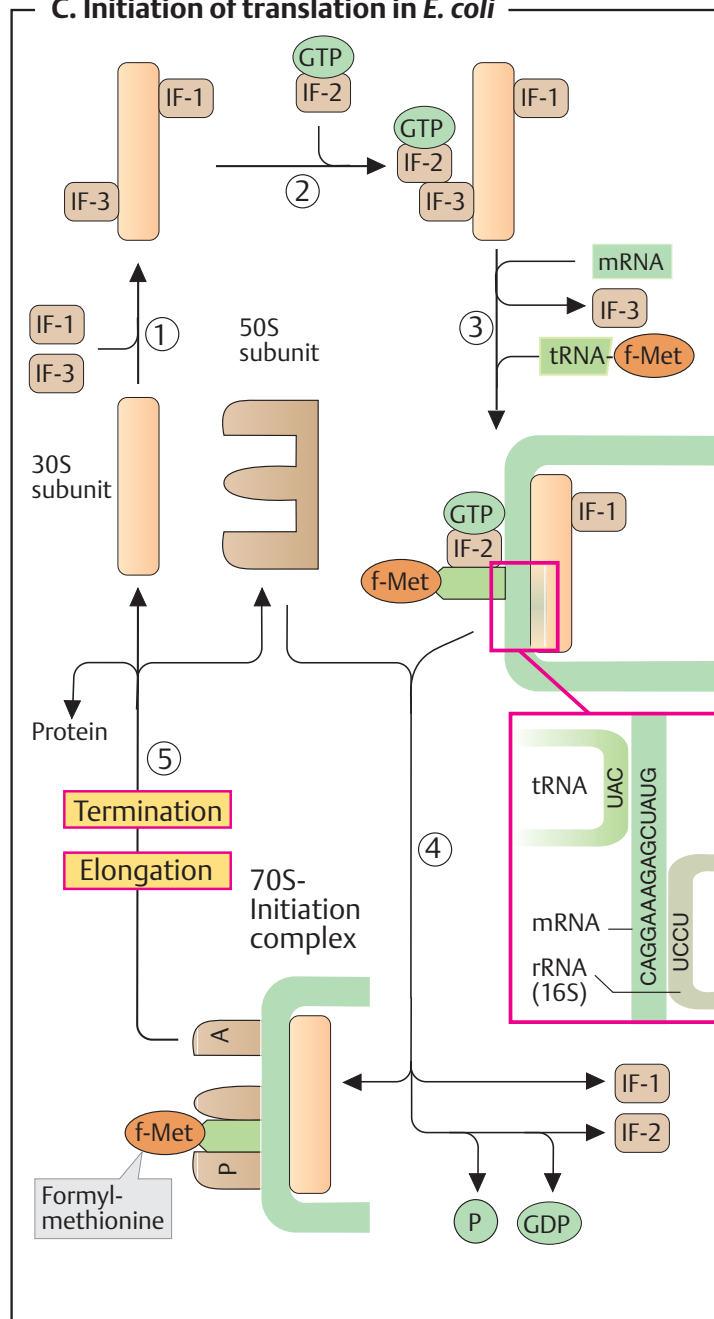
ing a single mRNA molecule simultaneously. The ribosome first binds near the **start codon** (AUG; see p. 248) at the 5' end of the mRNA (top). During translation, the ribosome moves in the direction of the 3' end until it reaches a **stop codon** (UAA, UAG, or UGA). At this point, the newly synthesized chain is released, and the ribosome dissociates again into its two subunits.

C. Initiation of translation in *E. coli* ●

Protein synthesis in prokaryotes is in principle the same as in eukaryotes. However, as the process is simpler and has been better studied in prokaryotes, the details involved in translation are discussed here and on p. 252 using the example of the bacterium *Escherichia coli*.

The first phase of translation, **initiation**, involves several steps. First, two proteins, **initiation factors** IF-1 and IF-3, bind to the 30 S subunit (1). Another factor, IF-2, binds as a complex with GTP (2). This allows the subunit to associate with the mRNA and makes it possible for a special tRNA to bind to the start codon (3). In prokaryotes, this starter tRNA carries the substituted amino acid *N*-**formylmethionine** (fMet). In eukaryotes, it carries an unsubstituted *methionine*. Finally, the 50 S subunit binds to the above complex (4). During steps 3 and 4, the initiation factors are released again, and the GTP bound to IF-2 is hydrolyzed to GDP and P_i.

In the **70 S initiation complex**, formylmethionine tRNA is initially located at a binding site known as the **peptidyl site (P)**. A second binding site, the **acceptor site (A)**, is not yet occupied during this phase of translation. Sometimes, a third tRNA binding site is defined as an *exit site (E)*, from which uncharged tRNAs leave the ribosome again (see p. 252; not shown).

A. Structure of eukaryotic ribosomes**B. Polysome****C. Initiation of translation in *E. coli***

Translation II: elongation and termination

After translation has been initiated (see p. 250), the peptide chain is extended by the addition of further amino acid residues (**elongation**) until the ribosome reaches a stop codon on the mRNA and the process is interrupted (**termination**).

A. Elongation and termination of protein biosynthesis in *E. coli* ①

Elongation can be divided into four phases:

[1] **Binding of aminoacyl tRNA.** First, the peptidyl site (P) of the ribosome is occupied by a tRNA that carries at its 3' end the complete peptide chain formed up to this point (top left). A second tRNA, loaded with the next amino acid (Val-tRNA^{Val} in the example shown), then binds via its complementary anticodon (see p. 82) to the mRNA codon exposed at the acceptor site (in this case GUG). The tRNA binds as a complex with a GTP-containing protein, the *elongation factor Tu* (EF-Tu) (**1a**). It is only after the bound GTP has been hydrolyzed to GDP and phosphate that EF-Tu dissociates again (**1b**). As the binding of the tRNA to the mRNA is still loose before this, GTP hydrolysis acts as a delaying factor, making it possible to check whether the correct tRNA has been bound. A further protein, the *elongation factor Ts* (EF-Ts), later catalyzes the exchange of GDP for GTP and in this way regenerates the EF-Tu-GTP complex. EF-Tu is related to the G proteins involved in signal transduction (see p. 384).

[2] **Synthesis of the peptide bond** takes place in the next step. Ribosomal *peptidyltransferase* catalyzes (without consumption of ATP or GTP) the transfer of the peptide chain from the tRNA at the P site to the NH₂ group of the amino acid residue of the tRNA at the A site. The ribosome's peptidyltransferase activity is not located in one of the ribosomal proteins, but in the 23 S rRNA. Catalytically active RNAs of this type are known as *ribozymes* (cf. p. 246). It is thought that the few surviving ribozymes are remnants of the "RNA world"—an early phase of evolution in which proteins were not as important as they are today.

[3] After the transfer of the growing peptide to the A site, the free tRNA at the P site dissociates and another GTP-containing elongation factor (EF-G-GTP) binds to the ribosome. Hydrolysis of the GTP in this factor provides the energy for **translocation** of the ribosome. During this process, the ribosome moves three bases along the mRNA in the direction of the 3' end. The tRNA carrying the peptide chain is stationary relative to the mRNA and reaches the ribosome's P site during translocation, while the next mRNA codon (in this case GUG) appears at the A site.

[4] The uncharged Val-tRNA then dissociates from the E site. The ribosome is now ready for the next elongation cycle.

When one of the three stop codons (UAA, UAG, or UGA) appears at the A site, **termination** starts.

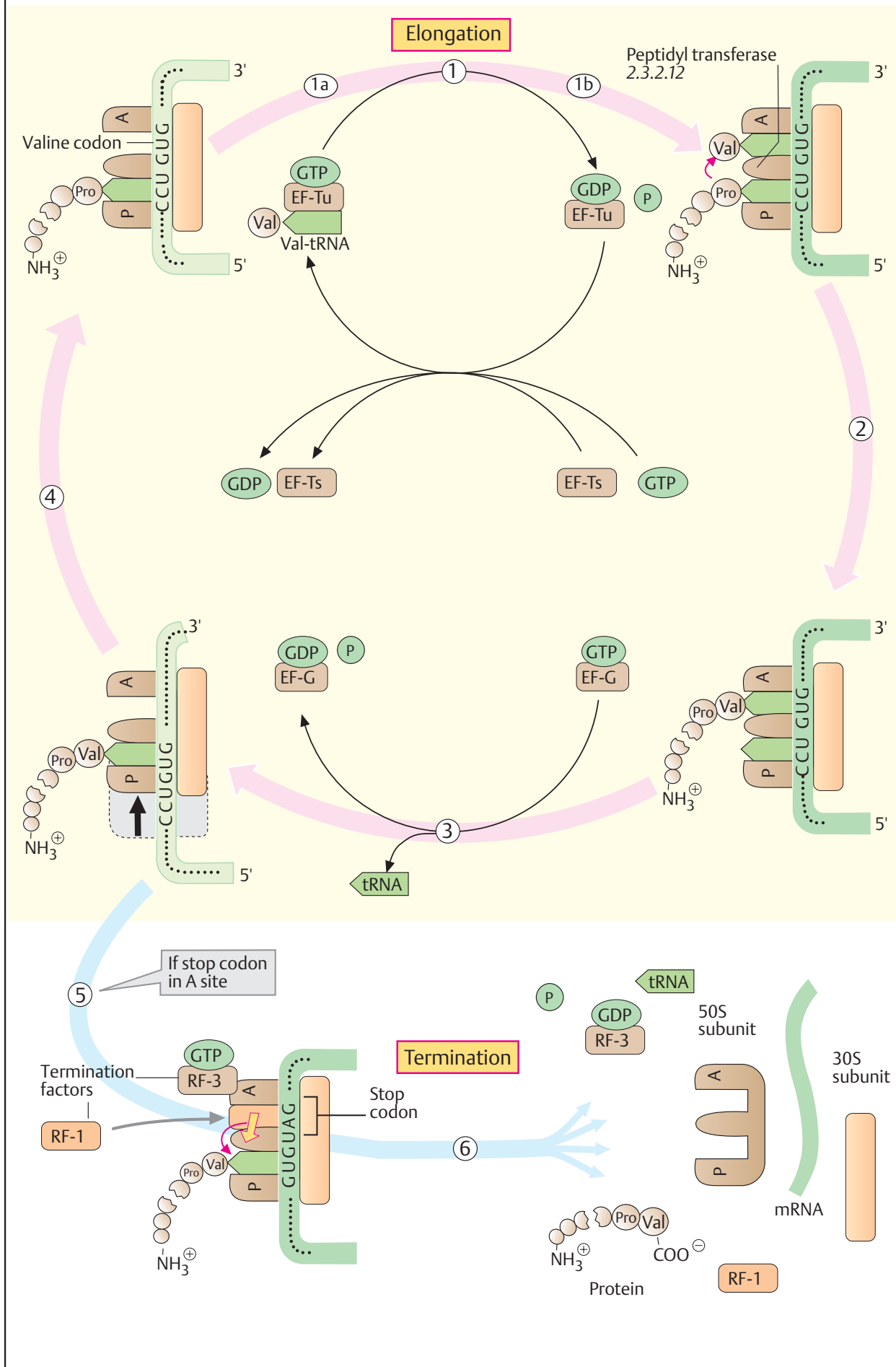
[5] There are no complementary tRNAs for the stop codons. Instead, two *releasing factors* bind to the ribosome. One of these factors (RF-1) catalyzes hydrolytic cleavage of the ester bond between the tRNA and the C-terminus of the peptide chain, thereby releasing the protein.

[6] Hydrolysis of GTP by factor RF-3 supplies the energy for the dissociation of the whole complex into its component parts.

Energy requirements in protein synthesis are high. Four energy-rich phosphoric acid anhydride bonds are hydrolyzed for each amino acid residue. Amino acid activation uses up two energy-rich bonds per amino acid (ATP → AMP + PP_i; see p. 248), and two GTPs are consumed per elongation cycle. In addition, initiation and termination each require one GTP per chain.

Further information

In eukaryotic cells, the number of initiation factors is larger and initiation is therefore more complex than in prokaryotes. The cap at the 5' end of mRNA and the polyA tail (see p. 246) play important parts in initiation. However, the elongation and termination processes are similar in all organisms. The individual steps of bacterial translation can be inhibited by antibiotics (see p. 254).

A. Elongation and termination of protein biosynthesis in *E. coli*

Antibiotics

A. Antibiotics: overview ●

Antibiotics are substances which, even at low concentrations, inhibit the growth and reproduction of bacteria and fungi. The treatment of infectious diseases would be inconceivable today without antibiotics. Substances that only restrict the reproduction of bacteria are described as having *bacteriostatic* effects (or *fungistatic* for fungi). If the target cells are killed, then the term *bactericidal* (or *fungicidal*) is used. Almost all antibiotics are produced by microorganisms—mainly bacteria of the genus *Streptomyces* and certain fungi. However, there are also synthetic antibacterial substances, such as sulfonamides and gyrase inhibitors.

A constantly increasing problem in antibiotic treatment is the development of resistant pathogens that no longer respond to the drugs available. The illustration shows a few of the therapeutically important antibiotics and their sites of action in the bacterial metabolism.

Substances known as **intercalators**, such as *rifamycin* and *actinomycin D* (bottom) are deposited in the DNA double helix and thereby interfere with replication and transcription (B). As DNA is the same in all cells, intercalating antibiotics are also toxic for eukaryotes, however. They are therefore only used as cytostatic agents (see p. 402). Synthetic inhibitors of DNA topoisomerase II (see p. 240), known as **gyrase inhibitors** (center), restrict replication and thus bacterial reproduction.

A large group of antibiotics attack bacterial ribosomes. These **inhibitors of translation** (left) include the *tetracyclines*—broad-spectrum antibiotics that are effective against a large number of different pathogens. The *aminoglycosides*, of which *streptomycin* is the best-known, affect all phases of translation. *Erythromycin* impairs the normal functioning of the large ribosomal subunit. *Chloramphenicol*, one of the few natural nitro compounds, inhibits ribosomal peptidyltransferase. Finally, *puromycin* mimics an aminoacyl tRNA and therefore leads to premature interruption of elongation.

The **β -lactam antibiotics** (bottom right) are also frequently used. The members of this group, the *penicillins* and *cephalosporins*, are

synthesized by fungi and have a reactive β -lactam ring. They are mainly used against Gram-positive pathogens, in which they inhibit cell wall synthesis (C).

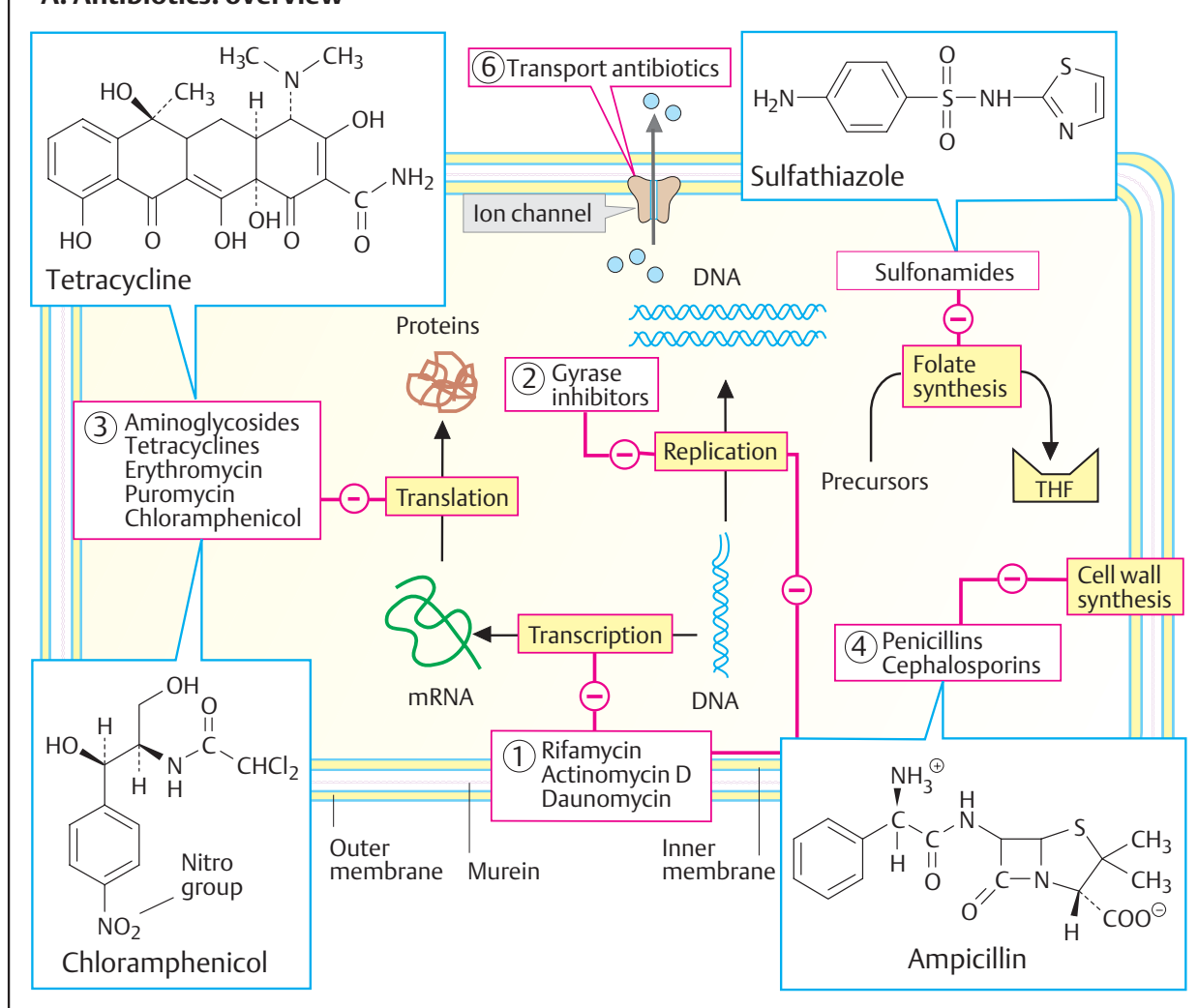
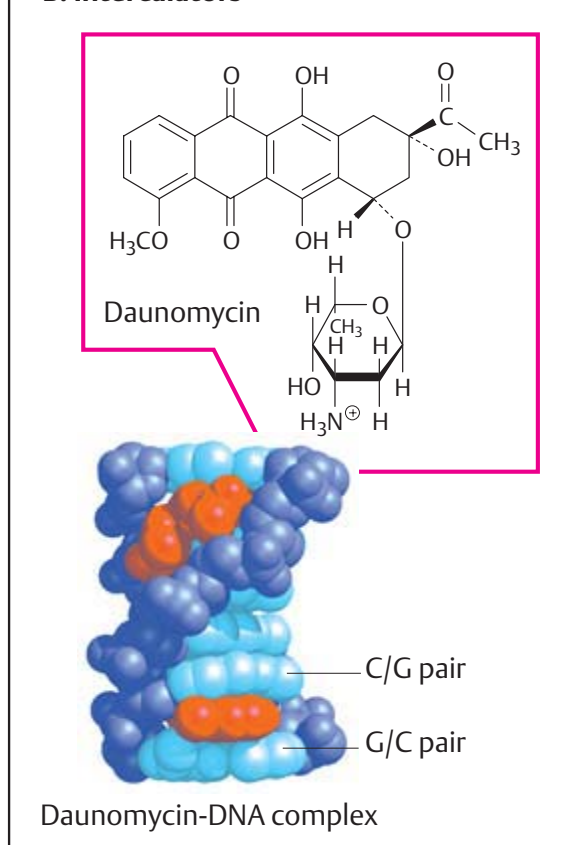
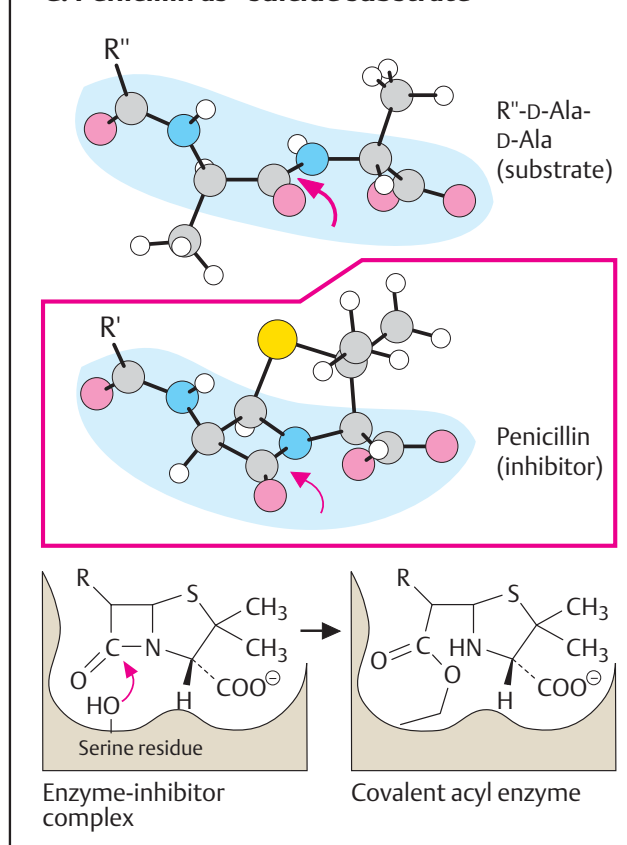
The first synthetic antibiotics were the **sulfonamides** (right). As analogues of p-aminobenzoic acid, these affect the synthesis of *folic acid*, an essential precursor of the coenzyme THF (see p. 108). **Transport antibiotics** (top center) have the properties of ion channels (see p. 222). When they are deposited in the plasma membrane, it leads to a loss of ions that damages the bacterial cells.

B. Intercalators ○

The effects of intercalators (see also p. 262) are illustrated here using the example of the **daunomycin-DNA complex**, in which two daunomycin molecules (red) are inserted in the double helix (blue). The antibiotic's ring system inserts itself between G/C base pairs (bottom), while the sugar moiety occupies the minor groove in the DNA (above). This leads to a localized change of the DNA conformation that prevents replication and transcription.

C. Penicillin as a “suicide substrate” ○

The site of action in the β -lactam antibiotics is *muramoylpentapeptide carboxypeptidase*, an enzyme that is essential for cross-linking of bacterial cell walls. The antibiotic resembles the substrate of this enzyme (a peptide with the C-terminal sequence D-Ala-D-Ala) and is therefore reversibly bound in the active center. This brings the β -lactam ring into proximity with an essential serine residue of the enzyme. Nucleophilic substitution then results in the formation of a stable covalent bond between the enzyme and the inhibitor, blocking the active center (see p. 96). In dividing bacteria, the loss of activity of the enzyme leads to the formation of unstable cell walls and eventually death.

A. Antibiotics: overview**B. Intercalators****C. Penicillin as "suicide substrate"**

Mutation and repair

Genetic information is set down in the base sequence of DNA. Changes in the DNA bases or their sequence therefore have *mutagenic* effects. Mutagens often also damage growth regulation in cells, and they are then also *carcinogenic* (see p. 400). Gene alterations (**mutations**) are one of the decisive positive factors in biological evolution. On the other hand, an excessive mutation frequency would threaten the survival of individual organisms or entire species. For this reason, every cell has **repair mechanisms** that eliminate most of the DNA changes arising from mutations (C).

A. Mutagenic agents ●

Mutations can arise as a result of physical or chemical effects, or they can be due to accidental errors in DNA replication and recombination.

The principal physical mutagen is **ionizing radiation** (α , β , and γ radiation, X-rays). In cells, it produces **free radicals** (molecules with unpaired electrons), which are extremely reactive and can damage DNA. Short-wavelength **ultraviolet light** (UV light) also has mutagenic effects, mainly in skin cells (sunburn). The most common chemical change due to UV exposure is the formation of **thymine dimers**, in which two neighboring thymine bases become covalently linked to one another (2). This results in errors when the DNA is read during replication and transcription.

Only a few examples of the group of **chemical mutagens** are shown here. *Nitrous acid* (HNO_2 ; salt: nitrite) and *hydroxylamine* (NH_2OH) both deaminate bases; they convert cytosine to uracil and adenine to inosine.

Alkylating compounds carry reactive groups that can form covalent bonds with DNA bases (see also p. 402). *Methylnitrosamines* (3) release the reactive methyl cation (CH_3^+), which methylates OH and NH_2 groups in DNA. The dangerous carcinogen *benzo[a]pyrene* is an aromatic hydrocarbon that is only converted into the active form in the organism (4; see p. 316). Multiple hydroxylation of one of the rings produces a reactive epoxide that can react with NH_2 groups in guanine residues, for example. Free radicals of benzo[a]pyrene also contribute to its toxicity.

B. Effects ●

Nitrous acid causes **point mutations** (1). For example, C is converted to U, which in the next replication pairs with A instead of G. The alteration thus becomes permanent. Mutations in which a number of nucleotides not divisible by three are inserted or removed lead to reading errors in whole segments of DNA, as they move the reading frame (**frame-shift mutations**). This is shown in Fig. 2 using a simple example. From the inserted C onwards, the resulting mRNA is interpreted differently during translation, producing a completely new protein sequence.

C. Repair mechanisms ○

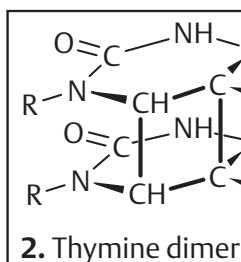
An important mechanism for the removal of DNA damage is **excision repair** (1). In this process, a specific *excision endonuclease* removes a complete segment of DNA on both sides of the error site. Using the sequence of the opposite strand, the missing segment is then replaced by a *DNA polymerase*. Finally, a *DNA ligase* closes the gaps again.

Thymine dimers can be removed by **photoreactivation** (2). A specific *photolyase* binds at the defect and, when illuminated, cleaves the dimer to yield two single bases again.

A third mechanism is **recombination repair** (3, shown in simplified form). In this process, the defect is omitted during replication. The gap is closed by shifting the corresponding sequence from the correctly replicated second strand. The new gap that results is then filled by polymerases and ligases. Finally, the original defect is corrected by excision repair as in Fig. 1 (not shown).

A. Mutagenic agents

1.



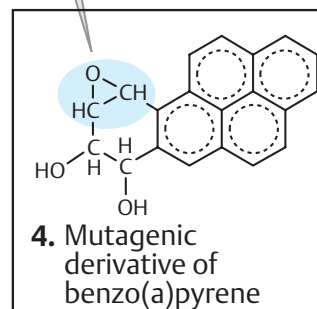
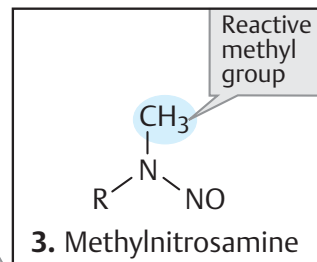
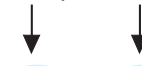
Deletions or insertions due to faulty recombination



HNO_2

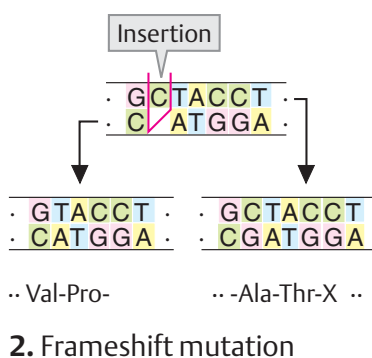
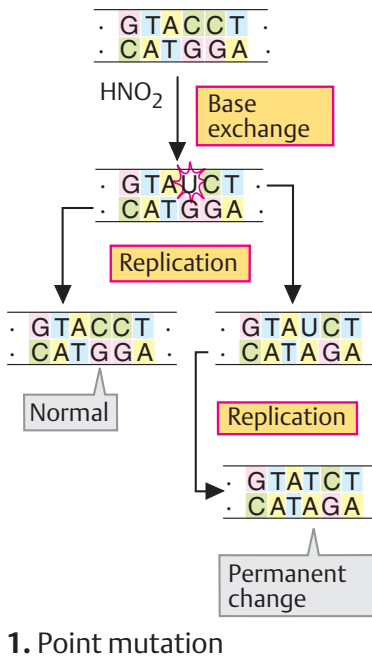


Alkylating compounds

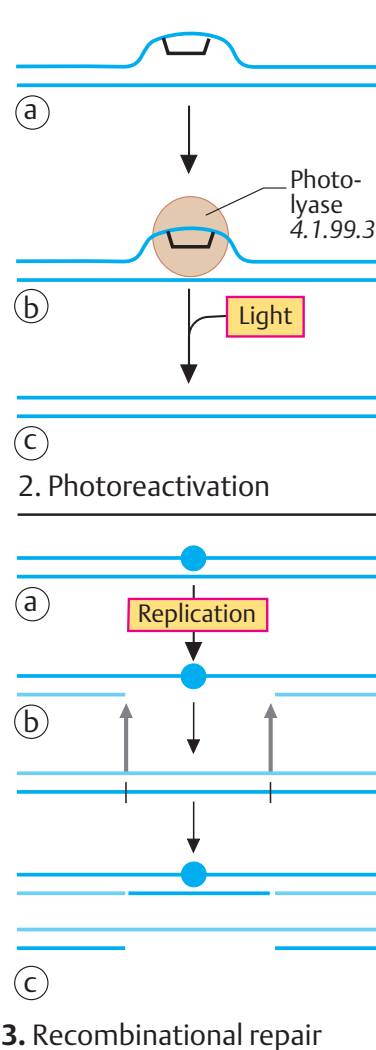
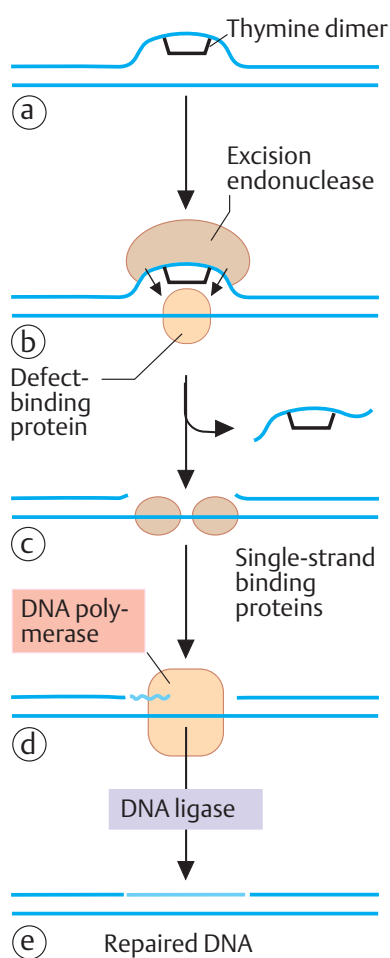


Formation of pyrimidine dimers
Base exchange
 $\text{C} \rightarrow \text{U}$
 $\text{A} \rightarrow \text{I}$
Spontaneous loss of bases
Chemical modification of bases

B. Effects



C. Repair mechanisms



DNA cloning

The growth of molecular genetics since 1970 has mainly been based on the development and refinement of methods of analyzing and manipulating DNA. **Genetic engineering** has practical applications in many fields. For example, it has provided new methods of diagnosing and treating diseases, and it is now also possible to create targeted changes in specific characteristics of organisms. Since biological risks cannot be completely ruled out with these procedures, it is particularly important to act responsibly when dealing with genetic engineering. A short overview of important methods involved in genetic engineering is provided here and on the following pages.

A. Restriction endonucleases ①

In many genetic engineering procedures, defined DNA fragments have to be isolated and then newly combined with other DNA segments. For this purpose, enzymes are used that can cut DNA and join it together again inside the cell. Of particular importance are *restriction endonucleases*—a group of bacterial enzymes that cleave the DNA double strand in a sequence-specific way. The numerous restriction enzymes known are named using abbreviations based on the organism from which they originate. The example used here is *EcoRI*, a nuclease isolated from the bacterium *Escherichia coli*.

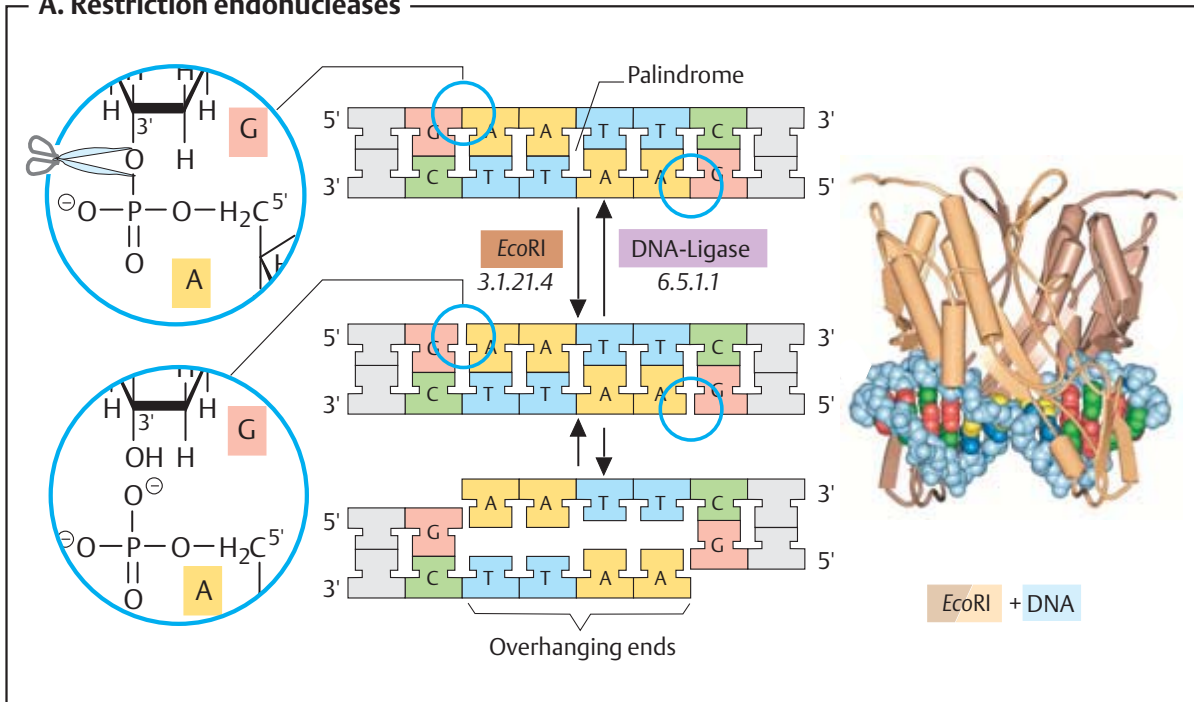
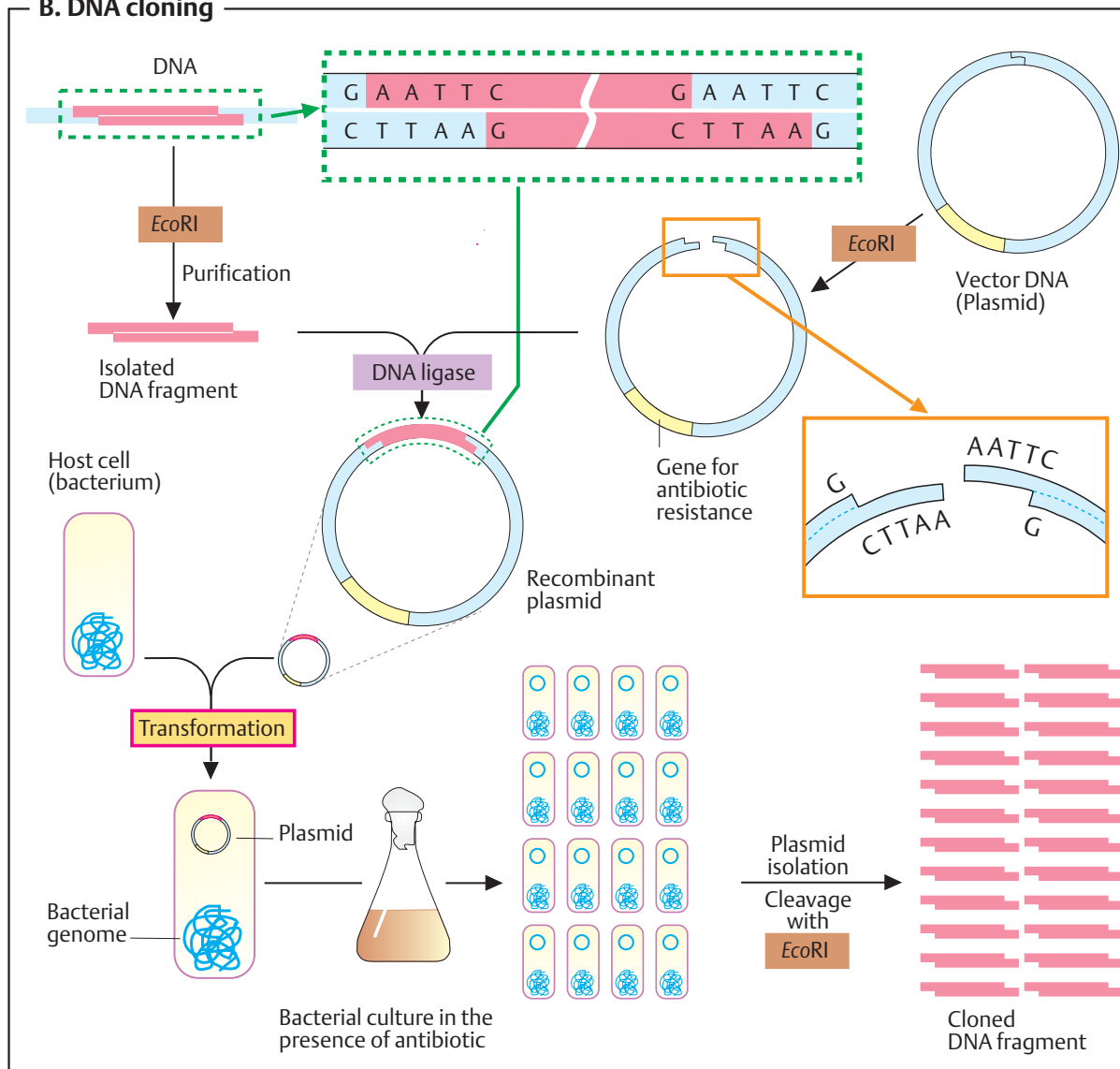
Like many other restriction endonucleases, *EcoRI* cleaves DNA at the site of a *palindrome*—i. e., a short segment of DNA in which both the strand and counter-strand have the same sequence (each read in the 5'→3' direction). In this case, the sequence is 5'-GAATTC-3'. *EcoRI*, a homodimer, cleaves the phosphoric acid diester bonds in both strands between G and A. This results in the formation of complementary *overhanging* or "*sticky*" ends (AATT), which are held together by base pairing. However, they are easily separated—e. g. by heating. When the fragments are cooled, the overhanging ends hybridize again in the correct arrangement. The cleavage sites can then be sealed again by a *DNA ligase*.

B. DNA cloning ○

Most DNA segments—e. g., genes—occur in very small quantities in the cell. To be able to work with them experimentally, a large number of identical copies ("**clones**") first have to be produced. The classic procedure for cloning DNA takes advantage of the ability of bacteria to take up and replicate short, circular DNA fragments known as **plasmids**.

The segment to be cloned is first cut out of the original DNA using restriction endonucleases (see above; for the sake of simplicity, cleavage using *EcoRI* alone is shown here, but in practice two different enzymes are usually used). As a vehicle ("*vector*"), a plasmid is needed that has only *one* *EcoRI* cleavage site. The plasmid rings are first opened by cleavage with *EcoRI* and then mixed with the isolated DNA fragments. Since the fragment and the vector have the same overhanging ends, some of the molecules will hybridize in such a way that the fragment is incorporated into the vector DNA. When the cleavage sites are now closed again using *DNA ligase*, a newly combined ("*recombinant*") *plasmid* arises.

By pretreating a large number of host cells, one can cause some of them to take up the plasmid (a process known as **transformation**) and replicate it along with their own genome when reproducing. To ensure that only host bacteria that contain the plasmid replicate, plasmids are used that give the host *resistance* to a particular antibiotic. When the bacteria are incubated in the presence of this antibiotic, only the cells containing the plasmid will replicate. The plasmid is then isolated from these cells, cleaved with *EcoRI* again, and the fragments are separated using agarose gel electrophoresis (see p. 262). The desired fragment can be identified using its size and then extracted from the gel and used for further experiments.

A. Restriction endonucleases**B. DNA cloning**

DNA sequencing

A. Gene libraries ○

It is often necessary in genetic engineering to isolate a DNA segment when its details are not fully known—e.g., in order to determine its nucleotide sequence. In this case, one can use what are known as **DNA libraries**. A DNA library consists of a large number of *vector DNA molecules* containing different fragments of *foreign DNA*. For example, it is possible to take all of the mRNA molecules present in a cell and transcribe them into DNA. These DNA fragments (known as copy DNA or **cDNA**) are then randomly introduced into vector molecules.

A library of genomic DNA can be established by cleaving the total DNA from a cell into small fragments using restriction endonucleases (see p. 258), and then incorporating these into vector DNA. Suitable vectors for gene libraries include **bacteriophages**, for example (“phages” for short). Phages are viruses that only infect bacteria and are replicated by them (see p. 404). Gene libraries have the advantage that they can be searched for specific DNA segments, using hybridization with oligonucleotides.

The first step is to strongly dilute a small part of the library (10^5 – 10^6 phages in a small volume), mix it with host bacteria, and plate out the mixture onto nutrient medium. The bacteria grow and form a continuous cloudy layer of cells. Bacteria infected by phages grow more slowly. In their surroundings, the bacterial “lawn” is less dense, and a clearer circular zone known as a **plaque** forms. The bacteria in this type of plaque exclusively contain the offspring of a single phage from the library.

The next step is to make an impression of the plate on a plastic foil, which is then heated. This causes the phage DNA to adhere to the foil. When the foil is incubated with a DNA fragment that hybridizes to the DNA segment of interest (a **gene probe**), the probe binds to the sites on the imprint at which the desired DNA is attached. Binding of the gene probe can be detected by prior radioactive or other labeling of the probe. Phages from the positive plaques in the original plate are then isolated and replicated. Restriction cleavage finally provides large amounts of the desired DNA.

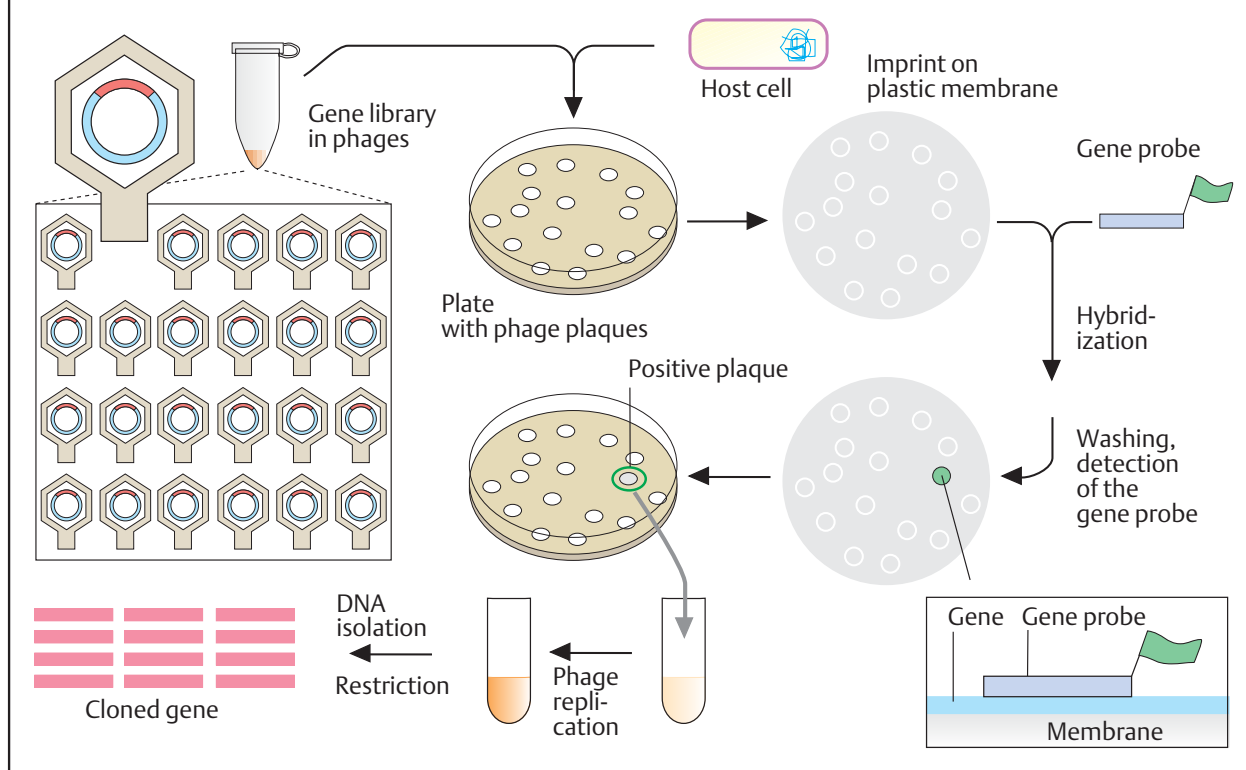
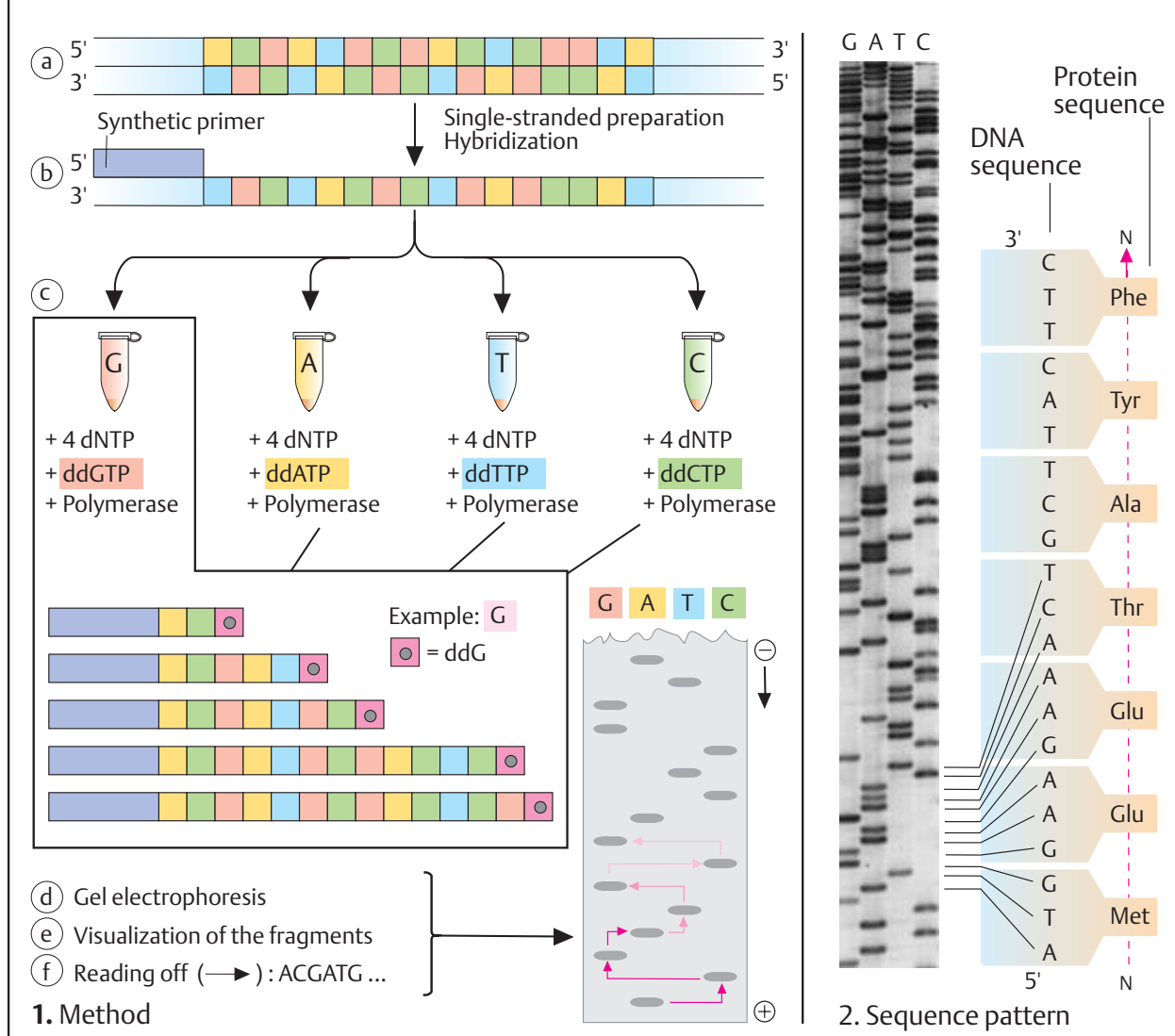
B. Sequencing of DNA ○

The nucleotide sequence of DNA is nowadays usually determined using the so-called **chain termination method**. In single-strand sequencing, the DNA fragment (**a**) is cloned into the DNA of **phage M13** (see p. 404), from which the coded single strand can be easily isolated. This is *hybridized* with a **primer**—a short, synthetically produced DNA fragment that binds to 3' end of the introduced DNA segment (**b**).

Based on this hybrid, the missing second strand can now be generated in the test tube by adding the four *deoxyribonucleoside triphosphates* (**dNTP**) and a suitable *DNA polymerase* (**c**). The trick lies in also adding small amounts of *dideoxynucleoside triphosphates* (**ddNTP**). Incorporating a ddNTP leads to the *termination of second-strand synthesis*. This can occur whenever the corresponding dNTP ought to be incorporated. The illustration shows this in detail using the example of ddGTP. In this case, fragments are obtained that each include the primer plus three, six, eight, 13, or 14 additional nucleotides. *Four separate reactions*, each with a different ddNTP, are carried out (**c**), and the products are placed side by side on a supporting material. The fragments are then separated by *gel electrophoresis* (see p. 76), in which they move in relation to their length.

Following *visualization* (**d**), the sequence of the fragments in the individual lanes is simply read from bottom to top (**e**) to directly obtain the nucleotide sequence. A detail from such a sequencing gel and the corresponding protein sequence are shown in Fig. 2.

In a more modern procedure, the four ddNTPs are covalently marked with fluorescent dyes, which produce a different color for each ddNTP on laser illumination. This allows the sequence in which the individual fragments appear at the lower end of the gel to be continuously recorded and directly stored in digital form.

A. Gene libraries**B. Sequencing of DNA**

PCR and protein expression

A. Polymerase chain reaction (PCR) ●

The polymerase chain reaction (PCR) is an important procedure in genetic engineering that allows any DNA segment to be replicated (**amplified**) without the need for restriction enzymes, vectors, or host cells (see p. 258). However, the nucleotide sequence of the segment has to be known. Two oligonucleotides (**primers**) are needed, which each hybridize with one of the strands at each end of the DNA segment to be amplified; also needed are sufficient quantities of the four **deoxyribonucleoside triphosphates** and a special heat-tolerant DNA polymerase. The primers are produced by chemical synthesis, and the polymerase is obtained from thermostable bacteria.

First, the starter is heated to around 90 °C to separate the DNA double helix into single strands (**a**; cf. p. 84). The mixture is then cooled to allow hybridization of the primers (**b**). Starting from the primers, complementary DNA strands are now synthesized in both directions by the polymerase (**c**). This cycle (cycle 1) is *repeated 20–30 times* with the same reaction mixture (cycle 2 and subsequent cycles). The cyclic heating and cooling are carried out by *computer-controlled thermostats*.

After only the third cycle, double strands start to form with a length equal to the distance between the two primers. The proportion of these approximately doubles during each cycle, until almost all of the newly synthesized segments have the correct length.

B. DNA electrophoresis ○

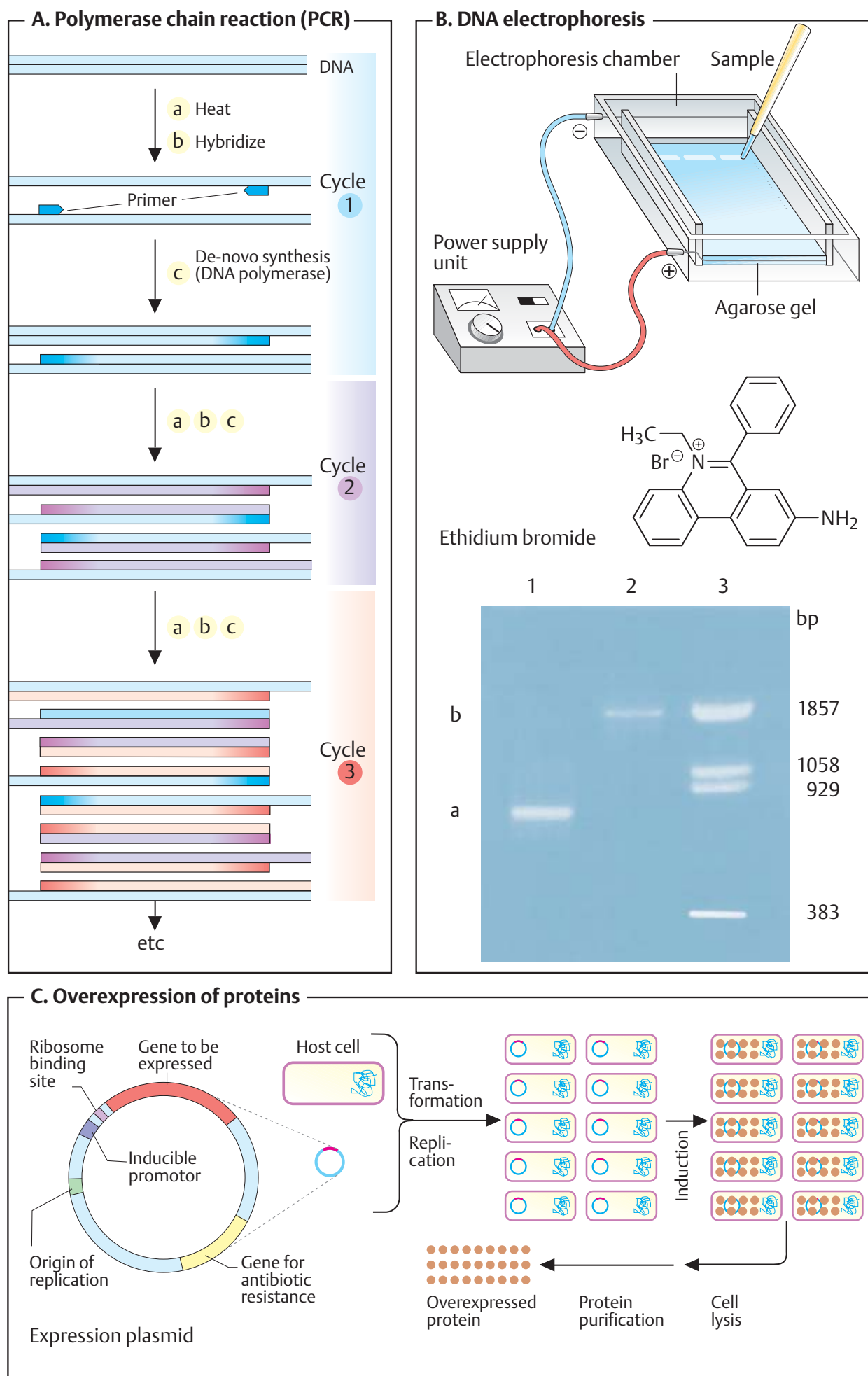
The separation of DNA fragments by electrophoresis is technically simpler than protein electrophoresis (see p. 78). The mobility of molecules in an electrical field of a given strength depends on the size and shape of the molecules, as well as their charge. In contrast to proteins, in which all three factors vary, the ratio of mass to charge in nucleic acids is constant, as all of the nucleotide components have similar masses and carry one negative charge. When electrophoresis is carried out in a wide-meshed support material that does not separate according to size and shape, the mobility of the molecules depends

on their mass alone. The supporting material generally used in genetic engineering is a gel of the polysaccharide **agarose** (see p. 40). Agarose gels are not very stable and are therefore poured horizontally into a plastic chamber in which they are used for separation (top).

To make the separated fragments visible, after running the procedure the gels are placed in solutions of **ethidium bromide**. This is an intercalator (see p. 254) that shows strong fluorescence in UV light after binding to DNA, although it barely fluoresces in an aqueous solution. The result of separating two PCR amplicates (lanes 1 and 2) is shown in the lower part of the illustration. Comparing their distances with those of polynucleotides of known lengths (lane 3; bp = base pairs) yields lengths of approximately 800 bp for fragment 1 and 1800 bp for fragment 2. After staining, the bands can be cut out of the gel and the DNA can be extracted from them and used for further experiments.

C. Overexpression of proteins ●

To treat some diseases, proteins are needed that occur in such small quantities in the organism that isolating them on a large scale would not be economically feasible. Proteins of this type can be obtained by *overexpression* in bacteria or eukaryotic cells. To do this, the corresponding *gene* is isolated from human DNA and cloned into an **expression plasmid** as described on p. 258. In addition to the gene itself, the plasmid also has to contain DNA segments that allow replication by the host cell and transcription of the gene. After *transformation* and *replication* of suitable host cells, **induction** is used in a targeted fashion to trigger efficient *transcription* of the gene. *Translation* of the mRNA formed in the host cell then gives rise to large amounts of the desired protein. Human insulin (see p. 76), plasminogen activators for dissolving blood clots (see p. 292), and the growth hormone somatotropin are among the proteins produced in this way.



Genetic engineering in medicine

Genetic engineering procedures are becoming more and more important in medicine for diagnostic purposes (A–C). New genetic approaches to the treatment of severe diseases are still in the developmental stage (“gene therapy,” D).

A. DNA fingerprinting ○

DNA fingerprinting is used to link small amounts of biological material—e.g., traces from the site of a crime—to a specific person. The procedure now used is based on the fact that the human genome contains non-coding repetitive DNA sequences, the length of which varies from individual to individual. **Short tandem repeats (STRs)** thus exist in which dinucleotides (e.g., -T-X-) are frequently repeated. Each STR can occur in five to 15 different lengths (alleles), of which one individual possesses only one or two. When the various allele combinations for several STRs are determined after PCR amplification of the DNA being investigated, a “genetic fingerprint” of the individual from whom the DNA originates is obtained. Using comparative material—e.g., saliva samples—definite identification is then possible.

B. Diagnosis of sickle-cell anemia using RFLP ○

This example illustrates a procedure for diagnosing a point mutation in the β -globin gene that leads to sickle-cell anemia (see p. 248). The mutation in the first exon of the gene destroys a cleavage site for the restriction endonuclease *MstII* (see p. 258). When the DNA of healthy and diseased individuals is cleaved with *MstII*, different fragments are produced in the region of the β -globin gene, which can be separated by electrophoresis and then demonstrated using specific probes (see p. 260). In addition, heterozygotic carriers of the sickle-cell gene can be distinguished from homozygotic ones.

C. Identification of viral DNA using RT-PCR ○

In viral infections, it is often difficult to determine the species of the pathogen precisely. **RT-PCR** can be used to identify RNA viruses. In this procedure, reverse transcriptase (see p. 404) is used to transcribe the viral RNA into dsDNA, and then PCR is employed to

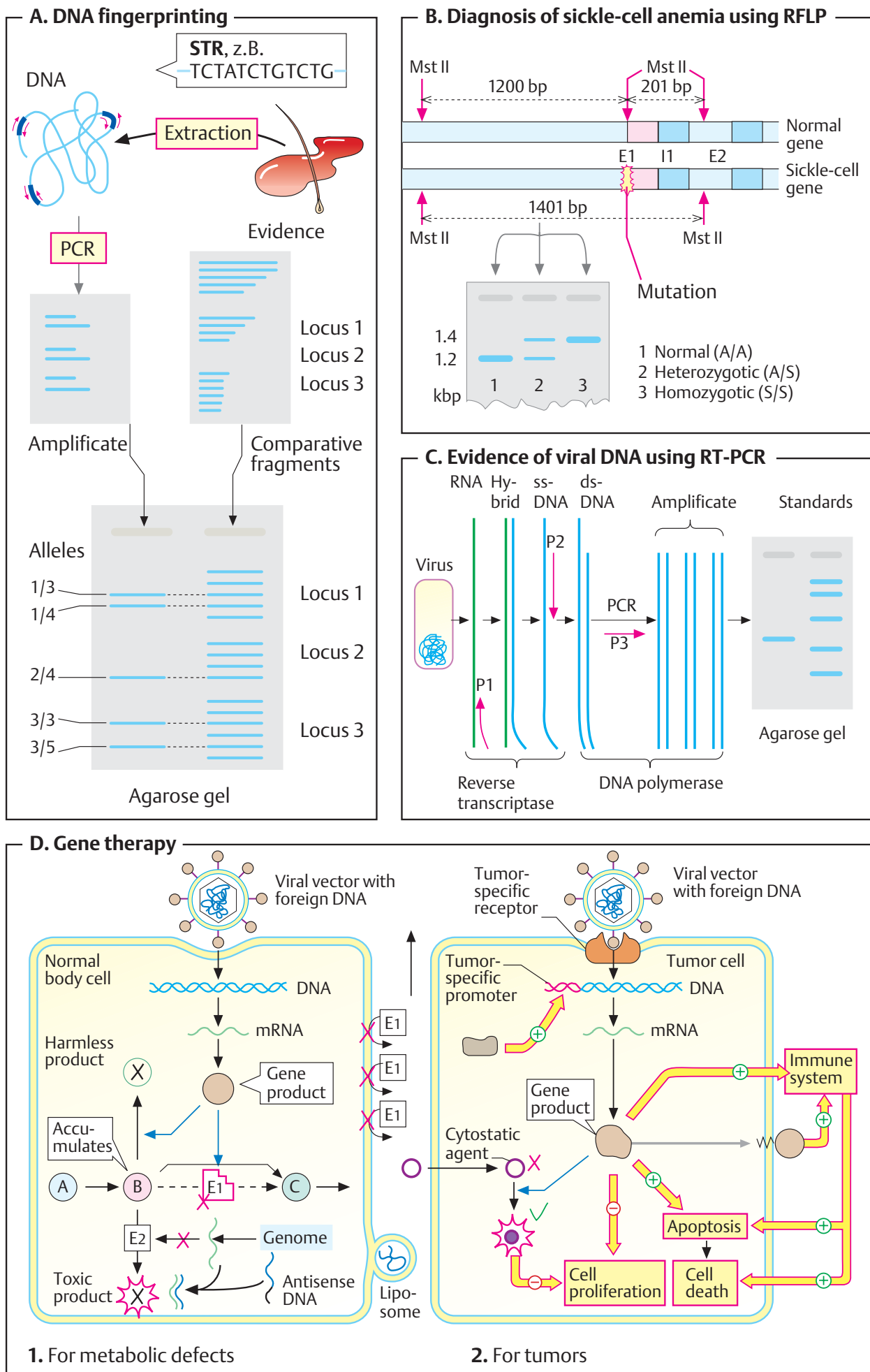
amplify a segment of this DNA with virus-specific primers. In this way, an amplificate with a characteristic length can be obtained for each pathogen and identified using gel electrophoresis as described above.

D. Gene therapy ●

Many diseases, such as hereditary metabolic defects and tumors, can still not be adequately treated. About 10 years ago, projects were therefore initiated that aimed to treat diseases of this type by transferring genes into the affected cells (gene therapy). The illustration combines conceivable and already implemented approaches to gene therapy for metabolic defects (left) and tumors (right). None of these procedures has yet become established in clinical practice.

If a mutation leads to failure of an enzyme E1 (left), its substrate B will no longer be converted into C and will accumulate. This can lead to cell damage by B itself or by a toxic product formed from it. Treatment with intact E1 is not possible, as the proteins are not capable of passing through the cell membrane. By contrast, it is in principle possible to introduce **foreign genes** into the cell using viruses as vectors (adenoviruses or retroviruses are mainly used). Their gene products could replace the defective E1 or convert B into a harmless product. Another approach uses the so-called **antisense DNA** (bottom right). This consists of polynucleotides that hybridize with the mRNA for specific cellular proteins and thereby prevent their translation. In the case shown, the synthesis of E2 could be blocked, for example.

The main problem in chemotherapy for **tumors** is the lack of tumor-specificity in the highly toxic cytostatic agents used (see p. 402). Attempts are therefore being made to introduce into tumor cells genes with products that are only released from a precursor to form active cytostatics once they have reached their target (left). Other gene products are meant to force the cells into apoptosis (see p. 396) or make them more susceptible to attack by the immune system. To steer the viral vectors to the tumor (targeting), attempts are being made to express proteins on the virus surface that are bound by tumor-specific receptors. Fusion with a tumor-specific promoter could also help limit the effect of the foreign gene to the tumor cells.



Digestion: overview

Most components of food (see p. 360) cannot be resorbed directly by the organism. It is only after they have been broken down into smaller molecules that the organism can take up the essential nutrients. *Digestion* refers to the mechanical and enzymatic breakdown of food and the resorption of the resulting products.

A. Hydrolysis and resorption of food components ●

Following mechanical fragmentation of food during chewing in the mouth, the process of enzymatic degradation starts in the stomach. For this purpose, the chyme is mixed with *digestive enzymes* that occur in the various digestive secretions or in membrane-bound form on the surface of the intestinal epithelium (see p. 268). Almost all digestive enzymes are *hydrolases* (class 3 enzymes; see p. 88); they catalyze the cleavage of composite bonds with the uptake of water.

Proteins are first denatured by the stomach's *hydrochloric acid* (see p. 270), making them more susceptible to attack by the *endopeptidases* (proteinases) present in gastric and pancreatic juice. The peptides released by endopeptidases are further degraded into amino acids by *exopeptidases*. Finally, the amino acids are resorbed by the intestinal mucosa in cotransport with Na^+ ions (see p. 220). There are separate transport systems for each of the various groups of amino acids.

Carbohydrates mainly occur in food in the form of polymers (starches and glycogen). They are cleaved by *pancreatic amylase* into oligosaccharides and are then hydrolyzed by *glycosidases*, which are located on the surface of the intestinal epithelium, to yield monosaccharides. Glucose and galactose are taken up into the enterocytes by secondary active cotransport with Na^+ ions (see p. 220). In addition, monosaccharides also have passive transport systems in the intestine.

Nucleic acids are broken down into their components by *nucleases* from the pancreas and small intestine (ribonucleases and deoxyribonucleases). Further breakdown yields the nucleobases (purine and pyrimidine derivatives), pentoses (ribose and deoxyribose),

phosphate, and nucleosides (nucleobase pentose). These cleavage products are resorbed by the intestinal wall in the region of the jejunum.

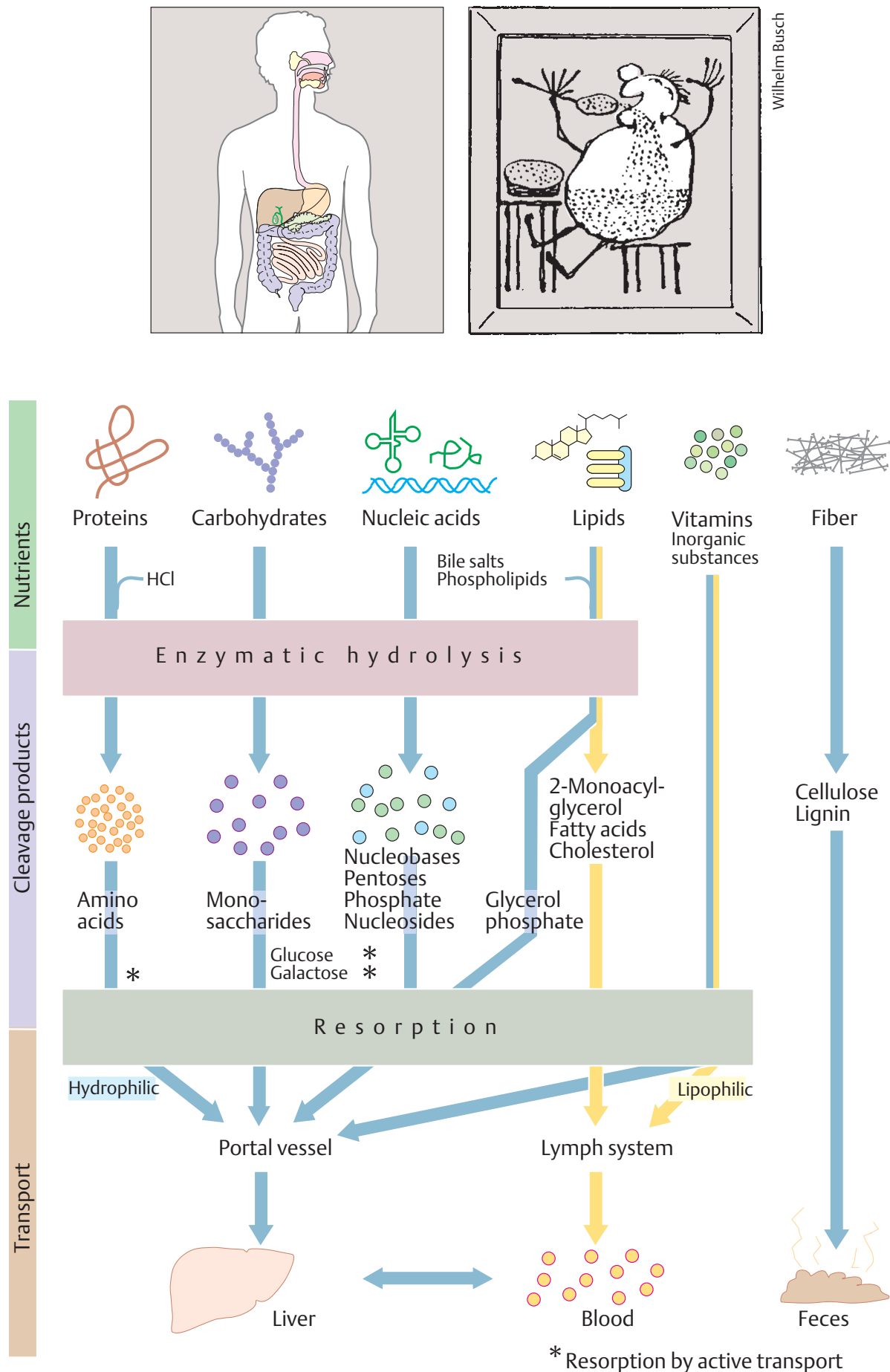
Lipids are a special problem for digestion, as they are not soluble in water. Before enzymatic breakdown, they have to be emulsified by *bile salts* and *phospholipids* in the bile (see p. 314). At the water–lipid interface, *pancreatic lipase* then attacks triacylglycerols with the help of *colipase* (see p. 270). The cleavage products include *fatty acids*, *2-monoacylglycerols*, *glycerol*, and *phosphate* from phospholipid breakdown. After resorption into the epithelial cells, fats are resynthesized from fatty acids, glycerol and 2-monoacylglycerols and passed into the lymphatic system (see p. 272). The lipids in milk are more easily digested, as they are already present in emulsion; on cleavage, they mostly provide short-chain fatty acids.

Inorganic components such as water, electrolytes, and *vitamins* are directly absorbed by the intestine.

High-molecular-weight indigestible components, such as the fibrous components of plant cell walls, which mainly consist of cellulose and lignin, pass through the bowel unchanged and form the main component of feces, in addition to cells shed from the intestinal mucosa. Dietary fiber makes a positive contribution to digestion as a *ballast material* by binding water and promoting intestinal peristalsis.

The food components resorbed by the epithelial cells of the intestinal wall in the region of the jejunum and ileum are transported directly to the liver via the *portal vein*. Fats, cholesterol, and lipid-soluble vitamins are exceptions. These are first released by the enterocytes in the form of *chylomicrons* (see p. 278) into the *lymph system*, and only reach the blood via the thoracic duct.

A. Hydrolysis and resorption of dietary constituents



Wilhelm Busch

Digestive secretions

A. Digestive juices ●

Saliva. The salivary glands produce a slightly alkaline secretion which—in addition to water and salts—contains *glycoproteins* (mucins) as lubricants, *antibodies*, and *enzymes*. α -Amylase attacks polysaccharides, and a lipase hydrolyzes a small proportion of the neutral fats. α -Amylase and *lysozyme*, a mu-rein-cleaving enzyme (see p. 40), probably serve to regulate the oral bacterial flora rather than for digestion (see p. 340).

Gastric juice. In the stomach, the chyme is mixed with gastric juice. Due to its hydrochloric acid content, this secretion of the gastric mucosa is strongly acidic (pH 1–3; see p. 270). It also contains *mucus* (mainly glycoproteins known as mucins), which protects the mucosa from the hydrochloric acid, *salts*, and *pepsinogen*—the proenzyme (“zymogen”) of the aspartate proteinase *pepsin* (see pp. 176, 270). In addition, the gastric mucosa secretes what is known as “*intrinsic factor*”—a glycoprotein needed for resorption of vitamin B₁₂ (“*extrinsic factor*”) in the bowel.

In the stomach, pepsin and related enzymes initiate the enzymatic digestion of proteins, which takes 1–3 hours. The acidic gastric contents are then released into the duodenum in batches, where they are neutralized by alkaline pancreatic secretions and mixed with cystic bile.

Pancreatic secretions. In the acinar cells, the pancreas forms a secretion that is alkaline due to its HCO₃[−] content, the buffer capacity of which is sufficient to neutralize the stomach’s hydrochloric acid. The pancreatic secretion also contains many *enzymes* that catalyze the hydrolysis of high-molecular-weight food components. All of these enzymes are hydrolases with pH optimums in the neutral or weakly alkaline range. Many of them are formed and secreted as proenzymes and are only activated in the bowel lumen (see p. 270).

Trypsin, *chymotrypsin*, and *elastase* are endopeptidases that belong to the group of serine proteinases (see p. 176). Trypsin hydrolyzes specific peptide bonds on the C side of the basic amino acids Arg and Lys, while chymotrypsin prefers peptide bonds of the apolar amino acids Tyr, Trp, Phe, and Leu (see p. 94).

Elastase mainly cleaves on the C side of the aliphatic amino acids Gly, Ala, Val, and Ile. Smaller peptides are attacked by *carboxypeptidases*, which as exopeptidases cleave individual amino acids from the C-terminal end of the peptides (see p. 176).

α -Amylase, the most important endoglycosidase in the pancreas, catalyzes the hydrolysis of $\alpha 1 \rightarrow 4$ bonds in the polymeric carbohydrates starch and glycogen. This releases maltose, maltotriose, and a mixture of other oligosaccharides.

Various pancreatic enzymes hydrolyze lipids, including *lipase* with its auxiliary protein *colipase* (see p. 270), *phospholipase A₂*, and *sterol esterase*. Bile salts activate the lipid-cleaving enzymes through micelle formation (see below).

Several hydrolases—particularly *ribonuclease* (RNAse) and *deoxyribonuclease* (DNAse)—break down the nucleic acids contained in food.

Bile. The liver forms a thin secretion (bile) that is stored in the gallbladder after water and salts have been extracted from it. From the gallbladder, it is released into the duodenum. The most important constituents of bile are *water* and inorganic *salts*, *bile acids* and *bile salts* (see p. 314), *phospholipids*, *bile pigments*, and *cholesterol*. Bile salts, together with phospholipids, emulsify insoluble food lipids and activate the lipases. Without bile, fats would be inadequately cleaved, if at all, resulting in “fatty stool” (steatorrhea). Resorption of fat-soluble vitamins would also be affected.

Small-intestinal secretions. The glands of the small intestine (the Lieberkühn and Brunner glands) secrete additional digestive enzymes into the bowel. Together with enzymes on the microvilli of the intestinal epithelium (peptidases, glycosidases, etc.), these enzymes ensure almost complete hydrolysis of the food components previously broken down by the endoenzymes.

A. Digestive juices

Saliva

Daily secretion 1.0–1.5 l
pH 7

Water	Moistens food
Salts	
Mucus	Lubricant
Antibodies	Bind to bacteria
α -Amylases (3.2.1.1)	Cleave starch $\downarrow$
Lysozyme (3.2.1.17)	Attacks bacterial cell walls $\downarrow$

Bile

Daily secretion 0.6 l
pH 6.9–7.7

Water	
HCO_3^-	Neutralizes gastric juice
Bile salts	Facilitate lipid digestion
Phospholipids	Facilitate lipid digestion
Bile pigments	Waste products
Cholesterol	Waste product

Gastric juice

Daily secretion 2–3 l
pH 1

Water	
Salts	
HCl	Denatures proteins, kills bacteria
Mucus	Protects stomach lining
Pepsins (3.4.23.1–3)	Cleave proteins $\downarrow$
Chymosin (3.4.23.4)	Precipitates casein $\downarrow$
Triacylglycerol lipase (3.1.1.3)	Cleaves fats
Intrinsic factor	Protects vitamin B ₁₂

$\downarrow$ Endoenzyme

$\leftarrow$ Exoenzyme

Components

Function or substrate

Pancreatic secretions

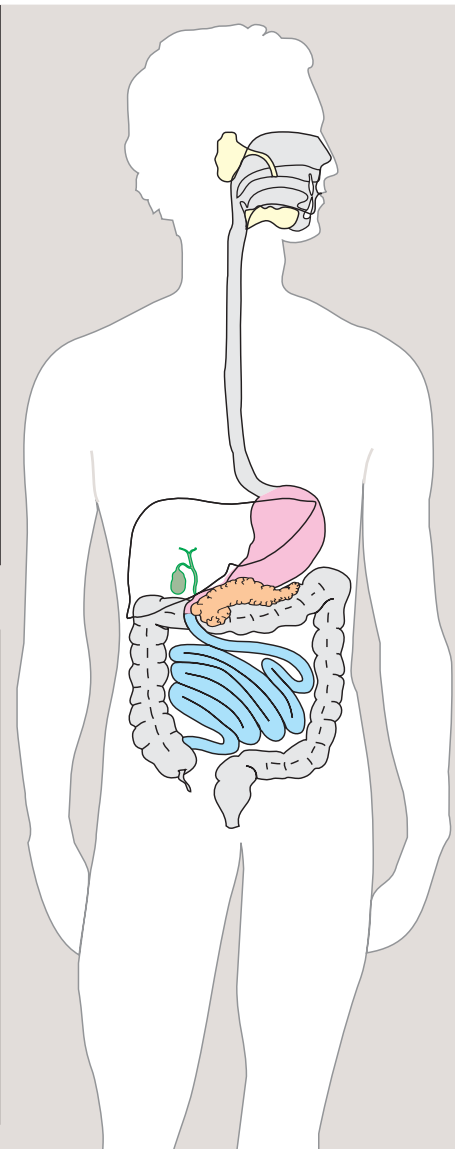
Daily secretion 0.7–2.5 l
pH 7.7 (7.5–8.8)

Water	
HCO_3^-	Neutralizes gastric juice
Trypsin (3.4.21.4)	Proteins $\downarrow$
Chymotrypsin (3.4.21.1)	Proteins $\downarrow$
Elastase (3.4.21.36)	Proteins $\downarrow$
Carboxypeptidases (3.4.n.n)	Peptides $\leftarrow$
α -Amylase (3.2.1.1)	Starch and glycogen $\downarrow$
Triacylglycerol lipase (3.1.1.3)	Fats
Co-Lipase	Cofactor for lipase
Phospholipase A ₂ (3.1.1.4)	Phospholipids
Sterol esterase (3.1.1.13)	Cholesterol esters
Ribonuclease (3.1.27.5)	RNA
Deoxyribonuclease I (3.1.21.1)	DNA

Secretions of the small intestine

Daily secretion unknown
pH 6.5–7.8

Aminopeptidases (3.4.11.n)	Peptides $\leftarrow$
Dipeptidases (3.4.13.n)	Dipeptides
α -Glucosidase (3.2.1.20)	Oligosaccharides $\downarrow$
Oligo-1,6-glucosidase (3.2.1.10)	Oligosaccharides $\downarrow$
β -Galactosidase (3.2.1.23)	Lactose
Sucrose α -glucosidase (3.2.1.48)	Sucrose
α , α -Trehalase (3.2.1.28)	Trehalose
Alkaline phosphatase (3.1.3.1)	Phosphoric acid esters
Polynucleotidases (3.1.3.n)	Nucleic acids, nucleotides $\downarrow$
Nucleosidases (3.2.2.n)	Nucleosides $\leftarrow$
Phospholipases (3.1.n.n)	Phospholipids



Digestive processes

Gastric juice is the product of several cell types. The *parietal cells* produce hydrochloric acid, *chief cells* release pepsinogen, and *accessory cells* form a mucin-containing mucus.

A. Formation of hydrochloric acid ①

The secretion of **hydrochloric acid** (H^+ and Cl^-) by the parietal cells is an active process that uses up ATP and takes place against a concentration gradient (in the gastric lumen, with a pH of 1, the H^+ concentration is some 10^6 times higher than in the parietal cells, which have a pH of 7).

The precursors of the exported H^+ ions are carbon dioxide (CO_2) and water (H_2O). CO_2 diffuses from the blood into the parietal cells, and in a reaction catalyzed by *carbonate dehydratase* (carbonic anhydrase [2]), it reacts with H_2O to form H^+ and hydrogen carbonate (HCO_3^-). The H^+ ions are transported into the gastric lumen in exchange for K^+ by a membrane-bound *H^+/K^+ -exchanging ATPase* [1] (a transport ATPase of the P type; see p. 220). The remaining hydrogen carbonate is released into the interstitium in electroneutral antiport in exchange for chloride ions (Cl^-), and from there into the blood. The Cl^- ions follow the secreted protons through a channel into the gastric lumen.

The hydrochloric acid in gastric juice is important for digestion. It activates pepsinogen to form pepsin (see below) and creates an optimal pH level for it to take effect. It also denatures food proteins so that they are more easily attacked by proteinases, and it kills micro-organisms.

Regulation. HCl secretion is stimulated by the peptide hormone *gastrin*, the mediator *histamine* (see p. 380), and—via the neurotransmitter *acetylcholine*—by the autonomous nervous system. The peptide *somatostatin* and certain *prostaglandins* (see p. 390) have inhibitory effects. Together with cholecystokinin, secretin, and other peptides, gastrin belongs to the group of **gastrointestinal hormones** (see p. 370). All of these are formed in the gastrointestinal tract and mainly act in the vicinity of the site where they are formed—i.e., they are paracrine hormones (see p. 372). While gastrin primarily enhances HCl secretion, *cholecystokinin* and *secretin* mainly

stimulate pancreatic secretion and bile release.

B. Zymogen activation ①

To prevent self-digestion, the pancreas releases most proteolytic enzymes into the duodenum in an inactive form as *proenzymes* (zymogens). Additional protection from the effects of premature activation of pancreatic proteinases is provided by *proteinase inhibitors* in the pancreatic tissue, which inactivate active enzymes by complex formation (right).

Trypsinogen plays a key role among the proenzymes released by the pancreas. In the bowel, it is proteolytically converted into active trypsin (see p. 176) by **enteropeptidase**, a membrane enzyme on the surface of the enterocytes. Trypsin then autocatalytically activates additional trypsinogen molecules and the other proenzymes (left).

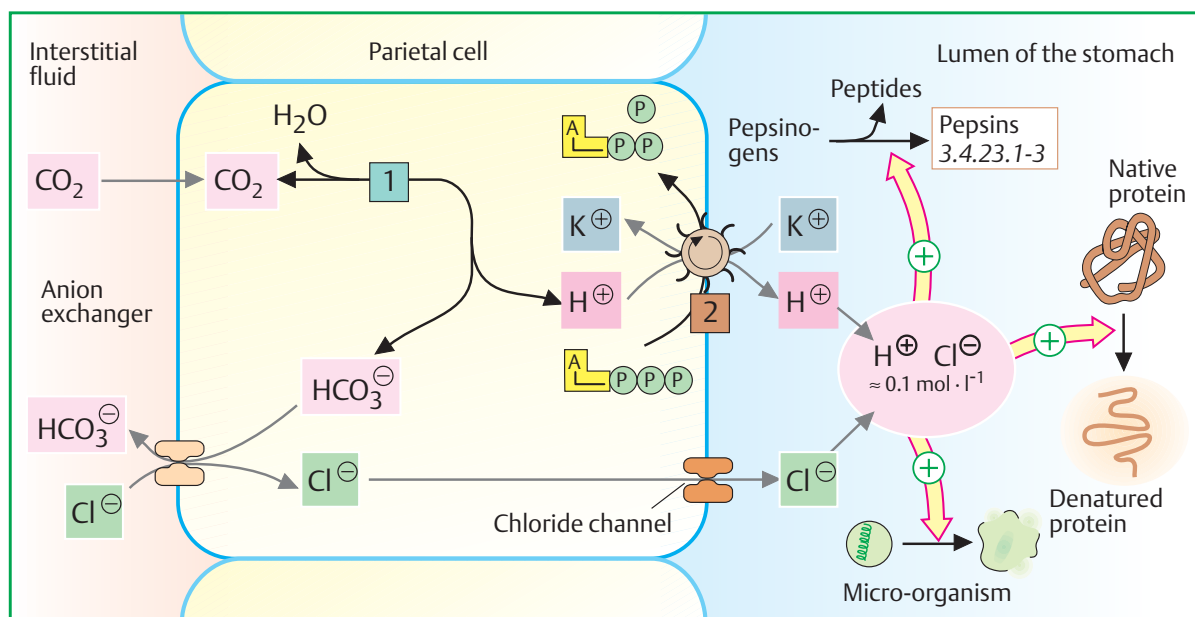
C. Fat digestion ①

Due to the “hydrophobic effect” (see p. 28), water-insoluble neutral fats in the aqueous environment of the bowel lumen would aggregate into drops of fat in which most of the molecules would not be accessible to pancreatic lipase. The amphipathic substances in bile (bile acids, bile salts, phospholipids) create an **emulsion** in which they occupy the surface of the droplets and thereby prevent them from coalescing into large drops. In addition, the bile salts, together with the auxiliary protein colipase, mediate binding of *triacylglycerol lipase* [1] to the emulsified fat droplets. Activation of the lipase is triggered by a conformation change in the C-terminal domain of the enzyme, which uncovers the active center.

During passage through the intestines, the active lipase breaks down the triacylglycerols in the interior of the droplets into free fatty acids and amphipathic monoacylglycerols. Over time, smaller **micelles** develop (see p. 28), in the envelope of which monoacylglycerols are present in addition to bile salts and phospholipids. Finally, the components of the micelles are resorbed by the enterocytes in ways that have not yet been explained.

Monoacylglycerols and fatty acids are reassembled into fats again (see p. 272), while the bile acids return to the liver (enterohepatic circulation; see p. 314).

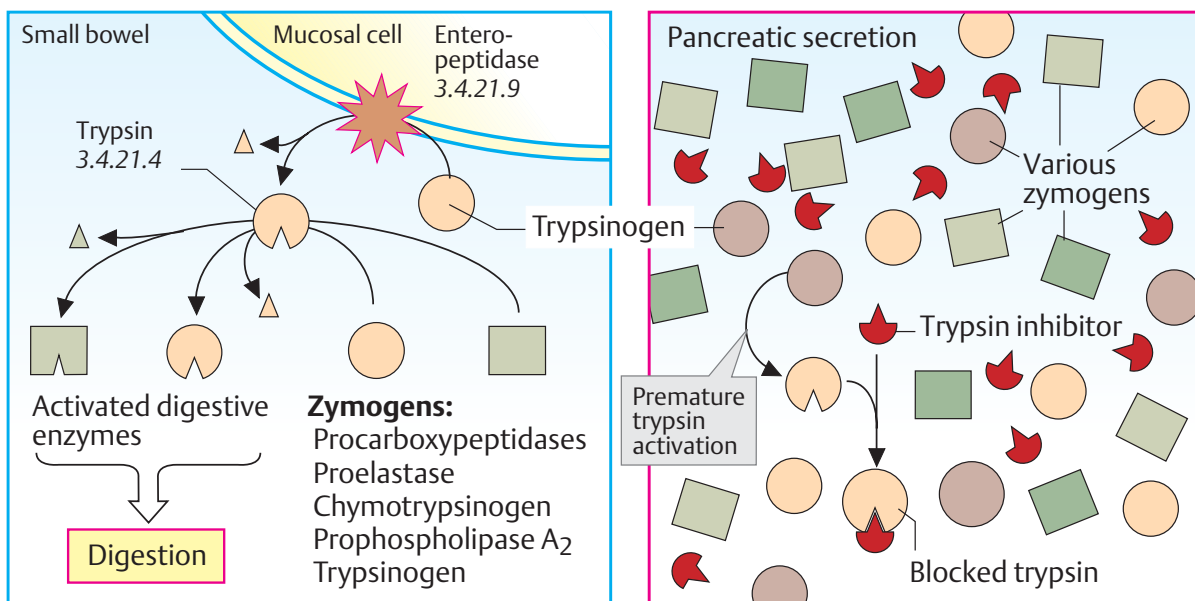
A. Formation of hydrochloric acid



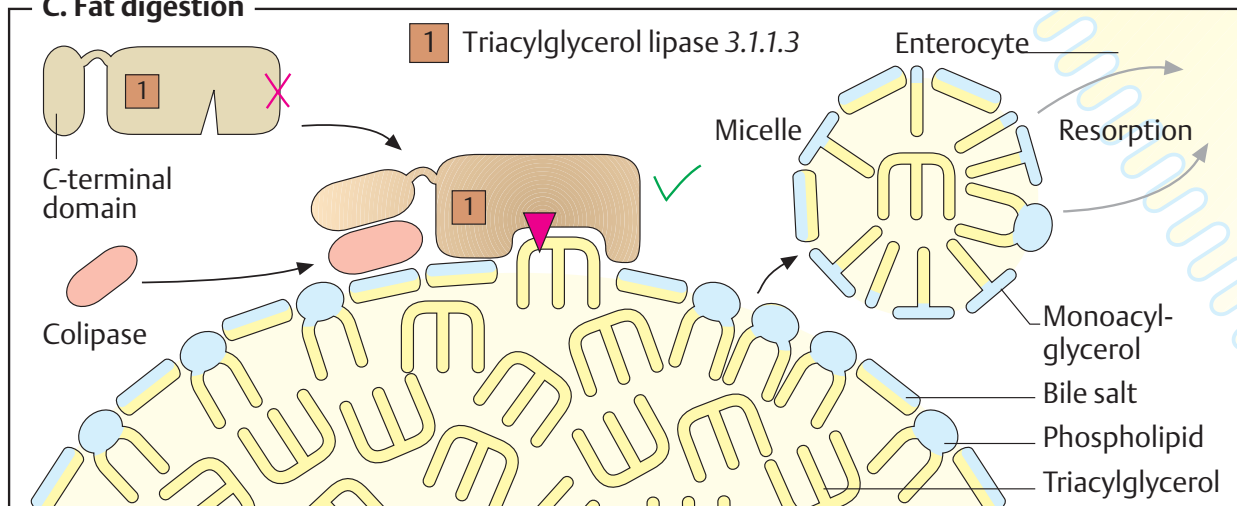
1 Carbonate dehydratase 4.2.1.1 [Zn^{2+}]

2 H^+/K^+ -exchanging ATPase 3.6.1.36

B. Zymogen activation



C. Fat digestion



Resorption

Enzymatic hydrolysis in the digestive tract breaks down foodstuffs into their resorbable components. *Resorption* of the cleavage products takes place primarily in the small intestine. Only ethanol and short-chain fatty acids are already resorbed to some extent in the stomach.

The resorption process is facilitated by the large inner surface of the intestine, with its brush-border cells. Lipophilic molecules penetrate the plasma membrane of the mucosal cells by simple diffusion, whereas polar molecules require *transporters* (facilitated diffusion; see p. 218). In many cases, carrier-mediated cotransport with Na^+ ions can be observed. In this case, the difference in the concentration of the sodium ions (high in the intestinal lumen and low in the mucosal cells) drives the import of nutrients against a concentration gradient (secondary active transport; see p. 220). Failure of carrier systems in the gastrointestinal tract can result in diseases.

A. Monosaccharides ●

The cleavage of polymeric carbohydrates by *~amylase* [1] leads to **oligosaccharides**, which are broken down further by *exoglycosidases* (oligosaccharidases and disaccharidases [2]) on the membrane surface of the brush border. The monosaccharides released in this way then pass with the help of various *sugar-specific transporters* into the cells of intestinal epithelium. *Secondary active transport* serves for the uptake of **glucose** and **galactose**, which are transported against a concentration gradient in cotransport with Na^+ . The Na^+ gradient is maintained on the basal side of the cells by *Na^+/K^+ -ATPase* [3]. Another passive transporter then releases glucose and galactose into the blood. **Fructose** is taken up by a special type of transporter using facilitated diffusion.

Amino acids (not illustrated)

Protein degradation is initiated by *proteinases*—by pepsins in the stomach and by trypsin, chymotrypsin, and elastase in the small intestine. The resulting peptides are then further hydrolyzed by various *peptidases* into amino

acids. Individual amino acid groups have *group-specific amino acid transporters*, some of which transport the amino acids into the enterocytes in cotransport with Na^+ ions (secondary active transport), while others transport them in an Na^+ -independent manner through facilitated diffusion. Small peptides can also be taken up.

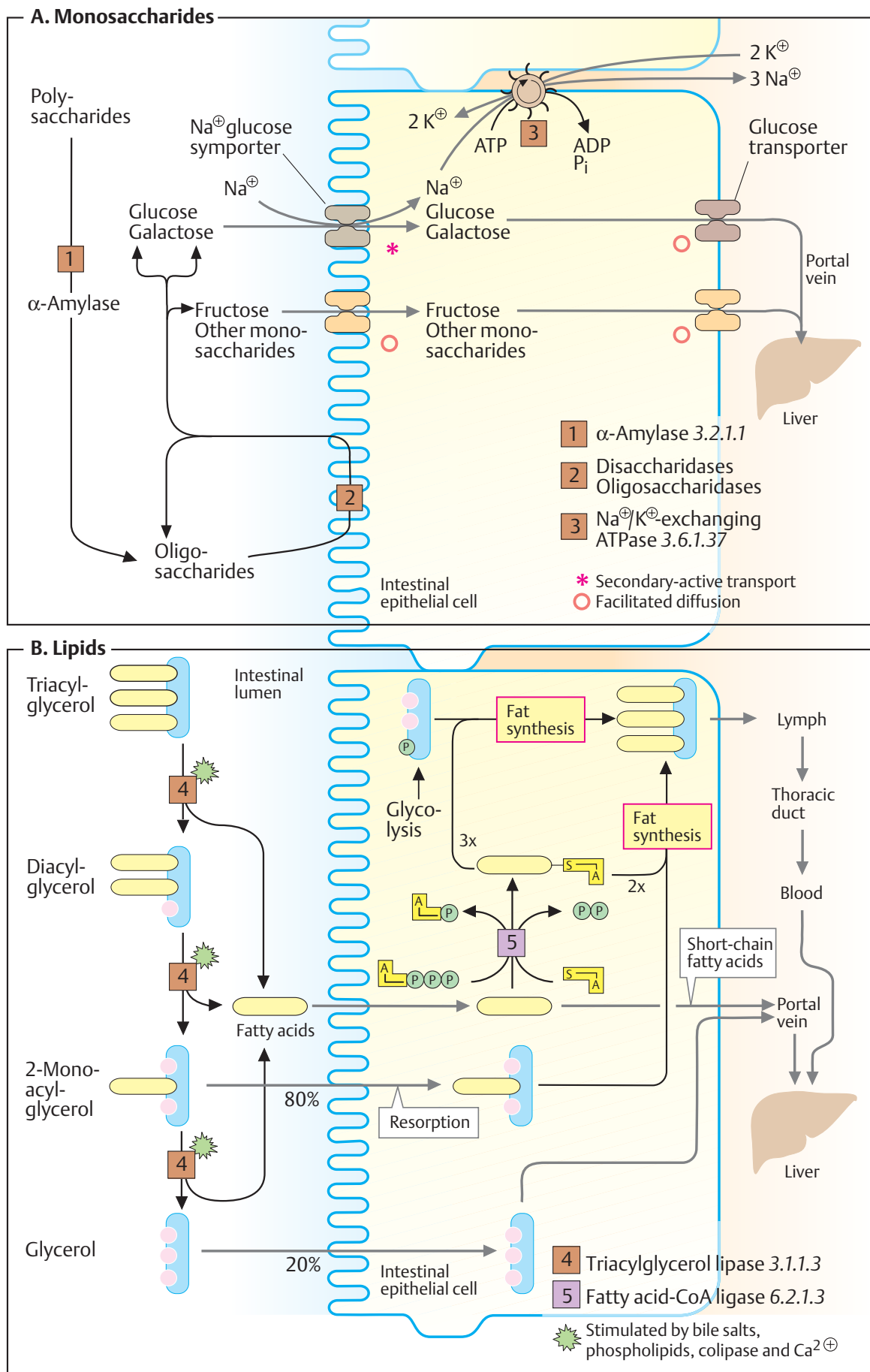
B. Lipids ●

Fats and other lipids are poorly soluble in water. The larger the accessible surface is—i.e., the better the fat is emulsified—the easier it is for enzymes to hydrolyze it (see p. 270). Due to the special properties of milk, milk fats already reach the gastrointestinal tract in emulsified form. Digestion of them therefore already starts in the oral cavity and stomach, where lipases in the saliva and gastric juice are available. Lipids that are less accessible—e.g., from roast pork—are emulsified in the small intestine by *bile salts* and *bile phospholipids*. Only then are they capable of being attacked by *pancreatic lipase* [4] (see p. 270).

Fats (triacylglycerols) are mainly attacked by pancreatic lipase at positions 1 and 3 of the glycerol moiety. Cleavage of two fatty acid residues gives rise to **fatty acids** and **2-monoacylglycerols**, which are quantitatively the most important products. However, a certain amount of **glycerol** is also formed by complete hydrolysis. These cleavage products are resorbed by a non-ATP-dependent process that has not yet been explained in detail.

In the mucosal cells, *long-chain fatty acids* are resynthesized by an ATP-dependent ligase [5] to form acyl-CoA and then triacylglycerols (fats; see p. 170). The fats are released into the lymph in the form of **chylomicrons** (see p. 278) and, bypassing the liver, are deposited in the thoracic duct—i.e., the blood system. *Cholesterol* also follows this route.

By contrast, *short-chain fatty acids* (with chain lengths of less than 12 C atoms) pass directly into the blood and reach the liver via the portal vein. Resorbed glycerol can also take this path.



Blood: composition and functions

Human blood constitutes about 8% of the body's weight. It consists of **cells** and cell fragments in an aqueous medium, the **blood plasma**. The proportion of cellular elements, known as *hematocrit*, in the total volume is approximately 45%.

A. Functions of the blood ●

The blood is the most important transport medium in the body. It serves to keep the “internal milieu” constant (homeostasis) and it plays a decisive role in defending the body against pathogens.

Transport. The *gases* oxygen and carbon dioxide are transported in the blood. The blood mediates *the exchange of substances between organs* and takes up *metabolic end products* from tissues in order to transport them to the lungs, liver, and kidney for excretion. The blood also distributes *hormones* throughout the organism (see p. 370).

Homeostasis. The blood ensures that a balanced distribution of water is maintained between the vascular system, the cells (intracellular space), and the extracellular space. The *acid–base balance* is regulated by the blood in combination with the lungs, liver, and kidneys (see p. 288). The regulation of *body temperature* also depends on the controlled transport of heat by the blood.

Defense. The body uses both non-specific and specific mechanisms to defend itself against pathogens. The defense system includes the *cells of the immune system* and certain *plasma proteins* (see p. 294).

Self-protection. To prevent blood loss when a vessel is injured, the blood has systems for stanching blood flow and coagulating the blood (hemostasis; see p. 290). The dissolution of blood clots (fibrinolysis) is also managed by the blood itself (see p. 292).

B. Cellular elements ●

The solid elements in the blood are the erythrocytes (red blood cells), leukocytes (white blood cells), and thrombocytes (platelets).

The **erythrocytes** provide for gas transport in the blood. They are discussed in greater detail on pp. 280–285.

The **leukocytes** include various types of granulocyte, monocyte, and lymphocyte. All of these have immune defense functions (see p. 294). The *neutrophil granulocytes*, *monocytes*, and the *macrophages* derived from monocytes are phagocytes. They can ingest and degrade invading pathogens. The *lymphocytes* are divided into two groups, B lymphocytes and T lymphocytes. B lymphocytes produce *antibodies*, while T lymphocytes regulate the immune response and destroy virus-infected cells and tumor cells. *Eosinophilic* and *basophilic granulocytes* have special tasks for defense against animal parasites.

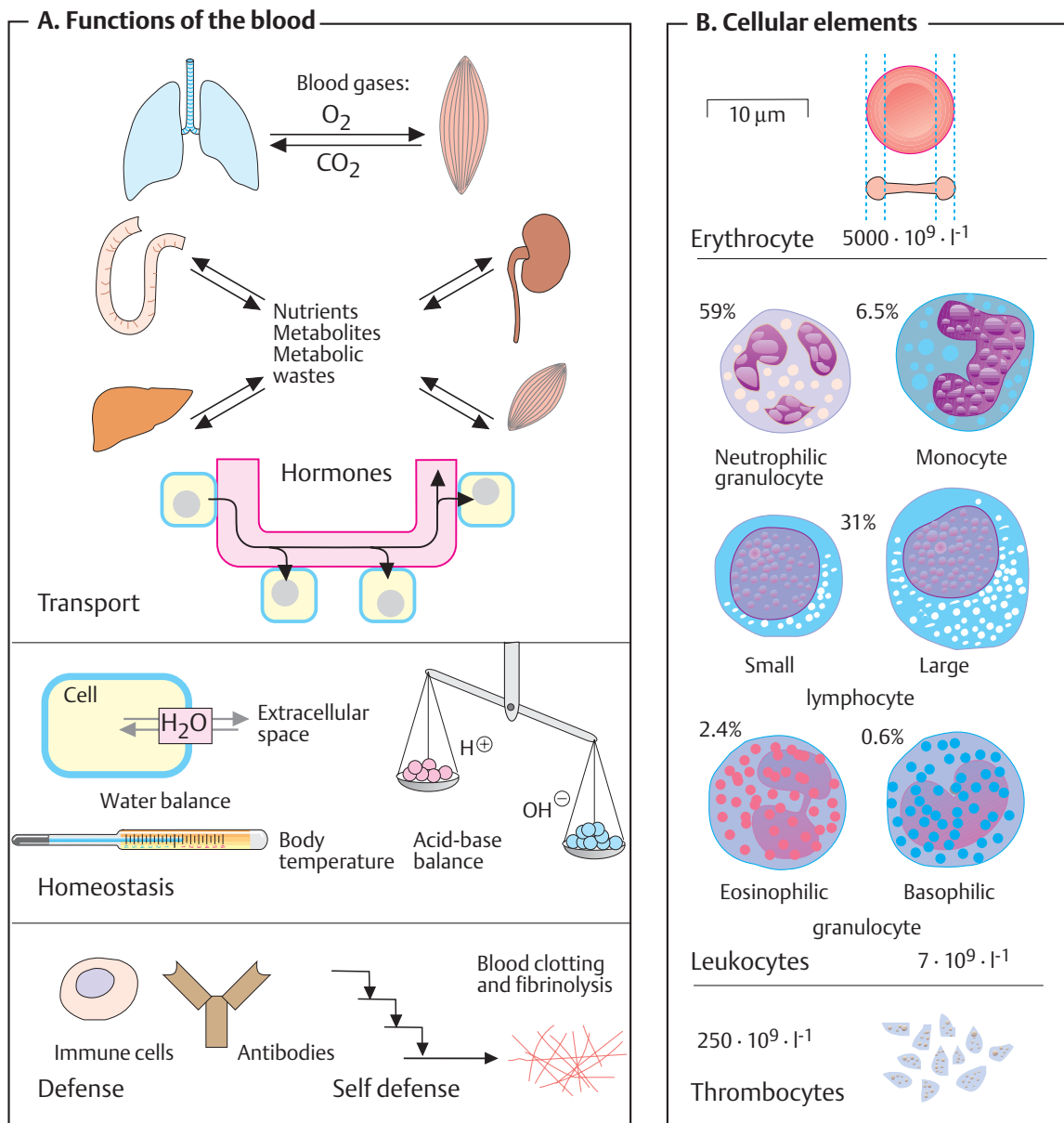
Thrombocytes are cell fragments that arise in the bone marrow from large precursor cells, the megakaryocytes. Their task is to promote hemostasis (see p. 290).

C. Blood plasma: composition ●

The **blood plasma** is an aqueous solution of electrolytes, nutrients, metabolites, proteins, vitamins, trace elements, and signaling substances. The fluid phase of coagulated blood is known as **blood serum**. It differs from the plasma in that it lacks fibrin and other coagulation proteins (see p. 290).

Laboratory assessment of the composition of the blood plasma is often carried out in clinical chemistry. Among the electrolytes, there is a relatively high concentration of Na^+ , Ca^{2+} , and Cl^- ions in the blood in comparison with the cytoplasm. By contrast, the concentrations of K^+ , Mg^{2+} , and phosphate ions are higher in the cells. Proteins also have a higher intracellular concentration. The electrolyte composition of blood plasma is similar to that of seawater, due to the evolution of early forms of life in the sea. The solution known as “*physiological saline*” (NaCl at a concentration of 0.15 mol L^{-1}) is almost isotonic with blood plasma.

A list of particularly important **metabolites** in the blood plasma is given on the right.



Plasma proteins

Quantitatively, proteins are the most important part of the soluble components of the blood plasma. With concentrations of between 60 and 80 g L⁻¹, they constitute approximately 4% of the body's total protein. Their tasks include transport, regulation of the water balance, hemostasis, and defense against pathogens.

A. Plasma proteins ①

Some 100 different proteins occur in human blood plasma. Based on their behavior during electrophoresis (see below), they are broadly divided into *five fractions*: **albumins** and α_1 -, α_2 -, β - and γ -**globulins**. Historically, the distinction between the albumins and globulins was based on differences in the proteins' solubility –albumins are soluble in pure water, whereas globulins only dissolve in the presence of salts.

The most frequent protein in the plasma, at around 45 g L⁻¹, is **albumin**. Due to its high concentration, it plays a crucial role in maintaining the blood's colloid osmotic pressure and represents an important amino acid reserve for the body. Albumin has binding sites for apolar substances and therefore functions as a transport protein for long-chain fatty acids, bilirubin, drugs, and some steroid hormones and vitamins. In addition, serum albumin binds Ca²⁺ and Mg²⁺ ions. It is the only important plasma protein that is not glycosylated.

The albumin fraction also includes *trans-thyretin* (prealbumin), which together with other proteins transports the hormone thyroxine and its metabolites.

The table also lists important **globulins** in blood plasma, with their mass and function. The α - and β -globulins are involved in the transport of lipids (lipoproteins; see p. 278), hormones, vitamins, and metal ions. In addition, they provide coagulation factors, protease inhibitors, and the proteins of the complement system (see p. 298). Soluble antibodies (immunoglobulins; see p. 300) make up the γ -globulin fraction.

Synthesis and degradation. Most plasma proteins are synthesized by the liver. Exceptions to this include the immunoglobulins, which are secreted by B lymphocytes known

as plasma cells (see p. 302) and peptide hormones, which derive from endocrine gland cells.

With the exception of albumin, almost all plasma proteins are *glycoproteins*. They carry oligosaccharides in *N*- and *O*-glycosidic bonds (see p. 44). *N*-acetylneuraminic acid (sialic acid; see p. 38) often occurs as a terminal carbohydrate among sugar residues. *Neuraminidases* (sialidases) on the surface of the vascular endothelia gradually cleave the sialic acid residues and thereby release galactose units on the surfaces of the proteins. These *asialoglycoproteins* ("asialo-" = without sialic acid) are recognized and bound by galactose receptors on hepatocytes. In this way, the liver takes up aged plasma proteins by endocytosis and breaks them down. The oligosaccharides on the protein surfaces thus determine the half-life of plasma proteins, which is a period of days to weeks.

In healthy individuals, the concentration of plasma proteins is constant. Diseases in organs that are involved in protein synthesis and breakdown can shift the protein pattern. For example, via cytokines (see p. 392), severe injuries trigger increased synthesis of *acute-phase proteins*, which include C-reactive protein, haptoglobin, fibrinogen, complement factor C-3, and others. The concentrations of individual proteins are altered in some diseases (known as *dysproteinemias*).

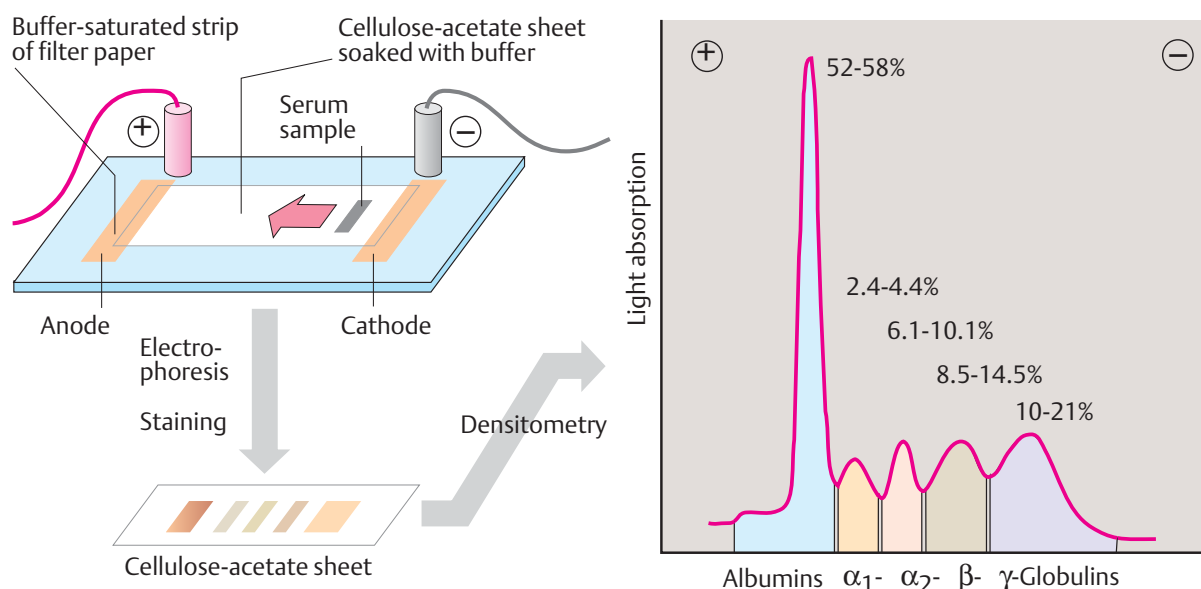
B. Carrier electrophoresis ①

Proteins and other electrically charged macromolecules can be separated using electrophoresis (see also pp. 78, 262). Among the various procedures used, **carrier electrophoresis** on cellulose acetate foil (CAF) is particularly simple. Using this method, serum proteins—which at slightly alkaline pH values all move towards the anode, due to their excess of negative charges—can be separated into the five fractions mentioned. After the proteins have been stained with dyes, the resulting bands can be quantitatively assessed using densitometry.

A. Plasma proteins

Group	Protein	M _r in kDa	Function
Albumins:	Transthyretin Albumin: 45 g · l ⁻¹	50-66 67	Transport of thyroxine and triiodothyronine Maintenance of osmotic pressure; transport of fatty acids, bilirubin, bile acids, steroid hormones, pharmaceuticals and inorganic ions.
α ₁ -Globulins:	Antitrypsin Antichymotrypsin Lipoprotein (HDL) Prothrombin Transcortin Acid glycoprotein Thyroxine-binding globulin	51 58-68 200-400 72 51 44 54	Inhibition of trypsin and other proteases Inhibition of chymotrypsin Transport of lipids Coagulation factor II, thrombin precursor (3.4.21.5) Transport of cortisol, corticosterone and progesterone Transport of progesterone Transport of iodothyronins
α ₂ -Globulins:	Ceruloplasmin Antithrombin III Haptoglobin Cholinesterase (3.1.1.8) Plasminogen Macroglobulin Retinol-binding protein Vitamin D-binding protein	135 58 100 ca. 350 90 725 21 52	Transport of copper ions Inhibition of blood clotting Binding of hemoglobin Cleavage of choline esters Precursor of plasmin (3.4.21.7), breakdown of blood clots Binding of proteases, transport of zinc ions Transport of vitamin A Transport of calcium
β-Globulins:	Lipoprotein (LDL) Transferrin Fibrinogen Sex hormone-binding globulin Transcobalamin C-reactive protein	2.000-4.500 80 340 65 38 110	Transport of lipids Transport of iron ions Coagulation factor I Transport of testosterone and estradiol Transport of vitamin B ₁₂ Complement activation
γ-Globulins:	IgG IgA IgM IgD IgE	150 162 900 172 196	Late antibodies Mucosa-protecting antibodies Early antibodies B-lymphocyte receptors Reagents

B. Electrophoresis



Lipoproteins

Most lipids are barely soluble in water, and many have amphipathic properties. In the blood, free triacylglycerols would coalesce into drops that could cause fat embolisms. By contrast, amphipathic lipids would be deposited in the blood cells' membranes and would dissolve them. Special precautions are therefore needed for lipid transport in the blood. While long-chain fatty acids are bound to albumin and short-chain ones are dissolved in the plasma (see p. 276), other lipids are transported in **lipoprotein complexes**, of which there are several types in the blood plasma, with different sizes and composition.

A. Composition of lipoprotein complexes ●

Lipoproteins are spherical or discoid aggregates of **lipids** and **apoproteins**. They consist of a nucleus of apolar lipids (triacylglycerols and cholesterol esters) surrounded by a single-layered shell approximately 2 nm thick of amphipathic lipids (phospholipids and cholesterol; the example shown here is LDL). The shell, in which the apoproteins are also deposited, gives the surfaces of the particles polar properties and thereby prevents them from aggregating into large particles. The larger the lipid nucleus of a lipoprotein is—i.e., the larger the number of apolar lipids it contains—the lower its density is.

Lipoproteins are classified into five groups. In order of decreasing size and increasing density, these are: *chylomicrons*, *VLDLs* (very-low-density lipoproteins), *IDLs* (intermediate-density lipoproteins), *LDLs* (low-density lipoproteins), and *HDLs* (high-density lipoproteins). The proportions of apoproteins range from 1% in chylomicrons to over 50% in HDLs. These proteins serve less for solubility purposes, but rather function as recognition molecules for the membrane receptors and enzymes that are involved in lipid exchange.

B. Transport functions

The classes of lipoproteins differ not only in their composition, but also in the ways in which they originate and function.

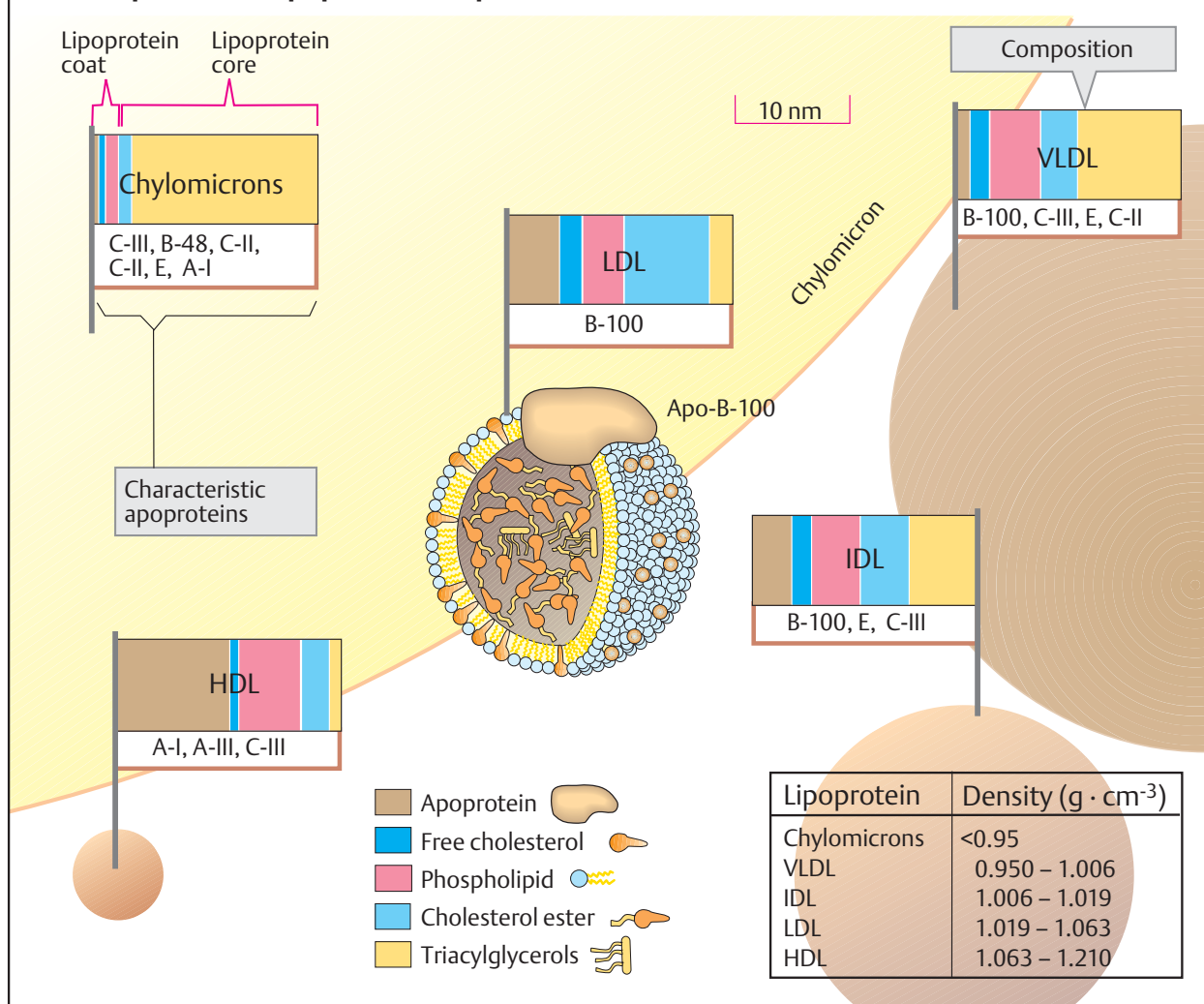
The **chylomicrons** take care of the transport of triacylglycerols from the intestine to the tissues. They are formed in the intestinal mu-

cosa and reach the blood via the lymphatic system (see p. 266). In the peripheral vessel—particularly in muscle and adipose tissue—*lipoprotein lipase* [1] on the surface of the vascular endothelia hydrolyzes most of the triacylglycerols. Chylomicron breakdown is activated by the transfer of apoproteins E and C from HDL. While the fatty acids released and the glycerol are taken up by the cells, the chylomicrons gradually become converted into **chylomicron remnants**, which are ultimately removed from the blood by the liver.

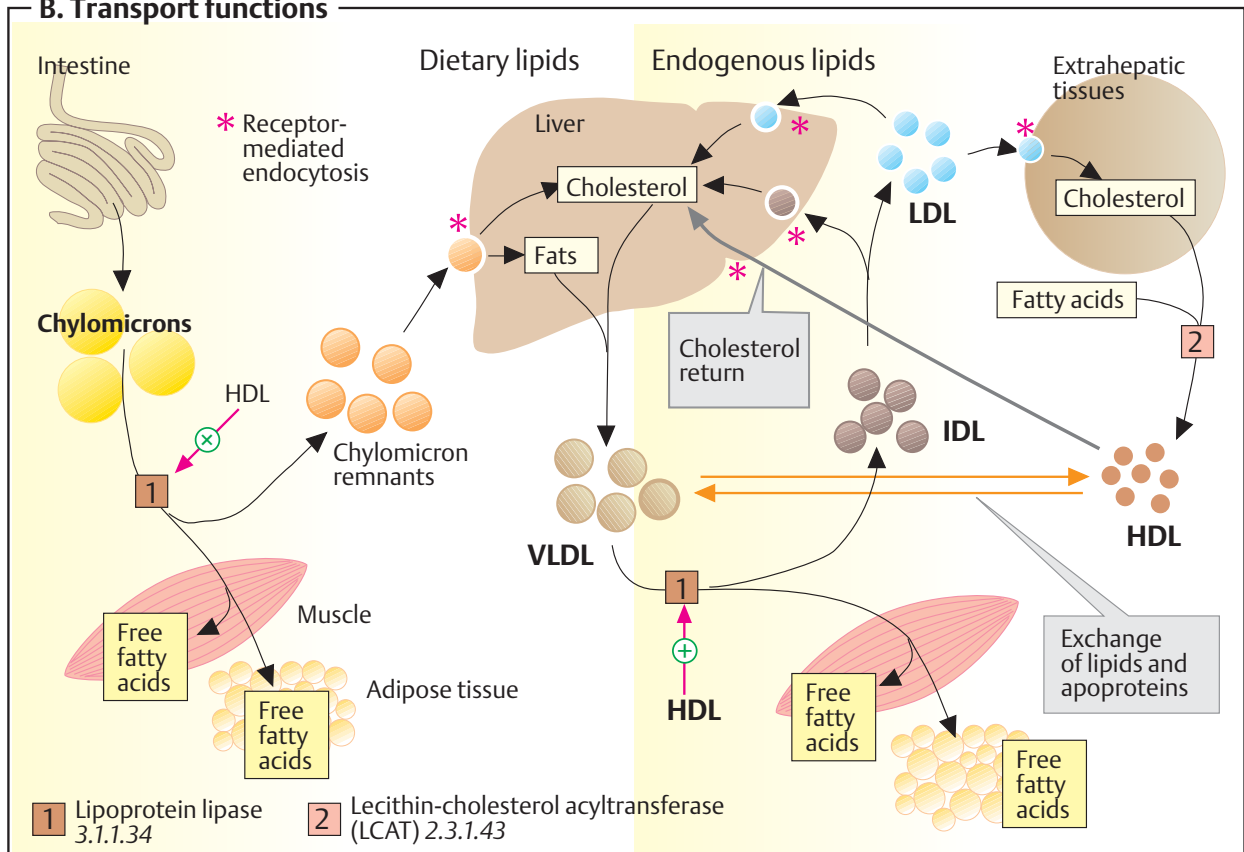
VLDLs, **IDLs**, and **LDLs** are closely related to one another. VLDLs formed in the liver (see p. 312) transport triacylglycerols, cholesterol, and phospholipids to other tissues. Like chylomicrons, they are gradually converted into IDL and LDL under the influence of *lipoprotein lipase* [1]. This process is also stimulated by HDL. Cells that have a demand for cholesterol bind LDL through an interaction between their **LDL receptor** and ApoB-100, and then take up the complete particle through **receptor-mediated endocytosis**. This type of transport is mediated by depressions in the membrane ("*coated pits*"), the interior of which is lined with the protein *clathrin*. After LDL binding, clathrin promotes invagination of the pits and pinching off of vesicles ("*coated vesicles*"). The clathrin then dissociates off and is reused. After fusion of the vesicle with lysosomes, the LDL particles are broken down (see p. 234), and cholesterol and other lipids are used by the cells.

The **HDLs** also originate in the liver. They return the excess cholesterol formed in the tissues to the liver. While it is being transported, cholesterol is acylated by *lecithin cholesterol acyltransferase* (LCAT). The cholesterol esters formed are no longer amphipathic and can be transported in the core of the lipoproteins. In addition, HDLs promote chylomicron and VLDL turnover by exchanging lipids and apoproteins with them (see above).

A. Composition of lipoprotein complexes



B. Transport functions



Hemoglobin

The most important task of the red blood cells (erythrocytes) is to **transport** molecular oxygen (O_2) from the lungs into the tissues, and carbon dioxide (CO_2) from the tissues back into the lungs. To achieve this, the higher organisms require a special transport system, since O_2 is *poorly soluble in water*. For example, only around 3.2 mL O_2 is soluble in 1 L blood plasma. By contrast, the protein **hemoglobin** (Hb), contained in the erythrocytes, can bind a maximum of 220 mL O_2 per liter—70 times the physically soluble amount.

The Hb content of blood, at 140–180 g L^{-1} in men and 120–160 g L^{-1} in women, is twice as high as that of the plasma proteins (50–80 g L^{-1}). Hb is therefore also responsible for the majority of the blood proteins' pH buffer capacity (see p. 288).

A. Hemoglobin: structure ●

In adults, hemoglobin (**HbA**; see below) is a *heterotetramer* consisting of two α -chains and two β -chains, each with masses of 16 kDa. The α - and β -chains have different sequences, but are similarly folded. Some 80% of the amino acid residues form α -*helices*, which are identified using the letters A–H.

Each subunit carries a **heme group** (formula on p. 106), with a central bivalent **iron ion**. When O_2 binds to the heme iron (**Oxygenation** of Hb) and when O_2 is released (**Deoxygenation**), the oxidation stage of the iron does *not* change. Oxidation of Fe^{2+} to Fe^{3+} only occurs occasionally. The oxidized form, *methemoglobin*, is then no longer able to bind O_2 . The proportion of Met-Hb is kept low by reduction (see p. 284) and usually amounts to only 1–2%.

Four of the six coordination sites of the iron in hemoglobin are occupied by the nitrogen atoms of the pyrrol rings, and another is occupied by a histidine residue of the globin (the *proximal histidine*). The iron's sixth site is coordinated with oxygen in oxyhemoglobin and with H_2O in deoxyhemoglobin.

B. Hemoglobin: allosteric effects ●

Like aspartate carbamoyltransferase (see p. 116), Hb can exist in two different states (*conformations*), known as the T form and

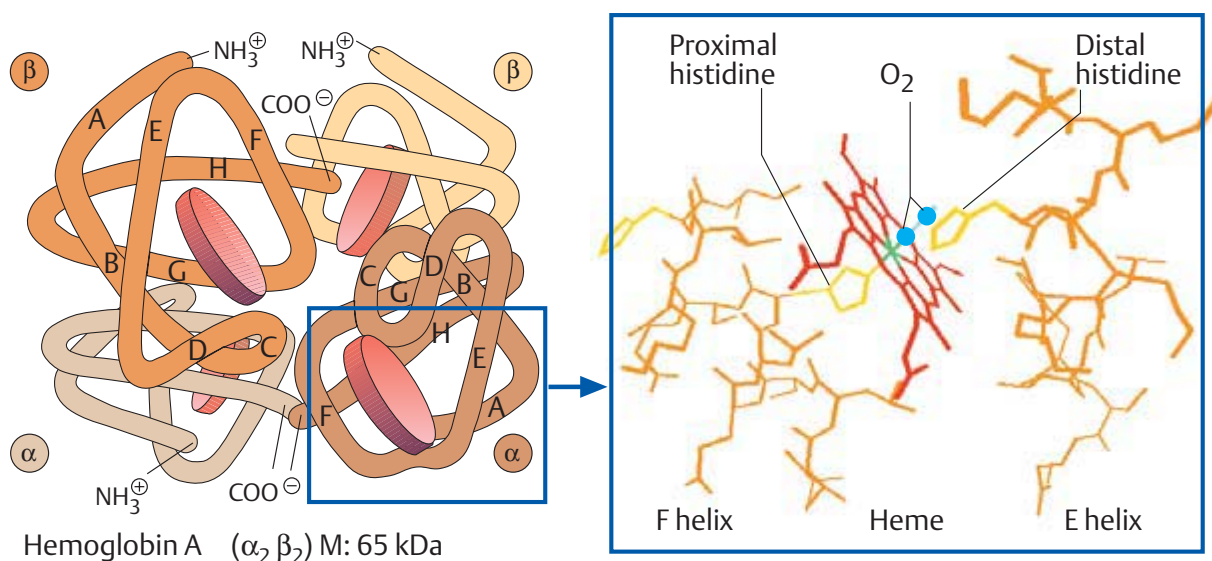
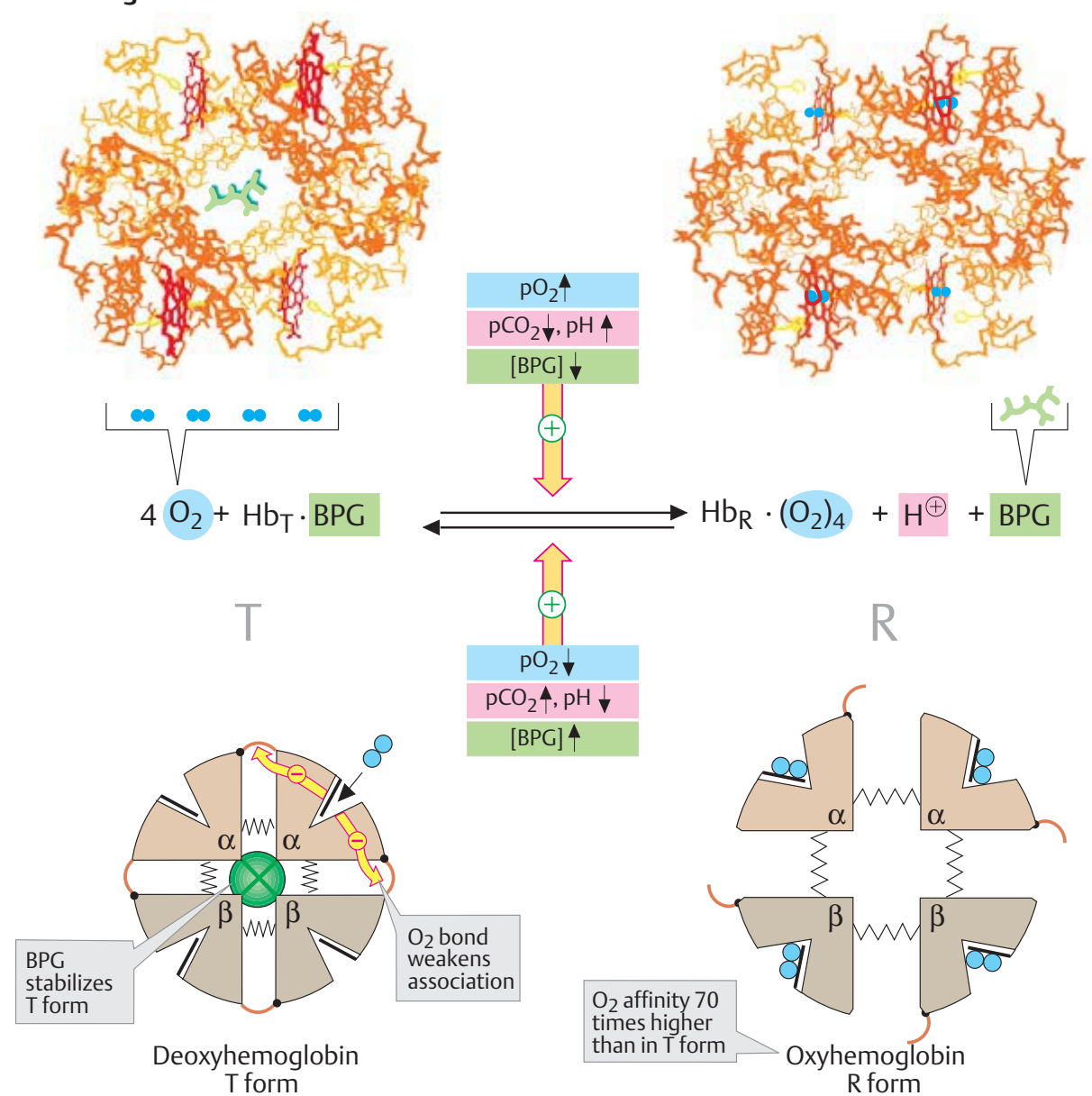
the R form. The **T form** (for *tense*; left) and has a much *lower* O_2 affinity than the R form (for *relaxed*; right).

Binding of O_2 to one of the subunits of the T form leads to a local conformational change that weakens the association between the subunits. Increasing O_2 partial pressure thus means that more and more molecules convert to the higher-affinity R form. This **cooperative interaction** between the subunits increases the O_2 affinity of Hb with increasing O_2 concentrations—i.e., the O_2 saturation curve is **sigmoidal** (see p. 282).

Various **allosteric effectors** influence the equilibrium between the T and R forms and thereby regulate the O_2 binding behavior of hemoglobin (yellow arrows). The most important effectors are CO_2 , H^+ , and 2,3-bisphosphoglycerate (see p. 282).

Further information

As mentioned above, hemoglobin in adults consists of two α - and two β -chains. In addition to this main form (**HbA₁**, $\alpha_2\beta_2$), adult blood also contains small amounts of a second form with a higher O_2 affinity in which the β -chains are replaced by δ -chains (**HbA₂**, $\alpha_2\delta_2$). Two other forms occur during embryonic and fetal development. In the first three months, **embryonic hemoglobins** are formed, with the structure $\zeta_2\varepsilon_2$ and $\alpha_2\varepsilon_2$. Up to the time of birth, **fetal hemoglobin** then predominates (HbF, $\alpha_2\gamma_2$), and it is gradually replaced by HbA during the first few months of life. Embryonic and fetal hemoglobins have higher O_2 affinities than HbA, as they have to take up oxygen from the maternal circulation.

A. Hemoglobin: structure**B. Hemoglobin: allosteric effects**

Gas transport

Most tissues are constantly dependent on a supply of molecular oxygen (O_2) to maintain their oxidative metabolism. Due to its poor solubility, O_2 is bound to hemoglobin for transport in the blood (see p. 280). This not only increases the oxygen transport capacity, but also allows regulation of O_2 uptake in the lungs and O_2 release into tissues.

A. Regulation of O_2 transport ①

When an enzyme reacts to effectors (substrates, activators, or inhibitors) with conformational changes that increase or reduce its activity, it is said to show *allosteric behavior* (see p. 116). Allosteric enzymes are usually oligomers with several subunits that mutually influence each other.

Although **hemoglobin** is not an enzyme (it releases the bound oxygen without changing it), it has all the characteristics of an allosteric protein. Its effectors include oxygen, which as a *positive homotropic effector* promotes its own binding. The **O_2 saturation curve** of hemoglobin is therefore markedly sigmoidal in shape (2, curve 2). The non-sigmoidal saturation curve of the muscular protein **myoglobin** is shown for comparison (curve 1). The structure of myoglobin (see p. 336) is similar to that of a subunit of hemoglobin, but as a monomer it does not exhibit any allosteric behavior.

CO_2 , H^+ , and a special metabolite of erythrocytes—**2,3-bisphosphoglycerate (BPG)**—act as *heterotropic effectors* of hemoglobin. BPG is synthesized from *1,3-bisphosphoglycerate*, an intermediate of glycolysis (see p. 150), and it can be returned to glycolysis again by breakdown into 2-phosphoglycerate (1), with loss of an ATP.

BPG binds selectively to *deoxy-Hb*, thereby increasing its amount of equilibrium. The result is *increased O_2 release* at constant pO_2 . In the diagram, this corresponds to a **right shift** of the saturation curve (2, curve 3). CO_2 and H^+ act in the same direction as BPG. Their influence on the position of the curve has long been known as the **Bohr effect**.

The effects of CO_2 and BPG are *additive*. In the presence of both effectors, the saturation curve of isolated Hb is similar to that of whole blood (curve 4).

B. Hemoglobin and CO_2 transport ①

Hemoglobin is also decisively involved in the transport of carbon dioxide (CO_2) from the tissues to the lungs.

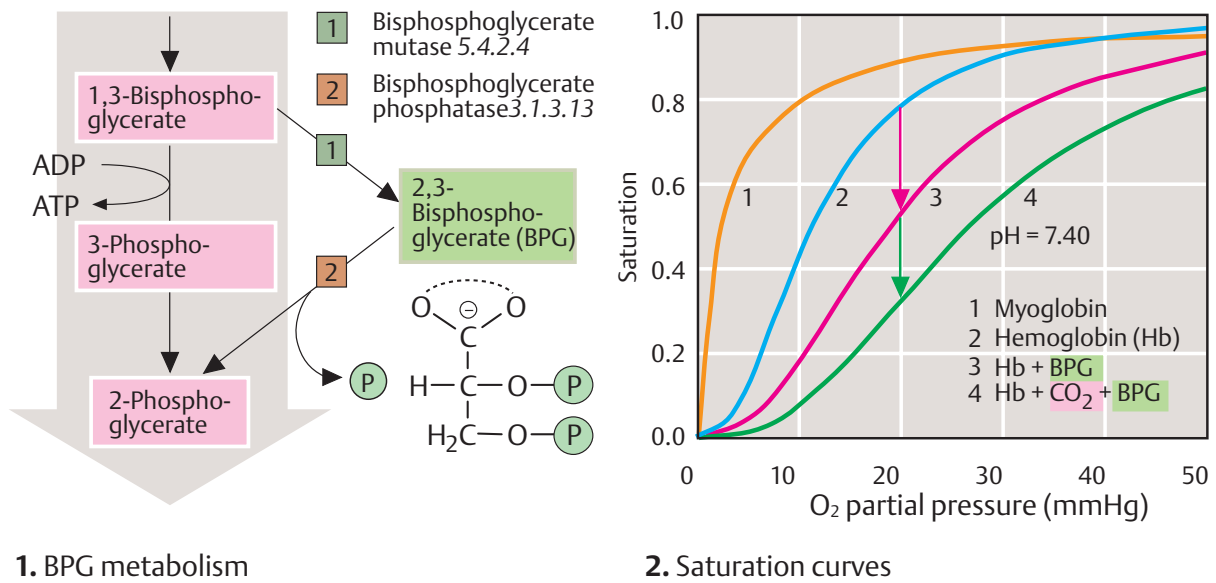
Some 5% of the CO_2 arising in the tissues is covalently bound to the N terminus of hemoglobin and transported as *carbaminohemoglobin* (not shown). About 90% of the CO_2 is first converted in the periphery into *hydrogen carbonate* (HCO_3^-), which is more soluble (bottom). In the lungs (top), CO_2 is regenerated again from HCO_3^- and can then be exhaled.

These two processes are coupled to the oxygenation and deoxygenation of Hb. *Deoxy-Hb* is a stronger base than oxy-Hb. It therefore binds additional protons (about $0.7 H^+$ per tetramer), which promotes the formation of HCO_3^- from CO_2 in the peripheral tissues. The resulting HCO_3^- is released into the plasma via an antiporter in the erythrocyte membrane in exchange for Cl^- , and passes from the plasma to the lungs. In the lungs, the reactions described above then proceed in reverse order: deoxy-Hb is oxygenated and releases protons. The protons shift the HCO_3^-/CO_2 equilibrium to the left and thereby promote CO_2 release.

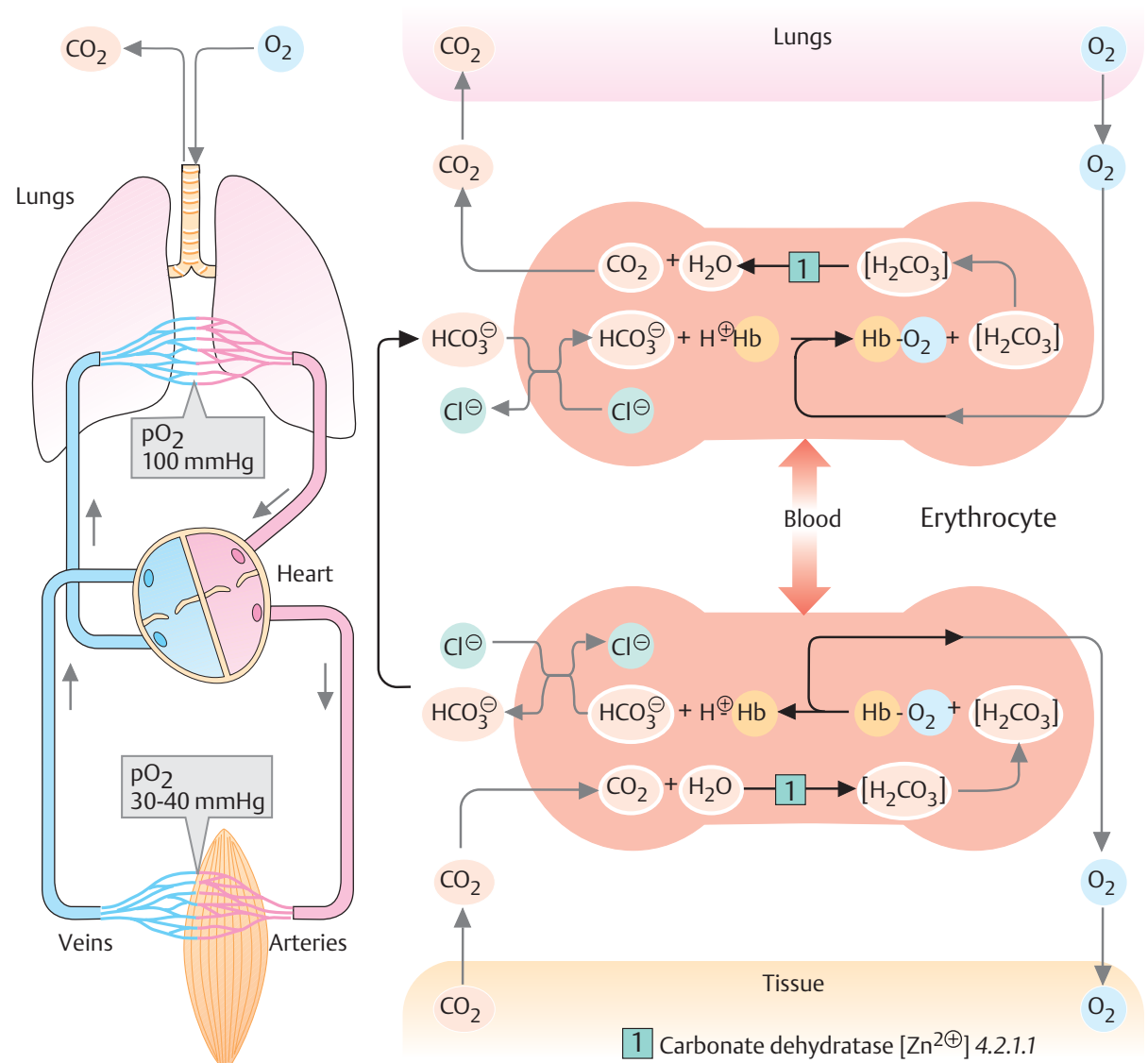
O_2 binding to Hb is regulated by H^+ ions (i.e., by the pH value) via the same mechanism. High concentrations of CO_2 such as those in tissues with intensive metabolism locally increase the H^+ concentration and thereby reduce hemoglobin's O_2 affinity (Bohr effect; see above). This leads to increased O_2 release and thus to an improved oxygen supply.

The adjustment of the equilibrium between CO_2 and HCO_3^- is relatively slow in the uncatalyzed state. It is therefore accelerated in the erythrocytes by *carbonate dehydratase* (carbonic anhydrase) [1])—an enzyme that occurs in high concentrations in the erythrocytes.

A. Regulation of O₂ transport



D. Hemoglobin and CO₂ transport



Erythrocyte metabolism

Cells living in aerobic conditions are dependent on molecular oxygen for energy production. On the other hand, O_2 constantly gives rise to small quantities of toxic substances known as **reactive oxygen species** (ROS). These substances are powerful *oxidation agents* or extremely reactive *free radicals* (see p. 32), which damage cellular structures and functional molecules. Due to their role in O_2 transport, the erythrocytes are constantly exposed to high concentrations of O_2 and are therefore particularly at risk from ROS.

A. Reactive oxygen species ●

The dioxygen molecule (O_2) contains two unpaired electrons—i. e., it is a *diradical*. Despite this, O_2 is relatively stable due to its special electron arrangement. However, if the molecule takes up an extra electron (**a**), the highly reactive **superoxide radical** (O_2^-) arises. Another reduction step (**b**) leads to the **peroxide anion** (O_2^{2-}), which easily binds protons and thus becomes **hydrogen peroxide** (H_2O_2). Inclusion of a third electron (**c**) leads to cleavage of the molecule into the ions O^{2-} and O^- . While O^{2-} can form **water** by taking up two protons, protonation of O^- provides the extremely dangerous **hydroxy radical** (OH). A fourth electron transfer and subsequent protonation also convert O^- into water.

The synthesis of ROS can be catalyzed by iron ions, for example. Reaction of O_2 with FMN or FAD (see p. 32) also constantly produces ROS. By contrast, reduction of O_2 by *cytochrome c-oxidase* (see p. 140) is “clean,” as the enzyme does not release the intermediates. In addition to antioxidants (**B**), **enzymes** also provide protection against ROS: *superoxide dismutase* [1] breaks down (“disproportionates”) two superoxide molecules into O_2 and the less damaging H_2O_2 . The latter is in turn disproportionated into O_2 and H_2O by heme-containing *catalase* [2].

B. Biological antioxidants ●

To protect them against ROS and other radicals, all cells contain **antioxidants**. These are *reducing agents* that react easily with oxidative substances and thus protect more important molecules from oxidation. Biological

antioxidants include vitamins C and E (see pp. 364, 368), coenzyme Q (see p. 104), and several carotenoids (see pp. 132, 364). Bilirubin, which is formed during heme degradation (see p. 194), also serves for protection against oxidation.

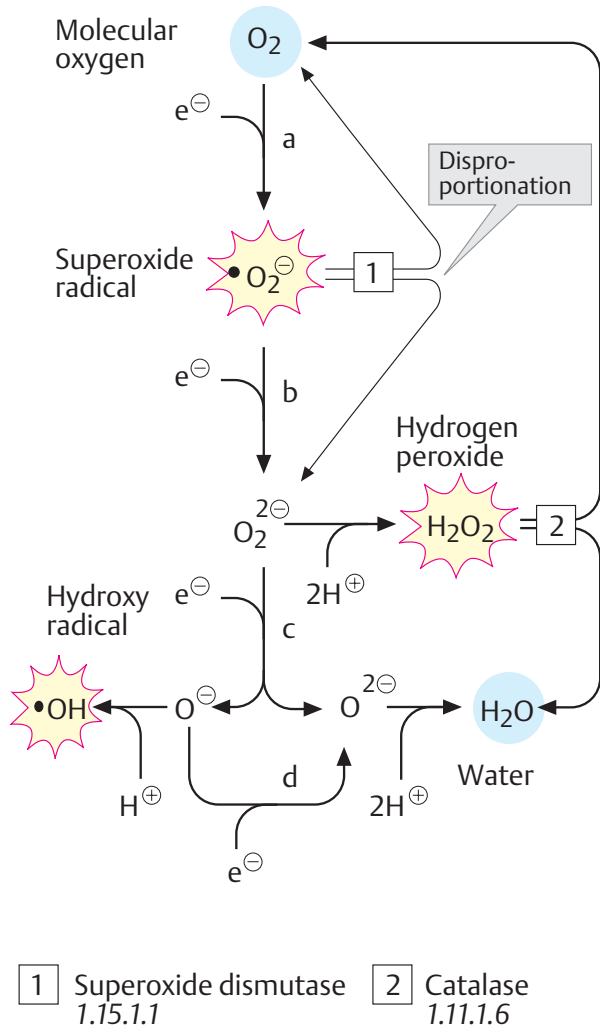
Glutathione, a tripeptide that occurs in high concentrations in almost all cells, is particularly important. Glutathione (sequence: Glu–Cys–Gly) contains an atypical γ -peptide bond between Glu and Cys. The thiol group of the cysteine residue is redox-active. Two molecules of the reduced form (GSH, top) are bound to the disulfide (GSSG, bottom) during oxidation.

C. Erythrocyte metabolism ●

Erythrocytes also have systems that can inactivate ROS (superoxide dismutase, catalase, GSH). They are also able to repair damage caused by ROS. This requires products that are supplied by the erythrocytes’ **maintenance metabolism**, which basically only involves anaerobic glycolysis (see p. 150) and the pentose phosphate pathway (PPP; see p. 152).

The **ATP** formed during glycolysis serves mainly to supply Na^+/K^+ -ATPase, which maintains the erythrocytes’ membrane potential. The allosteric effector **2,3-BPG** (see p. 282) is also derived from glycolysis. The PPP supplies **NADPH+H⁺**, which is needed to regenerate **glutathione** (GSH) from GSSG with the help of *glutathione reductase* [3]. GSH, the most important antioxidant in the erythrocytes, serves as a coenzyme for *glutathione peroxidase* [5]. This selenium-containing enzyme detoxifies H_2O_2 and hydroperoxides, which arise during the reaction of ROS with unsaturated fatty acids in the erythrocyte membrane. The reduction of methemoglobin ($Hb\ Fe^{3+}$) to $Hb\ (Hb\ Fe^{2+})$, [4]) is carried out by GSH or ascorbate by a non-enzymatic pathway; however, there are also NAD(P)H-dependent Met-Hb reductases.

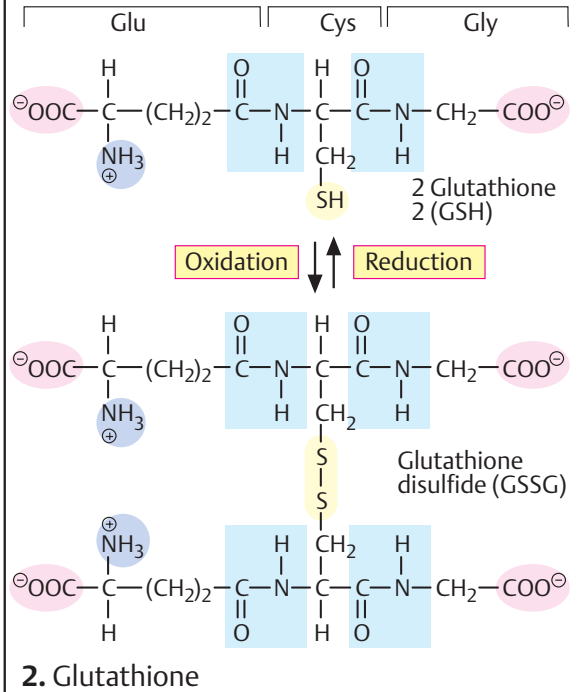
A. Reactive oxygen species



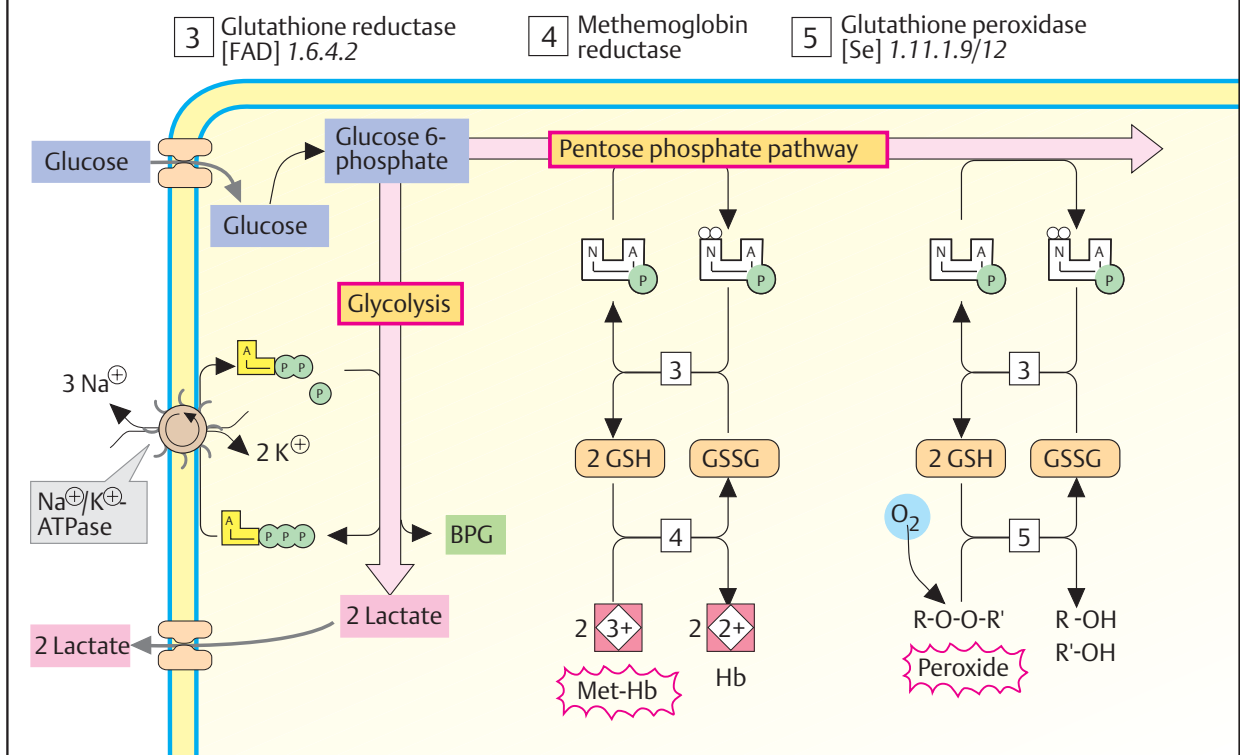
- B. Biological antioxidants

Quinols and enols	α -Tocopherol (vitamin E) Ubiquinol (coenzyme Q) Ascorbic acid (vitamin C)
Carotenoids	β -Carotin Lycopin
Others	Glutathione Bilirubin

1. Examples



C. Erythrocyte metabolism



Iron metabolism

A. Distribution of iron ●

Iron (Fe) is quantitatively the most important trace element (see p. 362). The human body contains 4–5 g iron, which is almost exclusively present in protein-bound form. Approximately three-quarters of the total amount is found in **heme proteins** (see pp. 106, 192), mainly hemoglobin and myoglobin. About 1% of the iron is bound in **iron-sulfur clusters** (see p. 106), which function as cofactors in the respiratory chain, in photosynthesis, and in other redox chains. The remainder consists of iron in transport and storage proteins (transferrin, ferritin; see B).

B. Iron metabolism ●

Iron can only be resorbed by the bowel in bivalent form (i.e., as Fe^{2+}). For this reason, reducing agents in food such as ascorbate (vitamin C; see p. 368) promote **iron uptake**. Via transporters on the luminal and basal side of the enterocytes, Fe^{2+} enters the blood, where it is bound by *transferrin*. Part of the iron that is taken up is stored in the bowel in the form of *ferritin* (see below). Heme groups can also be resorbed by the small intestine.

Most of the resorbed iron serves for the formation of red blood cells in the bone marrow (**erythropoiesis**, top). As discussed on p. 192, it is only in the final step of hem biosynthesis that Fe^{2+} is incorporated by *ferrochelatase* into the previously prepared tetrapyrrole framework.

In the blood, 2.5–3.0 g of hemoglobin iron circulates as a component of the erythrocytes (top right). Over the course of several months, the flexibility of the red blood cells constantly declines due to damage to the membrane and cytoskeleton. Old erythrocytes of this type are taken up by macrophages in the spleen and other organs and broken down. The organic part of the heme is oxidized into bilirubin (see p. 194), while the iron returns to the plasma pool. The quantity of heme iron recycled per day is much larger than the amount resorbed by the intestines.

Transferrin, a β -globulin with a mass of 80 kDa, serves to transport iron in the blood. This monomeric protein consists of two similar domains, each of which binds an Fe^{2+} ion

very tightly. Similar iron transport proteins are found in secretions such as saliva, tears, and milk; these are known as lactoferrins (bottom right). Transferrin and the lactoferrins maintain the concentration of *free* iron in body fluids at values below $10^{-10} \text{ mol L}^{-1}$. This low level prevents bacteria that require free iron as an essential growth factor from proliferating in the body. Like LDLs (see p. 278), transferrin and the lactoferrins are taken up into cells by *receptor-mediated endocytosis*.

Excess iron is incorporated into **ferritin** and stored in this form in the liver and other organs. The ferritin molecule consists of 24 subunits and has the shape of a hollow sphere (bottom left). It takes up Fe^{2+} ions, which in the process are oxidized to Fe^{3+} and then deposited in the interior of the sphere as *ferrihydrate*. Each ferritin molecule is capable of storing several thousand iron ions in this way. In addition to ferritin, there is another storage form, **hemosiderin**, the function of which is not yet clear.

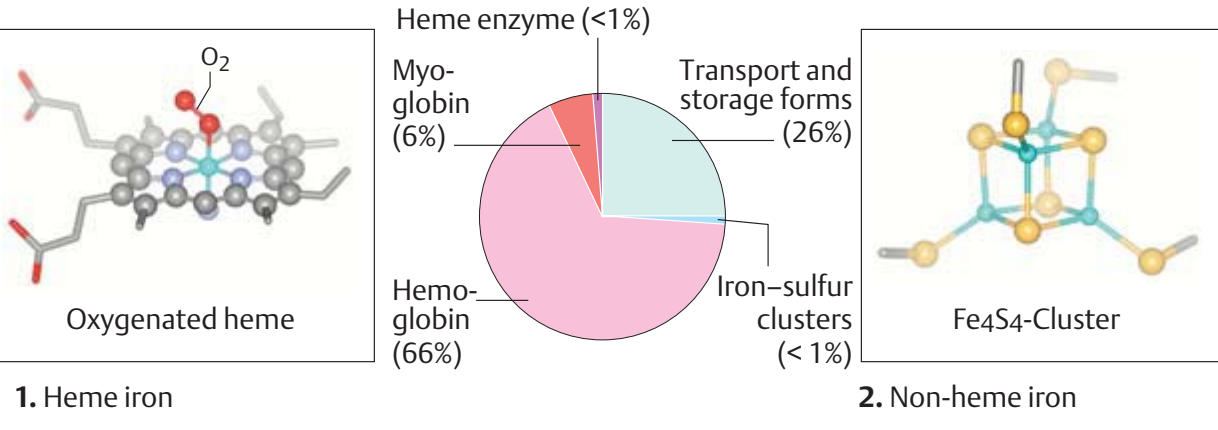
Further information

Disturbances of the iron metabolism are frequent and can lead to severe disease pictures.

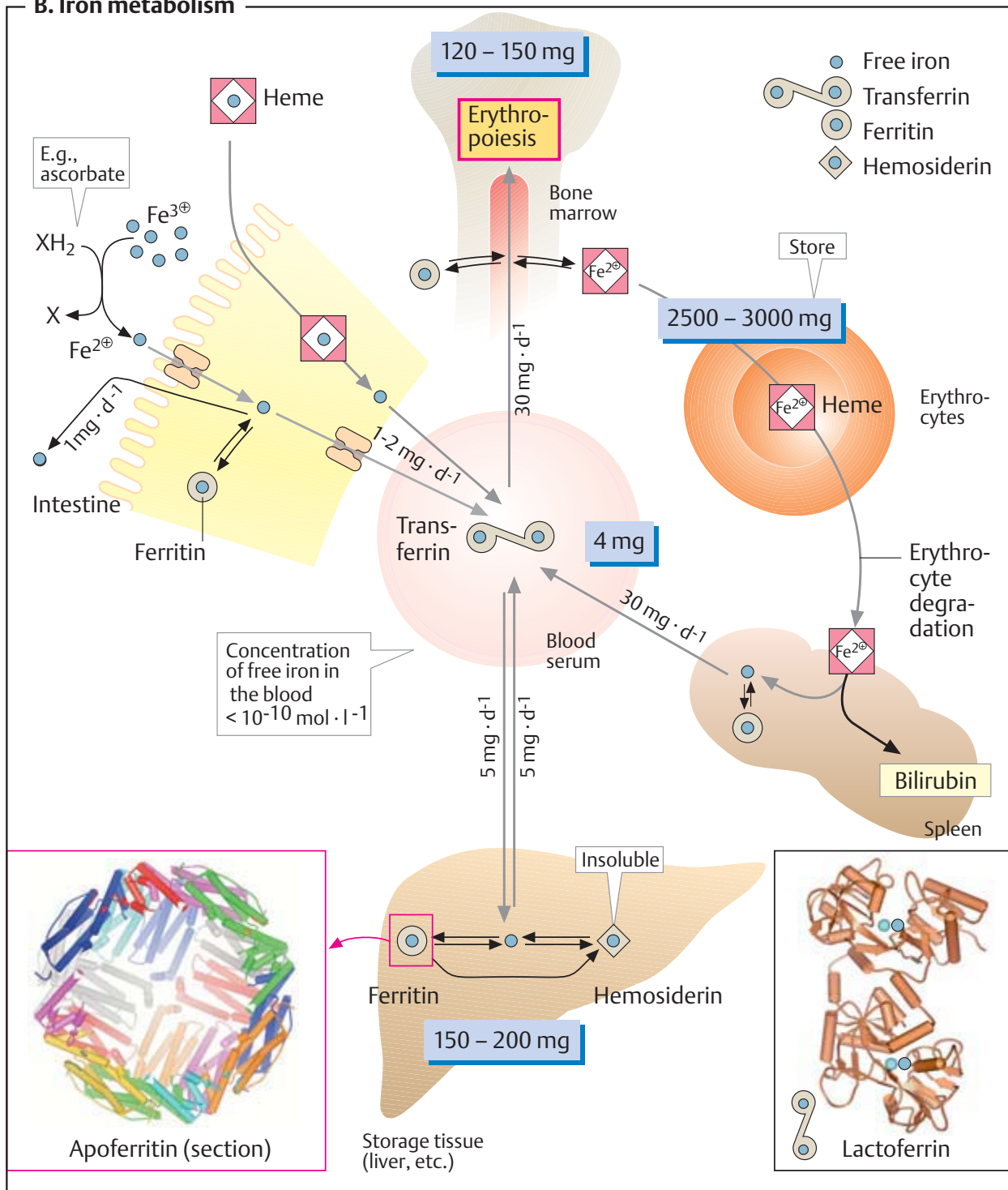
Iron deficiency is usually due to blood loss, or more rarely to inadequate iron uptake. During pregnancy, increased demand can also cause iron deficiency states. In severe cases, reduced hemoglobin synthesis can lead to **anemia** (“iron-deficiency anemia”). In these patients, the erythrocytes are smaller and have less hemoglobin. As their membrane is also altered, they are prematurely eliminated in the spleen.

Disturbances resulting from raised iron concentrations are less frequent. Known as **hemochromatoses**, these conditions can have genetic causes, or may be due to repeated administration of blood transfusions. As the body has practically no means of excreting iron, more and more stored iron is deposited in the organs over time in patients with untreated hemochromatosis, ultimately leading to severe disturbances of organ function.

A. Distribution of iron



B. Iron metabolism



Acid–base balance

A. Hydrogen ion concentration in the blood plasma ①

The H^+ concentration in the blood and extracellular space is approximately 40 nM ($4 \cdot 10^{-8} \text{ mol L}^{-1}$). This corresponds to a pH of 7.40. The body tries to keep this value constant, as large shifts in pH are incompatible with life.

The pH value is kept constant by **buffer systems** that cushion minor disturbances in the *acid–base balance* (C). In the longer term, the decisive aspect is maintaining a balanced equilibrium between H^+ production and uptake and H^+ release. If the blood's buffering capacity is not sufficient, or if the acid–base balance is not in equilibrium—e.g., in kidney disease or during *hypoventilation* or *hyperventilation*—shifts in the plasma pH value can occur. A reduction by more than 0.03 units is known as **acidosis**, and an increase is called **alkalosis**.

B. Acid–base balance ①

Protons are mainly derived from two sources—free acids in the diet and sulfur-containing amino acids. *Acids* taken up with food—e.g., citric acid, ascorbic acid, and phosphoric acid—already release protons in the alkaline pH of the intestinal tract. More important for proton balance, however, are the amino acids **methionine** and **cysteine**, which arise from protein degradation in the cells. Their S atoms are oxidized in the liver to form sulfuric acid, which supplies protons by dissociation into sulfate.

During anaerobic glycolysis in the muscles and erythrocytes, glucose is converted into **lactate**, releasing protons in the process (see p. 338). The synthesis of the **ketone bodies** acetoacetic acid and 3-hydroxybutyric acid in the liver (see p. 312) also releases protons. Normally, the amounts formed are small and of little influence on the proton balance. If acids are formed in large amounts, however (e.g., during *starvation* or in *diabetes mellitus*; see p. 160), they strain the buffer systems and can lead to a reduction in pH (**metabolic acidoses**; *lactacidosis* or *ketoacidosis*).

Only the **kidney** is capable of excreting protons in exchange for Na^+ ions (see p. 326). In

the urine, the H^+ ions are buffered by NH_3 and phosphate.

C. Buffer systems in the plasma ①

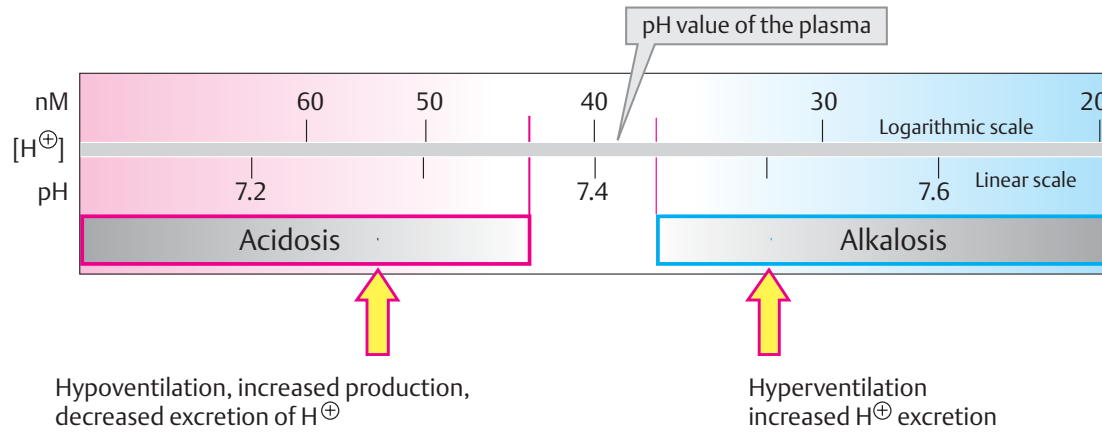
The **buffering capacity** of a buffer system depends on its concentration and its pK_a value. The strongest effect is achieved if the pH value corresponds to the buffer system's pK_a value (see p. 30). For this reason, weak acids with pK_a values of around 7 are best suited for buffering purposes in the blood.

The most important buffer in the blood is the **CO_2 /bicarbonate buffer**. This consists of water, carbon dioxide (CO_2 , the anhydride of carbonic acid H_2CO_3), and hydrogen carbonate (HCO_3^- , bicarbonate). The adjustment of the balance between CO_2 and HCO_3^- is accelerated by the zinc-containing enzyme *carbonate dehydratase* (carbonic anhydrase [1]; see also p. 282). At the pH value of the plasma, HCO_3^- and CO_2 are present in a ratio of about 20 : 1. However, the CO_2 in solution in the blood is in equilibrium with the gaseous CO_2 in the pulmonary alveoli. The CO_2/HCO_3^- system is therefore a powerful *open buffer system*, despite having a not entirely optimal pK_a value of 6.1. Faster or slower respiration increases or reduces CO_2 release in the lungs. This shifts the CO_2/HCO_3^- ratio and thus the plasma pH value (respiratory acidosis or alkalosis). In this way, respiration can compensate to a certain extent for changes in plasma pH values. However, it does *not* lead to the excretion of protons.

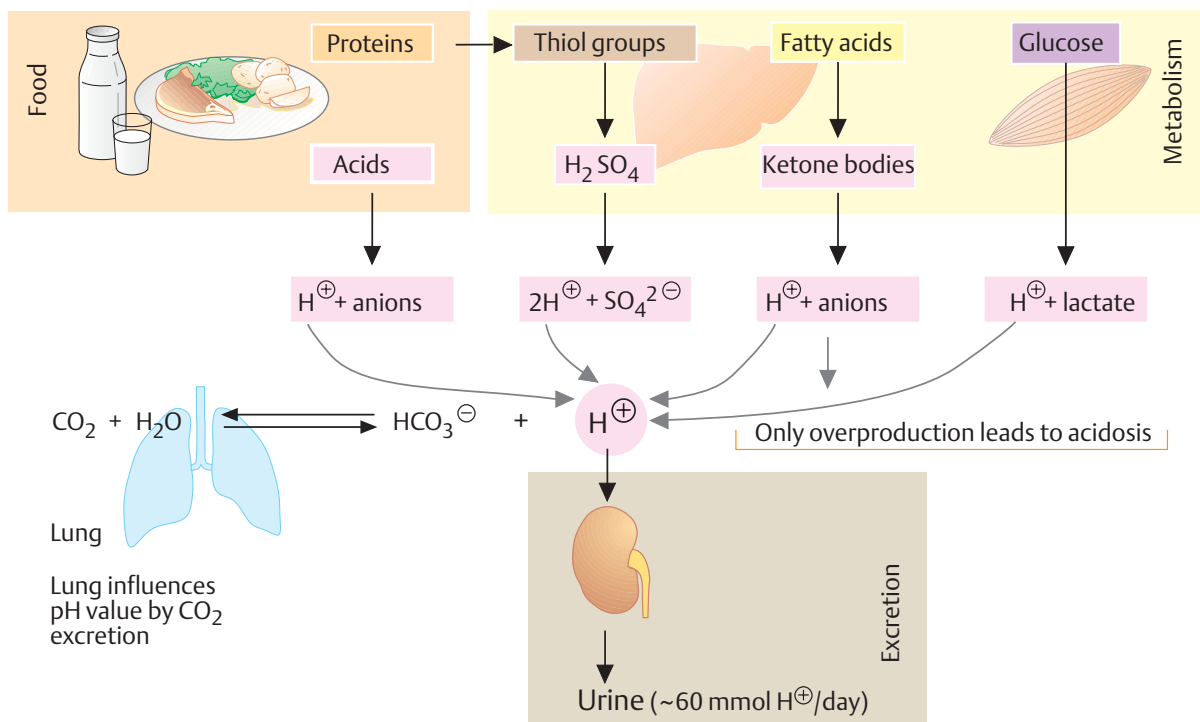
Due to their high concentration, **plasma proteins**—and **hemoglobin** in the erythrocytes in particular—provide about one-quarter of the blood's buffering capacity. The buffering effect of proteins involves contributions from all of the ionizable side chains. At the pH value of blood, the acidic amino acids (Asp, Glu) and histidine are particularly effective.

The second dissociation step in **phosphate** (H_2PO_4/HPO_4^{2-}) also contributes to the buffering capacity of the blood plasma. Although the pK_a value of this system is nearly optimal, its contribution remains small due to the low total concentration of phosphate in the blood (around 1 mM).

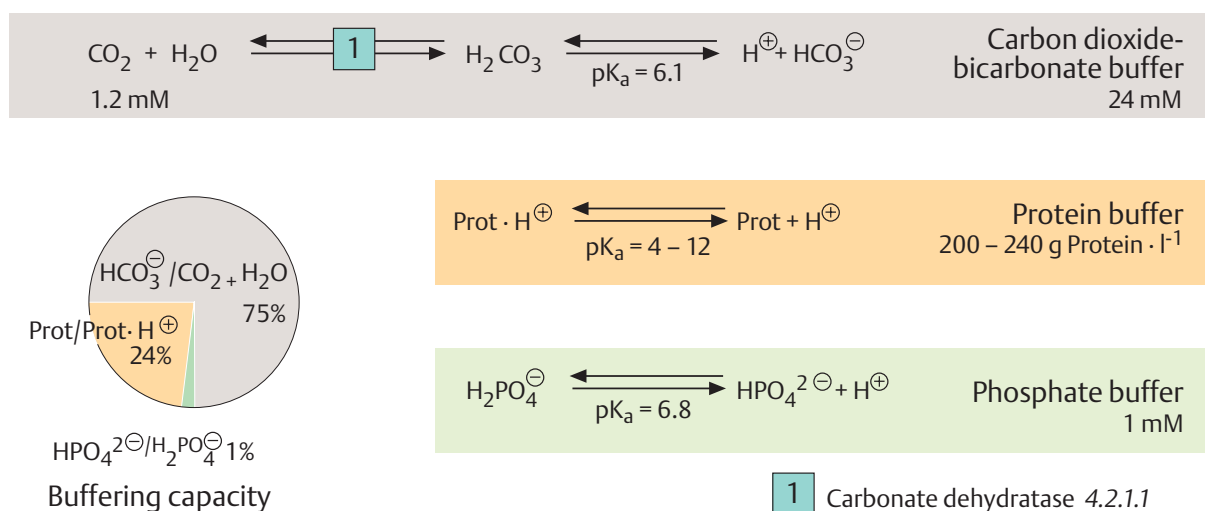
A. Hydrogen ion concentration in the blood plasma



B. Acid–base balance



C. Buffer systems in the plasma



Blood clotting

Following injury to blood vessels, **hemostasis** ensures that blood loss is minimized. Initially, thrombocyte activation leads to contraction of the injured vessel and the formation of a loose clot consisting of thrombocytes (**hemostasis**). Slightly later, the action of the enzyme *thrombin* leads to the formation and deposition in the thrombus of polymeric fibrin (**coagulation, blood clotting**). The coagulation process is discussed here in detail.

A. Blood clotting ●

The most important reaction in blood clotting is the conversion, catalyzed by *thrombin*, of the soluble plasma protein *fibrinogen* (factor I) into polymeric *fibrin*, which is deposited as a fibrous network in the primary thrombus. Thrombin (factor IIa) is a serine proteinase (see p. 176) that cleaves small peptides from fibrinogen. This exposes binding sites that spontaneously allow the fibrin molecules to aggregate into polymers. Subsequent covalent cross-linking of fibrin by a *transglutaminase* (factor XIII) further stabilizes the thrombus.

Normally, thrombin is present in the blood as an inactive proenzyme (see p. 270). Prothrombin is activated in two different ways, both of which represent cascades of enzymatic reactions in which inactive proenzymes (zymogens, symbol: circle) are proteolytically converted into active **proteinases** (symbol: sector of a circle). The proteinases activate the next proenzyme in turn, and so on. Several steps in the cascade require additional **protein factors** (factors III, Va and VIIIa) as well as anionic **phospholipids** (PL; see below) and Ca^{2+} ions. Both pathways are activated by injuries to the vessel wall.

In the **extravascular pathway** (right), *tissue thromboplastin* (factor III), a membrane protein in the deeper layers of the vascular wall, activates coagulation factor VII. The activated form of this (VIIa) autocatalytically promotes its own synthesis and also generates the active factors IXa and Xa from their precursors. With the aid of factor VIIIa, PL, and Ca^{2+} , factor IXa produces additional Xa, which finally—with the support of Va, PL, and Ca^{2+} —releases active thrombin.

The **intravascular pathway** (left) is probably also triggered by vascular injuries. It

leads in five steps via factors XIIa, XIa, IXa, and Xa to the activation of prothrombin. The significance of this pathway in vivo has been controversial since it was found that a genetic deficiency in factor XII does not lead to coagulation disturbances.

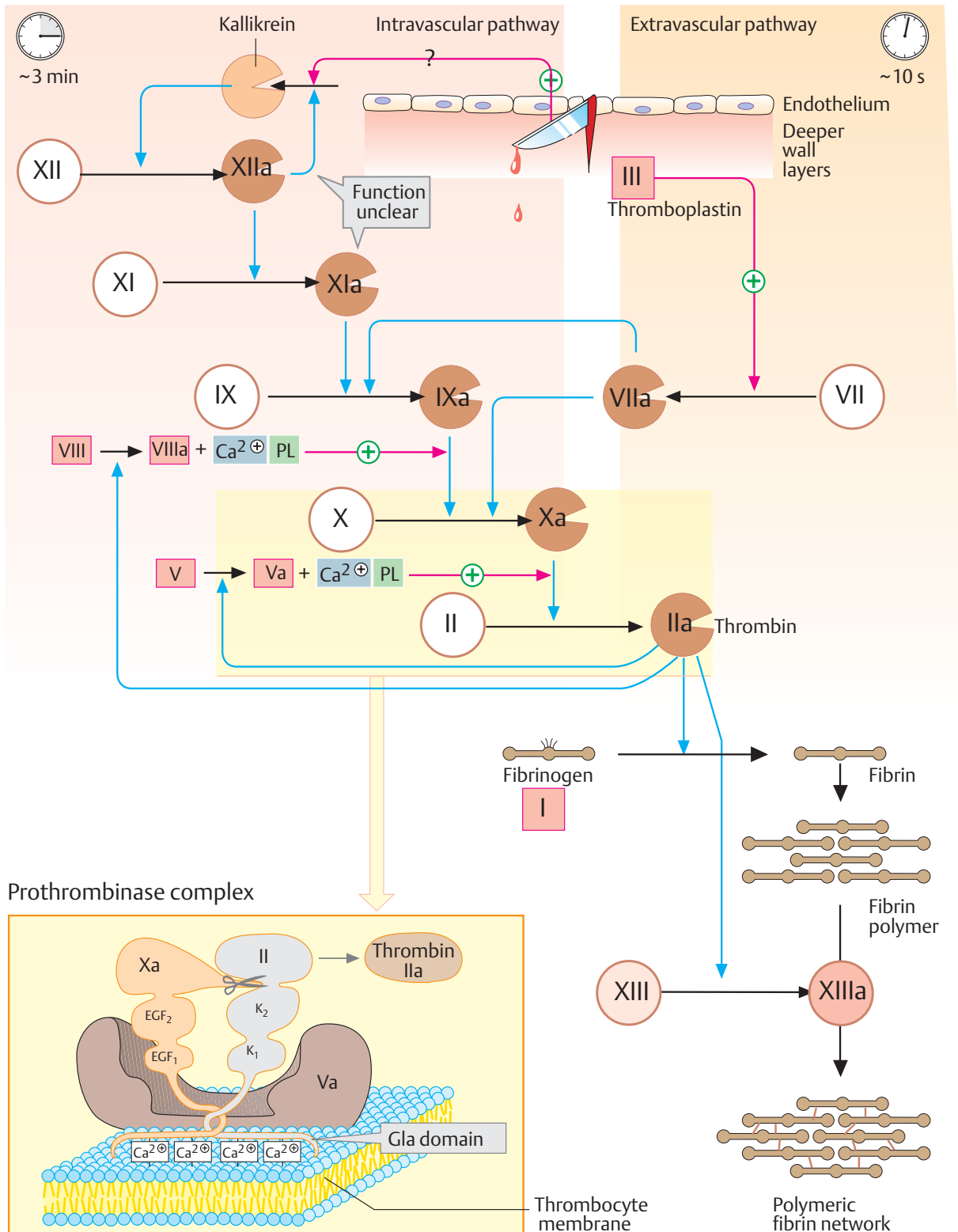
Both pathways depend on the presence of activated **thrombocytes**, on the surface of which several reactions take place. For example, the *prothrombinase complex* (left) forms when factors Xa and II, with the help of Va, bind via Ca^{2+} ions to anionic phospholipids in the thrombocyte membrane. For this to happen, factors II and X have to contain the non-proteinogenic amino acid **γ -carboxyglutamate** (Gla; see p. 62), which is formed in the liver by post-translational carboxylation of the factors. The Gla residues are found in groups in special domains that create contacts to the Ca^{2+} ions. Factors VII and IX are also linked to membrane phospholipids via Gla residues.

Substances that bind Ca^{2+} ions (e. g., *citrate*) prevent Gla-containing factors from attaching to the membrane and therefore inhibit coagulation. Antagonists of *vitamin K*, which is needed for synthesis of the Gla residues (see p. 364) also have anticoagulatory effects. These include dicumarol, for example.

Active **thrombin** not only converts fibrinogen into fibrin, but also indirectly promotes its own synthesis by catalyzing the activation of factors V and VIII. In addition, it catalyzes the activation of factor XIII and thereby triggers the cross-linking of the fibrin.

Regulation of blood clotting (not shown). To prevent the coagulation reaction from becoming excessive, the blood contains a number of anticoagulant substances, including highly effective proteinase inhibitors. For example, *antithrombin III* binds to various serine proteinases in the cascade and thereby inactivates them. *Heparin*, an anticoagulant glycosaminoglycan (see p. 346), potentiates the effect of antithrombin III. *Thrombomodulin*, which is located on the vascular endothelia, also inactivates thrombin. A glycoprotein known as *Protein C* ensures proteolytic degradation of factors V and VIII. As it is activated by thrombin, coagulation is shut down in this way.

A. Blood clotting



Fibrinolysis, blood groups

A. Fibrinolysis ●

The fibrin thrombus resulting from blood clotting (see p. 290) is dissolved again by *plasmin*, a serine proteinase found in the blood plasma. For this purpose, the precursor *plasminogen* first has to be proteolytically activated by enzymes from various tissues. This group includes the *plasminogen activator* from the kidney (*urokinase*) and *tissue plasminogen activator* (t-PA) from vascular endothelia. By contrast, the plasma protein α_2 -antiplasmin, which binds to active plasmin and thereby inactivates it, inhibits fibrinolysis.

Urokinase, t-PA, and streptokinase, a bacterial proteinase with similar activity, are used clinically to dissolve thrombi following *heart attacks*. All of these proteins are expressed recombinantly in bacteria (see p. 262).

B. Blood groups: the ABO system ●

During blood transfusions, immune reactions can occur that destroy the erythrocytes transfused from the donor. These reactions result from the formation of antibodies (see p. 300) directed to certain surface structures on the erythrocytes. Known as **blood group antigens**, these are *proteins* or *oligosaccharides* that can differ from individual to individual. More than 20 different blood group systems are now known. The ABO system and the Rh system are of particular clinical importance.

In the **ABO system**, the carbohydrate parts of glycoproteins or glycolipids act as antigens. In this relatively simple system, there are four *blood groups* (A, B, AB, and O). In individuals with blood groups A and B, the antigens consist of tetrasaccharides that only differ in their terminal sugar (galactose or *N*-acetylgalactosamine). Carriers of the AB blood group have both antigens (A and B). Blood group O arises from an oligosaccharide (the H antigen) that lacks the terminal residue of antigens A and B. The molecular causes for the differences between blood groups are mutations in the *glycosyl transferases* that transfer the terminal sugar to the core oligosaccharide.

Antibodies are only formed against antigens that the individual concerned does *not*

possess. For example, carriers of blood group A form antibodies against antigen B (“anti-B”), while carriers of group B form antibodies against antigen A (“anti-A”). Individuals with blood group O form both types, and those with blood group AB do not form any of these antibodies.

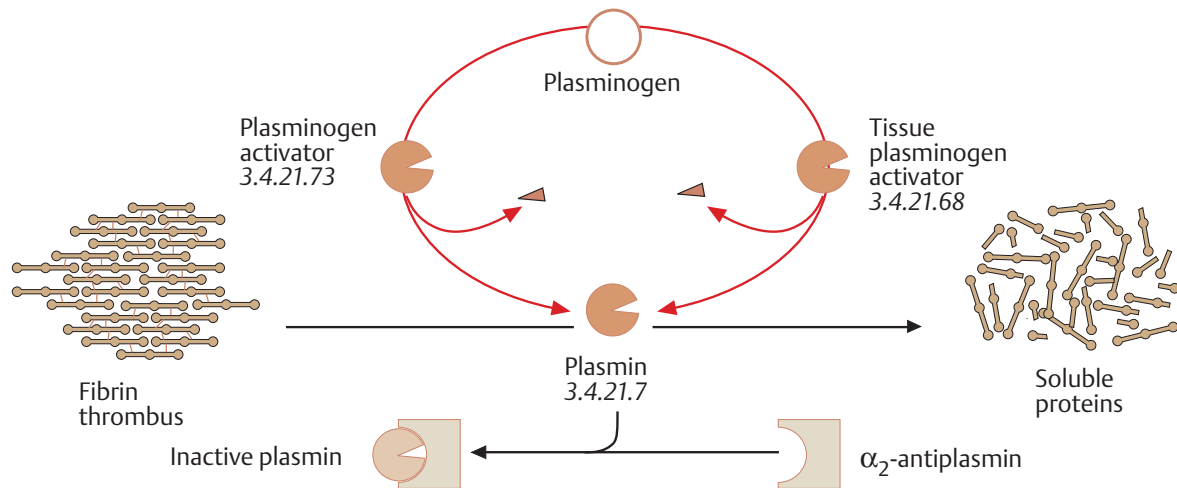
If blood from blood group A is transfused into the circulation of an individual with blood group B, for example, then the anti-A present there binds to the A antigens. The donor erythrocytes marked in this way are recognized and destroyed by the complement system (see p. 298). In the test tube, *agglutination* of the erythrocytes can be observed when donor and recipient blood are incompatible.

The recipient’s serum should not contain any antibodies against the donor erythrocytes, and the donor serum should not contain any antibodies against the recipient’s erythrocytes. Donor blood from blood group O is unproblematic, as its erythrocytes do not possess any antibodies and therefore do not react with anti-A or anti-B in the recipient’s blood. Conversely, blood from the AB group can only be administered to recipients with the AB group, as these are the only ones without antibodies.

In the **Rh system** (not shown), proteins on the surface of the erythrocytes act as antigens. These are known as “rhesus factors,” as the system was first discovered in rhesus monkeys.

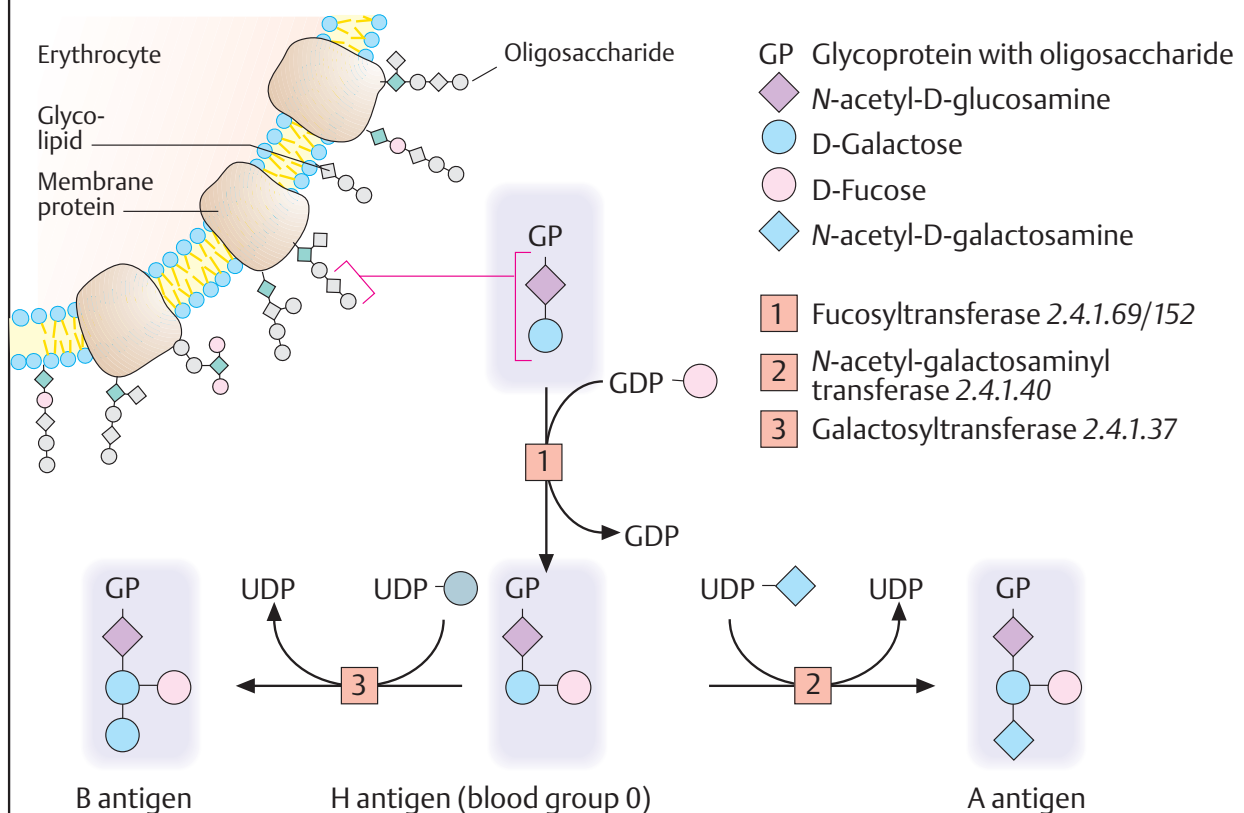
The rhesus D antigen occurs in 84% of all white individuals, who are therefore “*Rh-positive*.” If an Rh-positive child is born to an Rh-negative mother, fetal erythrocytes can enter the mother’s circulation during birth and lead to the formation of antibodies (IgG) against the D antigen. This initially has no acute effects on the mother or child. Complications only arise when there is a second pregnancy with an Rh-positive child, as maternal anti-D antibodies cross the placenta to the fetus even before birth and can trigger destruction of the child’s Rh-positive erythrocytes (*fetal erythroblastosis*).

A. Fibrinolysis



B. Blood groups: the ABO system

Blood group	A	B	AB	O
Genotypes	AA and AO	BB and BO	AB	OO
Antigens				
Antibodies in blood	anti-B	anti-A	—	anti-A anti-B
Frequency in central Europe	40%	16%	4%	40%



Immune response

Viruses, bacteria, fungi, and parasites that enter the body of vertebrates are recognized and attacked by the **immune system**. Endogenous cells that have undergone alterations—e.g., tumor cells—are also usually recognized as foreign and destroyed. The immune system is supported by physiological changes in infected tissue, known as **inflammation**. This reaction makes it easier for the immune cells to reach the site of infection.

Two different systems are involved in the immune response. The **innate immune system** is based on receptors that can distinguish between bacterial and viral surface structures or foreign proteins (known as *antigens*) and those that are endogenous. With the help of these receptors, *phagocytes* bind to the pathogens, absorb them by endocytosis, and break them down. The complement system (see p. 298) is also part of the innate system.

The **acquired** (adaptive) **immune system** is based on the ability of the *lymphocytes* to form highly specific antigen receptors “on suspicion,” without ever having met the corresponding antigen. In humans, there are several billion different lymphocytes, each of which carries a different antigen receptor. If this type of receptor recognizes “its” cognate antigen, the lymphocyte carrying it is activated and then plays its special role in the immune response.

In addition, a distinction is made between cellular and humoral immune responses. The *T lymphocytes* (T cells) are responsible for **cellular immunity**. They are named after the thymus, in which the decisive steps in their differentiation take place. Depending on their function, another distinction is made between *cytotoxic T cells* (green) and *helper T cells* (blue). **Humoral immunity** is based on the activity of the *B lymphocytes* (B cells, light brown), which mature in the bone marrow. After activation by T cells, B cells are able to release soluble forms of their specific antigen receptors, known as *antibodies* (see p. 300), into the blood plasma. The immune system’s “memory” is represented by memory cells. These are particularly long-lived cells that can arise from any of the lymphocyte types described.

A. Simplified diagram of the immune response ①

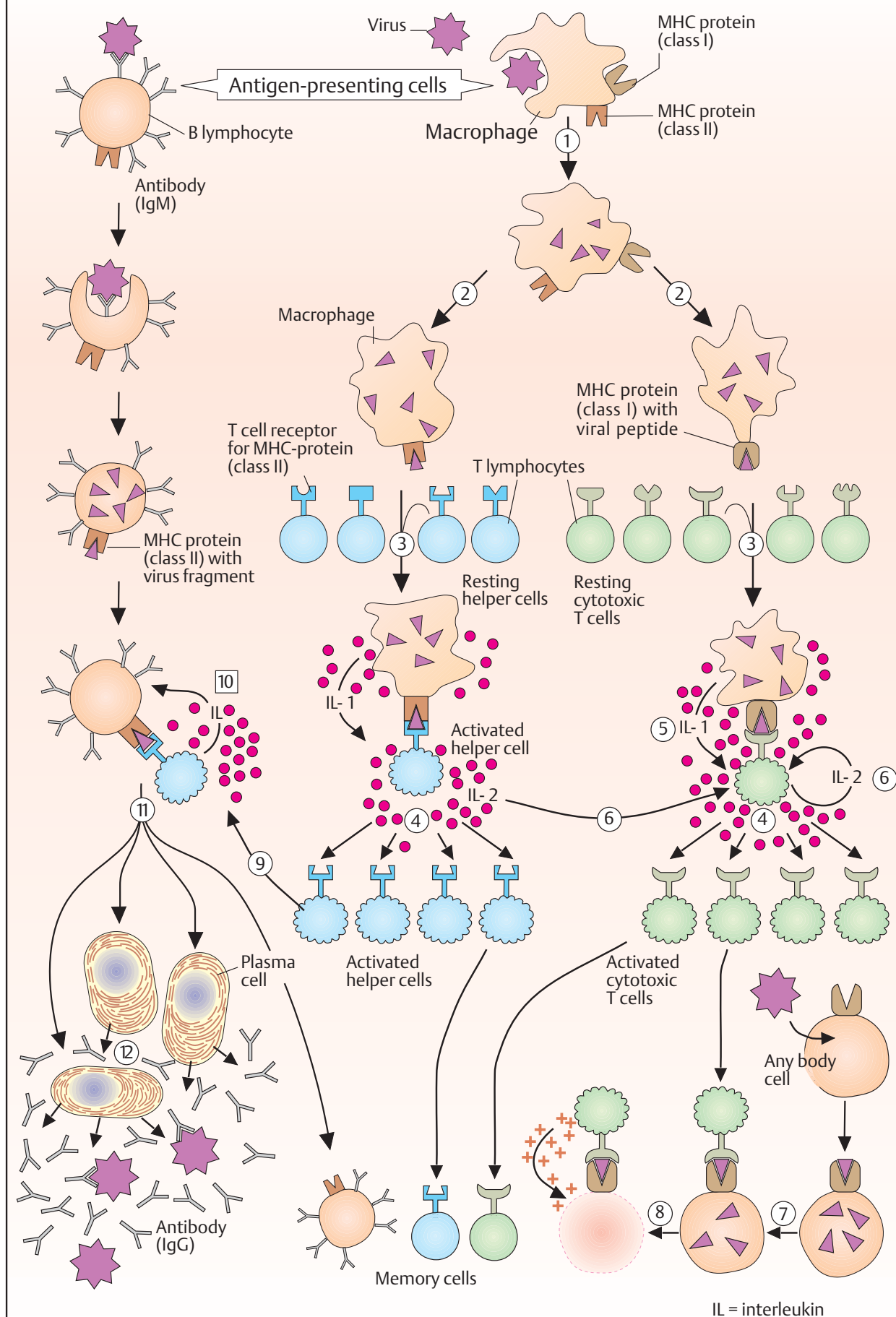
Pathogens that have entered the body—e.g., viruses (top)—are taken up by **antigen-presenting cells** (APCs) and proteolytically degraded (1). The viral fragments produced in this way are then presented on the surfaces of these cells with the help of special membrane proteins (MHC proteins; see p. 296) (2). The APCs include B lymphocytes, macrophages, and dendritic cells such as the skin’s Langerhans cells.

The complexes of MHC proteins and viral fragments displayed on the APCs are recognized by T cells that carry a receptor that matches the antigen (“T-cell receptors”; see p. 296) (3). Binding leads to activation of the T cell concerned and selective replication of it (4, “*clonal selection*”). The proliferation of immune cells is stimulated by *interleukins* (IL). These are a group of more than 20 signaling substances belonging to the cytokine family (see p. 392), with the help of which immune cells communicate with each other. For example, activated macrophages release IL-1 (5), while T cells stimulate their own replication and that of other immune cells by releasing IL-2 (6).

Depending on their type, activated T cells have different functions. **Cytotoxic T cells** (green) are able to recognize and bind virus-infected body cells or tumor cells (7). They then drive the infected cells into apoptosis (see p. 396) or kill them with *perforin*, a protein that perforates the target cell’s plasma membrane (8).

B lymphocytes, which as APCs present viral fragments on their surfaces, are recognized by **helper T cells** (blue) or their T cell receptors (9). Stimulated by interleukins, selective clonal replication then takes place of B cells that carry antigen receptors matching those of the pathogen (10). These mature into **plasma cells** (11) and finally secrete large amounts of soluble **antibodies** (12).

A. Simplified scheme of the immune response



T-cell activation

For the selectivity of the immune response (see p. 294), the cells involved must be able to recognize foreign antigens and proteins on other immune cells safely and reliably. To do this, they have antigen receptors on their cell surfaces and co-receptors that support recognition.

A. Antigen receptors ○

Many antigen receptors belong to the **immunoglobulin superfamily**. The common characteristic of these proteins is that they are made up from “immunoglobulin domains.” These are characteristically folded substructures consisting of 70–110 amino acids, which are also found in soluble immunoglobulins (Ig; see p. 300). The illustration shows schematically a few of the important proteins in the Ig superfamily. They consist of constant regions (brown or green) and variable regions (orange). Homologous domains are shown in the same colors in each case. All of the receptors have transmembrane helices at the C terminus, which anchor them to the membranes. Intramolecular and intermolecular disulfide bonds are also usually found in proteins belonging to the Ig family.

Immunoglobulin M (IgM), a membrane protein on the surface of B lymphocytes, serves to bind free antigens to the B cells. By contrast, **T cell receptors** only bind antigens when they are presented by another cell as a complex with an MHC protein (see below). Interaction between MHC-bound antigens and T cell receptors is supported by **co-receptors**. This group includes **CD8**, a membrane protein that is typical in cytotoxic T cells. T helper cells use **CD4** as a co-receptor instead (not shown). The abbreviation “CD” stands for “cluster of differentiation.” It is the term for a large group of proteins that are all located on the cell surface and can therefore be identified by antibodies. In addition to CD4 and CD8, there are many other co-receptors on immune cells (not shown).

The **MHC proteins** are named after the “*major histocompatibility complex*”—the DNA segment that codes for them. Human MHC proteins are also known as HLA antigens (“human leukocyte-associated” antigens). Their polymorphism is so large that it is unlikely

that any two individuals carry the same set of MHC proteins—except for monozygotic twins.

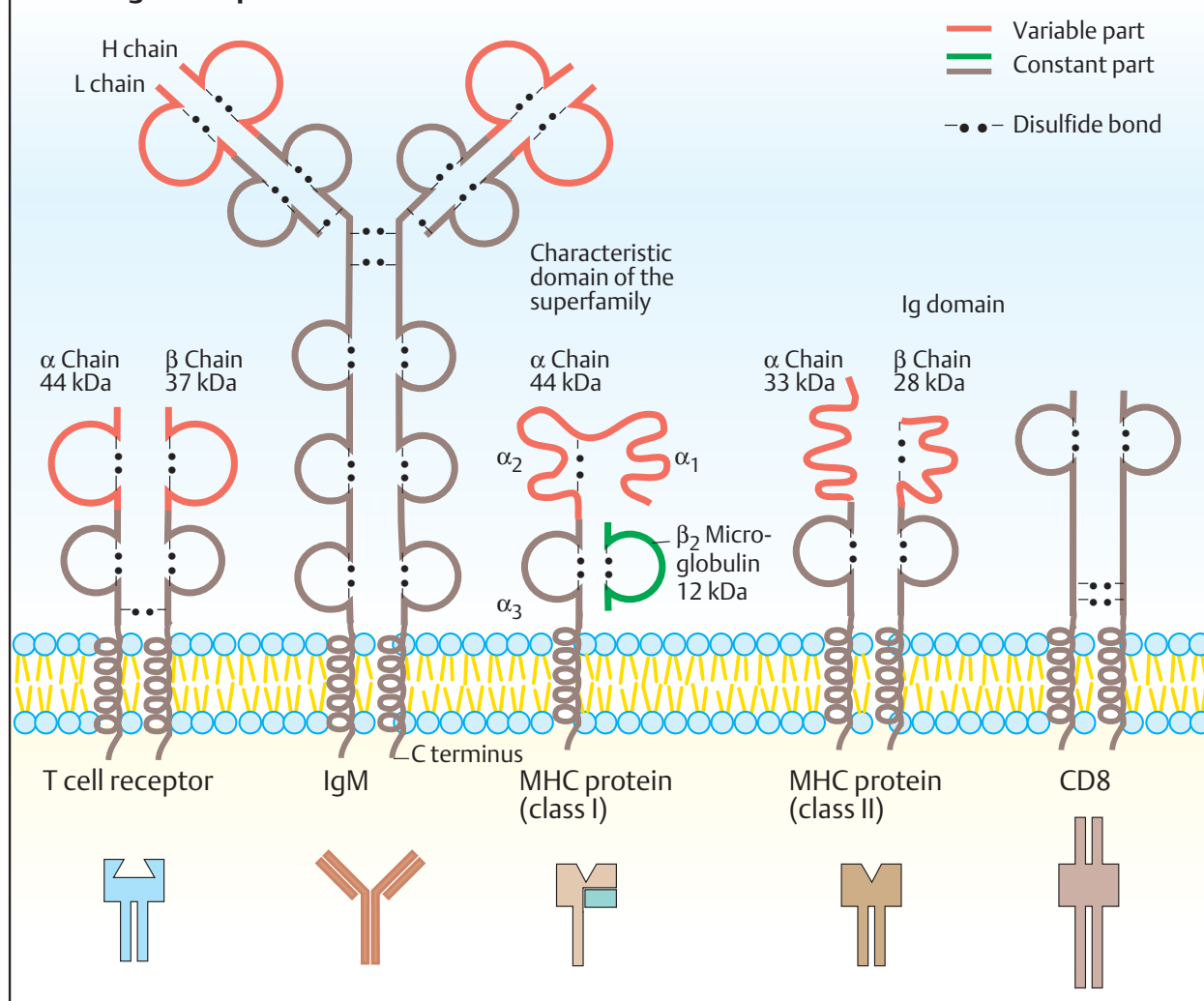
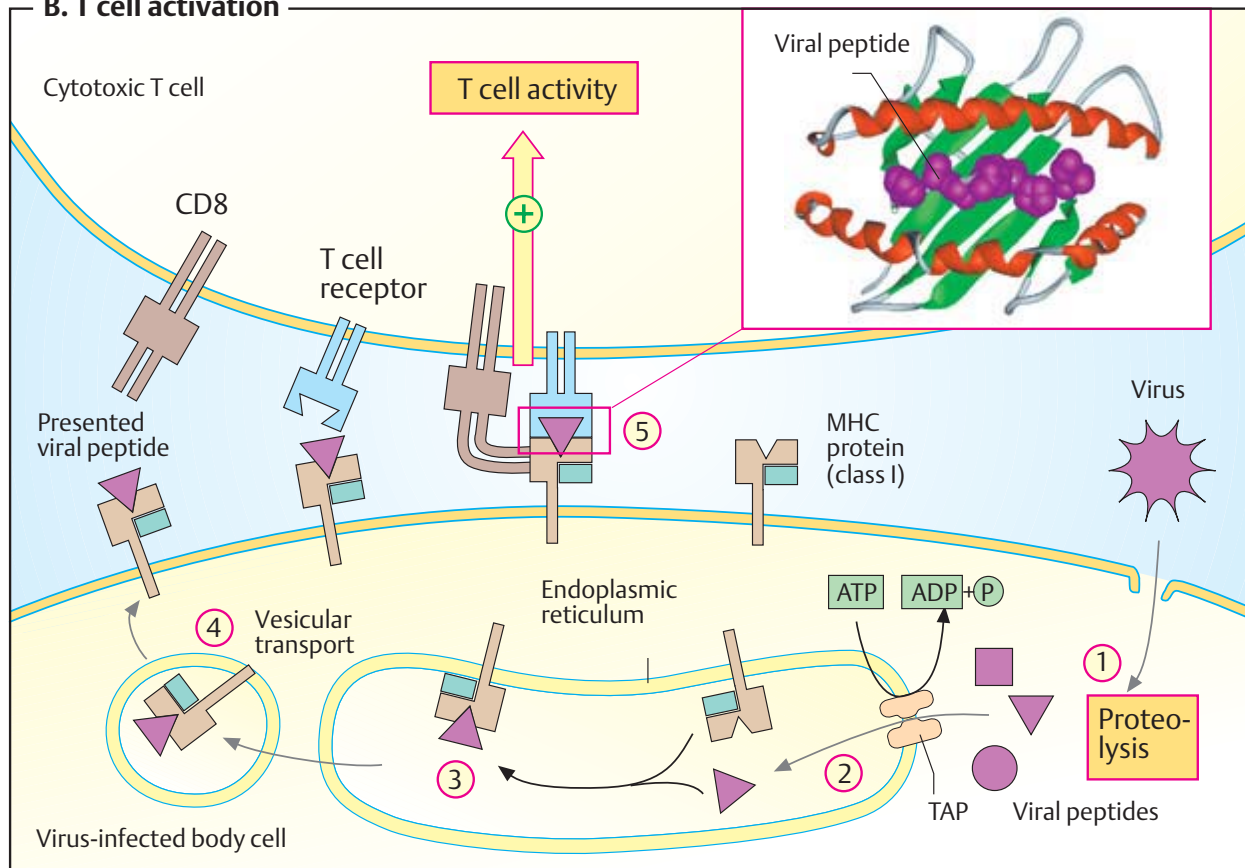
Class I MHC proteins occur in almost all nucleated cells. They mainly interact with cytotoxic T cells and are the reason for the rejection of transplanted organs. Class I MHC proteins are heterodimers ($\alpha\beta$). The β subunit is also known as β_2 -microglobulin.

Class II MHC proteins also consist of two peptide chains, which are related to each other. MHC II molecules are found on all antigen-presenting cells in the immune system. They serve for interaction between these cells and CD4-carrying T helper cells.

B. T-cell activation ●

The illustration shows an interaction between a virus-infected body cell (bottom) and a CD8-carrying cytotoxic T lymphocyte (top). The infected cell breaks down viral proteins in its cytoplasm (1) and transports the peptide fragments into the endoplasmic reticulum with the help of a special transporter (*TAP*) (2). Newly synthesized class I MHC proteins on the endoplasmic reticulum are loaded with one of the peptides (3) and then transferred to the cell surface by vesicular transport (4). The viral peptides are bound on the surface of the α_2 domain of the MHC protein in a depression formed by an insertion as a “floor” and two helices as “walls” (see smaller illustration).

Supported by CD8 and other co-receptors, a T cell with a matching T cell receptor binds to the MHC peptide complex (5; cf. p. 224). This binding activates protein kinases in the interior of the T cell, which trigger a chain of additional reactions (*signal transduction*; see p. 388). Finally, destruction of the virus-infected cell by the cytotoxic T lymphocytes takes place.

A. Antigen receptors**B. T cell activation**

Complement system

The complement system is part of the innate immune system (see p. 294). It supports *non-specific defense* against microorganisms. The system consists of some 30 different proteins, the “*complement factors*,” which are found in the blood and represent about 4% of all plasma proteins there. When inflammatory reactions occur, the complement factors enter the infected tissue and take effect there.

The complement system works in three different ways:

Chemotaxis. Various complement factors attract immune cells that can attack and phagocytose pathogens.

Opsonization. Certain complement factors (“opsonins”) bind to the pathogens and thereby mark them as targets for phagocytosing cells (e. g., macrophages).

Membrane attack. Other complement factors are deposited in the bacterial membrane, where they create pores that lyse the pathogen (see below).

A. Complement activation ❶

The reactions that take place in the complement system can be initiated in several ways. During the early phase of infection, lipopolysaccharides and other structures on the surface of the pathogens trigger the *alternative pathway* (right). If antibodies against the pathogens become available later, the antigen–antibody complexes formed activate the *classic pathway* (left). Acute-phase proteins (see p. 276) are also able to start the complement cascade (*lectin pathway*, not shown).

Factors C1 to C4 (for “complement”) belong to the classic pathway, while **factors B and D** form the reactive components of the alternative pathway. Factors **C5 to C9** are responsible for membrane attack. Other components not shown here regulate the system.

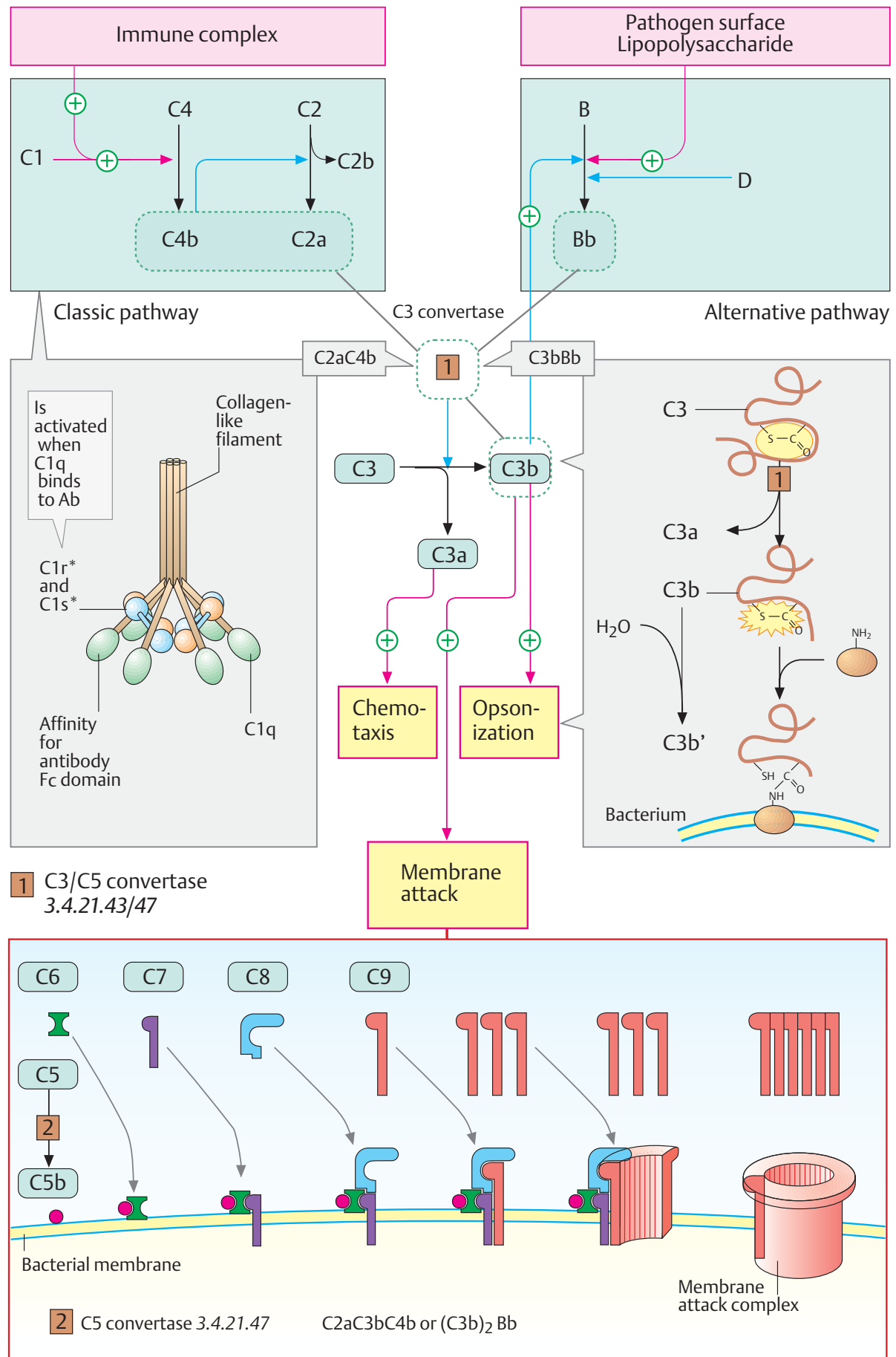
As in blood coagulation (see p. 290), the *early components* in the complement system are *serine proteinases*, which mutually activate each other through limited proteolysis. They create a self-reinforcing **enzyme cascade**. Factor **C3**, the products of which are involved in several functions, is central to the complement system.

The **classic pathway** is triggered by the formation of factor C1 at IgG or IgM on the surface of microorganisms (left). C1 is an 18-part molecular complex with three different components (C1q, C1r, and C1s). C1q is shaped like a bunch of tulips, the “flowers” of which bind to the F_c region of antibodies (left). This activates C1r, a *serine proteinase* that initiates the cascade of the classic pathway. First, C4 is proteolytically activated into C4b, which in turn cleaves C2 into C2a and C2b. C4b and C2a together form *C3 convertase* [1], which finally catalyzes the cleavage of C3 into C3a and C3b. Small amounts of C3b also arise from non-enzymatic hydrolysis of C3.

The **alternative pathway** starts with the binding of factors C3b and B to bacterial lipopolysaccharides (endotoxins). The formation of this complex allows cleavage of B by factor D, giving rise to a second form of *C3 convertase* (C3bBb).

Proteolytic cleavage of factor **C3** provides two components with different effects. The reaction exposes a highly *reactive thioester group* in C3b, which reacts with hydroxyl or amino groups. This allows C3b to bind covalently to molecules on the bacterial surface (*opsonization*, right). In addition, C3b initiates a chain of reactions leading to the formation of the *membrane attack complex* (see below). Together with C4a and C5a (see below), the smaller product C3a promotes the inflammatory reaction and has chemotactic effects.

The “late” factors C5 to C9 are responsible for the development of the **membrane attack complex** (bottom). They create an ion-permeable pore in the bacterial membrane, which leads to lysis of the pathogen. This reaction is triggered by *C5 convertase* [2]. Depending on the type of complement activation, this enzyme has the structure *C4b2a3b* or *C3bBb3b*, and it cleaves C5 into C5a and C5b. The complex of C5b and C6 allows deposition of C7 in the bacterial membrane. C8 and numerous C9 molecules—which form the actual pore—then bind to this core.

A. Complement activation

Structure of the membrane attack complex

Antibodies

Soluble antigen receptors, which are formed by activated B cells (plasma cells; see p. 294) and released into the blood, are known as **antibodies**. They are also members of the immunoglobulin family (Ig; see p. 296). Antibodies are an important part of the humoral immune defense system. They have no antimicrobial properties themselves, but support the cellular immune system in various ways:

1. They bind to antigens on the surface of pathogens and thereby prevent them from interacting with body cells (*neutralization*; see p. 404, for example).

2. They link single-celled pathogens into aggregates (immune complexes), which are more easily taken up by phagocytes (*agglutination*).

3. They activate the complement system (see p. 298) and thereby promote the innate immune defense system (*opsonization*).

In addition, antibodies have become indispensable aids in medical and biological diagnosis (see p. 304).

A. Domain structure of immunoglobulin G ●

Type G immunoglobulins (**IgG**) are quantitatively the most important antibodies in the blood, where they form the fraction of γ -globulins (see p. 276). IgGs (mass 150 kDa) are tetramers with two **heavy chains** (H chains; red or orange) and two **light chains** (L chains; yellow). Both H chains are glycosylated (violet; see also p. 43).

The proteinase *papain* cleaves IgG into two F_{ab} fragments and one F_c fragment. The F_{ab} ("antigen-binding") fragments, which each consist of one L chain and the N-terminal part of an H chain, are able to bind antigens. The F_c ("crystallizable") fragment is made up of the C-terminal halves of the two H chains. This segment serves to bind IgG to cell surfaces, for interaction with the complement system and antibody transport.

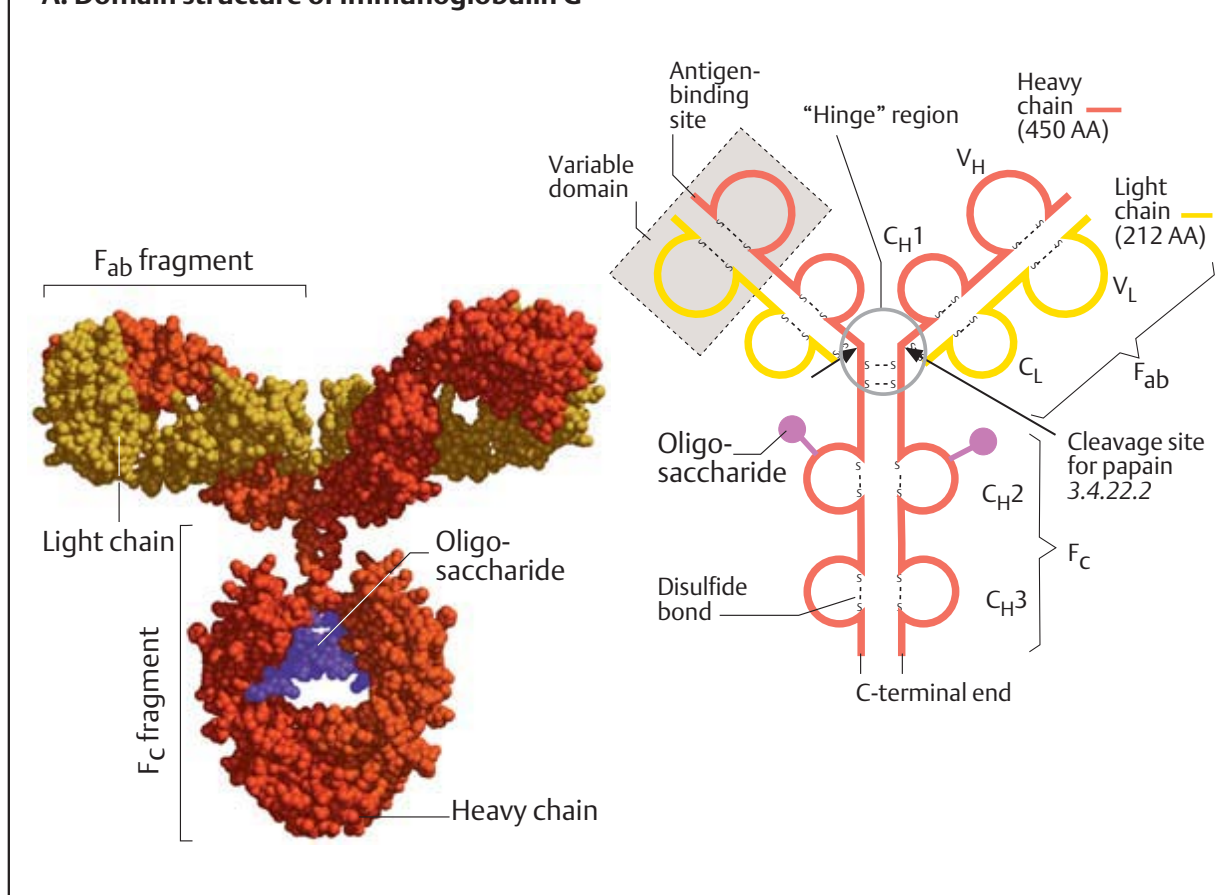
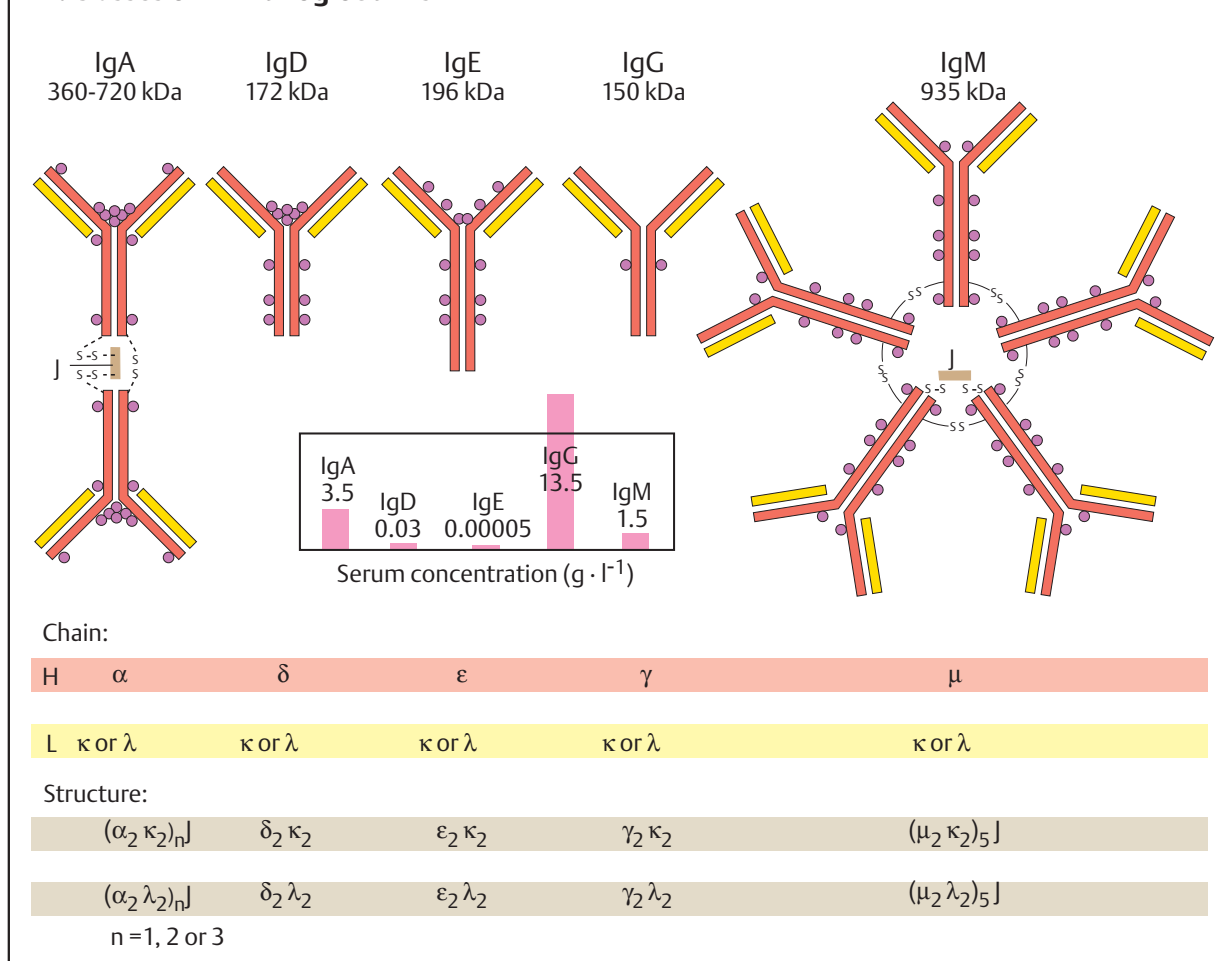
Immunoglobulins are constructed in a modular fashion from several **immunoglobulin domains** (shown in the diagram on the right in Ω form). The H chains of IgG contain four of these domains (V_H , C_{H1} , C_{H2} , and C_{H3}) and the L chains contain two (C_L and V_L). The letters C and V designate constant or variable regions.

Disulfide bonds link the two heavy chains to each other and also link the heavy chains to the light chains. Inside the domains, there are also disulfide bonds that stabilize the tertiary structure. The domains are approximately 110 amino acids (AA) long and are homologous with each other. The antibody structure evidently developed as a result of gene duplication. In its central region, known as the "hinge" region, the antibodies are highly mobile.

B. Classes of immunoglobulins ●

Human immunoglobulins are divided into five classes. **IgA** (with two subgroups), **IgD**, **IgE**, **IgG** (with four subgroups), and **IgM** are defined by their H chains, which are designated by the Greek letters α , δ , ϵ , γ , and μ . By contrast, there are only two types of **L chain** (κ and λ). IgD and IgE (like IgG) are tetramers with the structure H_2L_2 . By contrast, soluble IgA and IgM are multimers that are held together by disulfide bonds and additional **J peptides** (joining peptides).

The antibodies have different tasks. **IgMs** are the first immunoglobulins formed after contact with a foreign antigen. Their early forms are located on the surface of B cells (see p. 296), while the later forms are secreted from plasma cells as pentamers. Their action targets microorganisms in particular. Quantitatively, **IgGs** are the most important immunoglobulins (see the table showing serum concentrations). They occur in the blood and interstitial fluid. As they can pass the placenta with the help of receptors, they can be transferred from mother to fetus. **IgAs** mainly occur in the intestinal tract and in body secretions. **IgEs** are found in low concentrations in the blood. As they can trigger degranulation of mast cells (see p. 380), they play an important role in allergic reactions. The function of **IgDs** is still unexplained. Their plasma concentration is also very low.

A. Domain structure of immunoglobulin G**B. Classes of immunoglobulins**

Antibody biosynthesis

The acquired (adaptive) immune system (see p. 294) is based on the ability of the lymphocytes to keep an extremely large repertoire of antigen receptors and soluble antibodies ready for use, so that even infections involving new types of pathogen can be combated. The wide range of immunoglobulins (Ig) are produced by genetic recombination and additional mutations during the development and maturation of the individual lymphocytes.

A. Variability of immunoglobulins ○

It is estimated that more than 10^8 different antibody variants occur in every human being. This **variability** affects both the heavy and the light chains of immunoglobulins.

There are five different types of heavy (H) chain, according to which the antibody classes are defined (α , δ , ϵ , γ , μ), and two types of light (L) chain (κ and λ ; see p. 300). The various Ig types that arise from combinations of these chains are known as **isotypes**. During immunoglobulin biosynthesis, plasma cells can switch from one isotype to another (“gene switch”). **Allotypic variation** is based on the existence of various alleles of the same gene—i.e., genetic differences between individuals. The term **idiotypic variation** refers to the fact that the antigen binding sites in the F_{ab} fragments can be highly variable. Idiotypic variation affects the *variable domains* (shown here in pink) of the light and heavy chains. At certain sites—known as the *hypervariable regions* (shown here in red)—variation is particularly wide; these sequences are directly involved in the binding of the antigen.

B. Causes of antibody variety ○

There are three reasons for the extremely wide variability of antibodies:

1. **Multiple genes.** Various genes are available to code for the variable protein domains. Only one gene from among these is selected and expressed.

2. **Somatic recombination.** The genes are divided into several segments, of which there are various versions. Various (“untidy”) combinations of the segments during lymphocyte

maturation give rise to randomly combined new genes (“mosaic genes”).

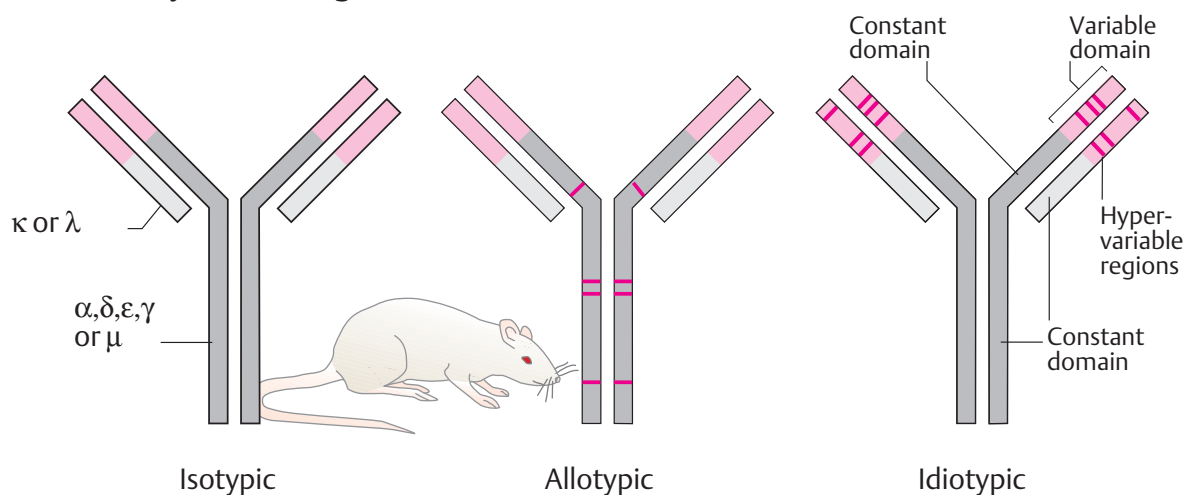
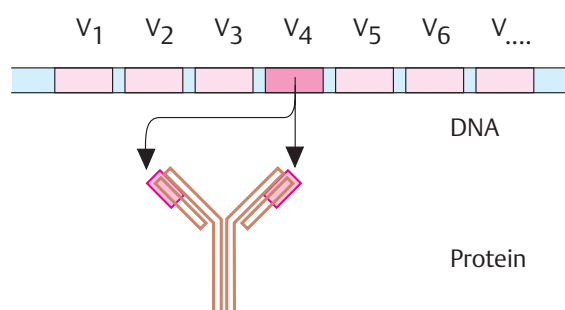
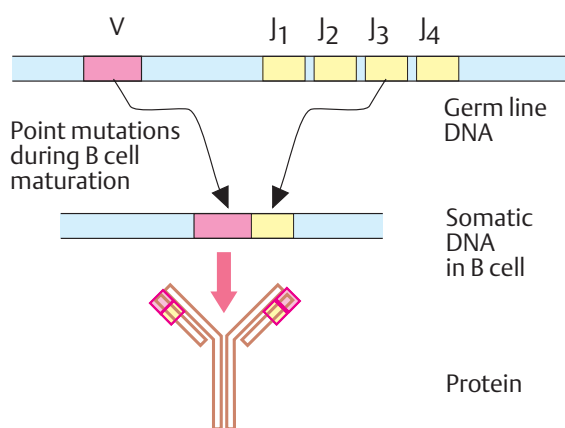
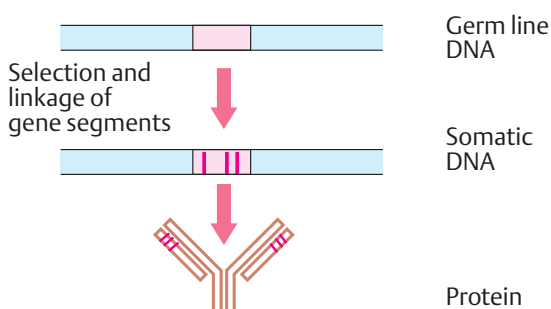
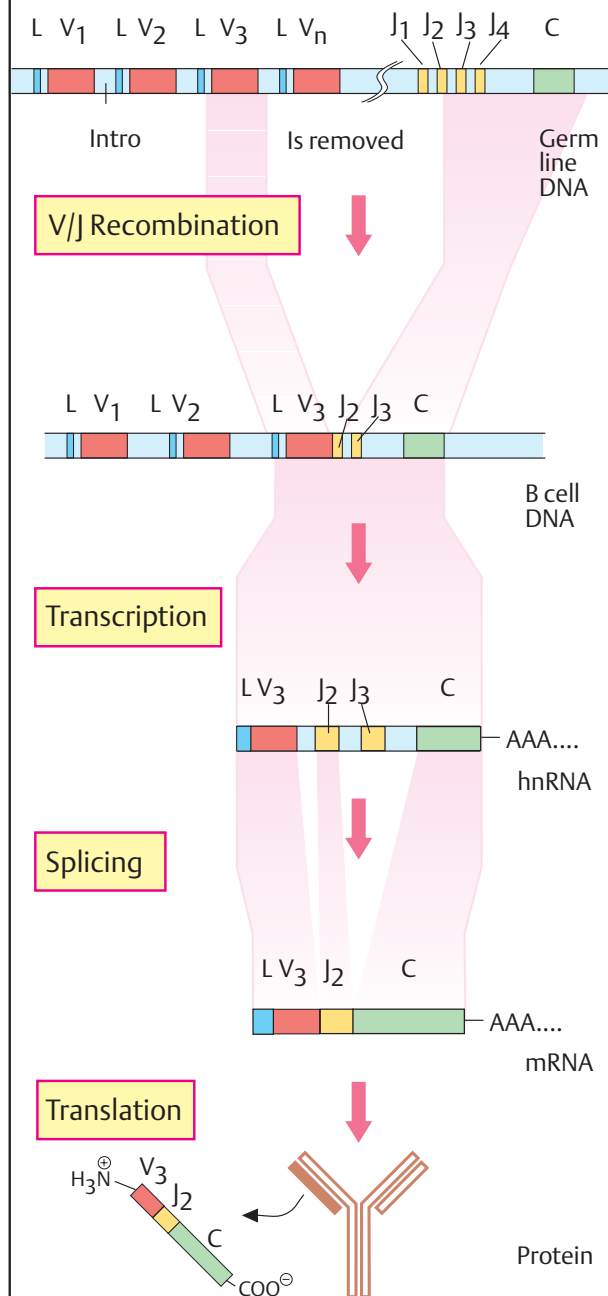
3. **Somatic mutation.** During differentiation of B cells into plasma cells, the coding genes mutate. In this way, the “primordial” *germ-line genes* can become different *somatic genes* in the individual B cell clones.

C. Biosynthesis of a light chain ○

We can look at the basic features of the genetic organization and synthesis of immunoglobulins using the biosynthesis of a mouse κ chain as an example. The gene segments for this light chain are designated L, V, J, and C. They are located on chromosome 6 in the **germ-line DNA** (on chromosome 2 in humans) and are separated from one another by introns (see p. 242) of different lengths.

Some 150 identical **L segments** code for the signal peptide (“leader sequence,” 17–20 amino acids) for secretion of the product (see p. 230). The **V segments**, of which there are 150 different variants, code for most of the variable domains (95 of the 108 amino acids). L and V segments always occur in pairs—in tandem, so to speak. By contrast, there are only five variants of the **J segments** (joining segments) at most. These code for a peptide with 13 amino acids that links the variable part of the κ chains to the constant part. A single **C segment** codes for the constant part of the light chain (84 amino acids).

During the differentiation of B lymphocytes, individual *V/J combinations* arise in each B cell. One of the 150 L/V tandem segments is selected and linked to one of the five J segments. This gives rise to a *somatic gene* that is much smaller than the germline gene. Transcription of this gene leads to the formation of the **hnRNA** for the κ chain, from which introns and surplus J segments are removed by splicing (see p. 246). Finally, the completed **mRNA** still contains one each of the L–V–J–C segments and after being transported into the cytoplasm is available for translation. The subsequent steps in Ig biosynthesis follow the rules for the synthesis of membrane-bound or secretory proteins (see p. 230).

A. Variability of immunoglobulins**B. Origins of antibody variety****1. Multiple genes****2. Somatic recombination****3. Somatic mutation****C. Biosynthesis of a light chain**

Monoclonal antibodies, immunoassay

A. Monoclonal antibodies ○

Monoclonal antibodies (MABs) are secreted by immune cells that derive from a single antibody-forming cell (from a single cell *clone*). This is why each MAB is directed against only one specific *epitope* of an immunogenic substance, known as an “*antigenic determinant*.” Large molecules contain several epitopes, against which various antibodies are formed by various B cells. An antiserum containing a mixture of all of these antibodies is described as being *polyclonal*.

To obtain MABs, **lymphocytes** isolated from the spleen of immunized mice (**1**) are fused with mouse tumor cells (**myeloma cells**, **2**). This is necessary because antibody-secreting lymphocytes in culture have a lifespan of only a few weeks. Fusion of lymphocytes with tumor cells gives rise to cell hybrids, known as **hybridomas**, which are potentially immortal.

Successful *fusion* (**2**) is a rare event, but the frequency can be improved by adding polyethylene glycol (PEG). To obtain only successfully fused cells, incubation is required for an extended period in a **primary culture** with *HAT medium* (**3**), which contains hypoxanthine, aminopterin, and thymidine. *Aminopterin*, an analogue of dihydrofolic acid, competitively inhibits *dihydrofolate reductase* and thus inhibits the synthesis of dTMP (see p. 402). As dTMP is essential for DNA synthesis, myeloma cells cannot survive in the presence of aminopterin. Although spleen cells are able to circumvent the inhibitory effect of aminopterin by using *hypoxanthine* and *thymidine*, they have a limited lifespan and die. Only hybridomas survive culture in HAT medium, because they possess both the immortality of the myeloma cells and the spleen cells' metabolic side pathway.

Only a few fused cells actually produce antibodies. To identify these cells, the hybridomas have to be isolated and replicated by **cloning** (**4**). After the clones have been tested for antibody formation, positive cultures are picked out and selected by further cloning (**5**). This results in hybridomas that synthesize *monoclonal antibodies*. Finally, MAB production is carried out in vitro using a bioreactor,

or in vivo by producing ascites fluid in mice (**6**).

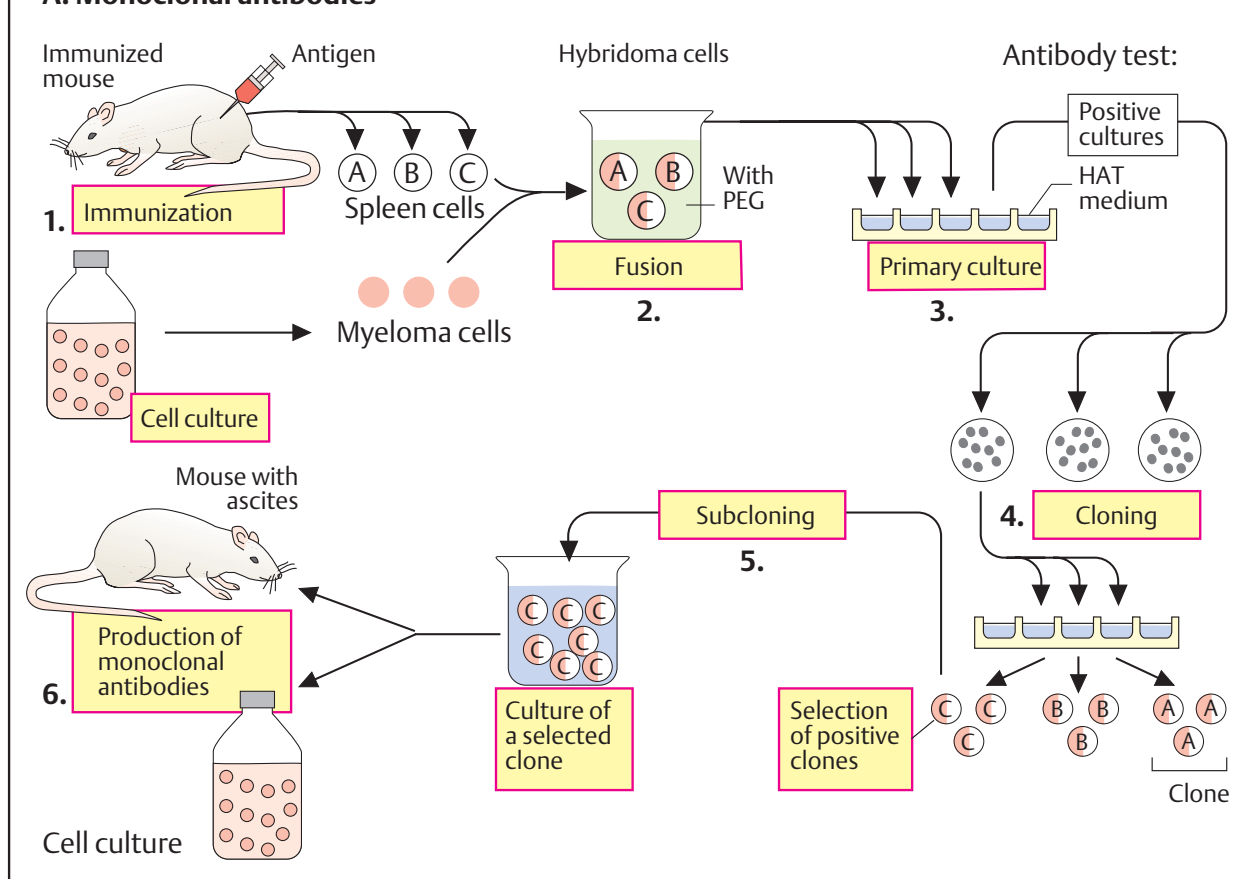
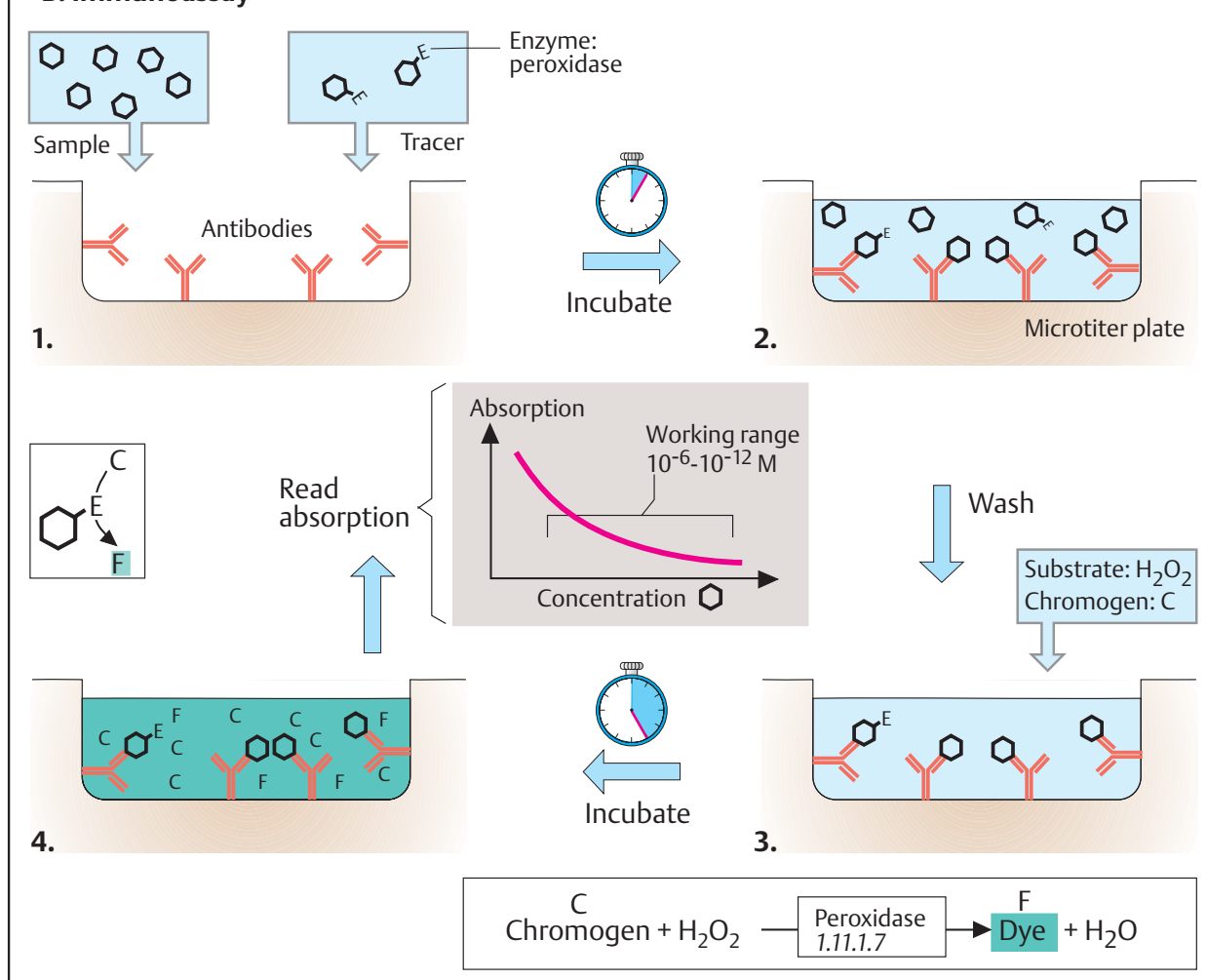
B. Immunoassay ○

Immunoassays are semiquantitative procedures for assessing substances with low concentrations. In principle, immunoassays can be used to assess any compound against which antibodies are formed.

The basis for this procedure is the *antigen–antibody “reaction”*—i. e., specific binding of an antibody to the molecule being assayed. Among the many different immunoassay techniques that have been developed—e. g., *radioimmunoassay* (RIA), and *chemoluminescence immunoassay* (CIA)—a version of the **enzyme-linked immunoassay** (EIA) is shown here.

The substance to be assayed—e. g., the hormone thyroxine in a serum sample—is pipetted into a microtiter plate (**1**), the walls of which are coated with *antibodies* that specifically bind the hormone. At the same time, a small amount of thyroxine is added to the incubation to which an enzyme known as the “*tracer*” (**1**) has been chemically coupled. The tracer and the hormone being assayed compete for the small number of antibody binding sites available. After binding has taken place (**2**), all of the unbound molecules are rinsed out. The addition of a substrate solution for the enzyme (a *chromogenic solution*) then triggers an indicator reaction (**3**), the products of which can be assessed using photometry (**4**).

The larger the amount of enzyme that can bind to the antibodies on the container's walls, the larger the amount of dye that is produced. Conversely, the larger the amount of the substance being assayed that is present in the sample, the *smaller* the amount of tracer that can be bound by the antibodies. Quantitative analysis can be carried out through parallel measurement using standards with a known concentration.

A. Monoclonal antibodies**B. Immunoassay**

Liver: functions

Weighing 1.5 kg, the liver is one of the largest organs in the human body. Although it only represents 2–3% of the body's mass, it accounts for 25–30% of oxygen consumption.

A. Diagram of a hepatocyte ●

The $3 \cdot 10^{11}$ cells in the liver—particularly the **hepatocytes**, which make up 90% of the cell mass—are the central location for the body's *intermediary metabolism*. They are in close contact with the blood, which enters the liver from the portal vein and the hepatic arteries, flows through capillary vessels known as sinusoids, and is collected again in the central veins of the hepatic lobes. Hepatocytes are particularly rich in endoplasmic reticulum, as they carry out intensive protein and lipid synthesis. The cytoplasm contains granules of insoluble glycogen. Between the hepatocytes, there are bile capillaries through which bile components are excreted.

B. Functions of the liver ●

The most important functions of the liver are:

1. **Uptake** of nutrients supplied by the intestines via the portal vein.
2. Biosynthesis of endogenous compounds and storage, conversion, and degradation of them into excretable molecules (**metabolism**). In particular, the liver is responsible for the biosynthesis and degradation of almost all plasma proteins.
3. **Supply** of the body with metabolites and nutrients.
4. **Detoxification** of toxic compounds by biotransformation.
5. **Excretion** of substances with the bile.

C. Hepatic metabolism ●

The liver is involved in the metabolism of practically all groups of metabolites. Its functions primarily serve to cushion fluctuations in the concentration of these substances in the blood, in order to ensure a constant supply to the peripheral tissues (*homeostasis*).

Carbohydrate metabolism. The liver takes up glucose and other monosaccharides from the plasma. Glucose is then either stored in

the form of the polysaccharide *glycogen* or converted into fatty acids. When there is a drop in the blood glucose level, the liver releases glucose again by breaking down glycogen. If the glycogen store is exhausted, glucose can also be synthesized by *gluconeogenesis* from lactate, glycerol, or the carbon skeleton of amino acids (see p. 310).

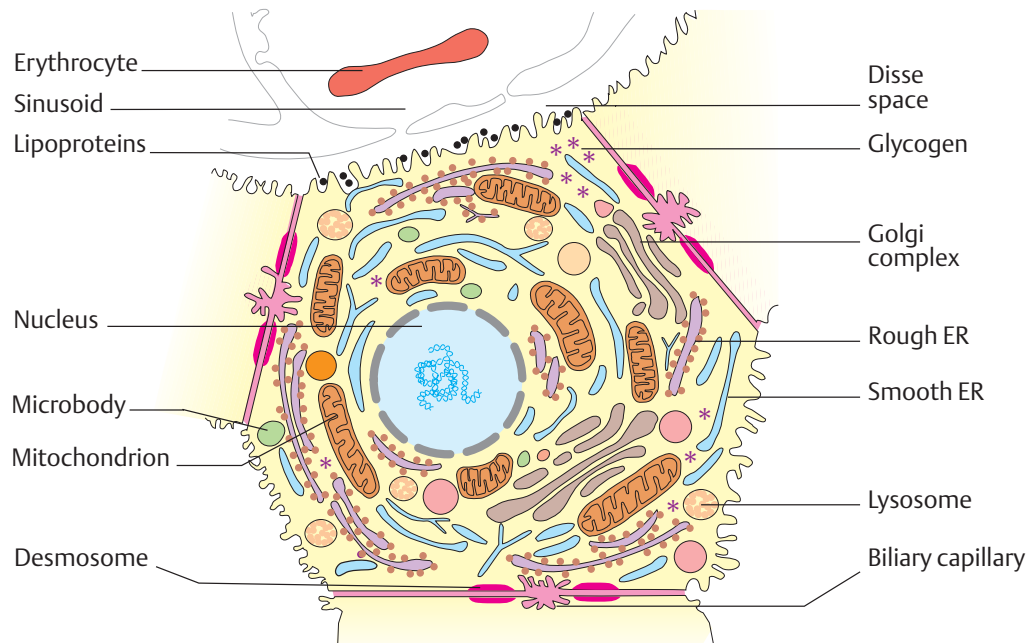
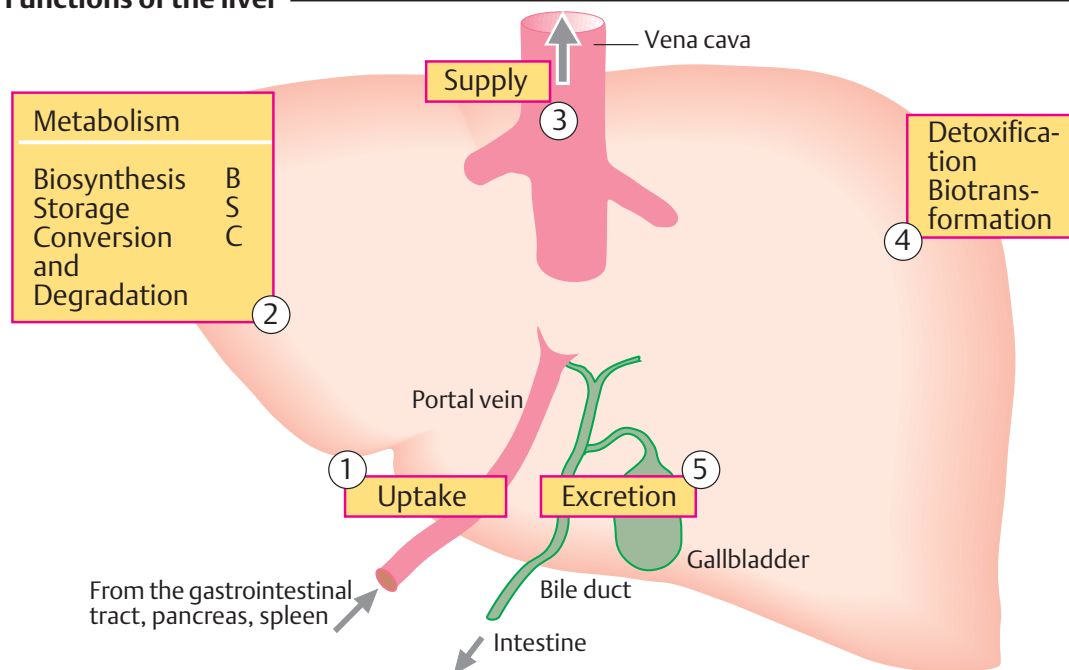
Lipid metabolism. The liver synthesizes fatty acids from acetate units. The fatty acids formed are then used to synthesize fats and phospholipids, which are released into the blood in the form of *lipoproteins*. The liver's special ability to convert fatty acids into *ketone bodies* and to release these again is also important (see p. 312).

Like other organs, the liver also synthesizes cholesterol, which is transported to other tissues as a component of lipoproteins. Excess cholesterol is converted into bile acids in the liver or directly excreted with the bile (see p. 314).

Amino acid and protein metabolism. The liver controls the plasma levels of the amino acids. Excess amino acids are broken down. With the help of the urea cycle (see p. 182), the nitrogen from the amino acids is converted into urea and excreted via the kidneys. The carbon skeleton of the amino acids enters the intermediary metabolism and serves for glucose synthesis or energy production. In addition, most of the plasma proteins are synthesized or broken down in the liver (see p. 276).

Biotransformation. Steroid hormones and bilirubin, as well as drugs, ethanol, and other xenobiotics are taken up by the liver and inactivated and converted into highly polar metabolites by conversion reactions (see p. 316).

Storage. The liver not only stores energy reserves and nutrients for the body, but also certain mineral substances, trace elements, and vitamins, including iron, retinol, and vitamins A, D, K, folic acid, and B₁₂.

A. Diagram of a hepatocyte**B. Functions of the liver****C. Liver metabolism**

Carbohydrate metabolism	Lipid metabolism	Amino acid metabolism	Biotransformation
Glucose BSC	Fatty acids BC	Amino acids C	Steroid hormones EC
Galactose C	Fats BC	Urea B	Bile pigments EC
Fructose C	Ketone bodies B		Ethanol C
Mannose C	Cholesterol BEC		Drugs EC
Pentoses BC	Bile acids BE		
Lactate C	Vitamins SC		
Glycerol BC			
Glycogen BSC			
		Plasma proteins	
		Lipoproteins BC	
		Albumin BC	
		Coagulation factors BC	
		Hormones C	
		Enzymes BC	

B Biosynthesis
C Conversion and degradation
E Excretion
S Storage

Buffer function in organ metabolism

All of the body's tissues have a constant requirement for energy substrates and nutrients. The body receives these metabolites with food, but the supply is irregular and in varying amounts. The liver acts here along with other organs, particularly adipose tissue, as a balancing *buffer* and *storage organ*.

In the metabolism, a distinction is made between the *absorptive state* (*well-fed state*) immediately after a meal and the *postabsorptive state* (*state of starvation*), which starts later and can merge into hunger. The switching of the organ metabolism between the two phases depends on the concentration of energy-bearing metabolites in the blood (plasma level). This is regulated jointly by hormones and by the autonomic nervous system.

A. Absorptive state ①

The absorptive state continues for 2–4 hours after food intake. As a result of food digestion, the plasma levels of glucose, amino acids, and fats (triacylglycerols) temporarily increase.

The endocrine **pancreas** responds to this by altering its hormone release—there is an increase in *insulin* secretion and a reduction in *glucagon* secretion. The increase in the insulin/glucagon quotient and the availability of substrates trigger an *anabolic phase* in the tissues—particularly liver, muscle, and adipose tissues.

The **liver** forms increased amounts of glycogen and fats from the substrates supplied. Glycogen is stored, and the fat is released into the blood in very low density lipoproteins (VLDLs).

Muscle also refills its glycogen store and synthesizes proteins from the amino acids supplied.

Adipose tissue removes free fatty acids from the lipoproteins, synthesizes triacylglycerols from them again, and stores these in the form of insoluble droplets.

During the absorptive state, the **heart** and **neural tissue** mainly use glucose as an energy source, but they are unable to establish any substantial energy stores. Heart muscle cells are in a sense “omnivorous,” as they can also use other substances to produce energy (fatty

acids, ketone bodies). By contrast, the central nervous system (CNS) is dependent on glucose. It is only able to utilize ketone bodies after a prolonged phase of hunger (**B**).

B. Postabsorptive state ②

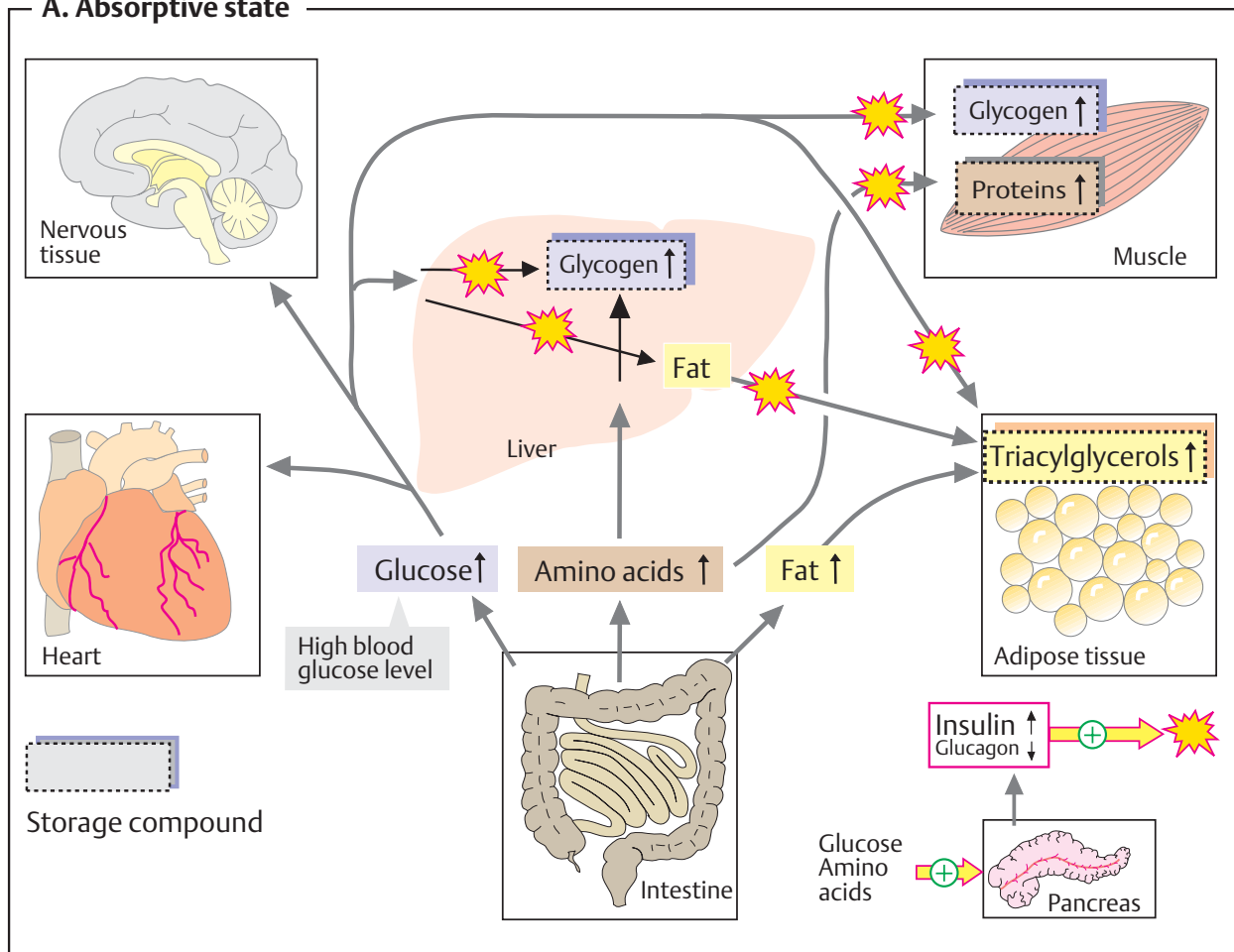
When the food supply is interrupted, the postabsorptive state quickly sets in. The pancreatic A cells now release increased amounts of *glucagon*, while the B cells reduce the amount of *insulin* they secrete. The reduced insulin/glucagon quotient leads to switching of the intermediary metabolism. The body now falls back on its energy reserves. To do this, it breaks down *storage substances* (glycogen, fats, and proteins) and shifts energy-supplying metabolites between the organs.

The **liver** first empties its glycogen store (*glycogenolysis*; see p.156). It does not use the released glucose itself, however, but supplies the other tissues with it. In particular, the brain, adrenal gland medulla, and erythrocytes depend on a constant supply of glucose, as they have no substantial glucose reserves themselves. When the liver's glycogen reserves are exhausted after 12–24 hours, *gluconeogenesis* begins (see p. 154). The precursors for this are derived from the musculature (amino acids) and adipose tissue (glycerol from fat degradation). From the fatty acids that are released (see below), the liver starts to form ketone bodies (*ketogenesis*; see p.312). These are released into the blood and serve as important energy suppliers during the hunger phase. After 1–2 weeks, the CNS also starts to use ketone bodies to supply part of its energy requirements, in order to save glucose.

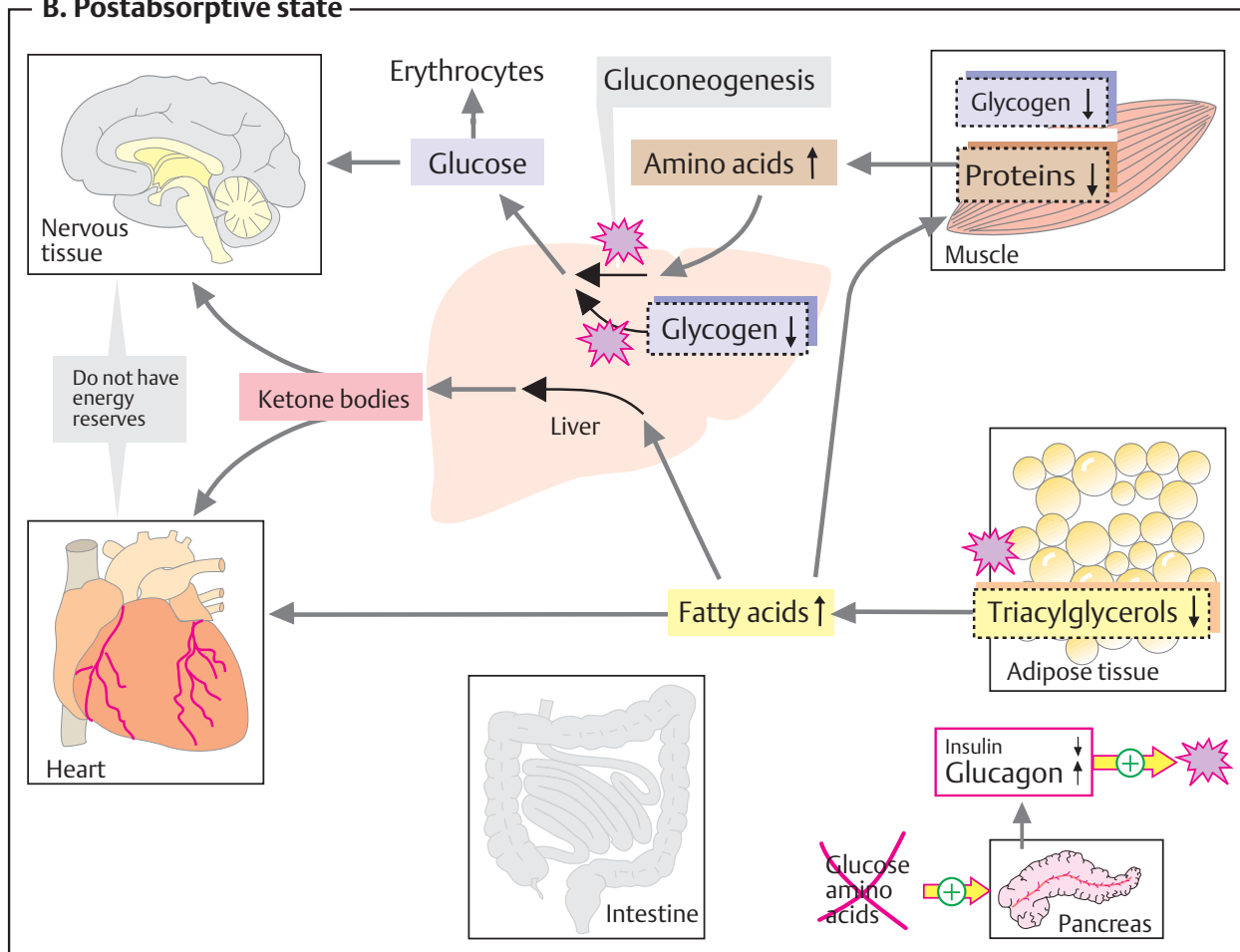
In **muscle**, the extensive glycogen reserves are exclusively used for the muscles' own requirements (see p. 320). The slowly initiated protein breakdown in muscle supplies amino acids for gluconeogenesis in the liver.

In **adipose tissue**, glucagon triggers *lipolysis*, releasing fatty acids and glycerol. The fatty acids are used as energy suppliers by many types of tissue (with the exception of brain and erythrocytes). An important recipient of the fatty acids is the liver, which uses them for ketogenesis.

A. Absorptive state



B. Postabsorptive state



Carbohydrate metabolism

Besides fatty acids and ketone bodies, glucose is the body's most important energy supplier. The concentration of glucose in the blood (the "*blood glucose level*") is maintained at 4–6 mM (0.8–1.0 g L⁻¹) by precise regulation of glucosesupplying and glucose-utilizing processes. Glucose suppliers include the intestines (glucose from food), liver, and kidneys. The liver plays the role of a "glucostat" (see p. 308).

The liver is also capable of forming glucose by converting other sugars—e. g., *fructose* and *galactose*—or by synthesizing from other metabolites. The conversion of lactate to glucose in the *Cori cycle* (see p. 338) and the conversion of alanine to glucose with the help of the *alanine cycle* (see p. 338) are particularly important for the supply of erythrocytes and muscle cells.

Transporters in the plasma membrane of hepatocytes allow insulin-independent transport of glucose and other sugars in both directions. In contrast to muscle, the liver possesses the enzyme *glucose-6-phosphatase*, which can release glucose from glucose-6-phosphate.

A. Gluconeogenesis: overview ❶

Regeneration of glucose (up to 250 g per day) mainly takes place in the liver. The tubule cells of the kidney are also capable of carrying out gluconeogenesis, but due to their much smaller mass, their contribution only represents around 10% of total glucose formation. Gluconeogenesis is regulated by hormones. *Cortisol*, *glucagon*, and *epinephrine* promote gluconeogenesis, while *insulin* inhibits it (see pp. 158, 244).

The main precursors of gluconeogenesis in the liver are *lactate* from anaerobically working muscle cells and from erythrocytes, glucogenic *amino acids* from the digestive tract and muscles (mainly alanine), and *glycerol* from adipose tissue. The kidney mainly uses amino acids for gluconeogenesis (Glu, Gln; see p. 328).

In mammals, fatty acids and other suppliers of acetyl CoA are not capable of being used for gluconeogenesis, as the acetyl residues formed during β -oxidation in the tricarboxylic acid cycle (see p. 132) are oxidized to CO₂

and therefore cannot be converted into oxaloacetic acid, the precursor for gluconeogenesis.

B. Fructose and galactose metabolism ❷

Fructose is mainly metabolized by the liver, which channels it into glycolysis (left half of the illustration).

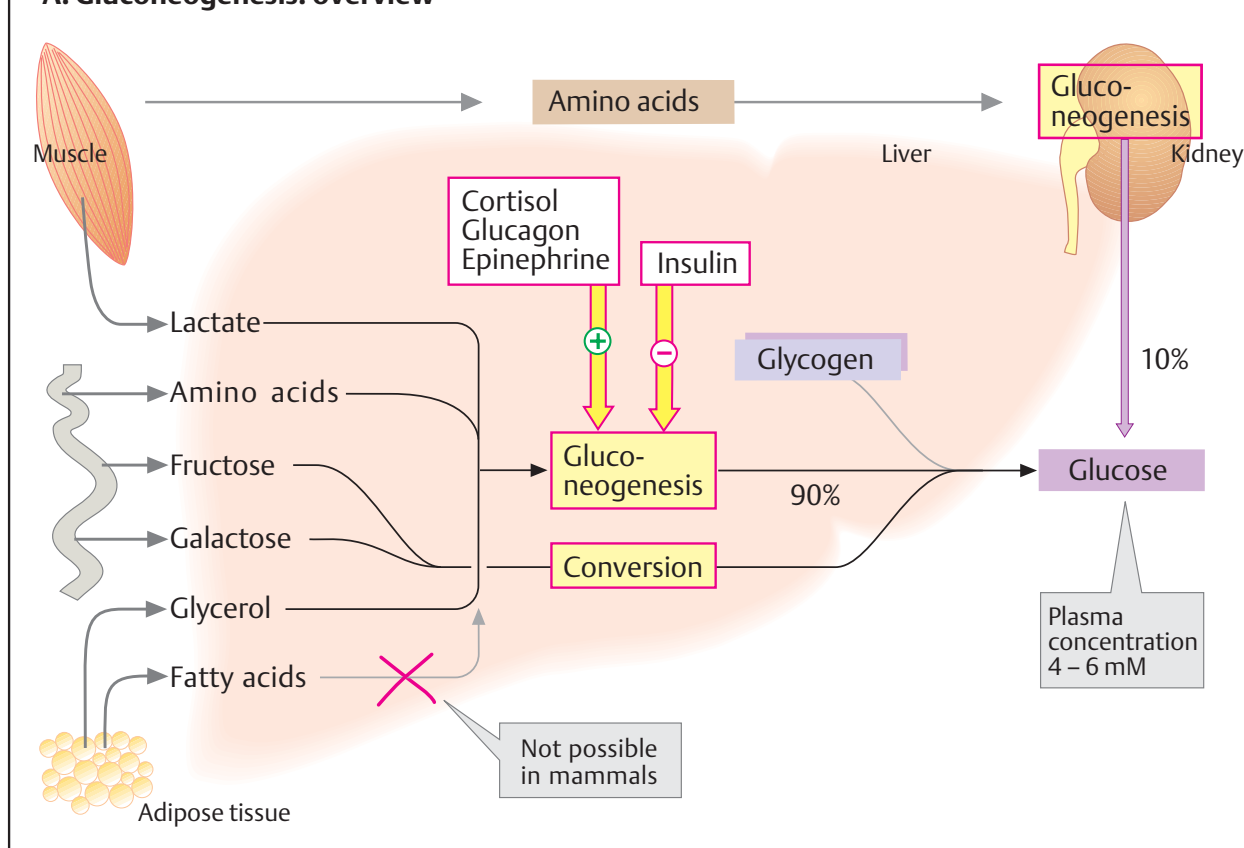
A special *ketoheokinase* [1] initially phosphorylates fructose into **fructose 1-phosphate**. This is then cleaved by an *aldolase* [2], which is also fructose-specific, to yield **glycerone 3-phosphate** (dihydroxyacetone phosphate) and **glyceraldehyde**. Glycerone 3-phosphate is already an intermediate of glycolysis (center), while glyceraldehyde can be phosphorylated into glyceraldehyde 3-phosphate by *triokinase* [3].

To a smaller extent, glyceraldehyde is also reduced to glycerol [4] or oxidized to glycerate, which can be channeled into glycolysis following phosphorylation (not shown). The reduction of glyceraldehyde [4] uses up NADH. As the rate of degradation of alcohol in the hepatocytes is limited by the supply of NAD⁺, fructose degradation accelerates alcohol degradation (see p. 320).

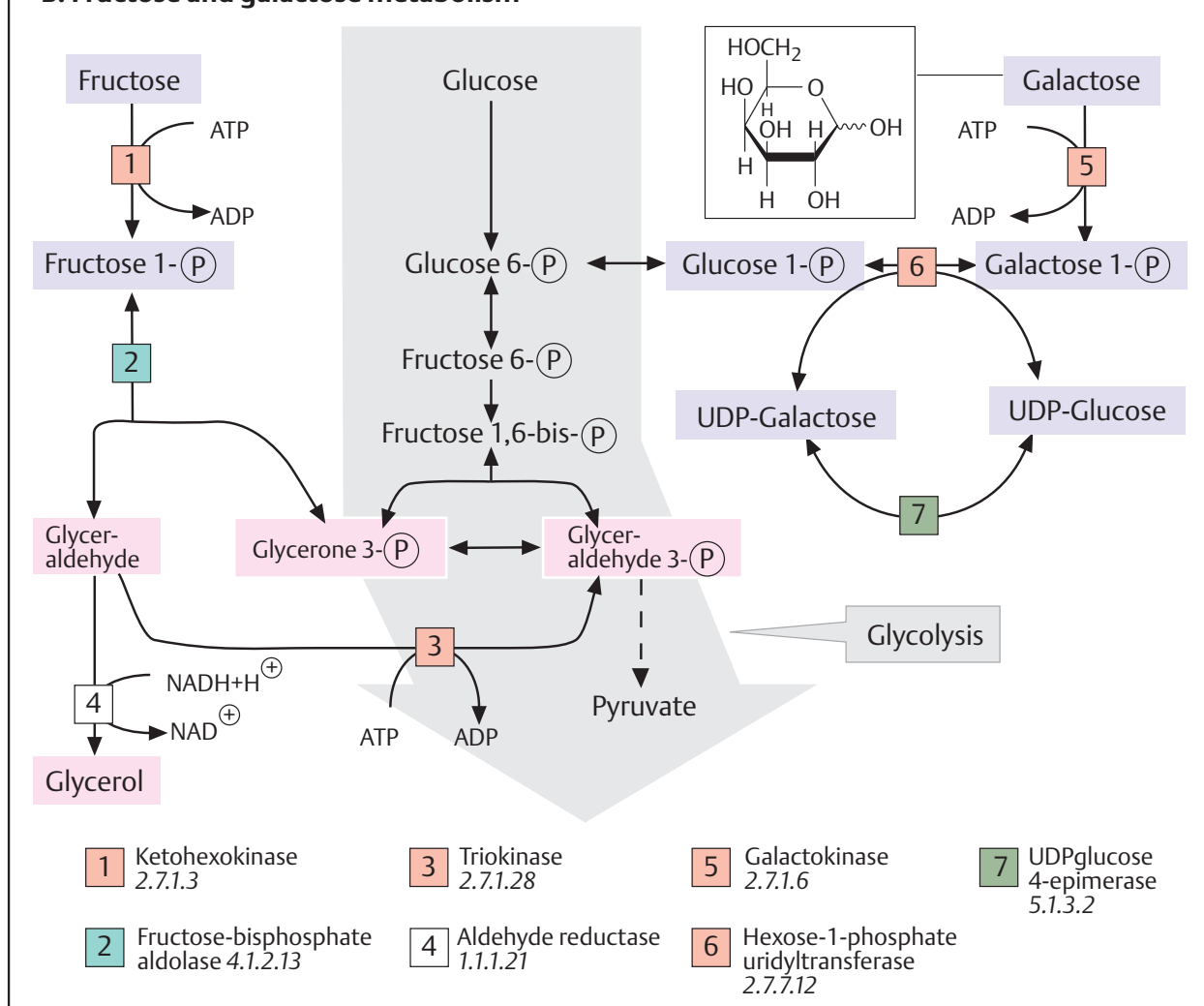
Outside of the liver, fructose is channeled into the sugar metabolism by reduction at C-2 to yield sorbitol and subsequent dehydration at C-1 to yield glucose (the *polyol pathway*; not shown).

Galactose is also broken down in the liver (right side of the illustration). As is usual with sugars, the metabolism of galactose starts with a phosphorylation to yield **galactose 1-phosphate** [5]. The connection to the glucose metabolism is established by C-4 epimerization to form **glucose 1-phosphate**. However, this does not take place directly. Instead, a *transferase* [6] transfers a uridine 5'-monophosphate (UMP) residue from uridine diphosphoglucose (UDPglucose) to galactose 1-phosphate. This releases glucose 1-phosphate, while galactose 1-phosphate is converted into uridine diphosphogalactose (UDPgalactose). This then is isomerized into UDPglucose. The *biosynthesis* of galactose also follows this reaction pathway, which is freely reversible up to reaction [5]. Genetic defects of enzymes [5] or [6] can lead to the clinical picture of *galactosemia*.

A. Gluconeogenesis: overview



B. Fructose and galactose metabolism



Lipid metabolism

The liver is the most important site for the formation of *fatty acids*, *fats* (*triacylglycerols*), *ketone bodies*, and *cholesterol*. Most of these products are released into the blood. In contrast, the triacylglycerols synthesized in adipose tissue are also stored there.

A. Lipid metabolism ①

Lipid metabolism in the liver is closely linked to the carbohydrate and amino acid metabolism. When there is a good supply of nutrients in the *resorptive (wellfed) state* (see p. 308), the liver converts glucose via acetyl CoA into **fatty acids**. The liver can also take up fatty acids from *chylomicrons*, which are supplied by the intestine, or from fatty acid–albumin complexes (see p. 162). Fatty acids from both sources are converted into **fats** and **phospholipids**. Together with apoproteins, they are packed into very-low-density lipoproteins (**VLDLs**; see p. 278) and then released into the blood by exocytosis. The VLDLs supply extrahepatic tissue, particularly adipose tissue and muscle.

In the *postresorptive state* (see p. 292)—particularly during fasting and starvation—the lipid metabolism is readjusted and the organism falls back on its own reserves. In these conditions, adipose tissue releases fatty acids. They are taken up by the liver and are mainly converted into **ketone bodies** (B).

Cholesterol can be derived from two sources—food or endogenous synthesis from acetyl-CoA. A substantial percentage of endogenous cholesterol synthesis takes place in the liver. Some cholesterol is required for the synthesis of **bile acids** (see p. 314). In addition, it serves as a building block for cell membranes (see p. 216), or can be esterified with fatty acids and stored in lipid droplets. The rest is released together into the blood in the form of lipoprotein complexes (VLDLs) and supplies other tissues. The liver also contributes to the cholesterol metabolism by taking up from the blood and breaking down lipoproteins that contain cholesterol and cholesterol esters (HDLs, IDLs, LDLs; see p. 278).

B. Biosynthesis of ketone bodies ①

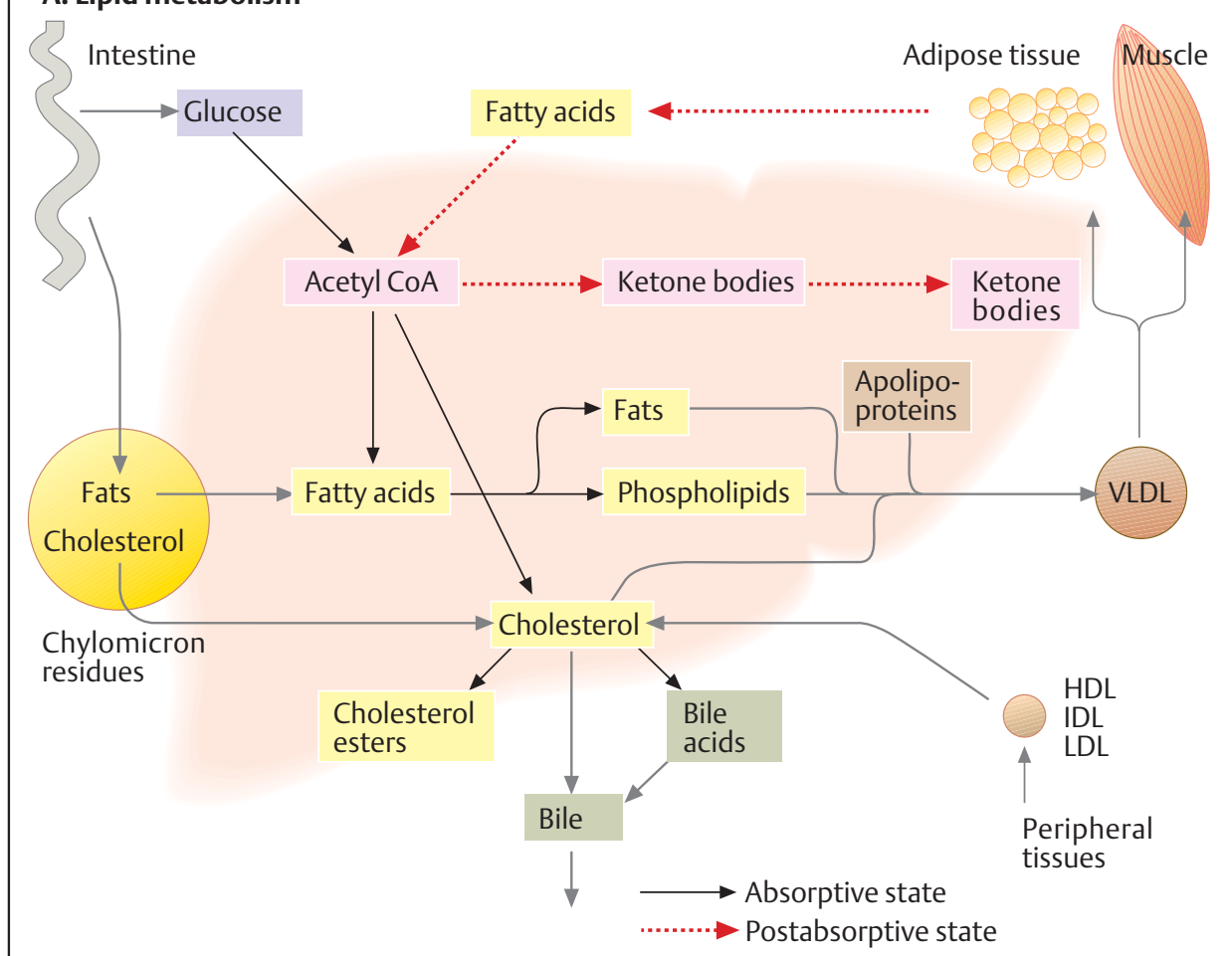
At high concentrations of acetyl-CoA in the liver mitochondria, two molecules condense to form **acetoacetyl CoA** [1]. The transfer of another acetyl group [2] gives rise to 3-hydroxy-3-methylglutaryl-CoA (**HMG CoA**), which after release of acetyl CoA [3] yields free **acetoacetate** (*Lynen cycle*). Acetoacetate can be converted to **3-hydroxybutyrate** by reduction [4], or can pass into **acetone** by nonenzymatic decarboxylation [5]. These three compounds are together referred to as “*ketone bodies*,” although in fact 3-hydroxybutyrate is not actually a ketone. As reaction [3] releases an H^+ ion, metabolic acidosis can occur as a result of increased ketone body synthesis (see p. 288).

The ketone bodies are released by the liver into the blood, in which they are easily soluble. Blood levels of ketone bodies therefore rise during periods of hunger. Together with free fatty acids, 3-hydroxybutyrate and acetoacetate are then the most important energy suppliers in many tissues (including heart muscle). Acetone cannot be metabolized and is exhaled via the lungs or excreted with urine.

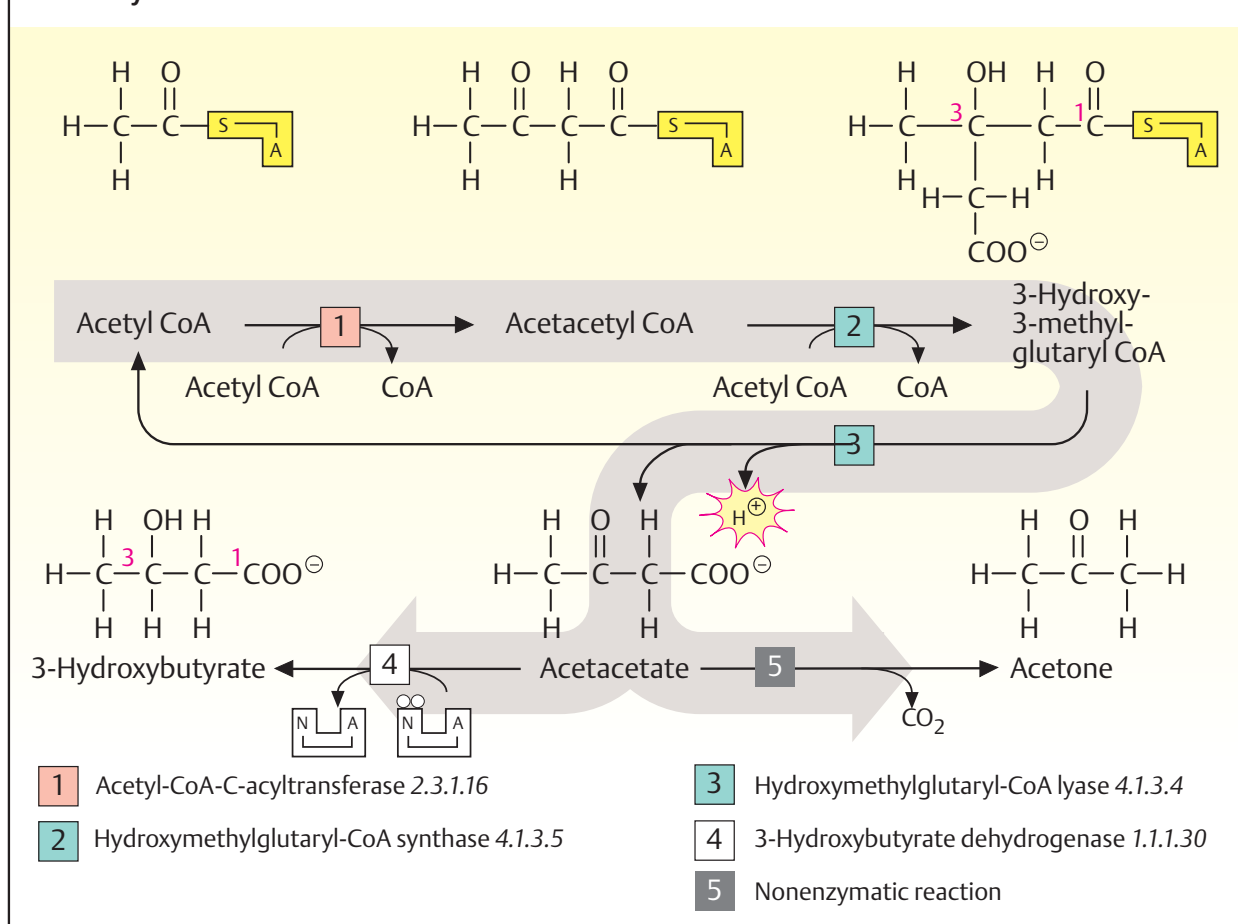
To channel ketone bodies into the energy metabolism, acetoacetate is converted with the help of succinyl CoA into succinic acid and acetoacetyl CoA, which is broken down by β -oxidation into acetyl CoA (not shown; see p. 180).

If the production of ketone bodies exceeds the demand for them outside the liver, there is an increase in the concentration of ketone bodies in the plasma (*ketonemia*) and they are also eventually excreted in the urine (*ketonuria*). Both phenomena are observed after prolonged starvation and in inadequately treated *diabetes mellitus*. Severe ketonuria with ketoacidosis can cause electrolyte shifts and loss of consciousness, and is therefore life-threatening (*ketoacidotic coma*).

A. Lipid metabolism



A. Biosynthesis of ketone bodies



Bile acids

Bile is an important product released by the hepatocytes. It promotes the digestion of fats from food by *emulsifying* them in the small intestine (see p. 2770). The emulsifying components of bile, apart from phospholipids, mainly consist of **bile acids** and **bile salts** (see below). The bile also contains free cholesterol, which is excreted in this way (see p. 312).

A. Bile acids and bile salts ①

Bile acids are steroids consisting of 24 C atoms carrying one carboxylate group and several hydroxyl groups. They are formed from cholesterol in the liver via an extensive reaction pathway (top). Cytochrome P450 enzymes in the sER of hepatocytes are involved in many of the steps (see p. 318). Initially, the cholesterol double bond is removed. Monooxygenases then introduce one or two additional OH groups into the sterane framework. Finally, the side chain is shortened by three C atoms, and the terminal C atom is oxidized to a carboxylate group.

It is important that the arrangement of the A and B rings is altered from *trans* to *cis* during bile acid synthesis (see p. 54). The result of this is that all of the hydrophilic groups in the bile acids lie on one side of the molecule. Cholesterol, which is weakly amphipathic (top), has a small polar “head” and an extended apolar “tail.” By contrast, the much more strongly amphipathic bile acid molecules (bottom) resemble disks with polar top sides and apolar bottom sides. At physiological pH values, the carboxyl groups are almost completely dissociated and therefore negatively charged.

Cholic acid and *chenodeoxycholic acid*, known as the **primary bile acids**, are quantitatively the most important metabolites of cholesterol. After being biosynthesized, they are mostly activated with coenzyme A and then conjugated with *glycine* or the non-proteinogenic amino acid *taurine* (see p. 62). The acid amides formed in this way are known as **conjugated bile acids** or **bile salts**. They are even more amphipathic than the primary products.

Deoxycholic acid and *lithocholic acid* are only formed in the intestine by enzymatic

cleavage of the OH group at C-7 (see B). They are therefore referred to as **secondary bile acids**.

B. Metabolism of bile salts ②

Bile salts are exclusively synthesized in the liver (see A). The slowest step in their biosynthesis is hydroxylation at position 7 by a 7- α -hydroxylase. Cholic acid and other bile acids inhibit this reaction (*end-product inhibition*). In this way, the bile acids present in the liver regulate the rate of cholesterol utilization.

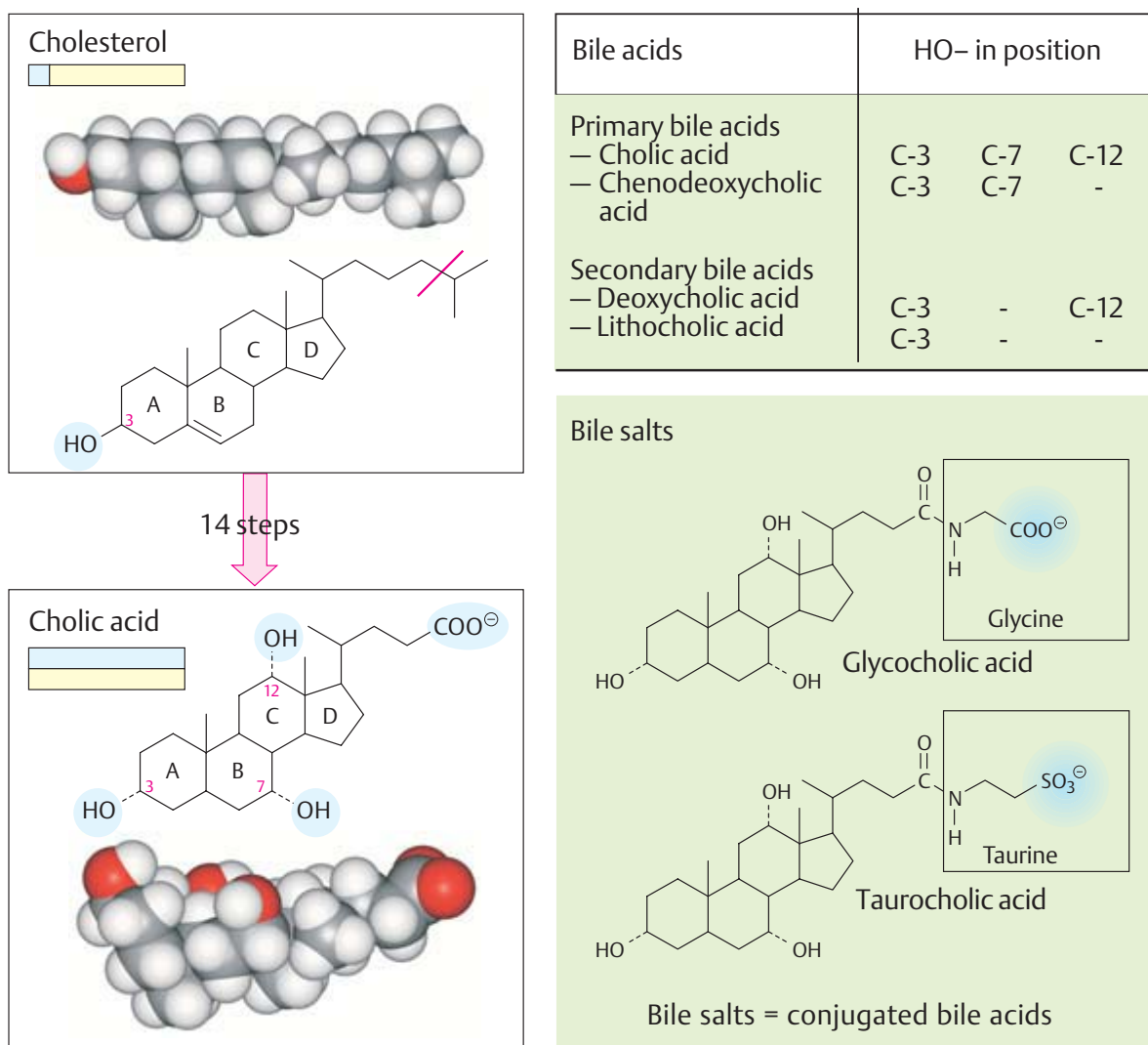
Before leaving the liver, a large proportion of the bile acids are activated with CoA and then conjugated with the amino acids *glycine* or *taurine* (2; cf. A). In this way, cholic acid gives rise to *glycocholic acid* and *taurocholic acid*. The *liver bile* secreted by the liver becomes denser in the gallbladder as a result of the removal of water (*bladder bile*; 3).

Intestinal bacteria produce enzymes that can chemically alter the bile salts (4). The acid amide bond in the bile salts is cleaved, and dehydroxylation at C-7 yields the corresponding secondary bile acids from the primary bile acids (5). Most of the intestinal bile acids are resorbed again in the ileum (6) and returned to the liver via the portal vein (*enterohepatic circulation*). In the liver, the secondary bile acids give rise to primary bile acids again, from which bile salts are again produced. Of the 15–30g bile salts that are released with the bile per day, only around 0.5g therefore appears in the feces. This approximately corresponds to the amount of daily de novo synthesis of cholesterol.

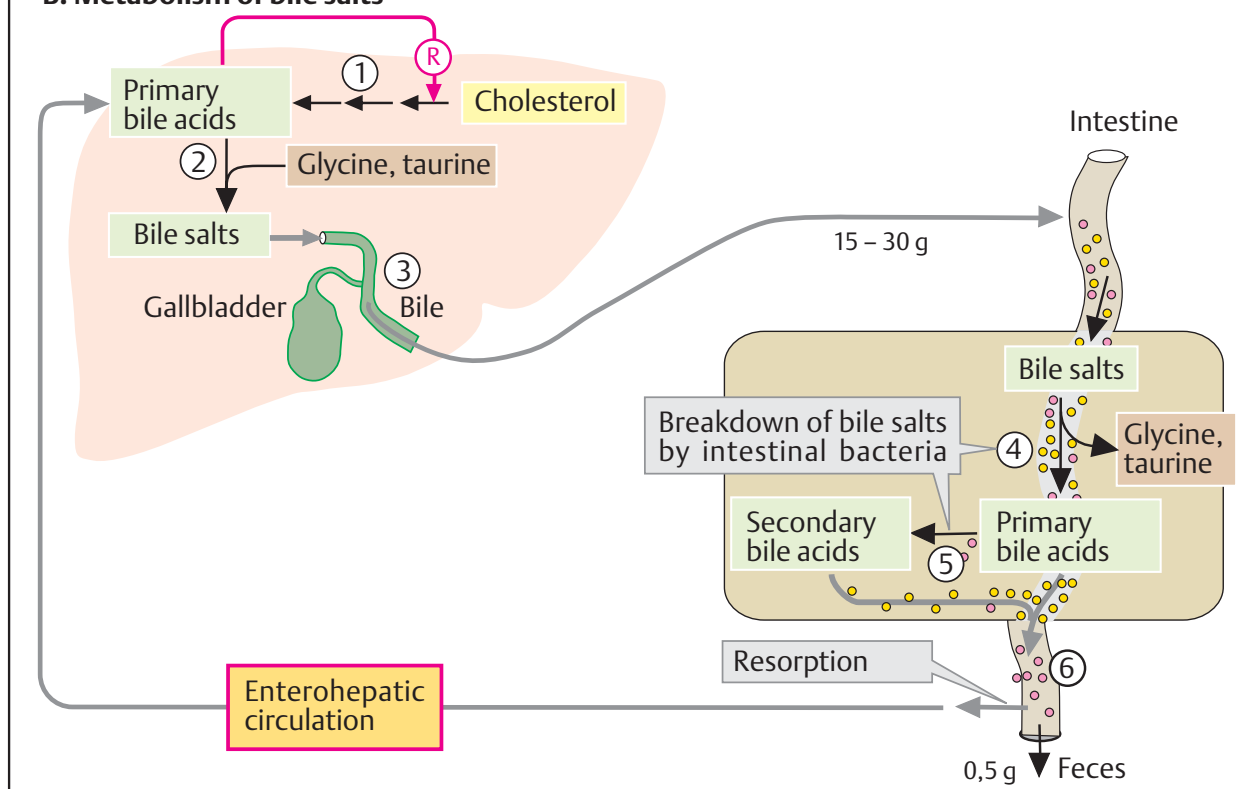
Further information

The cholesterol excreted with the bile is poorly water-soluble. Together with phospholipids and bile acids, it forms micelles (see p. 270), which keep it in solution. If the proportions of phospholipids, bile acids and cholesterol shift, *gallstones* can arise. These mainly consist of precipitated cholesterol (cholesterol stones), but can also contain Ca^{2+} salts of bile acids and bile pigments (pigment stones).

A. Bile acids and bile salts



B. Metabolism of bile salts



Biotransformations

The body is constantly taking up **foreign substances** (= *xenobiotics*) from food or through contact with the environment, via the skin and lungs. These substances can be natural in origin, or may have been synthetically produced by humans. Many of these substances are toxic, particularly at high concentrations. However, the body has effective mechanisms for inactivating and then excreting foreign substances through *biotransformations*. The mechanisms of biotransformation are similar to those with which **endogenous substances** such as bile pigments and steroid hormones are enzymatically converted. Biotransformations mainly take place in the *liver*.

A. Biotransformations ●

Phase I reactions (interconversion reactions). Type I reactions introduce functional groups into inert, apolar molecules or alter functional groups that are already present. In many cases, this is what first makes it possible for foreign substances to conjugate with polar molecules via phase II reactions (see below). Phase I reactions usually reduce the *biological activity* or *toxicity* of a substance (“detoxification”). However, some substances only become biologically active as a result of the interconversion reaction (see, for example, benzo[*a*]pyrene, p. 256) or become more toxic after interconversion than the initial substance (“toxicification”).

Important phase I biotransformation reactions include:

- **Hydrolytic cleavages** of ether, ester, and peptide bonds. Example (1) shows hydrolysis of the painkiller *acetylsalicylic acid*.
- **Oxidations**. Hydroxylations, epoxide formation, sulfoxide formation, dealkylation, deamination. For example, benzene is oxidized into phenol, and toluene (methylbenzene) is oxidized into benzoic acid.
- **Reductions**. Reduction of carbonyl, azo-, or nitro- compounds, dehalogenation.
- **Methylations**. Example (2) illustrates the inactivation of the catecholamine *norepinephrine* by methylation of a phenolic OH group (see p. 334).
- **Desulfurations**. The reactions take place in the hepatocytes on the smooth endoplasmic reticulum.

Most oxidation reactions are catalyzed by **cytochrome P450 systems** (see p. 318). These monooxygenases are induced by their substrates and show wide specificity. The substrate-specific enzymes of the steroid metabolism (see p. 376) are exceptions to this.

Phase II reactions (conjugate formation). Type II reactions couple their substrates (bilirubin, steroid hormones, drugs, and products of phase I reactions) via ester or amide bonds to highly polar negatively charged molecules. The enzymes involved are transferases, and their products are known as **conjugates**.

The most common type of conjugate formation is coupling with *glucuronate* (GlcUA) as an *O*- or *N*-glucuronide. The coenzyme for the reaction is uridine diphosphate glucuronate, the “*active glucuronate*” (see p. 110). Coupling with the polar glucuronate makes an apolar (hydrophobic) molecule more strongly polar, and it becomes sufficiently water-soluble and capable of being excreted. Example (3) shows the glucuronidation of *tetrahydrocortisol*, a metabolite of the glucocorticoid cortisol (see p. 374).

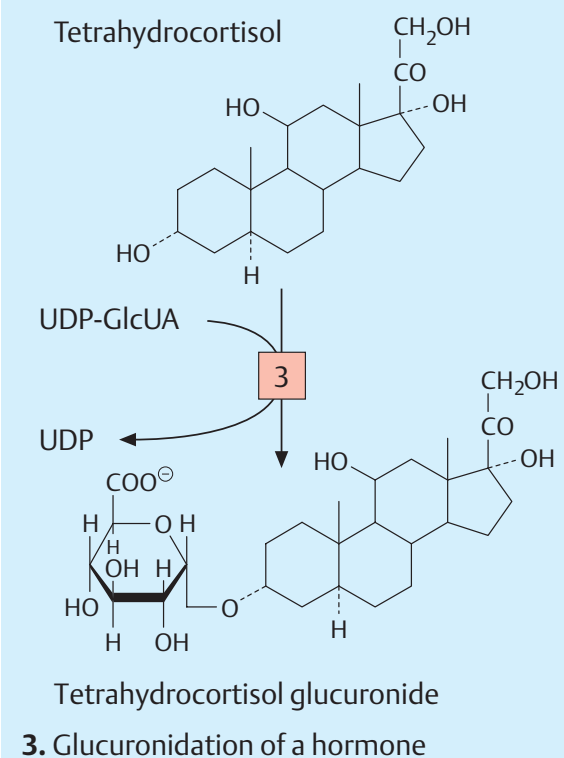
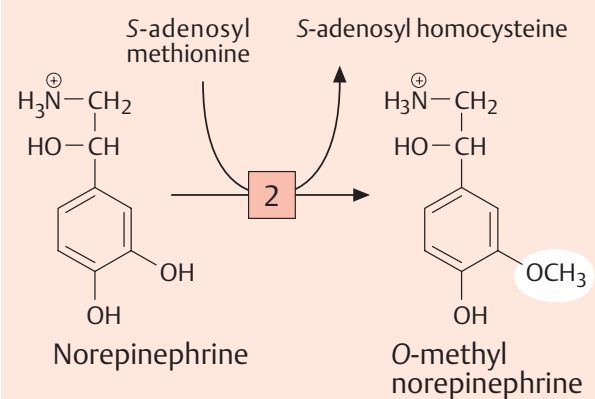
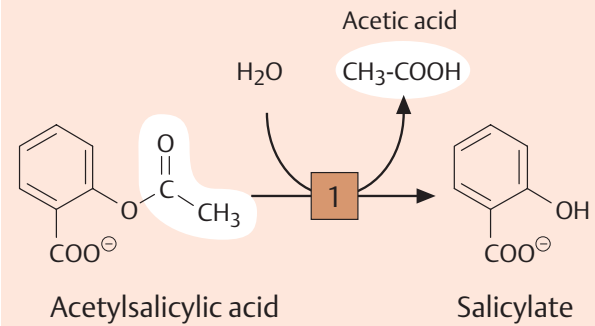
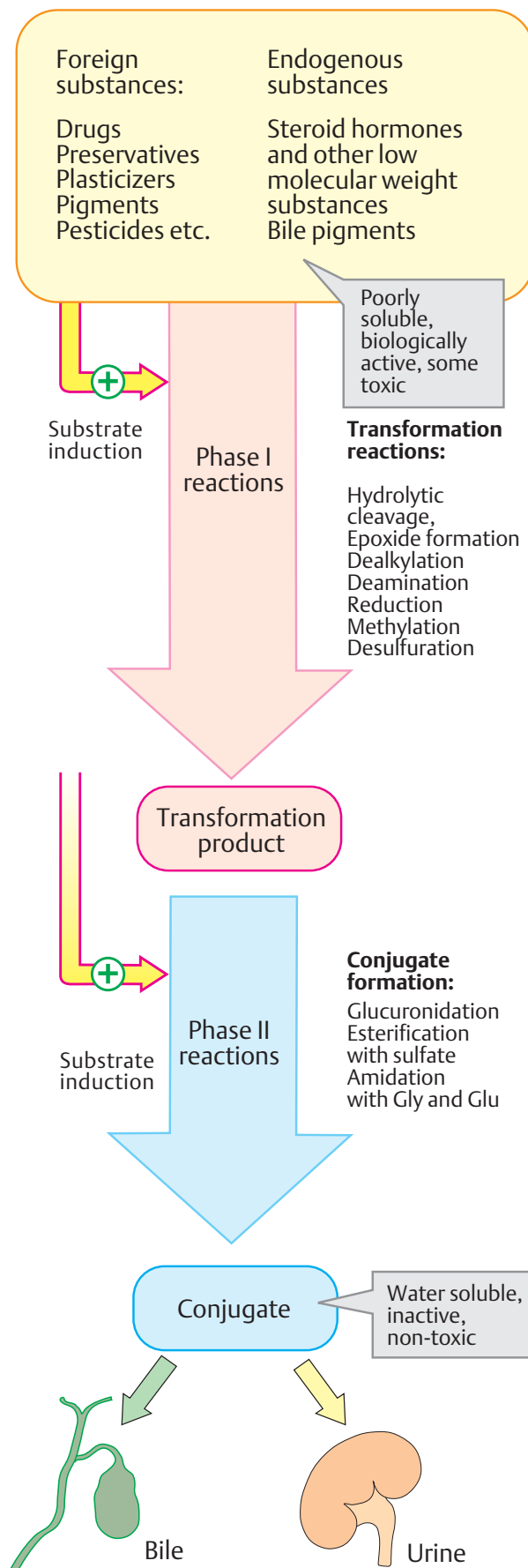
The biosynthesis of sulfate esters with the help of *phosphoadenosine phosphosulfate* (PAPS), the “*active sulfate*”, (see p. 110) and amide formation with *glycine* and *glutamine* also play a role in conjugation. For example, benzoic acid is conjugated with glycine to form the more soluble and less toxic *hippuric acid* (*N*-benzoylglycine; see p. 324).

In contrast with unconjugated compounds, the conjugates are much more water-soluble and capable of being excreted. The conjugates are eliminated from the liver either by the *biliary* route—i.e., by receptor-mediated excretion into the bile—or by the *renal* route, via the blood and kidneys by filtration.

Further information

To detoxify **heavy metals**, the liver contains *metallothioneins*, a group of cysteine-rich proteins with a high affinity for divalent metal ions such as Cd^{2+} , Cu^{2+} , Hg^{2+} , and Zn^{2+} . These metal ions also induce the formation of metallothioneins via a special metal-regulating element (MRE) in the gene's promoter (see p. 244).

A. Biotransformations



- 1 Arylesterase 3.1.1.2
- 2 Catechol O-methyltransferase 2.1.1.6
- 3 Glucuronosyltransferase 2.4.1.17

Cytochrome P450 systems

During the first phase of biotransformation in the liver, compounds that are weakly chemically reactive are enzymatically hydroxylated (see p. 316). This makes it possible for them to be conjugated with polar substances. The hydroxylating enzymes are generally *monooxygenases* that contain a **heme** as the redox-active coenzyme (see p. 106). In the reduced form, the heme can bind carbon monoxide (CO), and it then shows characteristic light absorption at 450 nm. This was what led to this enzyme group being termed **cytochrome P450** (Cyt P450).

Cyt P450 systems are also involved in many other metabolic processes—e.g., the biosynthesis of steroid hormones (see p. 172), bile acids (see p. 314), and eicosanoids (see p. 390), as well as the formation of unsaturated fatty acids (see p. 409). The liver's reddish-brown color is mainly due to the large amounts of P450 enzymes it contains.

A. Cytochrome P450-dependent mono oxygenases: reactions ●

Cyt P450-dependent monooxygenases catalyze reductive cleavage of molecular oxygen (O_2). One of the two oxygen atoms is transferred to the substrate, while the other is released as a water molecule. The necessary reducing equivalents are transferred to the actual monooxygenase by an FAD-containing auxiliary enzyme from the coenzyme $NADPH+H^+$.

Cyt P450 enzymes occur in numerous forms in the liver, steroid-producing glands, and other organs. The substrate specificity of liver enzymes is low. Apolar compounds containing aliphatic or aromatic rings are particularly easily converted. These include endogenous substances such as steroid hormones, as well as medical drugs, which are inactivated by phase I reactions. This is why Cyt P450 enzymes are of particular interest in pharmacology. The degradation of ethanol in the liver is also partly catalyzed by Cyt P450 enzymes (the “microsomal ethanol-oxidizing system”; see p. 304). As alcohol and drugs are broken down by the same enzyme system, the effects of alcoholic drinks and medical drugs can sometimes be mutually enhancing—even sometimes to the extent of becoming life-threatening.

Only a few examples of the numerous Cyt P450-dependent reactions are shown here. *Hydroxylation* of aromatic rings (**a**) plays a central part in the metabolism of medicines and steroids. Aliphatic methyl groups can also be oxidized to hydroxyl groups (**b**). *Epoxidation* of aromatics (**c**) by Cyt P450 yields products that are highly reactive and often toxic. For example, the mutagenic effect of benzo[*a*]pyrene (see p. 244) is based on this type of interconversion in the liver. In Cyt P450 dependent *dealkylations* (**d**), alkyl substituents of O, N, or S atoms are released as aldehydes.

B. Reaction mechanism ○

The course of Cyt P450 catalysis is in principle well understood. The most important function of the *heme group* consists of converting molecular oxygen into an especially reactive atomic form, which is responsible for all of the reactions described above.

[1] In the resting state, the heme iron is trivalent. Initially, the substrate binds near the heme group.

[2] Transfer of an electron from $FADH_2$ reduces the iron to the divalent form that is able to bind an O_2 molecule (2).

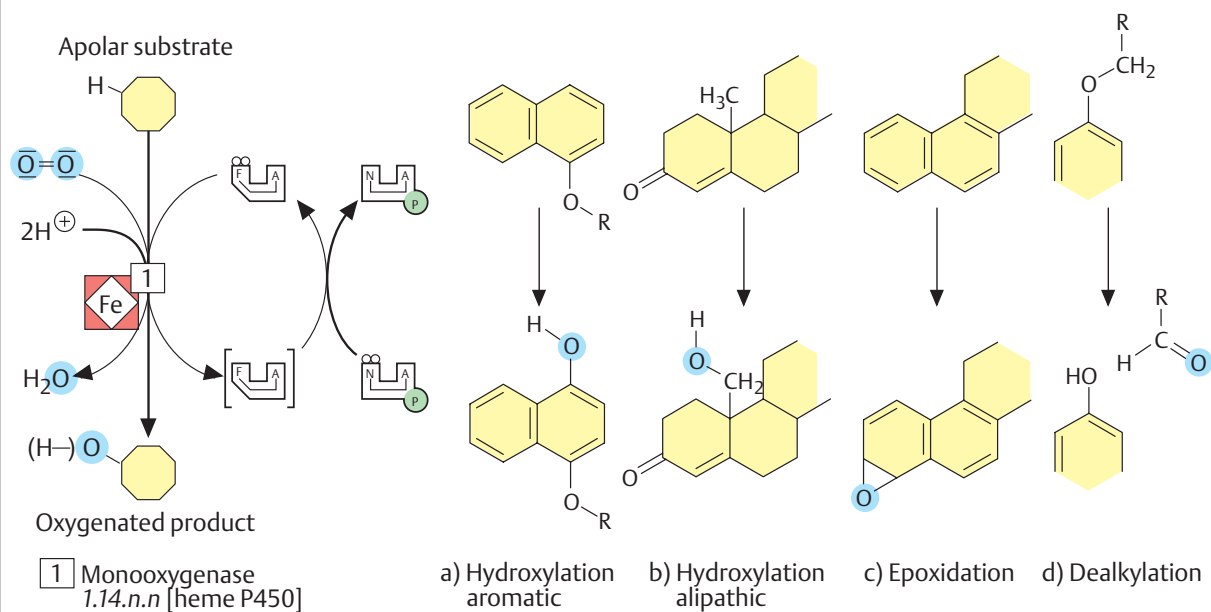
[3] Transfer of a second electron and a change in the valence of the iron reduce the bound O_2 to the peroxide.

[4] A hydroxyl ion is now cleaved from this intermediate. Uptake of a proton gives rise to H_2O and the reactive form of oxygen mentioned above. In this ferryl radical, the iron is formally tetravalent.

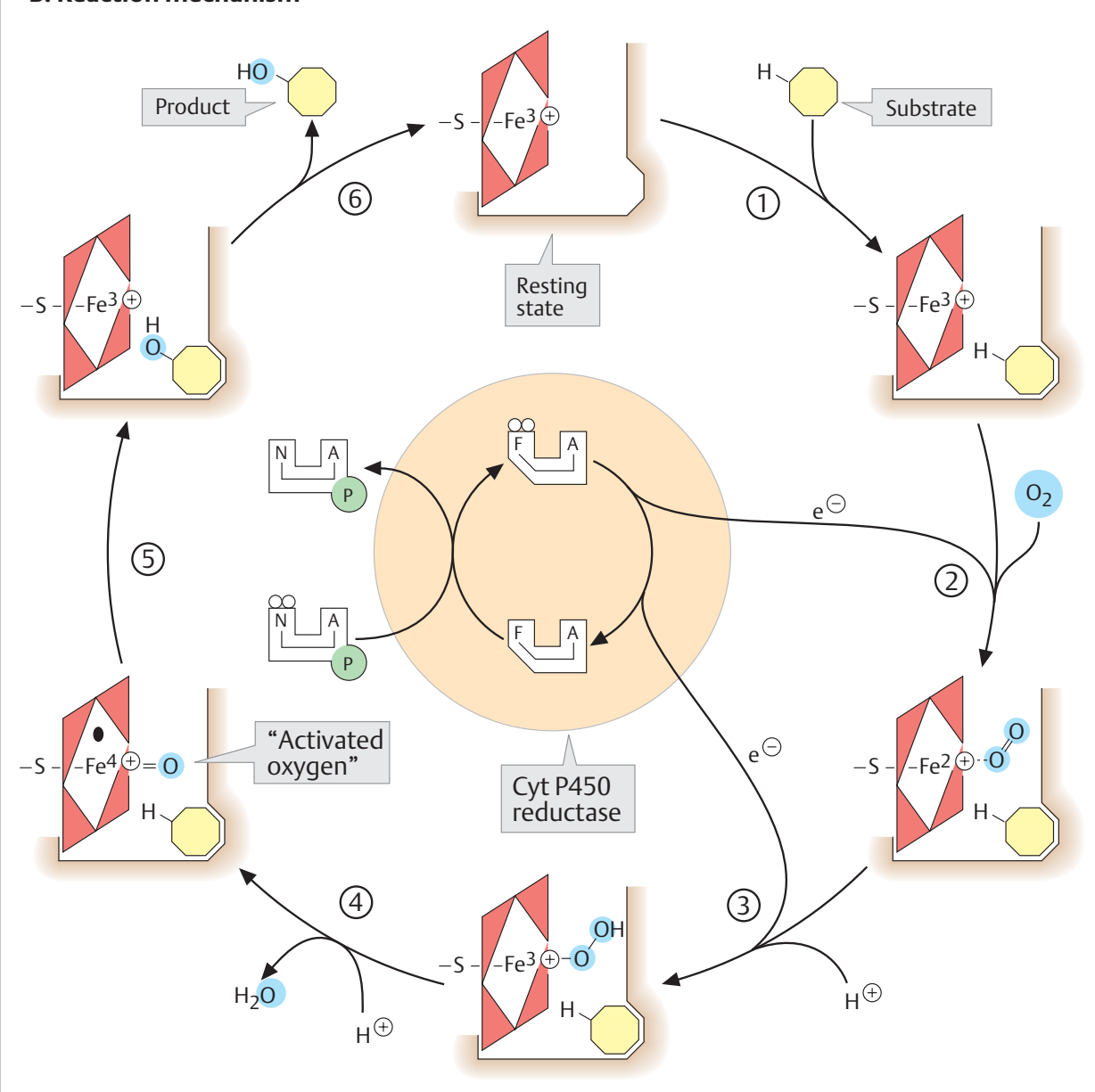
[5] The activated oxygen atom inserts itself into a C–H bond in the substrate, thereby forming an OH group.

[6] Dissociation of the product returns the enzyme to its initial state.

A. Cytochrome P450-dependent monooxygenases: reactions



B. Reaction mechanism



Ethanol metabolism

A. Blood ethanol level ●

Ethanol (EtOH, “alcohol”) naturally occurs in fruit in small quantities. Alcoholic drinks contain much higher concentrations. Their alcohol content is usually given as percent by volume. To estimate alcohol uptake and the blood alcohol level, it is useful to convert the amount to grams of ethanol (density 0.79 kg L^{-1}). For example, a bottle of beer (0.5 L at $4\% \text{ v/v}$ alcohol) contains $20 \text{ mL} = 16 \text{ g}$ of ethanol, while a bottle of wine (0.7 L at $12\% \text{ v/v}$ alcohol) contains $84 \text{ mL} = 66 \text{ g}$ ethanol.

Ethanol is membrane-permeable and is quickly resorbed. The maximum blood level is already reached within 60–90 min after drinking. The *resorption rate* depends on various conditions, however. An empty stomach, a warm drink (e.g., mulled wine), and the presence of sugar and carbonic acid (e.g., in champagne) promote ethanol resorption, whereas a heavy meal reduces it. Ethanol is rapidly distributed throughout the body. A large amount is taken up by the muscles and brain, but comparatively little by adipose tissue and bones. Roughly 70% of the body is accessible to alcohol. Complete resorption of the ethanol contained in one bottle of beer (16 g) by a person weighing 70 kg (distribution in $70 \text{ kg} \cdot 70/100 = 49 \text{ kg}$) leads to a blood alcohol level of 0.33 per thousand (7.2 mM). The lethal concentration of alcohol is approximately 3.5 per thousand (76 mM).

B. Ethanol metabolism ●

The major site of ethanol degradation is the liver, although the stomach is also able to metabolize ethanol. Most **ethanol** is initially oxidized by *alcohol dehydrogenase* to form **ethanal** (acetaldehyde). A further oxidation, catalyzed by *aldehyde dehydrogenase*, leads to acetate. Acetate is then converted with the help of *acetate-CoA ligase* to form **acetyl CoA**, using ATP and providing a link to the intermediary metabolism. In addition to cytoplasmic alcohol dehydrogenase, *catalase* and inducible *microsomal alcohol oxidase* (“MEOS”; see p. 318) also contribute to a lesser extent to ethanol degradation. Many of the enzymes mentioned above are induced by ethanol.

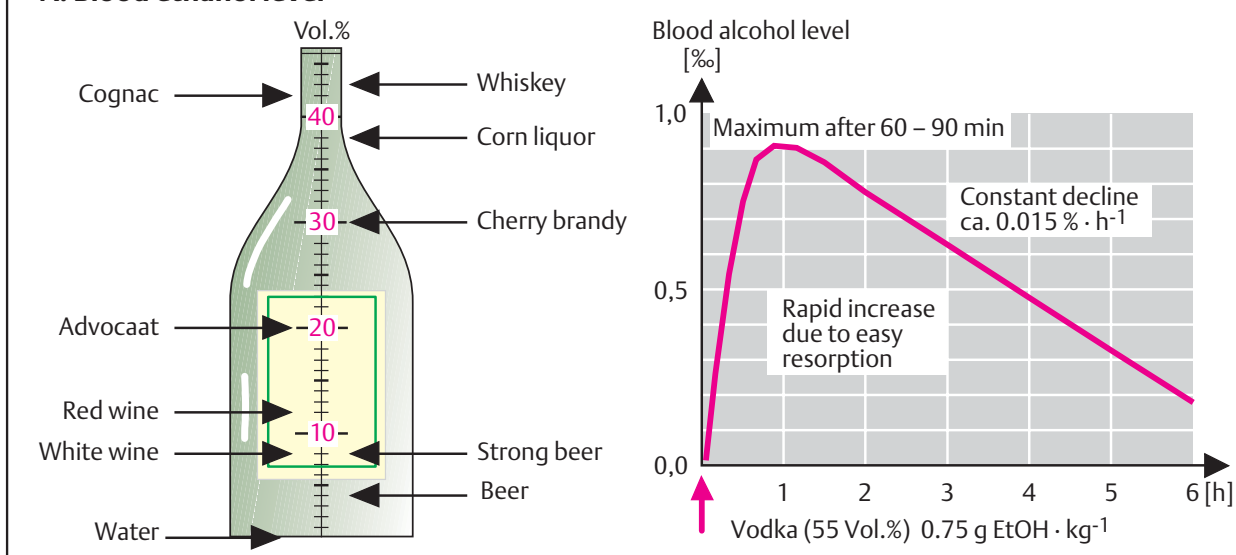
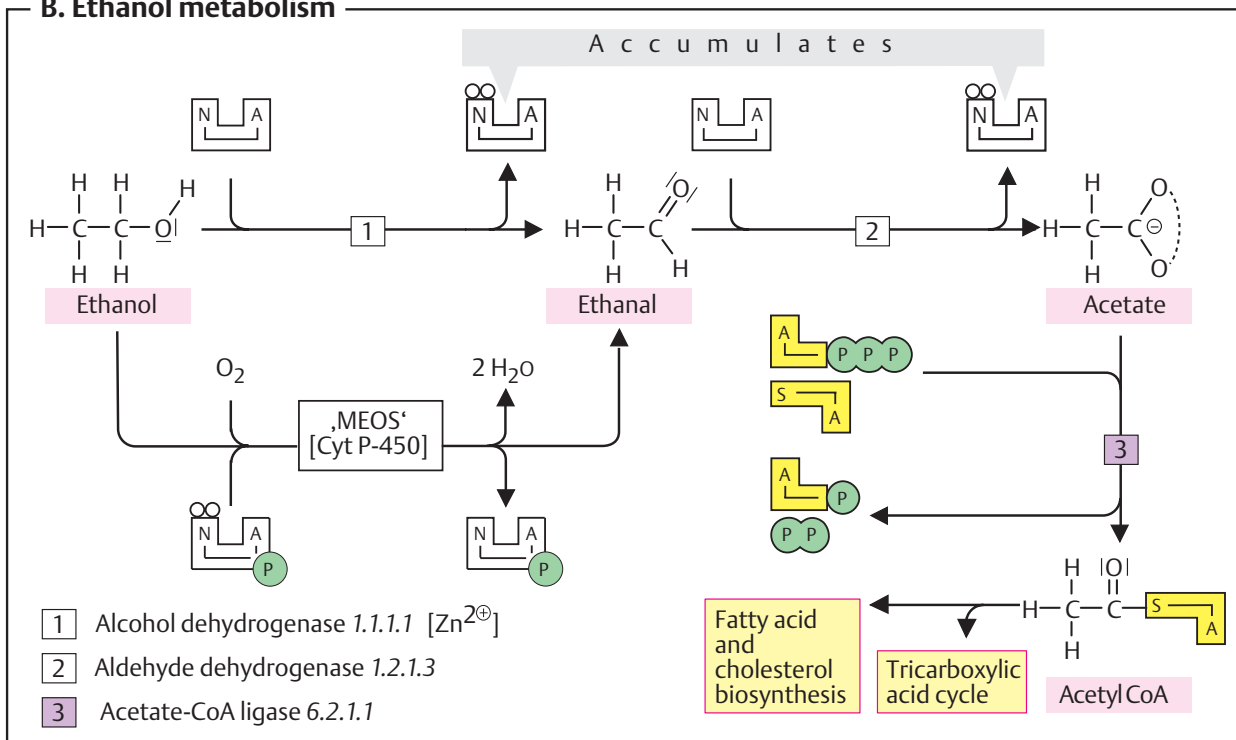
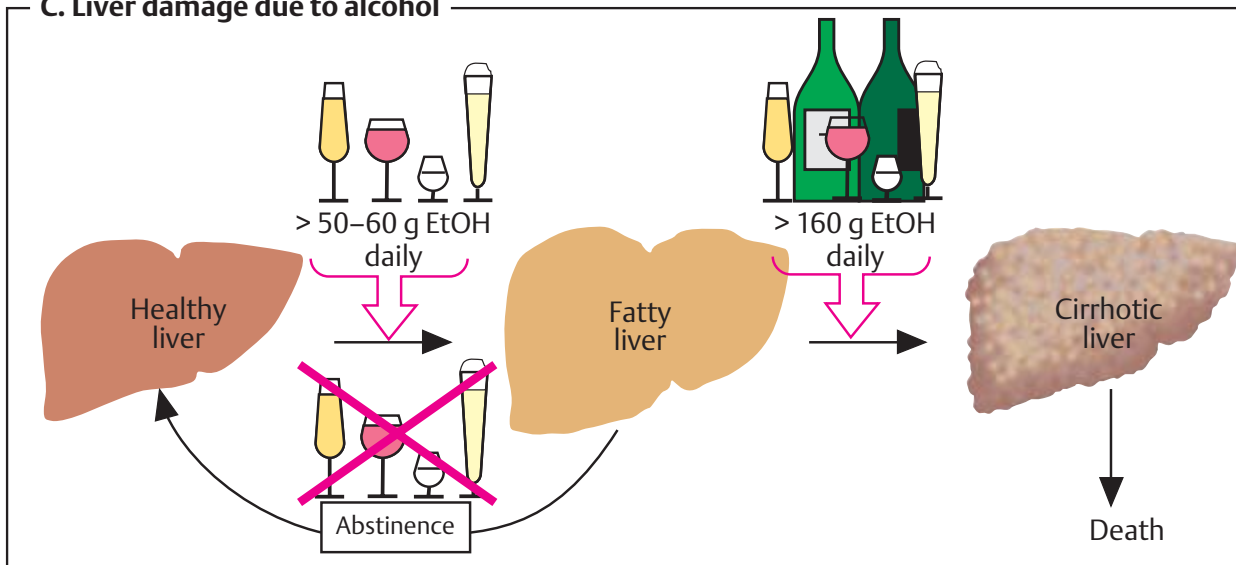
The rate of ethanol degradation in the liver is limited by alcohol dehydrogenase activity. The amount of NAD^+ available is the limiting factor. As the maximum degradation rate is already reached at low concentrations of ethanol, the ethanol level therefore declines at a constant rate (zero-order kinetics). The *calorific value* of ethanol is 29.4 kJ g^{-1} . Alcoholic drinks—particularly in alcoholics—can therefore represent a substantial proportion of dietary energy intake.

C. Liver damage due to alcohol ●

Alcohol is a socially accepted drug of abuse in Western countries. Due to the high potential for addiction to develop, however, it is actually a “hard” drug and has a much larger number of victims than the opiate drugs, for example. In the brain, ethanol is deposited in membranes due to its amphipathic properties, and it influences receptors for neurotransmitters (see p. 352). The effect of GABA is enhanced, while that of glutamate declines.

High ethanol consumption over many years leads to liver damage. For a healthy man, the limit is about 60 g per day, and for a woman about 50 g. However, these values are strongly dependent on body weight, health status, and other factors.

Ethanol-related high levels of $\text{NADH}+\text{H}^+$ and acetyl-CoA in the liver lead to increased synthesis of neutral fats and cholesterol. However, since the export of these in the form of VLDLs (see p. 278) is reduced due to alcohol, storage of lipids occurs (**fatty liver**). This increase in the fat content of the liver (from less than 5% to more than 50% of the dry weight) is initially reversible. However, in chronic alcoholism the hepatocytes are increasingly replaced by connective tissue. When **liver cirrhosis** occurs, the damage to the liver finally reaches an irreversible stage, characterized by progressive loss of liver functions.

A. Blood ethanol level**B. Ethanol metabolism****C. Liver damage due to alcohol**

Kidney: functions

A. Functions of the kidneys ●

The kidneys' main function is **excretion** of water and water-soluble substances (1). This is closely associated with their role in regulating the body's electrolyte and acid–base balance (**homeostasis**, 2; see pp. 326 and 328). Both excretion and homeostasis are subject to hormonal control. The kidneys are also involved in synthesizing several **hormones** (3; see p. 315). Finally, the kidneys also play a role in the **intermediary metabolism** (4), particularly in amino acid degradation and gluconeogenesis (see p. 154).

The kidneys are extremely well-perfused organs, with about 1500 L of blood flowing through them every day. Approximately 180 L of primary urine is filtered out of this. Removal of water leads to extreme concentration of the primary urine (to approximately one-hundredth of the initial volume). As a result, only a volume of 0.5–2.0 L of **final urine** is excreted per day.

B. Urine formation ●

The functional unit of the kidney is the **nephron**. It is made up of the Malpighian bodies or renal corpuscles (consisting of Bowman's capsules and the glomerulus), the proximal tubule, Henle's loop, and the distal tubule, which passes into a collecting duct. The human kidney contains around one million nephrons. The nephrons form urine in the following three phases.

Ultrafiltration. Ultrafiltration of the blood plasma in the glomerulus gives rise to *primary urine*, which is *isotonic* with plasma. The pores in the glomerular basal membrane, which are made up of type IV collagen (see p. 344), have an effective mean diameter of 2.9 nm. This allows all plasma components with a molecular mass of up to about 15 kDa to pass through unhindered. At increasing masses, molecules are progressively held back; at masses greater than 65 kDa, they are completely unable to enter the primary urine. This applies to almost all plasma proteins—which in addition, being anions, are repelled by the negative charge in the basal membrane.

Resorption. All low-molecular weight plasma components enter the primary urine via glomerular filtration. Most of these are

transported back into the blood by resorption, to prevent losses of valuable metabolites and electrolytes. In the proximal tubule, organic metabolites (e.g., glucose and other sugars, amino acids, lactate, and ketone bodies) are recovered by secondary active transport (see p. 220). There are several group-specific transport systems for resorbing amino acids, with which hereditary diseases can be associated (e.g., *cystinuria*, *glycinuria*, and *Hartnup's disease*). HCO_3^- , Na^+ , phosphate, and sulfate are also resorbed by ATP-dependent (active) mechanisms in the proximal tubule. The later sections of the nephron mainly serve for additional water recovery and regulated resorption of Na^+ and Cl^- (see pp. 326, 328). These processes are controlled by hormones (aldosterone, vasopressin).

Secretion. Some excretable substances are released into the urine by *active transport* in the renal tubules. These substances include H^+ and K^+ ions, urea, and creatinine, as well as drugs such as penicillin.

Clearance. Renal clearance is used as a quantitative measure of renal function. It is defined as the plasma volume cleared of a given substance per unit of time. *Inulin*, a fructose polysaccharide with a mass of ca. 6 kDa (see p. 40) that is neither actively excreted nor resorbed but is freely filtered, has a clearance of 120 mL min^{-1} in healthy individuals.

Further information

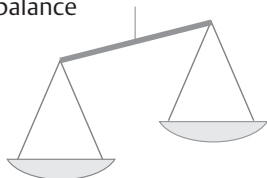
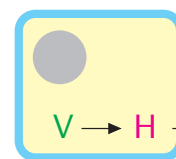
Concentrating urine and transporting it through membranes are processes that require large amounts of energy. The kidneys therefore have very high energy demands. In the proximal tubule, the ATP needed is obtained from oxidative metabolism of *fatty acids*, *ketone bodies*, and several *amino acids*. To a lesser extent, lactate, glycerol, and citric acid are also used. In the distal tubule and Henle's loop, *glucose* is the main substrate for the energy metabolism. The endothelial cells in the proximal tubule are also capable of *gluconeogenesis*. The substrates for this are mainly the carbohydrate skeletons of amino acids. Their amino groups are used as ammonia for buffering urine (see p. 311). Enzymes for peptide degradation and the amino acid metabolism occur in the kidneys at high levels of activity (e.g., amino acid oxidases, amine oxidases, glutaminase).

A. Functions of the kidneys**1. Excretion**

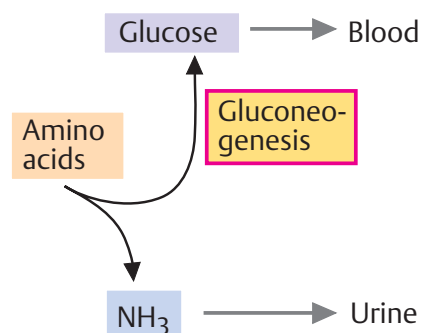
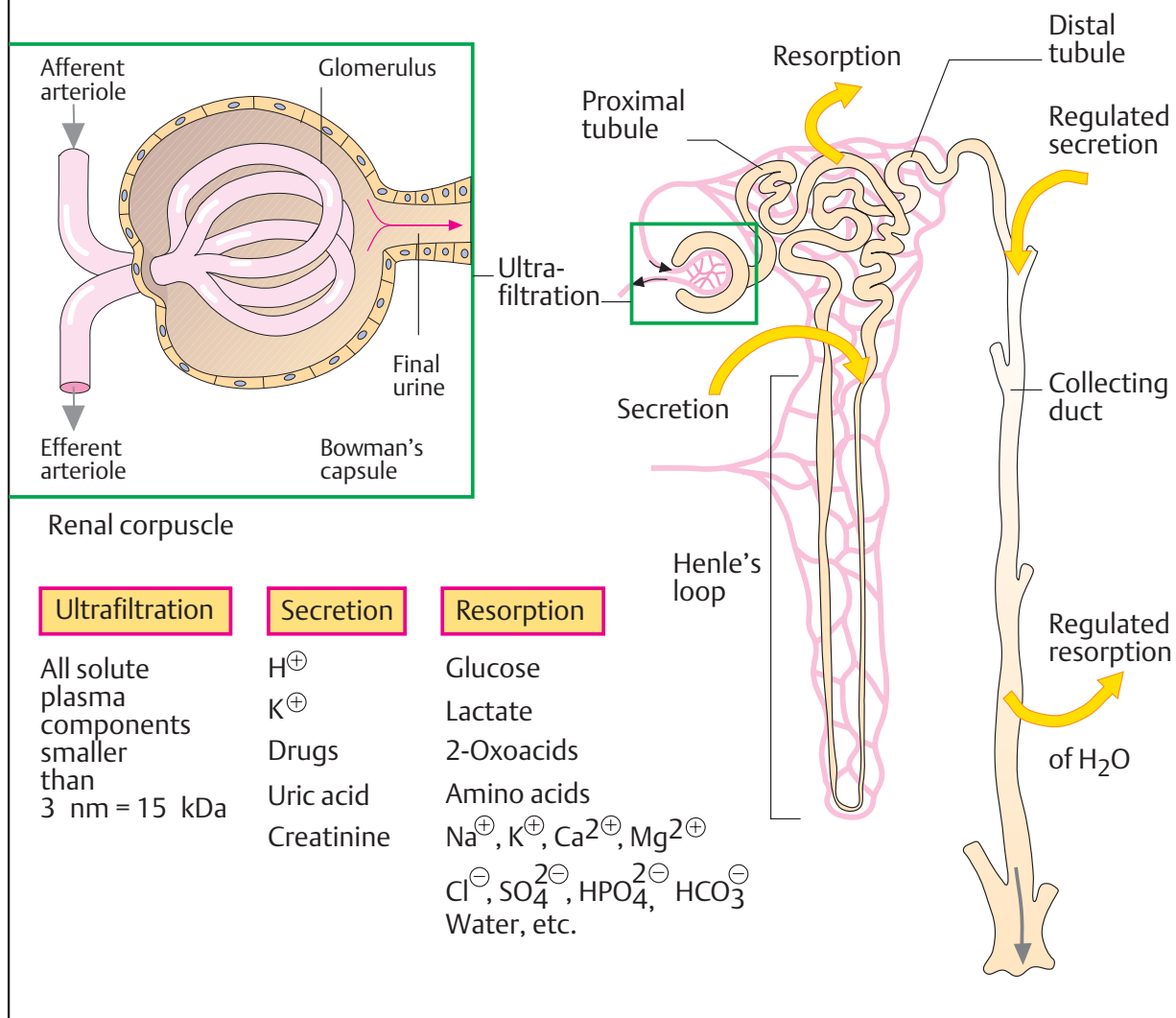
Water
Salts
Metabolic
wastes
Foreign
substances

**2. Homeostasis**

Acid-base balance
Electrolyte balance

**3. Hormone synthesis**

Erythropoietin
Calcitriol

4. Metabolism**B. Urine formation**

Urine

A. Urine ①

Water and water-soluble compounds are excreted with the urine. The volume and composition of urine are subject to wide variation and depend on food intake, body weight, age, sex, and living conditions such as temperature, humidity, physical activity, and health status. As there is a marked circadian rhythm in urine excretion, the amount of urine and its composition are usually given relative to a 24-hour period.

A human adult produces 0.5–2.0 L urine per day, around 95% of which consists of water. The urine usually has a slightly acidic pH value (around 5.8). However, the pH value of urine is strongly affected by metabolic status. After ingestion of large amounts of plant food, it can increase to over 7.

B. Organic components ①

Nitrogen-containing compounds are among the most important organic components of urine. **Urea**, which is mainly synthesized in the liver (urea cycle; see p. 182), is the form in which nitrogen atoms from amino acids are excreted. Breakdown of pyrimidine bases also produces a certain amount of urea (see p. 190). When the nitrogen balance is constant, as much nitrogen is excreted as is taken up (see p. 174), and the amount of urea in the urine therefore reflects protein degradation: 70 g protein in food yields approximately 30 g urea in the urine.

Uric acid is the end product of the purine metabolism. When uric acid excretion via the kidneys is disturbed, gout can develop (see p. 190). **Creatinine** is derived from the muscle metabolism, where it arises spontaneously and irreversibly by cyclization of creatine and creatine phosphate (see p. 336). Since the amount of creatinine an individual excretes per day is constant (it is directly proportional to muscle mass), creatinine as an endogenous substance can be used to measure the glomerular filtration rate. The amount of **amino acids** excreted in free form is strongly dependent on the diet and on the efficiency of liver function. Amino acid derivatives are also found in the urine (e.g., **hippurate**, a detoxification product of benzoic acid).

Modified amino acids, which occur in special proteins such as *hydroxyproline* in collagen and *3-methylhistidine* in actin and myosin, can be used as indicators of the degradation of these proteins.

Other components of the urine are conjugates with sulfuric acid, glucuronic acid, glycine, and other polar compounds that are synthesized in the liver by biotransformation (see p. 316). In addition, metabolites of many hormones (catecholamines, steroids, serotonin) also appear in the urine and can provide information about hormone production. The proteohormone *chorionic gonadotropin* (hCG, mass ca. 36 kDa), which is formed at the onset of pregnancy, appears in the urine due to its relatively small size. Evidence of hCG in the urine provides the basis for an immunological *pregnancy test*.

The yellow color of urine is due to *urochromes*, which are related to the bile pigments produced by hemoglobin degradation (see p. 194). If urine is left to stand long enough, oxidation of the urochromes may lead to a darkening in color.

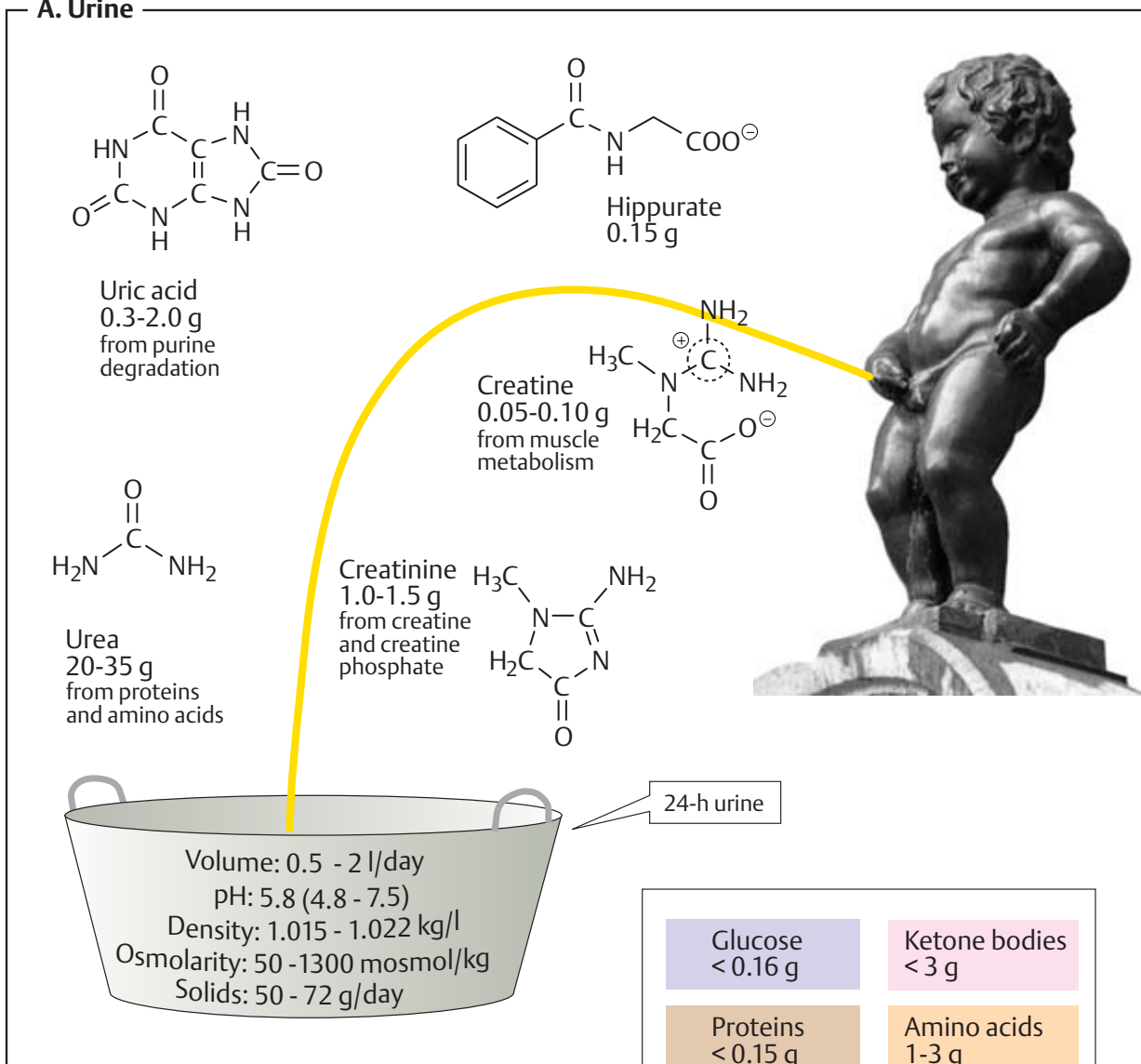
C. Inorganic components ①

The main inorganic components of the urine are the *cations* Na^+ , K^+ , Ca^{2+} , Mg^{2+} , and NH_4^+ and the *anions* Cl^- , SO_4^{2-} , and HPO_4^{2-} , as well as traces of other ions. In total, Na^+ and Cl^- represent about two-thirds of all the electrolytes in the final urine. Calcium and magnesium occur in the feces in even larger quantities. The amounts of the various inorganic components of the urine also depend on the composition of the diet. For example, in acidosis there can be a marked increase in the excretion of ammonia (see p. 326). Excretion of Na^+ , K^+ , Ca^{2+} , and phosphate via the kidneys is subject to hormonal regulation (see p. 330).

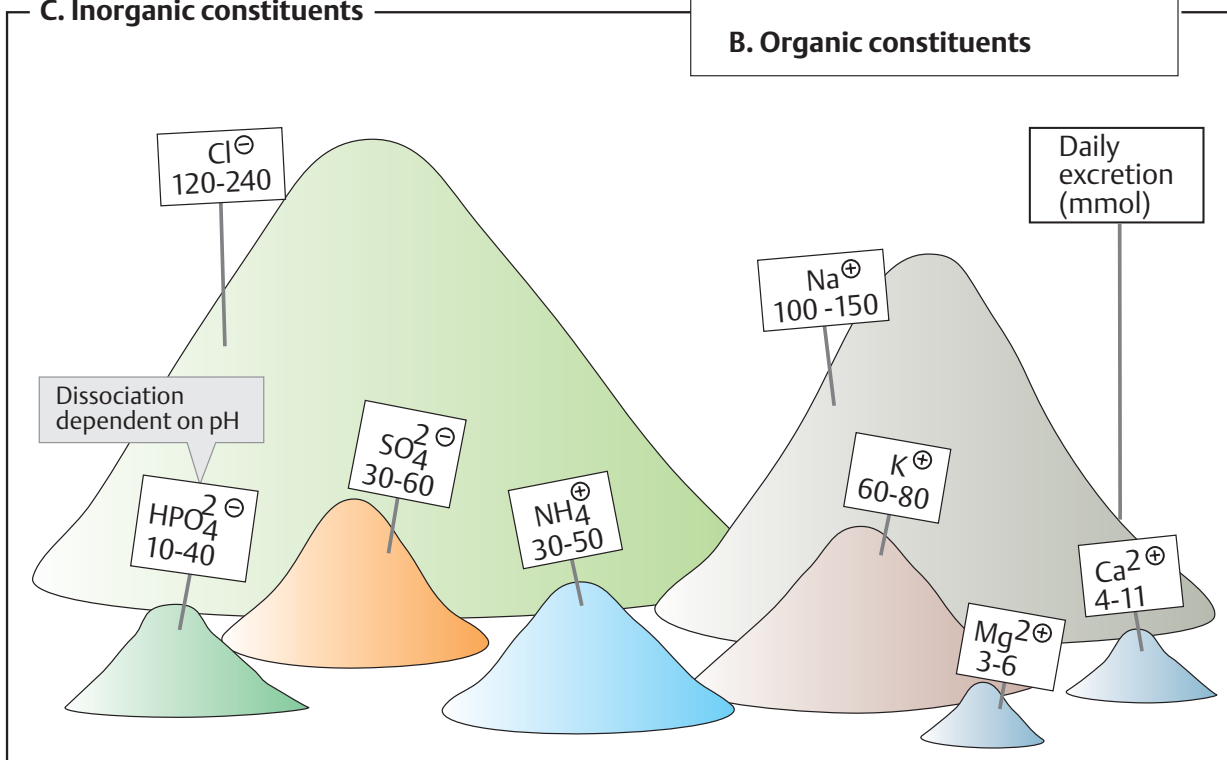
Further information

Shifts in the concentrations of the *physiological components* of the urine and the appearance of *pathological urine components* can be used to diagnose diseases. Important examples are glucose and ketone bodies, which are excreted to a greater extent in *diabetes mellitus* (see p. 160).

A. Urine



C. Inorganic constituents



Functions in the acid–base balance

Along with the lungs, the kidneys are particularly involved in keeping the pH value of the extracellular fluid constant (see p. 288). The contribution made by the kidneys particularly involves resorbing HCO_3^- and actively excreting protons.

A. Proton excretion ●

The renal tubule cells are capable of secreting protons (H^+) from the blood into the urine against a concentration gradient, despite the fact that the H^+ concentration in the urine is up to a thousand times higher than in the blood. To achieve this, carbon dioxide (CO_2) is taken up from the blood and—together with water (H_2O) and with the help of *carbonate dehydratase* (carbonic anhydrase, [1])—converted into hydrogen carbonate (“bicarbonate,” HCO_3^-) and one H^+ . Formally, this yields carbonic acid H_2CO_3 as an intermediate, but it is not released during the reaction.

The hydrogen carbonate formed in carbonic anhydrase returns to the plasma, where it contributes to the blood’s base reserve. The proton is exported into the urine by *secondary active transport* in antiport for Na^+ (bottom right). The driving force for **proton excretion**, as in other secondary active processes, is the Na^+ gradient established by the ATPase involved in the Na^+/K^+ exchange (“ Na^+/K^+ ATPase”, see p. 220). This integral membrane protein on the basal side (towards the blood) of tubule cells keeps the Na^+ concentration in the tubule cell low, thereby maintaining Na^+ inflow. In addition to this secondary active H^+ transport mechanism, there is a V-type H^+ -transporting ATPase in the distal tubule and collecting duct (see p. 220).

An important function of the secreted H^+ ions is to promote **HCO_3^- resorption** (top right). Hydrogen carbonate, the most important buffering base in the blood, passes into the primary urine quantitatively, like all ions. In the primary urine, HCO_3^- reacts with H^+ ions to form water and CO_2 , which returns by free diffusion to the tubule cells and from there into the blood. In this way, the kidneys also influence the $\text{CO}_2/\text{HCO}_3^-$ buffering balance in the plasma.

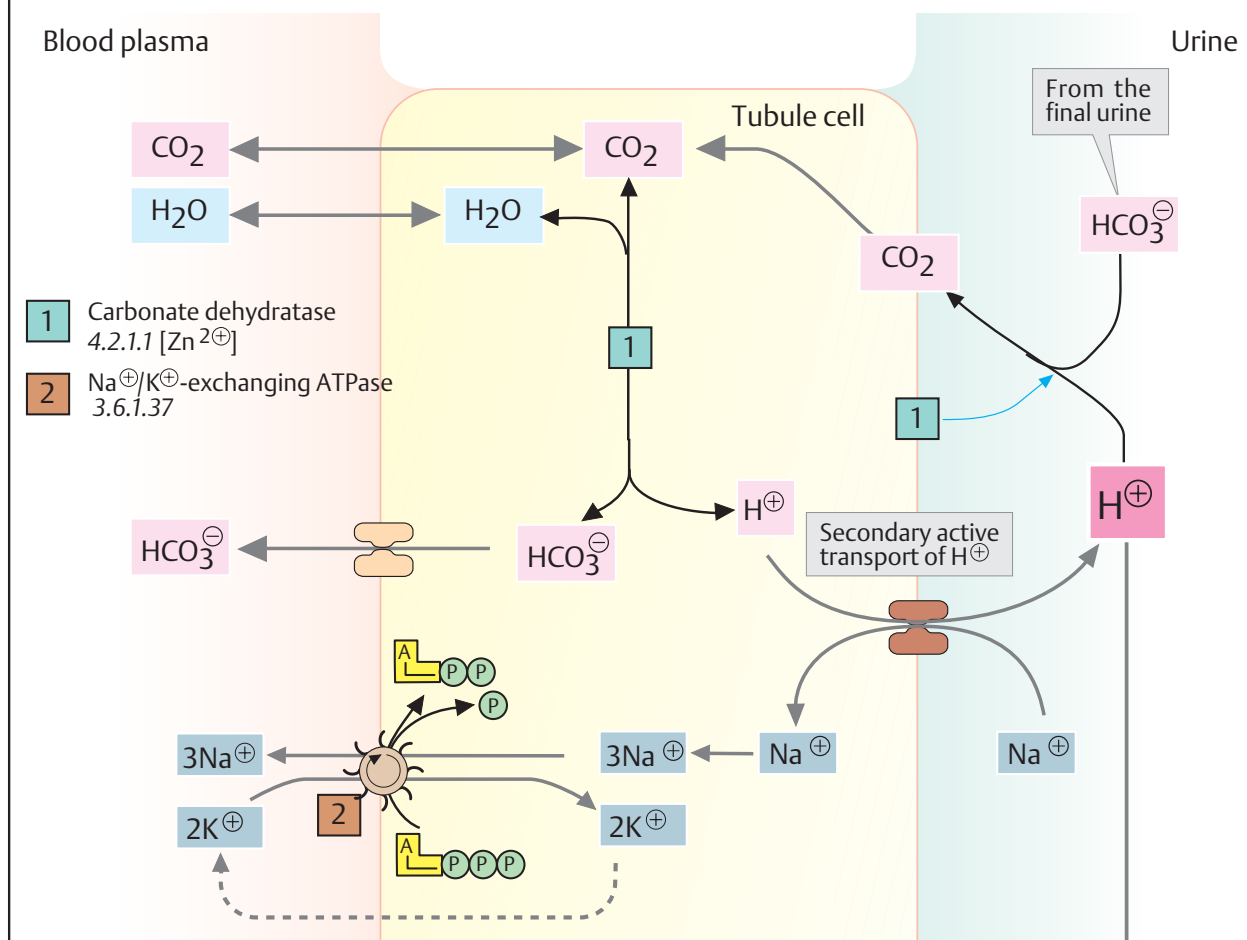
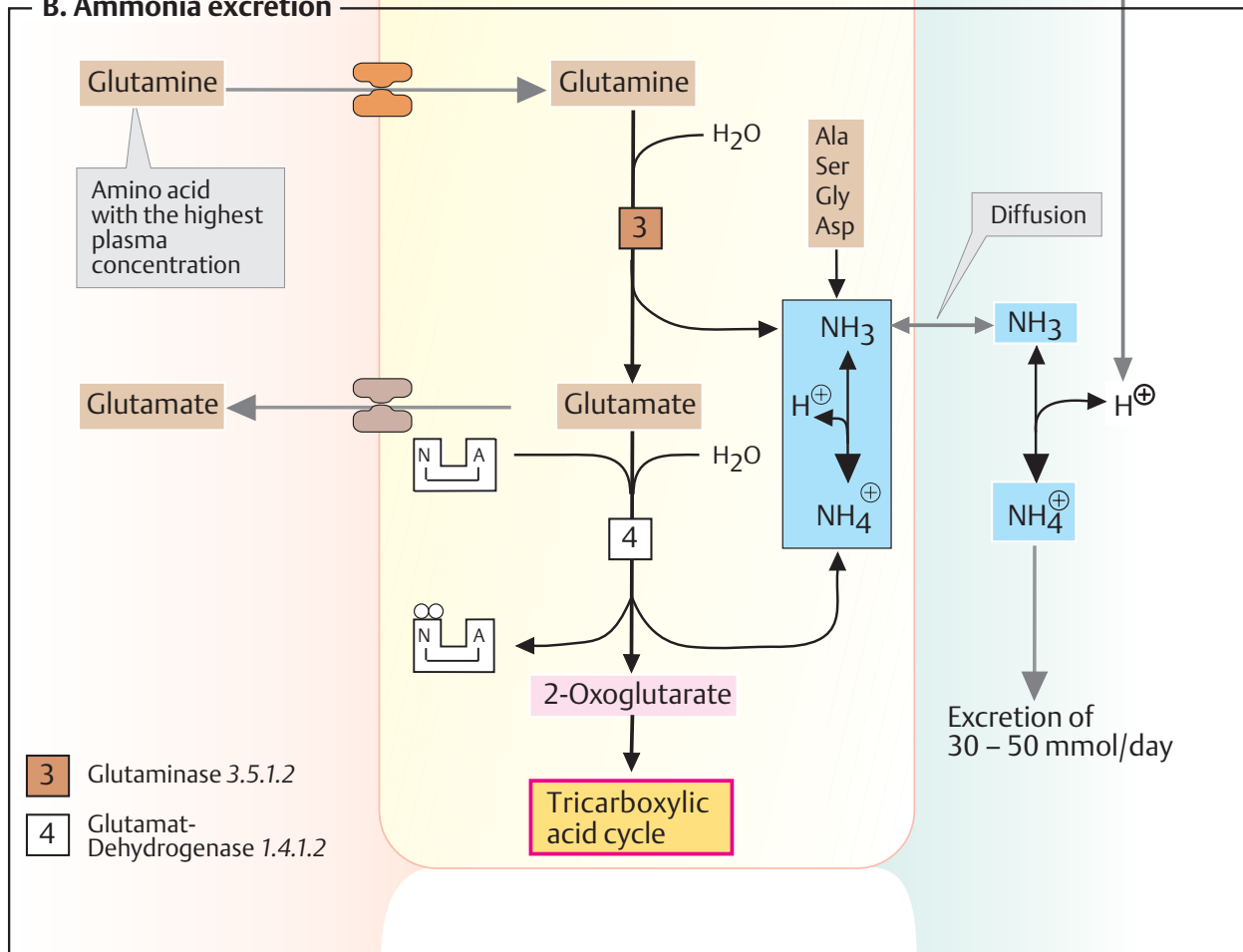
B. Ammonia excretion ●

Approximately 60 mmol of protons are excreted with the urine every day. Buffering systems in the urine catch a large proportion of the H^+ ions, so that the urine only becomes weakly acidic (down to about pH 4.8).

An important buffer in the urine is the *hydrogen phosphate/dihydrogen phosphate system* ($\text{HPO}_4^{2-}/\text{H}_2\text{PO}_4^-$). In addition, *ammonia* also makes a vital contribution to buffering the secreted protons.

Since plasma concentrations of *free ammonia* are low, the kidneys release NH_3 from **glutamine** and other amino acids. At 0.5–0.7 mM, glutamine is the most important amino acid in the plasma and is the preferred form for ammonia transport in the blood. The kidneys take up glutamine, and with the help of *glutaminase* [4], initially release NH_3 from the amide bond hydrolytically. From the **glutamate** formed, a second molecule of NH_3 can be obtained by oxidative deamination with the help of *glutamate dehydrogenase* [5] (see p. 178). The resulting **2-oxoglutarate** is further metabolized in the tricarboxylic acid cycle. Several other amino acids—*alanine* in particular, as well as *serine*, *glycine*, and *aspartate*—can also serve as suppliers of ammonia.

Ammonia can diffuse freely into the urine through the tubule membrane, while the **ammonium ions** that are formed in the urine are charged and can no longer return to the cell. Acidic urine therefore promotes ammonia excretion, which is normally 30–50 mmol per day. In metabolic **acidosis** (e.g., during fasting or in *diabetes mellitus*), after a certain time increased *induction of glutaminase* occurs in the kidneys, resulting in increased NH_3 excretion. This in turn promotes H^+ release and thus counteracts the acidosis. By contrast, when the plasma pH value shifts towards alkaline values (*alkalosis*), renal excretion of ammonia is reduced.

A. Proton secretion**B. Ammonia excretion**

Electrolyte and water recycling

A. Electrolyte and water recycling ❶

Electrolytes and other plasma components with low molecular weights enter the *primary urine* by ultrafiltration (right). Most of these substances are recovered by energy-dependent resorption (see p. 322). The extent of the resorption determines the amount that ultimately reaches the final urine and is excreted. The illustration does not take into account the *zoning* of transport processes in the kidney (physiology textbooks may be referred to for further details).

Calcium and phosphate ions. Calcium (Ca^{2+}) and phosphate ions are almost completely resorbed from the primary urine by active transport (i.e., in an ATP-dependent fashion). The proportion of Ca^{2+} resorbed is over 99%, while for phosphate the figure is 80–90%. The extent to which these two electrolytes are resorbed is regulated by the three hormones parathyrin, calcitonin, and calcitriol.

The peptide hormone *parathyrin* (PTH), which is produced by the parathyroid gland, stimulates Ca^{2+} resorption in the kidneys and at the same time inhibits the resorption of phosphate. In conjunction with the effects of this hormone in the bones and intestines (see p. 344), this leads to an *increase in the plasma level of Ca^{2+}* and a reduction in the level of phosphate ions.

Calcitonin, a peptide produced in the C cells of the thyroid gland, inhibits the resorption of both calcium and phosphate ions. The result is an overall *reduction in the plasma level* of both ions. Calcitonin is thus a parathyrin antagonist relative to Ca^{2+} .

The steroid hormone *calcitriol*, which is formed in the kidneys (see p. 304), stimulates the resorption of both calcium and phosphate ions and thus increases the plasma level of both ions.

Sodium ions. Controlled resorption of Na^+ from the primary urine is one of the most important functions of the kidney. Na^+ resorption is highly effective, with more than 97% being resorbed. Several mechanisms are involved: some of the Na^+ is taken up passively in the proximal tubule through the junctions between the cells (paracellularly). In addition, there is secondary active transport together

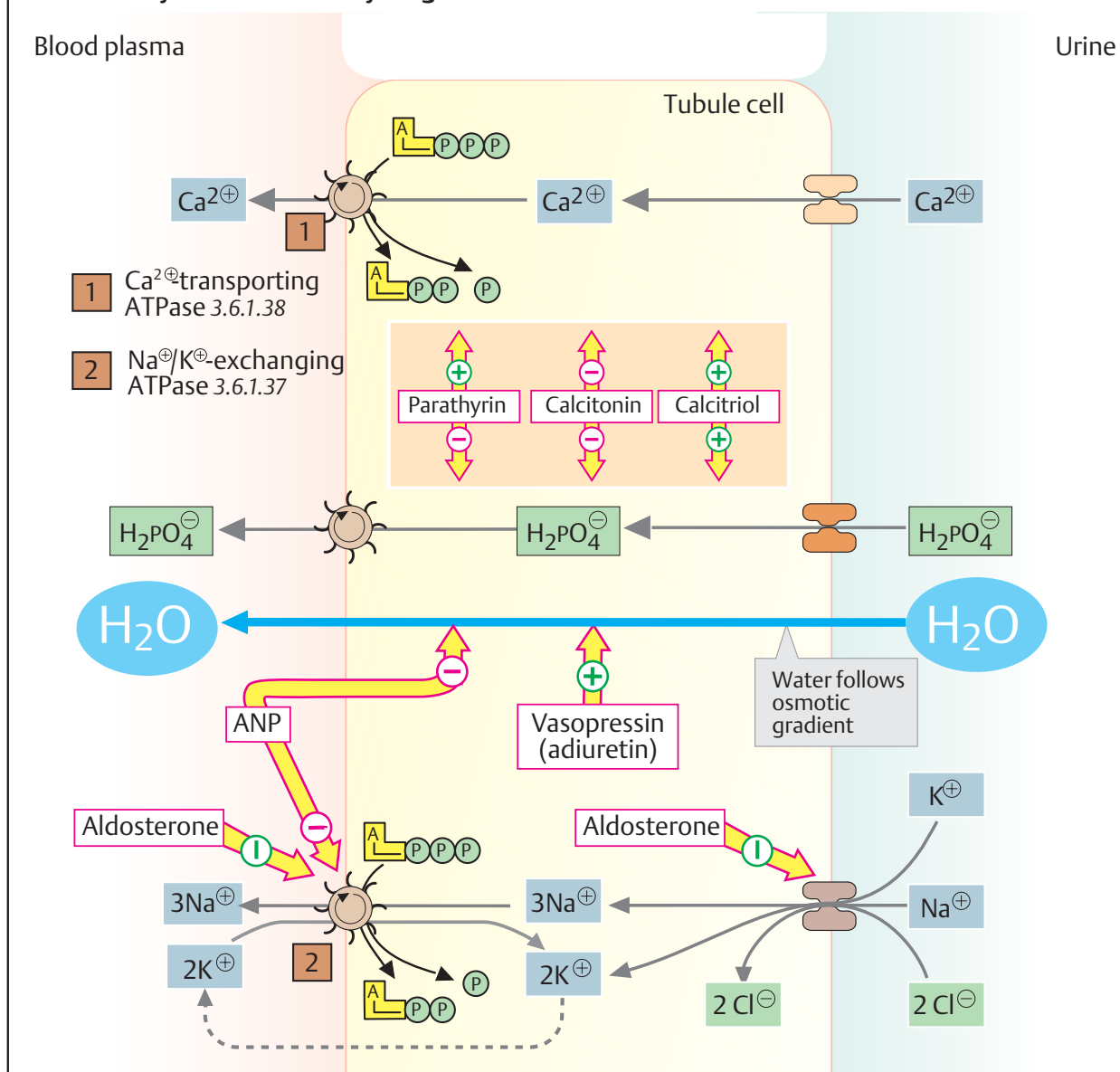
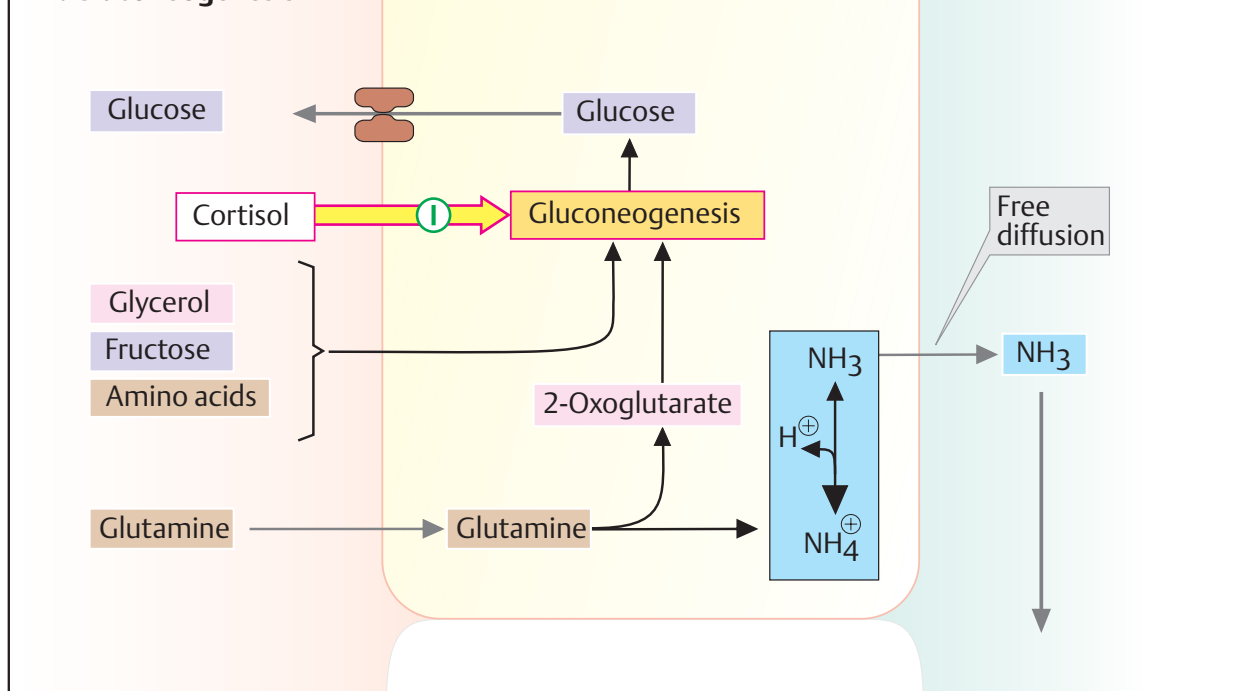
with glucose and amino acids (see p. 322). These two pathways are responsible for 60–70% of total Na^+ resorption. In the ascending part of Henle's loop, there is another transporter (shown at the bottom right), which functions electroneutrally and takes up one Na^+ ion and one K^+ ion together with two Cl^- ions. This symport is also dependent on the activity of Na^+/K^+ ATPase [2], which pumps the Na^+ resorbed from the primary urine back into the plasma in exchange for K^+ .

The steroid hormone *aldosterone* (see p. 55) increases Na^+ reuptake, particularly in the distal tubule, while *atrial natriuretic peptide* (ANP) originating from the cardiac atrium reduces it. Among other effects, aldosterone induces Na^+/K^+ ATPase and various Na^+ transporters on the luminal side of the cells.

Water. Water resorption in the proximal tubule is a *passive process* in which water follows the osmotically active particles, particularly the Na^+ ions. Fine regulation of water excretion (**diuresis**) takes place in the collecting ducts, where the peptide hormone *vasopressin* (antidiuretic hormone, ADH) operates. This promotes recovery of water by stimulating the transfer of aquaporins (see p. 220) into the plasma membrane of the tubule cells via V_2 receptors. A lack of ADH leads to the disease picture of *diabetes insipidus*, in which up to 30 L of final urine is produced per day.

B. Gluconeogenesis ❶

Apart from the liver, the kidneys are the only organs capable of producing glucose by neosynthesis (*gluconeogenesis*; see p.154). The main substrate for gluconeogenesis in the cells of the proximal tubule is **glutamine**. In addition, other amino acids and also **lactate**, **glycerol**, and **fructose** can be used as precursors. As in the liver, the key enzymes for gluconeogenesis are induced by *cortisol* (see p. 374). Since the kidneys also have a high level of glucose consumption, they only release very little glucose into the blood.

A. Electrolyte and water recycling**B. Gluconeogenesis**

Renal hormones

A. Renal hormones ❶

In addition to their involvement in excretion and metabolism, the kidneys also have endocrine functions. They produce the hormones **erythropoietin** and **calcitriol** and play a decisive part in producing the hormone **angiotensin II** by releasing the enzyme *renin*. Renal prostaglandins (see p. 390) have a local effect on Na^+ resorption.

Calcitriol (vitamin D hormone, $1\alpha,25$ -dihydroxycholecalciferol) is a hormone closely related to the steroids that is involved in Ca^{2+} homeostasis (see p. 342). In the kidney, it is formed from calcidiol by hydroxylation at C-1. The activity of *calcidiol-1-monooxygenase* [1] is enhanced by the hormone *parathyrin* (PTH).

Erythropoietin is a peptide hormone that is formed predominantly by the kidneys, but also by the liver. Together with other factors known as “*colony-stimulating factors*” (CSF; see p. 392), it regulates the differentiation of stem cells in the bone marrow.

Erythropoietin release is stimulated by hypoxia (low pO_2). Within hours, the hormone ensures that erythrocyte precursor cells in the bone marrow are converted to erythrocytes, so that their numbers in the blood increase. Renal damage leads to reduced erythropoietin release, which in turn results in *anemia*. Forms of anemia with renal causes can now be successfully treated using erythropoietin produced by genetic engineering techniques. The hormone is also administered to dialysis patients. Among athletes and sports professionals, there have been repeated cases of erythropoietin being misused for doping purposes.

B. Renin–angiotensin system ❷

The peptide hormone angiotensin II is not synthesized in a hormonal gland, but in the blood. The kidneys take part in this process by releasing the enzyme *renin*.

Renin [2] is an aspartate proteinase (see p. 176). It is formed by the kidneys as a precursor (prorenin), which is proteolytically activated into renin and released into the blood. In the blood plasma, renin acts on **angiotensinogen**, a plasma glycoprotein in

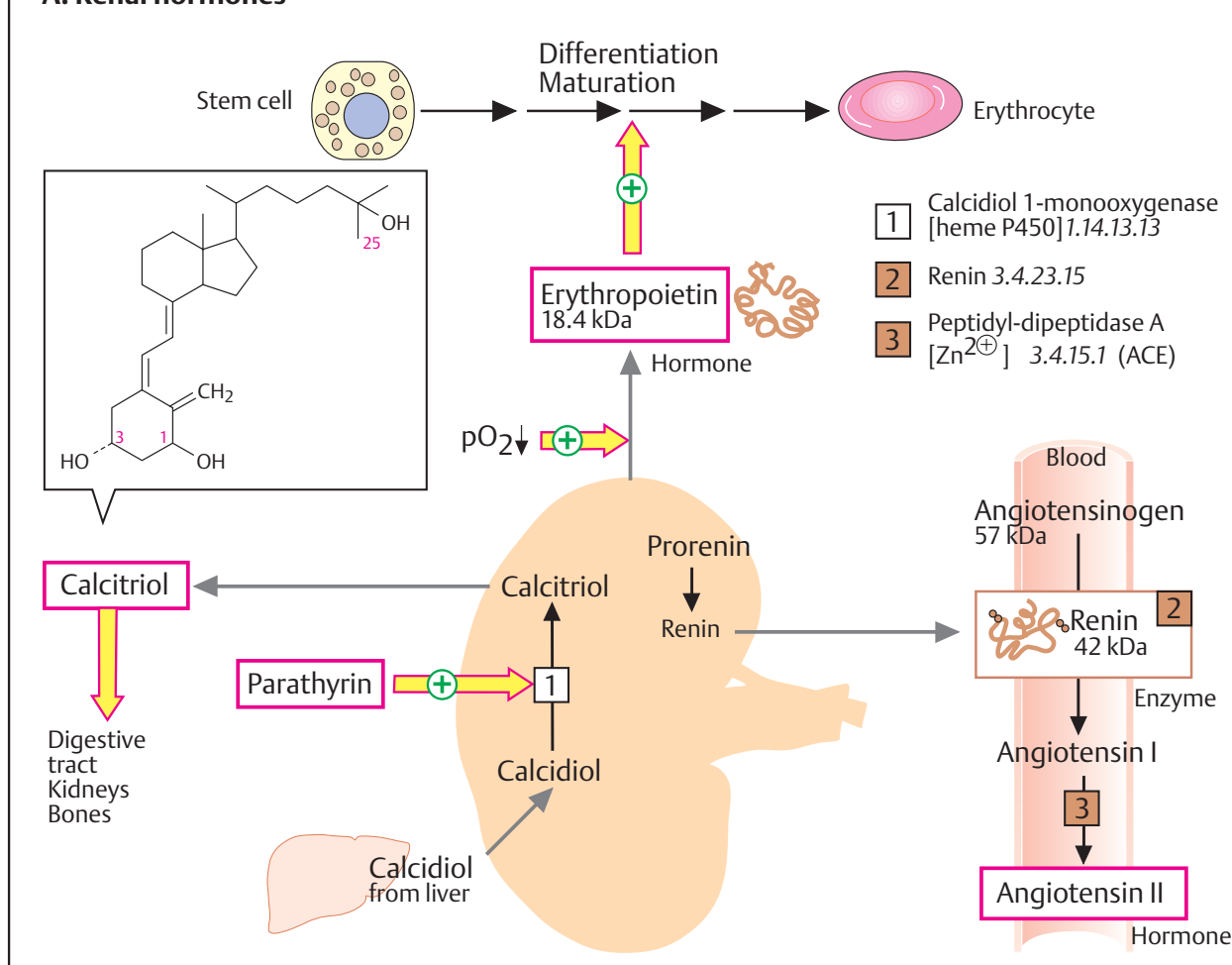
the α_2 -globulin group (see p. 276), which like almost all plasma proteins is synthesized in the liver. The decapeptide cleaved off by renin is called **angiotensin I**. Further cleavage by *peptidyl dipeptidase A* (*angiotensin-converting enzyme*, ACE), a membrane enzyme located on the vascular endothelium in the lungs and other tissues, gives rise to the octapeptide **angiotensin II** [3], which acts as a hormone and neurotransmitter. The lifespan of angiotensin II in the plasma is only a few minutes, as it is rapidly broken down by other peptidases (angiotensinases [4]), which occur in many different tissues.

The plasma level of angiotensin II is mainly determined by the rate at which renin is released by the kidneys. Renin is synthesized by juxtaglomerular cells, which release it when sodium levels decline or there is a fall in blood pressure.

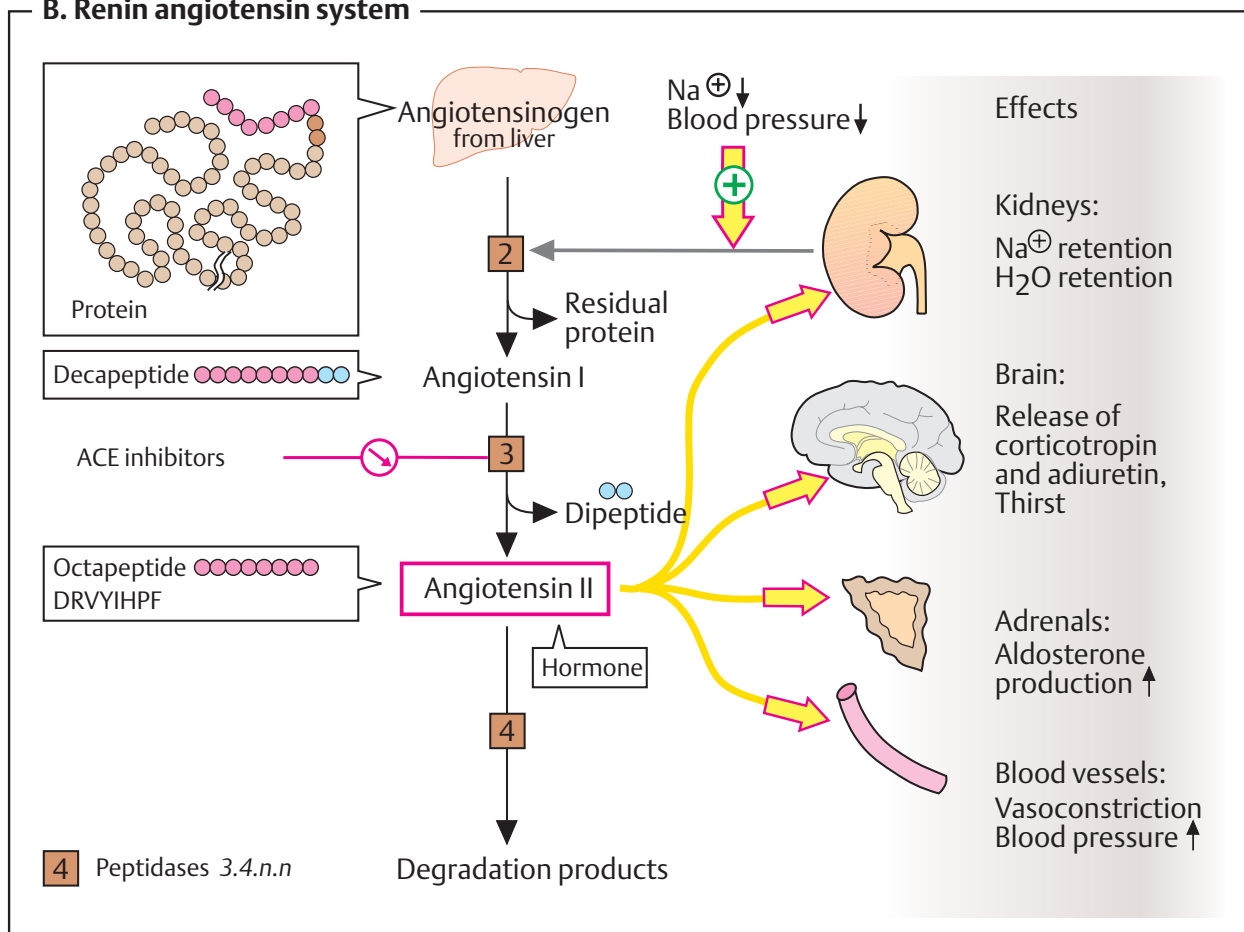
Effects of angiotensin II. Angiotensin II has effects on the kidneys, brain stem, pituitary gland, adrenal cortex, blood vessel walls, and heart via membrane-located receptors. It increases blood pressure by triggering *vasoconstriction* (narrowing of the blood vessels). In the kidneys, it promotes the *retention of Na^+* and water and reduces potassium secretion. In the brain stem and at nerve endings in the sympathetic nervous system, the effects of angiotensin II lead to increased tonicity (neurotransmitter effect). In addition, it triggers the *sensation of thirst*. In the pituitary gland, angiotensin II stimulates *vasopressin release* (antidiuretic hormone) and corticotropin (ACTH) release. In the adrenal cortex, it increases the *biosynthesis and release of aldosterone*, which promotes sodium and water retention in the kidneys. All of the effects of angiotensin II lead directly or indirectly to *increased blood pressure*, as well as increased *sodium and water retention*. This important hormonal system for blood pressure regulation can be pharmacologically influenced by *inhibitors* at various points:

- Using angiotensinogen analogs that inhibit renin.
- Using angiotensin I analogs that competitively inhibit the enzyme ACE [3].
- Using hormone antagonists that block the binding of angiotensin II to its receptors.

A. Renal hormones



B. Renin angiotensin system



Muscle contraction

The musculature is what makes movements possible. In addition to the **skeletal muscles**, which can be contracted voluntarily, there are also the autonomically activated **heart muscle** and **smooth muscle**, which is also involuntary. In all types of muscle, contraction is based on an interplay between the proteins actin and myosin.

A. Organization of skeletal muscle ①

Striated muscle consists of parallel bundles of **muscle fibers**. Each fiber is a single large multinucleate cell. The cytoplasm in these cells contains **myofibrils** 2–3 μm thick that can extend over the full length of the muscle fiber.

The *striation* of the muscle fibers is characteristic of skeletal muscle. It results from the regular arrangement of molecules of differing density. The repeating contractile units, the **sarcomeres**, are bounded by Z lines from which thin filaments of **F-actin** (see p. 204) extend on each side. In the A bands, there are also thick parallel filaments of **myosin**. The H bands in the middle of the A bands only contain myosin, while only actin is found on each side of the Z lines.

Myosin is quantitatively the most important protein in the myofibrils, representing 65% of the total. It is shaped like a golf club (bottom right). The molecule is a hexamer consisting of two identical *heavy chains* ($2 \times 223 \text{ kDa}$) and four *light chains* (each about 20 kDa). Each of the two heavy chains has a globular “head” at its amino end, which extends into a “tail” about 150 nm long in which the two chains are intertwined to form a superhelix. The small subunits are attached in the head area. Myosin is present as a bundle of several hundred stacked molecules in the form of a “*thick myosin filament*.” The head portion of the molecule acts as an *ATPase*, the activity of which is modulated by the small subunits.

Actin (42 kDa) is the most important component of the “*thin filaments*.” It represents ca. 20–25% of the muscle proteins. **F-actin** is also an important component of the cytoskeleton (see p. 204). This filamentous polymer is held in equilibrium with its monomer, **G-actin**. The other protein components of muscle include tropomyosin and troponin. **Tropomyosin**

(64 kDa) attaches to F-actin as a rod-like dimer and connects approximately seven actin units with each other. The heterotrimer **troponin** (78 kDa) is bound to one end of tropomyosin.

In addition to the above proteins, a number of other proteins are also typical of muscle—including *titin* (the largest known protein), α - and β -actinin, *desmin*, and *vimentin*.

B. Mechanism of muscle contraction ①

The *sliding filament model* describes the mechanism involved in muscle contraction. In this model, sarcomeres become shorter when the thin and thick filaments slide alongside each other and telescope together, with ATP being consumed. During contraction, the following reaction cycle is repeated several times:

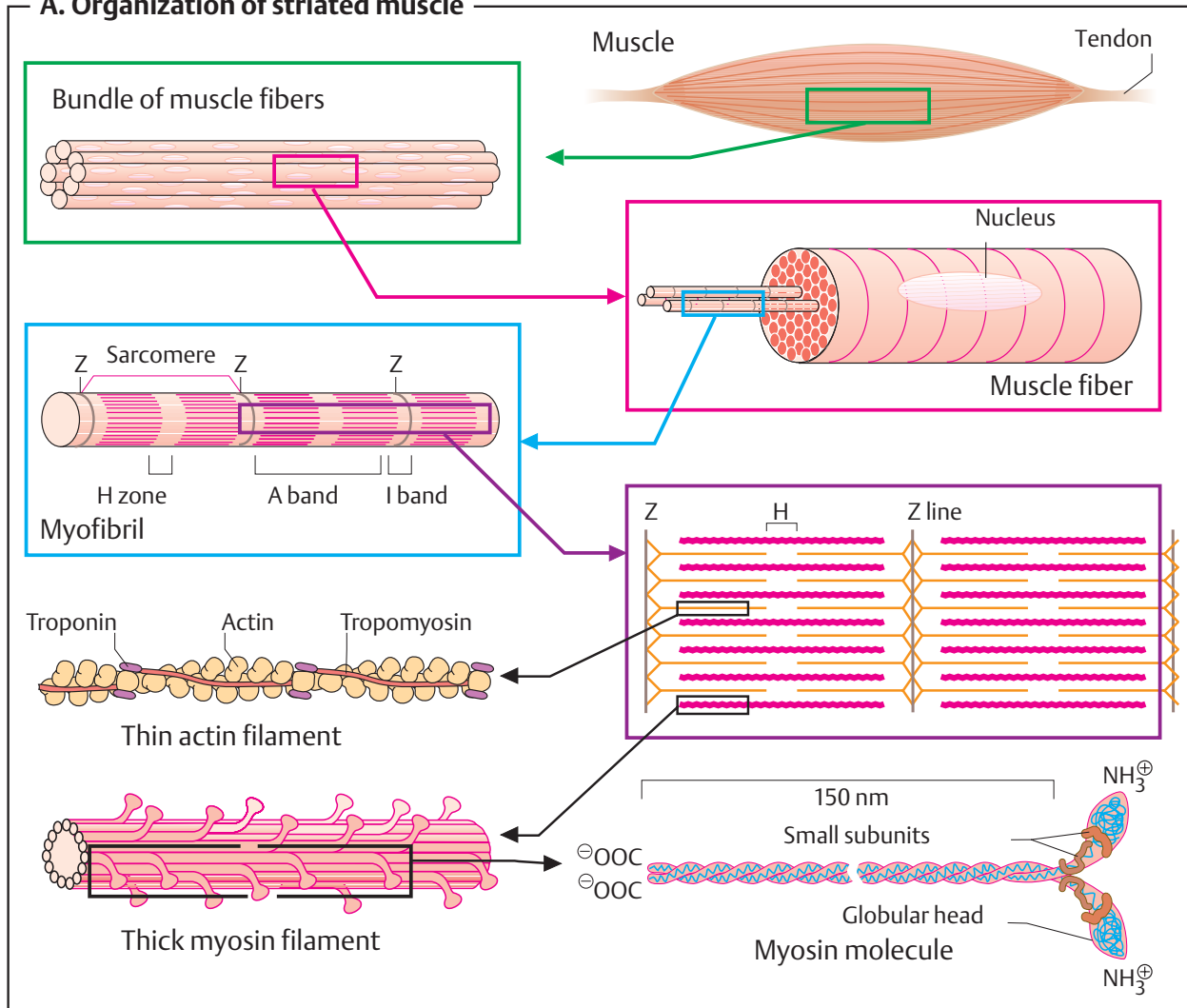
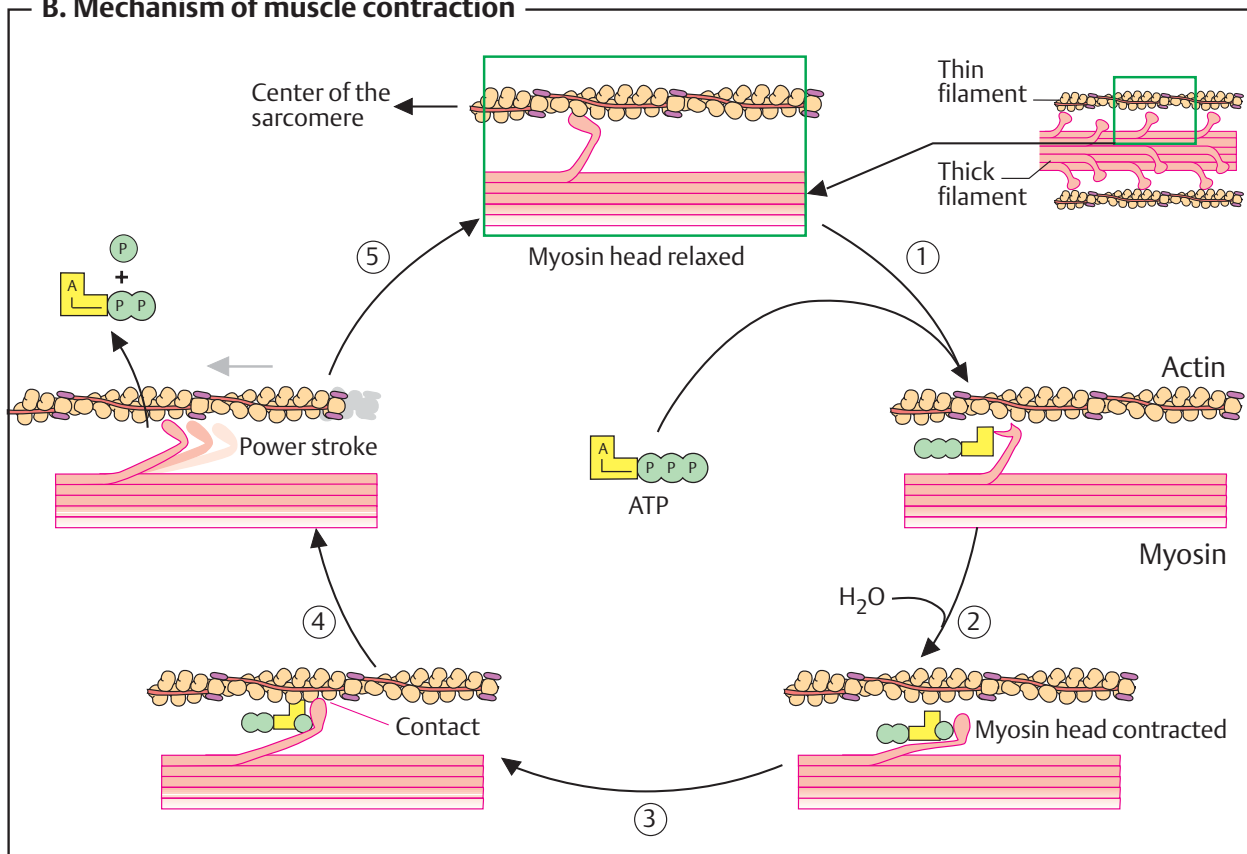
[1] In the initial state, the myosin heads are attached to actin. When ATP is bound, the heads detach themselves from the actin (the “plasticizing” effect of ATP).

[2] The myosin head hydrolyzes the bound ATP to ADP and P_i , but initially withholds the two reaction products. ATP cleavage leads to allosteric tension in the myosin head.

[3] The myosin head now forms a new bond with a neighboring actin molecule.

[4] The actin causes the release of the P_i , and shortly afterwards release of the ADP as well. This converts the allosteric tension in the myosin head into a conformational change that acts like a rowing stroke.

The cycle can be repeated for as long as ATP is available, so that the thick filaments are constantly moving along the thin filaments in the direction of the Z disk. Each rowing stroke of the 500 or so myosin heads in a thick filament produces a contraction of about 10 nm. During strong contraction, the process is repeated about five times per second. This leads to the whole complex of thin filaments moving together; the H band becomes shorter and the Z lines slide closer together.

A. Organization of striated muscle**B. Mechanism of muscle contraction**

Control of muscle contraction

A. Neuromuscular junction ●

Muscle contraction is triggered by *motor neurons* that release the neurotransmitter *acetylcholine* (see p. 352). The transmitter diffuses through the narrow synaptic cleft and binds to nicotinic *acetylcholine receptors* on the plasma membrane of the muscle cell (the *sarcolemma*), thereby opening the ion channels integrated into the receptors (see p. 222). This leads to an inflow of Na^+ , which triggers an *action potential* (see p. 350) in the sarcolemma. The action potential propagates from the end plate in all directions and constantly stimulates the muscle fiber. With a delay of a few milliseconds, the contractile mechanism responds to this by contracting the muscle fiber.

B. Sarcoplasmic reticulum (SR) ●

The action potential (A) produced at the neuromuscular junction is transferred in the muscle cell into a transient increase in the Ca^{2+} concentration in the cytoplasm of the muscle fiber (the *sarcoplasm*).

In the resting state, the Ca^{2+} level in the sarcoplasm is very low (less than 10^{-7} M). By contrast, the **sarcoplasmic reticulum** (SR), which corresponds to the ER, contains Ca^{2+} ions at a concentration of about 10^{-3} M. The SR is a branched organelle that surrounds the myofibrils like a net stocking inside the muscle fibers (illustrated at the top using the example of a heart muscle cell). The high Ca^{2+} level in the SR is maintained by Ca^{2+} -transporting ATPases (see p. 220). In addition, the SR also contains *calsequestrin*, a protein (55 kDa) that is able to bind numerous Ca^{2+} ions via acidic amino acid residues.

The transfer of the action potential to the SR is made possible by **transverse tubules** (T tubules), which are open to the extracellular space and establish a close connection with the SR. There is a structure involved in the contact between the T tubule and the SR that was formerly known as the “SR foot” (it involves parts of the *ryanodine receptor*; see p. 386).

At the point of contact with the SR, the action potential triggers the opening of the Ca^{2+} channels on the surface of the sarcolemma. Calcium ions then leave the SR and

enter the sarcoplasm, where they lead to a rapid increase in Ca^{2+} concentrations. This in turn causes the myofibrils to contract (C).

C. Regulation by calcium ions ●

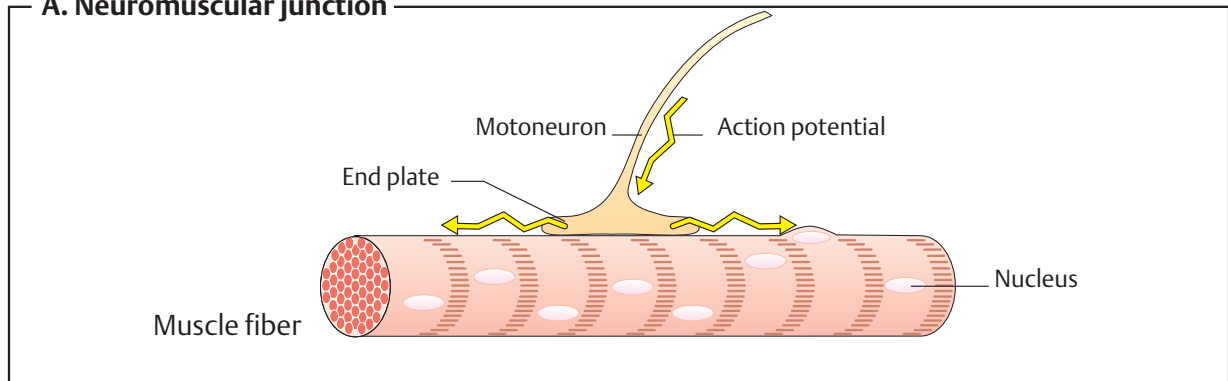
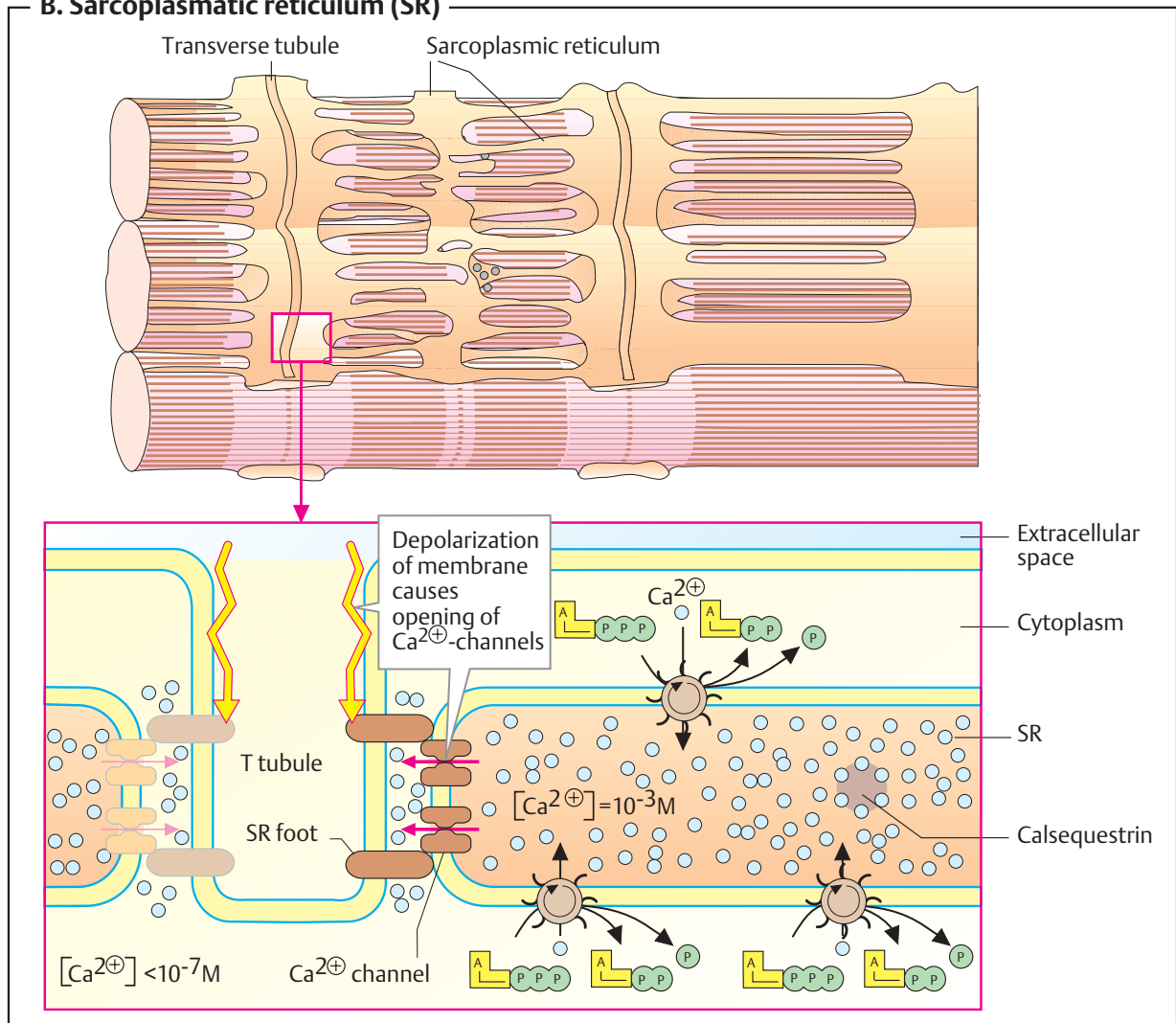
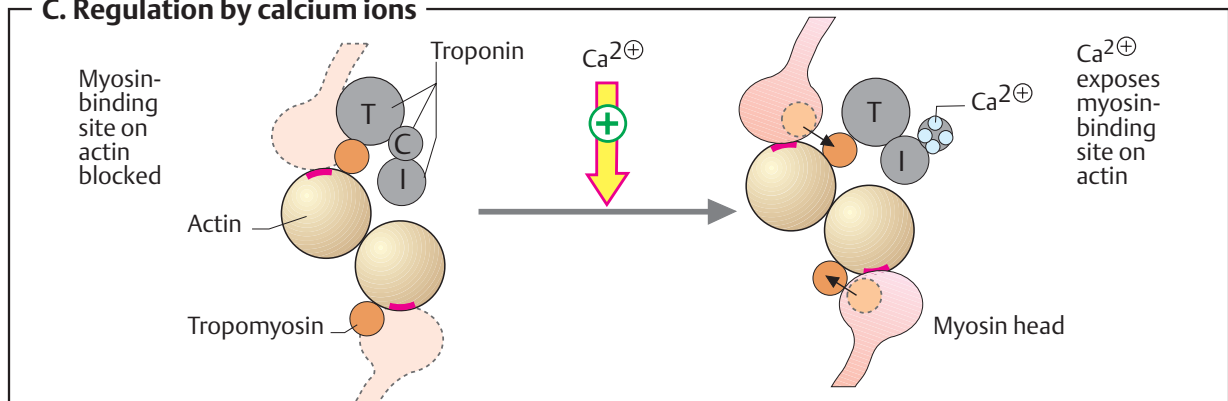
In relaxed skeletal muscle, the complex consisting of **troponin** and **tropomyosin** blocks the access of the myosin heads to actin (see p. 332). Troponin consists of three different subunits (T, C, and I). The rapid increase in cytoplasmic Ca^{2+} concentrations caused by opening of the calcium channels in the SR leads to binding of Ca^{2+} to the C subunit of troponin, which closely resembles calmodulin (see p. 386). This produces a conformational change in troponin that causes the whole troponin–tropomyosin complex to slip slightly and expose a binding site for myosin (red). This initiates the contraction cycle. After contraction, the sarcoplasmic Ca^{2+} concentration is quickly reduced again by active transport back into the SR. This results in troponin losing the bound Ca^{2+} ions and returning to the initial state, in which the binding site for myosin on actin is blocked. It is not yet clear whether the mechanism described above is the only one that triggers binding of myosin to actin.

When **triggering of contraction** in striated muscle occurs, the following sequence of processes thus takes place:

1. The sarcolemma is depolarized.
2. The action potential is signaled to Ca^{2+} channels in the SR.
3. The Ca^{2+} channels open and the Ca^{2+} level in the sarcoplasm increases.
4. Ca^{2+} binds to troponin C and triggers a conformational change.
5. Troponin causes tropomyosin to slip, and the myosin heads bind to actin.
6. The actin–myosin cycle takes place and the muscle fibers contract.

Conversely, at the **end of contraction**, the following processes take place:

1. The Ca^{2+} level in the sarcoplasm declines due to transport of Ca^{2+} back into the SR.
2. Troponin C loses Ca^{2+} and tropomyosin returns to its original position on the actin molecule.
3. The actin–myosin cycle stops and the muscle relaxes.

A. Neuromuscular junction**B. Sarcoplasmic reticulum (SR)****C. Regulation by calcium ions**

Muscle metabolism I

Muscle contraction is associated with a high level of ATP consumption (see p. 332). Without constant resynthesis, the amount of ATP available in the resting state would be used up in less than 1 s of contraction.

A. Energy metabolism in the white and red muscle fibers ●

Muscles contain two types of fibers, the proportions of which vary from one type of muscle to another. **Red fibers** (type I fibers) are suitable for prolonged effort. Their metabolism is mainly aerobic and therefore depends on an adequate supply of O₂. **White fibers** (type II fibers) are better suited for fast, strong contractions. These fibers are able to form sufficient ATP even when there is little O₂ available. With appropriate training, athletes and sports participants are able to change the proportions of the two fiber types in the musculature and thereby prepare themselves for the physiological demands of their disciplines in a targeted fashion. The expression of functional muscle proteins can also change during the course of training.

Red fibers provide for their ATP requirements mainly (but not exclusively) from **fatty acids**, which are broken down via β -oxidation, the tricarboxylic acid cycle, and the respiratory chain (right part of the illustration). The red color in these fibers is due to the monomeric heme protein **myoglobin**, which they use as an O₂ reserve. Myoglobin has a much higher affinity for O₂ than hemoglobin and therefore only releases its O₂ when there is a severe drop in O₂ partial pressure (cf. p. 282).

At a high level of muscular effort—e.g., during weightlifting or in very fast contractions such as those carried out by the eye muscles—the O₂ supply from the blood quickly becomes inadequate to maintain the aerobic metabolism. White fibers (left part of the illustration) therefore mainly obtain ATP from **anaerobic glycolysis**. They have supplies of **glycogen** from which they can quickly release glucose-1-phosphate when needed (see p. 156). By isomerization, this gives rise to glucose-6-phosphate, the substrate for glycolysis. The NADH+H⁺ formed during glycolysis has to be reoxidized into NAD⁺ in order to

maintain glucose degradation and thus ATP formation. If there is a lack of O₂, this is achieved by the formation of **lactate**, which is released into the blood and is resynthesized into glucose in the liver (Cori cycle; see p. 338).

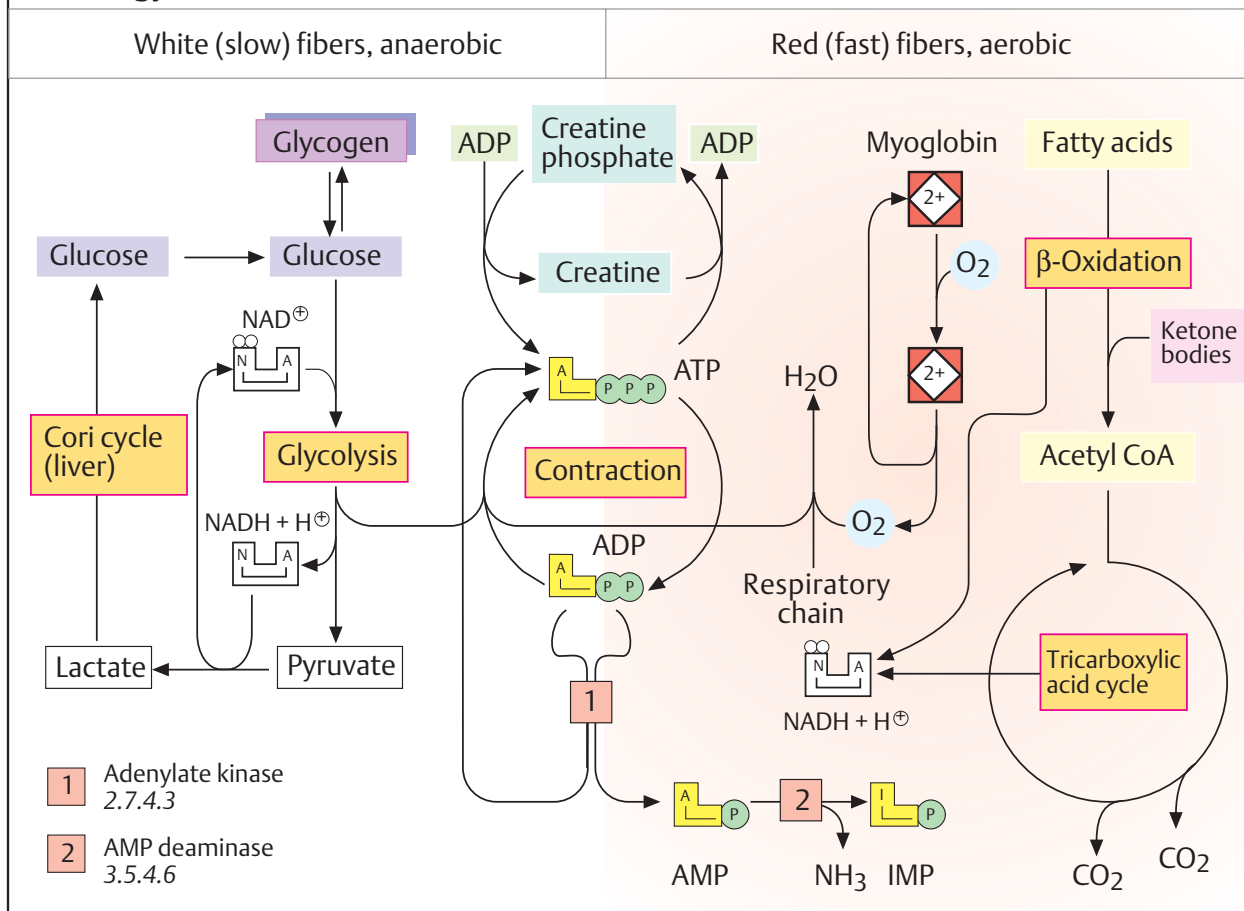
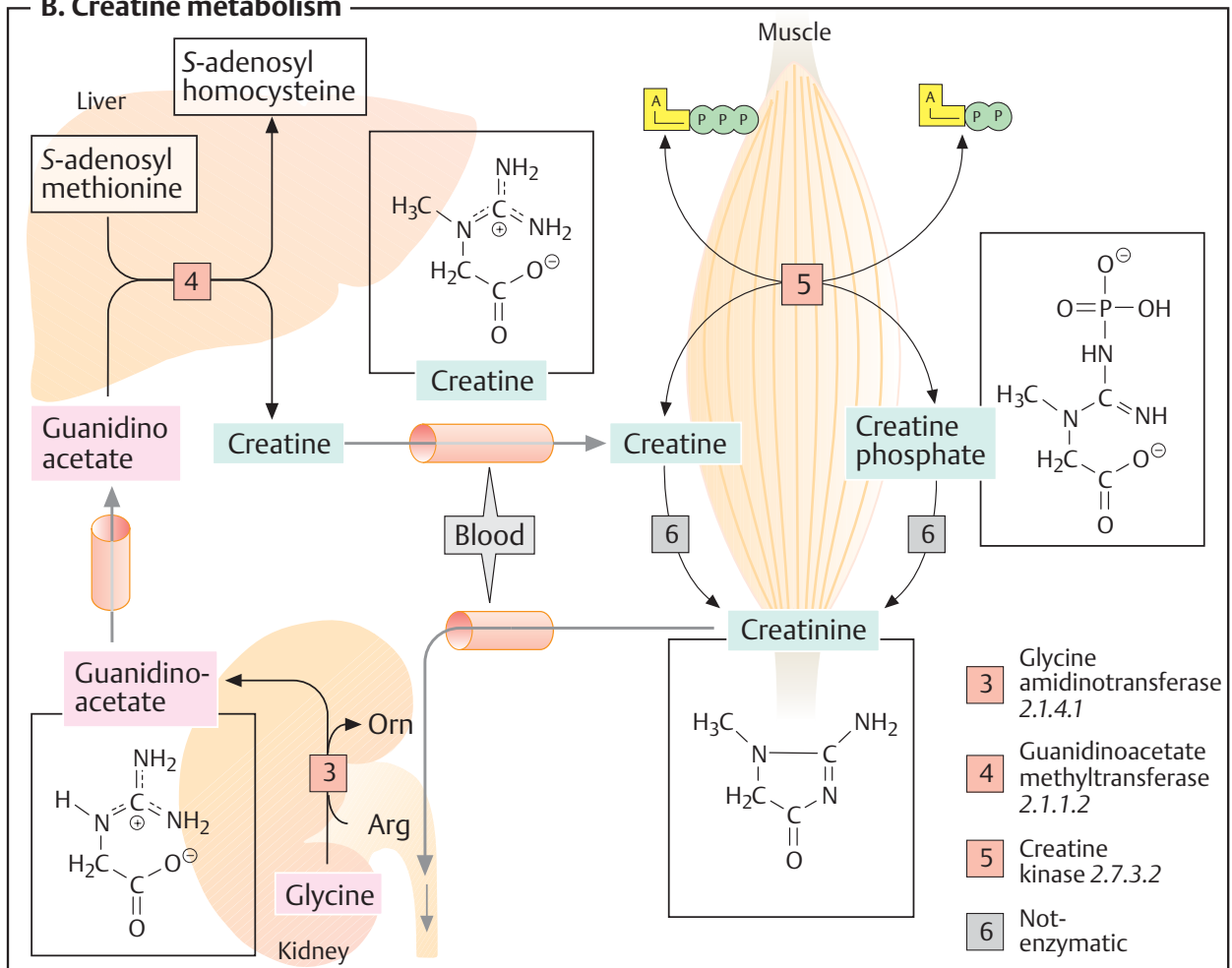
Muscle-specific auxiliary reactions for ATP synthesis exist in order to provide additional ATP in case of emergency. **Creatine phosphate** (see B) acts as a buffer for the ATP level. Another ATP-supplying reaction is catalyzed by *adenylate kinase* [1] (see also p. 72). This disproportionates two molecules of ADP into ATP and AMP. The AMP is deaminated into IMP in a subsequent reaction [2] in order to shift the balance of the reversible reaction [1] in the direction of ATP formation.

B. Creatine metabolism ●

Creatine (*N*-methylguanidoacetic acid) and its phosphorylated form **creatine phosphate** (a guanidophosphate) serve as an ATP buffer in muscle metabolism. In creatine phosphate, the phosphate residue is at a similarly high chemical potential as in ATP and is therefore easily transferred to ADP. Conversely, when there is an excess of ATP, creatine phosphate can arise from ATP and creatine. Both processes are catalyzed by *creatine kinase* [5].

In resting muscle, creatine phosphate forms due to the high level of ATP. If there is a risk of a severe drop in the ATP level during contraction, the level can be maintained for a short time by synthesis of ATP from creatine phosphate and ADP. In a nonenzymatic reaction [6], small amounts of creatine and creatine phosphate cyclize constantly to form **creatinine**, which can no longer be phosphorylated and is therefore excreted with the urine (see p. 324).

Creatine does not derive from the muscles themselves, but is synthesized in two steps in the kidneys and liver (left part of the illustration). Initially, the guanidino group of arginine is transferred to glycine in the kidneys, yielding **guanidino acetate** [3]. In the liver, *N*-methylation of guanidino acetate leads to the formation of creatine from this [4]. The coenzyme in this reaction is *S*-adenosyl methionine (SAM; see p. 110).

A. Energy metabolism in the white and red muscle fibers**B. Creatine metabolism**

Muscle metabolism II

A. Cori and alanine cycle ①

White muscle fibers (see p. 336) mainly obtain ATP from **anaerobic glycolysis**—i.e., they convert glucose into lactate. The **lactate** arising in muscle and, in smaller quantities, its precursor **pyruvate** are released into the blood and transported to the liver, where lactate and pyruvate are resynthesized into glucose again via **gluconeogenesis**, with ATP being consumed in the process (see p. 154). The glucose newly formed by the liver returns via the blood to the muscles, where it can be used as an energy source again. This circulation system is called the **Cori cycle**, after the researchers who first discovered it. There is also a very similar cycle for erythrocytes, which do not have mitochondria and therefore produce ATP by anaerobic glycolysis (see p. 284).

The muscles themselves are not capable of gluconeogenesis. Nor would this be useful, as gluconeogenesis requires much more ATP than is supplied by glycolysis. As O_2 deficiencies do not arise in the liver even during intensive muscle work, there is always sufficient energy there available for gluconeogenesis.

There is also a corresponding circulation system for the amino acid alanine. The **alanine cycle** in the liver not only provides alanine as a precursor for gluconeogenesis, but also transports to the liver the amino nitrogen arising in muscles during protein degradation. In the liver, it is incorporated into urea for excretion.

Most of the amino acids that arise in muscle during proteolysis are converted into glutamate and 2-oxo acids by **transamination** (not shown; cf. p. 180). Again by transamination, glutamate and pyruvate give rise to alanine, which after glutamine is the second important form of transport for amino nitrogen in the blood. In the liver, alanine and 2-oxoglutarate are resynthesized into pyruvate and glutamate (see p. 178). Glutamate supplies the urea cycle (see p. 182), while pyruvate is available for gluconeogenesis.

B. Protein and amino acid metabolism ①

The skeletal muscle is the most important site for degradation of the *branched-chain amino acids* (Val, Leu, Ile; see p. 414), but other amino acids are also broken down in the muscles. **Alanine** and **glutamine** are resynthesized from the components and released into the blood. They transport the nitrogen that arises during amino acid breakdown to the liver (*alanine cycle*; see above) and to the kidneys (see p. 328).

During periods of hunger, **muscle proteins** serve as an energy reserve for the body. They are broken down into amino acids, which are transported to the liver. In the liver, the carbon skeletons of the amino acids are converted into intermediates in the tricarboxylic acid cycle or into acetoacetyl-CoA (see p. 175). These amphibolic metabolites are then available to the energy metabolism and for gluconeogenesis. After prolonged starvation, the brain switches to using ketone bodies in order to save muscle protein (see p. 356).

The synthesis and degradation of muscle proteins are regulated by hormones. **Cortisol** leads to muscle degradation, while **testosterone** stimulates protein formation. Synthetic **anabolics** with a testosterone-like effect have repeatedly been used for doping purposes or for intensive muscle-building.

Further information

Smooth muscle differs from skeletal muscle in various ways. Smooth muscles—which are found, for example, in blood vessel walls and in the walls of the intestines—do not contain any muscle fibers. In smooth-muscle cells, which are usually spindle-shaped, the contractile proteins are arranged in a less regular pattern than in striated muscle. Contraction in this type of muscle is usually not stimulated by nerve impulses, but occurs in a largely spontaneous way. Ca^{2+} (in the form of Ca^{2+} -calmodulin; see p. 386) also activates contraction in smooth muscle; in this case, however, it does not affect troponin, but activates a protein kinase that phosphorylates the light chains in myosin and thereby increases myosin's ATPase activity. Hormones such as epinephrine and angiotensin II (see p. 330) are able to influence vascular tonicity in this way, for example.

Bone and teeth

The family of connective-tissue cells includes *fibroblasts*, *chondrocytes* (cartilage cells), and *osteoblasts* (bone-forming cells). They are specialized to secrete extracellular proteins, particularly collagens, and mineral substances, which they use to build up the *extracellular matrix* (see p. 346). By contrast, *osteoclasts* dissolve bone matter again by secreting H^+ and collagenases (see p. 342).

A. Bone ●

Bone is an extremely dense, specialized form of connective tissue. In addition to its supportive function, it serves to store calcium and phosphate ions. In addition, blood cells are formed in the bone marrow. The most important mineral component of bone is **apatite**, a form of crystalline *calcium phosphate*.

Apatites are complexes of cationic Ca^{2+} matched by HPO_4^{2-} , CO_3^{2-} , OH^- , or F^- as anions. Depending on the counter-ion, apatite can occur in the forms *carbonate apatite* $Ca_{10}(PO_4)_6CO_3$, as *hydroxyapatite* $Ca_{10}(PO_4)_6(OH)_2$, or *fluoroapatite* $Ca_{10}(PO_4)_6F_2$. In addition, alkaline earth carbonates also occur in bone. In adults, more than 1 kg calcium is stored in bone.

Osteoblast and *osteoclast* activity is constantly incorporating Ca^{2+} into bone and removing it again. There are various hormones that regulate these processes: *calcitonin* increases deposition of Ca^{2+} in the bone matrix, while *parathyroid hormone* (PTH) promotes the mobilization of Ca^{2+} , and *calcitriol* improves mineralization (for details, see p. 342).

The most important *organic components* of bone are **collagens** (mainly type I; see p. 344) and **proteoglycans** (see p. 346). These form the extracellular matrix into which the apatite crystals are deposited (*biomineralization*). Various proteins are involved in this not yet fully understood process of bone formation, including collagens and phosphatases. *Alkaline phosphatase* is found in osteoblasts and *acid phosphatase* in osteoclasts. Both of these enzymes serve as *marker enzymes* for bone cells.

B. Teeth ●

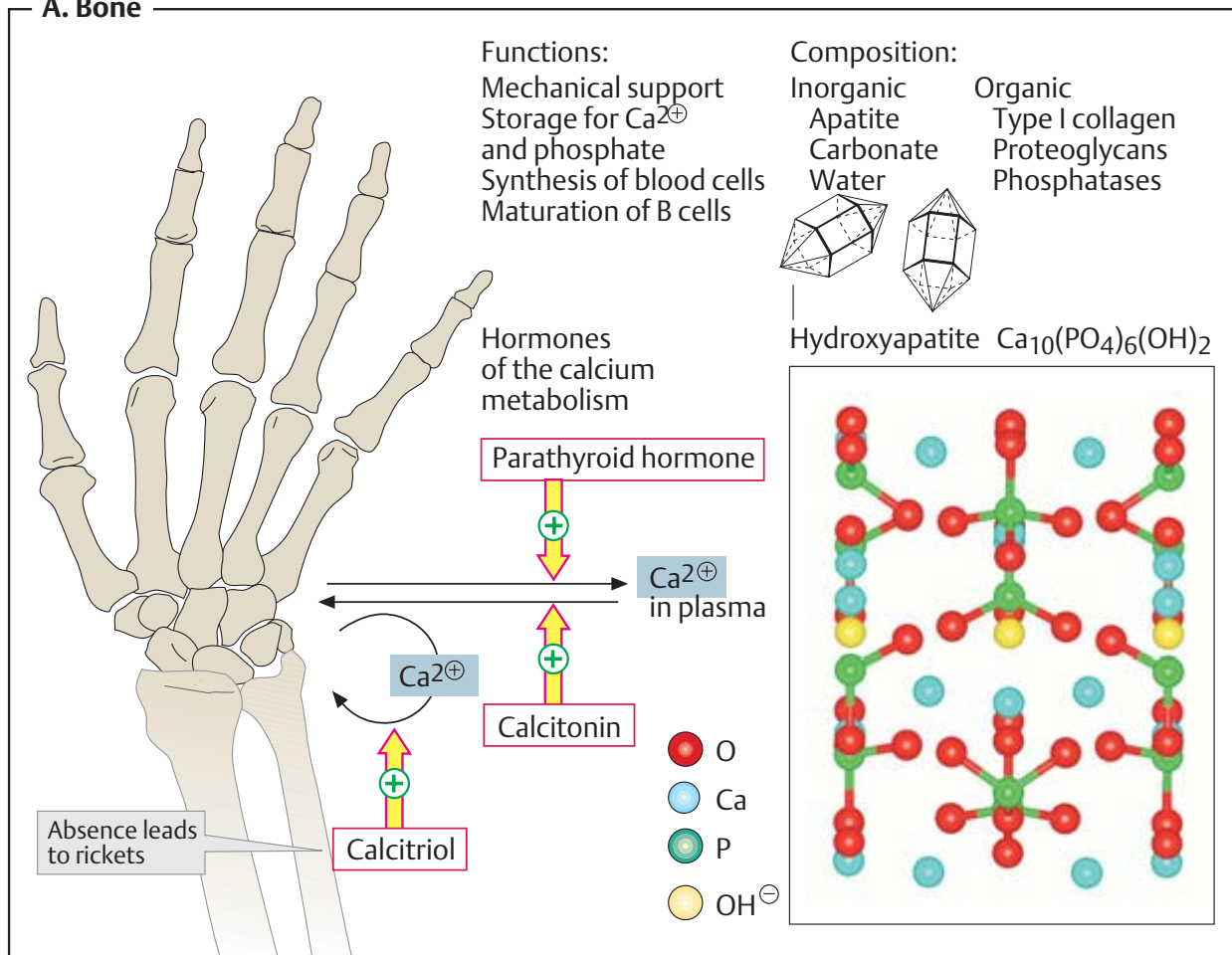
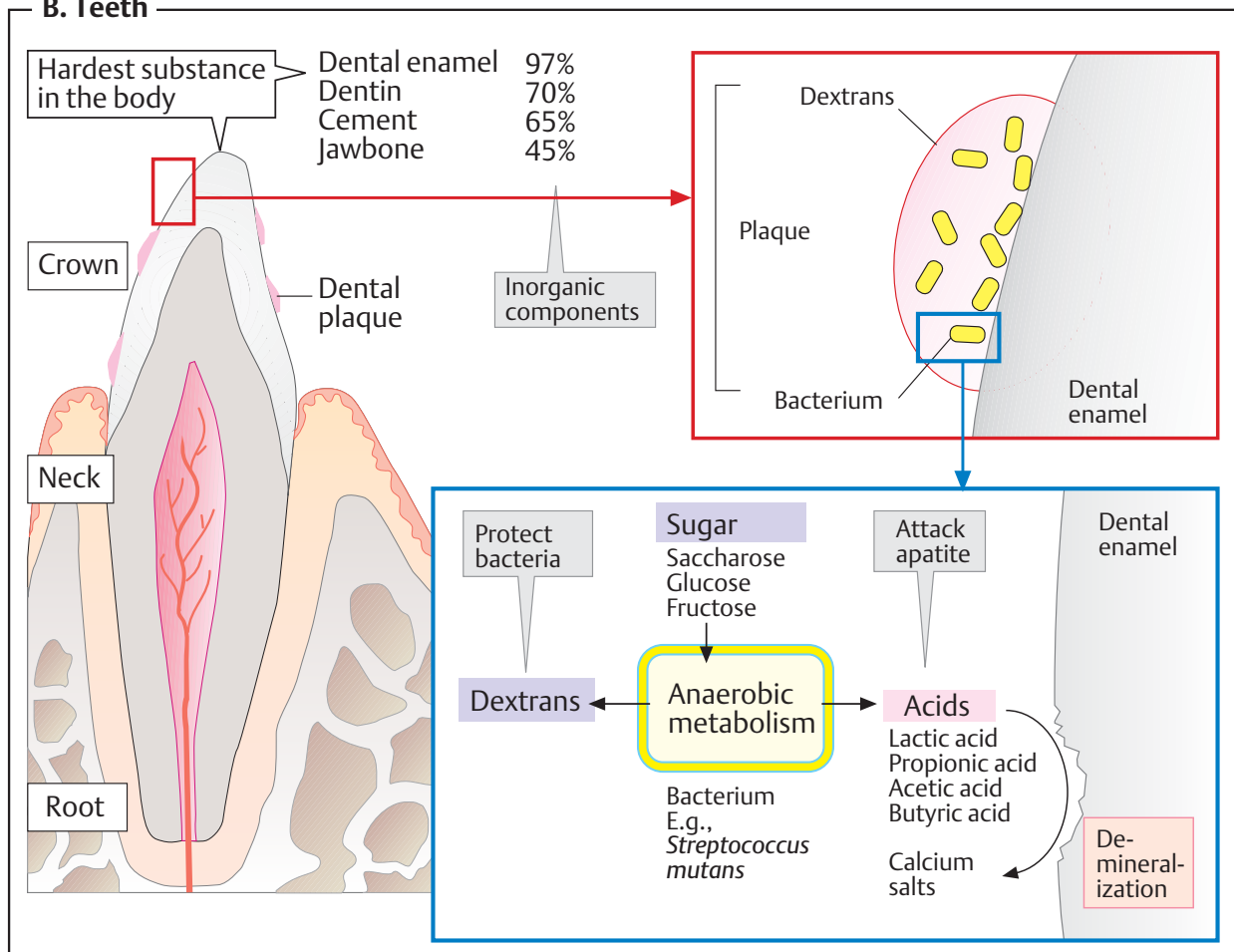
The illustration shows a longitudinal section through an incisor, one of the 32 permanent teeth in humans. The majority of the tooth consists of **dentine**. The crown of the tooth extends beyond the gums, and it is covered in **enamel**. By contrast, the root of the tooth is coated in dental **cement**.

Cement, dentin, and enamel are bone-like substances. The high proportion of inorganic matter they contain (about 97% in the dental enamel) gives them their characteristic hardness. The organic components of cement, dentin, and enamel mainly consist of *collagens* and *proteoglycans*; their most important mineral component is *apatite*, as in bone (see above).

A widespread form of dental disease, **caries**, is caused by acids that dissolve the mineral part of the teeth by neutralizing the negatively charged counter-ions in apatite (see A). Acids occur in food, or are produced by microorganisms that live on the surfaces of the teeth (e.g., *Streptococcus mutans*).

The main product of anaerobic degradation of sugars by these organisms is lactic acid. Other products of bacterial carbohydrate metabolism include extracellular dextrans (see p. 40)—insoluble polymers of glucose that help bacteria to protect themselves from their environment. Bacteria and dextrans are components of *dental plaque*, which forms on inadequately cleaned teeth. When Ca^{2+} salts and other minerals are deposited in plaque as well, *tartar* is formed.

The most important form of protection against caries involves avoiding sweet substances (foods containing saccharose, glucose, and fructose). Small children in particular should not have very sweet drinks freely available to them. Regular removal of plaque by cleaning the teeth and hardening of the dental enamel by fluoridization are also important. Fluoride has a protective effect because fluoroapatite (see A) is particularly resistant to acids.

A. Bone**B. Teeth**

Calcium metabolism

A. Functions of calcium ●

The human body contains 1–1.5 kg Ca^{2+} , most of which (about 98%) is located in the mineral substance of bone (see p. 362).

In addition to its role as a **bone component**, calcium functions as a **signaling substance**. Ca^{2+} ions act as *second messengers* in signal transduction pathways (see p. 386), they trigger *exocytosis* (see p. 228) and *muscle contraction* (see p. 334), and they are indispensable as cofactors in *blood coagulation* (see p. 290). Many *enzymes* also require Ca^{2+} for their activity. The intracellular and extracellular concentrations of Ca^{2+} are strictly regulated in order to make these functions possible (see **B**, **C**, and p. 388).

Proteins bind Ca^{2+} via oxygen ligands, particularly carboxylate groups and carbonyl groups of peptide bonds. This also applies to the structure illustrated here, in which a Ca^{2+} ion is coordinated by the oxygen atoms of carboxylate and acid amide groups.

B. Bone remodeling ●

Deposition of Ca^{2+} in bone (*mineralization*) and Ca^{2+} mobilization from bone are regulated by at least 15 hormones and hormone-like signaling substances. These mainly influence the maturation and activity of bone cells.

Osteoblasts (top) deposit collagen, as well as Ca^{2+} and phosphate, and thereby create new bone matter, while **osteoclasts** (bottom) secrete H^+ ions and collagenases that locally dissolve bone (*bone remodeling*). Osteoblasts and osteoclasts mutually activate each other by releasing **cytokines** (see p. 392) and **growth factors**. This helps keep bone formation and bone breakdown in balance.

The Ca^{2+} -selective hormones calcitriol, parathyroid hormone, and calcitonin influence this interaction in the bone cells. **Parathyroid hormone** promotes Ca^{2+} release by promoting the release of cytokines by osteoblasts. In turn, the cytokines stimulate the development of mature osteoclasts from precursor cells (bottom). **Calcitonin** inhibits this process. At the same time, it promotes the development of osteoblasts (top). *Osteoporosis*, which mainly occurs in women following the menopause, is based (at least in part) on a reduc-

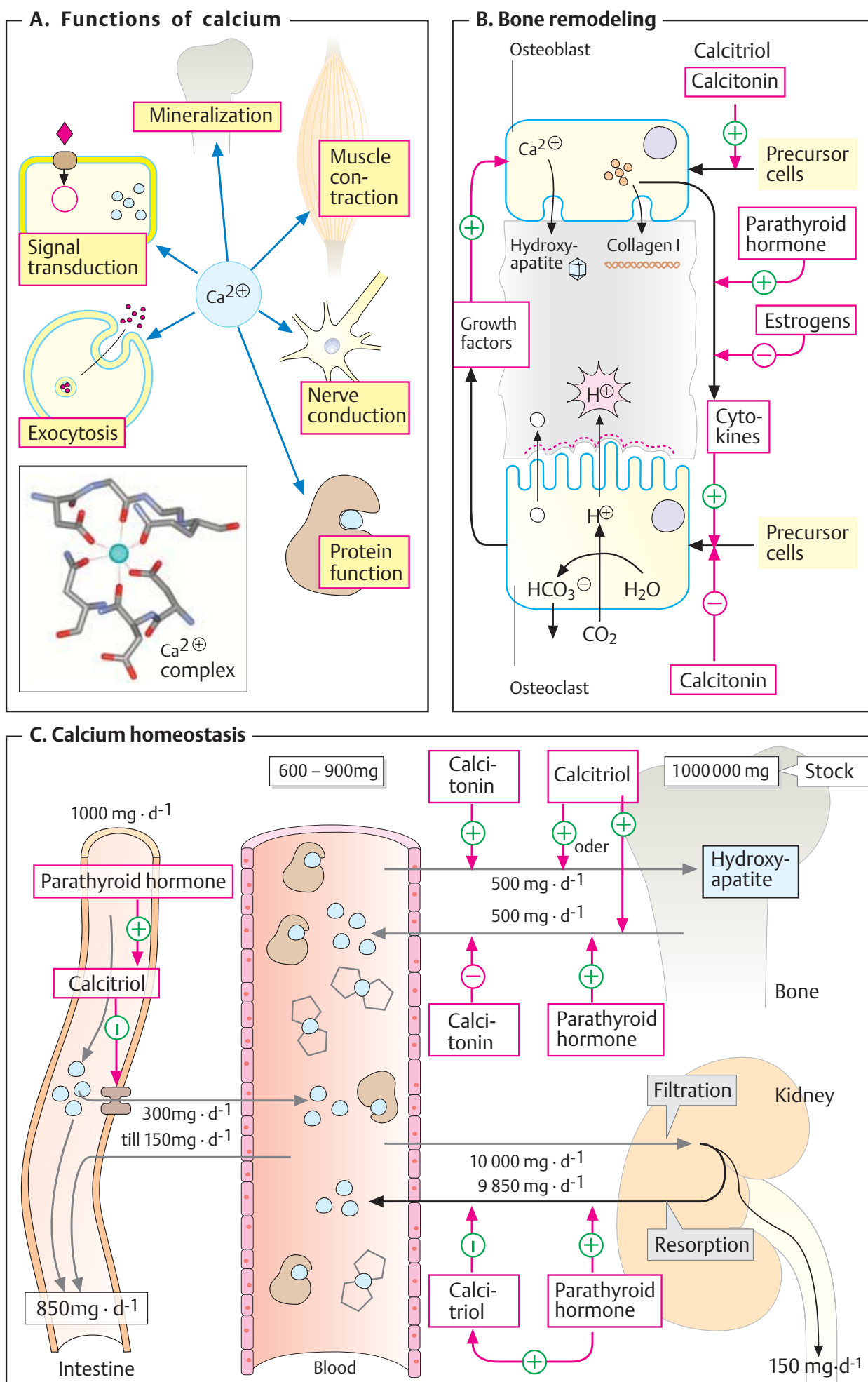
tion in **estrogen** levels. Estrogens normally inhibit the stimulation of osteoclast differentiation by osteoblasts. If the effects of estrogen decline, the osteoclasts predominate and excess bone removal occurs.

The effects of the steroid hormone **calcitriol** (see p. 330) in bone are complex. On the one hand, it promotes bone formation by stimulating osteoblast differentiation (top). This is particularly important in small children, in whom calcitriol deficiency can lead to mineralization disturbances (*rickets*; see p. 364). On the other hand, calcitriol increases blood Ca^{2+} levels through increased Ca^{2+} mobilization from bone. An overdose of vitamin D (cholecalciferol), the precursor of calcitriol, can therefore have unfavorable effects on the skeleton similar to those of vitamin deficiency (*hypervitaminosis*; see p. 364).

C. Calcium homeostasis ●

Ca^{2+} metabolism is balanced in healthy adults. Approximately 1 g Ca^{2+} is taken up per day, about 300 mg of which is resorbed. The same amount is also excreted again. The amounts of Ca^{2+} released from bone and deposited in it per day are much smaller. Milk and milk products, especially cheese, are particularly rich in calcium.

Calcitriol and parathyroid hormone, on the one hand, and calcitonin on the other, ensure a more or less constant level of Ca^{2+} in the blood plasma and in the extracellular space (80–110 mg 2.0–2.6 mM). The peptide **parathyroid hormone** (PTH; 84 AA) and the steroid **calcitriol** (see p. 374) promote direct or indirect processes that raise the Ca^{2+} level in blood. Calcitriol increases Ca^{2+} resorption in the intestines and kidneys by inducing transporters. Parathyroid hormone supports these processes by stimulating calcitriol biosynthesis in the kidneys (see p. 330). In addition, it directly promotes resorption of Ca^{2+} in the kidneys (see p. 328) and Ca^{2+} release from bone (see **B**). The PTH antagonist **calcitonin** (32 AA) counteracts these processes.



Collagens

Collagens are quantitatively the most abundant of animal proteins, representing 25% of the total. They form insoluble tensile fibers that occur as structural elements of the extracellular matrix and connective tissue throughout the body. Their name (which literally means “glue-producers”) is derived from the gelatins that appear as a decomposition product when collagen is boiled.

A. Structure of collagens ●

Nineteen different collagens are now known, and they are distinguished using roman numerals. They mostly consist of a dextrorotatory **triple helix** made up of three polypeptides (α -chains) (see p. 70).

The triplet **Gly-X-Y** is constantly repeated in the sequence of the triple-helical regions—i. e., every third amino acid in such sequences is a *glycine*. *Proline* (Pro) is frequently found in positions X or Y; the Y position is often occupied by *4-hydroxyproline* (4Hyp), although *3-hydroxyproline* (3Hyp) and *5-hydroxylysine* (5Hyl) also occur. These hydroxylated amino acids are characteristic components of collagen. They are only produced after protein biosynthesis by **hydroxylation** of the amino acids in the peptide chain (see p. 62).

The formation of Hyp and Hyl residues in procollagen is catalyzed by iron-containing *oxygenases* (“proline and lysine hydroxylase,” EC 1.14.11.1/2). *Ascorbate* is required to maintain their function. Most of the symptoms of the vitamin C deficiency disease *scurvy* (see p. 368) are explained by disturbed collagen biosynthesis.

The hydroxyproline residues stabilize the triple helix by forming hydrogen bonds between the α -chains, while the hydroxyl groups of hydroxylysine are partly **glycosylated** with a disaccharide (–Glc–Gal).

The various types of collagen consist of different combinations of α -chains ($\alpha 1$ to $\alpha 3$ and other subtypes). Types I, II, and III represent 90% of collagens. The **type I** collagen shown here has the structure $[\alpha 1(I)]_2\alpha 2(1)$.

Numerous **tropocollagen** molecules (mass 285 kDa, length 400 nm) aggregate extracellularly into a defined arrangement, forming cylindrical **fibrils** (20–500 nm in diameter). Under the electron microscope, these fibrils

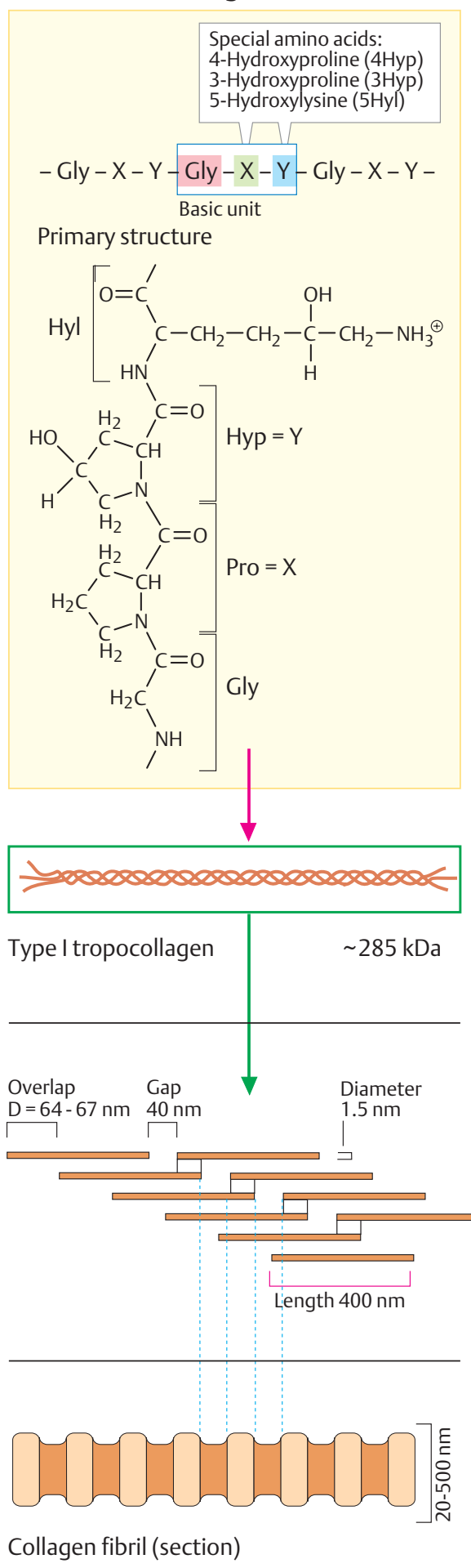
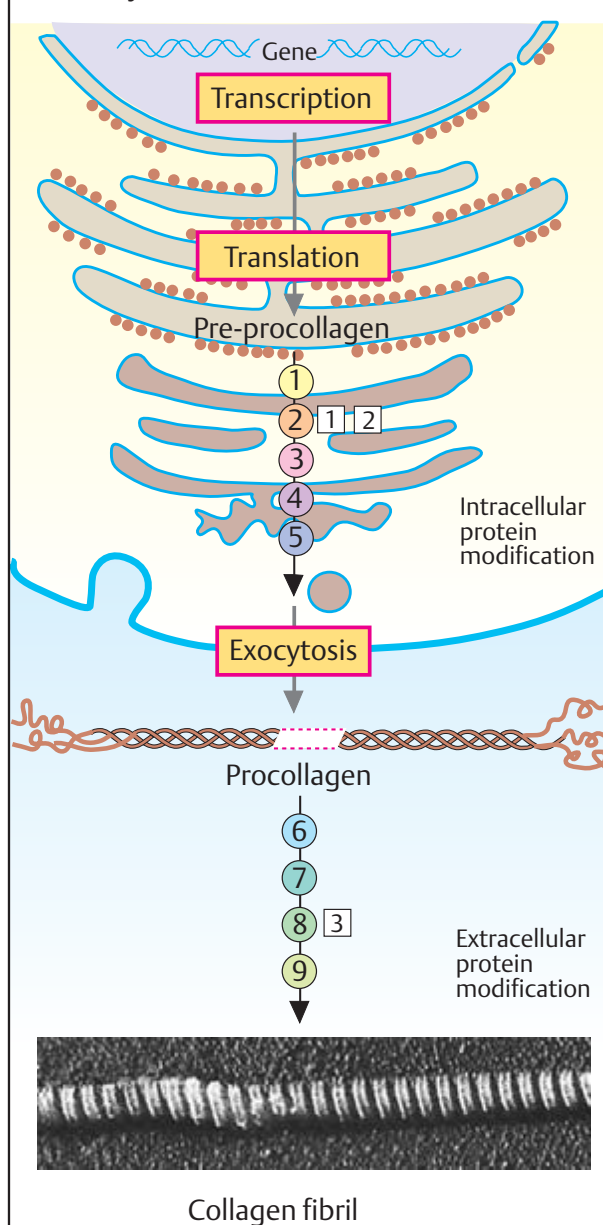
are seen to have a characteristic banding pattern of elements that are repeated every 64–67 nm.

Tropocollagen molecules are firmly linked together, particularly at their ends, by covalent networks of altered lysine side chains. The number of these links increases with age. **Type IV** collagens form networks with a defined mesh size. The size-selective filtering effect of the basal membranes in the renal glomeruli is based on this type of structure (see p. 322).

B. Biosynthesis ●

The precursor molecule of collagen (*preprocollagen*), formed in the rER, is subject to extensive **post-translational modifications** (see p. 232) in the ER and Golgi apparatus.

Cleavage of the signal peptide gives rise to **procollagen**, which still carries large *propeptides* at each end [1]. During this phase, most proline residues and some lysine residues of procollagen are hydroxylated [2]. The procollagen is then glycosylated at hydroxylysine residues [3]. Intramolecular and intermolecular disulfide bonds form in the propeptides [4], allowing correct positioning of the peptide strands to form a triple helix [5]. It is only after these steps have been completed that procollagen is secreted into the extracellular space by exocytosis. This is where the N- and C-terminal propeptides are removed proteolytically [6], allowing the staggered aggregation of the **tropocollagen** molecules to form fibrils [7]. Finally, several ϵ -amino groups in lysine residues are oxidatively converted into aldehyde groups [8]. Covalent links between the molecules then form as a result of condensation [9]. In this way, the fibrils reach their final structure, which is characterized by its high *tensile strength* and *proteinase resistance*.

A. Structure of collagens**B. Biosynthesis**

- ① Removal of the prepeptide
- ② Hydroxylation of Pro and Lys residues
- ③ Glycosylation of 5Hyl and Asn
- ④ Oxidation of Cys in propeptides
- ⑤ Assemblage to form triple helix
- ⑥ Removal of the propeptide
- ⑦ Staggered deposition to form fibrils
- ⑧ Oxidation of Lys and 5Hyl to aldehydes
- ⑨ Cross-linking to form supramolecules

- ① Procollagen-proline 4-dioxygenase 1.14.11.2 [ascorbate, Fe]
- ② Procollagen-lysine 5-dioxygenase 1.14.11.4 [ascorbate, Fe]
- ③ Protein-lysine 6-oxidase 1.4.3.13 [Cu]

Extracellular matrix

A. Extracellular matrix ①

The space between the cells (the interstitium) is occupied by a substance with a complex composition known as the **extracellular matrix** (ECM). In many types of tissue—e.g., muscle and liver—the ECM is only a narrow border between the cells, while in others it forms a larger space. In *connective tissue*, *cartilage*, and *bone*, the ECM is particularly strongly marked and is actually the functional part of the tissue (see p. 340). The illustration shows the three main constituents of the extracellular matrix in a highly schematic way: collagen fibers, network-forming adhesive proteins, and space-filling proteoglycans.

The ECM has a very wide variety of functions: it establishes mechanical connections between cells; it creates structures with special mechanical properties (as in bone, cartilage, tendons, and joints); it creates filters (e.g., in the basal membrane in the renal corpuscles; see p. 322); it separates cells and tissues from each other (e.g., to allow the joints to move freely); and it provides pathways to guide migratory cells (important for embryonic development). The chemical composition of the ECM is just as diverse as its functions.

Collagens (see p. 344), of which there are at least 19 different varieties, form fibers, fibrils, networks, and ligaments. Their characteristic properties are *tensile strength* and *flexibility*. **Elastin** is a fiber protein with a high degree of elasticity.

Adhesive proteins provide the connections between the various components of the extracellular matrix. Important representatives include **laminin** and **fibronectin** (see B). These multifunctional proteins simultaneously bind to several other types of matrix component. Cells attach to the cell surface receptors in the ECM with the help of the adhesive proteins.

Due to their polarity and negative charge, **proteoglycans** (see C) bind water molecules and cations. As a homogeneous “cement,” they fill the gaps between the ECM fibers.

B. Fibronectins ○

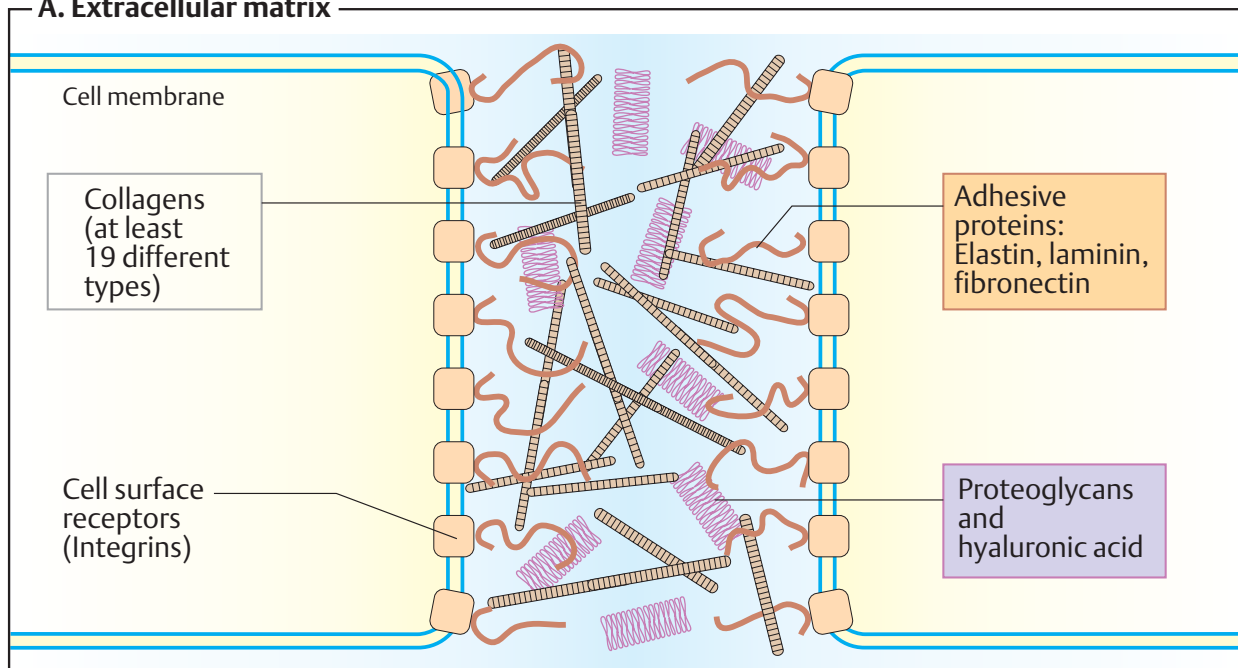
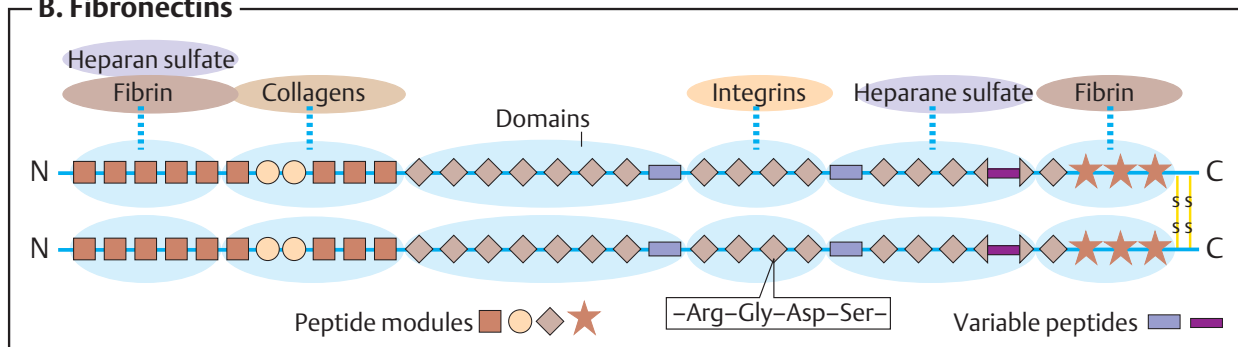
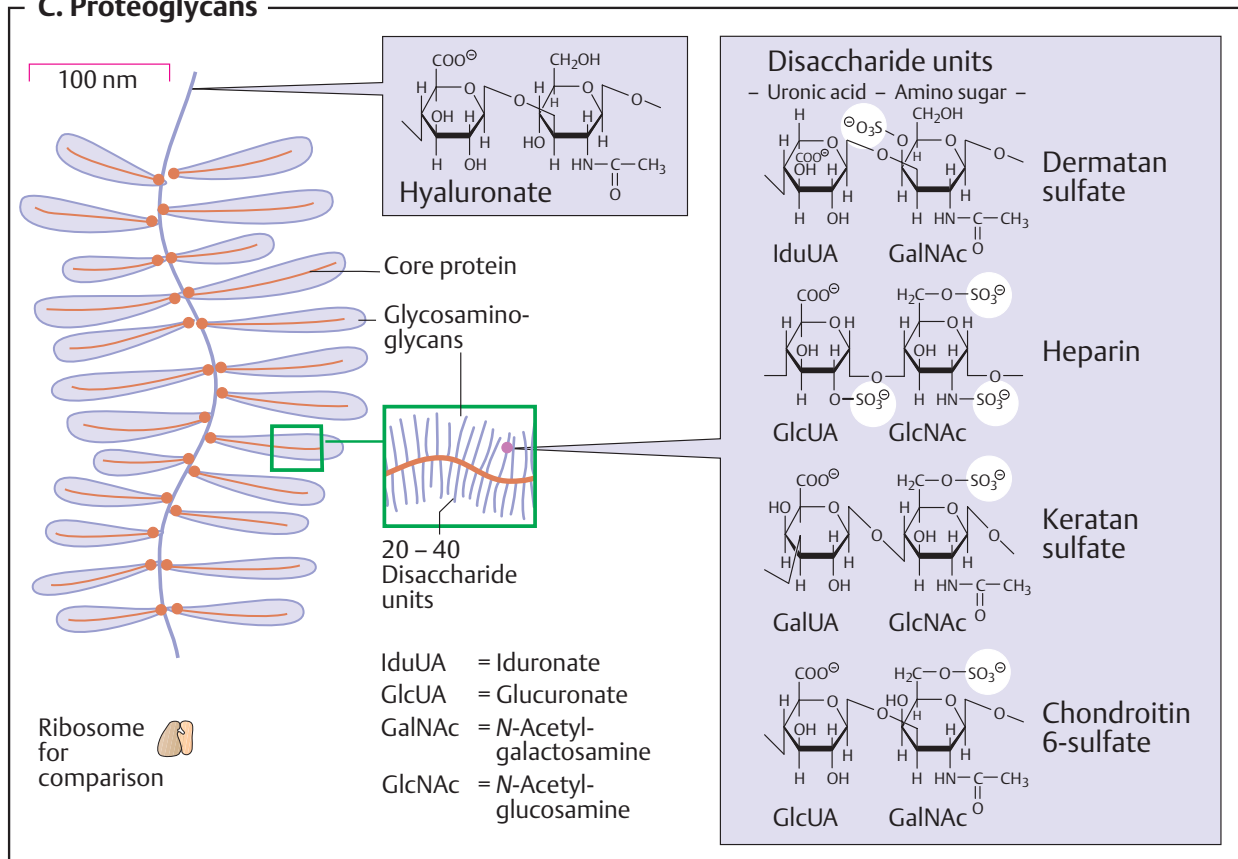
Fibronectins are typical representatives of adhesive proteins. They are filamentous dimers consisting of two related peptide chains (each with a mass of 250 kDa) linked to each other by disulfide bonds. The fibronectin molecules are divided into different *domains*, which bind to cell-surface receptors, collagens, fibrin, and various proteoglycans. This is what gives fibronectins their “*molecular glue*” characteristics.

The domain structure in fibronectins is made up of a few types of *peptide module* that are repeated numerous times. Each of the more than 50 modules is coded for by one exon in the fibronectin gene. *Alternative splicing* (see p. 246) of the hnRNA transcript of the fibronectin gene leads to fibronectins with different compositions. The module that causes adhesion to cells contains the characteristic amino acid sequence –Arg–Gly–Asp–Ser–. It is these residues that enable fibronectin to bind to cell-surface receptors, known as **integrins**.

C. Proteoglycans ①

Proteoglycans are giant molecule complexes consisting of carbohydrates (95%) and proteins (5%), with masses of up to $2 \cdot 10^6$ Da. Their bottlebrush-shaped structure is produced by an axis consisting of **hyaluronate**. This thread-like polysaccharide (see p. 44) has **proteins** attached to it, from which in turn long polysaccharide chains emerge. Like the central hyaluronate, these terminal polysaccharides belong to the glycosaminoglycan group (see p. 44).

The **glycosaminoglycans** are made up of repeating disaccharide units, each of which consists of one *uronic acid* (glucuronic acid or iduronic acid) and one *amino sugar* (N-acetylglucosamine or N-acetylgalactosamine) (see p. 38). Many of the amino sugars are also esterified with sulfuric acid (sulfated), further increasing their polarity. The proteoglycans bind large amounts of water and fill the gaps between the fibrillar components of the ECM in the form of a hydrated gel. This inhibits the spread of pathogens in the ECM, for example.

A. Extracellular matrix**B. Fibronectins****C. Proteoglycans**

Signal transmission in the CNS

A. Structure of nerve cells ●

Nerve cells (neurons) are easily excitable cells that produce electrical signals and can react to such signals as well. Their structure is markedly different from that of other types of cell. Numerous branching processes project from their **cell body** (soma). Neurons are able to receive signals via **dendrites** and to pass them on via **axons**. The axons, which can be up to 1 m long, are usually surrounded by *Schwann cells*, which cover them with a lipid-rich myelin sheath to improve their electrical insulation.

The transfer of stimuli occurs at the **synapses**, which link the individual neurons to each other as well as linking neurons functionally to muscle fibers. *Neurotransmitters* (see p. 352) are stored in the axonal nerve endings. These signaling substances are released in response to electrical signals in order to excite neighboring neurons (or muscle cells). It is estimated that each neuron in the brain is in contact via synapses with approximately 10 000 other neurons.

There is a noticeably high proportion of lipids in the composition of nerve cells, representing about 50% of their dry weight. In particular, there is a very wide variety of phospholipids, glycolipids, and sphingolipids (see p. 216).

B. Neurotransmitters and neurohormones ①

Neurosecretions are classed into two groups: **neurotransmitters** are released into the *synaptic cleft* in order to influence neighboring cells (C). They have a short range and a short lifespan. By contrast, **neurohormones** are released into the blood, allowing them to cover larger distances. However, the distinction between the two groups is a fluid one; some neurotransmitters simultaneously function as neurohormones.

C. Synaptic signal transmission ①

All chemical synapses function according to a similar principle. In the area of the synapse, the surface of the signaling cell (*presynaptic membrane*) is separated from the surface of the receiving cell (*postsynaptic membrane*)

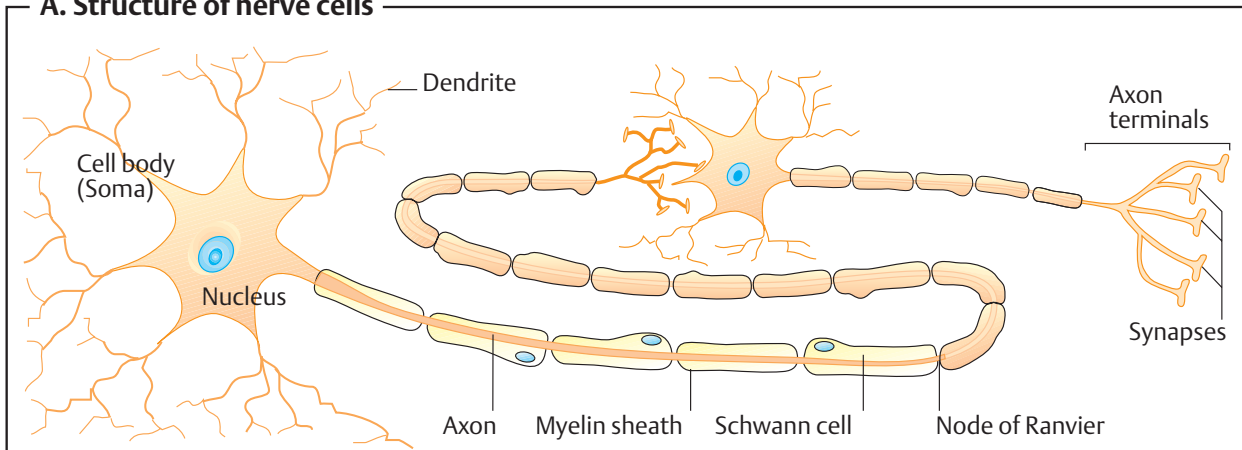
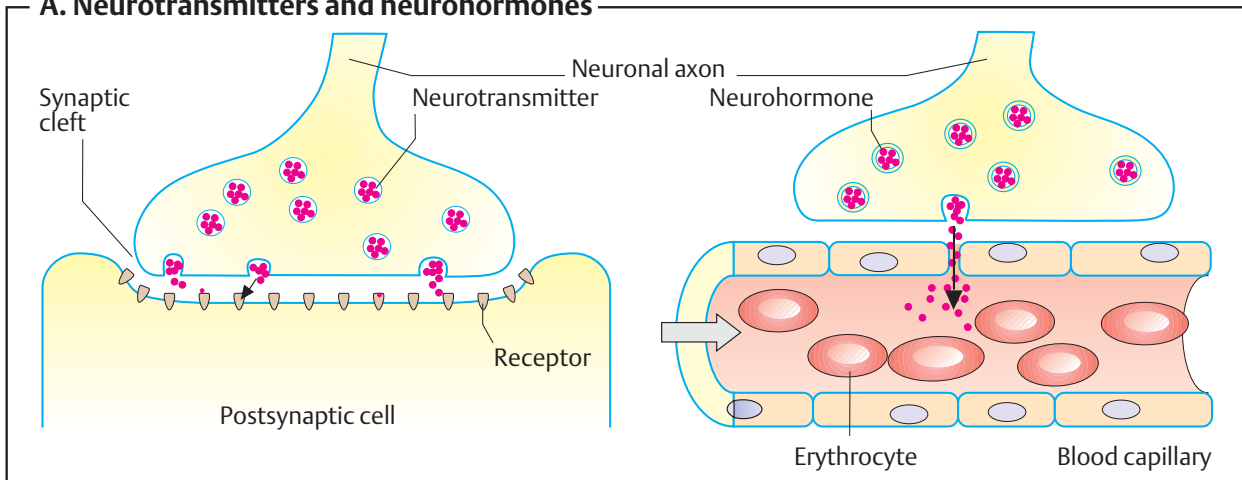
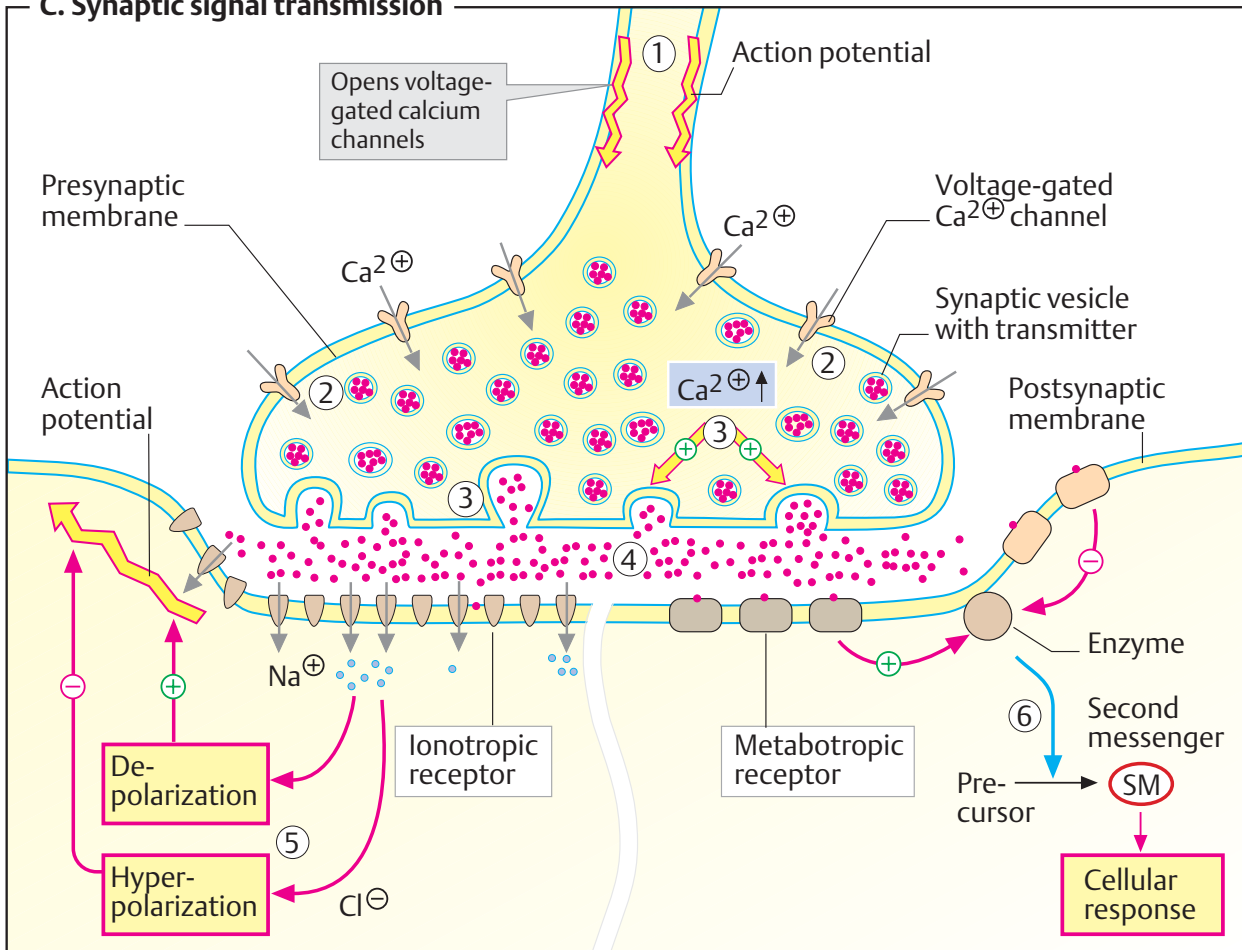
only by a narrow *synaptic cleft*. When an **action potential** (see p. 350) reaches the presynaptic membrane, *voltage-gated Ca^{2+} channels* integrated into the membrane open and trigger **exocytosis** of the neurotransmitter stored in the presynaptic cell (for details, see p. 228).

Each neuron usually releases *only one type* of **neurotransmitter**. Neurons that release dopamine are referred to as “dopaminergic,” for example, while those that release acetylcholine are “cholinergic,” etc. The transmitters that are released diffuse through the synaptic cleft and bind on the other side to **receptors** on the postsynaptic membrane. These receptors are integral membrane proteins that have binding sites for neurotransmitters on their exterior (see p. 224).

The receptors for neurotransmitters are divided into two large groups according to the effect produced by binding of the transmitter (for details, see p. 354).

Ionotropic receptors (bottom left) are *ligand-gated ion channels*. When they open as a result of the transmitter's influence, ions flow in due to the membrane potential (see p. 126). If the inflowing ions are cations (Na^+ , K^+ , Ca^{2+}), **depolarization** of the membrane occurs and an action potential is triggered on the surface of the postsynaptic cell. This is the way in which stimulatory transmitters work (e.g., acetylcholine and glutamate). By contrast, if anions flow in (mainly Cl^-), the result is **hyperpolarization** of the postsynaptic membrane, which makes the production of a postsynaptic action potential more difficult. The action of inhibitory transmitters such as glycine and GABA is based on this effect.

A completely different type of effect is observed in **metabotropic receptors** (bottom right). After binding of the transmitter, these interact on the inside of the postsynaptic membrane with *G proteins* (see p. 384), which in turn activate or inhibit the synthesis of **second messengers**. Finally, second messengers activate or inhibit **protein kinases**, which phosphorylate cellular proteins and thereby alter the behavior of the postsynaptic cells (*signal transduction*; see p. 386).

A. Structure of nerve cells**A. Neurotransmitters and neurohormones****C. Synaptic signal transmission**

Resting potential and action potential

A. Resting potential ①

A characteristic property of living cells is the uneven distribution of positively and negatively charged ions on the inside and outside of the plasma membrane. This gives rise to a **membrane potential** (see p. 126)—i. e., there is electrical voltage between the two sides of the membrane, which can only balance out when *ion channels* allow the unevenly distributed ions to move.

At rest, the membrane potential in most cells is -60 to -90 mV. It mainly arises from the activity of *Na⁺/K⁺ transporting ATPase* (“Na⁺/K⁺ ATPase”), which occurs on practically all animal cells. Using up ATP, this P-type enzyme (see p. 220) “pumps” three Na⁺ ions out of the cell in exchange for two K⁺ ions. Some of the K⁺ ions, following the concentration gradient, leave the cell again through *potassium channels*. As the protein anions that predominate inside the cell cannot follow them, and inflow of Cl[−] ions from the outside is not possible, the result is an excess of positive charges outside the cell, while anions predominate inside it.

An **equilibrium potential** exists for each of the ions involved. This is the value of the membrane potential at which there is no net inflow or outflow of the ions concerned. For K⁺ ions, the resting potential lies in the range of the membrane potential, while for Na⁺ ions it is much higher at $+70$ mV. At the first opportunity, Na⁺ ions will therefore spontaneously flow into the cell. The occurrence of action potentials is based on this (see B).

Nerve cell membranes contain **ion channels** for Na⁺, K⁺, Cl[−], and Ca²⁺. These channels are usually closed and open only briefly to let ions pass through. They can be divided into channels that are regulated by membrane potentials (“*voltage-gated*”—e. g., fast Na⁺ channels; see p. 222) and those regulated by ligands (“*ligand-gated*”—e. g., nicotinic acetylcholine receptors; see p. 222).

B. Action potential ①

Action potentials are special signals that are used to transmit information in the nervous system. They are triggered by chemical stim-

uli (or more rarely electrical stimuli). Binding of a neurotransmitter to an ionotropic receptor results in a brief local increase in the membrane potential from -60 mV to about $+30$ mV. Although the membrane potential quickly returns to the initial value within a few milliseconds (ms) at its site of origin, the depolarization is propagated because neighboring membrane areas are activated during this time period.

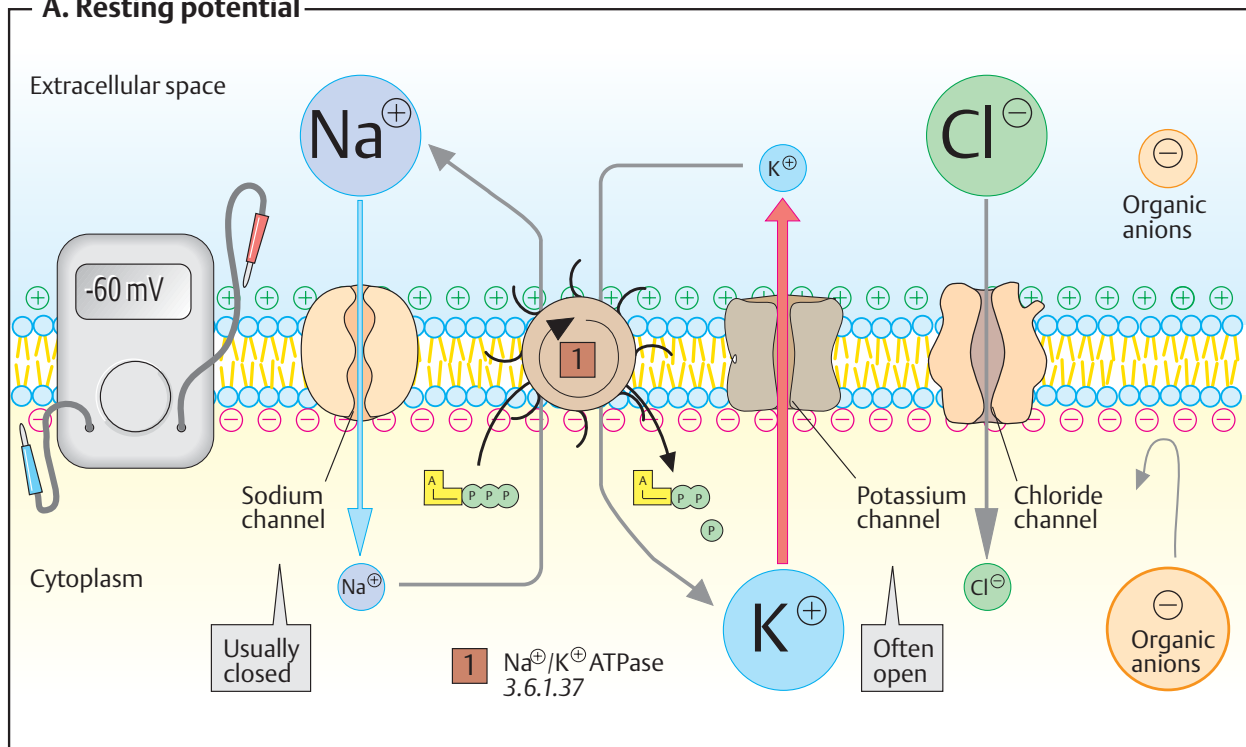
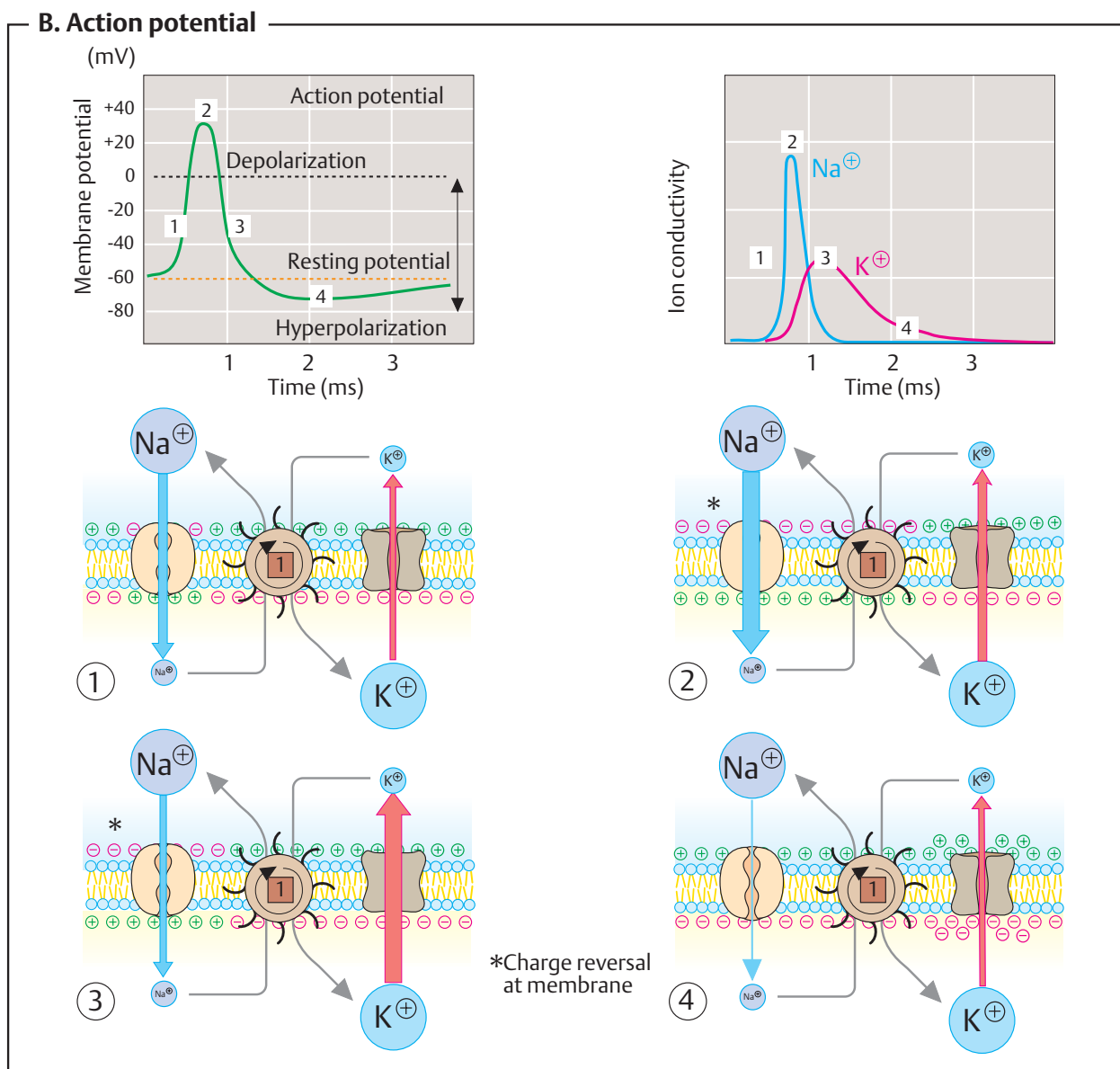
[1] The process starts with the opening of voltage-gated Na⁺ channels (see p. 222). Due to their high equilibrium potential (see A), Na⁺ ions flow into the cell and reverse the local membrane potential (**depolarization**).

[2] The Na⁺ channels immediately close again, so that the inflow of positive charges is only very brief.

[3] Due to the increase in the membrane potential, voltage-dependent K⁺ channels open and K⁺ ions flow out. In addition, Na⁺/K⁺ ATPase (see A) pumps the Na⁺ ions that have entered back out again. This leads to **repolarization** of the membrane.

[4] The two processes briefly lead to the charge even falling below the resting potential (**hyperpolarization**). The K⁺ channels also close after a few milliseconds. The nerve cell is then ready for re-stimulation.

Generally, it is always only a very small part of the membrane that is depolarized during an action potential. The process can therefore be repeated again after a short refractory period, when the nerve cell is stimulated again. Conduction of the action potential on the surface of the nerve cell is based on the fact that the local increase in the membrane potential causes neighboring voltage-gated ion channels to open, so that the membrane stimulation spreads over the whole cell in the form of a *depolarization wave*.

A. Resting potential**B. Action potential**

Neurotransmitters

Neurotransmitters in the strict sense are substances that are produced by neurons, stored in the synapses, and released into the synaptic cleft in response to a stimulus. At the post-synaptic membrane, they bind to special receptors and affect their activity.

A. Important neurotransmitters ❶

Neurotransmitters can be classified into several groups according to their chemical structure. The table lists the most important representatives of this family, which has more than 100 members.

Acetylcholine, the acetic acid ester of the cationic alcohol choline (see p. 50) acts at neuromuscular junctions, where it triggers muscle contraction (see p. 334), and in certain parts of the brain and in the autonomous nervous system.

Several proteinogenic **amino acids** (see p. 60) have neurotransmitter effects. A particularly important one is *glutamate*, which acts as a stimulatory transmitter in the CNS. More than half of the synapses in the brain are glutaminergic. The metabolism of glutamate and that of the amine GABA synthesized from it (see below) are discussed in more detail on p. 356. *Glycine* is an inhibitory neurotransmitter with effects in the spinal cord and in parts of the brain.

Biogenic amines arise from amino acids by decarboxylation (see p. 62). This group includes *4-aminobutyrate* (γ -aminobutyric acid, GABA), which is formed from glutamate and is the most important inhibitory transmitter in the CNS. The *catecholamines* norepinephrine and epinephrine (see **B**), *serotonin*, which is derived from tryptophan, and *histamine* also belong to the biogenic amine group. All of them additionally act as hormones or mediators (see p. 380).

Peptides make up the largest group among the neurosecretions. Many peptide hormones—e.g., thyrotiberin (TRH) and angiotensin II—simultaneously act as transmitters. Most neuropeptides are small (3–15 AA). At their N-terminus, many of them have a glutamate residue that has been cyclized to form *pyroglutamate* (5-oxoproline, <G>), while the C-terminus is often an acid amide ($-\text{NH}_2$).

This provides better protection against breakdown by peptidases.

Endorphins, *dynorphins*, and *enkephalins* are a particularly interesting group of neuropeptides. They act as “endogenous opiates” by producing analgetic, sedative, and euphoriant effects in extreme situations. Drugs such as morphine and heroin activate the receptors for these peptides (see p. 354).

Purine derivatives with neurotransmitter function are all derived from adenine-containing nucleotides or nucleosides. ATP is released along with acetylcholine and other transmitters, and among other functions it regulates the emission of transmitters from its synapse of origin. The stimulatory effect of *caffeine* is mainly based on the fact that it binds to adenosine receptors.

B. Biosynthesis of catecholamines ❷

The catecholamines are *biogenic amines* that have a catechol group. Their biosynthesis in the adrenal cortex and CNS starts from **tyrosine**.

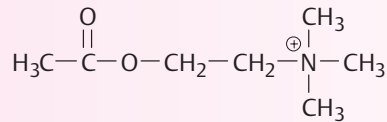
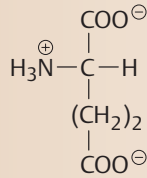
[1] Hydroxylation of the aromatic ring initially produces **dopa** (3,4-dihydroxyphenylalanine). This reaction uses the unusual coenzyme *tetrahydrobiopterin* (THB). Dopa (cf. p. 6) is also used in the treatment of Parkinson's disease.

[2] Decarboxylation of dopa yields **dopamine**, an important transmitter in the CNS. In dopaminergic neurons, catecholamine synthesis stops at this point.

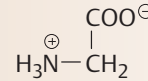
[3] The adrenal gland and adrenergic neurons continue the synthesis by hydroxylating dopamine into **norepinephrine** (noradrenaline). *Ascorbic acid* (vitamin C; see p. 368) acts as a hydrogen-transferring coenzyme here.

[4] Finally, *N*-methylation of norepinephrine yields **epinephrine** (adrenaline). The coenzyme for this reaction is *S*-adenosylmethionine (SAM; see p. 110).

The physiological effects of the catecholamines are mediated by a large number of different receptors that are of particular interest in pharmacology. Norepinephrine acts in the autonomic nervous system and certain areas of the brain. Epinephrine is also used as a transmitter by some neurons.

A. Important neurotransmittersAcetyl-
cholineAmino
acids:Glutamate
Glycine
Dopa

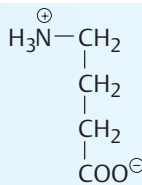
Glutamate



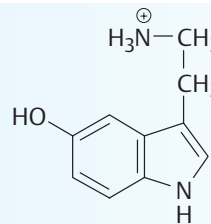
Glycine

Biogenic
amines:

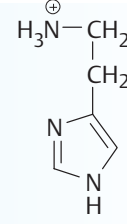
γ-Aminobutyrate (GABA)
Dopa
Dopamine
Norepinephrine
Epinephrine
Serotonin
Histamine



GABA



Serotonin



Histamine

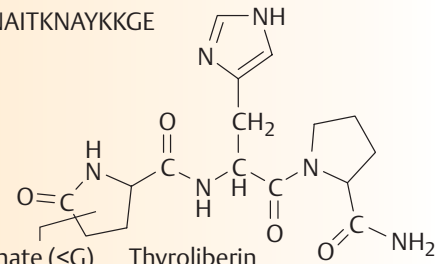
Peptides:

β-Endorphin
Met- and Leu-enkephalin

YGGFMTSEKSTPLVTLFKNAITKNAYKKGE
YGGFM und YGGFL

Thyroliberin (TRH)
Gonadoliberin (GnRH)
Substance P
Somatostatin
Angiotensin II
Cholecystokinin (CCK-4)
and many others

<GHP-NH₂
<GHWSYGLRPG-NH₂
RPKPQQFFGLM
AGCKNFFWKFTSC
DRVYIHPF
WMDF-NH₂

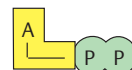


Pyroglutamate (<G)

Thyroliberin

Purine
derivatives:

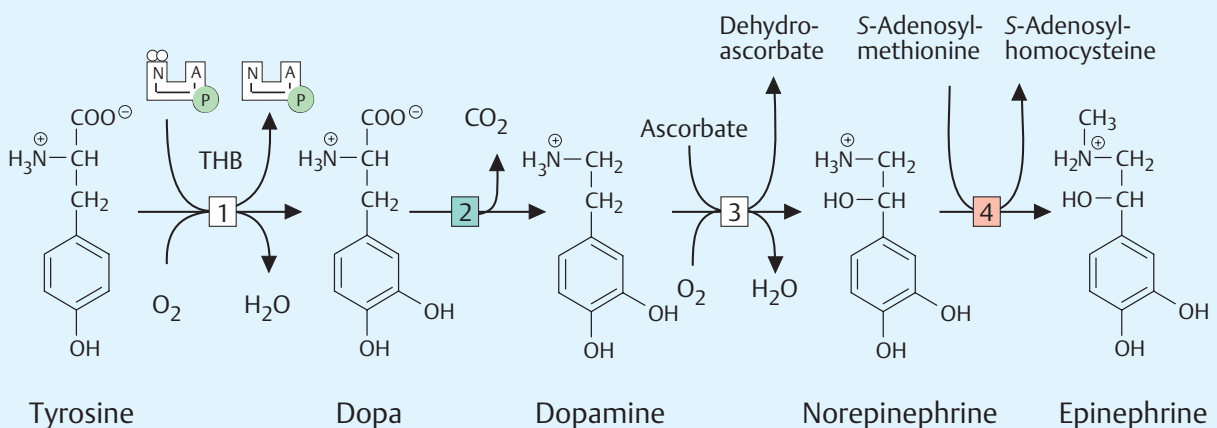
ATP
ADP
AMP
Adenosine

**B. Biosynthesis of the catecholamines**1 Tyrosine 3-monooxygenase [Fe²⁺, THB] 1.14.16.2

3 Dopamine β-monooxygenase [Cu] 1.14.17.1

2 Aromatic-L-amino-acid decarboxylase (Dopa decarboxylase) [PLP] 4.1.1.28

4 Phenylethanolamine N-methyltransferase 2.1.1.28



Receptors for neurotransmitters

Like all signaling substances, neurotransmitters (see p. 352) act via receptor proteins. The receptors for neurotransmitters are integrated into the membrane of the postsynaptic cell, where they trigger ion inflow or signal transduction processes (see p. 348).

A. Receptors for neurotransmitters ❶

A considerable number of receptors for neurotransmitters are already known and new ones are continuing to be discovered. The table only lists the most important examples. They are classified into two large groups according to their mode of action.

Ionotropic receptors are *ligand-gated ion channels* (left half of the table). The receptors for stimulatory transmitters (indicated in the table by a $\oplus$) mediate the inflow of cations (mainly Na^+). When these open after binding of the transmitter, local *depolarization* of the postsynaptic membrane occurs. By contrast, inhibitory neurotransmitters (GABA and glycine) allow Cl^- to flow in. This increases the membrane's negative resting potential and hinders the action of stimulatory transmitters (*hyperpolarization*, $\ominus$).

Metabotropic receptors (right half of the table) are coupled to G proteins (see p. 386), through which they influence the *synthesis of second messengers*. Receptors that work with type G_s proteins (see p. 386) increase the cAMP level in the postsynaptic cell ($[\text{cAMP}] \uparrow$), while those that activate G_i proteins reduce it ($[\text{cAMP}] \downarrow$). Via type G_q proteins, other receptors increase the intracellular Ca^{2+} concentration ($[\text{Ca}^{2+}] \uparrow$).

There are several **receptor subtypes** for most neurotransmitters. These are distinguished numerically (e.g., D_1 to D_5) or are named after their agonists—i.e., after molecules experimentally found to activate the receptor. For example, one specific subtype of glutamate receptors reacts to NMDA (*N*-methyl-D-aspartate), while another subtype reacts to the compound AMPA, etc.

B. Acetylcholine receptors ❶

Acetylcholine (ACh) was the neurotransmitter first discovered, at the beginning of the last century. It binds to two types of receptor.

The **nicotinic ACh receptor** responds to the alkaloid *nicotine* contained in tobacco (many of the physiological effects of nicotine are based on this). The nicotinic receptor is ionotropic. Its properties are discussed in greater detail on p. 222.

The **muscarinic ACh receptors** (of which there are at least five subtypes) are metabotropic. Their name is derived from the alkaloid *muscarine*, which is found in the fly agaric mushroom (*Amanita muscaria*), for example. Like ACh, muscarine is bound at the receptor, but in contrast to ACh (see C), it is not broken down and therefore causes permanent stimulation of muscle.

The muscarinic ACh receptors influence the cAMP level in the postsynaptic cells (M_1 , M_3 and M_5 increase it, while subtypes M_2 and M_4 reduce it).

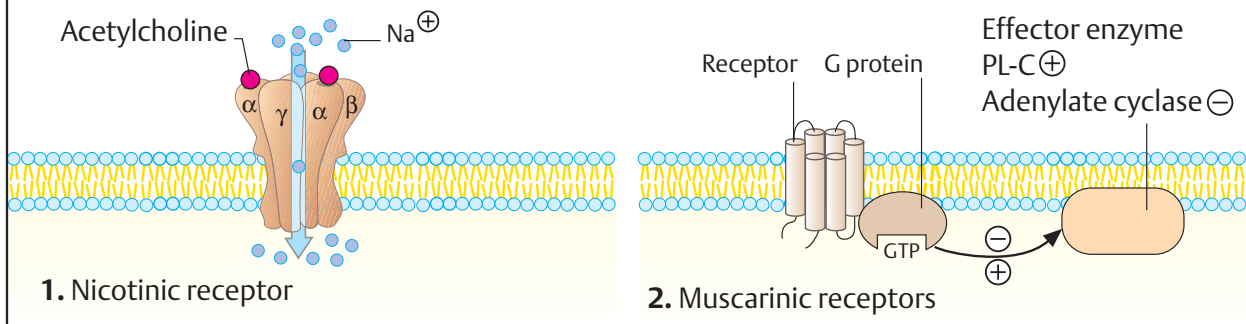
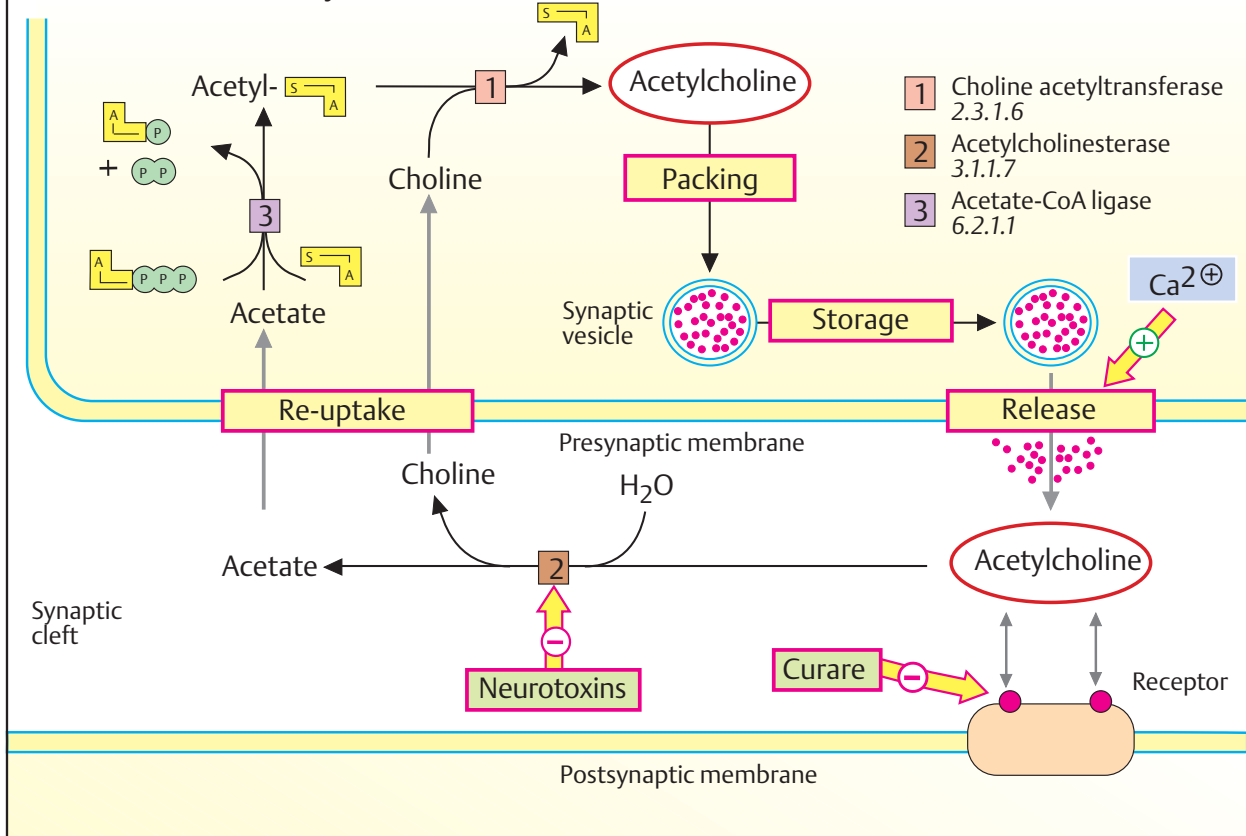
C. Metabolism of acetylcholine ❶

Acetylcholine is synthesized from acetyl-CoA and choline in the cytoplasm of the presynaptic axon [1] and is stored in **synaptic vesicles**, each of which contains around 1000–10 000 ACh molecules. After it is released by exocytosis (see p. 228), the transmitter travels by diffusion to the receptors on the postsynaptic membrane. Catalyzed by *acetylcholinesterase*, hydrolysis of ACh to acetate and choline immediately starts in the synaptic cleft [2], and within a few milliseconds, the ACh released has been eliminated again. The cleavage products **choline** and **acetate** are taken up again by the presynaptic neuron and reused for acetylcholine synthesis [3].

Substances that block the serine residue in the active center of acetylcholinesterase [2]—e.g., the neurotoxin E605 and other *organophosphates*—prevent ACh degradation and thus cause prolonged stimulation of the postsynaptic cell. This impairs nerve conduction and muscle contraction. *Curare*, a paralyzing arrow-poison used by South American Indians, competitively inhibits binding of ACh to its receptor.

A. Receptors for neurotransmitters

Ionotropic				Metabotropic		
Receptor	Transmitter	Ion(s)	Effect	Receptor	Transmitter	Effect
Acetylcholine (nicotinic)	Acetylcholine	Na^+	$\oplus$	Acetylcholine (muscarinic) M1, M3, M5, M2, M4	Acetylcholine	$[\text{Ca}^{2+}] \uparrow$ $[\text{cAMP}] \downarrow$
5HT ₃	Serotonin	Na^+	$\oplus$	5HT ₁ 5HT ₂ 5HT ₄	Serotonin "	$[\text{Ca}^{2+}] \uparrow$ $[\text{cAMP}] \uparrow$ $[\text{cAMP}] \downarrow$
GABA _A	GABA	Cl^-	$\ominus$	α_1 α_2	Norepinephrine "	$[\text{Ca}^{2+}] \uparrow$ $[\text{cAMP}] \uparrow$ $[\text{cAMP}] \downarrow$
Glycine	Glycine	Cl^-	$\ominus$	$\beta_1, \beta_2, \beta_3$	"	
AMPA NMDA Kainate	Glutamate Glutamate Glutamate	$\text{Na}^+ \text{K}^+$ $\text{Na}^+ \text{K}^+ \text{Ca}^{2+}$ $\text{Na}^+ \text{K}^+$	$\oplus$ $\oplus$ $\oplus$	D ₁ , D ₅ D ₂ , D ₃ , D ₄	Dopamine "	$[\text{cAMP}] \uparrow$ $[\text{cAMP}] \downarrow$
				δ, κ, μ	Opioids	$[\text{cAMP}] \downarrow$

B. Acetylcholine receptors**C. Metabolism of acetylcholine**

Metabolism

The brain and other areas of the central nervous system (CNS) have high ATP requirements. Although the brain only represents about 2% of the body's mass, it consumes around 20% of the metabolized oxygen and ca. 60% of the glucose. The neurons' high energy requirements are mainly due to ATP-dependent ion pumps (particularly Na^+/K^+ ATPase) and other active transport processes that are needed for nerve conduction (see p. 350).

A. Energy metabolism of the brain ●

Glucose is normally the only metabolite from which the brain is able to obtain adequate amounts of ATP through aerobic glycolysis and subsequent terminal oxidation to CO_2 and H_2O . Lipids are unable to pass the **blood-brain barrier**, and amino acids are also only available in the brain in limited quantities (see B). As neurons only have minor glycogen reserves, they are dependent on a constant supply of glucose from the blood. A severe drop in the blood glucose level—as can occur after insulin overdosage in diabetics, for example—rapidly leads to a drop in the ATP level in the brain. This results in loss of consciousness and neurological deficits that can lead to death. Oxygen deficiency (hypoxia) also affects the brain. The effects of a brief period of hypoxia are still reversible, but as time progresses irreversible damage increasingly occurs and finally complete loss of function (“brain death”).

During periods of starvation, the brain after a certain time acquires the ability to use **ketone bodies** (see p. 312) in addition to glucose to form ATP. In the first weeks of a starvation period, there is a strong increase in the activities of the enzymes required for this in the brain. The degradation of ketone bodies in the CNS saves glucose and thereby reduces the breakdown of muscle protein that maintains gluconeogenesis in the liver during starvation. After a few weeks, the extent of muscle breakdown therefore declines to one-third of the initial value.

B. Glutamate, glutamine, and GABA ●

The proteinogenic amino acid **glutamate** (Glu) and the biogenic amine **4-aminobutyrate** derived from it are among the most important neurotransmitters in the brain (see p. 352). They are both synthesized in the brain itself. In addition to the neurons, which use Glu or GABA as transmitters, *neuroglia* are also involved in the metabolism of these substances.

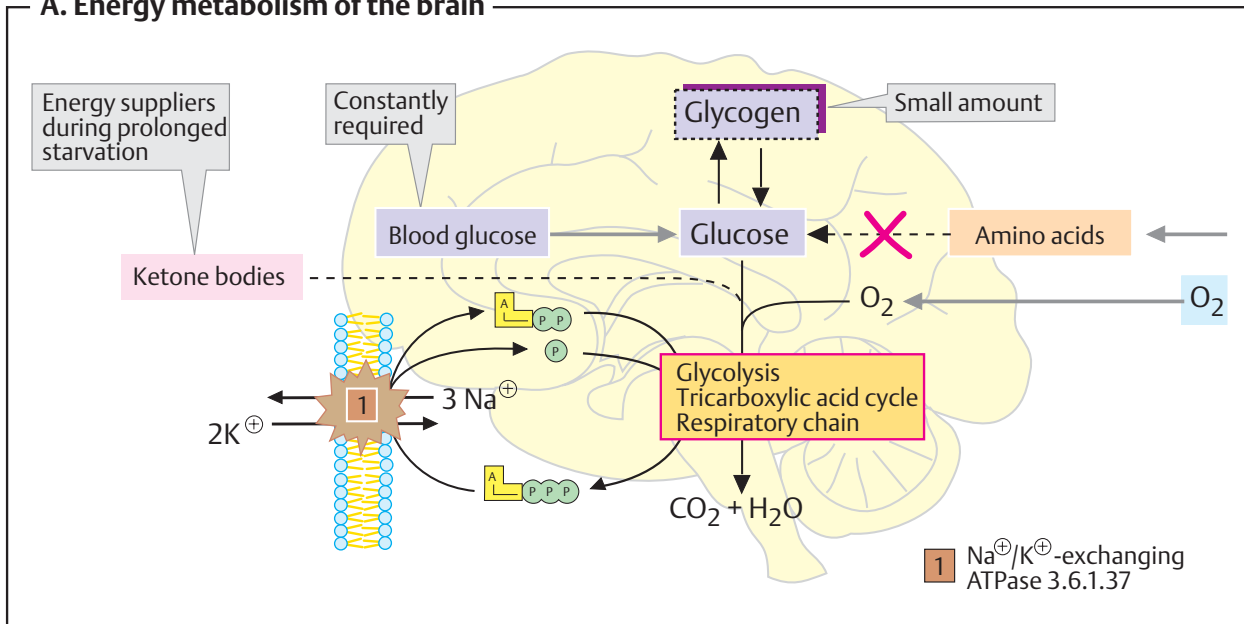
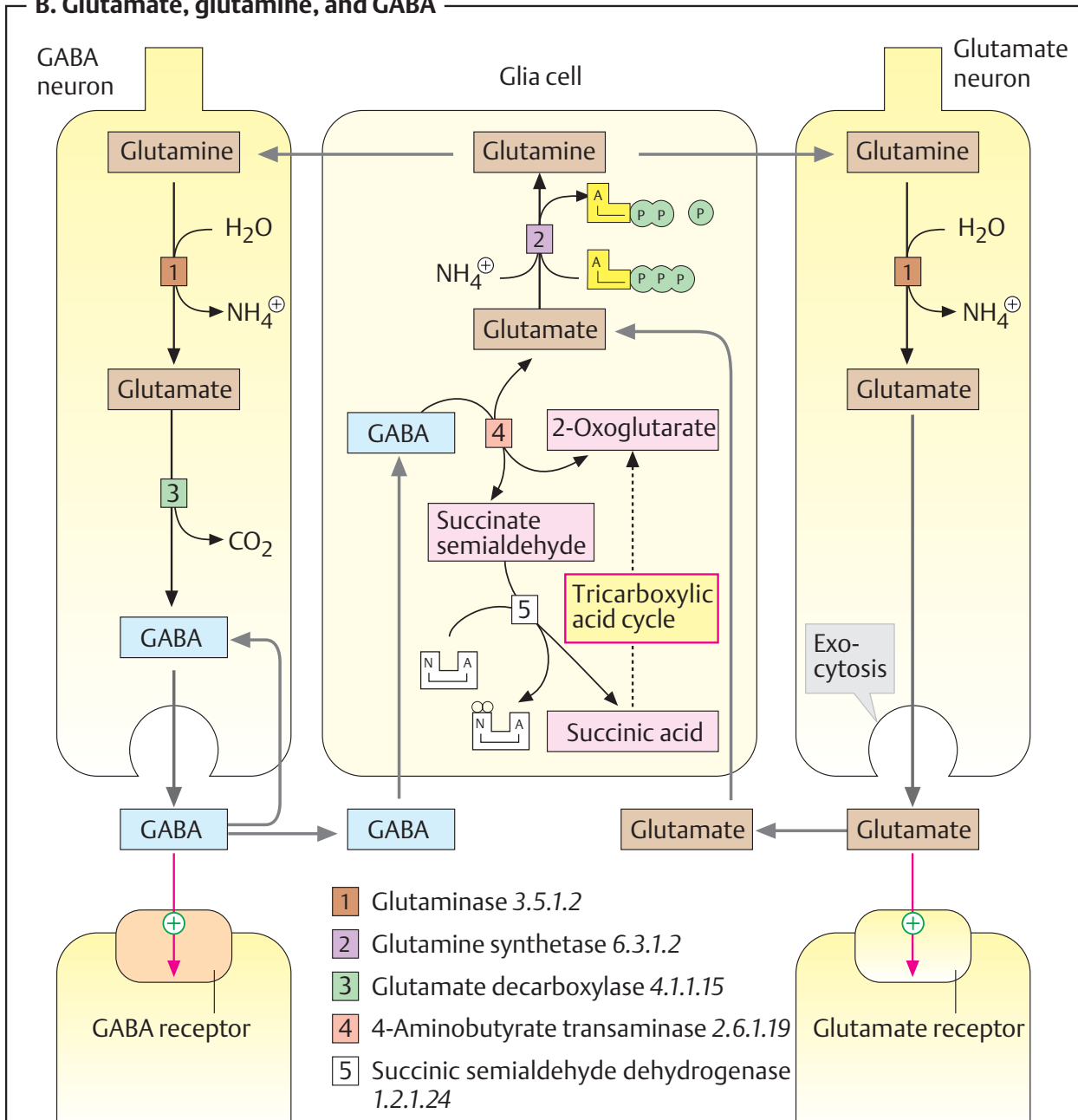
Since glutamate and GABA as transmitters must not appear in the extracellular space in an unregulated way, the cells of the neuroglia (center) supply “glutaminergic” and “GABAergic” neurons with the precursor **glutamine** (Gln), which they produce from glutamate with the help of *glutamine synthetase* [1].

GABA neurons (left) and glutamate neurons (right) initially hydrolyze glutamine with the help of *glutaminase* [1] to form glutamate again. The glutamate neurons store this in vesicles and release it when stimulated. The GABA neurons continue the degradation process by using *glutamate decarboxylase* [3] to convert glutamate into the transmitter GABA.

Both types of neuron take up their transmitter again. Some of it also returns to the neuroglia, where glutamate is amidated back into glutamine.

Glutamate can also be produced again from GABA. The reaction sequence needed for this, known as the **GABA shunt**, is characteristic of the CNS. A *transaminase* [4] first converts GABA and 2-oxoglutarate into glutamate and succinate semialdehyde ($-\text{OOC}-\text{CH}_2-\text{CH}_2-\text{CHO}$). In an NAD^+ -dependent reaction, the aldehyde is oxidized to succinic acid [5], from which 2-oxoglutarate can be regenerated again via tricarboxylic acid cycle reactions.

The function of glutamate as a stimulatory transmitter in the brain is the cause of what is known as the “Chinese restaurant syndrome.” In sensitive individuals, the monosodium glutamate used as a flavor enhancer in Chinese cooking can raise the glutamate level in the brain to such an extent that transient mild neurological disturbances can occur (dizziness, etc.).

A. Energy metabolism of the brain**B. Glutamate, glutamine, and GABA**

Sight

Two types of photoreceptor cell are found in the human retina—*rods* and *cones*. Rods are sensitive to low levels of light, while the cones are responsible for color vision at higher light intensities.

Signaling substances and many proteins are involved in visual processes. Initially, a **light-induced *cis*–*trans* isomerization** of the pigment retinal triggers a conformational change in the membrane protein *rhodopsin*. Via the G protein *transducin*, which is associated with rhodopsin, an enzyme is activated that breaks down the second messenger *cGMP*. Finally, the *cGMP* deficiency leads to *hyperpolarization* of the light-sensitive cell, which is registered by subsequent neurons as *reduced neurotransmitter release*.

A. Photoreceptor ❶

The cell illustrated opposite, a **rod**, has a structure divided by membrane discs into which the 7-helix receptor **rhodopsin** is integrated (see p. 224). In contrast to other receptors in the 7-helix class (see p. 384), rhodopsin is a light-sensitive *chromoprotein*. Its protein part, **opsin**, contains the aldehyde **retinal** (see p. 364)—an isoprenoid which is bound to the ϵ -amino group of a lysine residue as an *aldimine*.

The light absorption of rhodopsin is in the visible range, with a maximum at about 500 nm. The absorption properties of the visual pigment are thus optimally adjusted to the spectral distribution of sunlight.

Absorption of a photon triggers isomerization from the 11-*cis* form of retinal to all-*trans*-retinal (top right). Within milliseconds, this *photochemical process* leads to an allosteric conformational change in rhodopsin. The active conformation (**rhodopsin***) binds and activates the G protein **transducin**. The *signal cascade* (B) that now follows causes the rod cells to release less neurotransmitter (glutamate) at their synapses. The adjoining bipolar neurons register this change and transmit it to the brain as a signal for light.

There are several different rhodopsins in the **cones**. All of them contain retinal molecules as light-sensitive components, the absorption properties of which are modulated by the different proportions of opsin they

contain in such a way that colors can also be perceived.

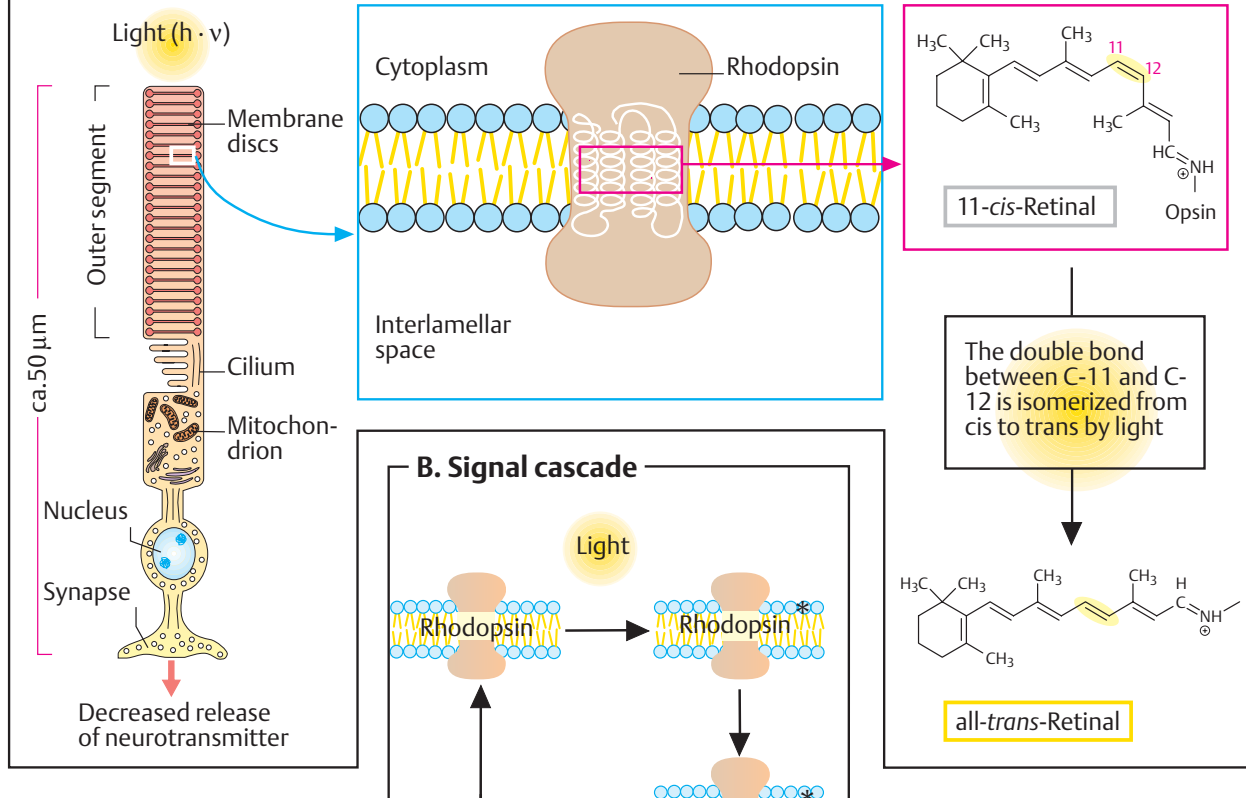
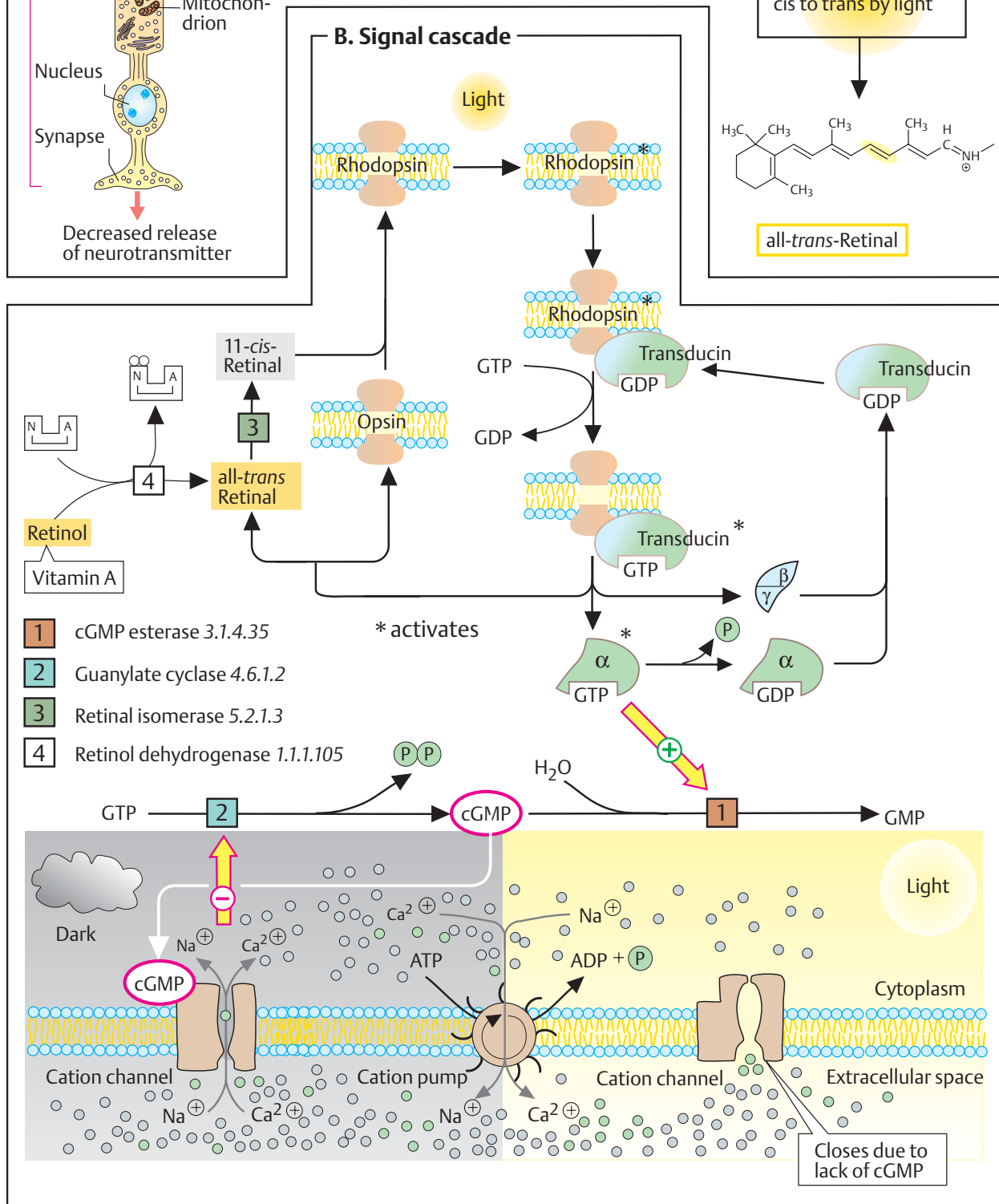
B. Signal cascade ❷

Dark (bottom left). Rod cells that are not exposed to light contain relatively high concentrations (70 μM) of the cyclic nucleotide **cGMP** (3',5'-cycloGMP; cf. cAMP, p.386), which is synthesized by a *guanylate cyclase* ([2], see p.388). The cGMP binds to an ion channel in the rod membrane (bottom left) and thus keeps it open. The inflow of cations (Na^+ , Ca^{2+}) depolarizes the membrane and leads to release of the neurotransmitter glutamate at the synapse (see p. 356).

Light (bottom right). When the G protein transducin binds to light-activated rhodopsin* (see A, on the structure of the complex; see p.224), it leads to the GDP that is bound to the transducin being exchanged for GTP. In transducin* that has been activated in this way, the GTP-containing α -subunit breaks off from the rest of the molecule and in turn activates a membrane *cGMP phosphodiesterase* [1]. This hydrolyzes cGMP to GMP and thus reduces the level of free cGMP within milliseconds. As a consequence, the cGMP bound at the ion channel dissociates off and the channel closes. As cations are constantly being pumped out of the cell, the membrane potential falls and **hyperpolarization** of the cell occurs, which interrupts glutamate release.

Regeneration. After exposure to light, several processes restore the initial conditions:

1. The α -subunit of transducin* inactivates itself by GTP hydrolysis and thus terminates the activation of cGMP esterase.
2. The reduced Ca^{2+} concentration causes activation of guanylate cyclase, which increases the cGMP level until the cation channels reopen.
3. An isomerase [3] transfers all-*trans* -retinal to the 11-*cis* -form, in which it is available for the next cycle. A dehydrogenase [4] can also allow retinal to be supplied from vitamin A (retinol).

A. Photoreceptor**B. Signal cascade**

Organic substances

A balanced human diet needs to contain a large number of different components. These include *proteins*, *carbohydrates*, *fats*, *minerals* (including water), and *vitamins*. These substances can occur in widely varying amounts and proportions, depending on the type of diet. As several components of the diet are *essential for life*, they have to be regularly ingested with food. Recommended daily minimums for nutrients have been published by the World Health Organization (WHO) and a number of national expert committees.

A. Energy requirement ●

The amount of energy required by a human is expressed in kJ d^{-1} (kilojoule per day). An older unit is the kilocalorie (kcal; $1 \text{ kcal} = 4.187 \text{ kJ}$). The figures given are recommended values for adults with a normal body weight. However, actual requirements are based on age, sex, body weight, and in particular on physical activity. In those involved in competitive sports, for example, requirements can increase from 12 000 to 17 000 kJ d^{-1} .

It is recommended that about half of the energy intake should be in the form of carbohydrates, a third at most in the form of fat, and the rest as protein. The fact that *alcoholic beverages* can make a major contribution to daily energy intake is often overlooked. Ethanol has a caloric value of about 30 kJ g^{-1} (see p. 320).

B. Nutrients ●

Proteins provide the body with amino acids, which are used for endogenous protein biosynthesis. Excess amino acids are broken down to provide energy (see p. 174). Most amino acids are *glucogenic*—i.e., they can be converted into glucose (see p. 180).

Proteins are essential components of the diet, as they provide **essential amino acids** that the human body is not capable of producing on its own (see the table). Some amino acids, including cysteine and histidine, are not absolutely essential, but promote growth in children. Some amino acids are able to substitute for each other in the diet. For example, humans can form tyrosine, which is actually essential, by hydroxylation from phenylalanine, and cysteine from methionine.

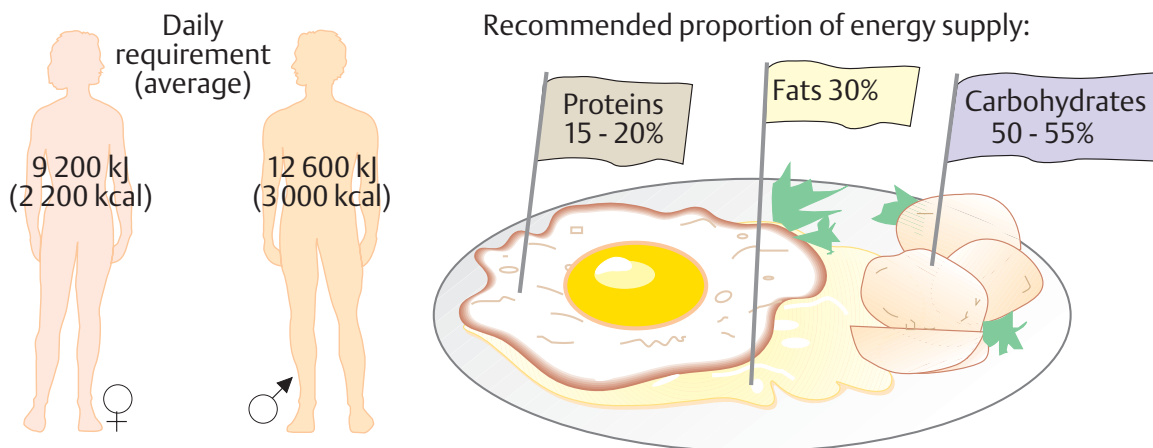
The *minimum daily requirement* of protein is 37 g for men and 29 g for women, but the recommended amounts are about twice these values. Requirements in pregnant and breast-feeding women are even higher. Not only the quantity, but also the quality of protein is important. Proteins that lack several essential amino acids or only contain small quantities of them are considered to be of low value, and larger quantities of them are therefore needed. For example, pulses only contain small amounts of methionine, while wheat and corn proteins are poor in lysine. In contrast to vegetable proteins, most animal proteins are high-value (with exceptions such as collagen and gelatin).

Carbohydrates serve as a general and easily available energy source. In the diet, they are present as *monosaccharides* in honey and fruit, or as *disaccharides* in milk and in all foods sweetened with sugar (sucrose). Metabolically usable *polysaccharides* are found in vegetable products (starch) and animal products (glycogen). Carbohydrates represent a substantial proportion of the body's energy supply, but they are not essential.

Fats are primarily important energy suppliers in the diet. Per gram, they provide more than twice as much energy as proteins and carbohydrates. Fats are essential as suppliers of *fat-soluble vitamins* (see p. 364) and as sources of *polyunsaturated fatty acids*, which are needed to biosynthesize eicosanoids (see pp. 48, 390).

Mineral substances and trace elements, a very heterogeneous group of essential nutrients, are discussed in more detail on p. 362. They are usually divided into *macro-minerals* and *microminerals*.

Vitamins are also indispensable components of the diet. The animal body requires them in very small quantities in order to synthesize coenzymes and signaling substances (see pp. 364–369).

A. Energy requirement**B. Nutrients**

	Quantity in body (kg)	Energy content $\text{kJ} \cdot \text{g}^{-1}$ ($\text{kcal} \cdot \text{g}^{-1}$)	Daily requirement (g) a b c	General function in metabolism	Essential constituents
Proteins	10	17 (4.1)	♂ 37 55 92 ♀ 29 45 75	Supplier of amino acids Energy source Daily requirement in mg per kg body weight	Essential amino acids: Val (14) Leu (16) Ile (12) Lys (12) Phe (16) Trp (3) Met (10) Thr (8) Cys and His stimulate growth
Carbohydrates	1	17 (4.1)	0 390 240-310	General source of energy (glucose) Energy reserve (glycogen) Roughage (cellulose) Supporting substances (bones, cartilage, mucus)	Non-essential nutritional constituent
Fats	10-15	39 (9.3)	10 80 130	General energy source Most important energy reserve Solvent for vitamins Supplier of essential fatty acids	Poly-unsaturated fatty acids: Linoleic acid Linolenic acid Arachidonic acid (together 10 g/day)
Water	35-40	0	2 400 - -	Solvent Cellular building block Dielectric Reaction partner Temperature regulator	
Minerals	3	0		Building blocks Electrolytes Cofactors of enzymes	Macrominerals Microminerals (trace elements)
Vitamins	-	-		Often precursors of coenzymes	Lipid-soluble vitamins Water-soluble vitamins
a: Minimum daily requirement			b: Recommended daily intake		c: Actual daily intake in industrialized nations

Minerals and trace elements

A. Minerals ●

Water is the most important essential inorganic nutrient in the diet. In adults, the body has a daily requirement of 2–3 L of water, which is supplied from drinks, water contained in solid foods, and from the *oxidation water* produced in the respiratory chain (see p. 140). The special role of water for living processes is discussed in more detail elsewhere (see p. 26).

The elements essential for life can be divided into **macroelements** (daily requirement > 100 mg) and **microelements** (daily requirement < 100 mg). The macroelements include the **electrolytes** sodium (Na), potassium (K), calcium (Ca), and magnesium (Mg), and the nonmetals chlorine (Cl), phosphorus (P), sulfur (S), and iodine (I).

The essential microelements are only required in trace amounts (see also p. 2). This group includes iron (Fe), zinc (Zn), manganese (Mn), copper (Cu), cobalt (Co), chromium (Cr), selenium (Se), and molybdenum (Mo). Fluorine (F) is not essential for life, but does promote healthy bones and teeth. It is still a matter of controversy whether vanadium, nickel, tin, boron, and silicon also belong to the essential trace elements.

The second column in the table lists the average **amounts** of mineral substances in the body of an adult weighing 65 kg. The **daily requirements** listed in the fourth column also apply to an adult, and are *average values*. Children, pregnant and breast-feeding women, and those who are ill generally have higher mineral requirements relative to body weight than men.

As the human body is able to store many minerals, deviations from the daily ration are balanced out over a given period of time. Minerals stored in the body include water, which is distributed throughout the whole body; calcium, stored in the form of apatite in the bones (see p. 340); iodine, stored as thyroglobulin in the thyroid; and iron, stored in the form of ferritin and hemosiderin in the bone marrow, spleen, and liver (see p. 286). The storage site for many trace elements is the liver. In many cases, the metabolism of minerals is regulated by *hormones*—for example, the uptake and excretion of H_2O , Na^+ ,

Ca^{2+} , and phosphate (see p. 328), and storage of Fe^{2+} and I^- .

Resorption of the required mineral substances from food usually depends on the body's requirements, and in several cases also on the composition of the diet. One example of dietary influence is calcium (see p. 342). Its resorption as Ca^{2+} is promoted by lactate and citrate, but phosphate, oxalic acid, and phytol inhibit calcium uptake from food due to complex formation and the production of insoluble salts.

Mineral deficiencies are not uncommon and can have quite a variety of causes—e.g., an unbalanced diet, resorption disturbances, and diseases. *Calcium deficiency* can lead to rickets, osteoporosis, and other disturbances. *Chloride deficiency* is observed as a result of severe Cl^- losses due to vomiting. Due to the low content of iodine in food in many regions of central Europe, *iodine deficiency* is widespread there and can lead to goiter. *Magnesium deficiency* can be caused by digestive disorders or an unbalanced diet—e.g., in alcoholism. Trace element deficiencies often result in a disturbed blood picture—i.e., forms of anemia.

The last column in the table lists some of the functions of minerals. It should be noted that almost all of the **macroelements** in the body function either as *nutrients* or *electrolytes*. Iodine (as a result of its incorporation into iodothyronines) and calcium act as *signaling substances*. Most **trace elements** are *cofactors for proteins*, especially for enzymes. Particularly important in quantitative terms are the *iron proteins* hemoglobin, myoglobin, and the cytochromes (see p. 286), as well as more than 300 different *zinc proteins*.

A. Minerals

Mineral	Content* (g)	Major source	Daily requirement (g)	Functions/Occurrence
Water	35 000-40 000	Drinks Water in solid foods From metabolism 300g	1200 900	Solvent, cellular building block, dielectric, coolant, medium for transport, reaction partner
Macroelements (daily requirement >100 mg)				
Na	100	Table salt	1.1-3.3	Osmoregulation, membrane potential, mineral metabolism
K	150	Vegetables, fruit, cereals	1.9-5.6	Membrane potential, mineral metabolism
Ca	1 300	Milk, milk products	0.8	Bone formation, blood clotting, signal molecule
Mg	20	Green vegetables	0.35	Bone formation, cofactor for enzymes
Cl	100	Table salt	1.7-5.1	Mineral metabolism
P	650	Meat, milk, cereals, vegetables	0.8	Bone formation, energy metabolism, nucleic acid metabolism
S	200	S-containing amino acids (Cys and Met)	0.2	Lipid and carbohydrate metabolism, conjugate formation
Microelements (trace elements)			(mg)	
Fe	4-5	Meat, liver, eggs, vegetables, potatoes, cereals	10	Hemoglobin, myoglobin, cytochromes, Fe/S clusters
Zn	2-3	Meat, liver, cereals	15	Zinc enzymes
Mn	0.02	Found in many foodstuffs	2-5	Enzymes
Cu	0.1-0.2	Meat, vegetables, fruit, fish	2-3	Oxidases
Co	<0.01	Meat	Traces	Vitamin B ₁₂
Cr	<0.01		0.05-0.2	Not clear
Mo	0.02	Cereals, nuts, legumes	0.15-0.5	Redox enzymes
Se		Vegetables, meat	0.05-0.2	Selenium enzymes
I	0.03	Seafood, iodized salt, drinking water	0.15	Thyroxin
Requirement not known				■ Metals ■ Non-metals
F		Drinking water (fluoridated), tea, milk	0.0015-0.004	Bones, dental enamel

* Content in the body of a 65 kg adult

Lipid-soluble vitamins

Vitamins are essential organic compounds that the animal organism is not capable of forming itself, although it requires them in small amounts for metabolism. Most vitamins are **precursors of coenzymes**; in some cases, they are also precursors of **hormones** or act as **antioxidants**. Vitamin requirements vary from species to species and are influenced by age, sex, and physiological conditions such as pregnancy, breast-feeding, physical exercise, and nutrition.

A. Vitamin supply ●

A healthy diet usually covers average daily vitamin requirements. By contrast, malnutrition, malnourishment (e.g., an unbalanced diet in older people, malnourishment in alcoholics, ready meals), or resorption disturbances lead to an inadequate supply of vitamins from which **hypovitaminosis**, or in extreme cases **avitaminosis**, can result. Medical treatments that kill the intestinal flora—e.g., antibiotics—can also lead to vitamin deficiencies (K, B₁₂, H) due to the absence of bacterial vitamin synthesis.

Since only a few vitamins can be stored (A, D, E, B₁₂), a lack of vitamins quickly leads to **deficiency diseases**. These often affect the skin, blood cells, and nervous system. The causes of vitamin deficiencies can be treated by improving nutrition and by administering vitamins in tablet form. An overdose of vitamins only leads to **hypervitaminoses**, with toxic symptoms, in the case of vitamins A and D. Normally, excess vitamins are rapidly excreted with the urine.

B. Lipid-soluble vitamins ●

Vitamins are classified as either lipid-soluble or water-soluble. The lipid-soluble vitamins include vitamins A, D, E, and K, all of which belong to the isoprenoids (see p. 52).

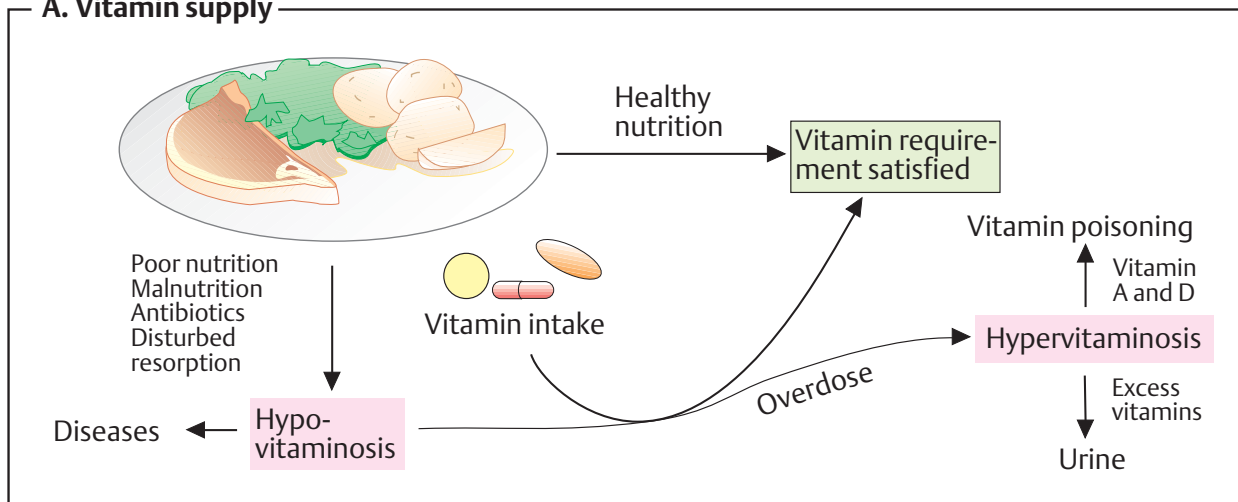
Vitamin A (retinol) is the parent substance of the *retinoids*, which include *retinal* and *retinoic acid*. The retinoids also can be synthesized by cleavage from the provitamin β -carotene. Retinoids are found in meat-containing diets, whereas β -carotene occurs in fruits and vegetables (particularly carrots). **Retinal** is involved in visual processes as the pigment of

the chromoprotein *rhodopsin* (see p. 358). **Retinoic acid**, like the steroid hormones, influences the transcription of genes in the cell nucleus. It acts as a differentiation factor in growth and development processes. Vitamin A deficiency can result in *night blindness*, *visual impairment*, and *growth disturbances*.

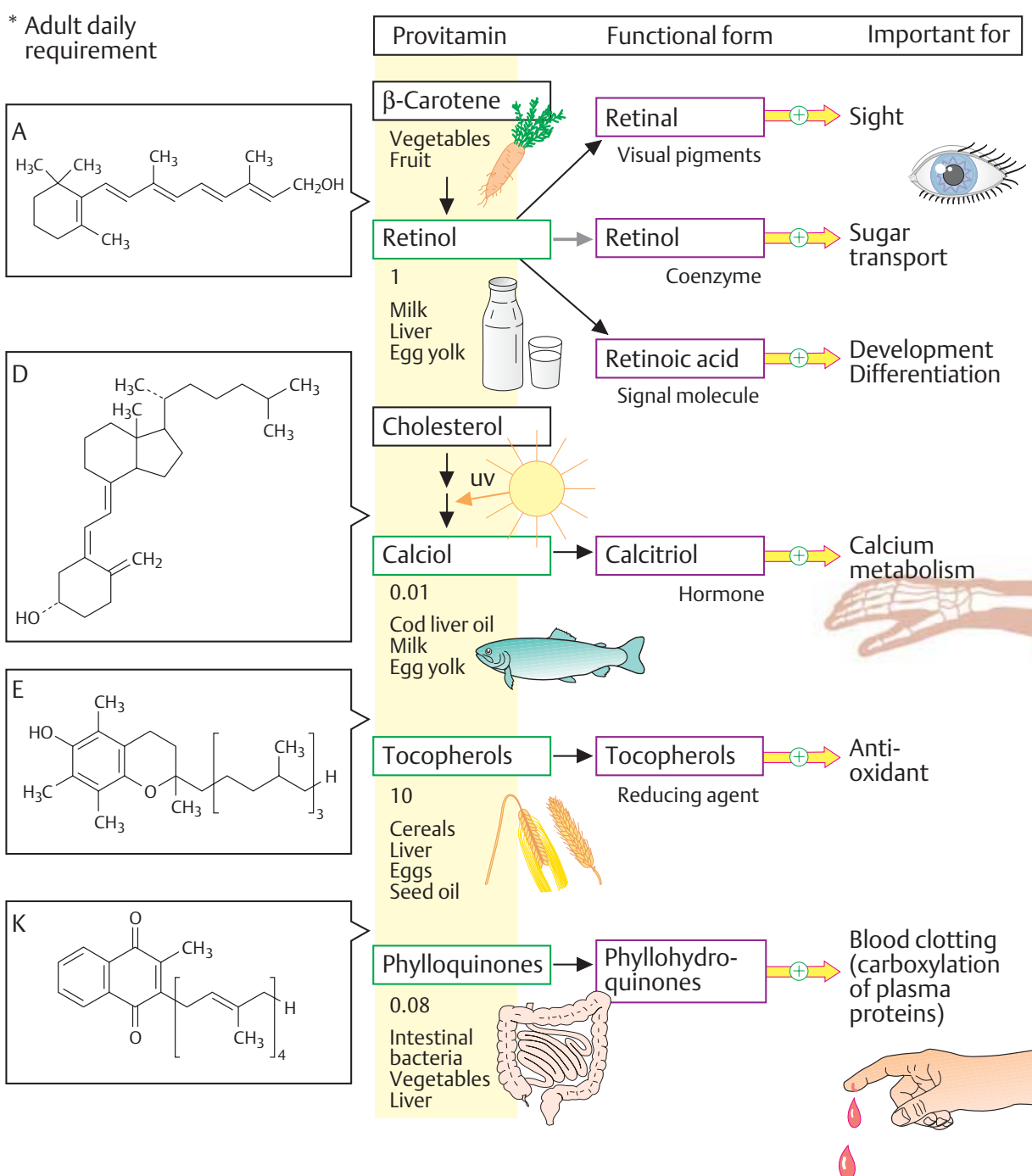
Vitamin D (calciol, cholecalciferol) is the precursor of the hormone *calcitriol* (1 α ,25-dihydroxycholecalciferol; see p. 320). Together with two other hormones (parathyrin and calcitonin), calcitriol regulates the calcium metabolism (see p. 342). Calciol can be synthesized in the skin from 7-dehydrocholesterol, an endogenous steroid, by a photochemical reaction. Vitamin D deficiencies only occur when the skin receives insufficient exposure to ultraviolet light and vitamin D is lacking in the diet. Deficiency is observed in the form of *rickets* in children and *osteomalacia* in adults. In both cases, bone mineralization is disturbed.

Vitamin E (tocopherol) and related compounds only occur in plants (e.g., wheat germ). They contain what is known as a *chroman ring*. In the lipid phase, vitamin E is mainly located in biological membranes, where as an *antioxidant* it protects unsaturated lipids against ROS (see p. 284) and other radicals.

Vitamin K (phylloquinone) and similar substances with modified side chains are involved in carboxylating glutamate residues of coagulation factors in the liver (see p. 290). The form that acts as a cofactor for carboxylase is derived from the vitamin by enzymatic reduction. Vitamin K antagonists (e.g., coumarin derivatives) inhibit this reduction and consequently carboxylation as well. This fact is used to inhibit blood coagulation in *prophylactic treatment against thrombosis*. Vitamin K deficiency occurs only rarely, as the vitamin is formed by bacteria of the intestinal flora.

A. Vitamin supply**B. Lipid-soluble vitamins**

* Adult daily requirement



Water-soluble vitamins I

The B group of vitamins covers water-soluble vitamins, all of which serve as precursors for coenzymes. Their numbering sequence is not continuous, as many substances that were originally regarded as vitamins were not later confirmed as having vitamin characteristics.

A. Water-soluble vitamins I

Vitamin B₁ (thiamine) contains two heterocyclic rings—a *pyrimidine ring* (a six-membered aromatic ring with two Ns) and a *thiazole ring* (a five-membered aromatic ring with N and S), which are joined by a methylene group. The active form of vitamin B₁ is **thiamine diphosphate** (TPP), which contributes as a coenzyme to the transfer of hydroxyalkyl residues (active aldehyde groups). The most important reactions of this type are oxidative decarboxylation of 2-oxoacids (see p. 134) and the transketolase reaction in the pentose phosphate pathway (see p. 152). Thiamine was the first vitamin to be discovered, around 100 years ago. Vitamin B₁ deficiency leads to *beriberi*, a disease with symptoms that include neurological disturbances, cardiac insufficiency, and muscular atrophy.

Vitamin B₂ is a complex of several vitamins: riboflavin, folate, nicotinate, and pantothenic acid.

Riboflavin (from the Latin *flavus*, yellow) serves in the metabolism as a component of the redox coenzymes flavin mononucleotide (FMN) and flavin adenine dinucleotide (FAD; see p. 104). As prosthetic groups, **FMN** and **FAD** are cofactors for various oxidoreductases (see p. 32). No specific disease due to a deficiency of this vitamin is known.

Folate, the anion of folic acid, is made up of three different components—a *pteridine derivative*, *4-aminobenzoate*, and one or more *glutamate* residues. After reduction to tetrahydrofolate (**THF**), folate serves as a coenzyme in the C₁ metabolism (see p. 418). Folate deficiency is relatively common, and leads to disturbances in nucleotide biosynthesis and thus cell proliferation. As the precursors for blood cells divide particularly rapidly, disturbances of the blood picture can occur, with increased amounts of abnormal precursors for megalocytes (*megaloblastic anemia*). Later, general damage ensues as phospholipid

synthesis and the amino acid metabolism are affected.

In contrast to animals, microorganisms are able to synthesize folate from their own components. The growth of microorganisms can therefore be inhibited by *sulfonamides*, which competitively inhibit the incorporation of 4-aminobenzoate into folate (see p. 254). Since folate is not synthesized in the animal organism, sulfonamides have no effect on animal metabolism.

Nicotinate and **nicotinamide**, together referred to as “niacin,” are required for biosynthesis of the coenzymes nicotinamide adenine dinucleotide (**NAD⁺**) and nicotinamide adenine dinucleotide phosphate (**NADP⁺**). These both serve in energy and nutrient metabolism as carriers of *hydride ions* (see pp. 32, 104). The animal organism is able to convert *tryptophan* into nicotinate, but only with a poor yield. Vitamin deficiency therefore only occurs when nicotinate, nicotinamide, and tryptophan are all simultaneously lacking in the diet. It manifests in the form of skin damage (*pellagra*), digestive disturbances, and depression.

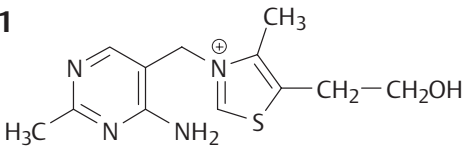
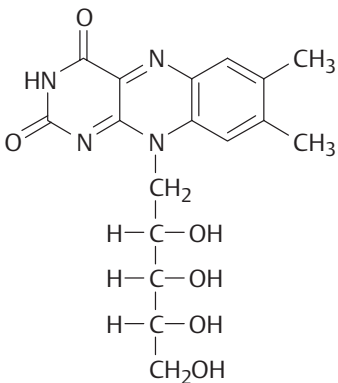
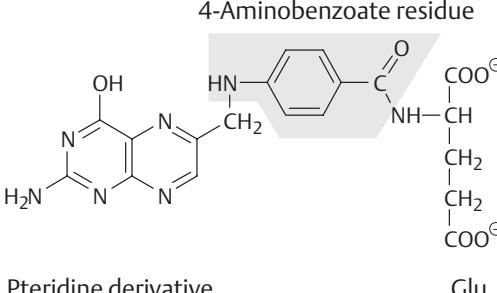
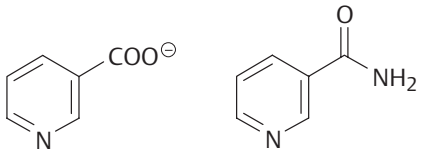
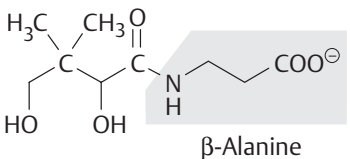
Pantothenic acid is an acid amide consisting of β -alanine and 2,4-dihydroxy-3,3'-dimethylbutyrate (pantoic acid). It is a precursor of *coenzyme A*, which is required for activation of acyl residues in the lipid metabolism (see pp. 12, 106). *Acyl carrier protein* (ACP; see p. 168) also contains pantothenic acid as part of its prosthetic group. Due to the widespread availability of pantothenic acid in food (Greek *pantothēn* = “from everywhere”), deficiency diseases are rare.

Further information

The requirement for vitamins in humans and other animals is the result of mutations in the enzymes involved in biosynthetic coenzymes. As intermediates of coenzyme biosynthesis are available in sufficient amounts in the diet of heterotrophic animals (see p. 112), the lack of endogenous synthesis did not have unfavorable effects for them. Microorganisms and plants whose nutrition is mainly autotrophic have to produce all of these compounds themselves in order to survive.

A. Water-soluble vitamins I

* Adult daily requirement

	Vitamin	Active form: coenzyme	Function in metabolism
B₁ 	Thiamine 1.5 mg * Grain Yeast products Pork	 TPP Thiamine diphosphate	Transfer of hydroxy-alkyl residues
B₂ 	Riboflavin 1.8 mg * Milk Eggs	 FMN FAD	Hydrogen transfer
 4-Aminobenzoate residue Pteridine derivative Glu	Folate 0.2 mg * Fresh green vegetables Liver	 THF Tetrahydrofolate	C ₁ -metabolism
 Nicotinate Nicotinamide	Nicotinate Nicotinamide 20 mg * (or 1.2 g tryptophan) Meat, yeast products Fruit and vegetables	 NADP ⁺ NAD ⁺	Hydride transfer
 Pantoate β-Alanine	Pantothenate 7 mg * Widely distributed	 CoA	Activation of carboxylic acids

Water-soluble vitamins II

A. Water-soluble vitamins II ●

Vitamin B₆ consists of three substituted pyridines—**pyridoxal**, **pyridoxol**, and **pyridoxamine**. The illustration shows the structure of pyridoxal, which carries an aldehyde group (–CHO) at C-4. Pyridoxol is the corresponding alcohol (–CH₂OH), and pyridoxamine the amine (–CH₂NH₂).

The active form of vitamin B₆, **pyridoxal phosphate**, is the most important coenzyme in the amino acid metabolism (see p. 106). Almost all conversion reactions involving amino acids require pyridoxal phosphate, including transaminations, decarboxylations, dehydrogenations, etc. *Glycogen phosphorylase*, the enzyme for glycogen degradation, also contains pyridoxal phosphate as a cofactor. Vitamin B₆ deficiency is rare.

Vitamin B₁₂ (cobalamin) is one of the most complex low-molecular-weight substances occurring in nature. The core of the molecule consists of a tetrapyrrol system (*corrin*), with cobalt as the central atom (see p. 108). The vitamin is exclusively synthesized by microorganisms. It is abundant in liver, meat, eggs, and milk, but not in plant products. As the intestinal flora synthesize vitamin B₁₂, strict vegetarians usually also have an adequate supply of the vitamin.

Cobalamin can only be resorbed in the small intestine when the gastric mucosa secretes what is known as *intrinsic factor*—a glycoprotein that binds cobalamin (the *extrinsic factor*) and thereby protects it from degradation. In the blood, the vitamin is bound to a special protein known as *transcobalamin*. The liver is able to store vitamin B₁₂ in amounts sufficient to last for several months. Vitamin B₁₂ deficiency is usually due to an absence of intrinsic factor and the resulting resorption disturbance. This leads to a disturbance in blood formation known as *pernicious anemia*.

In animal metabolism, derivatives of cobalamin are mainly involved in rearrangement reactions. For example, they act as coenzymes in the conversion of methylmalonyl-CoA to succinyl-CoA (see p. 166), and in the formation of methionine from homocysteine (see p. 418). In prokaryotes, cobalamin derivatives also play a part in the reduction of ribonucleotides.

Vitamin C is **L-ascorbic acid** (chemically: 2-oxogulonolactone). The two hydroxyl groups have acidic properties. By releasing a proton, ascorbic acid therefore turns into its anion, **ascorbate**. Humans, apes, and guinea pigs require vitamin C because they lack the enzyme *L-gulonolactone oxidase* (1.1.3.8), which catalyzes the final step in the conversion of glucose into ascorbate.

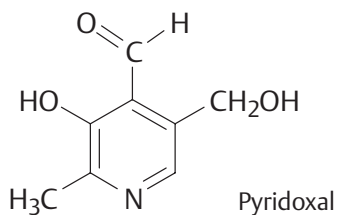
Vitamin C is particularly abundant in fresh fruit and vegetables. Many soft drinks and foodstuffs also have synthetic ascorbic acid added to them as an antioxidant and flavor enhancer. Boiling slowly destroys vitamin C. In the body, ascorbic acid serves as a reducing agent in various reactions (usually hydroxylations). Among the processes involved are *collagen synthesis*, *tyrosine degradation*, *catecholamine synthesis*, and *bile acid biosynthesis*. The daily requirement for ascorbic acid is about 60 mg, a comparatively large amount for a vitamin. Even higher doses of the vitamin have a protective effect against infections. However, the biochemical basis for this effect has not yet been explained. Vitamin C deficiency only occurs rarely nowadays; it becomes evident after a few months in the form of *scurvy*, with connective-tissue damage, bleeding, and tooth loss.

Vitamin H (biotin) is present in liver, egg yolk, and other foods; it is also synthesized by the intestinal flora. In the body, biotin is covalently attached via a lysine side chain to enzymes that catalyze carboxylation reactions. Biotin-dependent carboxylases include *pyruvate carboxylase* (see p. 154) and *acetyl-CoA carboxylase* (see p. 162). CO₂ binds, using up ATP, to one of the two N atoms of biotin, from which it is transferred to the acceptor (see p. 108).

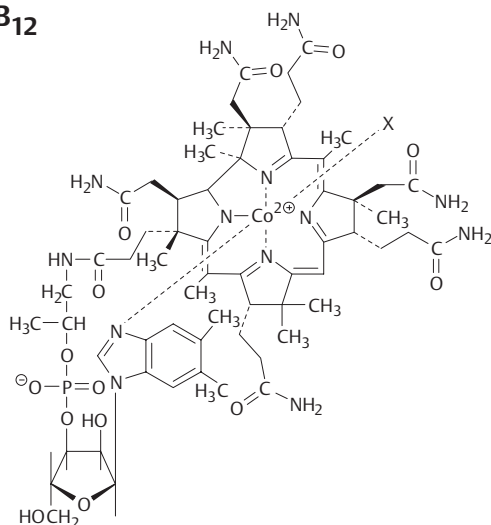
Biotin binds with high affinity ($K_d = 10^{-15}$ M) and specificity to *avidin*, a protein found in egg white. Since boiling denatures avidin, biotin deficiency only occurs when egg whites are eaten raw.

A. Water-soluble vitamins II

* Adult daily requirement

B₆

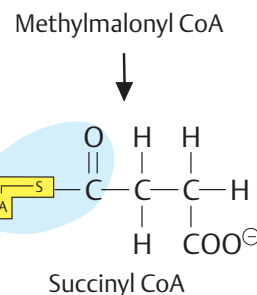
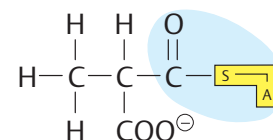
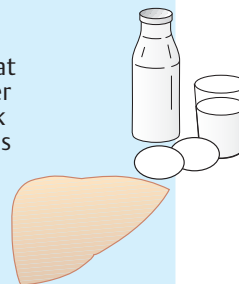
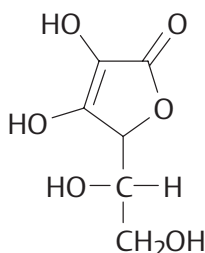
Vitamin

Active form:
coenzymeFunction in
metabolismPyridoxal
Pyridoxol
PyridoxaminePyridoxal
phosphateActivation
of amino
acids2 mg*
Meat
Vegetables
Grain products**B₁₂**

Cobalamin

5-Deoxy-
adenosyl
cobalaminIsomerization
e.g.

0.002 mg*

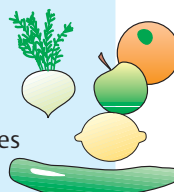
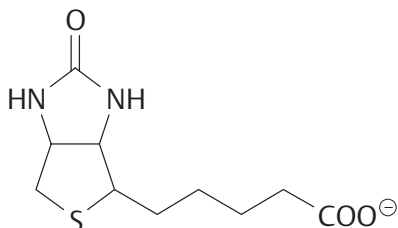
Meat
Liver
Milk
Eggs**C**

Ascorbic acid

Ascorbate

Stabilization
of enzyme
systems,
coenzyme,
antioxidant

60 mg*

Fruit
Vegetables**H**

Biotin



Biotin

Transfer
of carboxyl
groups

0.1 mg*

Yeast products
Legumes
Nuts

Basics

Hormones are *chemical signaling substances*. They are synthesized in specialized cells that are often associated to form *endocrine glands*. Hormones are released into the *blood* and transported with the blood to their *effector organs*. In the organs, the hormones carry out physiological and biochemical regulatory functions. In contrast to endocrine hormones, **tissue hormones** are only active in the immediate vicinity of the cells that secrete them.

The distinctions between hormones and other signaling substances (mediators, neurotransmitters, and growth factors) are fluid. **Mediators** is the term used for signaling substances that do not derive from special hormone-forming cells, but are formed by many cell types. They have hormone-like effects in their immediate surroundings. *Histamine* (see p. 352) and *prostaglandins* (see p. 390) are important examples of these substances. **Neurohormones** and **neurotransmitters** are signaling substances that are produced and released by nerve cells (see p. 348). **Growth factors** and **cytokines** mainly promote cell proliferation and cell differentiation (see p. 392).

A. Hormones: overview ●

The animal organism contains more than 100 hormones and hormone-like substances, which can be classified either according to their structure or according to their function. In chemical terms, most hormones are *amino acid derivatives*, *peptides* or *proteins*, or *steroids*. Hormones regulate the following processes:

- **Growth and differentiation of cells, tissues, and organs**

These processes include cell proliferation, embryonic development, and sexual differentiation—i.e., processes that require a prolonged time period and involve proteins de novo synthesis. For this reason, mainly steroid hormones which function via transcription regulation are active in this field (see p. 244).

- **Metabolic pathways**

Metabolic regulation requires rapidly acting mechanisms. Many of the hormones involved therefore regulate *interconversion* of enzymes (see p. 120). The main processes

subject to hormonal regulation are the uptake and degradation of storage substances (glycogen, fat), metabolic pathways for biosynthesis and degradation of central metabolites (glucose, fatty acids, etc.), and the supply of metabolic energy.

- **Digestive processes**

Digestive processes are usually regulated by locally acting peptides (paracrine; see p. 372), but mediators, biogenic amines, and neuropeptides are also involved (see p. 270).

- **Maintenance of ion concentrations (homeostasis)**

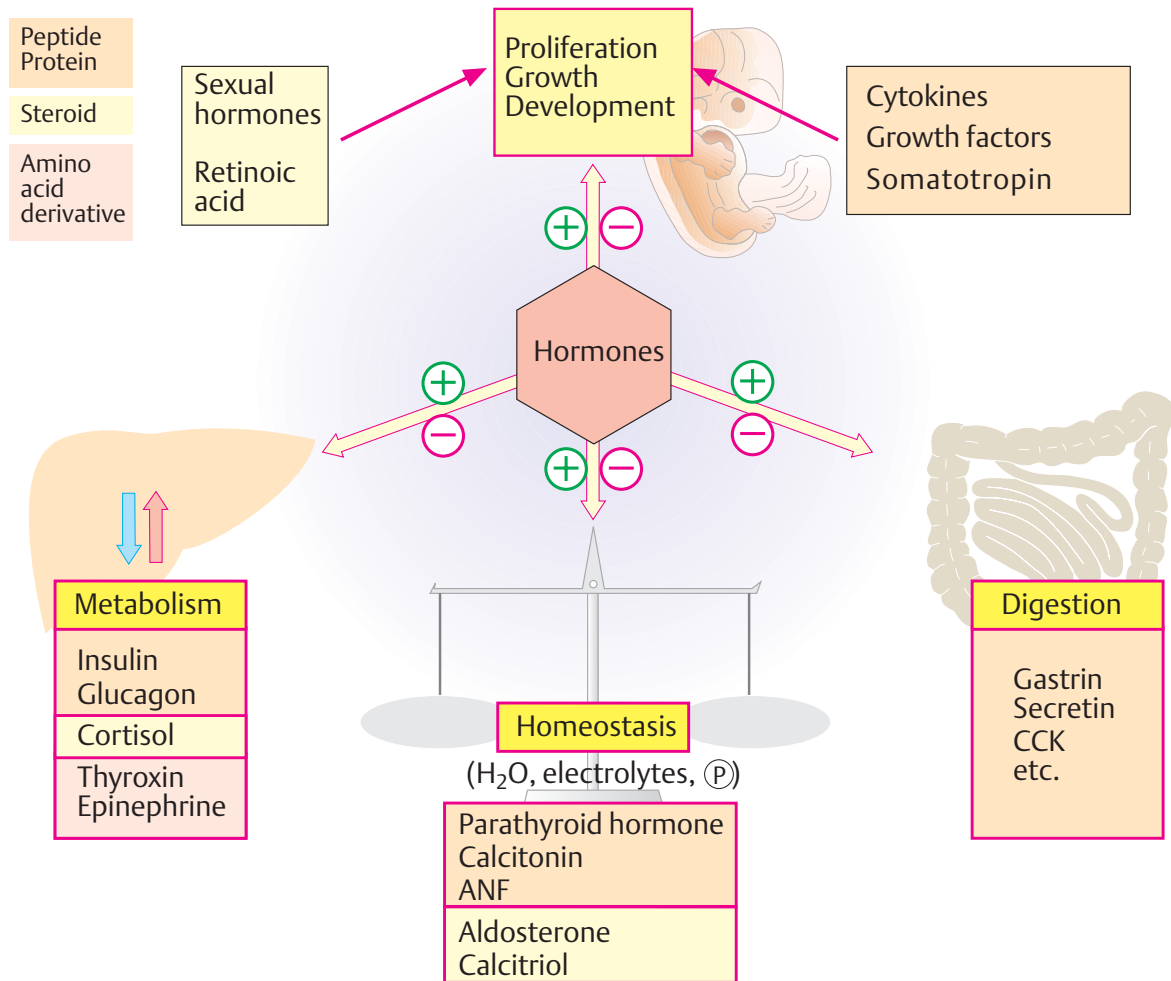
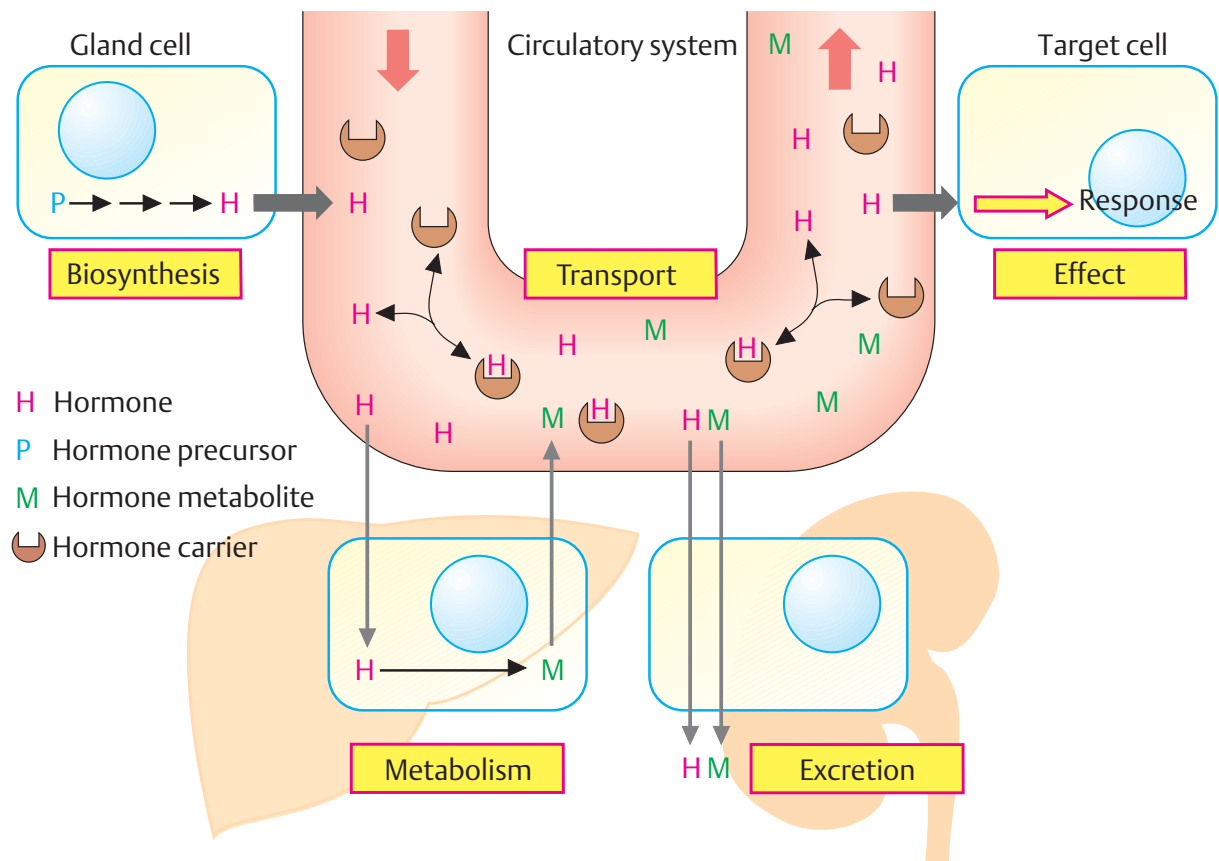
Concentrations of Na^+ , K^+ , and Cl^- in body fluids, and the physiological variables dependent on these (e.g. blood pressure), are subject to strict regulation. The principal site of action of the hormones involved is the kidneys, where hormones increase or reduce the resorption of ions and recovery of water (see pp. 326–331). The concentrations of Ca^{2+} and phosphate, which form the mineral substance of bone and teeth, are also precisely regulated.

Many hormones influence the above processes only indirectly by regulating the synthesis and release of other hormones (*hormonal hierarchy*; see p. 372).

B. Hormonal regulation system ●

Each hormone is the center of a hormonal regulation system. Specialized glandular cells synthesize the hormone from precursors, store it in many cases, and release it into the bloodstream when needed (**biosynthesis**). For **transport**, the poorly water-soluble lipophilic hormones are bound to plasma proteins known as hormone carriers. To stop the effects of the hormone again, it is inactivated by enzymatic reactions, most of which take place in the liver (**metabolism**). Finally, the hormone and its metabolites are expelled via the excretory system, usually in the kidney (**excretion**). All of these processes affect the concentration of the hormone and thus contribute to regulation of the hormonal signal.

In the effector organs, target cells receive the hormone's message. These cells have hormone receptors for the purpose, which bind the hormone. Binding of a hormone passes information to the cell and triggers a response (**effect**).

A. Hormones: overview**A. Hormonal regulation system**

Plasma levels and hormone hierarchy

A. Endocrine, paracrine, and autocrine hormone effects ●

Hormones transfer signals by migrating from their site of synthesis to their site of action. They are usually transported in the blood. In this case, they are said to have an **endocrine effect** (1; example: *insulin*). By contrast, *tissue hormones*, the target cells for which are in the immediate vicinity of the glandular cells that produce them, are said to have a **paracrine effect** (2; example: *gastrointestinal tract hormones*). When signal substances also pass effects back to the cells that synthesize them, they are said to have an **autocrine effect** (3; example: *prostaglandins*). Autocrine effects are often found in tumor cells (see p. 400), which stimulate their own proliferation in this way.

Insulin, which is formed in the B cells of the pancreas, has both endocrine and paracrine effects. As a hormone with endocrine effects, it regulates glucose and fat metabolism. Via a paracrine mechanism, it inhibits the synthesis and release of *glucagon* from the neighboring A cells.

B. Dynamics of the plasma level ●

Hormones circulate as signaling substances in the blood at very low concentrations (10^{-12} to between 10^{-7} mol L⁻¹). These values change periodically in rhythms that depend on the time of day, month, or year, or on physiological cycles.

The first example shows the **circadian** rhythm of the cortisol level. As an activator of gluconeogenesis (see p. 158), cortisol is mainly released in the early morning, when the liver's glycogen stores are declining. During the day, the plasma cortisol level declines.

Many hormones are released into the blood in a spasmodic and irregular manner. In this case, their concentrations change in an **episodic** or **pulsatile** fashion. This applies, for instance, to luteinizing hormone (LH, lutropin).

Concentrations of other hormones are **event-regulated**. For example, the body responds to increased blood sugar levels after meals by releasing *insulin*. Regulation of hor-

mone synthesis, release, and degradation allows the blood concentrations of hormones to be precisely adjusted. This is based either on simple feedback control or on hierarchically structured regulatory systems.

C. Closed-loop feedback control ●

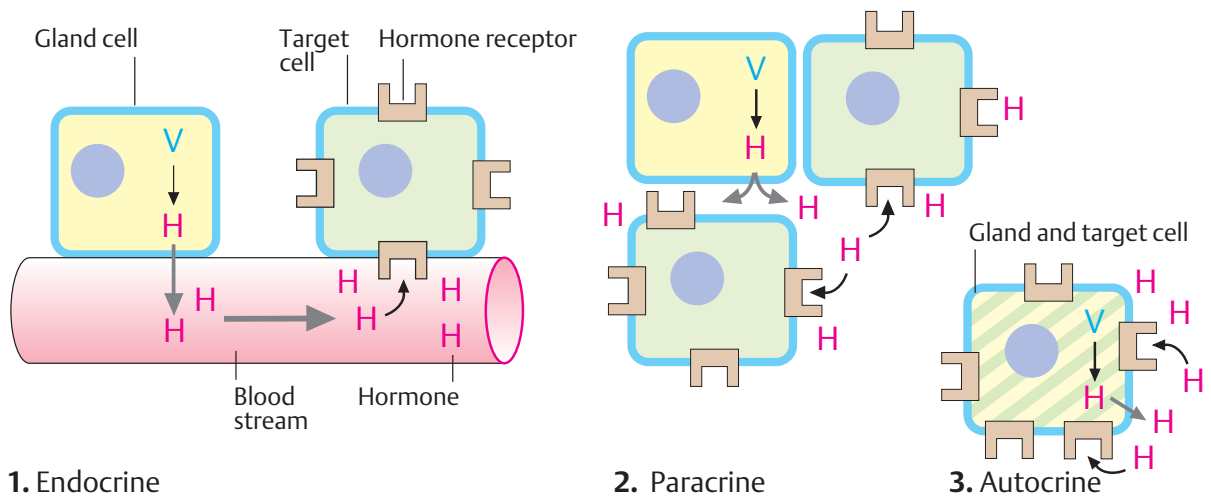
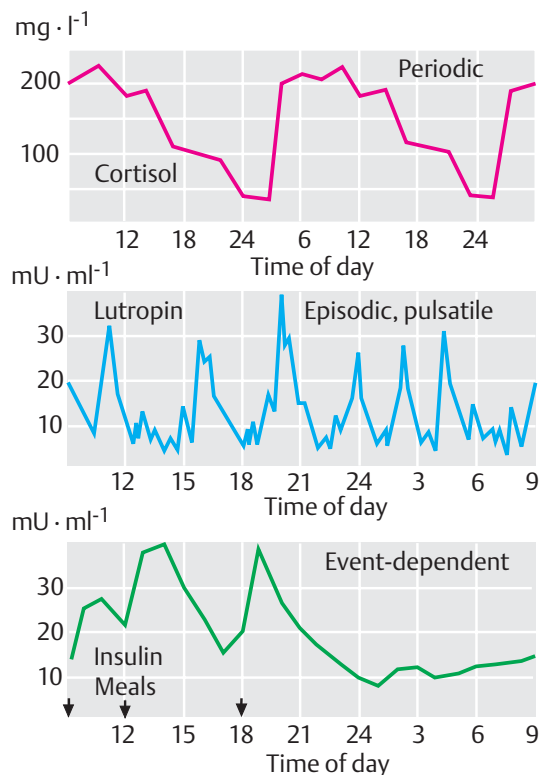
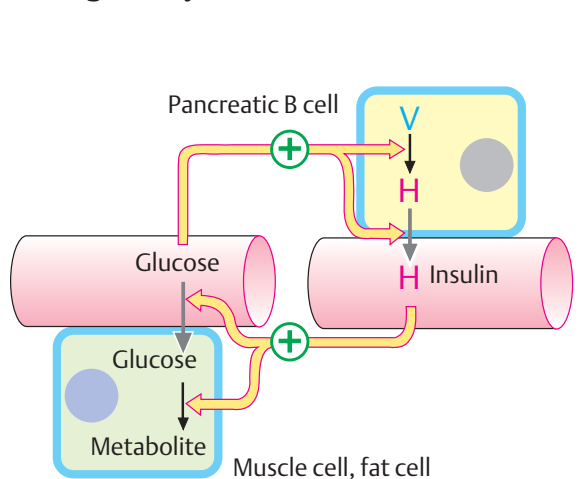
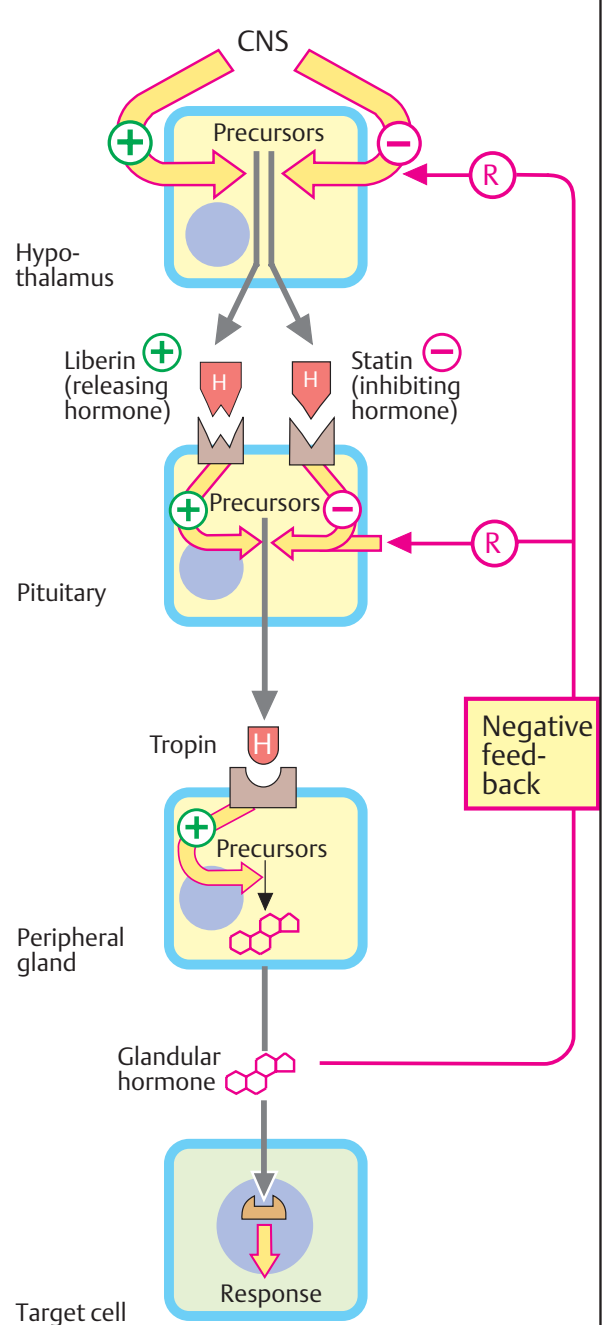
The biosynthesis and release of *insulin* by the pancreatic B cells (see p. 160) is stimulated by high blood glucose levels (> 5 mM). The insulin released then stimulates increased uptake and utilization of glucose by the cells of the muscle and adipose tissues. As a result, the blood glucose level falls back to its normal value, and further release of insulin stops.

D. Hormone hierarchy ●

Hormone systems are often linked to each another, giving rise in some cases to a hierarchy of higher-order and lower-order hormones. A particularly important example is the *pituitary-hypothalamic axis*, which is controlled by the central nervous system (CNS).

Nerve cells in the hypothalamus react to stimulatory or inhibitory signals from the CNS by releasing activating or inhibiting factors, which are known as **liberins** ("releasing hormones") and **statins** ("inhibiting hormones"). These neurohormones reach the adenohypophysis by short routes through the bloodstream. In the adenohypophysis, they stimulate (liberins) or inhibit (statins) the biosynthesis and release of tropines. **Tropines** (*glandotropic hormones*) in turn stimulate peripheral glands to synthesize glandular hormones. Finally, the **glandular hormone** acts on its target cells in the organism. In addition, it passes effects back to the higher-order hormone systems. This (usually negative) feedback influences the concentrations of the higher-order hormones, creating a feedback loop.

Many steroid hormones are regulated by this type of axis—e.g., thyroxine, cortisol, estradiol, progesterone, and testosterone. In the case of the glucocorticoids, the hypothalamus releases corticotropin-releasing hormone (CRH or corticoliberin, a peptide consisting of 41 amino acids), which in turn releases corticotropin (ACTH, 39 AAs) in the pituitary gland. Corticotropin stimulates synthesis and release of the glandular steroid hormone cortisol in the adrenal cortex.

A. Endocrine, paracrine and autocrine hormone effects**B. Plasma level dynamics****C. Regulatory circuit****D. Hormone hierarchy**

Lipophilic hormones

Classifying hormones into hydrophilic and lipophilic molecules indicates the chemical properties of the two groups of hormones and also reflects differences in their mode of action (see p. 120).

A. Lipophilic hormones ●

Lipophilic hormones, which include *steroid hormones*, *iodothyronines*, and *retinoic acid*, are relatively small molecules (300–800 Da) that are poorly soluble in aqueous media. With the exception of the iodothyronines, they are not stored by hormone-forming cells, but are released immediately after being synthesized. During transport in the blood, they are bound to specific carriers. Via intracellular receptors, they mainly act on transcription (see p. 358). Other effects of steroid hormones—e.g., on the immune system—are not based on transcriptional control. Their details have not yet been explained.

Steroid hormones

The most important steroid hormones in vertebrates are listed on p. 57. *Calcitriol* (vitamin D hormone) is also included in this group, although it has a modified steroid structure. The most important steroid hormone in invertebrates is *ecdysone*.

Progesterone is a female sexual steroid belonging to the progestin (*gestagen*) family. It is synthesized in the corpus luteum of the ovaries. The blood level of progesterone varies with the menstrual cycle. The hormone prepares the uterus for a possible pregnancy. Following fertilization, the placenta also starts to synthesize progesterone in order to maintain the pregnant state. The development of the mammary glands is also stimulated by progesterone.

Estradiol is the most important of the *estrogens*. Like progesterone, it is synthesized by the ovaries and, during pregnancy, by the placenta as well. Estradiol controls the menstrual cycle. It promotes proliferation of the uterine mucosa, and is also responsible for the development of the female secondary sexual characteristics (breast, fat distribution, etc.).

Testosterone is the most important of the male sexual steroids (*androgens*). It is synthesized in the Leydig interstitial cells of the testes, and controls the development and functioning of the male gonads. It also determines secondary sexual characteristics in men (muscles, hair, etc.).

Cortisol, the most important *glucocorticoid*, is synthesized by the adrenal cortex. It is involved in regulating protein and carbohydrate metabolism by promoting protein degradation and the conversion of amino acids into glucose. As a result, the blood glucose level rises (see p. 152). Synthetic glucocorticoids (e.g., dexamethasone) are used in drugs due to their anti-inflammatory and immunosuppressant effects.

Aldosterone, a *mineralocorticoid*, is also synthesized in the adrenal gland. In the kidneys, it promotes Na^+ resorption by inducing Na^+/K^+ ATPase and Na^+ channels (see p. 328). At the same time, it leads to increased K^+ excretion. In this way, aldosterone indirectly increases blood pressure.

Calcitriol is a derivative of vitamin D (see p. 364). On exposure to ultraviolet light, a precursor of the hormone can also arise in the skin. Calcitriol itself is synthesized in the kidneys (see p. 330). Calcitriol promotes the resorption of calcium in the intestine and increases the Ca^{2+} level in the blood.

Iodothyronines

The thyroid hormone **thyroxine** (tetraiodothyronine, T_4) and its active form **triiodothyronine** (T_3) are derived from the amino acid *tyrosine*. The iodine atoms at positions 3 and 5 of the two phenol rings are characteristic of them. Post-translational synthesis of thyroxine takes place in the thyroid gland from tyrosine residues of the protein *thyroglobulin*, from which it is proteolytically cleaved before being released. Iodothyronines are the only organic molecules in the animal organism that contain iodine. They increase the basal metabolic rate, partly by regulating mitochondrial ATP synthesis. In addition, they promote embryonic development.

A. Lipophilic hormones

Hormone	Site of formation	Sites of action	Actions
Progesterone	Ovaries	Uterus	Prepares uterus for pregnancy Promotes implantation of fertilized egg
Estradiol	Ovaries	Uterus and other organs	Stimulates proliferation of endometrium
Testosterone	Testes		Causes: Sexual differentiation to male phenotype Formation of ejaculate Spermatogenesis
Cortisol	Adrenal glands (cortex)	Proteins ↔ Amino acids → Glucose Immune system	Proteolysis ↑ Protein synthesis ↓ Gluconeogenesis ↑ Blut-Glucose ↑ Activity of the immune system ↓
Aldosterone	Adrenal glands (cortex)	Kidneys	ATP → ADP+P _i 3Na ⁺ / 2K ⁺ exchange
Calcitriol	Kidneys	Gut, Bones	Ca ²⁺ resorption ↑ Ca ²⁺ metabolism of bones ↑
Thyroxine	Thyroid gland	Embryo, Intermediary metabolism	O ₂ → H ₂ O, CO ₂ S → ATP, Heat Basal metabolic rate ↑ Heat generation ↑ O ₂ consumption ↑

Metabolism of steroid hormones

A. Biosynthesis of steroid hormones ○

All steroid hormones are synthesized from **cholesterol**. The gonane core of cholesterol consists of 19 carbon atoms in four rings (A–D). The D ring carries a side chain of eight C atoms (see p. 54).

The cholesterol required for biosynthesis of the steroid hormones is obtained from various sources. It is either taken up as a constituent of LDL lipoproteins (see p. 278) into the hormone-synthesizing glandular cells, or synthesized by glandular cells themselves from acetyl-CoA (see p. 172). Excess cholesterol is stored in the form of fatty acid esters in lipid droplets. Hydrolysis allows rapid mobilization of the cholesterol from this reserve again.

Biosynthetic pathways. Only an overview of the synthesis pathways that lead to the individual hormones is shown here. Further details are given on p. 410.

Among the reactions involved, hydroxylations (H) are particularly numerous. These are catalyzed by specific *monooxygenases* (“hydroxylases”) of the cytochrome P450 family (see p. 318). In addition, there are NADPH-dependent and NADP⁺-dependent *hydrogenations* (Y) and *dehydrogenations* (D), as well as *cleavage* and *isomerization reactions* (S, I). The estrogens have a special place among the steroid hormones, as they are the only ones that contain an aromatic A ring. When this is formed, catalyzed by *aromatase*, the angular methyl group (C-19) is lost.

Pregnenolone is an important intermediate in the biosynthesis of most steroid hormones. It is identical to cholesterol with the exception of a shortened and oxidized side chain. Pregnenolone is produced in three steps by hydroxylation and cleavage in the side chain. Subsequent dehydrogenation of the hydroxyl group at C-3 (b) and shifting of the double bond from C-5 to C-4 results in the gestagen **progesterone**.

With the exception of calcitriol, all steroid hormones are derived from progesterone. Hydroxylations of progesterone at C atoms 17, 21, and 11 lead to the glucocorticoid **cortisol**. Hydroxylation at C-17 is omitted during synthesis of the mineralocorticoid **aldosterone**. Instead, the angular methyl group (C-18) is oxidized to the aldehyde group. During syn-

thesis of the androgen **testosterone** from progesterone, the side chain is completely removed. Aromatization of the A ring, as mentioned above, finally leads to **estradiol**.

On the way to **calcitriol** (vitamin D hormone; see p. 342), another double bond in the B ring of cholesterol is first introduced. Under the influence of UV light on the skin, the B ring is then photochemically cleaved, and the secosteroid *cholecalciferol* arises (vitamin D₃; see p. 364). Two Cyt P450-dependent hydroxylations in the liver and kidneys produce the active vitamin D hormone (see p. 330).

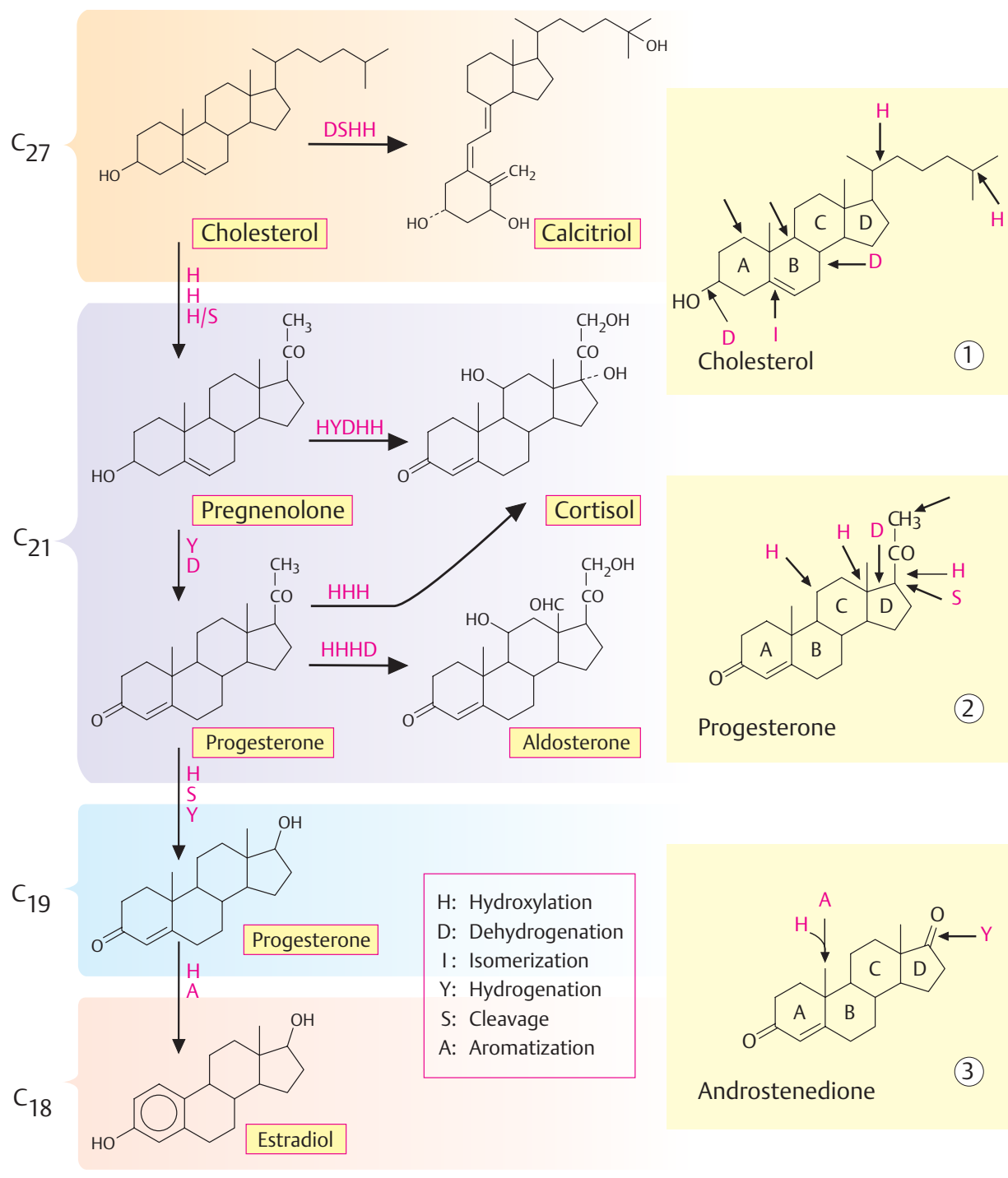
B. Inactivation of steroid hormones ○

The steroid hormones are mainly inactivated in the liver, where they are either reduced or further hydroxylated and then conjugated with *glucuronic acid* or *sulfate* for excretion (see p. 316). The *reduction reactions* attack oxo groups and the double bond in ring A. A combination of several inactivation reactions gives rise to many different steroid metabolites that have lost most of their hormonal activity. Finally, they are excreted with the *urine* and also partly via the *bile*. Evidence of steroids and steroid metabolites in the urine is used to investigate the hormone metabolism.

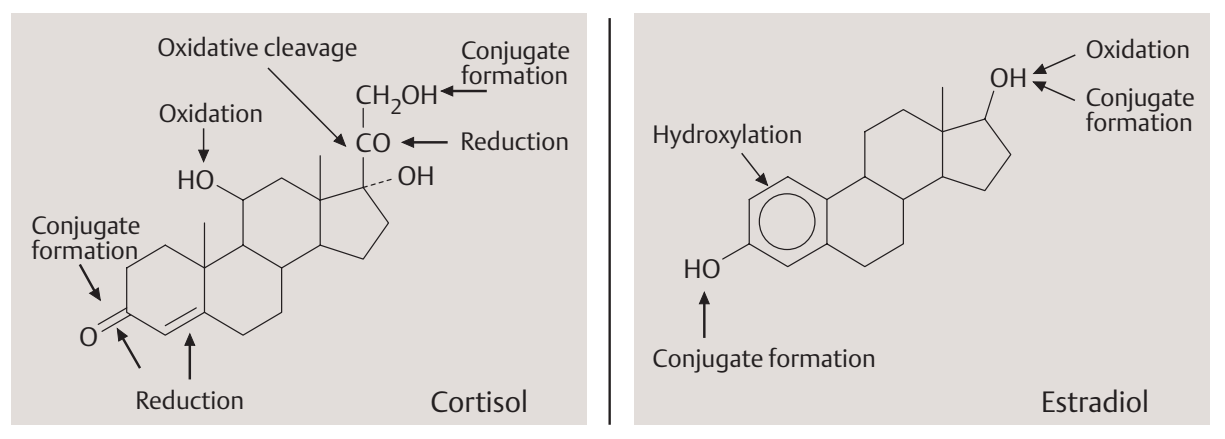
Further information

Congenital defects in the biosynthesis of steroid hormones can lead to severe developmental disturbances. In the *adrenogenital syndrome* (AGS), which is relatively common, there is usually a defect in *21-hydroxylase*, which is needed for synthesis of cortisol and aldosterone from progesterone. Reduced synthesis of this hormone leads to increased formation of testosterone, resulting in masculinization of female fetuses. With early diagnosis, this condition can be avoided by providing the mother with hormone treatment before birth.

A. Biosynthesis of steroid hormones



B. Inactivation of steroid hormones



Mechanism of action

A. Mechanism of action of lipophilic hormones ○

Lipophilic signaling substances include the *steroid hormones*, *calcitriol*, the *iodothyronines* (T_3 and T_4), and *retinoic acid*. These hormones mainly act in the *nucleus* of the target cells, where they regulate gene transcription in collaboration with their receptors and with the support of additional proteins (known as coactivators and mediators; see p. 244). There are several effects of steroid hormones that are not mediated by transcription control. These alternative pathways for steroid effects have not yet been fully explained.

In the blood, there are a number of transport proteins for lipophilic hormones (see p. 276). Only the free hormone is able to penetrate the membrane and enter the cell. The hormone encounters its receptor in the nucleus (and sometimes also in the cytoplasm).

The **receptors** for lipophilic hormones are rare proteins. They occur in small numbers (10^3 – 10^4 molecules per cell) and show marked *specificity* and high *affinity* for the hormone ($K_d = 10^{-8}$ – 10^{-10} M). After binding to the hormone, the steroid receptors are able to bind as homodimers or heterodimers to *control elements* in the promoters of specific genes, from where they can influence the transcription of the affected genes—i.e., they act as *transcription factors*.

The illustration shows the particularly well-investigated mechanism of action for **cortisol**, which is unusual to the extent that the hormone–receptor complex already arises in the cytoplasm. The free receptor is present in the cytoplasm as a monomer in complex with the chaperone **hsp90** (see p. 232). Binding of cortisol to the complex leads to an *allosteric conformational change* in the receptor, which is then released from the hsp90 and becomes capable of DNA binding as a result of *dimerization*.

In the nucleus, the hormone–receptor complex binds to nucleotide sequences known as **hormone response elements** (HREs). These are short palindromic DNA segments that usually promote transcription as enhancer elements (see p. 244). The illustration shows the HRE for glucocorticoids (GRE;

“n” stands for any nucleotide). Each hormone receptor only recognizes its “own” HRE and therefore only influences the transcription of genes containing that HRE. Recognition between the receptor and HRE is based on interaction between the amino acid residues in the DNA-binding domain (**B**) and the relevant bases in the HRE (emphasized in color in the structure illustrated).

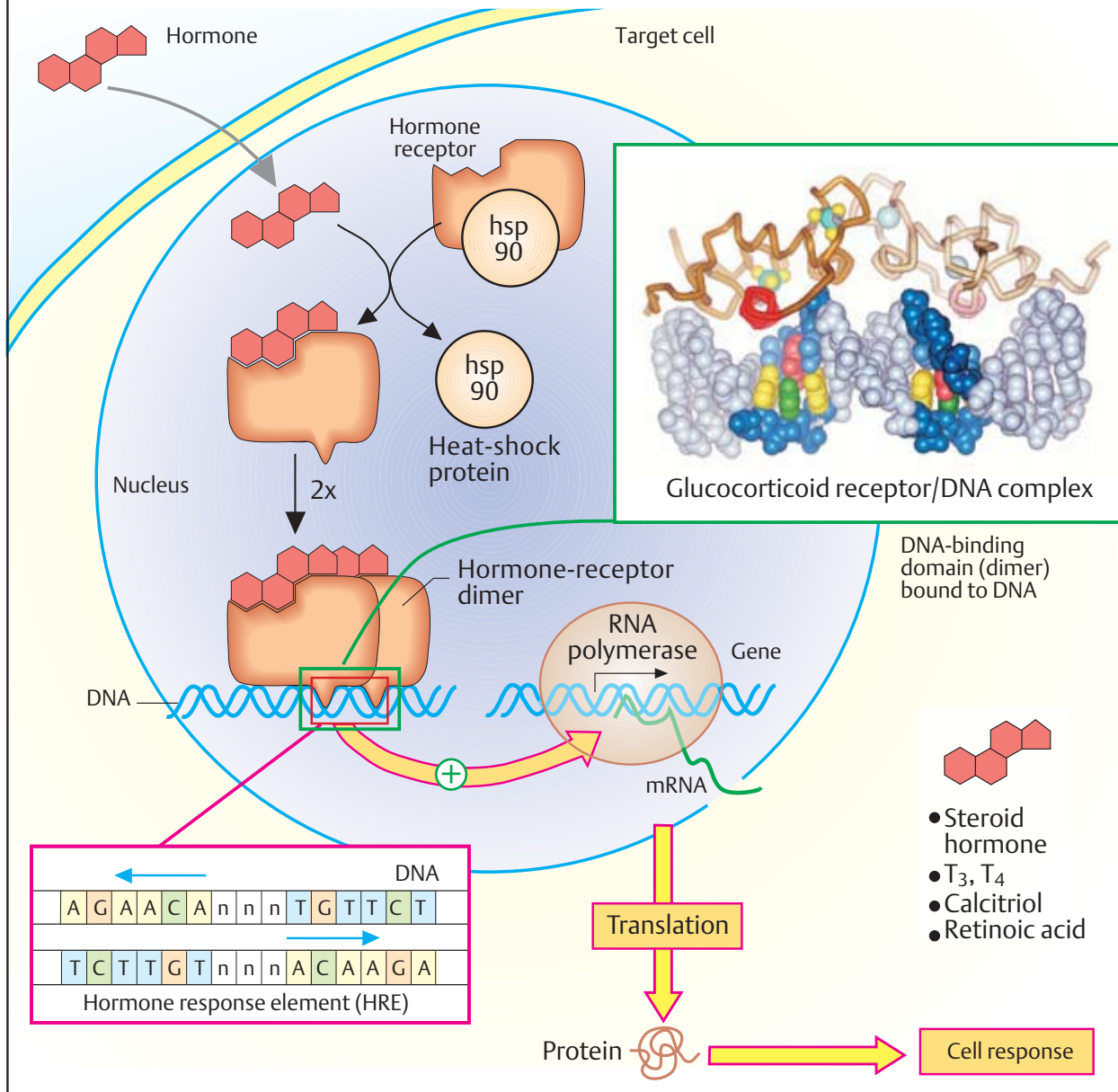
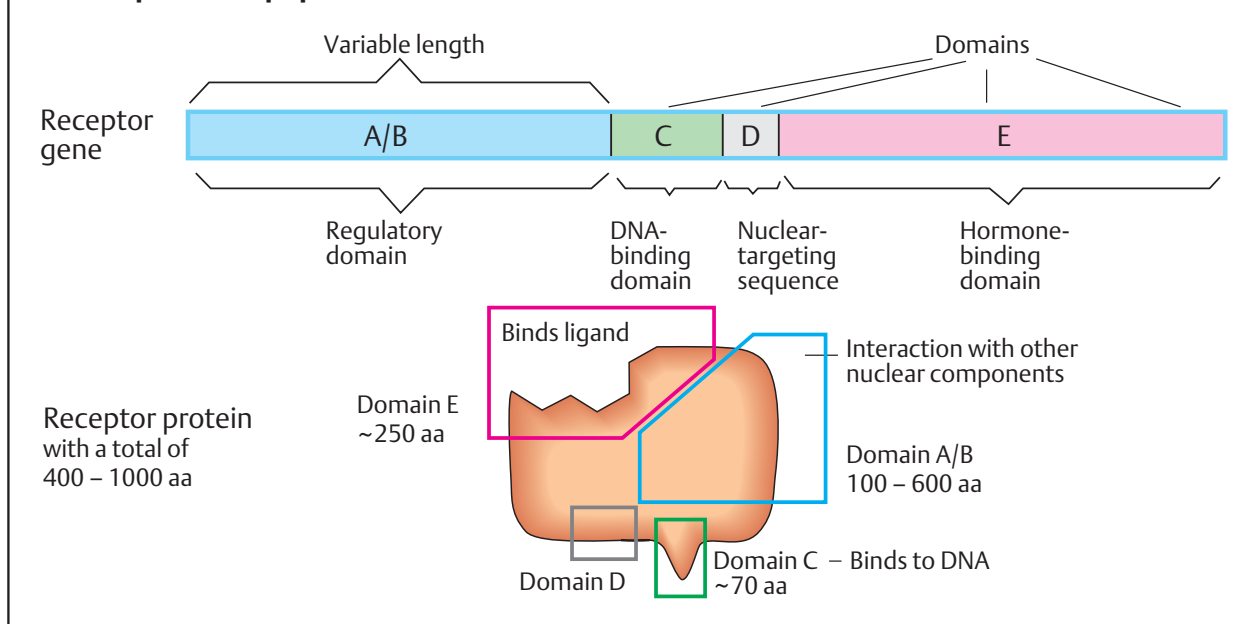
As discussed on p. 244, the hormone receptor does not interact directly with the RNA polymerase, but rather—along with other transcription factors—with a coactivator/mediator complex that processes all of the signals and passes them on to the polymerase. In this way, hormonal effects lead within a period of minutes to hours to altered levels of mRNAs for key proteins in cellular processes (“cellular response”).

B. Steroid receptors ○

The receptors for lipophilic signaling substances all belong to one *protein superfamily*. They are constructed in a modular fashion from **domains** with various lengths and functions. Starting from the N terminal, these are: the *regulatory domain*, the *DNA-binding domain*, a *nuclear localization sequence* (see p. 228), and the *hormone-binding domain* (see p. 73D).

The homology among receptors is particularly great in the area of the DNA-binding domain. The proteins have cysteine-rich sequences here that coordinatively bind zinc ions (**A**, Cys shown in yellow, Zn^{2+} in light blue). These centers, known as “zinc fingers” or “**zinc clusters**,” stabilize the domains and support their dimerization, but do not take part in DNA binding directly. As in other transcription factors (see p. 118), “recognition helices” are responsible for that.

In addition to the receptors mentioned in **A**, the family of steroid receptors also includes the product of the oncogene *erb-A* (see p. 398), the receptor for the environmental toxin *dioxin*, and other proteins for which a distinct hormone ligand has not been identified (known as “orphan receptors”). Several steroid receptors—e.g., the retinoic acid receptor—form functional heterodimers with orphan receptors.

A. Mechanism of action of lipophilic hormones**B. Receptors of lipophilic hormones**

Hydrophilic hormones

The hydrophilic hormones are derived from amino acids, or are peptides and proteins composed of amino acids. Hormones with endocrine effects are synthesized in glandular cells and stored there in vesicles until they are released. As they are easily soluble, they do not need carrier proteins for transport in the blood. They bind on the plasma membrane of the target cells to receptors that pass the hormonal signal on (signal transduction; see p. 384). Several hormones in this group have paracrine effects—i.e., they only act in the immediate vicinity of their site of synthesis (see p. 372).

A. Signaling substances derived from amino acids ❶

Histamine, serotonin, melatonin, and the catecholamines dopa, dopamine, norepinephrine, and epinephrine are known as “*biogenic amines*.” They are produced from amino acids by decarboxylation and usually act not only as hormones, but also as neurotransmitters.

Histamine, an important *mediator* (local signaling substance) and *neurotransmitter*, is mainly stored in tissue mast cells and basophilic granulocytes in the blood. It is involved in inflammatory and allergic reactions. “Histamine liberators” such as tissue hormones, type E immunoglobulins (see p. 300), and drugs can release it. Histamine acts via various types of receptor. Binding to H_1 receptors promotes contraction of smooth muscle in the bronchia, and dilates the capillary vessels and increases their permeability. Via H_2 receptors, histamine slows down the heart rate and promotes the formation of HCl in the gastric mucosa. In the brain, histamine acts as a neurotransmitter.

Epinephrine is a hormone synthesized in the adrenal glands from tyrosine (see p. 352). Its release is subject to neuronal control. This “emergency hormone” mainly acts on the blood vessels, heart, and metabolism. It constricts the blood vessels and thereby increases blood pressure (via α_1 and α_2 receptors); it increases cardiac function (via β_2 receptors); it promotes the degradation of glycogen into glucose in the liver and muscles (via β_2 receptors); and it dilates the bronchia (also via β_2 receptors).

B. Examples of peptide hormones and proteohormones ❷

Numerically the largest group of signaling substances, these arise by protein biosynthesis (see p. 382). The smallest peptide hormone, thyroliberin (362 Da), is a tripeptide. Proteohormones can reach masses of more than 20 kDa—e.g., thyrotropin (28 kDa). Similarities in the primary structures of many peptide hormones and proteohormones show that they are related to one another. They probably arose from common predecessors in the course of evolution.

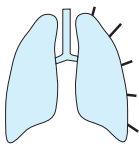
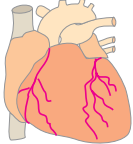
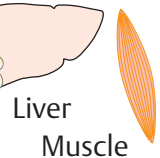
Thyroliberin (thyrotropin-releasing hormone, TRH) is one of the neurohormones of the hypothalamus (see p. 330). It stimulates pituitary gland cells to secrete thyrotropin (TSH). TRH consists of three amino acids, which are modified in characteristic ways (see p. 353).

Thyrotropin (thyroid-stimulating hormone, TSH) and the related hormones *lutropin* (luteinizing hormone, LH) and *folitropin* (follicle-stimulating hormone, FSH) originate in the adenohypophysis. They are all dimeric glycoproteins with masses of around 28 kDa. Thyrotropin stimulates the synthesis and secretion of thyroxin by the thyroid gland.

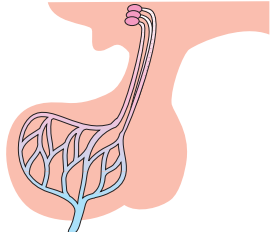
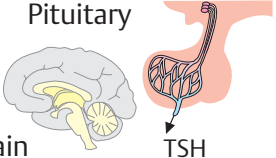
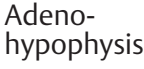
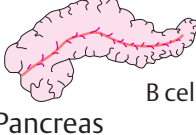
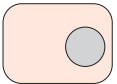
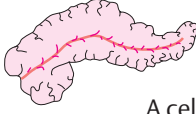
Insulin (for the structure, see p. 70) is produced and released by the B cells of the pancreas and is released when the glucose level rises. Insulin reduces the blood sugar level by promoting processes that consume glucose—e.g., glycolysis, glycogen synthesis, and conversion of glucose into fatty acids. By contrast, it inhibits gluconeogenesis and glycogen degradation. The transmission of the insulin signal in the target cells is discussed in greater detail on p. 388.

Glucagon, a peptide of 29 amino acids, is a product of the A cells of the pancreas. It is the antagonist of insulin and, like insulin, mainly influences carbohydrate and lipid metabolism. Its effects are each opposite to those of insulin. Glucagon mainly acts via the second messenger cAMP (see p. 384).

A. Signaling substances derived from amino acids

Hormone	Sites of formation	Sites of action	Actions
<chem>[NH3+]CCc1c[nH]cn1</chem> Histamine	 Mast cell  Basophilic granulocyte	Lungs  Stomach 	Width of bronchi ↓ Capillaries: width ↑ permeability ↑ Gastric acid secretion by parietal cells ↑
<chem>CC(C1=CC(=C(C=C1)O)O)C[C@H](O)[NH3+]</chem> Epinephrine	Adrenal glands (medulla) 	Heart  Adipose tissue  Liver Muscle 	Cardiac output ↑ Width of blood vessels ↓ Blood pressure ↑ Metabolism: Glycogenolysis ↑ Blood glucose ↑ Lipolysis ↑

B. Examples of peptide hormones and proteohormones

Thyroliberin (TRH) 3 AA	Hypothalamus 	Pituitary  Brain  TSH	Thyrotropin secretion ↑ Neurotransmitter action
Thyrotropin (TSH) α chain 92 AA β chain 112 AA	Adeno- hypophysis 	Thyroid gland  Thyroxine	Synthesis and secretion of thyroxine ↑
Insulin A chain 21 AA B chain 30 AA	Pancreas  B cells	Glucose →  Glycogen ↔ Glucose Proteins ↔ Amino acids Fats ↔ Fatty acids	Glucose uptake by cells ↑ Blood glucose ↓ Storage compounds: formation ↑ degradation ↓
Glucagon 29 AA	Pancreas  A cells	Glycogen → Glucose Amino acids → Glucose Fats → Fatty acids → Ketone bodies	Glycogenolysis ↑ Gluconeogenesis ↑ Blood glucose ↑ Ketone body formation ↑

Metabolism of peptide hormones

Hydrophilic hormones and other water-soluble signaling substances have a variety of biosynthetic pathways. Amino acid derivatives arise in special *metabolic pathways* (see p. 352) or through *post-translational modification* (see p. 374). Proteohormones, like all proteins, result from *translation* in the ribosome (see p. 250). Small peptide hormones and neuropeptides, most of which only consist of 3–30 amino acids, are released from precursor proteins by *proteolytic degradation*.

A. Biosynthesis ○

The illustration shows the synthesis and processing of the precursor protein **proopiomelanocortin** (POMC) as an example of the biosynthesis of small peptides with signaling functions. POMC arises in cells of the adenohypophysis, and after processing in the rER and Golgi apparatus, it supplies the opiate-like peptides *met-enkephalin* and *β-endorphin* (implying “opio-”; see p. 352), three *melanocyte-stimulating hormones* (α -, β - and γ -MSH, implying “melano-”), and the glandotropic hormone *corticotropin* (ACTH, implying “-cortin”). Additional products of POMC degradation include two *lipotropins* with catabolic effects in the adipose tissue (β - and γ -LPH).

Some of the peptides mentioned are overlapping in the POMC sequence. For example, additional cleavage of ACTH gives rise to α -MSH and corticotropin-like intermediary peptide (CLIP). Proteolytic degradation of β -LPH provides γ -LPH and β -endorphin. The latter can be further broken down to yield met-enkephalin, while γ -LPH can still give rise to β -MSH (not shown). Due to the numerous derivative products with biological activity that it has, POMC is also known as a **poly-protein**. Which end product is formed and in what amounts depends on the activity of the proteinases in the ER that catalyze the individual cleavages.

The principles underlying protein synthesis and protein maturation (see pp. 230–233) can be summed up once again using the example of POMC:

[1] As a result of *transcription* of the POMC gene and *maturation of the hnRNA*, a mature **mRNA** consisting of some 1100 nucleotides

arises, which is modified at both ends (see p. 246). This mRNA codes for prepro-POMC—i.e., a POMC protein that still has a signal peptide for the ER at the N terminus (see p. 230).

[2] **Prepro-POMC** arises through *translation* in the rough endoplasmic reticulum (rER). The growing peptide chain is introduced into the ER with the help of a signal peptide.

[3] Cleavage of the signal peptide and other modifications in the ER (formation of disulfide bonds, glycosylation, phosphorylation) give rise to the mature prohormone (“**pro-POMC**”).

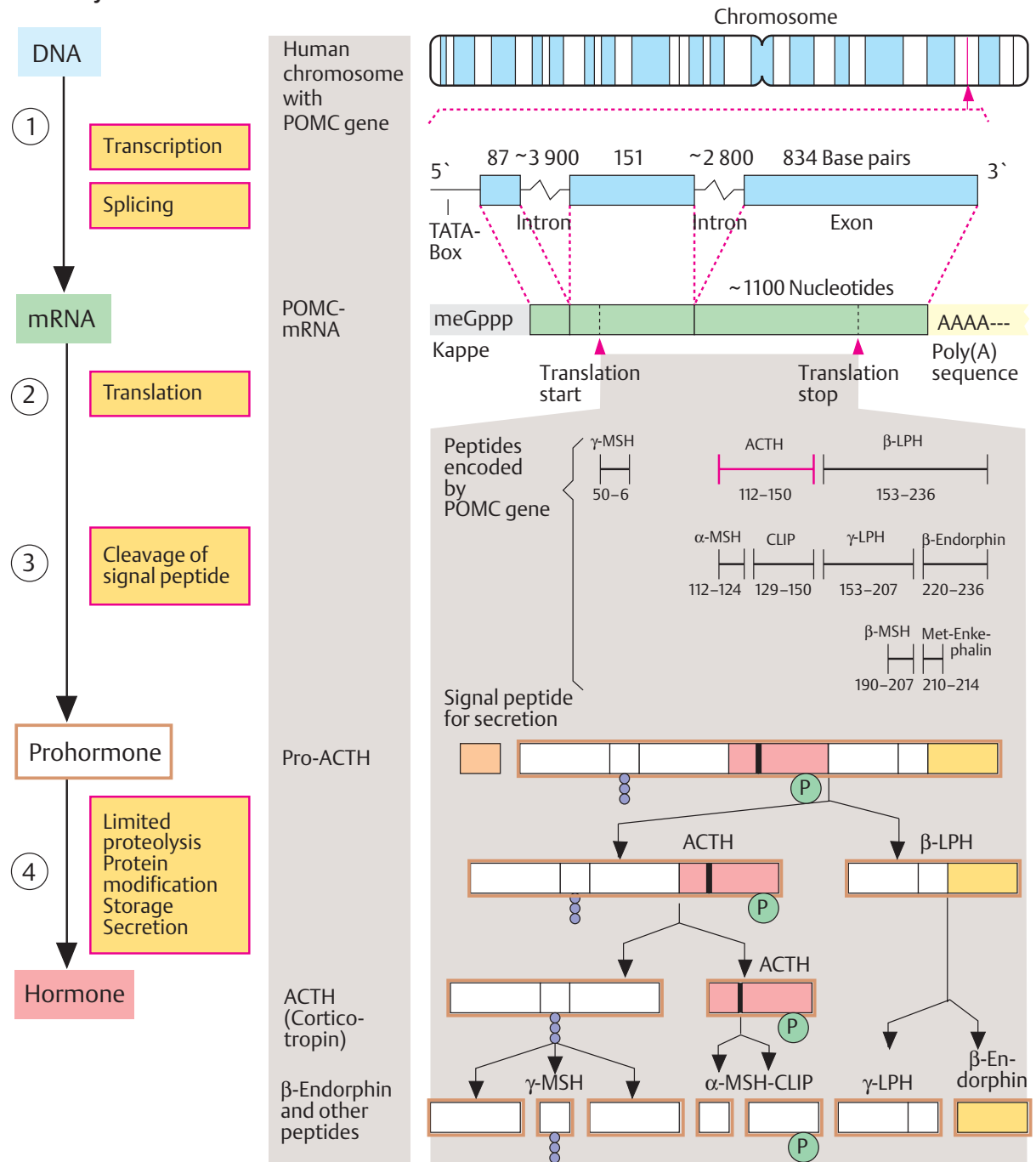
[4] The **neuropeptides** and **hormones** mentioned are now formed by *limited proteolysis* and stored in vesicles. Release from these vesicles takes place by exocytosis when needed.

The biosynthesis of peptide hormones and proteohormones, as well as their secretion, is controlled by higher-order regulatory systems (see p. 372). Calcium ions are among the substances involved in this regulation as *second messengers*; an increase in calcium ions stimulates synthesis and secretion.

B. Degradation and inactivation ●

Degradation of peptide hormones often starts in the blood plasma or on the vascular walls; it is particularly intensive in the kidneys.

Several peptides that contain disulfide bonds (e.g., insulin) can be inactivated by reductive *cleavage of the disulfide bonds* (1). Peptides and proteins are also cleaved by *peptidases*, starting from one end of the peptide by *exopeptidases* (2), or in the middle of it by proteinases (endopeptidases, 3). *Proteolysis* gives rise to a variety of hormone fragments, several of which are still biologically active. Some peptide hormones and proteohormones are removed from the blood by binding to their receptors with subsequent *endocytosis* of the hormone–receptor complex (4). They are then broken down in the lysosomes. All of the degradation reactions lead to amino acids, which become available to the metabolism again.

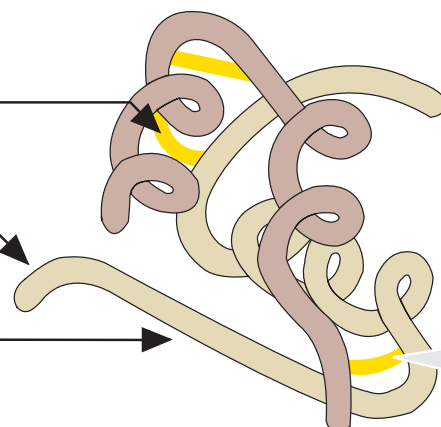
A. Biosynthesis**B. Degradation and inactivation**

Extracellular degradation:

1. Cleavage of disulfide bonds by reductases
2. Degradation by exopeptidases
3. Degradation by proteinases

Intracellular degradation:

4. Binding to membrane receptors
Endocytosis
Degradation in lysosomes



Mechanisms of action

The messages transmitted by hydrophilic signaling substances (see p. 380) are sent to the interior of the cell by *membrane receptors*. These bind the hormone on the outside of the cell and trigger a new second signal on the inside by altering their conformation. In the interior of the cell, this secondary signal influences the activity of enzymes or ion channels. Via further steps, *switching of the metabolism*, changes in the *cytoskeleton*, and activation or inhibition of *transcription factors* can occur ("signal transduction") can occur.

A. Mechanisms of action ❶

Receptors are classified into three different types according to their structure (see also p. 224):

1. 1-Helix receptors (left) are proteins that span the membrane with only one α -helix. On their inner (cytoplasmic) side, they have domains with *allosterically activatable enzyme activity*. In most cases, these are *tyrosine kinases*.

Insulins (see p. 388), *growth factors*, and *cytokines* (see p. 392), for example, act via 1-helix receptors. Binding of the signaling substance leads to activation of internal kinase activity (in some cases, dimerization of the receptor is needed for this). The activated kinase phosphorylates itself using ATP (*auto-phosphorylation*), and also phosphorylates tyrosine residues of other proteins (known as *receptor substrates*). Adaptor proteins that recognize the phosphotyrosine residues bind to the phosphorylated proteins (see pp. 388, 392). They pass the signal on to other protein kinases.

2. Ion channels (center). These receptors contain *ligand-gated ion channels*. Binding of the signaling substance opens the channels for ions such as Na^+ , K^+ , Ca^{2+} , and Cl^- . This mechanism is mainly used by *neurotransmitters* such as *acetylcholine* (nicotinic receptor; see p. 224) and *GABA* (A receptor; see p. 354).

3. 7-Helix receptors (serpentine receptors, right) represent a large group of membrane proteins that transfer the hormone or transmitter signal, with the help of *G proteins* (see below), to *effector proteins* that alter the concentrations of *ions* and *second messengers* (see B).

B. Signal transduction by G proteins ❷

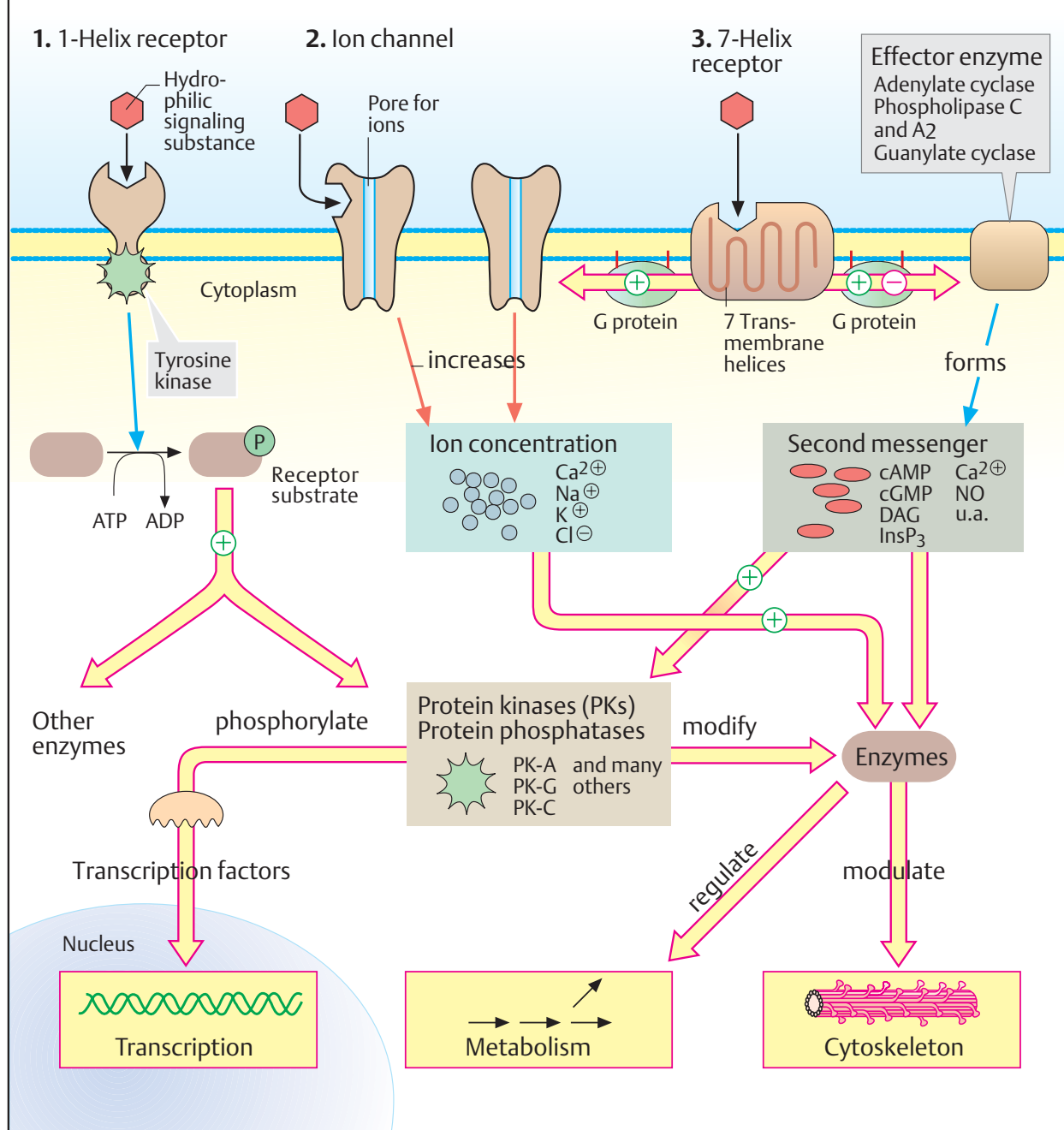
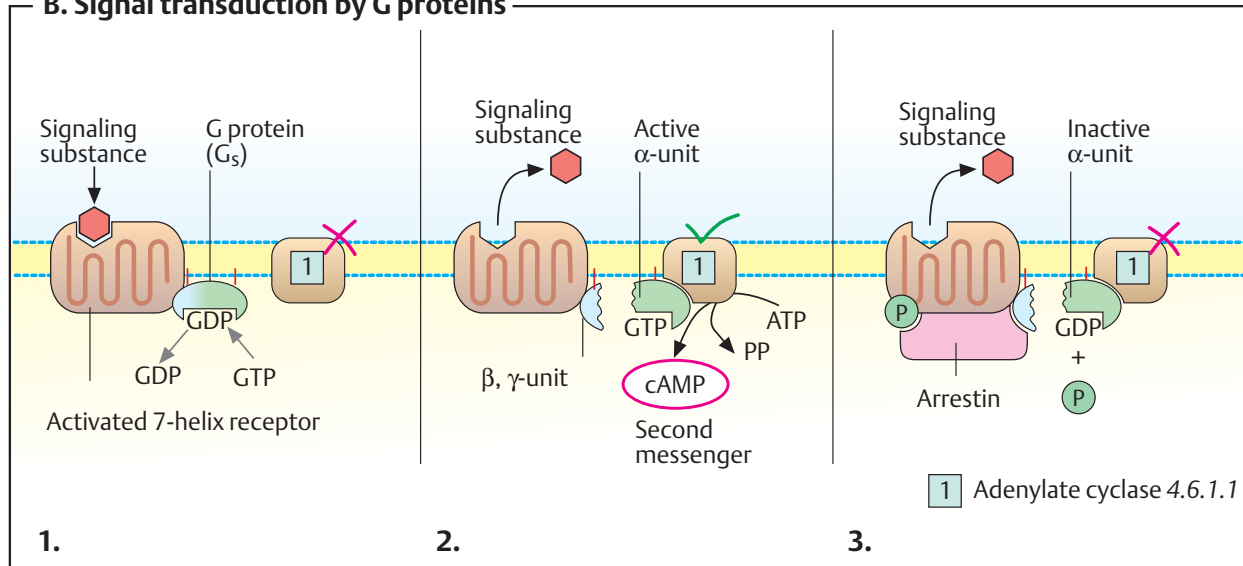
G proteins transfer signals from 7-helix receptors to effector proteins (see above). G protein are *heterotrimers* consisting of three different types of subunit (α , β , and γ ; see p. 224). The α -subunit can bind GDP or GTP (hence the name "G protein") and has *GTPase activity*. Receptor-coupled G proteins are related to other GTP-binding proteins such as *Ras* (see pp. 388, 398) and *EF-Tu* (see p. 252).

G proteins are divided into several types, depending on their effects. *Stimulatory G proteins* (G_s) are widespread. They activate adenylate cyclases (see below) or influence ion channels. *Inhibitory G proteins* (G_i) inhibit adenylate cyclase. G proteins in the G_q family activate another effector enzyme—phospholipase c (see p. 386).

Binding of the signaling substance to a 7-helix receptor alters the receptor conformation in such a way that the corresponding G protein can attach on the inside of the cell. This causes the α -subunit of the G protein to exchange bound GDP for GTP (1). The G protein then separates from the receptor and dissociates into an α -subunit and a $\beta\gamma$ -unit. Both of these components bind to other membrane proteins and alter their activity; *ion channels* are opened or closed, and *enzymes* are activated or inactivated.

In the case of the β_2 -catecholamine receptor (illustrated here), the α -subunit of the G_s protein, by binding to adenylate cyclase, leads to the synthesis of the **second messenger** cAMP. cAMP activates protein kinase A, which in turn activates or inhibits other proteins (2; see p. 120).

The $\beta\gamma$ -unit of the G protein stimulates a kinase (β ARK, not shown), which phosphorylates the receptor. This reduces its affinity for the hormone and leads to binding of the blocking protein *arrestin*. The internal GTPase activity of the α -subunit hydrolyzes the bound GTP to GDP within a period of seconds to minutes, and thereby terminates the action of the G protein on the adenylate cyclase (3).

A. Mechanisms of action**B. Signal transduction by G proteins**

Second messengers

Second messengers are *intracellular* chemical signals, the concentration of which is regulated by hormones, neurotransmitters, and other extracellular signals (see p. 384). They arise from easily available substrates and only have a short half-life. The most important second messengers are cAMP, cGMP, Ca^{2+} , inositol triphosphate (InsP_3), diacylglycerol (DAG), and nitrogen monoxide (NO).

A. Cyclic AMP ●

Metabolism. The nucleotide **cAMP** (adenosine 3',5'-cyclic monophosphate) is synthesized by membrane-bound *adenylate cyclases* [1] on the inside of the plasma membrane. The adenylyl cyclases are a family of enzymes that cyclize ATP to cAMP by cleaving diphosphate (PP_i). The degradation of cAMP to AMP is catalyzed by *phosphodiesterases* [2], which are inhibited by *methylxanthines* such as caffeine, for example. By contrast, *insulin* activates the esterase and thereby reduces the cAMP level (see p. 388).

Adenylyl cyclase activity is regulated by *G proteins* (G_s and G_i), which in turn are controlled by extracellular signals via *7-helix receptors* (see p. 384). Ca^{2+} -calmodulin (see below) also activates specific adenylyl cyclases.

Action. cAMP is an allosteric effector of *protein kinase A* (PK-A, [3]). In the inactive state, PK-A is a heterotetramer (C_2R_2), the catalytic subunits of which (C) are blocked by regulatory units (R; autoinhibition). When cAMP binds to the regulatory units, the C units separate from the R units and become enzymatically active. Active PK-A phosphorylates serine and threonine residues of more than 100 different proteins, enzymes, and transcription factors. In addition to cAMP, **cGMP** also acts as a second messenger. It is involved in sight (see p. 358) and in the signal transduction of NO (see p. 388).

B. Inositol 1,4,5-trisphosphate and diacylglycerol ●

Type G_q G proteins activate *phospholipase C* [4]. This enzyme creates two second messengers from the double-phosphorylated membrane phospholipid *phosphatidylinositol bisphosphate* (PIInsP_2), i.e., inositol 1,4,5-tris-

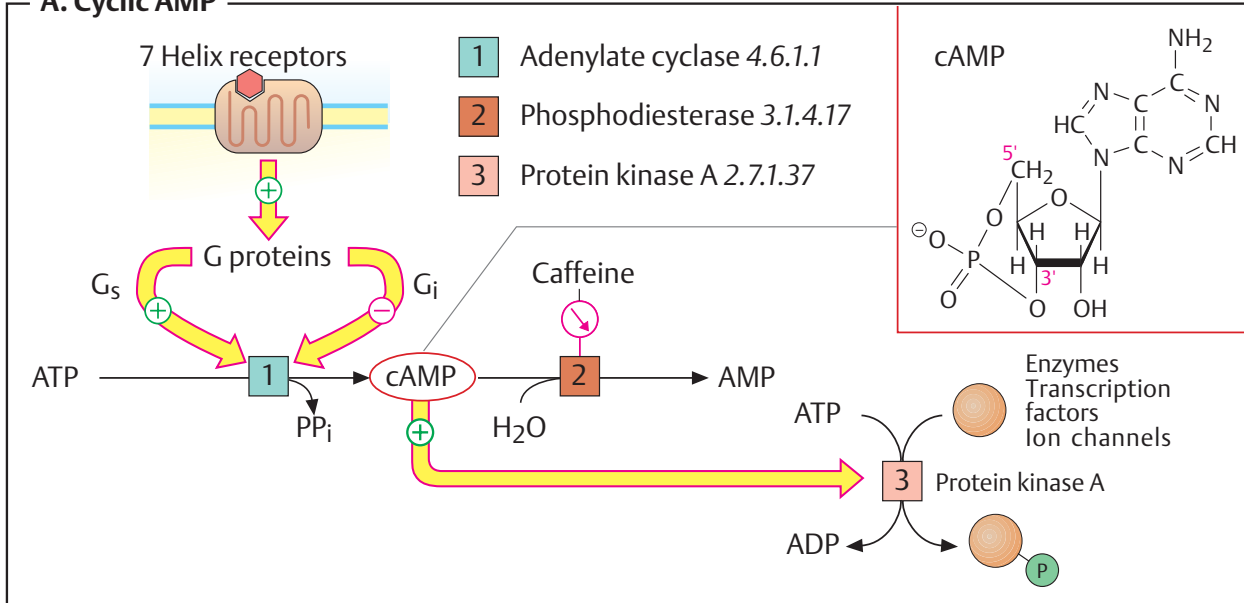
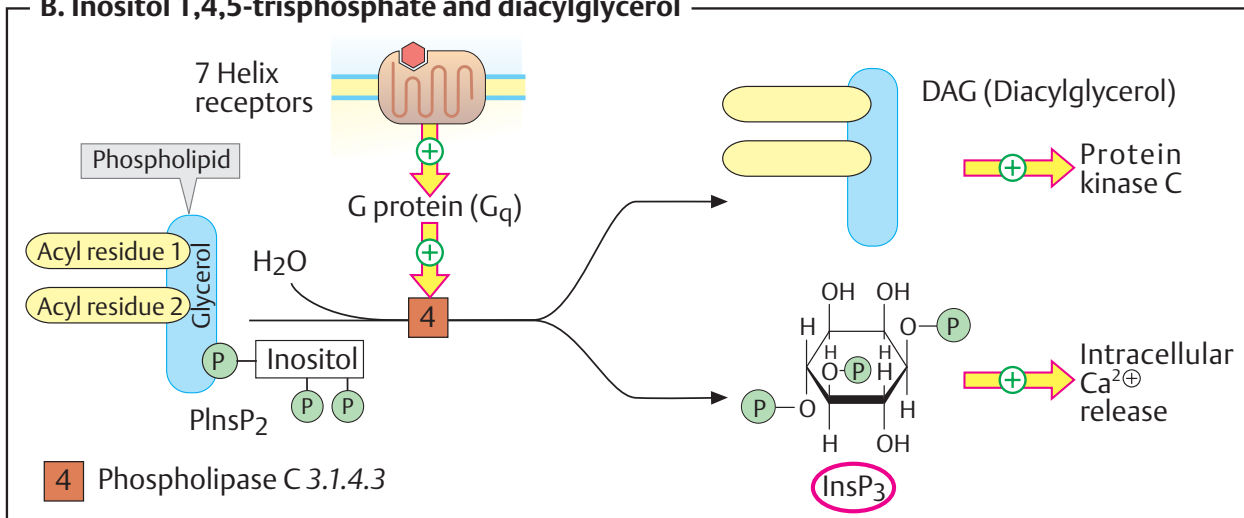
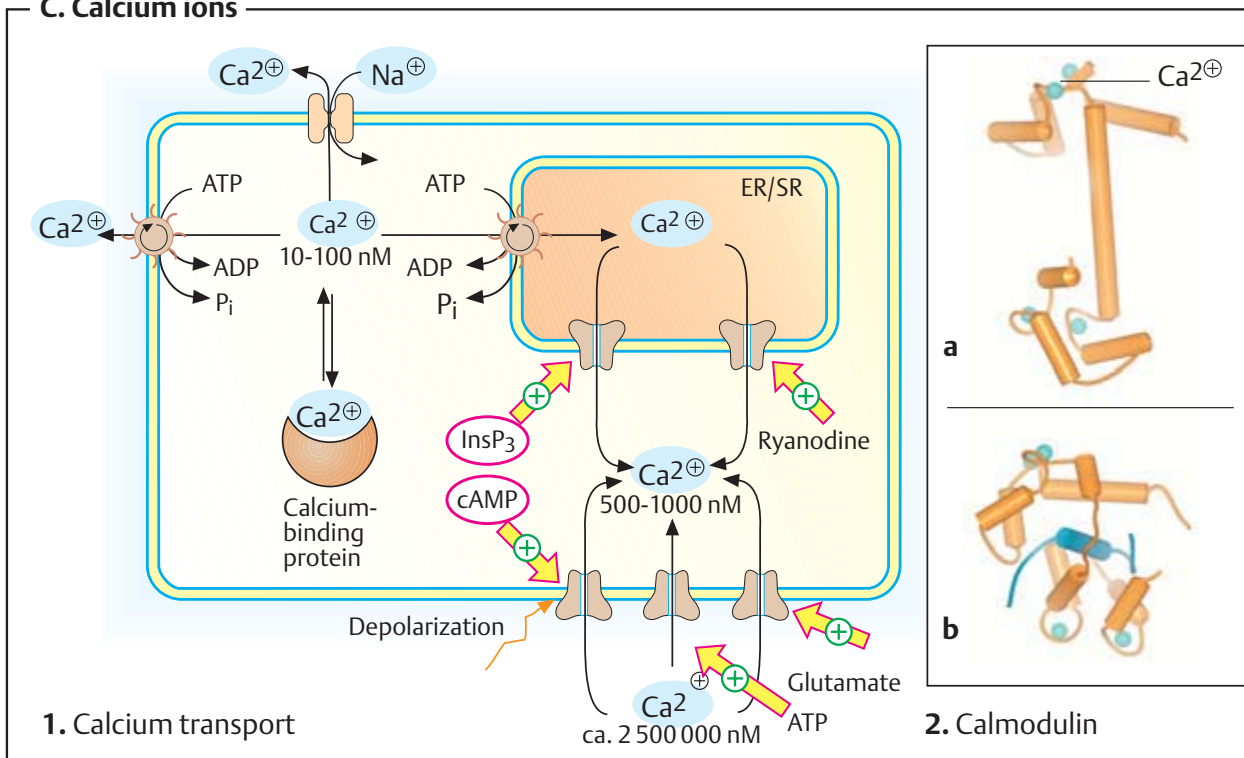
phosphate (**InsP₃**), which is soluble, and diacylglycerol (**DAG**). InsP_3 migrates to the endoplasmic reticulum (ER), where it opens Ca^{2+} channels that allow Ca^{2+} to flow into the cytoplasm (see C). By contrast, DAG, which is lipophilic, remains in the membrane, where it activates type C *protein kinases*, which phosphorylate proteins in the presence of Ca^{2+} ions and thereby pass the signal on.

C. Calcium ions ●

Calcium level. Ca^{2+} (see p. 342) is a signaling substance. The concentration of Ca^{2+} ions in the cytoplasm is normally very low (10–100 nM), as it is kept down by ATP-driven Ca^{2+} pumps and $\text{Na}^+/\text{Ca}^{2+}$ exchangers. In addition, many proteins in the cytoplasm and organelles bind calcium and thus act as Ca^{2+} buffers.

Specific signals (e.g., an action potential or second messenger such as InsP_3 or cAMP) can trigger a sudden increase in the cytoplasmic Ca^{2+} level to 500–1000 nM by opening Ca^{2+} channels in the plasma membrane or in the membranes of the *endoplasmic* or *sarcomplasmic reticulum*. *Ryanodine*, a plant substance, acts in this way on a specific channel in the ER. In the cytoplasm, the Ca^{2+} level always only rises very briefly (Ca^{2+} “spikes”), as prolonged high concentrations in the cytoplasm have cytotoxic effects.

Calcium effects. The biochemical effects of Ca^{2+} in the cytoplasm are mediated by special Ca^{2+} -binding proteins (“*calcium sensors*”). These include the *annexins*, *calmodulin*, and *troponin C* in muscle (see p. 334). **Calmodulin** is a relatively small protein (17 kDa) that occurs in all animal cells. Binding of four Ca^{2+} ions (light blue) converts it into a *regulatory element*. Via a dramatic conformational change (cf. 2a and 2b), Ca^{2+} -calmodulin enters into interaction with other proteins and modulates their properties. Using this mechanism, Ca^{2+} ions regulate the activity of enzymes, ion pumps, and components of the cytoskeleton.

A. Cyclic AMP**B. Inositol 1,4,5-trisphosphate and diacylglycerol****C. Calcium ions**

Signal cascades

The signal transduction pathways that mediate the effects of the metabolic hormone **insulin** are of particular medical interest (see **A**). The mediator **nitrogen monoxide** (NO) is also clinically important, as it regulates vascular caliber and thus the body's perfusion with blood (see **B**).

A. Insulin: signal transduction ○

The diverse effects of insulin (see p. 160) are mediated by protein kinases that mutually activate each other in the form of enzyme cascades. At the end of this chain there are kinases that influence gene transcription in the nucleus by phosphorylating target proteins, or promote the uptake of glucose and its conversion into glycogen. The signal transduction pathways involved have not yet been fully explained. They are presented here in a simplified form.

The **insulin receptor** (top) is a dimer with subunits that have activatable tyrosine kinase domains in the interior of the cell (see p. 224). Binding of the hormone increases the tyrosine kinase activity of the receptor, which then phosphorylates itself and other proteins (**receptor substrates**) at various tyrosine residues. **Adaptor proteins**, which conduct the signal further, bind to the phosphotyrosine residues.

The effects of insulin on transcription are shown on the left of the illustration. Adaptor proteins **Grb-2** and **SOS** ("son of sevenless") bind to the phosphorylated **IRS** (insulin-receptor substrate) and activate the G protein **Ras** (named after its gene, the oncogene *ras*; see p. 398). Ras activates the protein kinase **Raf** (another oncogene product). Raf sets in motion a phosphorylation cascade that leads via the kinases **MEK** and **ERK** (also known as MAPK, "mitogen-activated protein kinase") to the phosphorylation of transcription factors in the nucleus.

Some of the effects of insulin on the **carbohydrate metabolism** (right part of the illustration) are possible without protein synthesis. In addition to Grb-2, another dimeric adaptor protein can also bind to phosphorylated IRS. This adaptor protein thereby acquires *phosphatidylinositol-3-kinase* activity (**PI₃K**) and, in the membrane, phosphorylates phospholi-

pids from the phosphatidylinositol group (see p. 50) at position 3. Protein kinase **PK-B** binds to these reaction products, becoming activated itself and in turn activating protein kinase B (**PK-B**).

This has several effects. In a manner not yet fully understood, PK-B leads to the fusion with the plasma membrane of vesicles that contain the glucose transporter Glut-4. This results in inclusion of Glut-4 in the membrane and thus to increased glucose uptake into the muscles and adipose tissue (see p. 160). In addition, PK-B inhibits glycogen synthase kinase 3 (**GSK-3**) by phosphorylation. As GSK-3 in turn inhibits glycogen synthase by phosphorylation (see p. 120), its inhibition by PK-B leads to *increased* glycogen synthesis. Protein phosphatase-1 (**PP-1**) converts glycogen synthase into its active form by dephosphorylation (see p. 120). PP-1 is also activated by insulin.

B. Nitrogen monoxide (NO) as a mediator ○

Nitrogen monoxide (NO) is a short-lived radical that functions as a locally acting mediator (see p. 370).

In a complex reaction, NO arises from arginine in the endothelial cells of the blood vessels [1]. The trigger for this is Ca²⁺-calmodulin (see p. 386), which forms when there is an increase in the cytoplasmic Ca²⁺ level.

NO diffuses from the endothelium into the underlying vascular muscle cells, where it leads, as a result of activation of *guanylate cyclase* [2], to the formation of the second messenger **cGMP** (see pp. 358, 384). Finally, by activating a special protein kinase (**PK-G**), cGMP triggers relaxation of the smooth muscle and thus dilation of the vessels. The effects of atrionatriuretic peptide (**ANP**; see p. 328) in reducing blood pressure are also mediated by cGMP-induced vasodilation. In this case, cGMP is formed by the guanylate cyclase activity of the ANP receptor.

Further information

The drug *nitroglycerin* (glyceryl trinitrate), which is used in the treatment of *angina pectoris*, releases NO in the bloodstream and thereby leads to better perfusion of cardiac muscle.

A. Insulin: signal transduction

The diagram illustrates the signaling pathway initiated by insulin binding to its receptor. Insulin binds to the α subunits of the Insulin receptor, which is a dimer of α and β subunits. This activates the β subunits, which then phosphorylate IRS-1. Activated IRS-1 recruits Grb-2, which in turn activates Ras. Ras activates Raf, which activates MEK (MAPKK), which activates MAPK (ERK). Activated MAPK (ERK) enters the nucleus to activate transcription factors, leading to transcription. Another branch of the pathway involves PI-3-Kinase, which phosphorylates Phosphatidylinositols to form Phosphatidylinositol 3-phosphate. This activates PDK-1, which phosphorylates PK-B. Activated PK-B phosphorylates GSK-3, leading to its inactivation. Inactivated GSK-3 allows Glycogen synthase to become active, leading to Glycogen synthesis. Additionally, PI-3-Kinase activates Glut-4, leading to Glucose uptake. The diagram also shows that MEK (MAPKK) and MAPK (ERK) are inactive (marked with a red X) in the absence of insulin, while the other components are active (marked with a green checkmark).

Legend:

- Protein kinase (Green star)
- Protein phosphatase (Green diamond)
- Inactive (Red X)
- Active (Green checkmark)

B. Nitrogen monoxide (NO) as a mediator

The diagram illustrates the physiological effects of Nitrogen monoxide (NO) as a mediator, showing the signaling pathway from an endothelial cell to a vascular muscle cell.

Endothelial cell:

- A **Signaling substance** binds to a receptor, activating **InsP₃** and **Ca²⁺** release from the **ER**.
- Ca²⁺** release is stimulated by **InsP₃** and **Ca²⁺** itself (positive feedback).
- Ca²⁺** release leads to the activation of **NO synthase (1)**.
- NO synthase (1)** converts **Arginine** and **2 O₂** into **NO** and **Citrulline**, with the release of **3/2 H₂O**.
- Calmodulin** is also involved in the activation of **NO synthase (1)**.

Vascular muscle cell:

- NO** diffuses from the endothelial cell into the vascular muscle cell.
- NO** activates **Guanylate cyclase (2)**, which converts **GTP** into **cGMP**.
- cGMP** activates the **ANF receptor (3)**, which inhibits **PK-G**, leading to **Physiological effects**.
- The chemical structure of **Guanine** is shown, with the **5'** and **3'** positions labeled.

Legend:

- 1** NO synthase 1.14.13.39
- 2** Guanylate cyclase 4.6.1.2
- 3** ANF receptor 4.6.1.2

Eicosanoids

The eicosanoids are a group of signaling substances that arise from the C-20 fatty acid *arachidonic acid* and therefore usually contain 20 C atoms (Greek *eicosa* = 20). As mediators, they influence a large number of physiological processes (see below). Eicosanoid metabolism is therefore an important drug target. As short-lived substances, eicosanoids only act in the vicinity of their site of synthesis (paracrine effect; see p. 372).

A. Eicosanoids ○

Biosynthesis. Almost all of the body's cells form eicosanoids. Membrane phospholipids that contain the polyunsaturated fatty acid **arachidonic acid** (20:4; see p. 48) provide the starting material.

Initially, *phospholipase A₂* [1] releases the arachidonate moiety from these phospholipids. The activity of phospholipase A₂ is strictly regulated. It is activated by hormones and other signals via *G proteins*. The arachidonate released is a signaling substance itself. However, its metabolites are even more important.

Two different pathways lead from arachidonate to **prostaglandins**, **prostacyclins**, and **thromboxanes**, on the one hand, or **leukotrienes** on the other. The key enzyme for the first pathway is *prostaglandin synthase* [2]. Using up O₂, it catalyzes in a two-step reaction the cyclization of arachidonate to prostaglandin H₂, the parent substance for the prostaglandins, prostacyclins, and thromboxanes. Acetylsalicylic acid (aspirin) irreversibly acetylates a serine residue near the active center of prostaglandin synthase, so that access for substrates is blocked (see below).

As a result of the action of *lipxygenases* [3], hydroxyfatty acids and hydroperoxyfatty acids are formed from arachidonate, from which elimination of water and various conversion reactions give rise to the leukotrienes. The formulae only show one representative from each of the various groups of eicosanoids.

Effects. Eicosanoids act via membrane receptors in the immediate vicinity of their site of synthesis, both on the synthesizing cell itself (*autocrine* action) and on neighboring cells (*paracrine* action). Many of their effects are mediated by the second messengers cAMP and cGMP.

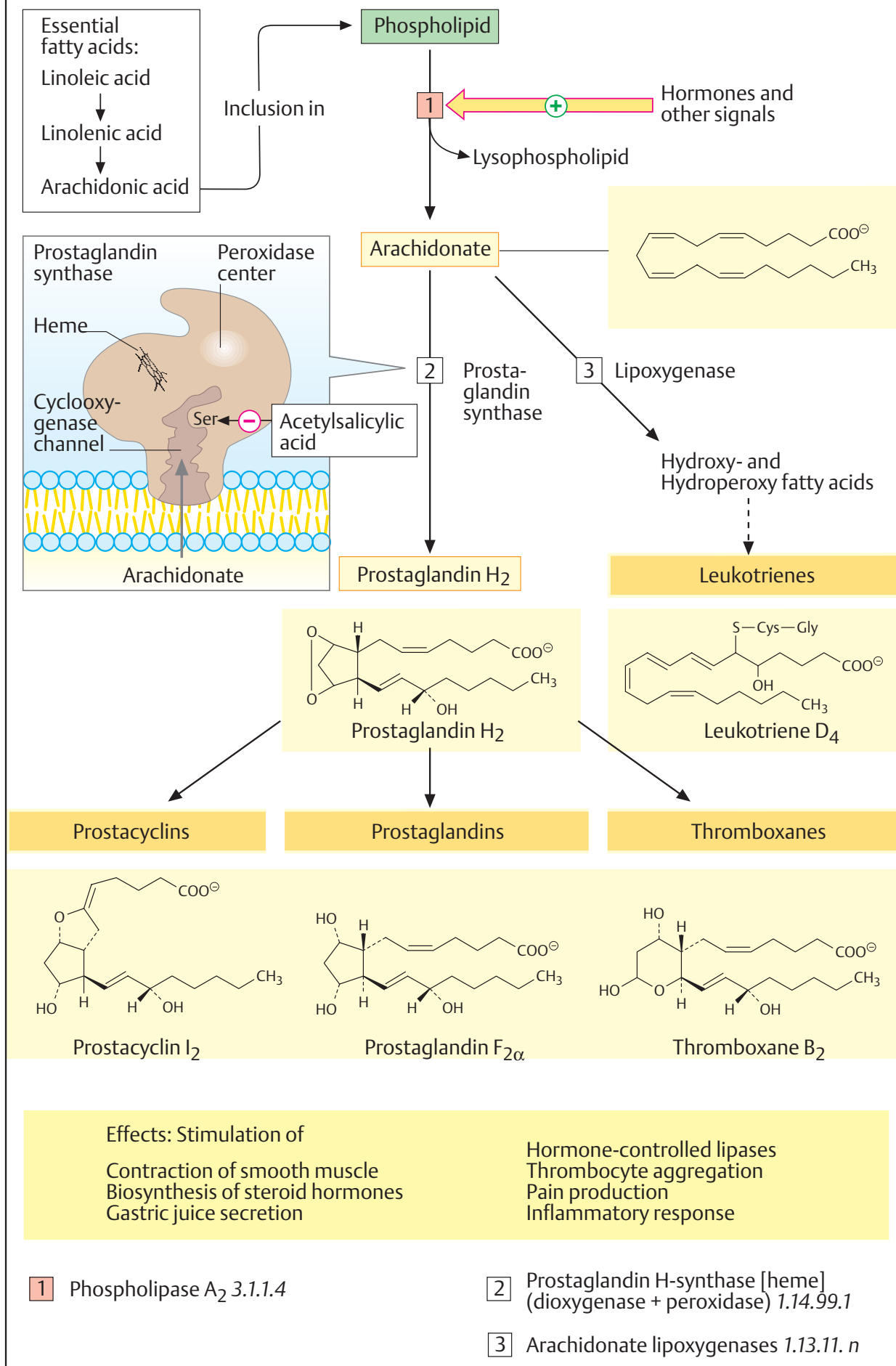
The eicosanoids have a very wide range of physiological effects. As they can stimulate or inhibit smooth-muscle contraction, depending on the substance concerned, they affect blood pressure, respiration, and intestinal and uterine activity, among other properties. In the stomach, prostaglandins inhibit HCl secretion via G_i proteins (see p. 270). At the same time, they promote mucus secretion, which protects the gastric mucosa against the acid. In addition, prostaglandins are involved in bone metabolism and in the activity of the sympathetic nervous system. In the immune system, prostaglandins are important in the inflammatory reaction. Among other things, they attract leukocytes to the site of infection. Eicosanoids are also decisively involved in the development of pain and fever. The thromboxanes promote thrombocyte aggregation and other processes involved in hemostasis (see p. 290).

Metabolism. Eicosanoids are inactivated within a period of seconds to minutes. This takes place by enzymatic reduction of double bonds and dehydrogenation of hydroxyl groups. As a result of this rapid degradation, their range is very limited.

Further information

Acetylsalicylic acid and related non-steroidal anti-inflammatory drugs (NSAIDs) selectively inhibit the *cyclooxygenase activity* of prostaglandin synthase [2] and consequently the synthesis of most eicosanoids. This explains their analgesic, antipyretic, and antirheumatic effects. Frequent side effects of NSAIDs also result from inhibition of eicosanoid synthesis. For example, they impair hemostasis because the synthesis of thromboxanes by thrombocytes is inhibited. In the stomach, NSAIDs increase HCl secretion and at the same time inhibit the formation of protective mucus. Long-term NSAID use can therefore damage the gastric mucosa.

A. Eicosanoids



Cytokines

A. Cytokines ●

Cytokines are hormone-like *peptides* and *proteins* with signaling functions, which are synthesized and released by cells of the immune system and other cell types. Their numerous biological functions operate in three areas: they regulate the *development and homeostasis of the immune system*; they control the *hematopoietic system*; and they are involved in *non-specific defense*, influencing inflammatory processes, blood coagulation, and blood pressure. In general, cytokines regulate the growth, differentiation, and survival of cells. They are also involved in regulating apoptosis (see p. 396).

There is an extremely large number of cytokines; only the most important representatives are listed opposite. The cytokines include *interleukins* (IL), *lymphokines*, *monokines*, *chemokines*, *interferons* (IFN), and *colony-stimulating factors* (CSF). Via interleukins, immune cells stimulate the proliferation and activity of other immune cells (see p. 294). Interferons are used medically in the treatment of viral infections and other diseases.

Although cytokines rarely show structural homologies with each other, their effects are often very similar. The cytokines differ from *hormones* (see p. 370) only in certain respects: they are released by many different cells, rather than being secreted by defined glands, and they regulate a wider variety of target cells than the hormones.

B. Signal transduction in the cytokines ○

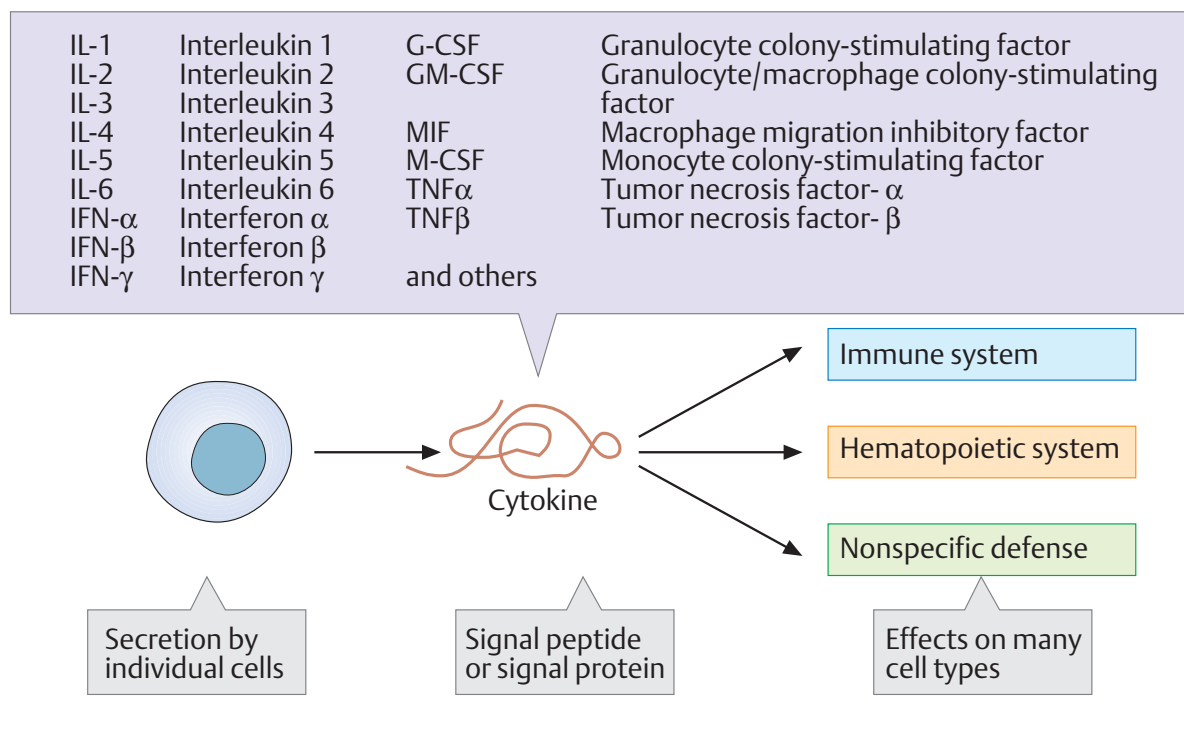
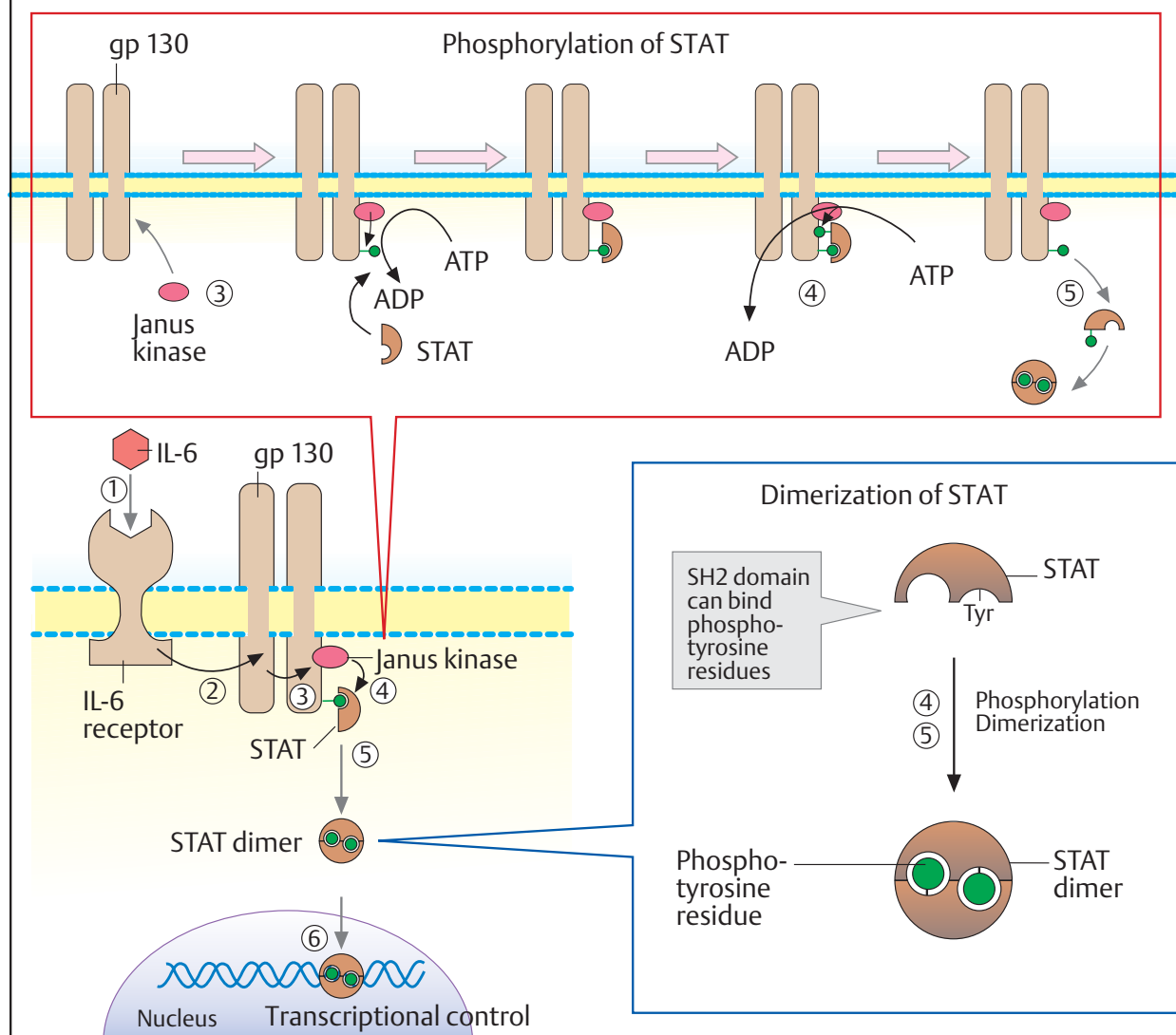
As peptides or proteins, the cytokines are **hydrophilic signaling substances** that act by binding to *receptors* on the cell surface (see p. 380). Binding of a cytokine to its receptor (1) leads via several intermediate steps (2–5) to the activation of transcription of specific genes (6).

In contrast to the receptors for insulin and growth factors (see p. 388), the **cytokine receptors** (with a few exceptions) have *no* tyrosine kinase activity. After binding of cytokine (1), they associate with one another to form homodimers, join together with other signal transduction proteins (STPs) to form dimers, or promote dimerization of other

STPs (2). Class I cytokine receptors interact with three different STPs (gp130, β_c , and γ_c). The STPs themselves do not bind cytokines, but conduct the signal to tyrosine kinases (3). The fact that different cytokines can activate the same STP via their receptors explains the overlapping biological activity of some cytokines.

As an example of the signal transduction pathway in cytokines, the illustration shows the way in which the **IL-6 receptor**, after binding its ligand **IL-6** (1), induces the dimerization of the STP **gp130** (2). The dimeric gp130 binds cytoplasmic *tyrosine kinases* from the Jak family ("**Janus kinases**," with two kinase centers) and activates them (3). The Janus kinases phosphorylate cytokine receptors, STPs, and various cytoplasmic proteins that conduct the signal further. In addition, they phosphorylate transcription factors known as **STATs** ("**signal transducers and activators of transcription**"). STATs are among the proteins that have an *SH2 domain* and are able to bind phosphotyrosine residues (see p. 388). They therefore bind to cytokine receptors that have been phosphorylated by Janus kinases. When STATs are then also phosphorylated themselves (4), they are converted into their active form and become dimers (5). After transfer to the nucleus, they bind—along with auxiliary proteins as transcription factors—to the promoters of inducible genes and in this way regulate their transcription (6).

The activity of the cytokine receptors is terminated by *protein phosphatases*, which hydrolytically cleave the phosphotyrosine residues. Several cytokine receptors are able to lose their ligand-binding extracellular domain by proteolysis (not shown). The extracellular domain then appears in the blood, where it competes for cytokines. This reduces the effective cytokine concentration.

A. Cytokines**B. Signal transduction in the cytokines**

Cell cycle

A. Cell cycle ●

Proliferating cells undergo a cycle of division (the cell cycle), which lasts approximately 24 hours in mammalian cells in cell culture. The cycle is divided into four different phases (G_1 , S, G_2 , and M—in that sequence).

Fully differentiated animal cells only divide rarely. These cells are in the so-called **G_0 phase**, in which they can remain permanently. Some G_0 cells return to the **G_1 phase** again under the influence of mitogenic signals (growth factors, cytokines, tumor viruses, etc.), and after crossing a *control point* (G_1 to S), enter a new cycle. DNA is replicated (see p. 240) during the **S phase**, and new chromatin is formed. Particularly remarkable in morphological terms is the actual mitosis (**M phase**), in which the chromosomes separate and two daughter cells are formed. The M and S phases are separated by two segments known as the **G_1** and **G_2** phases (the G stands for “gap”). In the G_1 phase, the duration of which can vary, the cell grows by de novo synthesis of cell components. Together, the G_1 , G_0 , S, and G_2 phases are referred to as the **interphase**, which alternates in the cell cycle with the short M phase.

B. Control of the cell cycle ○

The progression of the cell cycle is regulated by **interconversion processes**. In each phase, special *Ser/Thr-specific protein kinases* are formed, which are known as cyclin-dependent kinases (**CDKs**). This term is used because they have to bind an activator protein (**cyclin**) in order to become active. At each control point in the cycle, specific CDKs associate with equally phase-specific cyclins. If there are no problems (e.g., DNA damage), the CDK–cyclin complex is activated by phosphorylation and/or dephosphorylation. The activated complex in turn phosphorylates **transcription factors**, which finally lead to the formation of the proteins that are required in the cell cycle phase concerned (enzymes, cytoskeleton components, other CDKs, and cyclins). The activity of the CDK–cyclin complex is then terminated again by proteolytic **cyclin degradation**.

The above outline of cell cycle progression can be examined here in more detail using the G_2 –M transition as an example.

Entry of animal cells into mitosis is based on the “mitosis-promoting factor” (**MPF**). MPF consists of **CDK1** (*cdc2*) and **cyclin B**. The intracellular concentration of cyclin B increases constantly until mitosis starts, and then declines again rapidly (top left). MPF is initially inactive, because CDK1 is phosphorylated and cyclin B is dephosphorylated (top center). The M phase is triggered when a *protein phosphatase* [1] dephosphorylates the CDK while cyclin B is phosphorylated by a *kinase* [2]. In its active form, MPF phosphorylates various proteins that have functions in mitosis—e.g., *histone H1* (see p. 238), components of the cytoskeleton such as the *laminins* in the nuclear membrane, *transcription factors*, *mitotic spindle* proteins, and various *enzymes*.

When mitosis has been completed, cyclin B is marked with **ubiquitin** and broken down proteolytically by *proteasomes* (see p. 176). Protein phosphatases then regain control and dephosphorylate the proteins involved in mitosis. This returns the cell to the interphase.

Further information

The G_1 –S transition (not shown) is particularly important for initiating the cell cycle. It is triggered by the CDK4–cyclin D complex, which by phosphorylating the protein **pRb** releases the transcription factor E2F previously bound to pRb. This activates the transcription of genes needed for DNA replication.

If the DNA is damaged by mutagens or ionizing radiation, the protein **p53** initially delays entry into the S phase. If the DNA repair system (see p. 256) does not succeed in removing the DNA damage, p53 forces the cell into apoptosis (see p. 396). The genes coding for pRb and p53 belong to the **tumor-suppressor genes** (see p. 398). In many tumors (see p. 400), these genes are in fact damaged by mutation.

A. Cell cycle**M phase**

Mitosis
Chromosome
separation
Cell division

G₂ phase

Preparation
for mitosis

S phase

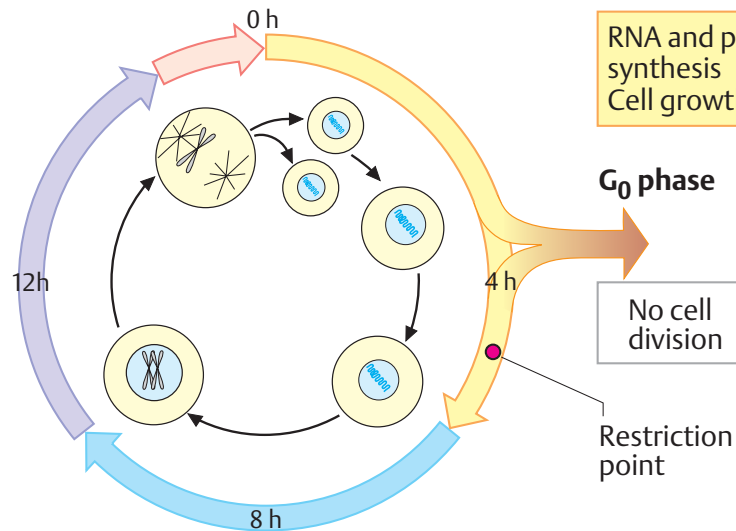
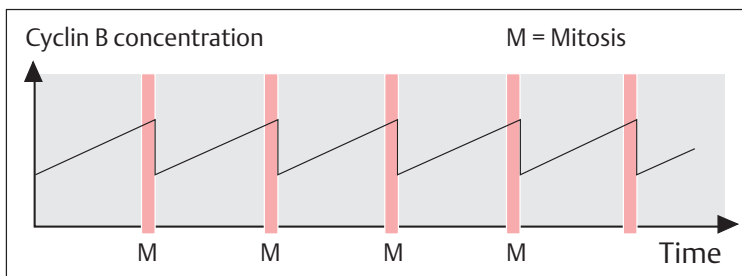
DNA replication
Histone synthesis
Centrosome formed
Chromosome
duplication

G₁ phase

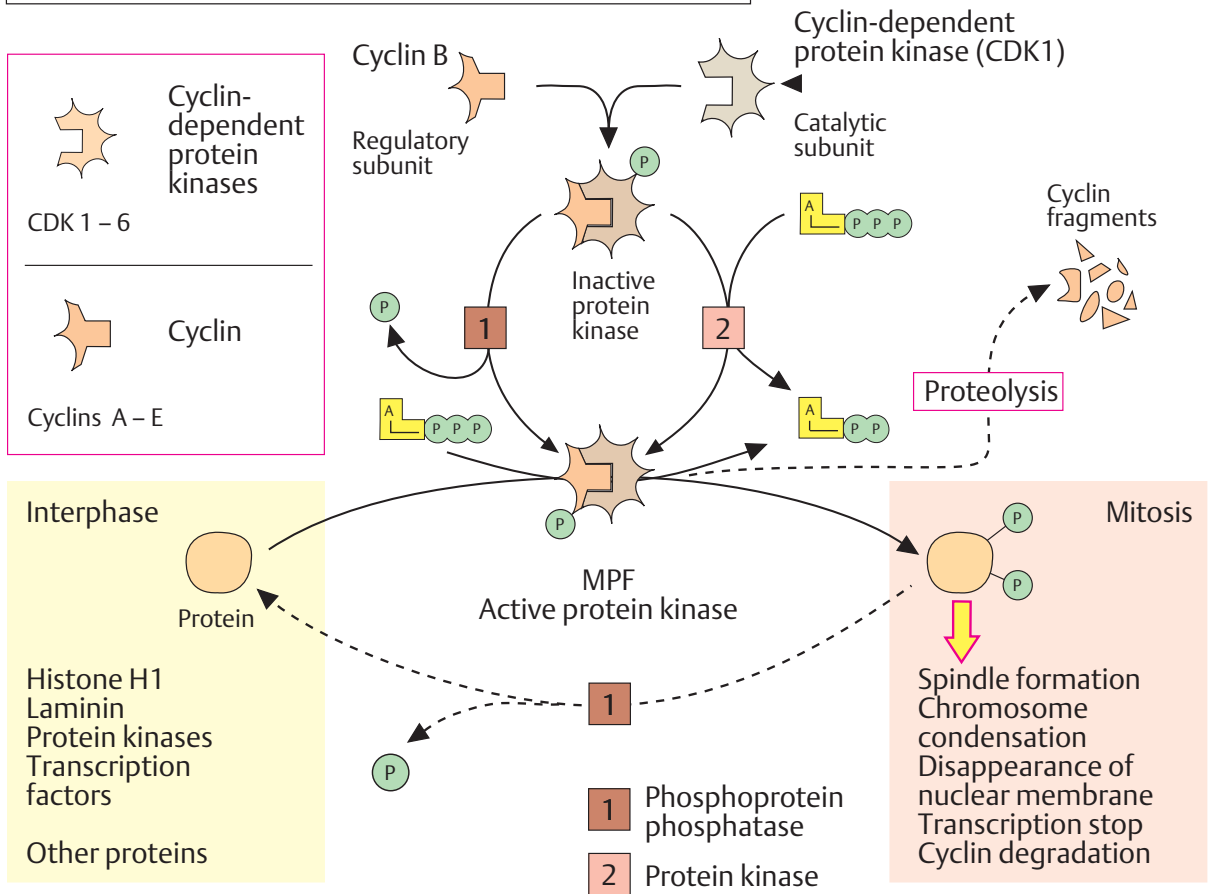
RNA and protein
synthesis
Cell growth

G₀ phase

No cell
division

**B. Control of the cell cycle**

← Reactions early in mitosis
← Reactions late in mitosis



Apoptosis

A. Cell proliferation and apoptosis ●

The number of cells in any tissue is mainly regulated by two processes—cell **proliferation** and *physiological cell death*, **apoptosis**. Both of these processes are regulated by stimulatory and inhibitory factors that act in solute form (growth factors and cytokines) or are presented in bound form on the surface of neighboring cells (see below).

Apoptosis is genetically programmed cell death, which leads to “tidy” breakdown and disposal of cells. Morphologically, apoptosis is characterized by changes in the cell membrane (with the formation of small blebs known as “apoptotic bodies”), shrinking of the nucleus, chromatin condensation, and fragmentation of DNA. *Macrophages* and other phagocytic cells recognize apoptotic cells and remove them by phagocytosis without inflammatory phenomena developing.

Cell necrosis (not shown) should be distinguished from apoptosis. In cell necrosis, cell death is usually due to physical or chemical damage. Necrosis leads to swelling and bursting of the damaged cells and often triggers an inflammatory response.

The growth of tissue (or, more precisely, the number of cells) is actually regulated by apoptosis. In addition, apoptosis allows the elimination of unwanted or superfluous cells—e.g., during embryonic development or in the immune system. The contraction of the uterus after birth is also based on apoptosis. Diseased cells are also eliminated by apoptosis—e.g., tumor cells, virus-infected cells, and cells with irreparably damaged DNA. An everyday example of this is the peeling of the skin after sunburn.

B. Regulation of apoptosis ○

Apoptosis can be triggered by a number of different signals that use various transmission pathways. Other signaling pathways prevent apoptosis.

At the center of the apoptotic process lies a group of specialized *cysteine-containing aspartate proteinases* (see p. 176), known as **caspases**. These mutually activate one another, creating an *enzyme cascade* resembling the cascade involved in blood coagulation (see

p. 290). Other enzymes in this group, known as **effector caspases**, cleave cell components after being activated—e.g., laminin in the nuclear membrane and snRP proteins (see p. 246)—or activate special DNases which then fragment the nuclear DNA.

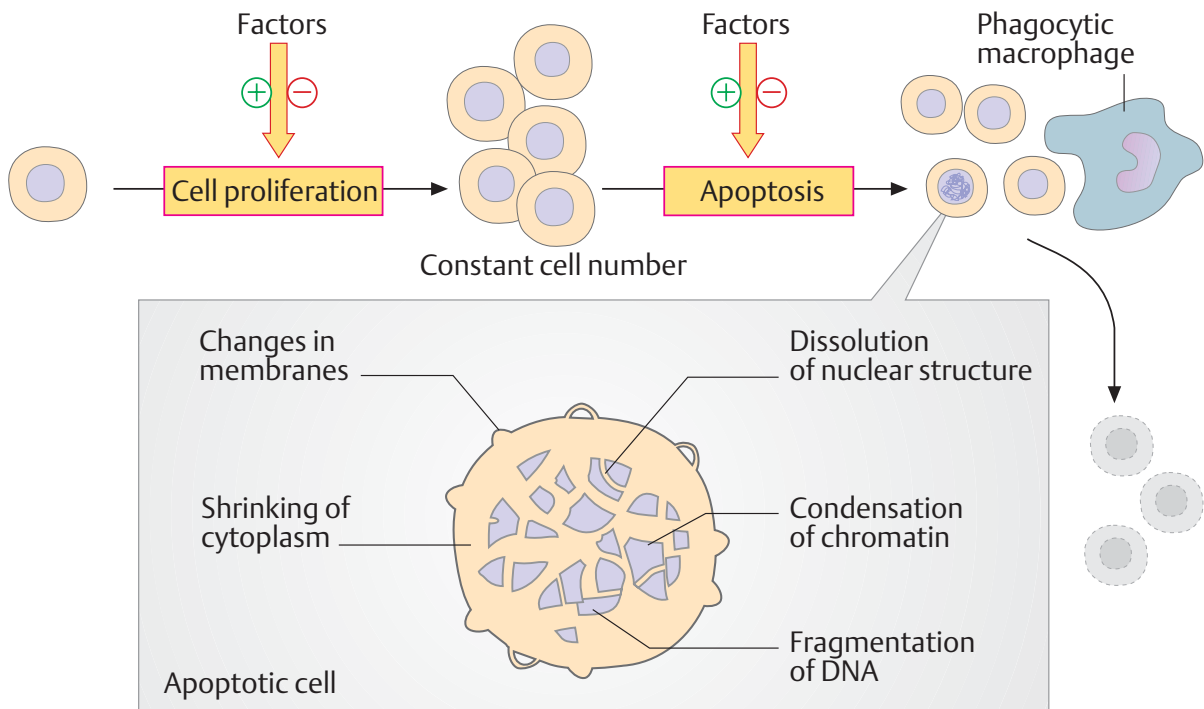
An important trigger for apoptosis is known as the **Fas system**. This is used by cytotoxic T cells, for example, which eliminate infected cells in this way (top left). Most of the body's cells have *Fas receptors* (CD 95) on their plasma membrane. If a T cell is activated by contact with an MHC presenting a viral peptide (see p. 296), binding of its *Fas ligands* occurs on the target cell's Fas receptors. Via the mediator protein FADD (“Fas-associated death domain”), this activates *caspase-8* inside the cell, setting in motion the apoptotic process.

Another trigger is provided by **tumor necrosis factor- α** (TNF- α), which acts via a similar protein (TRADD) and supports the endogenous defense system against tumors by inducing apoptosis.

Caspase-8 activates the effector caspases either directly, or indirectly by promoting the **cytochrome c** (see p. 140) from mitochondria. Once in the cytoplasm, cytochrome c binds to and activates the protein Apaf-1 (not shown) and thus triggers the caspase cascade. Apoptotic signals can also come from the cell nucleus. If irreparable DNA damage is present, the **p53 protein** (see p. 394)—the product of a *tumor suppressor gene*—promotes apoptosis and thus helps eliminate the defective cell.

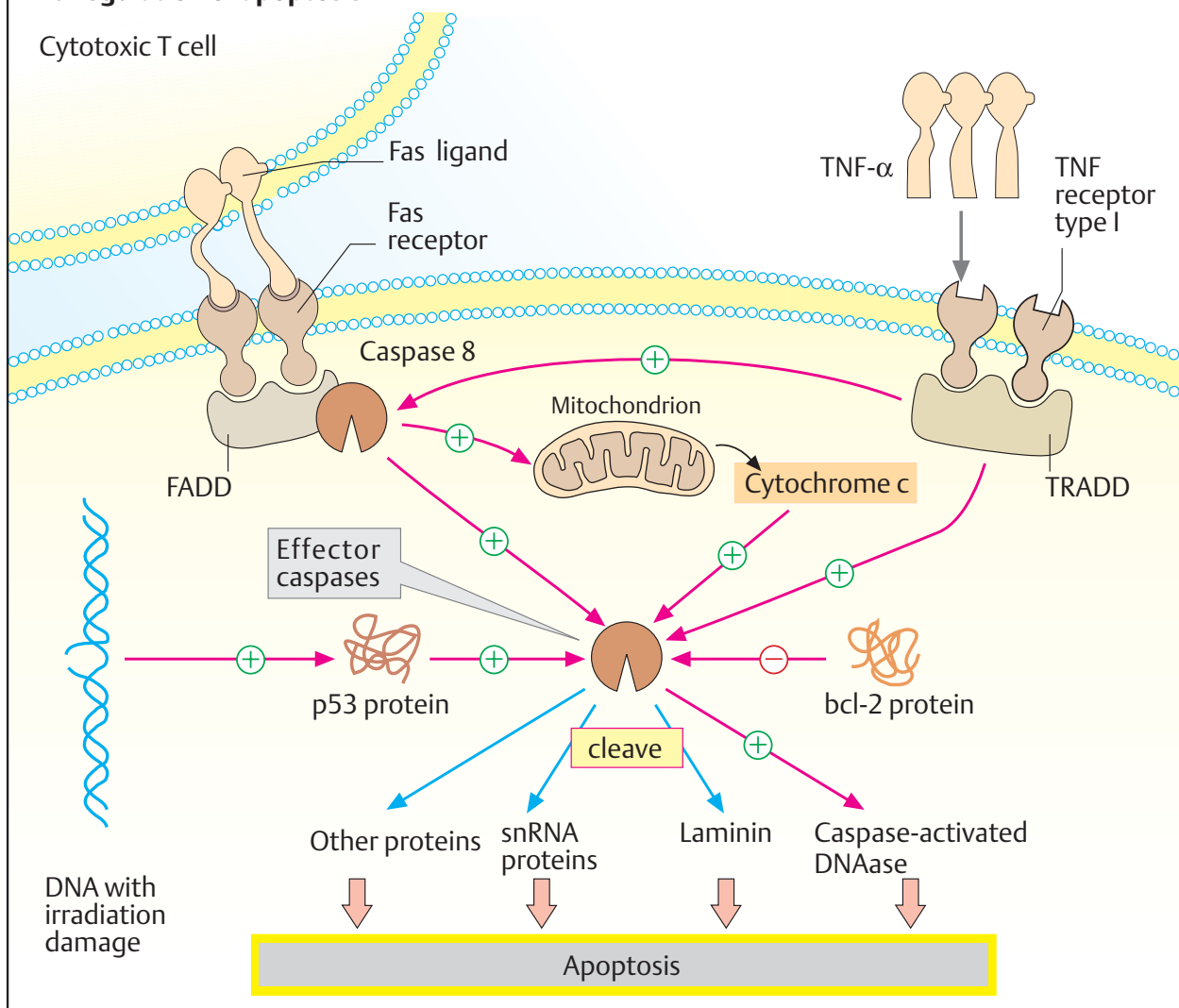
There are also inhibitory factors that oppose the signals that activate apoptosis. These include **bcl-2** and related proteins. The genomes of several viruses include genes for this type of protein. The genes are expressed by the host cell and (to the benefit of the virus) prevent the host cell from being prematurely eliminated by apoptosis.

A. Cell proliferation and apoptosis



B. Regulation of apoptosis

Cytotoxic T cell



Oncogenes

Oncogenes are cellular genes that can trigger uncontrolled cell proliferation if their sequence is altered or their expression is incorrectly regulated. They were first discovered as *viral (v-) oncogenes* in retroviruses that cause tumors (tumor viruses). Viruses of this type (see p. 404) sometimes incorporate genes from the host cell into their own genome. If these genes are reincorporated into the host DNA again during later infection, tumors can then be caused in rare cases. Although virus-related tumors are rare, research into them has made a decisive contribution to our understanding of oncogenes and their functioning.

A. Proto-oncogenes: biological role ●

The cellular form of oncogenes (known as *c-oncogenes* or **proto-oncogenes**) code for proteins involved in controlling growth and differentiation processes. They only become oncogenes if their sequence has been altered by *mutations* (see p. 256), *deletions*, and other processes, or when excessive amounts of the gene products have been produced as a result of *overexpression*.

Overexpression can occur when **amplification** leads to numerous functional copies of the respective gene, or when the gene falls under the influence of a highly active promoter (see p. 244). If the control of oncogene expression by **tumor suppressor genes** (see p. 394) is also disturbed, **transformation** and unregulated proliferation of the cells can occur. A single activated oncogene does not usually lead to a loss of growth control. It only occurs when over the course of time mutations and regulation defects accumulate in one and the same cell. If the immune system does not succeed in eliminating the transformed cell, it can over the course of months or years grow into a macroscopically visible **tumor**.

B. Oncogene products: biochemical functions ●

A feature common to all oncogenes is the fact that they code for proteins involved in *signal transduction processes*. The genes are designated using three-letter abbreviations that

usually indicate the origin of the viral gene and are printed in italics (e.g., *myc* for myelocytomatosis, a viral disease in birds). Oncogene products can be classified into the following groups according to their functions.

1. **Ligands** such as *growth factors* and *cytokines*, which promote cell proliferation.
2. **Membrane receptors** of the 1-helix type with tyrosine kinase activity, which can bind growth factors and hormones (see p. 394).
3. **GTP-binding proteins**. This group includes the G proteins in the strict sense and related proteins such as Ras (see p. 388), the product of the oncogene *c-ras*.
4. **Receptors for lipophilic hormones** mediate the effects of steroid hormones and related signaling substances. They regulate the transcription of specific genes (see p. 378). The products of several oncogenes (e.g., *erbA*) belong to this superfamily of *ligand-controlled transcription factors*.
5. **Nuclear tumor suppressors** inhibit return to the cell cycle in fully differentiated cells. The genes that code for these proteins are referred to as *anti-oncogenes* due to this function. On the role of p53 and pRb, see p. 394.
6. **DNA-binding proteins**. A whole series of oncogenes code for *transcription factors*. Particularly important for cell proliferation are *myc*, as well as *fos* and *jun*. The protein products of the latter two genes form the transcription factor AP-1 as a heterodimer (see p. 244).
7. **Protein kinases** play a central role in intracellular signal transduction. By phosphorylating proteins, they bring about alterations in biological activity that can only be reversed again by the effects of *protein phosphatases*. The interplay between protein phosphorylation by protein kinases and dephosphorylation by protein phosphatases (*interconversion*) serves to regulate the cell cycle (see p. 394) and other important processes. The protein kinase Raf is also involved in the signal transduction of insulin (see p. 388).

Tumors

A. Division behavior of cells ●

The body's cells are normally subject to strict "social" control. They only divide until they come into contact with neighboring cells; cell division then ceases due to *contact inhibition*. Exceptions to this rule include embryonic cells, cells of the intestinal epithelium (where the cells are constantly being replaced), cells in the bone marrow (where formation of blood cells takes place), and **tumor cells**. *Uncontrolled cell proliferation* is an important indicator of the presence of a tumor. While normal cells in cell culture only divide 20–60 times, tumor cells are potentially immortal and are not subject to contact inhibition.

In medicine, a distinction is made between *benign* and *malignant tumors*. Benign tumors consist of slowly growing, largely differentiated cells. By contrast, malignant tumors show rapid, invasive growth and tend to form *metastases* (dissemination of daughter lesions). The approximately 100 different types of tumor that exist are responsible for more than 20% of deaths in Europe and North America.

B. Transformation ●

The transition of a normal cell into a tumor cell is referred to as **transformation**.

Normal cells have all the characteristics of fully differentiated cells specialized for a particular function. Their division is inhibited and they are usually in the G_0 phase of the cell cycle (see p. 394). Their external shape is variable and is determined by a strongly structured cytoskeleton.

In contrast, *tumor cells* divide without inhibition and are often de-differentiated—i.e., they have acquired some of the properties of embryonic cells. The surface of these cells is altered, and this is particularly evident in a disturbance of contact inhibition by neighboring cells. The cytoskeleton of tumor cells is also restructured and often reduced, giving them a rounded shape. The nuclei of tumor cells can be atypical in terms of shape, number, and size.

Tumor markers are clinically important for detecting certain tumors. These are proteins that are formed with increasing frequency by

tumor cells (group 1) or are induced by them in other cells (group 2). Group 1 tumor markers include *tumor-associated antigens*, secreted hormones, and enzymes. The table lists a few examples.

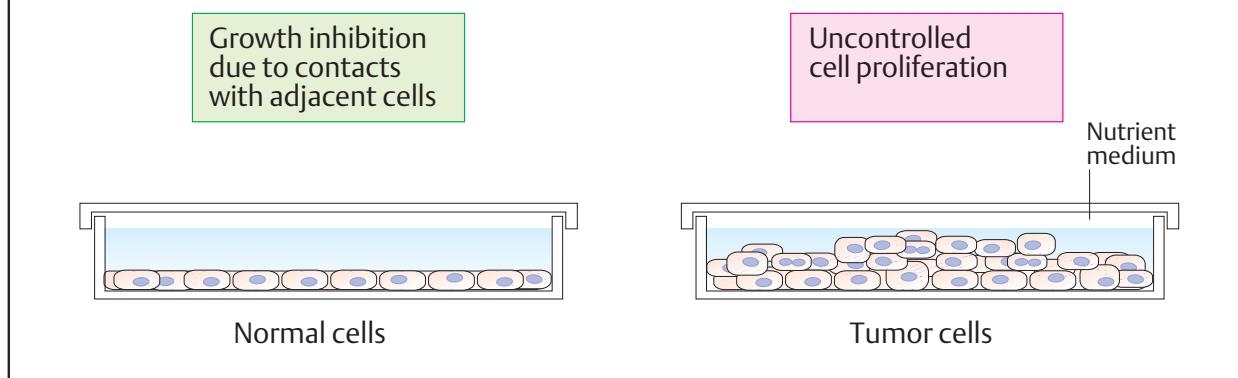
The transition from a normal to a transformed state is a process involving several steps.

1. Tumor initiation. Almost every tumor begins with damage to the DNA of an individual cell. The genetic defect is almost always caused by environmental factors. These can include tumor-inducing chemicals (*carcinogens*—e.g., components of tar from tobacco), physical processes (e.g., UV light, X-ray radiation; see p. 256), or in rare cases tumor viruses (see p. 398). Most of the approximately 10^{14} cells in the human body probably suffer this type of DNA damage during the average lifespan, but it is usually repaired again (see p. 256). It is mainly defects in **proto-oncogenes** (see p. 398) that are relevant to tumor initiation; these are the decisive cause of *transformation*. Loss of an **anti-oncogene** (a tumor-suppressor gene) can also contribute to tumor initiation.

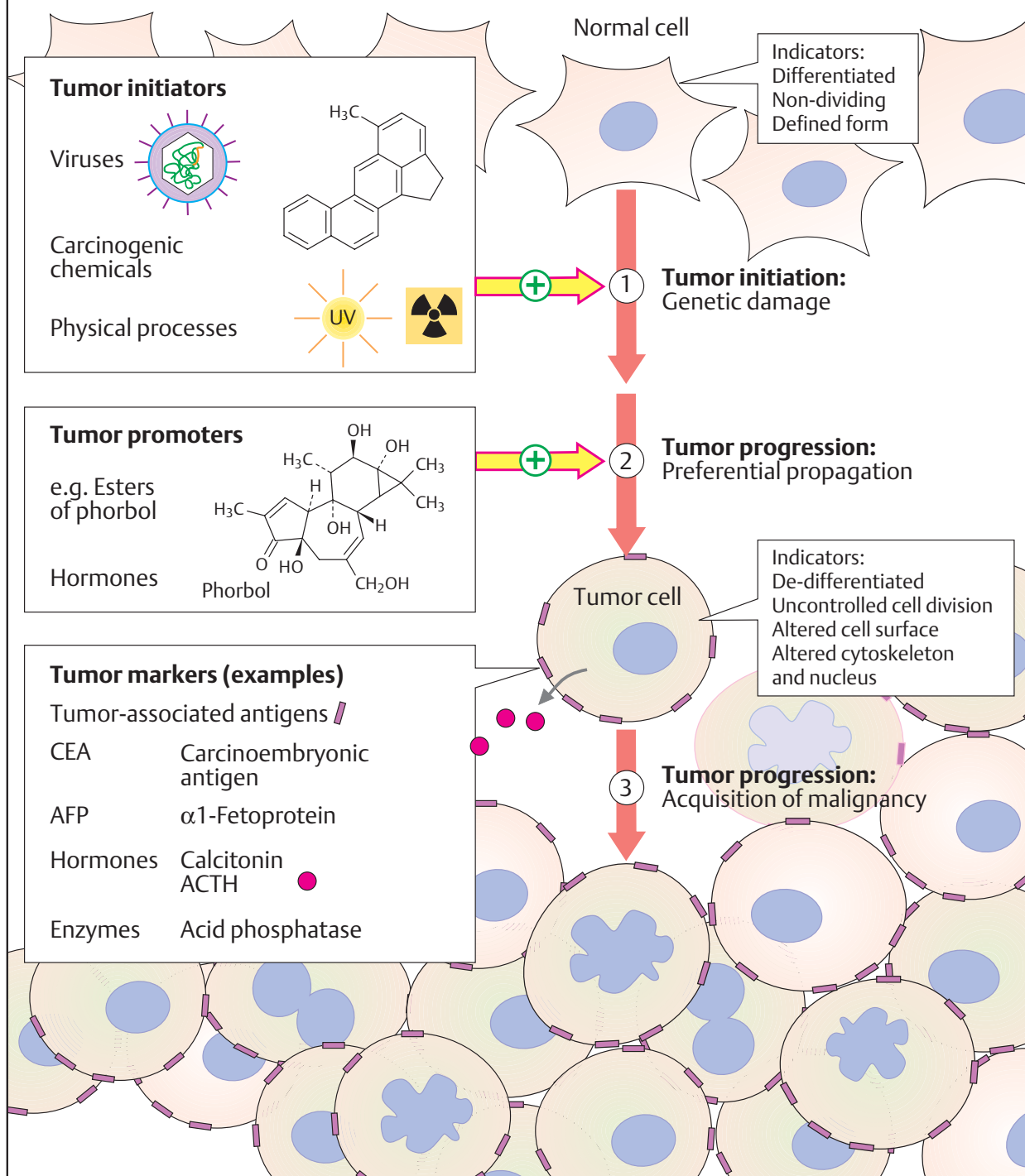
2. Tumor promotion is preferential proliferation of a cell damaged by transformation. It is a very slow process that can take many years. Certain substances are able to strongly accelerate it—e.g., *phorbol esters*. These occur in plants (e.g., *Euphorbia* species) and act as activators of protein kinase C (see p. 386).

3. Tumor progression finally leads to a macroscopically visible tumor as a result of growth. When solid tumors of this type exceed a certain size, they form their own vascular network that supplies them with blood (angiogenesis). *Collagenases* (matrix metalloproteinases, MMPs) play a special role in the metastatic process, by loosening surrounding connective tissue and thereby allowing tumor cells to disseminate and enter the bloodstream. New approaches to combating tumors have been aimed at influencing tumor angiogenesis and metastatic processes.

A. Division behavior of cells



B. Transformation



Cytostatic drugs

Tumors (see p. 400) arise from degenerated (transformed) cells that grow in an uncontrolled way as a result of genetic defects. Most transformed cells are recognized by the immune system and eliminated (see p. 294). If endogenous defense is not sufficiently effective, rapid tumor growth can occur. Attempts are then made to inhibit growth by physical or chemical treatment.

A frequently used procedure is targeted irradiation with γ -rays, which block cell reproduction due their mutagenic effect (see p. 256). Another approach is to inhibit cell growth by chemotherapy. The growth-inhibiting substances used are known as **cytostatic drugs**. Unfortunately, neither radiotherapy nor chemotherapy act selectively—i.e., they damage normal cells as well, and are therefore often associated with severe side effects.

Most cytostatic agents directly or indirectly inhibit DNA replication in the S phase of the cell cycle (see p. 394). The first group (**A**) lead to chemical changes in cellular DNA that impede transcription and replication. A second group of cytostatic agents (**B**) inhibit the synthesis of DNA precursors.

A. Alkylating agents, anthracyclines ○

Alkylating agents are compounds capable of reacting covalently with DNA bases. If a compound of this type contains *two* reactive groups, intramolecular or intermolecular *crosslinking* of the DNA double helix and “bending” of the double strand occurs. Examples of this type shown here are **cyclophosphamide** and the inorganic complex **cisplatin**. Anthracyclines such as **doxorubicin** (adriamycin) insert themselves non-covalently between the bases and thus lead to local alterations in the DNA structure (see p. 254 B).

B. Antimetabolites ○

Antimetabolites are enzyme inhibitors (see p. 96) that selectively block metabolic pathways. The majority of clinically important cytostatic drugs act on *nucleotide biosynthesis*. Many of these are modified nucleobases or nucleotides that *competitively* inhibit their target enzymes (see p. 96). Many are also incorporated into the DNA, thereby preventing replication.

The cytostatic drugs administered (indicated by a syringe in the illustration) are often not active themselves but are only converted into the actual active agent in the metabolism. This also applies to the adenine analogue **6-mercaptopurine**, which is initially converted to the mononucleotide tIMP (thioinosine monophosphate). Via several intermediate steps, tIMP gives rise to tdGTP, which is incorporated into the DNA and leads to cross-links and other anomalies in it. The second effective metabolite of 6-mercaptopurine is S-methylated tIMP, an inhibitor of *amidophosphoribosyl transferase* (see p. 188).

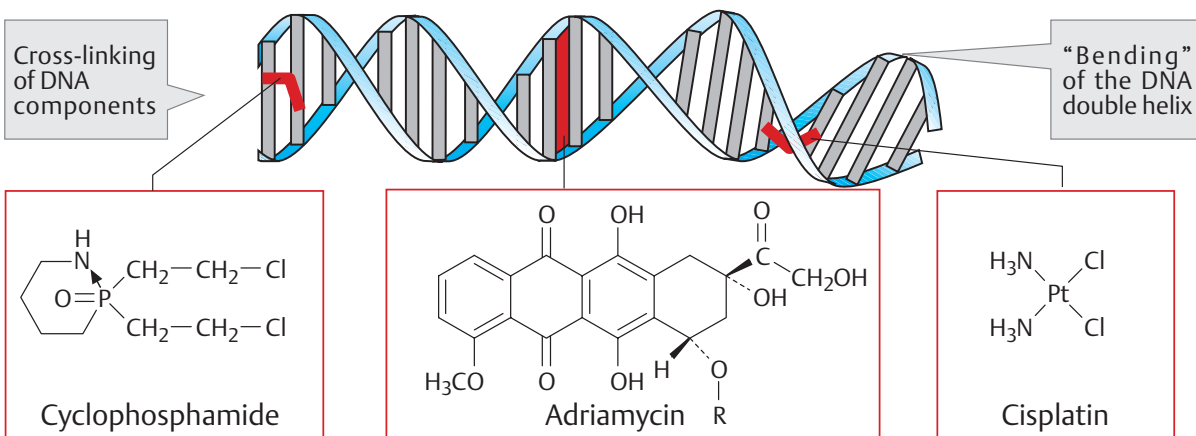
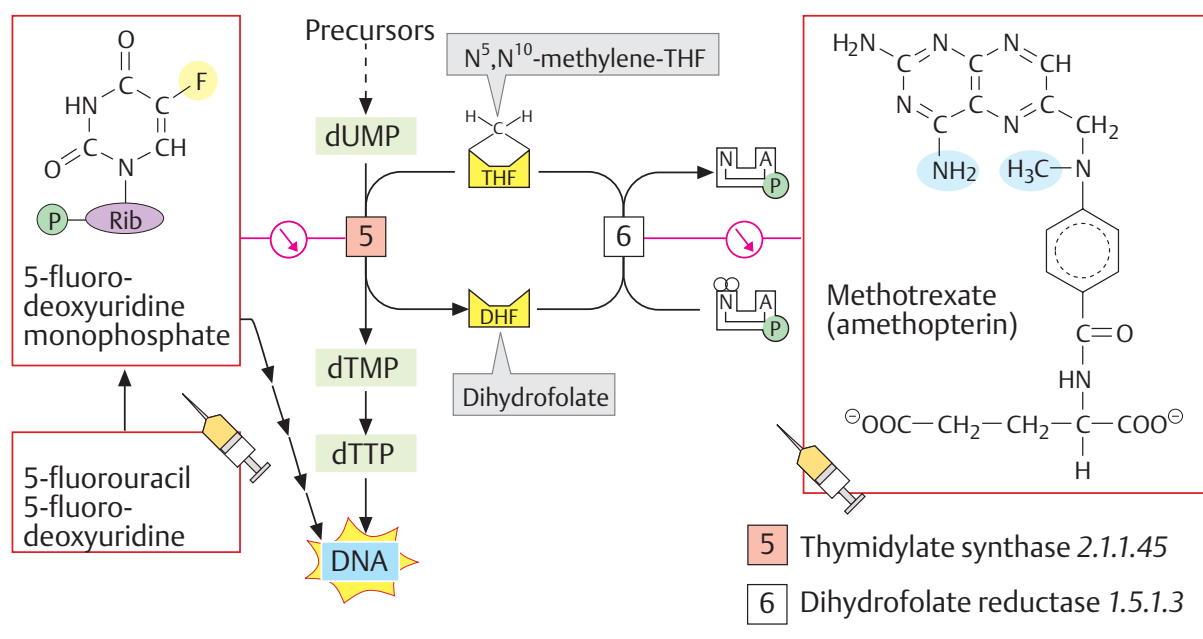
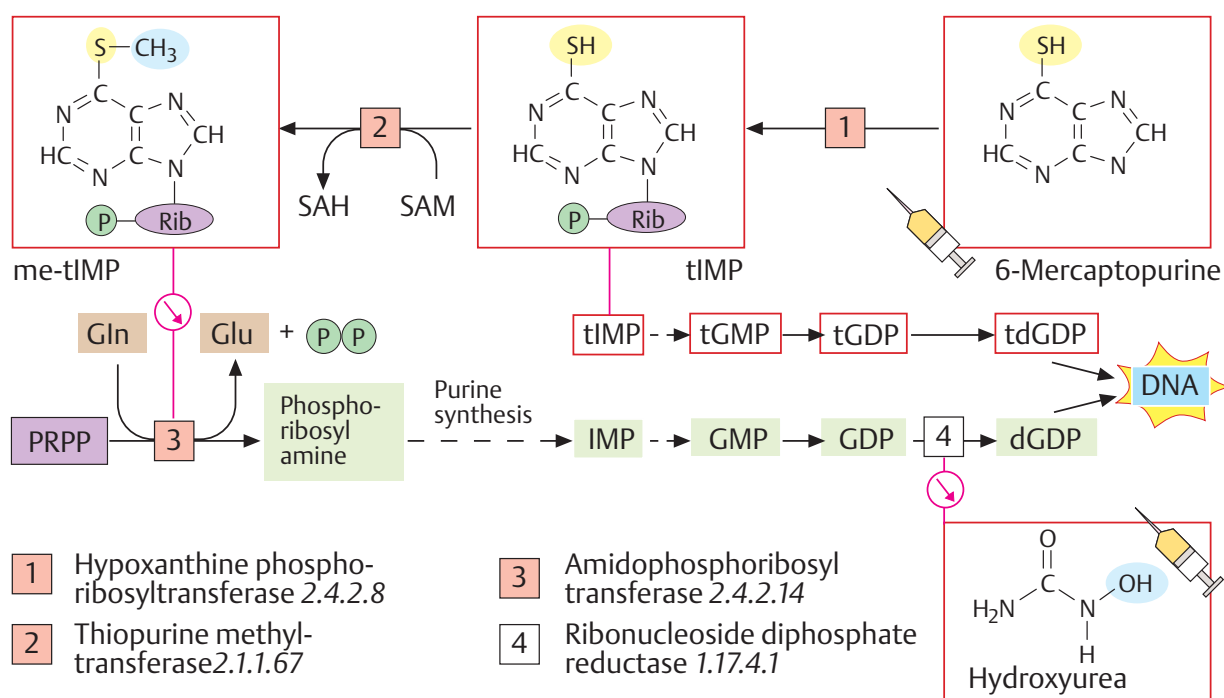
Hydroxyurea selectively inhibits *ribonucleotide reductase* (see p. 190). As a radical scavenger, it removes the tyrosine radicals that are indispensable for the functioning of the reductase.

Two other important cytostatic agents target the synthesis of DNA-typical thymine, which takes place at the level of the deoxymononucleotide (see p. 190). The deoxymononucleotide formed by **5-fluorouracil** or the corresponding nucleoside inhibits *thymidylate synthase*. This inhibition is based on the fact that the fluorine atom in the pyrimidine ring cannot be substituted by a methyl group. In addition, the fluorine analogue is also incorporated into the DNA.

Dihydrofolate reductase acts as an auxiliary enzyme for thymidylate synthase. It is involved in the regeneration of the coenzyme N^5,N^{10} -methylene-THF, initially reducing DHF to THF with NADPH as the reductant (see p. 418). The folic acid analogue **methotrexate**, a frequently used cytostatic agent, is an extremely effective competitive inhibitor of dihydrofolate reductase. It leads to the depletion of N^5,N^{10} -methylene-THF in the cells and thus to cessation of DNA synthesis.

Further information

To reduce the side effects of cytostatic agents, new approaches are currently being developed on the basis of **gene therapy** (see p. 264). Attempts are being made, for example, to administer drugs in the form of precursors (known as prodrugs), which only become active in the tumor itself (“tumor targeting”).

A. Alkylating agents, anthracyclines**B. Antimetabolites**

Viruses

Viruses are *parasitic nucleoprotein complexes*. They often consist of only a single nucleic acid molecule (DNA or RNA, never both) and a protein coat. Viruses have *no metabolism of their own*, and can therefore only replicate themselves with the help of host cells. They are therefore not regarded as independent organisms. Viruses that damage the host cell when they replicate are *pathogens*. Diseases caused by viruses include AIDS, rabies, poliomyelitis, measles, German measles, smallpox, influenza, and the common cold.

A. Viruses: examples ●

Only a few examples from the large number of known viruses are illustrated here. They are all shown on the same scale.

Viruses that only replicate in bacteria are known as **bacteriophages** (or “phages” for short). An example of a phage with a simple structure is **M13**. It consists of a single-stranded DNA molecule (ssDNA) of about 7000 bp with a coat made up of 2700 helically arranged protein subunits. The coat of a virus is referred to as a *capsid*, and the complete structure as a *nucleocapsid*. In genetic engineering, M13 is important as a *vector* for foreign DNA (see p. 258).

The phage **T4** (bottom left), one of the largest viruses known, has a much more complex structure with around 170 000 base pairs (bp) of double-stranded DNA (dsDNA) contained within its “head.”

The **tobacco mosaic virus** (center right), a plant pathogen, has a structure similar to that of M13, but contains ssRNA instead of DNA. The **poliovirus**, which causes poliomyelitis, is also an RNA virus. In the **influenza virus**, the pathogen that causes viral flu, the nucleocapsid is additionally surrounded by a *coat* derived from the plasma membrane of the host cell (C). The coat carries viral proteins that are involved in the infection process.

B. Capsid of the rhinovirus ○

Rhinoviruses cause the common cold. In these viruses, the capsid is shaped like an *icosahedron*—i. e., an object made up of 20 equilateral triangles. Its surface is formed from three different proteins, which associate with one an-

other to form pentamers and hexamers. In all, 60 protein molecules are involved in the structure of the capsid.

C. Life cycle of HIV ●

The *human immunodeficiency virus (HIV)* causes the immunodeficiency disease known as **AIDS** (*acquired immune deficiency syndrome*). The structure of this virus is similar to that of the influenza virus (A).

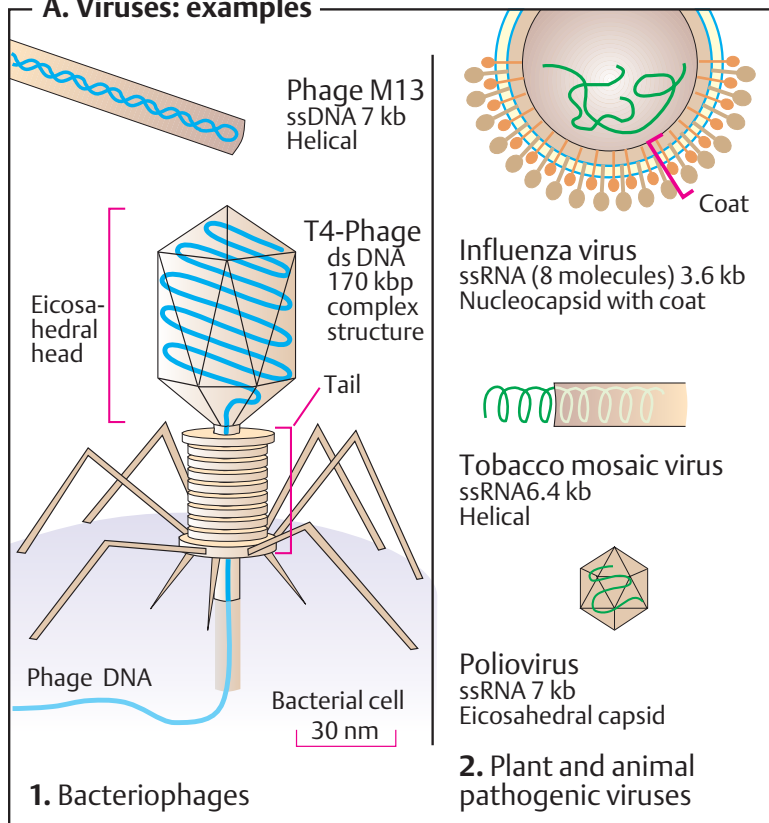
The HIV genome consists of two molecules of ssRNA (each 9.2 kb). It is enclosed by a double-layered capsid and a protein-containing coating membrane. HIV mainly infects T helper cells (see p. 294) and can thereby lead to failure of the immune system in the longer term.

During infection (1), the virus’s coating membrane fuses with the target cell’s plasma membrane, and the core of the nucleocapsid enters the cytoplasm (2). In the cytoplasm, the viral RNA is initially transcribed into an RNA/DNA hybrid (3) and then into dsDNA (4). Both of these reactions are catalyzed by *reverse transcriptase*, an enzyme deriving from the virus. The dsDNA formed is integrated into the host cell genome (5), where it can remain in an inactive state for a long time.

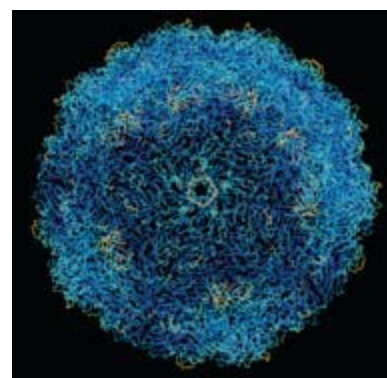
When viral replication occurs, the DNA segment corresponding to the viral genome is first transcribed by host cell enzymes (6). This gives rise not only to viral ssRNA, but also to transcription of mRNAs for precursors of the viral proteins (7). These precursors are integrated into the plasma membrane (8, 9) before undergoing proteolytic modification (10). The cycle is completed by the release of new virus particles (11).

The group of RNA viruses to which HIV belongs are called **retroviruses**, because DNA is produced from RNA in their replication cycle—the reverse of the usual direction of transcription (DNA → RNA).

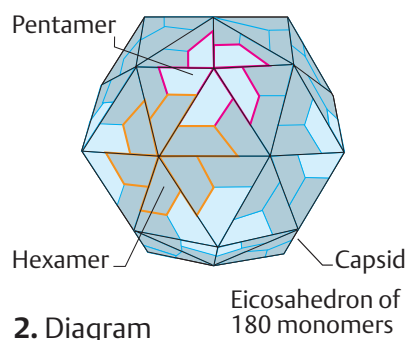
A. Viruses: examples



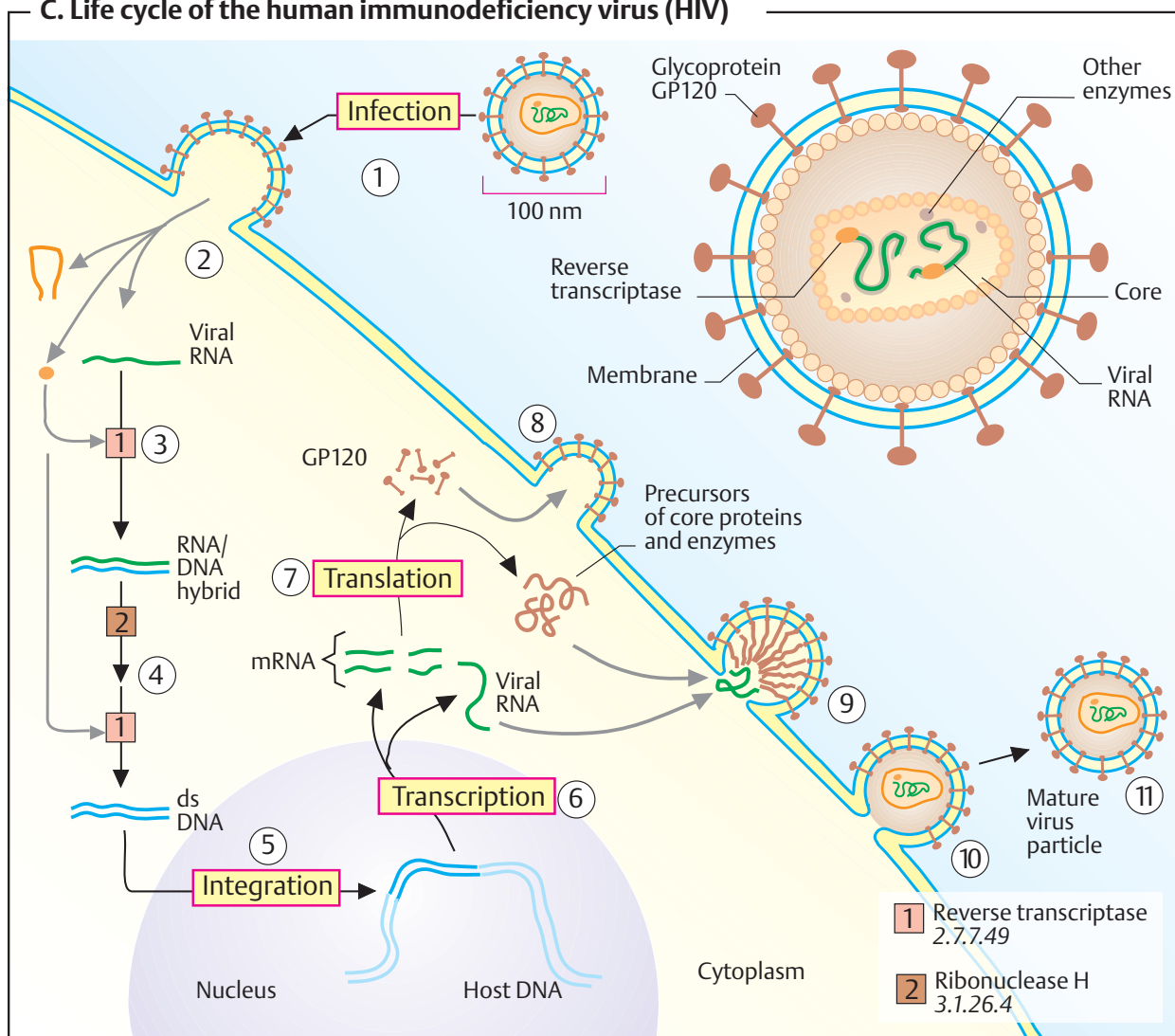
B. Capsid of the rhinovirus



1. Structure



C. Life cycle of the human immunodeficiency virus (HIV)



Metabolic charts

Explanations

The following 13 plates (pp. 407–419) provide a concise schematic overview of the most important metabolic pathways. Explanatory text is deliberately omitted from them.

These “charts”:

- Contain details of metabolic pathways that are only shown in outline in the main text for reasons of space. This applies in particular to the synthesis and degradation of the amino acids and nucleotides, and for some aspects of carbohydrate and lipid metabolism.
- Offer a quick overview of a specific pathway, the metabolites that arise in it, and the enzymes involved.
- Can be used for reference purposes and for revising material previously learned.

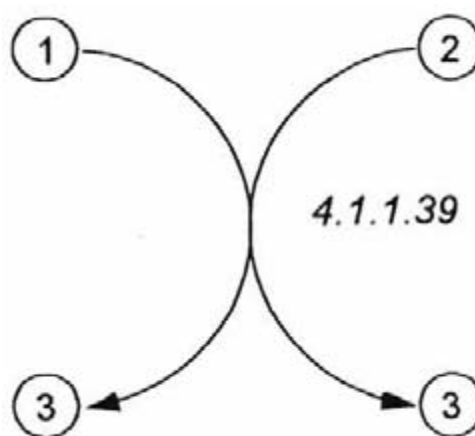
The most important **intermediates** are shown with numbers in the charts. The corresponding compounds can be identified using the table on the same page.

In addition, at each step the four-figure EC number (see p. 88) for the **enzyme** responsible for a reaction is given in *italics*. The enzyme name and its systematic classification in the system used by the *Enzyme Catalogue* are available in the following **annotated enzyme list** (pp. 420–430), in which all of the enzymes mentioned in this book are listed according to their EC number. The book's index is helpful when looking for a specific enzyme in the text.

In reactions that involve **coenzymes**, the coenzyme names are also given (sometimes in simplified form). Particularly important starting, intermediate, or end products are given with the full name, or as formulae.

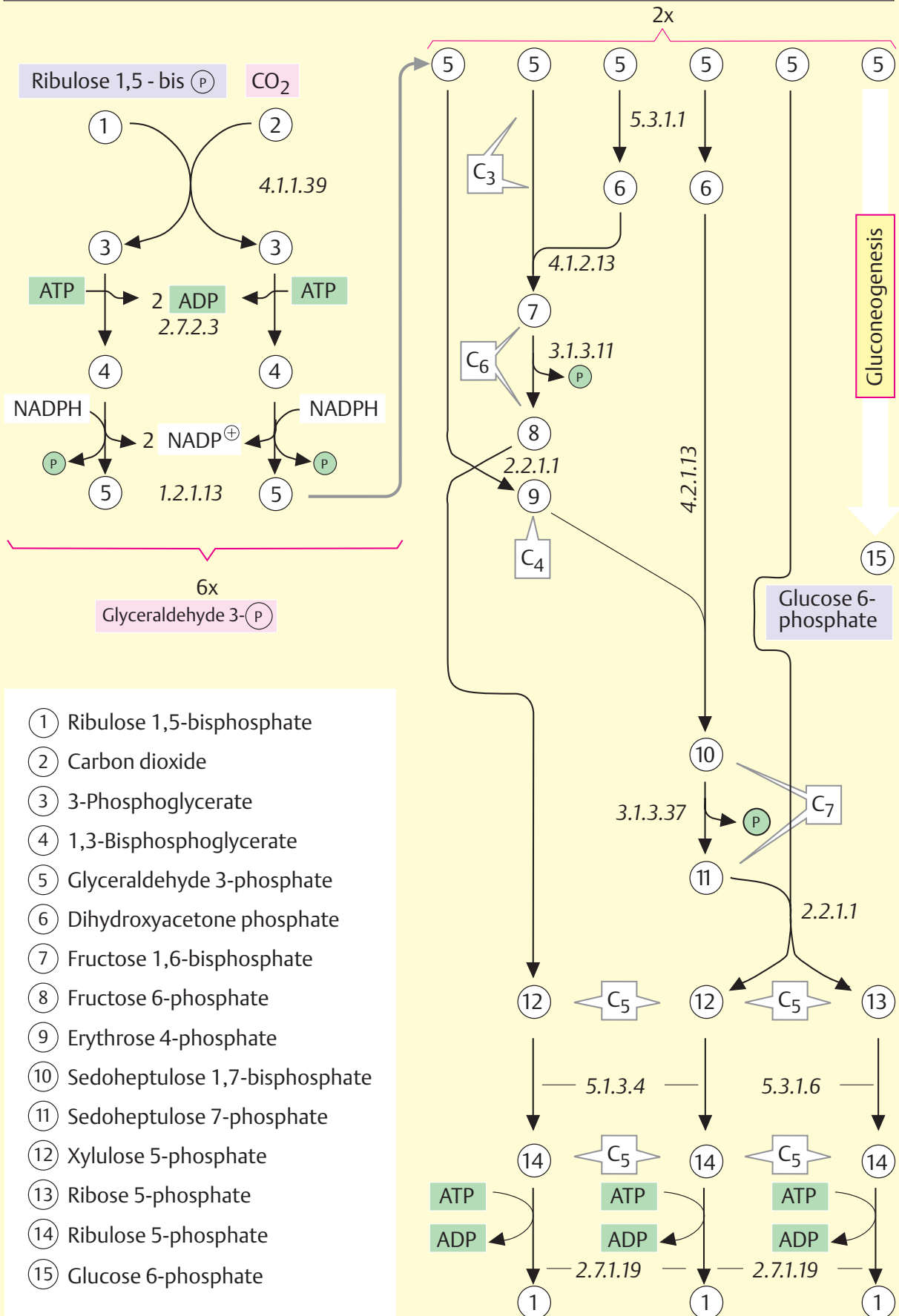
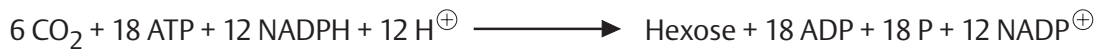
Example

On p. 407, the initial step of the dark reactions in plant photosynthesis (in the Calvin cycle) is shown at the top left.

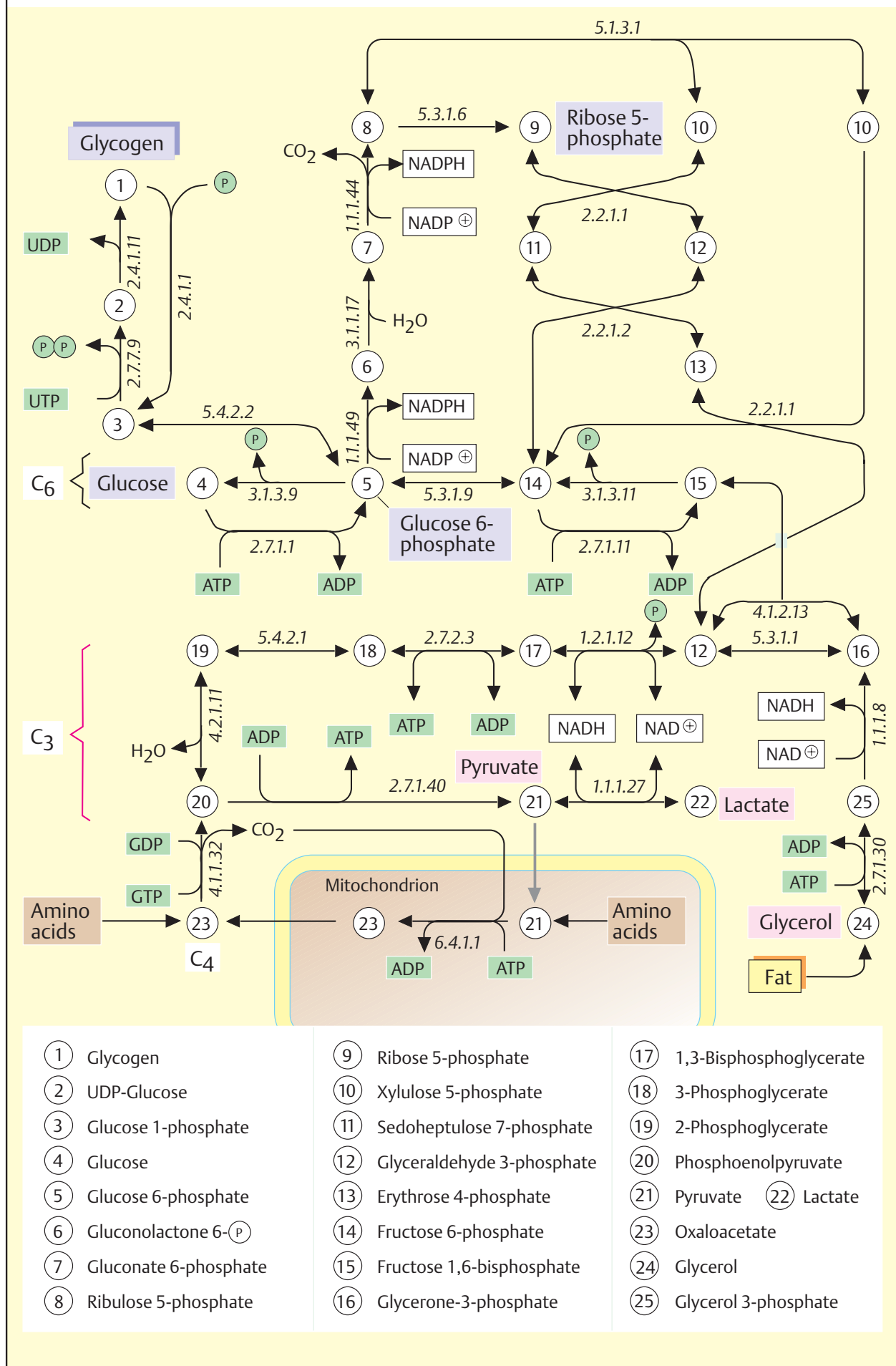


In this reaction, one molecule of ribulose-1,5-bisphosphate (metabolite 1) and one molecule of CO₂ (metabolite 2) give rise to two molecules of 3-phosphoglycerate (metabolite 3). The enzyme responsible has the EC number *4.1.1.39*. The annotated enzyme list shows that this refers to *ribulose biphosphate carboxylase* (“rubisco” for short). Rubisco belongs to enzyme class 4 (the lyases) and, within that group, to subclass 4.1 (the carboxy-lyases). It contains copper as a cofactor ([Cu]).

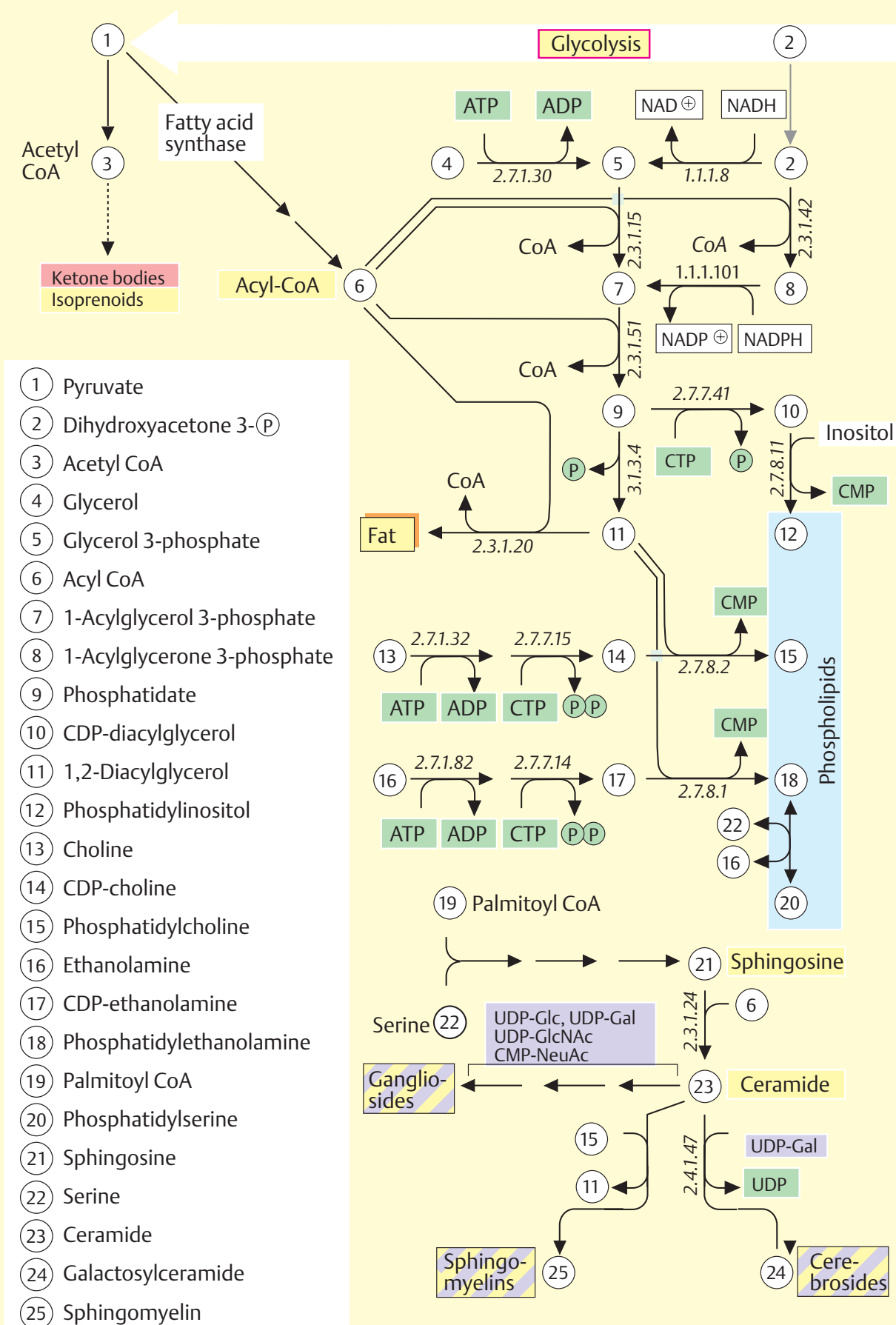
A. Calvin cycle (plant chloroplasts)



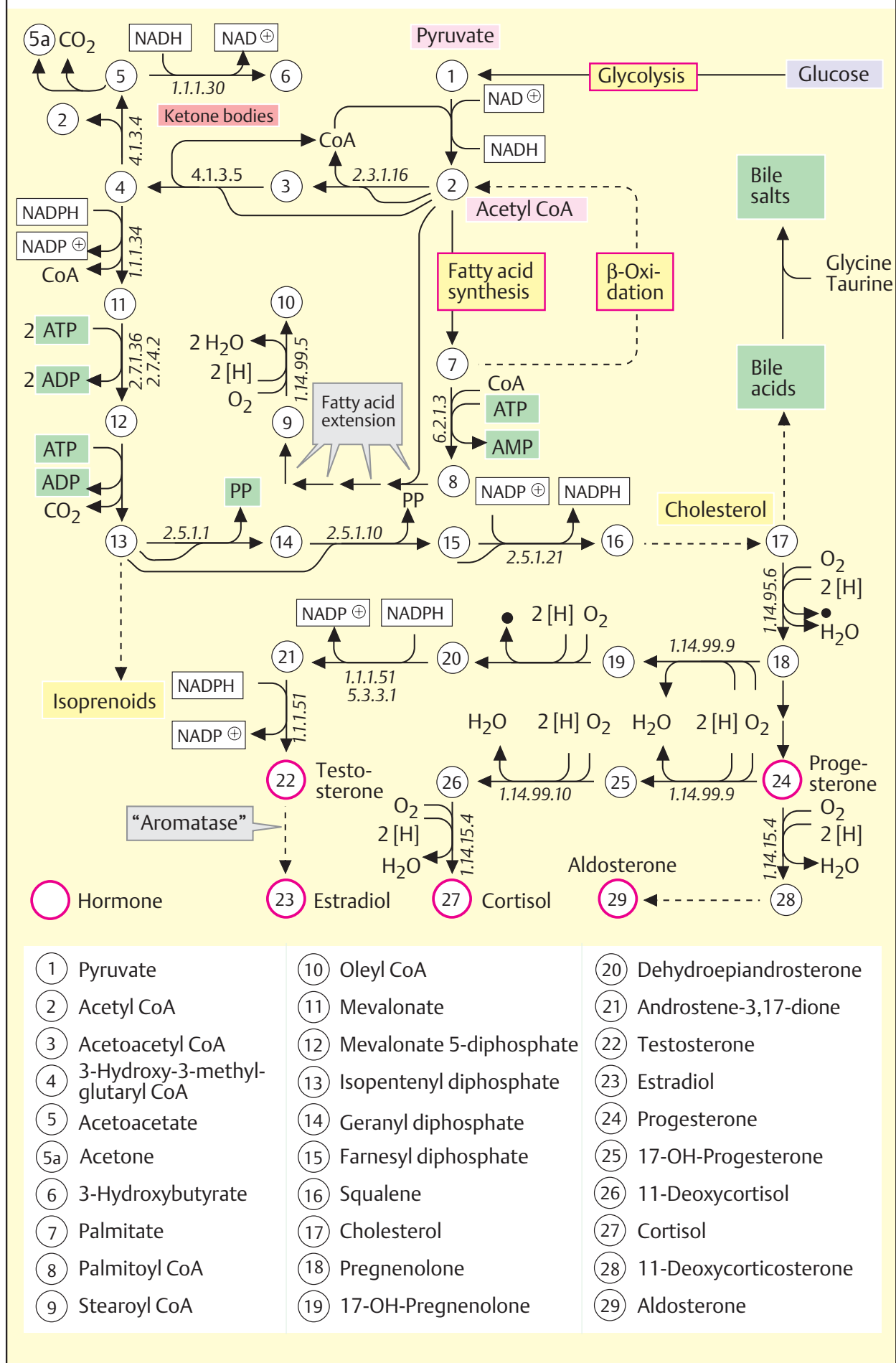
A. Carbohydrate metabolism



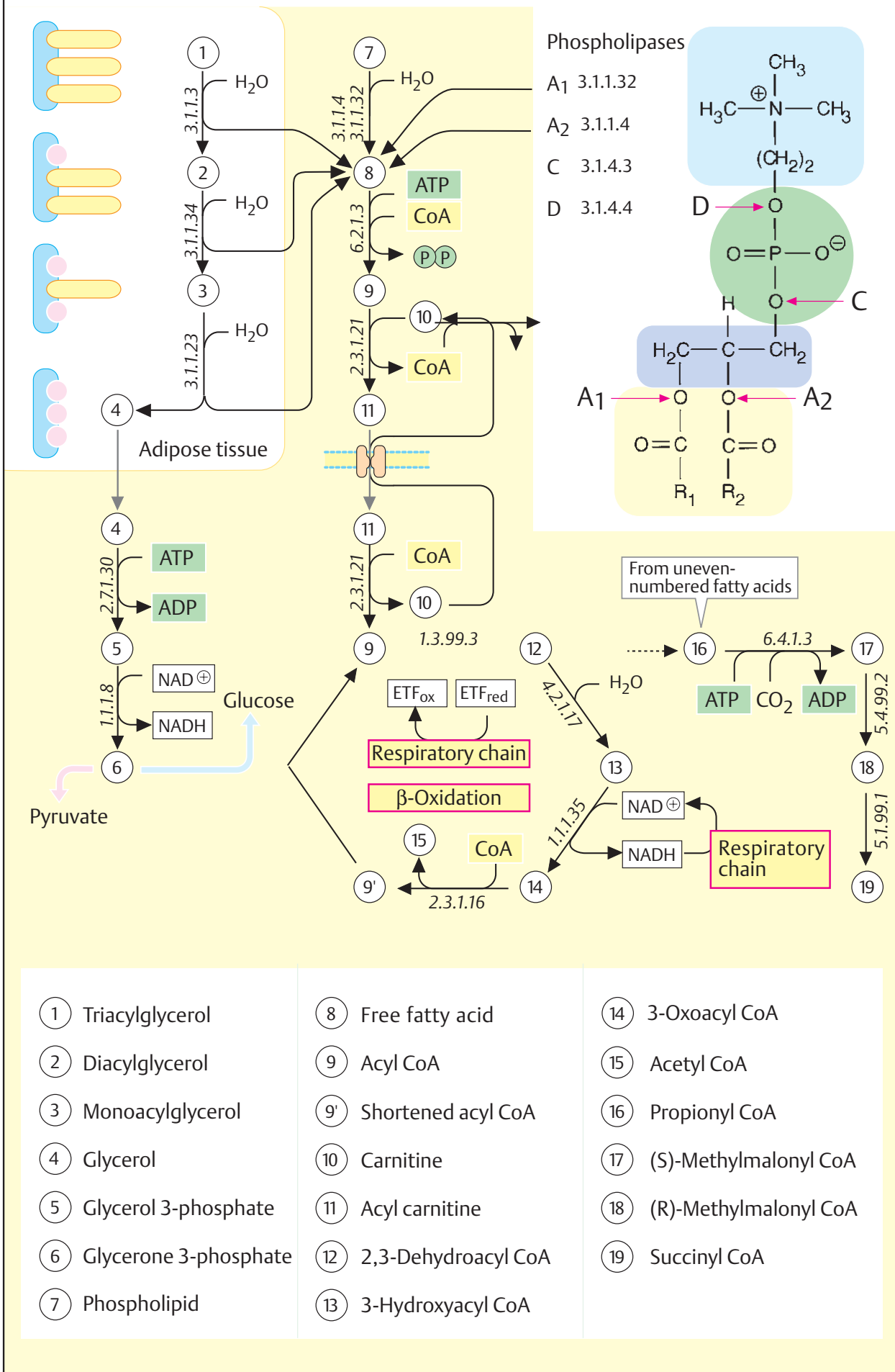
A. Biosynthesis of fats and membrane lipids



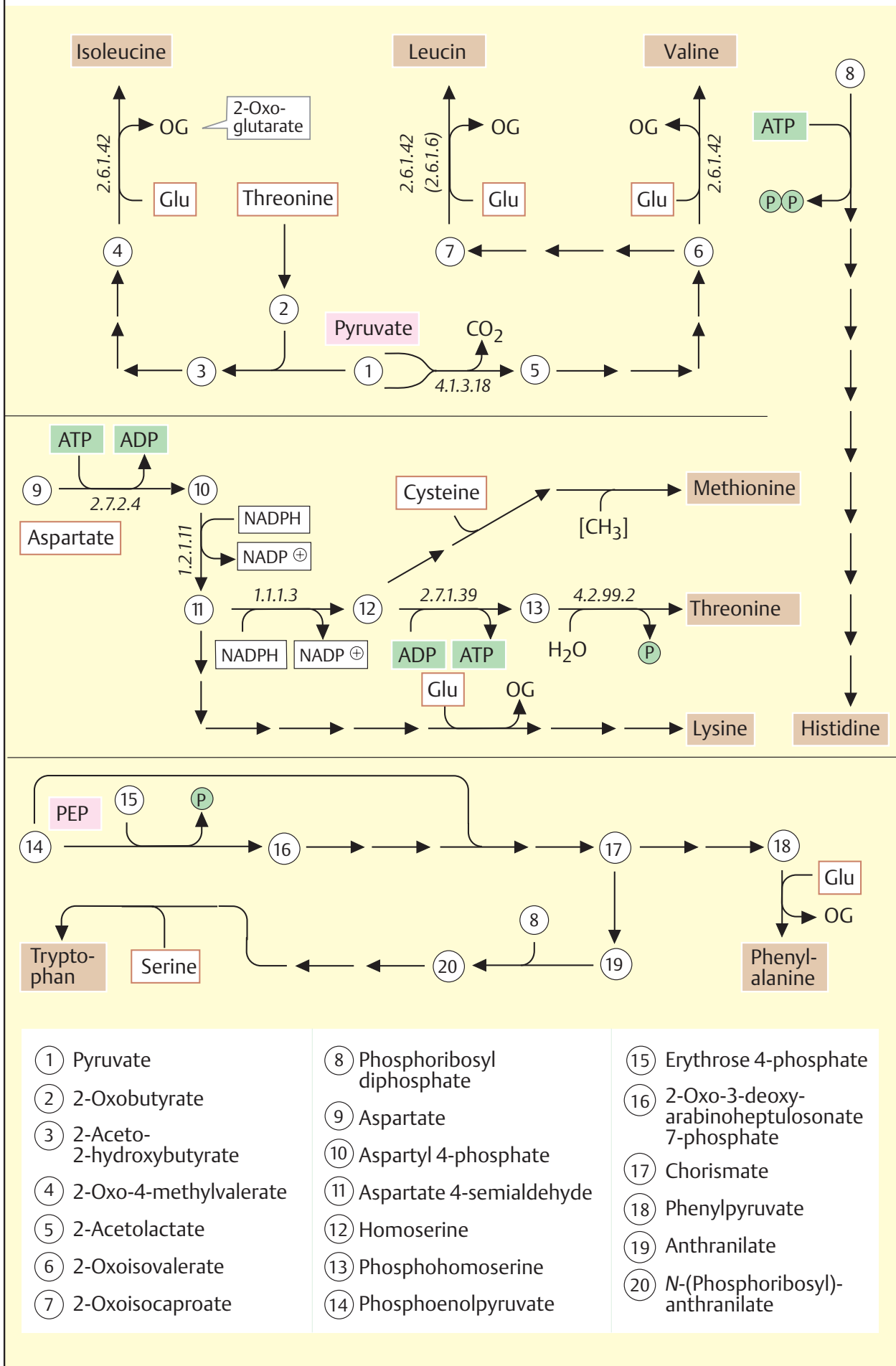
A. Synthesis of ketone bodies and steroids



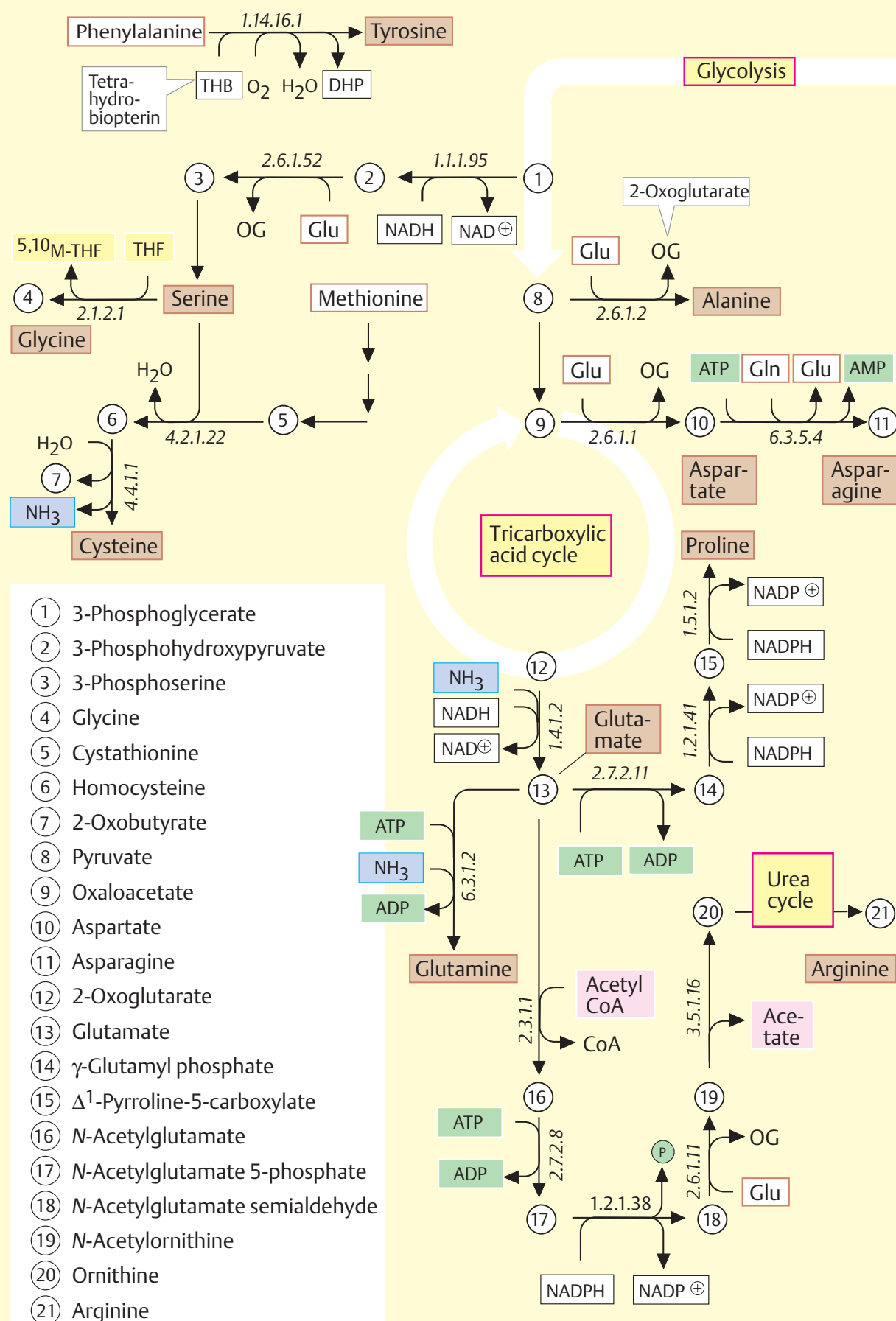
A. Degradation of fats and phospholipids



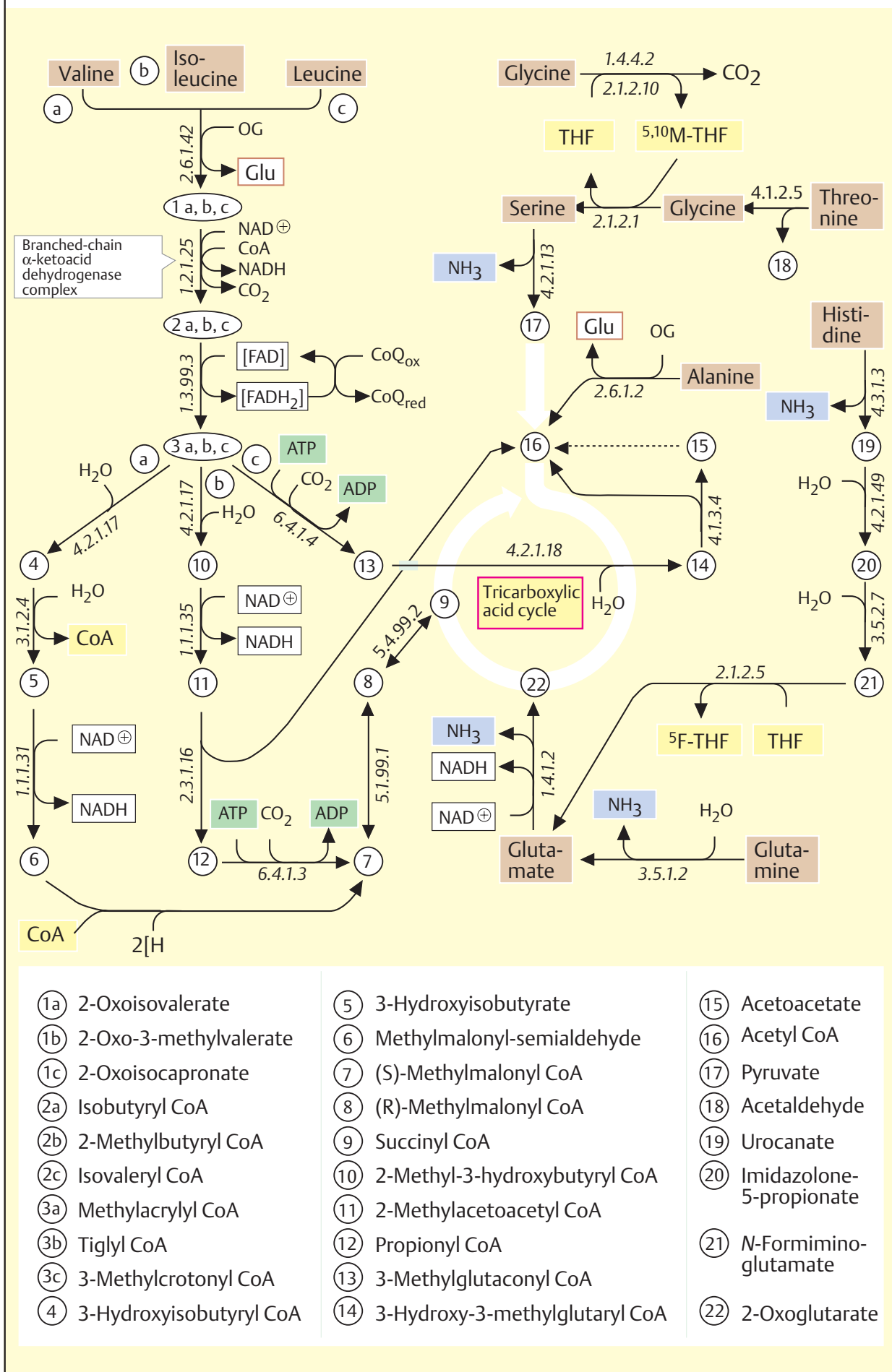
A. Biosynthesis of the essential amino acids



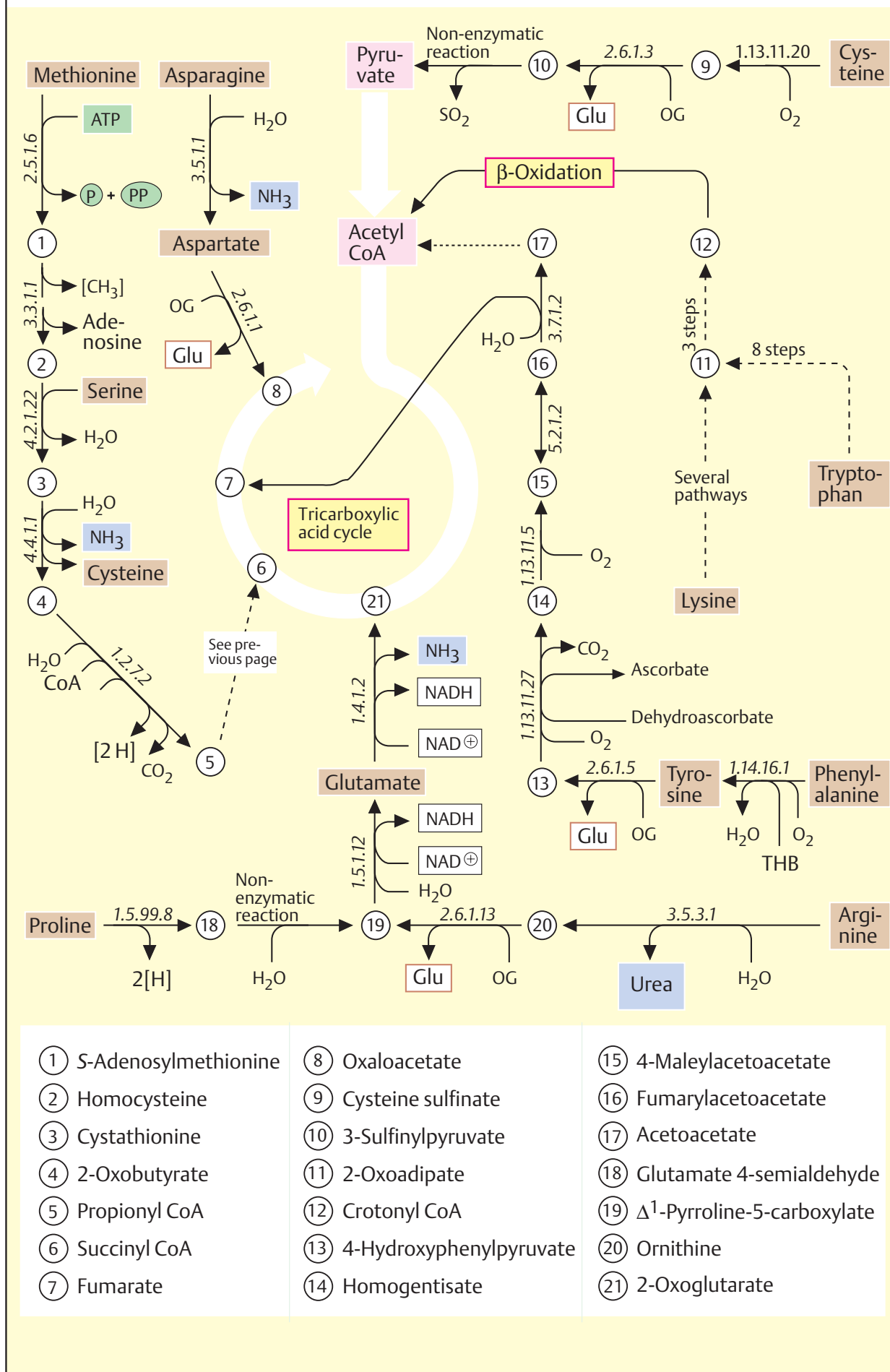
A. Biosynthesis of the non-essential amino acids



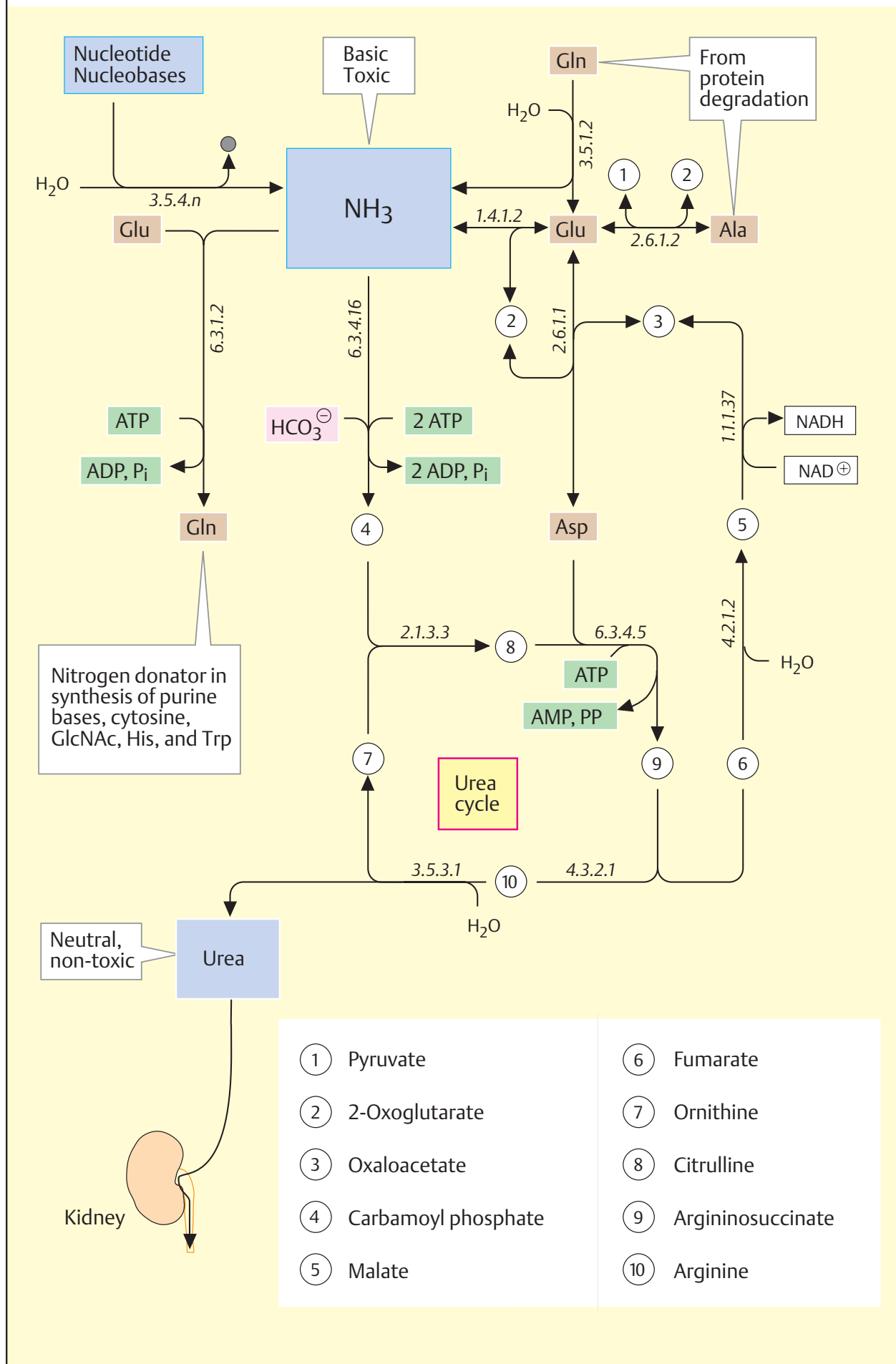
A. Amino acid degradation I



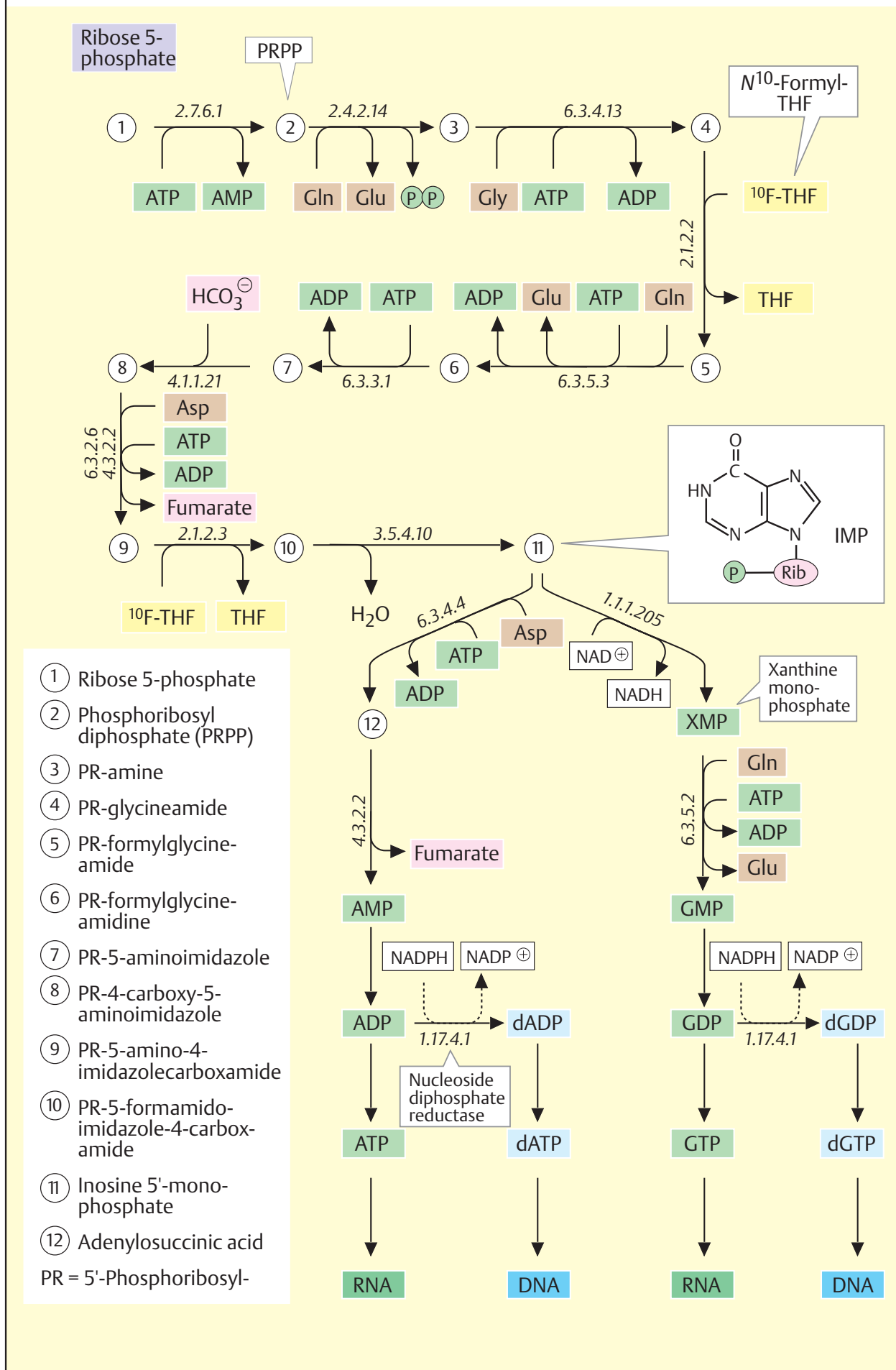
A. Amino acid degradation II



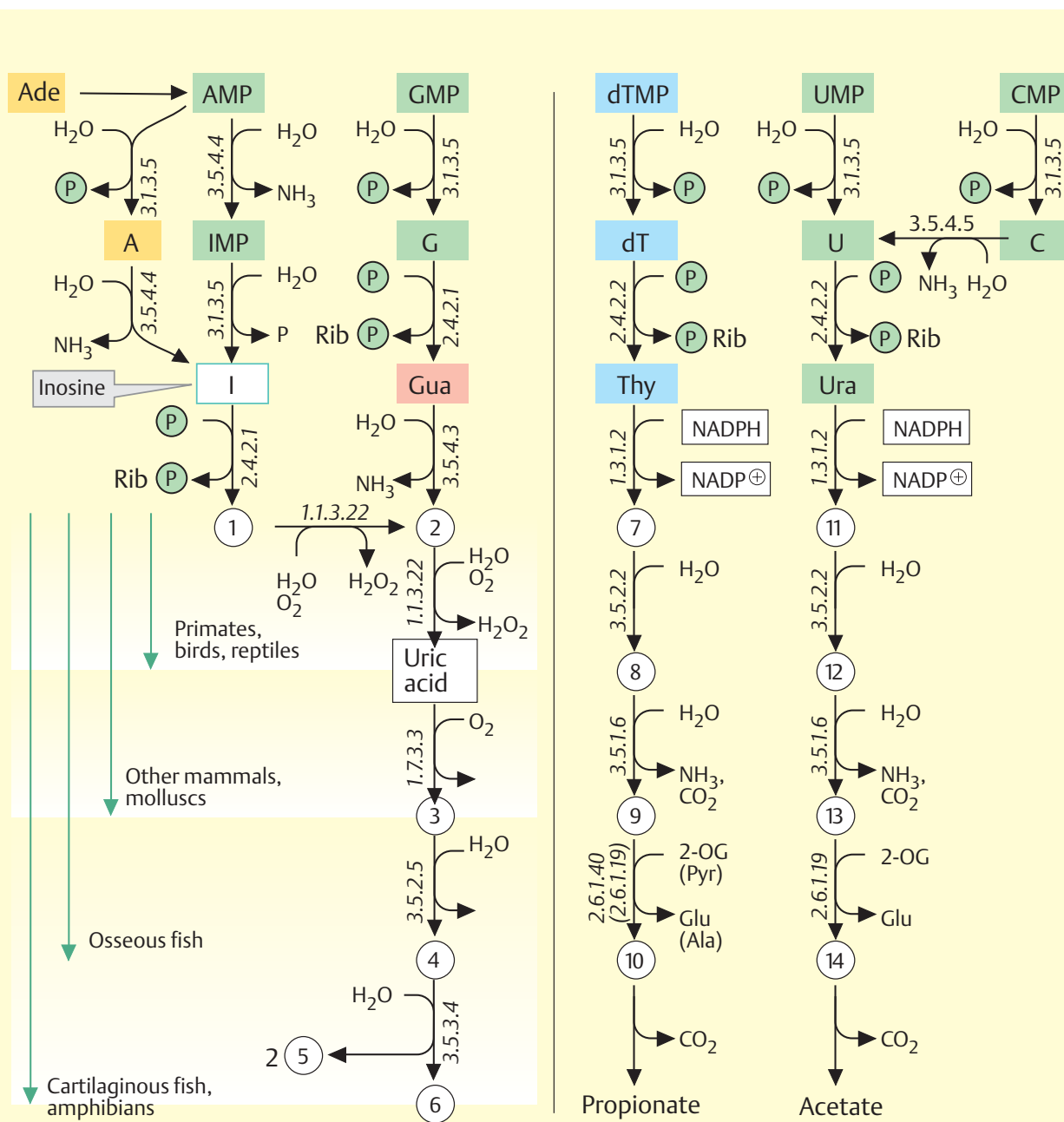
A. Ammonia metabolism



A. Biosynthesis of purine nucleotides



A. Nucleotide degradation



Annotated enzyme list

Only the enzymes mentioned in this atlas are listed here, from among the more than 2000 enzymes known. The enzyme names are based on the IUBMB's official *Enzyme nomenclature* 1992. The additions shown in round brackets belong to the enzyme name, while prosthetic groups and other cofactors are enclosed in square brackets. Common names of enzyme groups are given in italics, and trivial names are shown in quotation marks.

Class 1: Oxidoreductases (catalyze reduction-oxidation reactions)

Subclass 1.n: What is the electron donor?

Sub-subclass 1.n.n: What is the electron acceptor?

1.1 A –CH–OH group is the donor

1.1.1 NAD(P)⁺ is the acceptor (*dehydrogenases, reductases*)

- 1.1.1.1 Alcohol dehydrogenase [Zn²⁺]
- 1.1.1.3 Homoserine dehydrogenase
- 1.1.1.8 Glycerol 3-phosphate dehydrogenase (NAD⁺)
- 1.1.1.21 Aldehyde reductase
- 1.1.1.27 Lactate dehydrogenase
- 1.1.1.30 3-Hydroxybutyrate dehydrogenase
- 1.1.1.31 3-Hydroxyisobutyrate dehydrogenase
- 1.1.1.34 Hydroxymethylglutaryl-CoA reductase (NADPH)
- 1.1.1.35 3-Hydroxyacyl-CoA dehydrogenase
- 1.1.1.37 Malate dehydrogenase
- 1.1.1.40 Malate dehydrogenase (oxaloacetate-decarboxylating, NADP⁺)—"malic enzyme"
- 1.1.1.41 Isocitrate dehydrogenase (NAD⁺)
- 1.1.1.42 Isocitrate dehydrogenase (NADP⁺)
- 1.1.1.44 Phosphogluconate dehydrogenase (decarboxylating)
- 1.1.1.49 Glucose 6-phosphate 1-dehydrogenase
- 1.1.1.51 3(or 17)β-Hydroxysteroid dehydrogenase
- 1.1.1.95 Phosphoglycerate dehydrogenase
- 1.1.1.100 3-Oxoacyl-[ACP] reductase
- 1.1.1.101 Acylglycerone phosphate reductase
- 1.1.1.105 Retinol dehydrogenase
- 1.1.1.145 3β-Hydroxy-Δ⁵-steroid dehydrogenase
- 1.1.1.205 IMP dehydrogenase

1.1.3 Molecular oxygen is the acceptor (*oxidases*)

- 1.1.3.4 Glucose oxidase [FAD]
- 1.1.3.8 L-Gulonolactone oxidase
- 1.1.3.22 Xanthine oxidase [Fe, Mo, FAD]
- 1.1.99.5 Glycerol-3-phosphate dehydrogenase (FAD)

1.2 An aldehyde or keto group is the donor

1.2.1 NAD(P)⁺ is the acceptor (*dehydrogenases*)

- 1.2.1.3 Aldehyde dehydrogenase (NAD⁺)
- 1.2.1.11 Aspartate semialdehyde dehydrogenase
- 1.2.1.12 Glyceraldehyde 3-phosphate dehydrogenase
- 1.2.1.13 Glyceraldehyde 3-phosphate dehydrogenase (NADP⁺) (phosphorylating)
- 1.2.1.24 Succinate semialdehyde dehydrogenase
- 1.2.1.25 2-Oxoisovalerate dehydrogenase (acylating)
- 1.2.1.38 N-Acetyl-γ-glutamylphosphate reductase
- 1.2.1.41 Glutamylphosphate reductase

- 1.2.4 A disulfide is the acceptor**
 1.2.4.1 Pyruvate dehydrogenase (lipoamide) [TPP]
 1.2.4.2 Oxoglutarate dehydrogenase (lipoamide) [TPP]
- 1.2.7 An Fe/S protein is the acceptor**
 1.2.7.2 2-Oxobutyrate synthase
- 1.3 A –CH–CH– group is the donor**
 1.3.1.10 Enoyl-[ACP] reductase (NADPH)
 1.3.1.24 Biliverdin reductase
 1.3.1.34 2,4-Dienoyl-CoA reductase
 1.3.5.1 Succinate dehydrogenase (ubiquinone) [FAD, Fe₂S₂, Fe₄S₄], “complex II”
 1.3.99.3 Acyl-CoA dehydrogenase [FAD]
 1.3.99.11 Dihydroorotate dehydrogenase [FMN]
- 1.4 A –CH–NH₂ group is the donor**
 1.4.1.2 Glutamate dehydrogenase
 1.4.3.4 Amine oxidase [FAD], “monoamine oxidase (MAO)”
 1.4.3.13 Protein lysine 6-oxidase [Cu]
 1.4.4.2 Glycine dehydrogenase (decarboxylating) [PLP]
- 1.5 A –CH–NH group is the donor**
 1.5.1.2 Pyrroline-5-carboxylate reductase
 1.5.1.3 Dihydrofolate reductase
 1.5.1.5 Methylenetetrahydrofolate dehydrogenase (NADP⁺)
 1.5.1.12 1-Pyrroline-5-carboxylate dehydrogenase
 1.5.1.20 Methylenetetrahydrofolate reductase (NADPH) [FAD]
 1.5.5.1 Electron-transferring flavoprotein (ETF) dehydrogenase [Fe₄S₄]
 1.5.99.8 Proline dehydrogenase [FAD]
- 1.6 NAD(P)H is the donor**
 1.6.4.2 Glutathione reductase (NADPH) [FAD]
 1.6.4.5 Thioredoxin reductase (NADPH) [FAD]
 1.6.5.3 NADH dehydrogenase (ubiquinone) [FAD, Fe₂S₂, Fe₄S₄]—“complex I”
- 1.8 A sulfur group is the donor**
 1.8.1.4 Dihydrolipoamide dehydrogenase [FAD]
- 1.9 A heme group is the donor**
 1.9.3.1 Cytochrome c oxidase [heme, Cu, Zn] – “cytochrome oxidase,” “complex IV”
- 1.10 A diphenol is the donor**
 1.10.2.2 Ubiquinol cytochrome c reductase [heme, Fe₂S₂]—“complex III”
- 1.11 A peroxide is the acceptor (*peroxidases*)**
 1.11.1.6 Catalase [heme]
 1.11.1.7 Peroxidase [heme]
 1.11.1.9 Glutathione peroxidase [Se]
 1.11.1.12 Lipid hydroperoxide glutathione peroxidase [Se]
- 1.13 Molecular oxygen is incorporated into the electron donor (*oxygenases*)**
1.13.11 One donor, both O atoms are incorporated (*dioxygenases*)
 1.13.11.5 Homogentisate 1,2-dioxygenase [Fe]
 1.13.11.20 Cysteine dioxygenase [Fe]
 1.13.11.27 4-Hydroxyphenylpyruvate dioxygenase [ascorbate]
 1.13.11.n Arachidonate lipoxygenases

- 1.14 Two donors, one O atom is incorporated into both** (*monooxygenases, hydroxylases*)
- 1.14.11.2 Procollagen proline 4-dioxygenase [Fe, ascorbate]—"proline hydroxylase"
 - 1.14.11.4 Procollagen lysine 5-dioxygenase [Fe, ascorbate]—"lysine hydroxylase"
 - 1.14.13.13 Calcidol 1-monooxygenase [heme]
 - 1.14.15.4 Steroid 11 β -monooxygenase [heme]
 - 1.14.15.6 Cholesterol monooxygenase (side-chain-cleaving) [heme]
 - 1.14.16.1 Phenylalanine 4-monooxygenase [Fe, tetrahydrobiopterin]
 - 1.14.16.2 Tyrosine 3-monooxygenase [Fe, tetrahydrobiopterin]
 - 1.14.17.1 Dopamine β -monooxygenase [Cu]
 - 1.14.99.1 Prostaglandin H-synthase [heme]
 - 1.14.99.3 Heme oxygenase (decyclizing) [heme]
 - 1.14.99.5 Stearoyl-CoA desaturase [heme]
 - 1.14.99.9 Steroid 17 α -monooxygenase [heme]
 - 1.14.99.10 Steroid 21-monooxygenase [heme]
- 1.15 A superoxide radical is the acceptor**
- 1.15.1.1 Superoxide dismutase
- 1.17 A -CH₂ group is the donor**
- 1.17.4.1 Ribonucleoside diphosphate reductase [Fe]—"ribonucleotide reductase"
- 1.18 Reduced ferredoxin is the donor**
- 1.18.1.2 Ferredoxin-NADP⁺ reductase [FAD]
 - 1.18.6.1 Nitrogenase [Fe, Mo, Fe₄S₄]

Class 2: Transferases (catalyze the transfer of groups from one molecule to another)

Subclass 2.n: Which group is transferred?

- 2.1 A C₁ group is transferred**
- 2.1.1 A methyl group**
- 2.1.1.2 Guanidinoacetate N-methyltransferase
 - 2.1.1.6 Catechol O-methyltransferase
 - 2.1.1.13 5-Methyltetrahydrofolate-homocysteine S-methyltransferase
 - 2.1.1.28 Phenylethanolamine N-methyltransferase
 - 2.1.1.45 Thymidylate synthase
 - 2.1.1.67 Thiopurine methyltransferase
- 2.1.2 A formyl group**
- 2.1.2.1 Glycine hydroxymethyltransferase [PLP]
 - 2.1.2.2 Phosphoribosylglycinamide formyltransferase
 - 2.1.2.3 Phosphoribosylaminoimidazolecarboxamide formyltransferase
 - 2.1.2.5 Glutamate formiminotransferase [PLP]
 - 2.1.2.10 Aminomethyltransferase
- 2.1.3 A carbamoyl group**
- 2.1.3.2 Aspartate carbamoyltransferase [Zn²⁺]
 - 2.1.3.3 Ornithine carbamoyltransferase
- 2.1.4 An amidino group**
- 2.1.4.1 Glycine amidinotransferase
- 2.2 An aldehyde or ketone residue is transferred**
- 2.2.1.1 Transketolase [TPP]
 - 2.2.1.2 Transaldolase

2.3 An acyl group is transferred**2.3.1 With acyl-CoA as donor**

- 2.3.1.1 Amino acid *N*-acetyltransferase
- 2.3.1.6 Choline *O*-acetyltransferase
- 2.3.1.12 Dihydrolipoamide acetyltransferase [lipoamide]
- 2.3.1.15 Glycerol 3-phosphate *O*-acyltransferase
- 2.3.1.16 Acetyl-CoA acyltransferase
- 2.3.1.20 Diacylglycerol *O*-acyltransferase
- 2.3.1.21 Carnitine *O*-palmitoyltransferase
- 2.3.1.22 Acylglycerol *O*-palmitoyltransferase
- 2.3.1.24 Sphingosine *N*-acyltransferase
- 2.3.1.37 5-Aminolevulinate synthase [PLP]
- 2.3.1.38 [ACP] *S*-acetyltransferase
- 2.3.1.39 [ACP] *S*-malonyltransferase
- 2.3.1.41 3-Oxoacyl-[ACP] synthase
- 2.3.1.42 Glycerone phosphate *O*-acyltransferase
- 2.3.1.43 Phosphatidylcholine-sterol acyltransferase—"lecithin-cholesterol acyltransferase (LCAT)"
- 2.3.1.51 Acylglycerol-3-phosphate *O*-acyltransferase
- 2.3.1.61 Dihydrolipoamide succinyltransferase
- 2.3.1.85 Fatty-acid synthase

2.3.2 An aminoacyl group is transferred

- 2.3.2.2 γ -glutamyltransferase
- 2.3.2.12 Peptidyltransferase (*a ribozyme*)
- 2.3.2.13 Protein-glutamine γ -glutamyltransferase [Ca]—"fibrin-stabilizing factor"

2.4 A glycosyl group is transferred**2.4.1 A hexose residue**

- 2.4.1.1 Phosphorylase [PLP]—"glycogen (starch) phosphorylase"
- 2.4.1.11 Glycogen (starch) synthase
- 2.4.1.17 Glucuronosyltransferase
- 2.4.1.18 1,4- α -Glucan branching enzyme
- 2.4.1.25 4- α -Glucanotransferase
- 2.4.1.47 *N*-Acylsphingosine galactosyltransferase
- 2.4.1.119 Protein glycotransferase

2.4.2 A pentose residue

- 2.4.2.7 Adenine phosphoribosyltransferase
- 2.4.2.8 Hypoxanthine phosphoribosyltransferase
- 2.4.2.10 Orotate phosphoribosyltransferase
- 2.4.2.14 Amidophosphoribosyl transferase

2.5 An alkyl or aryl group is transferred

- 2.5.1.1 Dimethylallyltransferase
- 2.5.1.6 Methionine adenosyltransferase
- 2.5.1.10 Geranyltransferase
- 2.5.1.21 Farnesyl diphosphate farnesyltransferase

2.6 A nitrogen-containing group is transferred**2.6.1 An amino group (*transaminases*)**

- 2.6.1.1 Aspartate transaminase [PLP]—"GOT"
- 2.6.1.2 Alanine transaminase [PLP]—"GPT"

- 2.6.1.3 Cysteine transaminase [PLP]
- 2.6.1.5 Tyrosine transaminase [PLP]
- 2.6.1.6 Leucine transaminase (PLP)
- 2.6.1.11 Acetylornithine transaminase [PLP]
- 2.6.1.13 Ornithine transaminase [PLP]
- 2.6.1.19 4-Aminobutyrate transaminase [PLP]
- 2.6.1.42 Branched-chain amino acid transaminase [PLP]
- 2.6.1.52 Phosphoserine transaminase [PLP]
- 2.7 A phosphorus-containing group is transferred (*kinases*)**
 - 2.7.1 With –CH–OH as acceptor**
 - 2.7.1.1 Hexokinase
 - 2.7.1.3 Ketohexokinase
 - 2.7.1.6 Galactokinase
 - 2.7.1.11 6-Phosphofructokinase
 - 2.7.1.19 Phosphoribulokinase
 - 2.7.1.28 Triokinase (triosekinase)
 - 2.7.1.30 Glycerol kinase
 - 2.7.1.32 Choline kinase
 - 2.7.1.36 Mevalonate kinase
 - 2.7.1.37 Protein kinase
 - 2.7.1.38 Phosphorylase kinase
 - 2.7.1.39 Homoserine kinase
 - 2.7.1.40 Pyruvate kinase
 - 2.7.1.67 1-Phosphatidylinositol-4-kinase
 - 2.7.1.68 1-Phosphatidylinositol 4-phosphate kinase
 - 2.7.1.82 Ethanolamine kinase
 - 2.7.1.99 [Pyruvate dehydrogenase] kinase
 - 2.7.1.105 6-Phosphofructo-2-kinase
 - 2.7.1.112 Protein tyrosine kinase
 - 2.7.2 With –CO–OH as acceptor**
 - 2.7.2.3 Phosphoglycerate kinase
 - 2.7.2.4 Aspartate kinase
 - 2.7.2.8 Acetylglutamate kinase
 - 2.7.2.11 Glutamate 5-kinase
 - 2.7.3 With a nitrogen-containing group as acceptor**
 - 2.7.3.2 Creatine kinase
 - 2.7.4 With a phosphate group as acceptor**
 - 2.7.4.2 Phosphomevalonate kinase
 - 2.7.4.3 Adenylate kinase
 - 2.7.4.4 Nucleoside phosphate kinase
 - 2.7.4.6 Nucleoside diphosphate kinase
 - 2.7.6 A diphosphate residue is transferred**
 - 2.7.6.1 Ribose phosphate pyrophosphokinase
 - 2.7.7 A nucleotide is transferred**
 - 2.7.7.6 DNA-directed RNA polymerase—“RNA polymerase”
 - 2.7.7.7 DNA-directed DNA polymerase—“DNA polymerase”
 - 2.7.7.9 UTP-glucose-1-phosphate uridyltransferase
 - 2.7.7.12 Hexose-1-phosphate uridyltransferase
 - 2.7.7.14 Ethanolamine phosphate cytidyltransferase

- 2.7.7.15 Choline phosphate cytidyltransferase
- 2.7.7.41 Phosphatidate cytidyltransferase
- 2.7.7.49 RNA-directed DNA polymerase—"reverse transcriptase"

2.7.8 Another substituted phosphate is transferred

- 2.7.8.1 Ethanolaminephosphotransferase
- 2.7.8.2 Diacylglycerol cholinephosphotransferase
- 2.7.8.11 CDPdiacylglycerol-inositol 3-phosphatidyltransferase
- 2.7.8.16 1-Alkyl-2-acetylgllycerol cholinephosphotransferase
- 2.7.8.17 N-Acetylglucosaminephosphotransferase

Class 3: Hydrolases (catalyze bond cleavage by hydrolysis)

Subclass 3.n: What kind of bond is hydrolyzed?

3.1 An ester bond is hydrolyzed (*esterases*)

3.1.1 In carboxylic acid esters

- 3.1.1.2 Arylesterase
- 3.1.1.3 Triacylglycerol lipase
- 3.1.1.4 Phospholipase A₂
- 3.1.1.7 Acetylcholinesterase
- 3.1.1.13 Cholesterol esterase
- 3.1.1.17 Gluconolactonase
- 3.1.1.32 Phospholipase A₁
- 3.1.1.34 Lipoprotein lipase, diacylglycerol lipase

3.1.2 In thioesters 3.1.2.4

- 3-Hydroxyisobutyryl-CoA hydrolase
- 3.1.2.14 Acyl-[ACP] hydrolase

3.1.3 In phosphoric acid monoesters (*phosphatases*)

- 3.1.3.1 Alkaline phosphatase [Zn²⁺]
- 3.1.3.2 Acid phosphatase
- 3.1.3.4 Phosphatidate phosphatase
- 3.1.3.9 Glucose 6-phosphatase
- 3.1.3.11 Fructose biphosphatase
- 3.1.3.13 Bisphosphoglycerate phosphatase
- 3.1.3.16 Phosphoprotein phosphatase
- 3.1.3.37 Sedoheptulose biphosphatase
- 3.1.3.43 [Pyruvate dehydrogenase] phosphatase
- 3.1.3.46 Fructose-2,6-bisphosphate 2-phosphatase
- 3.1.3.n Polynucleotidases

3.1.4 In phosphoric acid diesters (*phosphodiesterases*)

- 3.1.4.1 Phosphodiesterase
- 3.1.4.3 Phospholipase C
- 3.1.4.4 Phospholipase D
- 3.1.4.17 3',5'-cNMP phosphodiesterase
- 3.1.4.35 3',5'-cGMP phosphodiesterase
- 3.1.4.45 N-Acetylglucosaminyl phosphodiesterase

3.1.21 In DNA

- 3.1.21.1 Deoxyribonuclease I
- 3.1.21.4 Site-specific deoxyribonuclease (type II)—"restriction endonuclease"

3.10.26–7 In RNA

3.1.26.4 Ribonuclease H

3.1.27.5 Pancreatic ribonuclease

3.2 A glycosidic bond is hydrolyzed (*glycosidases*)**3.2.1 In O-glycosides**3.2.1.1 α -Amylase

3.2.1.10 Oligo-1,6-glucosidase

3.2.1.17 Lysozyme

3.2.1.18 Neuraminidase

3.2.1.20 α -Glucosidase3.2.1.23 β -Galactosidase3.2.1.24 α -Mannosidase3.2.1.26 β -Fructofuranosidase—"saccharase," "invertase"3.2.1.28 α,α -Trehalase

3.2.1.33 Amylo-1,6-glucosidase

3.2.1.48 Sucrose α -glucosidase3.2.1.52 β -N-Acetylhexosaminidase

3.2.2.n Nucleosidases

3.3 An ether bond is hydrolyzed

3.3.1.1 Adenosylhomocysteinase

3.4 A peptide bond is hydrolyzed (*peptidases*)**3.4.11 Aminopeptidases (*N*-terminal exopeptidases)**3.4.11.n Various aminopeptidases [Zn^{2+}]**3.4.13 Dipeptidases (act on dipeptides only)**3.4.13.n Various dipeptidases [Zn^{2+}]**3.4.15 Peptidyl dipeptidases (C-terminal exopeptidases, releasing dipeptides)**3.4.15.1 Peptidyl-dipeptidase A [Zn^{2+}]"angiotensin-converting enzyme (ACE)"**3.4.17 Carboxypeptidases (C-terminal exopeptidases)**3.4.17.1 Carboxypeptidase A [Zn^{2+}]3.4.17.2 Carboxypeptidase B [Zn^{2+}]

3.4.17.8 Muramoylpentapeptide carboxypeptidase

3.4.21 Serine proteinases (endopeptidases)

3.4.21.1 Chymotrypsin

3.4.21.4 Trypsin

3.4.21.5 Thrombin

3.4.21.6 Coagulation factor Xa—"Stuart–Prower factor"

3.4.21.7 Plasmin

3.4.21.9 Enteropeptidase—"enterokinase"

3.4.21.21 Coagulation factor VIIa—"proconvertin"

3.4.21.22 Coagulation factor IXa—"Christmas factor"

3.4.21.27 Coagulation factor XIa—"plasma thromboplastin antecedent"

3.4.21.34 Plasma kallikrein

3.4.21.35 Tissue kallikrein

3.4.21.36 Elastase

3.4.21.38 Coagulation factor XIIa—"Hageman factor"

3.4.21.43 C3/C5 convertase (complement—classical pathway)

3.4.21.47 C3/C5 convertase (complement—alternative pathway)

- 3.4.21.68 Plasminogen activator (tissue)—“tissue plasminogen activator (t-PA)”
 3.4.21.73 Plasminogen activator (urine)—“urokinase”
- 3.4.22 Cysteine proteinases (endopeptidases)**
 3.4.22.2 Papain
- 3.4.23 Aspartate proteinases (endopeptidases)**
 3.4.23.1 Pepsin A
 3.4.23.2 Pepsin B
 3.4.23.3 Gastricsin (pepsin C)
 3.4.23.4 Chymosin
 3.4.23.15 Renin
- 3.4.24 Metalloproteinases (endopeptidases)**
 3.4.24.7 Collagenase
- 3.4.99 Other peptidases**
 3.4.99.36 Signal peptidase
- 3.5 Another amide bond is hydrolyzed (amidases)**
 3.5.1.1 Asparaginase
 3.5.1.2 Glutaminase
 3.5.1.16 Acetylornithine deacetylase [Zn^{2+}]
 3.5.2.3 Dihydroorotase
 3.5.2.7 Imidazolonepropionase
 3.5.3.1 Arginase
 3.5.4.6 AMP deaminase
 3.5.4.9 Methylenetetrahydrofolate cyclohydrolase
 3.5.4.10 IMP cyclohydrolase
- 3.6 An anhydride bond is hydrolyzed**
 3.6.1.6 Nucleoside diphosphatase
 3.6.1.32 Myosin ATPase
 3.6.1.34 H^+ -transporting ATP synthase—“ATP synthase,” “complex V”
 3.6.1.35 H^+ -transporting ATPase
 3.6.1.36 H^+/K^+ -exchanging ATPase
 3.6.1.37 Na^+/K^+ -exchanging ATPase—“ Na^+/K^+ -ATPase”
 3.6.1.38 Ca^{2+} -transporting ATPase
- 3.7 A C–C bond is hydrolyzed**
 3.7.1.2 Fumarylacetoacetase

Class 4: Lyases (cleave or form bonds without oxidative or hydrolytic steps)

Subclass 4.n: What kind of bond is formed or cleaved?

- 4.1 A C–C bond is formed or cleaved**
- 4.1.1 Carboxy-lyases (carboxylases, decarboxylases)**
 4.1.1.1 Pyruvate decarboxylase [TPP]
 4.1.1.15 Glutamate decarboxylase [PLP]
 4.1.1.21 Phosphoribosylaminoimidazole carboxylase
 4.1.1.23 Orotidine-5'-phosphate decarboxylase
 4.1.1.28 Aromatic L-amino acid decarboxylase [PLP]
 4.1.1.32 Phosphoenolpyruvate carboxykinase (GTP)
 4.1.1.39 Ribulose biphosphate carboxylase [Cu]—“rubisco”

4.1.2 Acting on aldehydes or ketones

- 4.1.2.5 Threonine aldolase [PLP]
- 4.1.2.13 Fructose bisphosphate aldolase—"aldolase"
- 4.1.3.4 Hydroxymethylglutaryl-CoA lyase
- 4.1.3.5 Hydroxymethylglutaryl-CoA synthase
- 4.1.3.7 Citrate synthase
- 4.1.3.8 ATP-citrate lyase
- 4.1.3.18 Acetolactate synthase [TPP, flavin]

4.1.99 Other C–C lyases

- 4.1.99.3 Deoxyribodipyrimidine photolyase [FAD]—"photolyase"

4.2 A C–O bond is formed or cleaved**4.2.1 Hydrolyases (*hydratases, dehydratases*)**

- 4.2.1.1 Carbonate dehydratase [Zn^{2+}]—"carbonic anhydrase"
- 4.2.1.2 Fumarate hydratase—"fumarase"
- 4.2.1.3 Aconitate hydratase [Fe_4S_4]—"aconitase"
- 4.2.1.11 Phosphopyruvate hydratase—"enolase"
- 4.2.1.13 Serine dehydratase
- 4.2.1.17 Enoyl-CoA hydratase
- 4.2.1.18 Methylglutaconyl-CoA hydratase
- 4.2.1.22 Cystathionine β -synthase [PLP]
- 4.2.1.24 Porphobilinogen synthase
- 4.2.1.49 Urocanate hydratase
- 4.2.1.61 3-Hydroxypalmitoyl-[ACP] dehydratase
- 4.2.1.75 Uroporphyrinogen III synthase

4.2.99 Other C–O lyases

- 4.2.99.2 Threonine synthase [PLP]

4.3 A C–N bond is formed or cleaved**4.3.1 Ammonia lyases 4.3.1.3**

- Histidine ammonia lyase
- 4.3.1.8 Hydroxymethylbilane synthase

4.3.2 Amidine lyases 4.3.2.1

- Argininosuccinate lyase
- 4.3.2.2 Adenylosuccinate lyase

4.4 A C–S bond is formed or cleaved

- 4.4.1.1 Cystathionine γ -lyase [PLP]

4.6 A P–O bond is formed or cleaved

- 4.6.1.1 Adenylate cyclase
- 4.6.1.2 Guanylate cyclase

Class 5: Isomerases (catalyze changes within one molecule)**Subclass 5.n: *What kind of isomerization is taking place?*****5.1 A racemization or epimerization (*epimerases*)**

- 5.1.3.1 Ribulose phosphate 3-epimerase
- 5.1.3.2 UDPglucose 4-epimerase
- 5.1.3.4 L-Ribulose phosphate 4-epimerase
- 5.1.99.1 Methylmalonyl-CoA epimerase

- 5.2 A *cis*–*trans* isomerization**
 5.2.1.2 Maleylacetoacetate isomerase
 5.2.1.3 Retinal isomerase
 5.2.1.8 Peptidyl proline *cis*–*trans*-isomerase
- 5.3 An intramolecular electron transfer**
 5.3.1.1 Triose phosphate isomerase
 5.3.1.6 Ribose 5-phosphate isomerase
 5.3.1.9 Glucose 6-phosphate isomerase
 5.3.3.1 Steroid Δ -isomerase
 5.3.3.8 Enoyl-CoA isomerase
 5.3.4.1 Protein disulfide isomerase
- 5.4 An intramolecular group transfer (*mutases*)**
 5.4.2.1 Phosphoglycerate mutase
 5.4.2.2 Phosphoglucomutase
 5.4.2.4 Bisphosphoglycerate mutase
 5.4.99.2 Methylmalonyl-CoA mutase [cobamide]
- 5.99 Another kind of isomerization**
 5.99.1.2 DNA topoisomerase (type I)—“DNA helicase”
 5.99.1.3 DNA topoisomerase (ATP-hydrolyzing, type II)—“DNA gyrase”

Class 6: Ligases (join two molecules with hydrolysis of an “energy-rich” bond)

Subclass 6.n: What kind of bond is formed?

- 6.1 A C–O bond is formed**
 6.1.1.n (Amino acid)-tRNA ligases (*aminoacyl-tRNA synthetases*)
- 6.2 A C–S bond is formed**
 6.2.1.1 Acetate-CoA ligase
 6.2.1.3 Long-chain fatty-acid-CoA ligase
 6.2.1.4 Succinate-CoA ligase (GDP-forming)—“thiokinase”
- 6.3 A C–N bond is formed**
 6.3.1.2 Glutamate-NH₃ ligase—“glutamine synthetase”
 6.3.2.6 Phosphoribosylaminoimidazolesuccinocarboxamide synthase (*sorry!*)
 6.3.3.1 Phosphoribosylformylglycinamidine cycloligase
 6.3.3.2 5-Formyltetrahydrofolate cycloligase
 6.3.4.2 CTP synthase
 6.3.4.4 Adenylosuccinate synthase
 6.3.4.5 Argininosuccinate synthase
 6.3.4.13 Phosphoribosylamine glycine ligase
 6.3.4.16 Carbamoylphosphate synthase (NH₃)
 6.3.5.2 GMP synthase (glutamine-hydrolyzing)
 6.3.5.3 Phosphoribosylformylglycinamidine synthase
 6.3.5.4 Asparagine synthase (glutamine-hydrolyzing)
 6.3.5.5 Carbamoylphosphate synthase (glutamine-hydrolyzing)
- 6.4 A C–C bond is formed**
 6.4.1.1 Pyruvate carboxylase [biotin]
 6.4.1.2 Acetyl-CoA carboxylase [biotin]
 6.4.1.3 Propionyl-CoA carboxylase [biotin]
 6.4.1.4 Methylcrotonyl-CoA carboxylase [biotin]
- 6.5 A P–O bond is formed**
 6.5.1.1 DNA ligase (ATP)

Abbreviations

Abbreviations for amino acids, p. 60

For bases and nucleosides, p. 80

For monosaccharides, p. 38

AA	Amino acid	FAD	Flavin adenine dinucleotide
ACE	Angiotensin-converting enzyme (peptidyl-dipeptidase A)	Fd	Ferredoxin
ACP	Acyl carrier protein	FFA	Free fatty acid
ACTH	Adrenocorticotrophic hormone (corticotropin)	fMet	<i>N</i> -formylmethionine
ADH	Antidiuretic hormone (adiuretin, vasopressin)	FMN	Flavin mononucleotide
ADP	Adenosine 5'-diphosphate	Fp	Flavoprotein (containing FMN or FAD)
AIDS	Acquired immunodeficiency syndrome	GABA	γ -Aminobutyric acid
ALA	5-Aminolevulinic acid	GDP	Guanosine 5'-diphosphate
AMP	Adenosine 5'-monophosphate	Glut	Glucose transporter
ANF	Atrial natriuretic factor	GMP	Guanosine 5'-monophosphate
ANP	Atrial natriuretic peptide (= ANF)	GSH	Reduced glutathione
ATP	Adenosine 5'-triphosphate	GSSG	Oxidized glutathione
AVP	Arginine vasopressin	GTP	Guanosine 5'-triphosphate
b	Base	h	hour
bp	Base pair	HAT medium	Medium containing hypoxanthine, aminopterin, and thymidine
BPG	2,3-Bisphosphoglycerate	Hb	Hemoglobin
cAMP	3',5'-Cyclic AMP	HDL	High-density lipoprotein
CAP	Catabolite activator protein	HIV	Human immunodeficiency virus
CDK	Cyclin-dependent protein kinase (in cell cycle)	HLA	Human leukocyte-associated antigen
cDNA	Complementary DNA	HMG-CoA	3-Hydroxy-3-methylglutaryl-CoA
CDP	Cytidine 5'-diphosphate	HMP	Hexose monophosphate pathway
cGMP	3',5'-Cyclic GMP	hnRNA	Heterogeneous nuclear ribonucleic acid
CIA	Chemoluminescence immunoassay	HPLC	High-performance liquid chromatography
CMP	Cytidine 5'-monophosphate	hsp	Heat-shock protein
CoA	Coenzyme A	IDL	Intermediate-density lipoprotein
CoQ	Coenzyme Q (ubiquinone)	IF	Intermediary filament
CSF	colony-stimulating factor	IFN	Interferon
CTP	Cytidine 5'-triphosphate	Ig	Immunoglobulin
d	Deoxy-	IL	Interleukin
Da	Dalton (atomic mass unit)	InsP ₃	Inositol 1,4,5-trisphosphate
DAG	Diacylglycerol	IPTG	Isopropylthiogalactoside
dd	Dideoxy-	IRS	Insulin-receptor substrate
DH	Dehydrogenase	kDa	Kilodalton (10 ³ atomic mass units)
DNA	Deoxyribonucleic acid	K _m	Michaelis constant
dsDNA	Double-stranded DNA		
EA	Ethanolamine		
EIA	Enzyme-linked immunoassay		
ER	Endoplasmic reticulum		

LDH	Lactate dehydrogenase	QH ₂	Reduced coenzyme Q (ubiquinol)
LDL	Low-density lipoprotein	R	Gas constant
M	Molarity (mol L ⁻¹)	rER	Rough endoplasmic reticulum
Mab	Monoclonal antibody	RES	Reticuloendothelial system
MAP kinase	Mitogen-activated protein kinase	RFLP	Restriction fragment length polymorphism
MHC	Major histocompatibility complex	RIA	Radioimmunoassay
MPF	Maturation-promoting factor	RNA	Ribonucleic acid
mRNA	Messenger ribonucleic acid	ROS	Reactive oxygen species
N	Nucleotide with any base	RP	Reversed phase (of silica gel)
NAD ⁺	Oxidized nicotinamide adenine dinucleotide	rRNA	Ribosomal ribonucleic acid
NADH	Reduced nicotinamide adenine dinucleotide	S	Svedberg (unit of sedimentation coefficient)
NADP ⁺	Oxidized nicotinamide adenine dinucleotide phosphate	SAH	S-adenosyl L-homocysteine
NADPH	Reduced nicotinamide adenine dinucleotide phosphate	SAM	S-adenosyl L-methionine
NeuAc	N-acetylneuraminic acid	SDS	Sodium dodecylsulfate
nm	Nanometer (10 ⁻⁹ m)	sER	Smooth endoplasmic reticulum
ODH	2-Oxoglutarate dehydrogenase	sn	Stereospecific numbering
PAGE	Polyacrylamide gel electrophoresis	snRNA	Small nuclear ribonucleic acid
Pan	Pantetheine	SR	Sarcoplasmic reticulum
PAPS	Phosphoadenosine phosphosulfate	ssDNA	Single-stranded DNA
PCR	Polymerase chain reaction	TBG	Thyroxine-binding globulin
PDH	Pyruvate dehydrogenase	THB	Tetrahydrobiopterin
PEG	Polyethylene glycol	THF	Tetrahydrofolate
PEP	Phosphoenolpyruvate	TLC	Thin-layer chromatography
pH	pH value	TPP	Thiamine diphosphate
P _i	Inorganic phosphate	TRH	Thyrotropin-releasing hormone (thyroliberin)
PK	Protein kinase	Tris	Tris(hydroxymethyl)aminomethane
PLP	Pyridoxal phosphate	tRNA	Transfer ribonucleic acid
PP	Protein phosphatase	TSH	Thyroid-stimulating hormone (thyrotropin)
PPP	Pentose phosphate pathway	UDP	Uridine 5'-diphosphate
PQ	Plastoquinone	UMP	Uridine 5'-monophosphate
PRPP	5-Phosphoribosyl 1-diphosphate	UTP	Uridine 5'-triphosphate
PS	Photosystem	UV	Ultraviolet radiation
PTH	Parathyroid hormone	V _{max} , V	Maximal velocity (of an enzyme)
Q	Oxidized coenzyme Q (ubiquinone)	VLDL	Very-low-density lipoprotein

Quantities and units

1. SI base units

Quantity	SI unit	Symbol	Remarks
Length	Meter	m	1 yard (yd) = 0.9144 m 1 inch (in) = 0.0254 m 1 Å = 10^{-10} m = 0.1 nm
Mass	Kilogram	kg	1 pound (lb) = 0.4536 kg
Time	Second	s	
Current strength	Ampere	A	
Temperature	Kelvin	K	°C (degree Celsius) = K – 273.2 Fahrenheit: °C = 5/9 (°F – 32)
Light	Candela	Cd	
Amount of substance	Mol	mol	

2 Derived units

Quantity	Unit	Symbol	Derivation	Remarks
Frequency	Hertz	Hz	s^{-1}	
Volume	Liter	L	$10^{-3} m^3$	1 U.S. gallon (gal) = 3.785 L
Force	Newton	N	$kg m s^{-2}$	
Pressure	Pascal	Pa	$N m^{-2}$	1 bar = 10^5 Pa 1 mmHg = 133.3 Pa
Energy, work, heat	Joule	J	$N m$	1 calorie (cal) = 4.1868 J
Power	Watt	W	$J s^{-1}$	
Electrical charge	Coulomb	C	$A s$	
Voltage	Volt	V	$W A^{-1}$	
Concentration	Molarity	M	$mol L^{-1}$	
Molecular mass	Dalton	Da	$1.6605 \cdot 10^{-24} g$	
Molar mass	–	–	g	
Molecular weight	–	M_r	–	Nondimensional
Reaction rate		v	$mol s^{-1}$	
Catalytic activity	Katal	kat	$mol s^{-1}$	1 unit (U) = $1.67 \cdot 10^{-8}$ kat
Specific activity	–	–	$kat (kg enzyme)^{-1}$	Usually: U $(mg enzyme)^{-1}$
Sedimentation coef cient	Svedberg	S	$10^{-13} s$	
Radioactivity	Becquerel	Bq	Decays s^{-1}	1 curie (Ci) = $3.7 \cdot 10^{10}$ Bq

3 Multiples and fractions

Factor	Prefix	Sym- bol	Example
10^9	Giga	G	GHz = 10^9 hertz
10^6	Mega	M	MPa = 10^6 pascal
10^3	Kilo	k	kJ = 10^3 joule
10^{-3}	Milli	m	mM = 10^{-3} mol L^{-1}
10^{-6}	Micro	μ	μV = 10^{-6} volt
10^{-9}	Nano	n	nkat = 10^{-9} katal
10^{-12}	Pico	p	pm = 10^{-12} meter

4 Important constants

General gas constant, R	$R = 8.314 J mol^{-1} K^{-1}$
Loschmidt (Avogadro) number, N	$N = 6.0225 \cdot 10^{23}$
(number of particles per mol)	
Faraday constant F	$F = 96480 C mol^{-1}$

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207	C	Stryer L. Biochemistry. New York: Freeman, 1988, p. 945, Fig. 36–47
207	C	Alberts B, et al. The molecular biology of the cell. New York: Garland, 1989, p. 663, Fig. 11–73B and p. 634, Fig. 11–36
279	A, B	Voet D, Voet JG. Biochemistry. New York: Wiley, 1990, p. 305, Fig. 11–45 and p. 306, Fig. 11–47
295	A	Voet D, Voet JG. Biochemistry. New York: Wiley, 1990, p. 1097, Fig. 34–13
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297	A	Janeway CA, Travers P. Immunology. Heidelberg, Germany: Spektrum, 1994, p. 164, Fig. 4.3.c
333	B	Voet D, Voet JG. Biochemistry. New York: Wiley, 1990, p. 1126, Fig. 34–55
335	B	Darnell J, et al. Molecular cell biology. 2nd ed. New York: Freeman, 1990, p. 923, Fig. 23–26

Index

Numbers in *italics* indicate figures.

A

- A-DNA 86, 87
 AA *see* amino acids
 ABO blood groups 292, 293
 absorption
 A 102
 light 102, 103, 128, 129
 absorptive state 308
 acceptors 6
 electrons 88
 site (A) 250
 ACE *see* angiotensin-converting enzyme
 acetaldehyde *see* ethanal
 acetate 13, 354
 acetate-CoA ligase 320, 321, 354, 355, 429
 acetic acid 49, 52, 53
 activated *see* acetyl-CoA
 acetoacetate 180
 acetoacetic acid 160
 acetoacetone 312
 acetoacetyl-CoA 312
 acetolactate synthase 428
 acetylcholine 270
 N-acetyl glutamate 182, 183
 acetyl lipoamide 135
 acetyl-CoA 12, 52, 53, 110, 163
 acyltransferase 165, 312, 313, 423
 carboxylase 160, 161, 162, 163, 168, 429
 cholesterol 172, 173
 ethanol metabolism 320
 fatty acid degradation 164, 165, 167
 pyruvate kinase inhibitor 158
 tricarboxylic acid cycle 136, 137, 138, 139
 van der Waals model 13
 N-acetyl-galactosaminyl transferase 292, 293
 N-acetyl- γ -glutamylphosphate reductase 420
 acetylated amino sugars 38
 see also individual sugars
 acetylation 62, 244
 acetylcholine 352
 receptors 222, 335, 354
 acetylcholinesterase 354, 355, 425
 N-acetylglucosamine (GlcNAc) 40, 44a
 phosphoglycosidase 234, 235
 phosphotransferase 234, 235, 425
 acetylglutamate kinase 424
 β -N-acetylhexosaminidase 426
 N-acetylmuraminic acid (MurNAc) 40
 N-acetylneuraminic acid (NeuAc) 38
 acetylornithine
 deacetylase 427
 transaminase 424
 acetylsalicylic acid 317
 acid amide bonds 12
 acid anhydrides 10
 acid phosphatase 425
 acid–amide bonds 13
 acid–anhydride bonds 150
 acid–base
 balance 274, 288–9, 289, 326–7
 catalysis 90
 reactions 14, 15, 18, 19
 acidic amino acids 60
 acidic monosaccharides 38, 39
 see also N-acetylneuraminic acid; D-galacturonic acid; D-glucuronic acid; iduronic acid
 acidoses, metabolic 160, 161, 288, 326
 acids 15, 18, 30–1, 31
 aconitate 9
 hydratase (aconitase) 8, 9, 106, 136, 137, 428
 ACP *see* acyl carrier protein
 acquired immune system 294
 acquired immunodeficiency syndrome (AIDS) 404
 ACTase *see* aspartate carbamoyltransferase
 ACTH *see* adrenocorticotrophic hormone (corticotropin)
 actin 204, 205, 332
 F-actin 64, 65, 332
 G-actin 332
 actin-associated proteins 204
 actinomycin D 254, 255
 action potential 222, 348, 350, 351
 activating enzymes 114
 activation, T-cell 296–7
 activation energy (E_a) 22, 23, 25, 90
 active centers 74, 90
 enzymes 95
 essential amino acids 99
 lactate dehydrogenase 100
 trypsin 177
 active transport 218, 220
 acyl carnitine 164, 165
 acyl carrier proteins (ACP)
 S-acetyltransferase 168, 169, 423
 acyl carrier proteins (ACP), S-malonyltransferase 168, 169, 423
 S-ferase 423
 acyl-[ACP] hydrolase 168, 169, 425
 acyl-CoA 52, 141, 164, 168, 170
 dehydrogenase 164, 165, 421
 fat biosynthesis 170, 171
 fatty acid degradation 162, 163
 acylation 62, 380

- acylglycerol O-palmitoyl transferase 170, 171, 423
acylglycerol-3-phosphate O-acyltransferase 170, 171, 423
acylglycerone phosphate reductase 420
N-acylsphingosine galactosyltransferase 423
adaptive immune system 294
adaptor proteins 388
 Grb-2 388
 SOS 388
additions 14, 15
adenine 12, 13, 80, 81
 phosphoribosyltransferase 187, 423
adenosine 80, 81, 123
 5'-diphosphate (ADP) *see* ADP
 5'-monophosphate (AMP) *see* AMP
 5'-triphosphate (ATP) *see* ATP
S-adenosyl-L-homocysteine (SAH) 111
S-adenosyl-L-methionine (SAM) 110, 111
adenosylcobalamin 108
adenosylhomocysteinase 426
adenylate
 cyclase 120, 121, 384, 385, 386, 387, 428
 kinase 72, 73, 336, 337, 424
adenylosuccinate
 lyase 428
 synthase 429
ADH *see* antidiuretic hormone
adhesive proteins 346
adipocytes 46
adipose tissues 160, 161, 162, 308
adiuretin *see* antidiuretic hormone
ADP 190, 191, 212
adrenaline *see* epinephrine
adrenocorticotrophic hormone (ACTH)
adriamycin *see* doxorubicin
aerobic glycolysis 150
aerobic metabolism 122
aerobic oxidation 146, 147
agarose 40, 41, 262
agents, alkylating 256, 402, 403
AIDS 404
air 29
ALA *see* 5-aminolevulinic acid
alanine 60, 61, 178, 184, 185
 β -alanine 12, 13, 62, 63, 106
 degradation 186
 cycle 338, 339
 seryl 65, 66, 67
 transaminase 178, 179, 181, 423
albumins 194, 195, 277
alcohol 10
 blood 321
 dehydrogenase 64, 420
 fermentation 148, 149
 liver damage 320, 321
 primary 11
 secondary 11
alcohol dehydrogenase 65, 320, 321
 fermentation 148, 149
alcohols 15
 sugar 38, 39
aldehyde 11
 dehydrogenase 320, 321, 420
 reductase 310, 311, 420
aldehydes 10, 15
aldimes 10
aldimine 10, 108, 178, 179
aldohexoses 34, 38
 D-glucose *see* D-glucose
aldolase *see* fructose biphosphate adolase
aldopentoses 38
 see also L-arabinose; D-ribose; D-xylose
aldoses 39
aldosterone 56, 57, 374, 376
aliphatic amino acids 60
alkaline phosphatase 425
alkanes 46, 47
alkanols 46
alkenes 15
 1-alkyl-2-acetylglycerol choline phosphotransferase 170, 171, 425
alkylating agents 256, 402, 403
allantoic acid 187
allantoin 186, 187
allolactose 119
allopurinol 186, 187
allosteric effects 2-3, 114, 115, 116
 BPG 282, 284
 hemoglobin 116, 280
allosteric enzymes 92, 93, 96, 116
allosteric inhibitors 96
allosteric proteins 72
allotypic variation 302
 α -amanitin 242
 α_1 -globulins 276
 α_2 -globulins 276
 α -keratin 71
alternative pathway, complement 298
Amanita phalloides 242
amidases 89, 427
amidine lyases 428
amidophosphoribosyl transferase 402, 403, 423
amine oxidase 62, 421
amines 10
 biogenic 62, 63, 352
 primary 11
 secondary 11
 tertiary 11
Aminita phalloides 204
amino acid N-acetyltransferase 423
D-amino acid oxidase 58
amino acid tRNA ligases 248
amino acids (AA) 35, 58-63, 146
 activation 236, 248-9, 249
 aromatic 60, 184
 biosynthesis 184-5, 185, 412-13
 branched chain 184, 185
 charge 58
 degradation 180-1, 181, 414-15
 essential *see* essential amino acids
 excretion 324
 function 58, 59
 glucogenic 180
 gluconeogenesis 154
 ketogenic 180
 metabolism 306, 338
 neurotransmitters 352
 optical activity 58, 59

- proteinogenic 58, 60-1, 61
non-proteinogenic 62-3
resorption 272
signaling 380, 381
translation 60
- amino sugars 39, 44
- aminoacyl
adenylate 248, 249
tRNAs 248, 249, 252
- 4-aminobutyrate transaminase 356, 357, 424
- aminodipeptidases 425
- aminoglycosides 254
- β -aminoisobutyrate 186, 187
- 5-aminolevulinate 192, 193
synthase 192, 193, 423
- aminomethyltransferase 422
- aminopeptidases 176, 177, 426
- aminopropanol 63
- aminopterin 304
- aminotransferases 89
- ammonia 11
bases 30, 31
degradation 112
excretion 326, 327
lyases 428
metabolism 178, 179, 416
urea cycle 182, 183
- ammonium ions (NH_4^+) 31, 182, 183, 326
- ammonotelic animals 182
- AMP 158, 186, 190, 191
deaminase 336, 337, 427
- amphibolic function 136
- amphibolic pathways 112, 113
- amphipathic molecules 28, 29
- amphiphilic molecules 28
- ampicillin 255
- amplification 398
DNA 262
- amylases
 α -amylase 268, 269, 272, 273, 426
brewing 148, 149
- amylase-1,6-glucosidase 156, 157, 426
- amylpectin 41, 42, 43
- amyloplasts 42
- amylose 41, 42, 43
- anabolic pathways 112, 113, 138, 139, 338
- anaerobic conditions 112, 146
- anaerobic glycolysis 150, 336, 338
- anaerobic oxidation 146, 147
- analogs
substrates 96, 97
transition states 97
- analysis
analyzers 36
biochemical 102, 103
proteins 78
- anaplerotic effect 180
- anaplerotic pathways 139
- anchors, lipid 214
- anemia 286
sickle-cell 249, 264, 265
- ANF *see* atrial natriuretic factor
- angiotensin 330, 331
- angiotensin-converting enzyme (ACE) 426
- anhydrides
mixed 11
phosphoric acid 11
- animals
cells 197
types of 182
- anions 15, 27
- anomers 36
- ANP *see* atrial natriuretic peptide
- antenna pigments 132
- anthracyclines 402, 403
- anti-oncogenes 400
- antibiotics 225, 254-5, 255
- antibodies 300-3, 303
monoclonal 304-5, 305
plasma cells 294
- anticodons 82, 86, 236, 248, 249
- antidiuretic hormone (ADH) 220
- antigen-presenting cells (APCs) 294
- antigens
blood groups 292
receptors 296, 297
- antihemophilic factor A 291
- antimetabolites 402, 403
- antioxidants 104, 284, 285, 364
- antiparallel pleated sheet 68
- antiport 212, 218
- antisense DNA 264
- apatites 340, 341
- APC *see* antigen-presenting cells
- apoferritin 287
- apolar molecules 28
- apoproteins 278
- apoptosis 210, 396-7, 397
- aquaporin-1 220, 221
- arabinan 41
- L-arabinose 38, 39, 42
- arachidonate lipoxygenases 390, 391, 421
- arachidonic acid 48, 49, 390
- arginase 183, 427
- arginine 60, 61, 182, 183
lactate dehydrogenase 100
vasopressin (AVP) *see* antidiuretic hormone
- argininosuccinate 182, 183
lyase 183, 428
synthase 183, 429
- aromatic amino acids 60, 184
L-amino acid decarboxylase 352, 353, 427
- aromatic family 185
- aromatic ring systems 4
- arylesterase 316, 317, 425
- ascorbates 104, 368
- L-ascorbic acid *see* vitamin C
- asialoglycoproteins 276
- Asp-tRNA ligase 248, 249
- asparaginase 427
- asparagine 60, 61
synthase 429
- aspartate 63, 178, 182, 183
carbamoyltransferase (ACTase) 116, 117, 188, 422
family 184, 185
kinase 424
kinetics 117
proteinases 177, 427
purine ring 188, 189
pyrimidine ring 188, 189
semialdehyde dehydrogenase 420
transaminase 178, 179, 183, 212, 213, 423
- aspartic acid 60, 61
- atomic oxygen 25

- ATP 35, 112, 122-3, 123
 ATP, charge density 123
 dATP 240
 energetic coupling 16, 122, 124
 exchange 212
 formation 122, 123
 glucose oxidation 147
 hydrolysis 122, 123
 nucleotide biosynthesis 190, 191
 oxidative phosphorylation 123
 phosphate transfer 123
 structure 122
 synthase *see* ATP synthase
 synthesis 112, 122, 142-5, 210
 NADH+H⁺ 142, 143
 proton driven 126, 127
 ATP synthase 122, 126, 142, 143, 210
 catalytic units 142, 143
 H⁺-transporting 140, 141, 143, 427
 light reactions 128, 129
 proton channels 142, 143
 respiratory control 144, 145
 ATP-citrate lyase 428
 ATP-dependent phosphorylation 114
 ATPases
 Ca²⁺ 220
 F type 220
 H⁺-transporting 427
 H⁺/K⁺-exchanging 427
 P type 220
 V type 220
 atrial natriuretic factor (ANF), receptor 388, 389
 atrial natriuretic peptide (ANP) 388, 389
 autocrine effects
 eicosanoids 390
 hormones 372, 373
 autotrophs 112
 AVP *see* arginine vasopressin
 axial (a) position 34
 axons 348
- B**
- B cell 161, 274, 275, 294
 B-DNA 84, 87
 bacteria
 carbohydrates 35
 cell components 202-3, 203
 cyanobacteria 132
 lactic acid fermentation 148, 149
 nitrogen fixation 184, 185
 polysaccharides 41
 symbiosis 184
 bacteriophages 260, 404
 bacteriorhodopsin 126, 127
 bacteroids 184, 185
 ball-and-stick models 6
 basal transcription complex 242, 244, 245
 base pairs (bp) 84
 bases (b) 30-1, 31, 80-1
 ammonia 30, 31, 182
 chloride ions 31
 conjugated 18, 30
 DNA ratios 84
 hydroxyl ions 31
 water 31
 basic amino acids 60, *see also* individual amino acids
 bcl-2 396
 beer 148, 149
 Beer–Lambert law 102
 behenic acid 49
 bell-shaped curves 94, 95
 benzene 4, 5
 benzo(a)pyrenes 256, 257
 p-benzoquinone 33
 β-granules 160, 161
 β turns 69
 β-oxidation 210
 β-configurations 12
 β-globulins 276, 286
 bicarbonate, resorption 326
 bile 194, 268, 269
 pigments 194
 salts 314, 315
 bile acids 56, 57, 172, 312
 conjugated 314
 primary 56, 314-15
 secondary 56, 314-15
 bilirubin 194, 195, 287
 diglucuronides 194, 195
 monoglucuronides 194
 biliverdin 194, 195
 reductase 194, 195, 421
 binding sites 250
 aminoacyl tRNA 252
 biogenic amines 12, 62, 63, 352
 biomolecules 10-13
 biosynthesis
 heme 192-3, 193, 210
 NAD⁺ 208
 biotin *see* vitamin H
 biotransformations 226, 306, 316-17, 317
 BiP 233
 1,3-bisphosphoglycerate 124, 125, 150, 151, 155, 282, 283
 2,3-bisphosphoglycerate *see* BPG
 bisphosphoglycerate
 mutase 283, 429
 phosphatase 283, 425
 bisubstrate kinetics 95
 blood
 alcohol 321
 cells 274, 275
 clotting 290-1, 291
 composition 274
 ethanol 320, 321
 function 274, 275
 groups 292-3, 293
 metabolites 274
 plasma 31
 serum 274
 blood–brain barrier 356
 boat conformation 54, 55
 bodies, residual 234
 Bohr effect 282
 bonds
 acid–amide 13
 acid–anhydride 150
 angles 7
 chemical *see* chemical bonds
 disulfides 62, 72, 73, 106, 190
 double 4
 electron 6
 hydrogen 72
 lengths 7
 peptide 66-7
 phosphoric acid 122

polarity 6, 7
single 4

bone 340, 341
calcium 342
remodeling 342, 343

Bowman's capsule 323

BPG 282, 283

brain 356, 357
gluconeogenesis 154

branched-chain amino acid
transaminase 424

brewing 149

Brunner glands 268

buffers 30, 31
blood plasma 288, 289
capacity 30, 288
dialysis 78, 79
function, organ metabolism 308-9

building blocks, lipids 46

1,3-butadiene 4, 5

butyric acid 49

C

C peptide, insulin 76, 77,
160, 161

C segment 302

C state 116

C terminus 66

c-oncogenes *see* proto-oncogenes

C-peptides 76

C₁
metabolism 418
transferases 89

caesium chloride 201

calcidol 1-monooxygenase
330, 331, 422

calciol *see* vitamin D

calcitonin 342, 343

calcitriol 56, 57, 330, 342,
374, 376

calcium
ATPase 220
bone 342
functions 342, 343
homeostasis 342, 343
ions 328, 334, 335, 386,
387
metabolism 342-3
reservoir 210
sarcoplasmic pump 220
signaling 342

store 226

calcium-transporting ATP-
ase 427

calmodulin 386, 387

calorimetry 20, 21

Calvin cycle 128, 129, 130,
131, 407

cAMP 120, 380, 384, 386,
387
lactose operon 119
phosphodiesterase 120,
121

camphor 52, 53

CAP *see* catabolite activator
protein

cap, RNA modification 246

capacity factor 16

capric acid 49

caproic acid 49

capsids, rhinovirus 404

N-carbamoyl aspartate 188,
189

carbamoyl phosphate 182,
183, 188, 189
synthase 182, 183, 188,
429

carbohydrate metabolism
160, 306, 310-11, 388
metabolic charts 408, 409
regulation 158-9, 159

carbohydrates 34-45, 35
bacteria 35
disaccharides *see* disac-
charides
hydrolysis 266
membranes 216
metabolism *see* carbohy-
drate metabolism
monosaccharides *see*
monosaccharides
nutrients 360
oligosaccharides *see* oli-
gosaccharides
polysaccharides *see* poly-
saccharides
reserve 156
resorption 266
see also glycolipids; gly-
coproteins

carbamoyl phosphate 116,
182, 183, 188

carbon dioxide 35, 108
bicarbonate buffer 288
degradation 112
fixation 130
transport 282-3, 283

hemoglobin 280, 282

carbonate dehydratase 270,
271, 283, 289, 428
proton secretion 326, 327

carbonic anhydrase *see* car-
bonate dehydratase

carbonium ions 15

carbonyl groups 11

carboxy-lyases 427

N-carboxybiotin 108

γ-carboxyglutamate 62, 290

carboxyl groups 11

carboxylases *see* carboxy-
lyases

γ-carboxylation 62, 290

carboxylic acid 10, 11, 15, 48,
49
activated 106
amides 10, 11
ester 10, 11

carboxypeptidases 176, 177,
425-6

caries 340

carnitine 164, 165
acyltransferase 164, 165
O-palmitoyltransferase
163, 165, 423
shuttle 212

carotenoids 46, 47
β-carotene 132, 133

carrageenan 40, 41

carrier electrophoresis 276

caspases 396

catabolic pathways 112, 113,
138, 139

catabolite activator protein
(CAP) 64, 65, 118, 119

catalase 24, 25, 166, 284,
285, 421

catalysis 24-5, 25
acid-base 90
covalent 90
cycles 100, 143, 221
enzymes 90-1
hydrogen peroxide 25
iodide 25
lactate dehydrogenase
100
membranes 216

catalysts 88, 92, *see also* en-
zymes

catalytic units, ATP synthase
142, 143

catechol O-methyltransfer-
ase 316, 317, 422

catecholamines 352, 353

- cathepsins 176
 cations 27
 CD4 296
 CD8 296
 cdc2 *see* cyclin-dependent protein kinase 1 (CDK1)
 CDK *see* cyclin-dependent protein kinase
 cDNA 260
 CDP 190, 191
 choline 110, 111
 diacylglycerol 170, 171
 diacylglycerol-inositol 3-phosphatidyl transferase 170, 171, 425
 cell bodies 348
 cell cycle 394-5, 395
 cells
 animal 196
 antigen-presenting (APCs) 294
 division 400, 401
 fractionation 198-9
 inhibition 402
 interaction 216
 movement 206
 necrosis 396
 organelles 196, 198
 plant 196
 proliferation 396, 397
 structure 196-7
 transformation 401
 transport 206
 vacuoles 234
 cellular immunity 294
 cellulose 34, 41, 42, 43
 hemicellulose 43
 cement 340
 centers
 active 74, 90
 reaction 132
 central nervous system *see* CNS
 centrifugation 201
 density gradient 200, 201
 equipment 200
 isopyknic 200
 principles 200-1
 steps 198
 zonal 200
 centrosome 206
 cephalin *see* phosphatidylethanolamine
 cephalosporins 254, 255
 cerebrosides 46, 47, 51
 galactosyl 51
 glucosylceramide 50, 51
 cGMP 386
 esterase 358, 359
 phosphodiesterase 425
 photoreceptors 358
 chain termination method 260
 chains
 respiratory 106, 112, 126, 140-1, 210
 side 76, 77
 chair conformation 34, 54, 55
 channels
 ion 126, 218, 222-4, 350, 384
 K⁺ 222
 ligand-gated 222
 passive 222
 protein 218
 voltage-gated Na⁺ 222
 chaperones 232, 233
 chaperonins 232, 233
 charge 59
 chemical bonds 4-7, 5
 chemical energy 112, 122
 chemical potential 12, 16, 22
 chemical reactions 14-15
 chemical work 17
 chemokines 392
 chemoluminescent immunoassay (CIA)
 chemotaxis 298
 chenodeoxycholic acid 56, 57, 314
 chiral center 8, 12, 35, 49
 chitin 40, 41
 chloramphenicol 254, 255
 chloride ions, bases 31
 chlorophylls 128, 129, 132, 133
 chloroplasts 196, 407, 407
 lumen 128, 129
 cholecalciferol *see* vitamin D
 cholecystokinin 270
 cholestane 54, 55
 cholesterol 54, 55
 cholesterol 52, 53, 56, 57, 65, 312, 315
 biosynthesis 172, 173
 esterase 425
 membranes 214, 215
 monooxygenase 422
 van der Waals models 54, 55
 cholic acid 56, 57, 314, 315
 choline 50, 51, 141, 354
 kinase 424
 O-acetyltransferase 354, 355, 423
 phosphate cytidyltransferase 425
 chondroitin 6-sulfate 347
 Christmas factor 291
 chromatin 208, 238, 239
 chromatography 55, 78
 chromosomes 238
 chylomicrons 163, 272, 278
 chymosin 427
 chymotrypsin 94, 95, 268, 269, 426
 CIA *see* chemoluminescence immunoassay
 circadian rhythm 372
 cirrhosis 320
 cis-trans isomerases 89
 cis-trans isomers 8, 9, 54, 55, 429
 cisplatin 402
 citrate 9, 137, 158
 fat metabolism 163
 lyase 139
 synthase 136, 137, 138, 428
 tricarboxylic acid cycle 144, 145
 citric acid 136
 cycle 141, 147
 citronellal 52, 53
 citrulline 62, 63, 182, 183
 classic pathway, complement 298
 clathrate, structure 28, 29
 cleavage, homolytic 108
 clones 258, 304
 closed-loop feedback control 372
 clotting 290-1, 291
 CMP 187
 3',5'-cNMP phosphodiesterase 425
 CNS, signal transmission 348-9
 co-lipase 268, 269
 co-receptors 296
 CO₂ *see* carbon dioxide
 CoA *see* acetyl CoA; coenzyme A
 coactivator/mediator com-

- plex 244
- coagulation factors 291, 426
- cobalamine *see* vitamin B₁₂
- cobalt 108, 109
- coding strands 85
- codogenic strands 85
- codons 236
 - anticodons 82, 86, 236, 248, 249
 - start 248, 250
 - stop 248, 250
- coenzyme A (CoA) 106, 107, 134
 - acetyl CoA *see* acetyl CoA
- coenzyme Q (CoQ) 104, 136, 140, 212
 - ATP synthesis 142, 143
 - isoprenoid anchors 52, 53
 - oxidized 212
 - redox systems 105
 - reduced 147
 - cytochrome *c* reductase complex 141, 421
- coenzymes 104–11, 105
 - availability 114
 - coenzyme A *see* coenzyme A
 - coenzyme Q *see* coenzyme Q
 - definition 104
 - energetic coupling 106, 107
 - flavin *see* flavin
 - group-transferring 106–7, 107, 108–9, 109
 - heme 106, 107
 - NAD *see* NAD
 - nucleotide triphosphates 110, 122–3
 - redox 104, 105, 106–7, 107
 - soluble 104
 - vitamins 364
- cofactors 46, 104
- collagenase 427
- collagens 64, 70, 71, 346
 - biosynthesis 344, 345
 - bone 340
 - helix 65, 68, 69, 344
 - procollagen 344
 - structure 344, 345
 - tropocollagen 344
- collision complexes 90
- colony-stimulating factors (CSF) 392
- combustion 21
- competitive inhibitors 96, 97
- complement 298–9
 - activation 298, 299
 - alternative pathway 298
 - C1–C9 298, 299
 - C3/C5 convertase 298, 299, 426
 - C5 convertase 298, 299
 - classic pathway 298
 - factors 298
- complementary DNA (cDNA) *see* cDNA
- complex glycoproteins 45
- complex I 140
- complex II *see* succinate dehydrogenase
- complex III 140
- complex IV 140
- complex V *see* ATP synthase, H⁺-transporting
- complexes
- complex lipids 170–1
- complex oligosaccharides 44, 230
- complex
 - basal transcription 242, 244, 245
 - coactivator/mediator 244
 - daunomycin–DNA 254
 - enzyme–substrate 92
 - enzymes 91, 95
 - fatty acid–albumin 163
 - formation 72
 - initiation, 70S 250
 - lipoprotein 312
 - membrane attack 298
 - multienzyme 134
 - oxoglutarate dehydrogenase 144, 145
 - pores 208
 - prothrombinase 291
 - repressor–allolactose 119
 - translocator 232
 - vitamin B₂ 366
- compounds 11
- concentration
 - gradients 126, 127
 - substrates 97, 116
 - and velocity 93, 97
- condensation, chromosomal 238
- cones 358
- conformation 8, 9, 35
 - A 86
 - cis*–*trans* 8, 9, 54, 55, 429
 - enzymes 92
 - native 72, 74
- R 117, 280, 281
- ring 54, 55
- space 66
- stabilizing interactions 72, 73
- T 117, 280, 281
- zig-zag (Z) 86
- conjugates 316
 - steroid hormones 110
- conservation
 - differences 98
 - energy 106, 126–7, 127
- constitution 6
- contraction
 - muscles 332–3
 - control 334–5
- control elements 118
 - transcription 244
- copper ions (Cu) 132, 133, 142, 143
- coproporphyrinogen III 192, 193
- CoQ *see* coenzyme Q
- core oligosaccharide 230
- core structures 44
- Cori cycle 338, 339
- corrin 108
- corticotropin *see* adrenocorticotrophic hormone
- cortisol 56, 57, 159, 338, 374, 376, 378
 - gluconeogenesis 120, 121, 154
- cosubstrates 104
- cotton fibers 42
- coupling
 - energetic 16, 18, 106, 112
 - ATP 16, 122, 124, 125
- covalency 6, 90, 108
- creatine 324
 - kinase 98, 336, 337, 424
 - metabolism 336, 337
 - phosphate 122, 336
- creatinine 336
- cristae 210
- CSF *see* colony-stimulating factor
- CTP 190, 191
 - dCTP 190, 191, 240
 - synthase 190, 191, 429
- cultures, primary 304
- curves, bell-shaped 94, 95
- cyanobacteria, photosystem (PS) I 132
- cycles, catalytic 100, 143
- 3'5'-cyclic AMP *see* cAMP

3'5'-cyclic GMP *see* cGMP
 cyclic phosphorylation 130, 131
 cyclin 394
 B 394
 cyclin-dependent protein kinases (CDK) 394
 CDK1 394
 cyclophosphamide 402
 cystathionine
 β -syntase 428
 γ -lyase 428
 cysteamine 12, 13, 62, 63
 cysteine 60, 61, 63
 acid–base balance 288
 dioxygenase 421
 proteinases 177, 427
 transaminase 424
 cystinuria 322
 Cyt P450 *see* cytochrome P450
 cytidine 80, 81
 5'-diphosphate (CDP) *see* CDP
 5'-monophosphate (CMP) *see* CMP
 5'-triphosphate (CTP) *see* CTP
 cytochalasins 204
 cytochrome c 140, 396
 oxidase 132, 133, 140, 141, 421
 proton pumps 126, 127
 cytochromes 106
 b/f complex 128, 129
 c *see* cytochrome c
 P450 172, 316, 318–19, 319
 cytokines 342, 392–3, 393
 cytoplasm 196, 202
 biochemical functions 203
 nuclear interactions 208, 209
 pH 31
 cytoplasmic pathway 228
 cytosine 80, 81
 cytoskeleton 196, 204–7, 216
 cytosol, fractionation 198
 cytostatic drugs 402–3
 cytotoxic T cells 294

D series monosaccharides 34
 DAG *see* diacylglycerol
 dATP 240
 daunomycin 254, 255
 dCDP 190
 dCTP 190, 191, 240
 ddNTPs 260, 261
 deamination 178–80, 179, 180, 181
 decarboxylases *see* carboxylases
 decarboxylation 62, 63, 108
 oxidative 134
 defense, blood 274
 deficiency
 minerals 362
 vitamins 364
 genetic code 248
 degradation 112
 amino acids 180
 cyclin 394
 nucleotides 186–7
 dehydration 108
 dehydrogenases (DH) 89, 104, 145
 dehydroascorbic acid 104
 denaturation 72
 dendrites 348
 density gradient
 centrifugation 201
 dentine 340
 2-deoxy-D-ribose 39
 deoxyaldoses 38, 39
 see also 2-deoxy-D-ribose; L-fucose
 deoxycholic acid 56, 57, 314, 315
 deoxyhemoglobin 281
 deoxyribodipyrimidine
 photolyase 428
 deoxyribonuclease I 425
 deoxyribonucleic acid *see* DNA
 deoxyribonucleosides 80
 triphosphates *see* dNTPs, ddNTPs
 2-deoxythymidine-5'-monophosphate *see* dTMP
 dephosphorylation 114
 depolarization 222
 channels 350
 membranes 348
 dermatan sulfate 347
 desmin 332

desulfurations 316
 detoxification 306
 development, cytokines 392
 dextrans 40, 41
 dGTP 240
 DH *see* dehydrogenases
 diabetes mellitus 76, 160–1
 ketone bodies 312
 diacylglycerol (DAG) 48, 49, 170, 171, 386, 387
 O-acyltransferase 170, 171, 423
 choline phosphotransferase 425
 diacylglycerols (DAG), lipase 425
 dialysis 78, 79
 2,4-dienoyl-CoA reductase 167, 421
 differentiation, hormones 370
 diffusion
 facilitated 218
 free 218
 digestion 266–73, 370
 digestive juices 268, 269
 dihydrofolate reductase 242, 304, 402, 403
 dihydrogen phosphate 11
 dihydrolipoamide
 acetyltransferase 134, 135, 423
 dehydrogenase 1, 34, 135, 421
 succinyltransferase 423
 dihydroorotate 427
 dihydroorotate 141, 188, 189
 dehydrogenase 421
 dihydrouridine 82, 83
 dihydroxyacetone
 phosphate *see* glyceraldehyde-1-phosphate
 3,4-dihydroxyphenylalanine *see* dopa
 dimers 116, 117
 dimethylallyl
 diphosphate 173
 transferase 423
 dimethylbenzimidazole 108
 2,4-dinitrophenol (DNP) *see* DNP
 dioxygenases 89
 dipeptidases 176, 177, 425–6
 dipeptides 66
 disaccharides 38, 39, 272,

D

273
 disruption 198
 dissociation curves 58
 dissolution 20, 21
 disulfides 10-11
 bonds 62, 72, 73, 106, 190, 300
 dithiol 106
 diuresis 328
 DNA 80, 84-5
 amplification 262
 antisense 264
 B-DNA 84, 86
 cDNA 260
 cloning 258-9, 259
 conformation 86-7, 87
 electrophoresis 262, 263
 fingerprinting 264, 265
 genetic coding 84
 germ-line 302
 gyrase *see* DNA topoisomerase (ATP-hydrolyzing, type II)
 helicase *see* DNA topoisomerase (type I)
 libraries 260, 261
 ligase 256, 257, 429
 nucleus 208
 polymerases 240, 241
 see also individual polymerases
 primers 240
 proteins 65
 sequencing 260-1, 261
 structure 84, 85
 templates 240
 zig-zag 86
 Dna K 232
 DNA polymerases 240, 241
 repair 256, 257
 DNA topoisomerases 240
 (ATP-hydrolyzing, type II) 240, 429
 (type I) 240, 245, 429
 DNA-binding proteins, oncogenes 398
 DNA-dependent DNA polymerases 240, 242
 DNA-directed DNA polymerases 424
 DNA-directed RNA polymerases 240, 242, 243, 245, 424
 DNP 144
 dNTPs 260, 261, 262

dolichol 53, 53
 diphosphate 230
 domains
 dimers 116
 receptors 224
 SH2 392
 steroid receptors 378
 donors 6
 electrons 88
 dopa 62, 63, 352
 L-dopa 6, 7, 63
 dopamine 62, 63, 352
 β -monooxygenase 352, 353, 422
 double helix 84, 86
 double stranded (dsDNA)
 see dsDNA
 doxorubicin 402
 drugs, cytostatic 402-3
 dTMP 80, 81, 187, 191
 dTTP 190, 191, 240
 dUMP 190
 dynamics, proteins 72, 73

E

e^- *see* electrons
 EA *see* ethanolamine
 Eadie-Hofstee plots 96, 97
 EC number 88
 ecdysone 56, 57
 ECM *see* extracellular matrix
 EcoRI sites 258, 259
 effectors
 allosteric 114, 115, 116
 caspases 396
 domain 224
 EIA *see* enzyme-linked immunoassay
 eicosanoids 46, 47, 390-1, 391
 autocrine effects 390
 paracrine effects 390
 signaling 46, 390
 elastase 268, 269, 426
 electrical dipole 26
 electrical work 17
 electrochemical gradients 126, 127, 128
 electrolytes, recycling 328-9, 329
 electron-transferring flavoprotein (ETF) 164
 dehydrogenase 164, 165, 421
 electronegativity 6
 electroneutral processes 218
 electrons 14
 acceptors 88
 configurations 2, 3
 donors 88
 excitation 128, 129
 oxidized forms 32
 p 4
 pairs 6
 potential 18, 19
 reduced forms 32
 reducing equivalent 33
 singlet state 130
 transfers 33
 transport chain 128, 140
 electrophoresis 263, 277
 carrier 276
 DNA 262, 263
 lactate dehydrogenase (LDH) 99
 SDS gel 78, 79
 elementary fibrils 42, 43
 elements 3
 eliminations 14, 15
 eliminative deamination 180
 elongation 242
 Escherichia coli 251
 factors 252
 elution, proteins 78, 79
 embryo, hemoglobin 280
 emulsions 28, 270
 enamel 340
 enantiomers 8
 enclosure, membranes 216
 endergonic processes 16, 17, 112
 endergonic transfer 18
 endocrine effects 372, 373
 endocytosis 234, 235
 receptor-mediated 278
 endonucleases, restriction 258, 259, 425
 endopeptidases *see* metalloproteinases
 endoplasmic reticulum (ER) 196, 226, 227
 gluconeogenesis 154
 protein folding 232, 233
 protein synthesis 231
 rough (rER) 226, 232

- smooth (sER) 172, 173, 226, 227
- endosomes 196, 234
- endosymbiotic theory 210
- endothermic reactions 20
- energetic coupling 16, 17, 18, 112, 124-5, 125
ATP 16, 122, 124
coenzymes 106, 107
- energetics 16-17, 17
coupling *see* energetic coupling
protein folding 74, 75
- energy
activation 25
chemical 112, 122
conservation 106, 126-7, 127
definition 16
metabolism 336, 356
requirements 360, 361
- engineering, genetic 258
- enhancers 118
- enol phosphate 124
- enolase *see* phosphopyruvate hydratase
- enoyl-[ACP] reductase 168, 169, 421
- enoyl-CoA
hydratase 165, 166, 428
isomerase 166, 167, 429
- enterohepatic circulation 194, 314
- enteropeptidase (enterokinase) 176, 270, 426
- enthalpy 20-1, 21
folding 74
free 142, 143
- entropy 20-1, 21
- envelope conformation 54, 55
- enzyme-substrate complexes 92
- enzyme-catalyzed reactions 90, 93
- enzyme-linked immunoassay (EIA) 304
- enzymes
activating 114
active centers 95
activity 88, 89
allosteric 92, 93
analysis 102-3
basics 88-9
catalysis 90-1, 216
classes 88, 89
see also individual classes
complement cascade 298
complexes 91, 95
conformations 92
glucose determination 102, 103
inhibitors *see* inhibitors
interconversion 114
isoenzymes 98, 99
isosteric 92, 93
kinetics 92-5
markers 198
nomenclature 420-30
pH 94, 95
proteolytic 177
reaction specificity 88
regulatory mechanisms 114
serum diagnosis 98, 178
substrate specificity 88
temperature 94, 95
turnover 88, 89
watersplitting 130
see also individual enzymes
- epimers 36
epimerases 89, 428
epimerization 36, 37
- epinephrine 159, 163, 352, 380
- episodic release 372
- equilibrium 16, 17, 18-19
constant (K) 18, 19
nitrogen 174, 175
potential 126, 350
- equatorial (e) position 34
- ER *see* endoplasmic reticulum
- ergosterol 56, 57
- ERK *see* MAPK
- erucic acid 49
- erythrocytes 274
gluconeogenesis 154
metabolism 284-5, 285
- erythromycin 254, 255
- erythropoiesis 286
- erythropoietin 330
- erythrose-4-phosphate 153
- Escherichia coli*
aspartate carbamoyltransferase 116
cell components 202-3
elongation 251
lactose operon 118
protein biosynthesis 251
pyruvate dehydrogenase 134, 135
replication 240, 241
termination 251
translation 251
- essential amino acids (EAA) 60, 174, 360
active centers 99
biosynthesis 184, 185, 410, 412
non-essential 184, 413
- essential fatty acids 46, 48
- ester bonds 12
- esterases 89
- esterification 36, 37
- esters 15, 47
- estradiol 56, 57, 374, 376
- estrogen 342
receptor 72, 73
- ethanal 320
- ethanol 27, 320-1, 321
- ethanolamine (EA) 50, 51, 62, 63
kinase 424
phosphate cytidyltransferase 424
phosphotransferase 425
- ethers 10, 11
- ethidium bromide 262, 263
- euchromatin 208, 238
- eukaryotes 196, 197, 250, 251
- event-regulated hormones 372
- excision repair 256
- excitation, electrons 128, 129
- excretion
ammonia 326, 327
hormones 370
kidneys 322
liver 306
protons 326
- exergonic processes 16, 17, 18, 112
- exocytosis 226, 228, 229
neurotransmitters 348
- exons 242, 243
- exopeptidases 176, 177
- exosomes 196
- exothermic reactions 20
- expression plasmids 262

extensin 42, 43
extracellular matrix (ECM)
346-7, 347

F

- F type ATPase 220
F-actin 64, 65, 332
Fabales 184
facilitated diffusion 218
factors, intrinsic 268, 269
FAD *see* flavin adenine dinucleotide
Faraday constant 19
farnesol 52, 53
farnesyl diphosphate 173
farnesyltransferase 423
Fas system 396
fat-soluble vitamins 46
fats 47, 48-9
 biosynthesis 171, 409
 brown 144, 145
 degradation 410-11
 digestion 270, 271
 metabolism 162-3, 163, 312
 nutrients 360
 structure 48, 49, 51
 synthesis, liver 162
 uncouplers 144, 145
fatty acid synthase 12, 168, 169, 423
 fat metabolism 162, 163
fatty acid-albumin complex 163
fatty acid-CoA ligase 163, 272, 273
fatty acids 46, 47, 48-9, 146, 312
 degradation 163, 164-5, 165, 166-7
 essential 46, 48
 liver 162
 long 166
 muscle fibers 336
 odd-numbered 166, 167
 resorption 272
 synthesis 113, 163, 168-9, 169
 transport 164, 165
 unsaturated 48, 166, 167
fatty liver 320
Fd *see* ferredoxin
Fe protein 184-5
Fe-Mo protein 184, 185
Fe-protoporphyrin IX 192
feedback inhibition 114, 115, 158
 purines 188
fermentation 112, 146, 148-9, 149
ferredoxin 184, 185
ferredoxin-NADP⁺ reductase 422
ferritins 286, 287
ferredoxin (Fd) 128, 129
fetal hemoglobin 280
FFA *see* free fatty acids
fibers
 cotton 42
 muscle 332, 337
 red muscle 336
 resorption 266
 white muscle 336
fibrils 42, 344
 elementary 42, 43
fibrin stabilizing factor 291
fibrin-stabilizing factor *see* protein-glutamine γ -glutamyltransferase
fibrinogen 291
fibrinolysis 292-3, 293
fibroin 70
fibronectins 346, 347
filaments
 intermediate 70, 71, 204, 205, 206, 207
 keratin 70, 71, 206
 microfilaments 204, 206, 207
 protofilaments 71, 204
films
 amphipathic 28
 surface 29
filtration
 cell 198
 ultrafiltration 322
first-order reactions 22, 23
Fischer projections 9, 35, 59
fixation, carbon dioxide 130
flavin 104, 140
 redox systems 33
flavin adenine dinucleotide (FAD) 32, 81, 104
 pyruvate dehydrogenase 134
flavin mononucleotide (FMN) 32, 104, 105
 ATP synthesis 142, 143
flavodoxin 72, 73
flavoprotein (Fp)
5-fluorouracil 402
fMet *see* N-formylmethionine
folate 366
folic acid 254, 255
food 34, 266, 360
foreign genes 264
forks, replication 240
formate 4, 5
formic acid 49
N-formylmethionine (fMet) 250, 251
5-formyltetrahydrofolate cycloligase 429
Fp *see* flavoprotein
fragments, Okazaki 240
frameshift mutation 257
free diffusion 218
free electron pairs 6
free enthalpy (G) 16, 142, 143
free fatty acids (FFA) 48
free radicals 32, 190, 256
fructose-1,6-bisphosphate 151
 β -fructofuranosidase 38, 426
fructose
 -1,6-bisphosphatase 154, 155
 bisphosphate 425
 1,6-bisphosphate 150, 154, 159
 2,6-bisphosphate 158, 159
 2,6-bisphosphate 2-phosphatase 425
 bisphosphate, adolase 310, 311, 428
 D-fructose 36, 37, 39
 1-phosphate 38, 310
 6-phosphate 150, 151, 153, 155, 159
fructoses
 gluconeogenesis 328
 metabolism 310, 311
 uptake 272
fucose 34, 44
 L-fucose 39
fucosyltransferase 292, 293
fuel, lipids 46
fumarate 136, 137, 182, 183, 189
 dehydrogenase 136, 137
 hydratase (fumarase) 183,

428
 fumaric acid 8, 9
 fumarylacetoacetase 427
 functional groups 10
 furanose 34

G

G proteins
 Ras 388
 signal transduction 384, 385
 G₀ phase 394
 G₁ phase 394
 GABA 63, 356, 357
 galactokinase 310, 311, 424
 galactose 44
 D-galactose 38, 39, 44
 metabolism 310, 311
 uptake 272
 galactose 1-phosphate 310
 galactosidase 119
 β-galactosidase 426
 galactosylceramide 50, 51
 galactosyltransferase 292, 293
 gallbladder 195
 gallstones 314
 GalNAc *see* N-acetyl-D-galactosamine
 γ-aminobutyric acid *see* GABA
 γ-carboxyglutamate 62, 290
 γ-globulins 276
 γ-glutamyltransferase 423
 gangliosides 46, 47, 50, 51, 214, 215
 gas constant (R)
 gases, transport 282-3
 gastric juice 268, 269
 pH 31
 gastricsin (pepsin C) 427
 gastrin 270
 gastrointestinal hormones 270
 GDP 190, 191
 gels 79
 electrophoresis
 DNA 263
 lactate dehydrogenase (LDH) 99
 SDS 78, 79
 filtration 78, 79
 particles 78, 79

permeation chromatography 78
 genes
 expression 236-7
 foreign 264
 genetic information 236, 237
 libraries 260, 261
 multiple 302
 probes 260
 regulation 118
 structural 118
 therapy 264, 265, 402
 genetic code 85, 248, 249
 genetic engineering 258, 264-5
 genetics, molecular 236-65
 genomes 236, 238-9
 geometric isomers 8
 geraniol 52, 53
 geranyl diphosphate 173
 geranyltransferase 423
 germ-line DNA 302
 Gibbs—Helmholtz equation 20, 21
 glandular hormones 372
 GlcNAc *see* N-acetyl-D-glucosamine; N-acetylglucosamine
 Glc_A *see* D-glucuronic acid
 globular proteins 72-3
 globulins 276, 277
 β-globulins 286
 glucagon 163, 308, 380
 gluconeogenesis 154, 158, 159
 glucagon receptor 121
 1,4-α-glucan, branching enzyme 157, 423
 α-1,4-glucan, chain 157
 4-α-glucanotransferase 157, 423
 D-glucofuranose 35
 glycogen 120
 gluconeogenesis 35, 113, 310, 311
 amino acids 138, 154, 155, 180
 cortisol 120, 121
 glucagon 154, 158, 159
 insulin 160, 161
 recycling 328, 329
 gluconic acid 36
 gluconolactonase 153, 425

gluconolactone 37, 103
 D-glucopyranose 34, 35
 glucose 34, 35, 65, 146, 162
 adipose tissue 160, 161
 brain 356
 chain 35
 D-glucose 34, 36, 39, 44
 enzymic determination 102, 103
 gluconeogenesis 154, 155
 glycogen 156, 157
 glycolysis 150, 151
 insulin 121, 158, 159, 160, 161
 kidney 160, 161
 lactose operon 118
 liver 160, 161
 metabolism 120, 121
 muscle 160, 161
 oxidase 102, 103, 420
 oxidation 36, 146, 147
 reduction 36
 transporter Glut-1 220, 221
 uptake 272
 glucose oxidation, hexokinase 147
 glucose 1-phosphate 156, 157, 159, 310
 glucose 6-phosphatase 154, 155, 157, 425
 glucose 6-phosphate 37, 150, 151, 152, 157, 159
 dehydrogenase 152, 153, 420
 gluconeogenesis 154, 155
 isomerase 151, 429
 pentose phosphate pathway 153
 α-glucosidase 224
 glucoside formation 36
 glucosuria 160
 glucosylceramide 50
 glucuronate 37
 glucuronic acid 36, 44
 β-glucuronidases 194
 glucuronosyltransferase 194, 195, 316, 317, 423
 glutamate 178, 326, 357
 5-kinase 424
 decarboxylase 356, 357, 427
 dehydrogenase 178, 179, 181, 183, 326, 327, 421
 family 184, 185

- formiminotransferase 422
 neurotransmitters 356
 glutamate-NH₃ ligase 429
 glutamic acid 60, 61
 glutaminase 178, 179, 181, 327, 356, 357, 427
 glutamine 60, 61, 178, 326, 357
 brain 356
 gluconeogenesis 328
 muscle metabolism 338
 purine biosynthesis 188, 189
 pyrimidine biosynthesis 188, 189
 synthase 64
 synthetase 65, 124, 125, 356, 357, 429
 glutamylphosphate reductase 420
 glutathione 106, 284, 285
 disulfide bonds 72
 oxidized
 peroxidase 284, 285, 421
 reductase 284, 285, 421
 glyceral-3-phosphate dehydrogenase 212, 213
 glyceral 5-phosphate 153
 glyceraldehyde 310
 D-glyceraldehyde 34
 glyceraldehyde 3-phosphate 150, 151, 153, 155
 glyceraldehyde-3-phosphate dehydrogenase 124, 125, 420
 glucose oxidation 147
 glycerophosphate shuttle 212, 213
 glycolysis 150, 151
 glycerol 27, 48, 49, 154, 170, 171
 gluconeogenesis 328
 kinase 155, 170, 171, 424
 glycerol 3-phosphate 154, 155, 170, 171
 dehydrogenase 155, 170, 171, 212, 213, 420
 O-acyltransferase 170, 171, 423
 glycerone phosphate O-acyltransferase 423
 glycerone 1-phosphate 310
 glycerone 3-phosphate 151, 154, 170, 171
 α-glycerophosphate 141
 glycerophosphate shuttle 212, 213
 glycine 60, 61, 192, 193, 314
 amidinotransferase 336, 337, 422
 collagen 70
 dehydrogenase 421
 hydroxymethyltransferase 422
 glycinuria 322
 glycosaminoglycans 110
 glycocalyx 214, 230
 glycocholic acid 314, 315
 glycogen 34, 35, 40, 41, 120, 121
 balance 157
 liver 156
 metabolism 156-7, 157
 muscle 156, 336
 polymer 41
 (starch) phosphorylase *see* phosphorylase
 (starch) synthase 423
 synthase 120, 121, 156, 157, 158, 159
 kinase 3 (GSK-3) 388
 glycogenin 34, 35, 156
 glycolipids 35, 46, 47, 50-1
 membranes 214
 structure 50, 51
 glycolysis 35, 113, 146, 150-9, 151, 336, 338
 glycoproteins 35, 230
 complex 45
 forms 45, 45
 mannose-rich 45
 O-linked 45
 glycosaminoglycans 44-5, 346
 N-glycosidic bonds 80
 glycosidases 89, 426
 O-glycosidases 231
 N-glycosides 36, 37
 N-glycosidic bonds 12, 13, 44, 123
 N-glycosidic links 44
 O-glycosidic links 44
 glycosylation 62, 230
 collagen 344
 proteins 230, 231
 sequence 230
 glycosylceramide 50, 51
 glycosyltransferases 89, 231
 glyoxylic acid cycle 138
 GMP 186, 187, 190, 191
 synthase 429
 Golgi apparatus 196, 226, 227
 gonane 54, 55
 GOT *see* aspartate transaminase
 gout 186, 187
 gp130 392
 GPT *see* alanine transaminase
 grana 128, 129
 granulocytes 274, 275
 Grb-2 388
 GroEL 232, 233
 GroES 232, 233
 grooves, strands 86
 group transfer
 coenzymes 106-7, 107, 108-9, 109
 potential 12, 19
 reactions 18, 19
 group X 108
 groups
 hydroxyalkyl 106
 leaving 14
 prosthetic 104, 134, 142
 growth
 factors 342
 hormones 370
 GSH *see* glutathione, reduced
 GSK-3 *see* glycogen synthase kinase-3
 GSSG *see* glutathione, oxidized
 GTP 136, 190, 191
 dGTP 240
 GTP-binding proteins 398
 guanidine
 2'-O-methylguanidine 82, 83
 7-methylguanidine 82, 83
 guanidinoacetate 336
 N-methyltransferase 336, 337, 422
 guanine 80, 81, 186, 187
 guanosine 80, 81, 186
 5'-diphosphate (GDP) *see* GDP
 5'-monophosphate (GMP) *see* GMP

5'-triphosphate (GTP) *see* GTP
 guanylate cyclase 358, 359, 388, 389, 428
 L-gulonolactone oxidase 420
 gyrase 240
 inhibitors 254

H

H⁺ *see* hydronium ions (H₃O⁺); protons
 H⁺-transporting ATP synthase *see* ATP synthase, H⁺-transporting
 H⁺-transporting ATPase 427
 H⁺/K⁺-exchanging ATPase 270, 271, 427
 Hageman factor 291
 hair 70
 Hartnup's disease 322
 HAT medium 304
 Haworth projections 34, 35
 Hb *see* hemoglobin
 HCO₃⁻, resorption 326
 HDL 278
 heart
 absorptive state 308
 muscle 332
 heat
 loss 20
 of reaction 20, 21
 heat-shock proteins (hsp) 232, 378
 heavy chains, IgG 300
 helicase 240
 helices
 1-helix receptors 384
 7-helix receptors 225, 384
 α-helix 67, 68, 69, 71
 α-keratin 70, 71
 collagen 68, 70, 344
 DNA 84, 86
 double 84, 86
 left-handed 86
 pitch 86
 superhelix 70, 71
 triple helix 70, 71, 344
 helper T cells 294
 heme
 ATP synthesis 142, 143
 b 192
 biosynthesis 192-3, 193,

210
 catalase 25
 coenzymes 106, 107, 140, 141
 degradation 194-5, 195
 groups 280
 iron 286
 monooxygenases 318
 oxygenase 194, 195, 422
 oxygenated 287
 hemiacetals 10, 11, 34
 hemicellulose 42
 hemochromatosis 286
 hemoglobin (Hb) 64, 65, 280-1, 281
 allosteric effect 116, 280
 buffers 288
 transport 282, 283
 hemosiderin 286, 287
 Henderson–Hasselbalch equation 18, 19
 heparin 290, 347
 hepatic glycogen 156
 hepatic metabolism 306
 hepatocytes 163, 306, 307
 heterochromatin 208, 238
 heterogenous nuclear ribonucleic acid (hnRNA) *see* hnRNA
 heteroglycans 40
 heterotrophs 112, 128
Hevea brasiliensis 52
 hexokinase 147, 159, 424
 hexose monophosphate pathway (HMP) *see* pentose phosphate pathway
 hexose-1-phosphate uridylyltransferase 310, 311, 424
 hexoses 39
 high-density lipoprotein (HDL) *see* HDL
 high-performance liquid chromatography (HPLC)
 Hill coefficient (H) 116
 hippurate 324
 histamine 270, 380
 histidine 59, 60, 61, 63, 184, 185
 ammonia lyase 428
 dissociation curves 58
 lactate dehydrogenase 100
 histone
 acetyl transferase 244, 245

deacetylase 244, 245
 histones 64, 238, 239
 non-histone proteins 238
 phosphorylation 238
 HIV 404, 405
 HMG-CoA 172, 173, 312
 dehydrogenase 420
 lyase 312, 313, 428
 reductase 172, 173
 synthase 312, 313, 428
 HMP *see* hexose monophosphate pathway
 hnRNA 236, 242
 immunoglobulins 302
 splicing 246, 247
 homeostasis 322, 326, 328
 blood 274
 calcium 342, 343
 cytokines 392
 hormones 370
 see also acid–base balance
 homocysteine 110
 homogenization 198
 homogentisate 1,2-dioxygenase 421
 homoglycans 40
 homoserine
 dehydrogenase 420
 kinase 424
 honey 38
 hops 148
 hormone response elements (HREs) 245, 378
 hormone-sensitive lipase 162, 163
 hormones 315, 322, 370-1, 371
 action 120, 121, 378-9
 biosynthesis 370, 383
 carbohydrate metabolism 158
 closed-loop feedback control 372
 degradation 383
 differentiation 370
 digestive processes 370
 effects 370, 372, 373
 episodic release 372
 event-regulated 372
 excretion 370
 gastrointestinal 270
 glandular 372
 glucose metabolism 121
 growth 370

- hierarchy 372, 373
 - homeostasis 370
 - hydrophilic 120, 121, 380-9
 - inactivation 383
 - ion concentrations 370
 - juvenile 52, 53
 - lipophilic 120, 374-5, 375, 379
 - lipophilic *see* lipophilic hormones
 - metabolism 120-1, 370
 - neurohormones 348, 349
 - peptide 380, 381, 382-3
 - plasma levels 372-3, 373
 - proteohormones 380, 381
 - proteolysis 382
 - pulsatile release 372
 - receptors 120
 - regulation 120, 370, 371
 - renal 330-1, 331
 - second messengers 386-7
 - signal cascades 388-9
 - steroid *see* steroid hormones
 - tissue 370
 - transport 370
 - vitamins 364
 - HPLC *see* high performance liquid chromatography
 - HRE *see* hormone response elements
 - hsp *see* heat-shock proteins
 - human
 - body 31
 - immunodeficiency virus (HIV) *see* HIV
 - humoral immunity 294
 - hyaluronate 346, 347
 - hyaluronic acid 41, 44, 45
 - hybridomas 304
 - hydration 26, 27, 44
 - shell 90
 - hydride ions 32, 33, 104
 - hydrocarbons 46, 47
 - hydrochloric acid 266, 270, 271
 - hydrogen
 - atoms 33
 - bonds 6, 7, 26, 68, 72
 - carbonate 182
 - chloride 30, 31
 - ions 288, 289
 - peroxide 24, 186, 284
 - catalysis 25
 - redox systems 33
 - reducing equivalents 33
 - sulfide 11
 - hydrolases 88, 89, 234, 425
 - hydrolysis 266, 267
 - G of 18
 - hydrolytic cleavage 316
 - hydrolytic deamination 179, 180
 - hydrolyzable lipids 46
 - non-hydrolyzable 46
 - hydronium ions (H_3O^+) 30, 31
 - hydroperoxyl radical 33
 - hydrophilic hormones 120, 121, 380-9, 385
 - hydrophilic molecules 26, 28
 - hydrophilic signaling substances 392
 - hydrophobic effects 72
 - hydrophobic molecules 26, 28
 - hydroquinone 33, 104
 - hydroxy radicals 284
 - 3-hydroxy-3-methylglutaryl-CoA *see* HMG-CoA
 - 3 β -hydroxy- Δ^5 -steroid dehydrogenase 420
 - β -hydroxyacyl-CoA 164
 - 3-hydroxyacyl-CoA dehydrogenase 165, 420
 - hydroxyalkyl groups 106
 - hydroxyalkyl thiamine diphosphate 135
 - 3-hydroxybutyrate 141, 312
 - dehydrogenase 420
 - β -hydroxybutyric acid 160
 - 3-hydroxyisobutyrate dehydrogenase 312, 313, 420
 - 3-hydroxyisobutyryl-CoA hydrolase 425
 - hydroxyl ions (OH^-) 30, 31
 - 3-hydroxylacyl-CoA 141
 - hydroxylation 62, 70
 - collagen 344
 - hydroxymethylbilane synthase 192, 193, 428
 - hydroxymethylglutaryl-CoA *see* HMG-CoA
 - 3-hydroxypalmitoyl-[ACP] dehydratase 168, 169, 428
 - 4-hydroxyphenylpyruvate dioxygenase 421
 - hydroxyproline, collagen 70
 - 3 β -hydroxysteroid dehydrogenase 420
 - 17 β -hydroxysteroid dehydrogenase 420
 - 5-hydroxytryptophan 63
 - hydroxyurea 402
 - hyperbilirubinemia 194
 - hyperbolic plots 93, 97
 - hyperglycemia 160, 161
 - hyperlipidemia 160, 161
 - hyperpolarization 348
 - hyperuricemia 186, 187
 - hypervitaminosis 364
 - hypoiodide 25
 - hypovitaminosis 364
 - hypoxanthine 186, 187, 304
 - phosphoribosyltransferase 186, 187, 402, 403, 423
- I**
- ice 26, 27
 - idiotypic variation 302
 - IDL 278
 - iduronic acid 44
 - IF *see* intermediary filaments
 - IFN *see* interferons
 - Ig *see* immunoglobulins
 - IL *see* interleukins
 - imidazolonepropionase 427
 - imino acids 179
 - immune response 294-5, 295
 - immune system
 - acquired 294
 - cytokines 392
 - innate 294
 - immunoassays 304, 305
 - immunoglobulins (Ig)
 - classes 301
 - domains 300
 - heavy chains 302
 - IgA 300
 - IgD 300
 - IgE 300
 - IgG 44, 45, 64, 65
 - domains 300, 301
 - IgM 296, 300
 - light chains 302, 303
 - superfamily 296
 - variability 302, 303
 - IMP 186, 187
 - biosynthesis 188, 189,

- 190, 191
- cyclohydrolase 427
- dehydrogenase 420
- innate immune system 294
- inducers 118
 - catabolite activator protein (CAP) 118
- induction, transcription 114, 262
- influenza virus 404
- inhibition, feedback 114, 115
- inhibitors 96-7, 97
 - allosteric 96
 - competitive 96, 97
 - gyrase 254
 - irreversible 96
 - kinetics 96, 97
 - reversible 96
 - translation 254
- initiation
 - complexes, 70 S 250, 251
 - factors 250
 - transcription 242, 244, 245
- inorganic components
 - food resorption 266
 - urine 324
- inorganic ions 2, 202
- inorganic phosphate (P_i) 125, 150
- inosine 5'-monophosphate (IMP) *see* IMP
- inositol 1, 4, 5-trisphosphate ($InsP_3$) 170, 171, 386, 387
- insulation
 - lipids 46, 47
 - membranes 216
- insulin 64, 70, 76, 77, 160, 161, 380
 - A chain 76, 77
 - B chain 76, 77
 - biosynthesis 151, 160-1
 - C peptide 76, 77, 160, 161
 - circuit 373
 - deficiency 160, 161
 - gluconeogenesis 160, 161
 - glucose 121, 158, 159, 160, 161
 - human 76
 - monomer 76
 - overexpression 76
 - pentose phosphate pathway 152
 - postabsorptive state 308, 309
 - preproinsulin 160, 161
 - proinsulin 76, 160, 161
 - receptor *see* insulin receptor
 - signal transduction 388, 389
- insulin receptor 121, 224, 225, 388
 - substrate (IRS) 388
- intensity factor 16
- intercalators 225, 254
 - daunomycin-DNA complex 254
- interconversion 134
 - enzymes 114, 115
 - processes 144, 394
 - pyruvate dehydrogenase 134
 - regulation 118, 120
- interferons (IFN) 392
- interleukins (IL) 392
 - IL-6 receptor 392
- intermediary metabolism 112, 322
- intermediate filaments (IF) 204, 205, 206, 207
 - α -keratin 70, 71
- intermediate-density lipoprotein *see* IDL
- intermediates 142, 143
- intermembrane space 210
- international units (U) 88
- interphase 394
- intestines, pH 31
- intramolecular transferases 89
- intrinsic factor 268, 269
- introns 242, 243
- inulin 40, 41
- invertase *see* β -fructofuranosidase
- iodide
 - catalysis 25
 - hypoiodide 25
- iodothyronines 374
- ion channels 65, 126, 218, 222-4, 350, 384
 - see also* individual receptors
- ionic strength 78
- ionizing radiation 256
- ionotropic receptors 348, 354
- ions
 - calcium 328, 386
 - hormone concentrations 370
- inorganic 202
- products 30
- IPTG *see* isopropylthiogalactoside
- iron
 - deficiency 286
 - distribution 287
 - ion, hemoglobin 280
 - metabolism 286-7, 287
 - protein 184-5
 - uptake 286
- iron-sulfur centers 140, 141, 142, 143
- iron-sulfur clusters 106, 107, 286, 287
- iron-protoporphyrin IX 192
- IRS *see* insulin-receptor substrate
- islets of Langerhans 160
- isoalloxazine *see* flavin
- (2R,3S)-isocitrate 9, 127
- isocitrate 136, 141
 - dehydrogenase 136, 137, 420
 - glucose oxidation 147
 - tricarboxylic cycle 144, 145
- isoelectric point 58, 59
- isoenzymes 99
 - pattern 98
- isolation, proteins 78
- isoleucine 60, 61
- isomerases 88, 89, 428
 - cis-trans*-isomerases 89
 - coenzymes 108
- isomerization, light-induced 358
- isomers
 - geometric 8
 - isomerism 8-9
- isopentenyl diphosphate 52, 53, 172, 173
- isopentenyl-AMP 53
- isoprenoids 52-3, 53, 172
- isoprenylation 52
- isopropylthiogalactoside (IPTG)
- isosteric enzymes 92, 93
- isotypes 302
- isopyknic centrifugation 200

J

J segments 302
 Janus kinases (JAK) 392
 jaundice 194
 joules 16, 17
 juvenile hormone 52, 53

K

K⁺ channels 223
 K systems 116
 katal 88
 keratan sulfate 247
 keratin
 filaments 206
 α -keratin 70, 71
 kernicterus 194
 ketamine 179
 ketoacidosis 288
 ketoacidotic coma 312
 β -ketoacyl-CoA 164
 ketogenic amino acids 180
 α -ketoglutarate *see* 2-oxo-glutarate
 ketohexokinase 310, 311, 424
 ketohexoses 38
 D-fructose *see* D-fructose
 ketone bodies 160, 161, 162, 163, 172
 acid–base balance 288
 biosynthesis 312, 313
 brain 356
 synthesis 410
 ketonemia 312
 ketones 10, 11, 162
 ketonuria 160, 312
 ketopentoses 38
 D-ribulose *see* D-ribulose
 ketoses 39
 kidney, glucose 160, 161
 kidneys 154, 288, 322–3, 323
 kinetics
 allosteric enzymes 116
 aspartate 117
 bisubstrate 94, 95
 enzymes 92–3
 inhibition 96, 97
 see also reaction kinetics
 K_m *see* Michaelis constant
 knall-gas 20
 reaction *see* oxyhydrogen reaction

L

L segments 302
 lac-repressor 118, 119
 lactacidosis 288
 β -lactam antibiotics 254
 lactate 98, 99, 154, 155, 288
 dehydrogenase *see* lactate dehydrogenase
 gluconeogenesis 328
 muscle 338
 redox reactions 19
 lactate dehydrogenase (LDH) 99, 420
 activity 102, 103
 catalytic cycle 100, 101
 Cori cycle 338, 339
 fermentation 148, 149
 gel electrophoresis 99
 gluconeogenesis 155
 glucose oxidation 147
 isoenzymes 98
 structure 98
 lactic acid 8, 9, 149
Lactobacillus 148, 149
 lactoferrin 286, 287
 lactose 38, 39
 allolactose 119
 lactose operon 118
 operon 118, 119
 laminin 346
 lanosterol 173
 lattices
 hexagonal 27
 ice 26
 lauric acid 49
 LDH *see* lactate dehydrogenase
 LDL 278
 receptor 278
 leaving groups 14
 lecithin *see* phosphatidylcholine
 lecithin-cholesterol acyltransferase (LCAT) *see* phosphatidylcholine-sterol acyltransferase
 lefthanded α -helix (α_L) 68
 Lesch–Nyhan syndrome 186
 leucine 60, 61
 transaminase 424
 leukocytes 274
 leukotrienes 390
 liberins 372

libraries

 DNA 260, 261
 genes 260, 261
 Lieberkühn glands 268
 life cycle, HIV 404
 ligand-gated channels 222
 nicotinic acetylcholine receptor 222
 ligands 118, 119
 modulation 114, 115
 oncogenes 398
 ligases 88, 89, 106, 429
 light
 absorption 102, 103, 128, 129
 monochromatic 102
 ultraviolet 256
 light chains
 biosynthesis 302
 IgG 300
 light-induced isomerization 358
 lignoceric acid 49
 Lineweaver–Burk plots 92, 93, 96
 linoleic acid 48, 49
 linolenic acid 48, 49
 lipase 268, 269
 lipid 46–57
 alcohols 46
 anchors 214
 classification 47
 complex 170–1
 hydrolysis 266
 hydroperoxide glutathione peroxidase 421
 lipoproteins 278
 membranes 214, 215, 216
 metabolism 160, 306, 312–13, 313
 resorption 272, 273
 role 46, 47
 synthesis 226
 lipid-soluble vitamins 46, 364–5, 365
 lipoamide 106, 107, 134
 arm 134, 135
 lipogenesis 162
 lipoic acid 62, 106
 lipolysis 162
 lipophilic hormones 120, 374–5, 375, 378, 379
 receptors 378, 398
 lipoprotein 278–9, 279
 lipase 162, 163, 278, 279,

425
 lithocholic acid 56, 57, 314
 liver
 absorptive state 308
 alcohol 320
 cirrhosis 320
 damage, alcohol 321
 liver, fat synthesis 162
 fatty 320
 fatty acids 162
 functions 306-7, 307
 glucose 160, 161
 glycogen 156
 metabolism *see* liver metabolism
 postabsorptive state 308
 postresorption 309
 resorption 309
 urea 182
 liver metabolism 306, 307
 ammonia 178, 179
 glucose 120, 121
 long-chain fatty-acid-CoA ligase 429
 loops
 chromosomes 238
 mobile 99
 peptides 98
 low-density lipoprotein *see* LDL
 lumen, chloroplasts 128, 129
 lyases 88, 89, 427-8
 lymphocytes 161, 274, 275, 294
 monoclonal antibodies 304
 lymphokines 392
 Lynen cycle 312
 lysine 60, 61, 184
 hydroxylase *see* procollagen lysine 5-dioxygenase
 lysines, acetylation 238
 lysophosphatidate 170, 171
 lysophospholipid 50, 51
 lysosomes 196, 234-5, 235
 pH 31
 lysozyme 42

M

M phase 394
 M13 phage 260, 404
 mAb *see* monoclonal anti-

bodies
 macroelements 2, 363
 macromolecules 202
 macrophages 274, 275
 major histocompatibility complex *see* MHC
 malate 137, 141
 dehydrogenase 136, 137, 138, 139, 420
 gluconeogenesis 154, 155
 glucose oxidation 147
 urea cycle 183
 shuttle 212, 213
 maleic acid 8, 9
 maleylacetoacetate isomerase 429
 malic enzyme 139
 malonyl-CoA 162, 163
 malt 148
 maltose 38, 39
 brewing 148, 149
 D-mannitol 39
 mannose 44
 D-mannose 36, 37, 38, 39, 44
 mannose 6-phosphate 234
 mannose-rich glycoproteins 45
 mannose-rich oligosaccharides 44, 230
 α -mannosidase 426
 MAPK (MAP kinase) 388
 markers
 enzymes 198
 molecules 198, 199
 tumors 400
 mass action, law of 18, 19
 matrix
 extracellular 346-7
 space 210
 maturation, RNA 208, 236, 242, 243
 maturation-promoting factor (MPF)
 maximum velocity ($V_{\max}$) 92, 93
 mechanical damage 144
 mechanical work 17
 mechanisms
 ping-pong 90, 91, 94
 regulatory 114-21
 sequential 90, 91, 94, 95
 mediators 224, 370, 388
 MEK 388

membrane receptors 224-5, 398
 membranes
 attack complex 298
 carbohydrates 216
 cell interaction 216
 composition 216
 cytoskeletal anchors 216
 depolarization 348
 electrochemical gradients 126
 energy conservation 126-7
 functions 216-17
 hyperpolarization 348
 lipids 214, 216, 409
 micelles 28
 mitochondrial 210
 nuclear 208
 permeability 218, 219
 plasma 196, 214-15
 potential 126, 127, 350
 proteins 214, 216
 proton gradients 126, 127
 receptors *see* membrane receptors
 semipermeable 78
 transport 216-19
 uncouplers 144, 145
 vesicles 28, 198
 menaquinone 52, 53
 menthol 52, 53
 6-mercaptapurine 402
 messenger RNA *see* mRNA
 messengers, second *see* second messengers
 metabolic acidoses 160, 161, 288, 326
 metabolic charts 406-10
 metabolic defects, gene therapy 265
 metabolic pathways 88
 hormones 370
 metabolism 356-7
 acetylcholine 354, 355
 antimetabolites 402, 403
 buffer function 308-9
 calcium 342-3
 carbohydrate 388
 creatine 336
 energy, brain 356
 glycogen 157
 hormones 120-1, 370, 376-7
 intermediary 112, 113, 322

- liver 306
- muscle 336-7
- regulation 115, 120-1
- steroids 376-7
- metabolites 96, 158
 - activated 110-11, 111
 - liver 306
- metabotropic receptors 348, 354
- metalloproteinases 176, 177, 427
- metals
 - complexes 33
 - heavy 316
 - nonmetals 2
- methane 26, 27, 28, 29
- methemoglobin reductase 284, 285
- methionine 60, 61, 110, 184, 250, 251
 - acid-base balance 288
 - adenosyltransferase 423
- methotrexate 402
- methylation 110, 316
- methylcobalamin 108
- methylcrotonyl-CoA carboxylase 429
- methylenetetrahydrofolate dehydrogenase 421
- reductase 421
- α -methylglucoside 37
- methylglutaconyl-CoA hydratase 428
- 7-methylguanidine 82, 83
- 2'-O-methylguanidine 82, 83
- methylmalonyl-CoA 166, 167
 - epimerase 428
 - mutase 167, 429
- methylnitrosamines 256, 257
- O-methylnorepinephrine 317
- methyltetrahydrofolate cyclohydrolase 427
- 5-methyltetrahydrofolate-homocysteine S-methyltransferase 422
- methyltransferases 108
- mevalonate 172, 173
 - kinase 424
- mevalonyl diphosphate 173
- MHC proteins 224, 296
- micelles 28, 29, 270
- Michaelis constant (K_m) 92, 93, 96
- Michaelis-Menten equation 92, 93
- kinetics 92, 93
- model 96
- microelements 362
- microfibrils 42, 43
- microfilaments 204, 206, 207
- microtubules 204, 206, 207
- milk products, fermentation 148, 149
- minerals 360, 362-3, 363
- minibands, chromosomes 238
- mitochondria 196
 - fatty acid degradation 163
 - fractionation 198
 - gluconeogenesis 154
 - membrane 210
 - metabolic functions 210-11
 - protein import 232, 233
 - structure 210-11, 211
 - transport 212-13, 213
- mitogen-activated protein kinase *see* MAPK
- mitosis-promoting factor (MPF) 394
- mobile loops 99
- mobile phase 54
- modulation, ligands 114
- molecular genetics 236-65
- molecular models 76-7
- p-molecular orbitals 4
- molecular oxygen 186
- molecular structure 6
- molecules 7, 198, 202, 218
- monoacylglycerols 48, 49, 170, 171
- 2-monoacylglycerols 272
- monoamine oxidase (MAO) *see* amine oxidase
- monochromatic light 102
- monoclonal antibodies (mAb) 304-5, 305
- monocytes 274, 275
- monokines 392
- monomers, insulin 76
- monooxygenases 89, 104, 106, 166, 318, 319
 - cytochrome P450-dependent 318, 319
- monosaccharides 34-5, 35
 - acidic 38, 39
 - reactions 36-9, 37, 39
- resorption 272, 273
- motor end plate 335
- MPF *see* maturation-promoting factor; mitosis-promoting factor
- mRNA 82, 236, 246
 - 3' modification 246, 247
 - 5' modification 246, 247
- immunoglobulins 302
- POMC 382
- MstII 264, 265
- mucus 268
- multienzyme complexes 134
- muramoylpentapeptide carboxypeptidase 254, 426
- murein 34, 35
 - polymers 41
- MurNAc *see* N-acetylmuramic acid
- muscle
 - absorptive state 308
 - amino acid metabolism 338, 339
 - calcium ions 335
 - contraction *see* muscle contraction
 - contraction
 - fibers *see* muscle fibers
 - glucose 160, 161
 - glycogen 156, 336
 - heart 332
 - metabolism *see* muscle metabolism
 - metabolism
 - motor end plate 335
 - postabsorptive state 308
 - protein 338, 339
 - sarcoplasmic reticulum (SR) 335
 - skeletal 332
 - smooth 332, 338
 - striated 333
- muscle contraction 332-3, 333, 334-5
- muscle fibers 332, 336, 337
- muscle metabolism 336-9, 339
- mutarotation 36, 37
- mutases 429
- mutation 256-7
 - frameshift 256, 257
 - mutagens 256, 257
 - point 248, 256, 257
 - repair 256, 257
 - somatic 302
- myelomas 304
- myo-inositol 50, 51

myofibrils 332
 myoglobin 72, 73, 282, 336
 myosin 64, 65, 332
 ATPase 427
 tropomyosin 64, 65, 332, 334
 myristic acid 49

N

N terminus 66
 N-bound oligosaccharides 230
 Na⁺/K⁺-ATPase 126, 127, 272, 273, 326, 327, 350, 351, 427
 NAD⁺ 99, 101, 104, 366
 /NADH ratio 144
 biosynthesis 208
 gluconeogenesis 155
 redox reactions 19
 NADH 101
 dehydrogenase 141, 421
 glucose oxidation 147
 NADH+H⁺
 ATP synthesis 142, 143
 gluconeogenesis 155
 NADP⁺ 33, 104, 105, 366
 photosynthesis 128, 129, 131
 NADPH+H⁺ 33, 152, 190, 191, 284
 photosynthesis 128, 129
 native conformation 72
 nephron 322
 Nernst equation 126, 127
 nerve cells 348, 349
 nervonic acid 49
 net charge, amino acids 59
 net outcome 136
 NeuAc *see* N-acetylneuraminic acid
 neural synapses 349
 neural tissue 308
 neuraminidases 276, 426
 neurohormones 348, 349
 neuropeptides, proteolysis 382
 neurotransmitters 62, 63, 348, 349, 352-3, 353
 GABA 62, 63, 356
 glutamate 356
 receptors 354-5, 355
 neutral amino acids 60

urea 182
 niacin 366
 nicotinamide 366
 adenine dinucleotide
 oxidized *see* NAD⁺
 reduced *see* NADH
 adenine dinucleotide
 phosphate
 oxidized *see* NADP⁺
 reduced *see* NADPH
 nicotinate 366
 nicotinic acetylcholine
 receptor 223
 nitric oxide synthase 388, 389
 nitrogen 11
 balance 174
 fixation 184, 185
 monoxide (NO) 388, 389
 nitrogenase 184, 185, 422
 norepinephrine 145, 163, 317, 352
 nuclear localization sequence 208
 nuclear pores 208, 209
 nuclear tumor suppressors 398
 nucleic acids
 bases 80, 81
 genetic information 236
 hydrolysis 266
 see also individual nucleic acids
 nucleobases
 components 188, 189
 see also purine ring;
 pyrimidine ring
 nucleolus 208
 nucleophilic substitutions 14, 15
 nucleosidases 426
 nucleoside 80, 81
 diphosphatase 427
 diphosphate kinase 190, 191, 424
 phosphate 106, 107
 kinase 190, 191, 424
 nucleoside diphosphate reductase 190, 191
 nucleosomes 65, 238
 nucleotides 80-1, 81
 biosynthesis 190-1, 191, 417-18
 degradation 186-7, 187,

419

 dinucleotides 80
 nucleotide triphosphates 110, 122
 see also individual nucleotides
 nucleus 196, 208-9, 209
 and cytoplasm 208, 209
 fractionation 198
 localization sequence 208
 membrane 208
 pores 208, 209
 scaffolding 238
 tumor suppressors 398
 nutrients 112, 306, 360, 361

O

O₂ *see* oxygen
 octet rule 2
 ODH *see* 2-oxoglutarate dehydrogenase
 OH- *see* hydroxyl ions (OH-)
 oil
 drop effect 28, 29
 emulsions 28
 Okazaki fragments 240
 oleic acid 49
 oligo-1,6-glucosidase 426
 oligonucleotides 80, 81
 oligosaccharides 44, 45, 230, 272, 273
 OMP 188, 189
 oncogenes 398-9, 399, 400
 one-helix receptors 224
 operators 118
 opsin 358
 opsonization 298
 optical activity 8
 optical isomers 8, 9
 optical rotation 36
 orbitals 2, 4, 5
 organ metabolism 308-9
 organelles 196, 198, 199
 organic compounds 2
 organic substances 360-1
 organisms
 heterotrophic 128
 phototrophic 128
 orientation, substrates 90, 100
 origin of replication 240
 ornithine 62, 63, 182, 183
 carbamoyltransferase

183, 422
 orotate phosphoribosyl-
 transferase 423
 orotidine 5'-
 monophosphate *see* OMP
 orotidine-5'-phosphate de-
 carboxylase 427
 osteoblasts 342
 osteoclasts 342
 overexpression, insulin 76
 oxaloacetate 137, 155
 oxaloacetic acid 136, 154
 oxidases 89, 104
 oxidation 10, 11, 316
 α -oxidation 166
 β -oxidation 113, 141, 210
 oxidation, β -oxidation, fat
 metabolism 162, 163
 fatty acid degradation
 164, 165, 166, 167
 glucose 36, 37
 ω -oxidation 166
 terminal 136
 oxidative deamination 179-
 80
 oxidative decarboxylation
 134
 oxidative phosphorylation
 112, 140, 210
 ATP 122, 123
 oxidized forms, electrons 32
 oxidizing agents 32
 oxidoreductases 88, 89
 oxoacid 179
 oxoacid dehydrogenases
 134-5
 3-oxoacid-CoA transferase
 180
 3-oxoacyl-[ACP] reductase
 168, 169, 420
 3-oxoacyl-[ACP] synthase
 168, 169, 423
 2-oxobutyrate synthase 421
 2-oxoglutarate 136, 178
 dehydrogenase (ODH)
 144, 145, 147, 421
 glucose oxidation 147
 2-oxoisovalerate dehydro-
 genase 420
 oxygen 11
 atomic 25

photosynthesis 128, 129
 redox systems 33
 oxygen transport 282-3
 hemoglobin 280, 282
 saturation curve 280, 282
 2-oxyglutarate 137, 326
 dehydrogenase 134, 135
 complex 136, 137
 oxyhydrogen reaction 20, 21

P

P type ATPase 220
 P/O quotient 146
 p53 protein 394, 396
 PAGE *see* polyacrylamide gel
 electrophoresis
 palmitate 162, 163, 169
 palmitic acid 49, 168
 Pan *see* pantetheine
 pancreas 268, 269, 308
 pancreatic ribonuclease 426
 pantetheine 106, 169
 pantoic acid 12, 13, 106
 pantothenate 12
 pantothenic acid 106, 366
 papain 300, 427
 PAPS *see* phosphoadenosine
 phosphosulfate
 paracrine effects 372, 373,
 390
 parallel pleated sheet 68
 parathyroid hormones
 (PTH) 342
 Parkinson's disease 62
 passive channels 222
 passive transport 218
 pathways
 amphibolic 112, 113
 anabolic 112, 113, 138,
 139, 338
 anaplerotic 139
 catabolic 112, 113, 138,
 139
 cytoplasmic 228
 pentose phosphate (PPP)
 113, 152-3, 153
 secretory 228
 PCR 262-3, 263
 deoxyribonucleoside tri-
 phosphates 262
 primers 262
 RT-PCR 264, 265
 PDH *see* pyruvate dehydro-

genase
 PDK-1 388
 pectins 42, 43
 PEG *see* polyethylene glycol
 penicillins 40, 254, 255
 pentose phosphate pathway
 (PPP) 113, 152-3, 153
 and amino acid
 biosynthesis 185
 pentoses 39
 PEP 124, 125, 150, 154, 155
 carboxykinase 139, 154,
 155
 carboxylase 158, 159
 PEP-CK 242, 427
 gene 243
 transcription 245
 pepsin 94, 176, 268, 269,
 270
 A 427
 B 427
 pepsinogen 268
 peptidases 89, 176, 330, 331,
 426
 peptide hormones 380, 381,
 382-3
 biosynthesis 382
 degradation 382
 hormone-like 392-3
 inactivation 382
 peptides
 bonds 66-7, 67, 252
 C-peptides 76
 chains 66
 conformation 66, 67
 hormones *see* peptide
 hormones
 loops 98
 neurotransmitters 352
 nomenclature 66, 67
 sequence 66
 signal 228, 230
 peptidoglycan 35
 peptidyl dipeptidases 330,
 331, 425
 peptidyl proline *cis-trans*-
 isomerase 233, 429
 peptidyl site (P) 250, 252
 peptidyl-dipeptidase A *see*
 angiotensin-converting
 enzyme
 peptidyltransferase 423
 perinuclear space 208
 periodic table 2, 3
 peroxisomes 196

- periplasm 35
 permeability, membranes 218, 219
 permease 119
 peroxidases 89, 102, 103, 106, 305, 421
 peroxide anions 284
 pH 18
 cells 30, 31
 enzymes 94, 95
 fermentation 148
 pH, gradient 127
 membrane difference 126
 neutral 30
 phages *see* bacteriophages
 phagocytosis 234, 235
 phalloidin 204
 phenobarbital 194
 phenylalanine 60, 61
 4-monooxygenase 422
 phenylethanolamine *N*-methyltransferase 352, 353, 422
 pheophytin 130, 131
 phosphates 13
 ATP transfer 123
 buffers 288
 ions 328
 residues 12
 transport 212
 phosphatidates 47, 50, 51, 170, 171
 cytidyltransferase 170, 171, 425
 lysophosphatidate 170, 171
 phosphatase 170, 171, 425
 phosphatides 46, 47, 51
 phosphatidic acids 46
 phosphatidylcholine 50, 51, 170, 171, 212, 213
 sterol acyltransferase 278, 279, 423
 phosphatidylethanolamine 50, 170, 171, 212, 213
 phosphatidylinositol 50, 170, 171, 212, 213
 phosphatidylinositol 4, 5-bisphosphate (PIP₂) 170, 171
 phosphatidylinositol-4-phosphate kinase 170, 171
 phosphatidylserine 50, 170, 171, 212, 213
 phosphoadenosine phos-
 phosulfate (PAPS) 110, 111
 phosphodiesterase 120, 121, 386, 387, 425
 phosphoenolpyruvate *see* PEP
 phosphoenolpyruvate, carboxykinase *see* PEP-CK
 6-phosphofructo-2-kinase 424
 6-phosphofructokinase 158, 159
 glucose oxidation 147
 glycolysis 150, 151
 phosphoglucomutase 429
 phosphogluconate
 6-phosphogluconate 152, 153
 dehydrogenase 153, 420
 6-phosphogluconolactone 152, 153
 phosphoglycerate
 2-phosphoglycerate 125, 150, 151, 155
 3-phosphoglycerate 150, 151, 155
 dehydrogenase 420
 kinase 131, 147, 424
 glycolysis 150, 151
 mutase 151, 429
 phospholipases
 A₁ 425
 A₂ 268, 390, 391, 425
 C 386, 387, 425
 D 425
 phospholipids 46, 47, 50-1, 51
 biosynthesis 171
 blood clotting 290
 degradation 411
 lipid metabolism 312
 lysophospholipids 50
 membranes 214
 structure 50, 51
 phosphomevalonate kinase 424
 4'-phosphopantetheine 12
 phosphoprotein
 phosphatase 394, 395, 425
 phosphates 120, 121
 phosphopyruvate hydratase 124, 428
 5-phosphoribosyl 1-diphosphate (PRPP) 188, 189
 phosphoribosylamine gly-
 cine ligase 429
 phosphoribosylaminoimidazole
 carboxamide formyltransferase 422
 carboxylase 427
 succinocarboxamide synthase 429
 phosphoribosylformylglycinamidine synthase 429
 phosphoribosylglycinamide formyltransferase 422
 phosphoribulokinase 131
 phosphoric acid, esters 10, 11, 80
 phosphoric acid—anhydride 11, 80
 bonds 12, 13, 122, 123
 phosphoric acid—ester, bonds 13, 122, 123
 phosphorus 11
 phosphorylase 120, 121, 423
 carbohydrate metabolism 159
 glycogen metabolism 156, 157
 kinase 424
 phosphorylation 62, 106
 ATP-dependent 114
 cyclic 130, 131
 dephosphorylation 114
 histones 238
 oxidative 112, 122, 140, 210
 protein kinase A 120, 121
 substrate-level 122, 124, 150
 phosphoserine transaminase 424
 phosphatidylinositols
 1-phosphatidylinositol-4-kinase 424
 1-phosphatidylinositol-4-phosphate kinase 424
 phosphatidylinositol (PtdIns) 170
 phosphatidylinositol-3-kinase (PI3K) 388
 phosphatidylinositol-4,5-bisphosphate (PIP₂, Ptd InsP₂) 170
 phosphotransferases 89
 photolyase *see* deoxyribodipyrimidine photolyase

- photoreactivation 256
 photoreceptors 358, 359
 photosynthesis 106, 112,
 128-9, 129, 130-1
 photosystems (PS) 128
 I 131, 132, 133
 II 130, 131, 133
 phototrophs 128
 phyloquinone *see* vitamin K
 phytol 52, 53, 128, 129
 phytyl 52, 53
 P_i *see* inorganic phosphate
 π electrons 4
 PI3K *see* phosphatidylinosi-
 tol-3-kinase
 pigments, antenna 132
 ping-pong mechanisms 90,
 91, 94
 pitch 68
 helix 86
 PK *see* protein kinase
 PK-B *see* protein kinase B
 pK_a value 18
 plants 42-3
 plaques 260
 plasma 274, 275
 buffers 288, 289
 cells 294
 hormones 372-3, 373
 hydrogen ions 288, 289
 kallikrein 426
 membranes *see* plasma
 membranes
 pH 31
 proteins 276-7, 277, 288
 thromboplastin
 antecedent *see* coagu-
 lation factor XIa
 plasma membranes 196
 composition 217
 functions 217
 lipids 215
 permeability 219
 proteins 215
 structure 214-15, 215
 transport 219, 221
 plasma thromboplastin
 antecedent 291
 plasmids 258, 259, 262, 263
 plasmin 426
 plasminogen activator 427
 plastocyanin 128, 129
 plastoquinone (PQ) 52, 53
 photosystem I 133
 photosystem II 128, 129,
 131
 Q cycle 126, 127
 reduced (QH₂) 126, 127
see also coenzyme Q, re-
 duced
 pleated sheets
 α -pleated sheet 67
 β -pleated sheet 68, 69
 plots
 hyperbolic 93, 97
 Lineweaver–Burk 92, 93
 PLP *see* pyridoxal phosphate
 point mutations 248, 257
 polar molecules 28
 polarimetry 36, 37
 polarity 218
 polarization 6
 depolarization 222, 348,
 350
 hyperpolarization 348,
 350
 membranes 348
 repolarization 350
 polarizers 36
 poliovirus 404
 polyacrylamide gel electro-
 phoresis (PAGE)
 polyadenylate
 polymerase 246
 tail 243, 246, 247
 polyethylene glycol (PEG)
 polymerase chain reaction
see PCR
 polynucleotidases 425
 polynucleotides 80, 81
 polysaccharides 40-5, 41,
 42-3
 polysomes 250, 251
 POMC 382
 mRNA 382
 prepro-POMC 382
 pro-POMC 382
 pores 208, 209
 porins 210
 porphobilinogen 192, 193
 synthase 192, 193, 428
 porphyrias 192
 porphyrin systems 194
 post-translational modifica-
 tion 62, 63, 344
 postabsorptive state 308
 potassium channels 222, 233
 potential
 action 222, 348, 350, 351
 difference (ψ) 16
 electron transfer 19
 equilibrium 126, 350
 group transfer 19
 membrane 350
 normal 130, 131
 proton transfer 19
 resting 126, 350, 351
 PPP *see* protein phosphate
 pathway
 PQ *see* plastoquinone
 pRb 394
 prealbumin 64, 65
 precipitation 79
 precursors 114, 138
 pregnenolone 376
 prenylation 62, 214
 preproinsulin 160, 161
 preprotein 230
 primary structure 76
 primase 240
 primers
 DNA 240
 PCR 262
 sequencing 260
 proaccelerin 291
 probes, gene 260
 processes, electroneutral
 218
 procollagen 344
 lysine 5-dioxygenase
 344, 345, 422
 proline 4-dioxygenase
 344, 345, 422
 proconvertin 291
 proconvertin *see* coagula-
 tion factor VIIa
 product 22
 inhibition 144
 proenzymes 176
 progesterone 56, 57, 374,
 376
 proinsulin 76, 160, 161
 prokaryotes 196, 197
 proline 60, 61
 collagen 70
 dehydrogenase 421
 hydroxylase *see* procolla-
 gen proline 4-dioxyge-
 nase
 promoters 114, 118
 proopiomelanocortin *see*
 POMC
Propionibacterium 148, 149
 propionic acid 49
 fermentation 148, 149

- propionyl-CoA 166, 167
 carboxylase 166, 167, 429
 proprotein 230
 prostacyclins 390
 prostaglandins 390
 prostaglandin H-synthase 390, 391, 422
 prosthetic groups 104, 134, 142
 proteasomes 176, 177
 protein
 disulfide isomerase 233, 429
 glycosyltransferase 231, 423
 lysine 6-oxidase 344, 345, 421
 phosphatases 114, 144, 145, 392
 tyrosine kinase 424
 see also proteins
 protein folding 72, 74-5, 75, 174
 endoplasmic reticulum 233
 energetics 74
 rough endoplasmic reticulum 232
 protein kinases (PK) 114, 394, 395, 424
 ERK 388
 inactivating 144
 MEK 388
 oncogenes 398
 PDK-1 388
 PK-A 120, 121, 386, 387
 PK-B 388
 Raf 388
 second messengers 348
 protein metabolism 174-5, 175, 306
 muscle 338, 339
 protein-glutamine γ -glutamyltransferase 423
 proteinases 176, 290
 proteinogenic amino acids 60-1, 61
 non-proteinogenic 62-3
 proteins 64-5, 65
 adaptor 388
 adhesive 346
 allosteric 72
 analysis 78
 biosynthesis 174
 catalysis 64
 channels 218
 clotting factors 290
 defense 64
 denaturation 74, 75
 dynamics 72, 73
 elution 78
 Escherichia coli 251
 folding *see* protein folding
 globular 72-3
 glycosylation 230, 231
 gradients 127
 heat-shock (hsp) 232
 hyaluronate 346
 hydrolysis 266
 isolation 78
 lysosomal 234, 235
 marking 174
 maturation 230-3
 mediator 224
 membranes 214, 215, 216
 metabolism *see* protein metabolism
 mitochondrial import 232, 233
 motive force 127
 movement 64
 non-histone 238
 nutrients 360
 overexpression 262, 263
 plasma 277
 polyproteins 382
 post-translational modification 63, 174
 preprotein 230
 proprotein 230
 regulation 64
 regulatory 114, 119
 resorption 266
 ribosomal 250
 single-strand-binding 240
 sorting 174, 228-9, 229
 stabilization 74
 storage 64
 structure 64, 68-71
 synthesis 230-1, 231
 transport 64, 220-1
 proteoglycans 34, 35, 340, 346, 347
 proteohormones 380, 381
 proteolysis 174, 176-7
 enzymes 177
 hormones 382
 neuropeptides 382
 prothrombin 291
 prothrombinase complex 291
 proto-oncogenes 398, 399, 400
 protofilaments 70, 204
 proton channels, ATP synthase 142, 143
 proton gradients 126, 140, 210, 212
 membranes 126, 127
 protons
 excretion 326
 gradient *see* proton gradient
 motive force (P) 126
 pumps 126, 127
 secretion 327
 transfer 14, 18, 19
 protoporphyrin IX 192, 193
 protoporphyrinogen IX 192, 193
 proximity, substrates 90, 100
 PRPP *see* 5-phosphoribosyl-1-diphosphate
 pseudouridine 82, 83
 PTH *see* parathyroid hormone
 pulsatile release, hormones 372
 purines 80, 81
 biosynthesis 188-9, 189, 190, 417
 degradation 186, 187
 derivatives 352
 ring 188, 189
 pyranoses 34
 pyridine nucleotides 32, *see also* NAD⁺; NADP⁺
 pyridoxal 368
 phosphate (PLP) 62, 178, 358, 368
 transamination 108, 109
 see also vitamin B₆ complex
 pyridoxamine 179, 368
 phosphate 108, 178, 179
 pyridoxol 368
 pyrimidines 80, 81
 biosynthesis 188-9, 189, 190, 418
 degradation 186, 187
 ring 188, 189

pyrrolines
 1-pyrroline-5-carboxylate dehydrogenase 421
 pyrroline-5-carboxylate reductase 421
 pyruvate 35, 99, 154, 155, 338
 2-oxoglutarate 141
 carboxylase 159, 429
 decarboxylase 427
 dehydrogenase *see* pyruvate dehydrogenase
 glycolysis 150, 151
 kinase 151, 159, 424
 redox reactions 19, 101
 transport 212
 pyruvate carboxylase 138, 139, 155
 pyruvate decarboxylase, fermentation 148, 149
 pyruvate dehydrogenase (PDH) 210, 421
 -phosphatase 425
Escherichia coli 134, 135
 glucose oxidation 147
 kinase 424
 tricarboxylic cycle 144, 145
 pyruvate kinase 147
 glucagon 158, 159
 glycolysis 150, 151

Q

Q *see* coenzyme Q, oxidized (ubiquinone)
 Q cycle 126, 127
 QH₂ *see* coenzyme Q, reduced (ubiquinol); plastoquinone
 quaternary structure 76
 quinol system 32
 quinones 104, 131
 hydroquinone 33, 104
 redox systems 33
 semiquinone radical 32, 33, 104
 system 32
see also plastoquinone

R

R conformation 117
 hemoglobin 280, 281
 R state 116
 R/S system 8
 radiation 256
 radicals
 free 32, 256
 hydroperoxyl 33
 hydroxyl 33, 284
 semiquinone 32, 33, 130, 131
 superoxide 284
 tyrosine 190, 191
 radioimmunoassays (RIA) 388
 Raf 388
 Ramachandran plot 66
 Ras 388
 rate constants 22, 23
 reaction
 centers 132, 133
 P₆₈₀ 130, 131
 heat of 20, 21
 reactions
 catalyzed 25
 endergonic 112
 enzyme specificity 88, 89
 enzyme-catalyzed 90, 93
 exergonic 112
 first-order 22, 23
 kinetics 22-3
 order 22, 23
 rates 22, 23, 88
 second-order 22, 23
 specificity 88, 89
 spontaneous 18
 substrate specificity 88, 89
 uncatalyzed 90
 reactive oxygen species (ROS) 33, 284, 285
 reagents
 enzymes 102, 103
 modifying 97
 rearrangements 14
see also isomerizations
 receptor signaling 216
 signal recognition particle (SRP) 230
 signal transfer 224
 receptor-mediated endocytosis 278
 receptors
 1-helix 224, 384
 7-helix 224, 384
 acetylcholine 354

antigen 296, 297
 co-receptors 296
 cytokines 392
 domain 224
 effects of 225
 glucagon 121
 hormones 120
 insulin 224, 225, 388
 ion channels 126, 218, 222-4, 350, 384
 ionotropic 348, 354
 lipophilic hormones 378, 398
 membrane 216, 224-5
 metabotropic 348, 354
 neurotransmitters 354-5, 355
 nicotinic acetylcholine 222
 oncogenes 398
 photoreceptors 358, 359
 signaling *see* receptor signaling
 steroid 378
 substrates 388
 synapses 348
 T-cell 224
see also individual receptors
 reciprocal velocity 93
 recombination
 repair 256
 somatic 302, 303
 recycling 328-9, 329
 redox
 coenzymes 104, 105, 106-7, 107, 140
see also flavins; heme groups; iron-sulfur centers
 potential (E) 18, 19, 32
 processes 32-3
 reactions *see* redox reactions
 series 32, 130, 131, 142
 systems *see* redox systems
 redox reactions 14, 15, 32, 33
 lactate 19, 100, 101
 NAD⁺ 18, 19, 101
 pyruvate 19
 redox systems 32
 biological 32, 33
 coenzyme Q (CoQ) 105

- flavin 33
- hydrogen peroxide 33
- respiratory chain 142, 143
- reduced forms, electrons 32
- reducing agents 32
- reductases 89
- reduction 316
 - equivalents 32, 33, 212
 - glucose 36, 37
- Refsum disease 166
- regions, signal 228
- regulation 114-21, 144-5
 - allosteric 116
 - carbohydrate metabolism 158-9
 - gene 118
 - glycogen metabolism 156
 - interconversion 120
 - transcription 118-20, 244
 - tricarboxylic cycle 144
- regulatory proteins 114, 118, 119
- relative velocity 25
- renal hormones 330-1, 331
- renin 330, 331, 427
- renin-angiotensin system 330, 331
- repair, mutations 256, 257
- replication 208
 - DNA polymerases 240
 - Escherichia coli* 240, 241
 - forks 240
 - genetic information 236, 240-1
 - origin of 240
- repolarization 350
- repression, transcription 114
- repressors 118
- rER *see* rough endoplasmic reticulum
- RES *see* reticuloendothelial system
- residual bodies 234
- resonance 4, 5, 32, 66, 67
- resorption 266, 267, 272-3, 273
 - HCO₃⁻ 326
 - liver 309
 - minerals 362
 - urine 322
- respiration 144, 145, 146-7
- respiratory chain 106, 112, 126, 140-1, 210
 - ATP synthesis 142
 - components 132, 140, 141
 - redox systems 142, 143
- respiratory control, ATP synthase 144, 145
- resting potential 126, 350, 351
- restriction endonucleases 258, 259, 264, 425
- restriction fragment length polymorphisms *see* RFLP
- reticuloendothelial system (RES)
- retinal 62, 358, 364
 - isomerase 358, 359, 429
- retinoate 52, 53
- retinoic acid 364
- retinol *see* vitamin A
- retinol, dehydrogenase 358, 359, 420
- retroviruses 404
- reverse transcriptase *see* RNA-directed DNA polymerase
- RFLP 265
 - sickle-cell anemia 262, 264
- rhesus (Rh) blood groups 292
- rhinovirus 404, 405
- Rhizobium* 184
- rhodopsin 224, 358
- RIA *see* radioimmunoassay
- riboflavin 366
- ribonucleic acid, transcription 243
- ribonuclease
 - A 74, 75
 - H 404, 405, 426
 - pancreatic 426
- ribonucleic acid (RNA) *see* RNA
- ribonucleoside diphosphate reductase 190, 191, 403, 422
- ribonucleotide
 - reductase *see* ribonucleoside diphosphate reductase
 - reduction 191
- ribose 12, 13, 123
 - phosphate pyrophosphokinase 424
 - D-ribose 34, 38, 39
- ribose 5-phosphate 152, 153, 429
- ribosomal RNA *see* rRNA
- ribosomes 82, 236
 - eukaryotes 250, 251
 - fractionation 198
 - proteins 250
- ribozymes 24, 88
- ribuloses
 - D-ribulose 38, 39, 128, 152
 - ribulose phosphate 3-epimerase 428
 - L-ribulose phosphate 4-epimerase 428
 - ribulose 5-phosphate 152, 153
 - ribulose biphosphate, carboxylase 130, 131, 427
- Richardson diagrams 72, 73
- rifamycin 254, 255
- righthanded helix
 - α -helix (α_R) 68
 - double 86
- RNA 80, 81, 82-3, 83, 87
 - conformation 86
 - hnRNA 236, 246, 247, 302
 - immunoglobulins 302
 - maturation 208, 236, 242, 243, 246-7
 - transcription 242, 243
 - translation 83
 - see also* mRNA; rRNA; snRNA; tRNA
- RNA polymerase *see* DNA-directed RNA polymerase
- RNA-binding proteins 82
- RNA-directed DNA polymerase 404, 405, 425
- rods 358
- root nodules 184, 185
- ROS *see* reactive oxygen species
- rotation 67
- rough endoplasmic reticulum (rER) 226, 227, 231, 232
- roughage 42, 273
- rRNA 82
 - 5S-rRNA 86, 87
- RT-PCR 264, 265
- rubber 52, 53
- rubisco *see* ribulase biphosphate carboxylase

- S *see* Svedberg units
 S phase 394
 saccharase *see* β -fructofuranosidase
Saccharomyces cerevisiae 148, 149
 SAH *see* S-adenosyl-L-homocysteine
 salicylate 317
 saliva 268, 269
 salt, precipitation 78, 79
 SAM *see* S-adenosyl-L-methionine
 sarcomeres 332
 sarcoplasmic Ca^{2+} pump 220, 221
 sarcoplasmic reticulum (SR) 334, 335
 scaffolding
 cytoskeletal 206
 nuclear 238
 Schiff's base 108
 SDS gel electrophoresis 78, 79
 second messengers 224, 348
 glycolipids 50
 hormones 120, 121, 384, 386-7
 see also individual messengers
 second-order reactions 22, 23
 secondary structure 68-9, 76
 secretin 270
 secretions
 kidney 322
 pancreas 268-9
 small intestines 268-9
 secretory pathway 228
 sedimentation coefficient 200
 sedoheptulose biphosphatase 425
 sedoheptulose 153
 selenocysteine 6, 63
 self-protection, blood 274
 semiquinone radical 32, 130, 131
 sense strands 84
 separation, spontaneous 29
 sequences
 glycosylation 230
 nuclear localization 208
 peptides 66
 sequencing 261
 sequential mechanisms 90, 91, 94, 95
 sER *see* smooth endoplasmic reticulum
 serine 50, 51, 60, 61, 63
 dehydratase 181, 428
 family 184, 185
 proteases 176, 177, 298
 proteinases 177, 425
 serotonin 62, 63
 serum 274
 enzyme diagnosis 98, 178
 seryl alanine 65, 66, 67
 seven-helix receptors 224
 SH2 domain 392
 short tandem repeats (STRs) 264
 shuttles 212
 sialic acid *see* N-acetylneuraminic acid
 sickle-cell anemia 249, 264, 265
 sight 358-9
 sigmoidal substrate saturation curves 116
 signal
 cascades *see* signal cascades
 peptidase 427
 peptides 228, 230, 231
 receptor transfer 224
 regions 228
 signaling *see* signaling
 SRP *see* signal recognition particle
 stop-transfer 230
 transduction *see* signal transduction
 translocation 228, 229
 transmission *see* signal transmission
 signal cascades
 hormones 388-9
 photoreceptors 358, 359
 signal recognition particle (SRP) 230, 231
 receptor 230
 signal transducers and activators of transcription *see* STAT
 signal transduction 120
 cytokines 392, 393
 G proteins 384, 385
 insulin 388, 389
 oncogenes 398
 proteins (STPs) 392
 signal transmission
 CNS 348-9
 synaptic 348
 signaling
 amino acids 380, 381
 calcium 342
 cytokines 392-3
 eicosanoids 46, 390-1
 hydrophilic substances 392
 membranes 216
 second messengers 224
 steroids 46
 silencers 118
 silk 70, 71
 single-strand-binding proteins 240
 single-stranded DNA (ssDNA)
 site-specific
 deoxyribonuclease *see* restriction endonucleases
 β -sitosterol 56, 57
 size, molecular 218
 skeletal muscles 332
 small intestines 31, 268, 269
 small nuclear ribonucleoprotein particles (snRNPs) 246
 small nuclear RNA (snRNA) 82
 smooth endoplasmic reticulum (sER) 172, 173, 226, 227
 smooth muscle 332, 338
 sn *see* stereospecific numbering
 SNAREs 228, 229
 snRNA, splicing 83
 snRNPs (snurps) 246
 soap bubbles 28, 29
 sodium
 chloride dissolution 21
 dodecylsulfate *see* SDS
 ions 328
 voltage-gated channel 222, 223
 solenoids 238
 solubility 28, 29
 soluble coenzymes 104
 solvents 26-7, 54
 somatic mutation 302

- somatic recombination 302
 somatotropin 64, 65
 receptor 64
 son of sevenless *see* SOS
 sorbitol 36, 37
 D-sorbitol 39
 sorting, proteins 174, 228-9, 229
 SOS, adaptor proteins 388
 space
 space, intermembrane 210
 matrix 210
 spectrin 206
 spectrophotometry 102, 103
 sphingolipids 46, 47, 51, 170
 sphingomyelin 214, 215
 sphingophospholipid 51
 sphingosine 51
 N-acetyltransferase 423
 spliceosomes 246, 247
 splicing
 hnRNA 246, 247
 snRNA 83
 spontaneous separation 29
 squalene 52, 53, 172, 173
 SR *see* sarcoplasmic reticulum
 ssDNA *see* single-stranded DNA
 stabilization
 catalysis 90
 transition states 100, 101
 starches 40, 42, 43
 brewing 148, 149
 start codons 248, 250
 STAT 392
 statins 372
 stationary phase 54
 stearic acid 49
 stearyl-CoA desaturase 422
 stercobilin 195
 stercobilinogen 194, 195
 stereospecific numbering (sn)
 stereospecificity 8
 steroid alcohols *see* sterols
 steroid hormones 56, 57, 172, 374, 377
 biosynthesis 376
 conjugates 110
 inactivation 376
 metabolism 376-7, 377
 see also individual
 hormones
 steroids 46, 52, 53, 54-7, 55
 hormones *see* steroid hormones
 hormones
 receptors 378
 signaling 46
 steroid 11 β -monooxygenase 422
 steroid 17 α -monooxygenase 422
 steroid 21-monooxygenase 422
 steroid -isomerase 429
 structure 54
 synthesis 410
 sterol esterase 268, 269
 sterol esters 47
 sterols 46, 54, 55, 56, 57
 see also cholesterol;
 ergosterol; β -sitosterol; stigmasterol
 stearyl glycerols,
 tristearyl glycerol 49
 stigmasterol 56, 57
 stop codons 248, 250
 stop-transfer signal 230
 storage, liver 306
 STP *see* signal transduction protein
 strands
 double 85
 grooves 86
 matrix 85
 sense 85
Streptococcus 148, 149
Streptomyces lividans 222, 223
 streptomycin 254
 striated muscle 333
 stroma 128, 129
 STRs *see* short tandem repeats
 structural genes 118
 structure 6
 dimers 116
 isomers 8
 polysaccharides 40
 primary 76
 proteins 68, 70-1
 secondary 76
 tertiary 76
 quaternary 76
 Stuart-Prower factor *see* coagulation factor Xa
 Stuart-Prower factor 291
 substitution, nucleophilic 14
 substrate-level phosphorylation 150
 substrates
 activity 95
 analogs 96, 97
 bisubstrate kinetics 94, 95
 coenzymes 105
 concentration 97, 116
 cosubstrates 104
 enzymes 94
 insulin 388
 orientation 90, 100
 phosphorylation 122, 124, 125
 proximity 90, 100
 receptors 388
 specificity 88, 89, 94
 suicide 96, 97
 succinate 136, 137
 dehydrogenase 136, 137, 140, 421
 glucose oxidation 147
 semialdehyde dehydrogenase 356, 357, 420
 succinate-CoA ligase 136, 137, 147
 (GDP-forming) 429
 succinic acid 8, 9
 succinyl phosphate 125
 succinyl-CoA 136, 137, 166, 167
 heme biosynthesis 192, 193
 ligase 124, 125
 sucrose 38, 39
 α -glucosidase 426
 density gradients 201
 sugar 38
 alcohols 38, 39
 see also mannitol;
 sorbitol
 amino 44
 suicide substrates 96, 97
 sulfathiazole 255
 sulfatides 50, 51
 sulfides, disulfide 11
 sulfonamides 254, 255
 sulfur 11
 sulfur-containing amino acids 60
 superoxide
 dismutase 284, 285, 422
 radicals 284
 surface films 29
 Svedberg unit (S)
 sweat 31

symbiosis
 endosymbiotic theory 210
 nitrogen fixation 184, 185
 symport 212, 218, 220
 synapses 348
 receptors 348
 signal transmission 348, 349
 vesicles 354
 synaptobrevin 228, 229
 synaptotagmin 228, 229
Synechococcus elongatus 130, 132
 syntaxin 228, 229
 synthases *see* lyases
 synthetases *see* ligases

T

T cell 274, 275, 294
 activation 296-7, 297
 receptor 224, 225
 T form 117
 hemoglobin 280, 281
 T state 116
 T4 phase 404
 TAFs *see* TATA box, binding protein, associated factors (TAFs)
 tail, polyadenylate 246
 TATA box 242, 243
 binding protein (TBP) 244, 245
 associated factors (TAFs) 244, 245
 taurine 314
 taurocholic acid 314, 315
 TBG *see* thyroxine-binding globulin
 TBP *see* TATA box, binding protein
 teeth 340, 341
 temperature, enzymes 94, 95
 templates 84, 240
 terminal oxidation 136
 termination, *Escherichia coli* 251
 terpenes 52, 53
 tertiary structure 76
 testosterone 56, 57, 338, 374, 376
 tetracyclines 254
 tetrahydrobiopterin (THB)
 tetrahydrocortisol 317
 tetrahydrofolate (THF) 108, 110
 N⁵-formyl-THF 108
 N⁵-methyl-THF 108
 N⁵-methylene-THF 108
 N¹⁰-formyl-THF 108
 tetraiodothyronine *see* thyroxine
 tetramers 99
 tetrapyrroles 128, 129
 THB *see* tetrahydrobiopterin
 Thermodynamics, Second Law of 20
 thermogenin (UCP-1) 144, 145
 THF *see* tetrahydrofolate
 thiamine *see* vitamin B₁
 thiamine
 diphosphate (TPP) 106, 107, 366
 pyruvate dehydrogenase 134, 135
see also vitamin B₁
 thiazole ring 106
 thin-layer chromatography (TLC) 54, 55
 thioesters 10, 11, 106, 125
 bonds 12, 13
 thioethers 10
 thiokinase *see* succinate-CoA ligase (GDP-forming); succinyl-CoA ligase
 thiols 10, 11
 thiopurine methyltransferase 402, 403, 422
 thioredoxin 190, 191
 reductase 190, 191, 421
 threonine 60, 61, 63
 aldolase 428
 synthase 428
 thrombin 290, 426
 thrombocytes 274, 290
 thromboplastin 291
 thromboxanes 390
 thylakoids 128, 129, 133
 thymidine 80, 81, 82, 304
 thymidylate synthase 190, 191, 402, 403, 422
 thymine 81, 84, 186, 187
 dimers 256
 thyroid-stimulating hormone (TSH) *see* thyrotropin
 thyroliiberin 380
 thyrotropin 380
 thyrotropin-releasing hormone (TRH) *see* thyroliiberin
 thyroxine 62, 374
 thyroxine-binding globulin (TBG)
 TIM 232
 tissue kallikrein 426
 tissue plasminogen activator (tPa) *see* plasminogen activator (tissue)
 tissues
 disruption 198
 hormones 370
 titin 332
 titration curves 30
 TLC *see* thin-layer chromatography
 TMP, dTMP 80, 81, 187
 TNF- α *see* tumor necrosis factor- α
 tobacco mosaic virus 404
 tocopherol *see* vitamin E
 TOM 232
 topoisomerases 240, 245, 429
 toxicity, urea 182
 TPP *see* thiamine diphosphate
 trace elements 2, 360, 362-3
 TRADD 396
 transducin 224
 transferrin 286
 transacetylase 119
 transaldolase 152, 153, 422
 transaminases 94, 176, 177
 transamination 134, 178-9, 179, 180
 pyridoxal phosphate 108, 109
 transcription 208, 242-5, 243
 control 114, 115, 118-20, 244-5
 factors 114, 118, 119, 392, 394
 genetic information 236
 initiation 244, 245
 PEP-CK 244, 245
 regulation 244
 RNA 242, 243
 transducin 72, 73, 358
 transfer RNA *see* tRNA
 transferases 88, 89

C₁-transferases 89
 intermolecular 89
 transferrin 287
 transformation 258, 398, 401
 bio 226, 317
 tumors 400
 transition states 15, 90, 100
 analogs 97
 stabilization 100, 101
 transketolase 152, 153, 422
 translation 174, 208
 amino acids 60
 elongation 252-3
 Escherichia coli 250, 251, 252, 253
 genetic information 236, 248, 250-3
 inhibitors 254
 initiation 250-1
 RNA 83
 termination 252-3
 translocation 252
 signals 228, 229
 translocator complexes
 TIM 232
 TOM 232
 translocon 230
 transmembrane helix 214
 transport
 active 218
 antibiotics 254
 antiport 212, 218
 blood 274
 CO₂ 282, 283
 hemoglobin 282, 283
 hormones 370
 lipoproteins 279
 mechanisms 220
 membrane regulation 216
 mitochondrial 212-13
 O₂ 282-3, 283
 passive 218
 plasma membranes 219
 processes 218-19
 proteins 220-1
 secondary 220
 symport 212, 218, 220
 uniport 212, 218
 transporters 35, 210, 212, 218, 220
 transthyretin 64, 65, 276
 transverse tubules 334
 α, α-trehalase 426

triacylglycerol 48, 49, 163, 170, 171
 lipase 270, 271, 272, 273, 425
 tricarboxylic acid cycle 113, 136-9, 137, 139, 210
 and amino acid
 biosynthesis 185
 regulation 144, 145
 triiodothyronine 374
 see also thyroxine
 triokinase 310, 311, 424
 triose phosphate isomerase 150, 151, 429
 Tris(hydroxymethyl)amino-methane
 tristearylglycerol, van der Waals model 49
 tRNA^{Phe} 82, 83, 86, 87
 aminoacyl 248
 genetic information 236
 tropines 372
 tropomyosin 64, 65, 332, 334
 troponin 332, 334
 trypsin 94, 95, 177, 268, 269, 426
 trypsinogen 176, 177, 270
 tryptophan 60, 61
 TSH see thyroid-stimulating hormone
 tubules 210, 221
 microtubules 204, 206, 207
 transverse 334
 tubulins 204, 205
 tumor necrosis factor-α (TNF-α) 396
 tumor-suppressor genes 394, 398
 tumors 400-1, 401
 gene therapy 264, 265
 monoclonal antibodies 302
 turnover
 enzymes 88, 89
 number 92
 β-turns 68
 type 1 diabetes 160
 type 2 diabetes 160
 tyrosine 60, 61, 352
 kinases 224, 392
 radical 190, 191
 transaminase 424
 tyrosine-3-monooxygenase

352, 353, 422

U

ubiquinol see coenzyme Q, reduced
 ubiquinone see coenzyme Q, oxidized
 ubiquitin 62, 176, 177, 394
 UCP-1 see thermogenin
 UDP glucose 110, 111, 156
 4-epimerase 310, 311, 428
 UDP glucose 1-phosphate
 uridyltransferase 157
 UDP glucuronic acid 194
 UDP-glucuronosyltransferase 194
 ultrafiltration 322
 ultraviolet radiation (UV) 256
 UMP 187, 188, 189, 190
 uncouplers 144, 145
 uniport 212, 218
 units, international (U) 88
 unsaturated fatty acids 48
 uracil 80, 81, 84, 186, 187
 urea 324
 cycle 113, 182-3, 183
 degradation 112
 liver 182
 ureotelic animals 182
 uric acid 186, 187, 324
 uricase 186
 uricotelic animals 182
 uridine 80, 81
 5'-diphosphate see UDP
 5'-monophosphate see UMP
 5'-triphosphate see UTP
 dihydrouridine 82, 83
 pseudouridine 82, 83
 urine 31, 322, 323, 324-5, 325
 urobilin 194, 195
 urobilinogen 194, 195
 urocanate hydratase 428
 urokinase see plasminogen activator (urine)
 uroporphyrinogen III 192, 193
 synthase 192, 193, 428
 UTP 190
 UTP-glucose-1-phosphate
 uridyltransferase 424

UV *see* ultraviolet radiation

V

V segments 302
 V systems 116
 V type ATPase 220
 V/J recombination 302, 303
 vacuoles 43, 196, 234
 valerianic acid 49
 valine 60, 61
 van der Waals models 6, 7
 acetyl-CoA 13
 cholesterol 54, 55
 tristearylglycerol 49
 variability, immunoglobulins 302
 variation
 allotypic 302
 idiotypic 302
 vasopressin *see* antidiuretic hormone (ADH)
 vectors 258, 259
 velocity
 and concentration 93, 97
 maximum 92, 93
 reciprocal 93
 relative 25
 very-low-density lipoprotein *see* VLDL
 vesicles 28, 29
 fractionation 198
 synaptic 354
 vimentin 332
 viruses 404-5, 405
 DNA 264, 265
 rhinovirus 404, 405

vitamins 365
 A (retinol) 52, 53, 364, 365
 antioxidants 364
 B₁ (thiamine) 366, 367
 B₂ complex 366, 367
 B₆ complex 368, 369
 B₁₂ (cobalamine) 108, 368, 369
 C (L-ascorbic acid) 104, 105, 368, 369
 coenzymes 364
 D (calcitriol, cholecalciferol) 52, 53, 364, 365
 deficiency diseases 364
 E (tocopherol) 32, 52, 53, 104, 364, 365
 H (biotin) 62, 108, 109, 368, 369
 hormones 364
 K (phyloquinone) 52, 53, 104, 364, 365
 lipid-soluble 46, 364-5, 365
 nutrients 360
 water-soluble 366-9, 367, 369
 VLDL 163, 278, 312
 V_{max} *see* maximum velocity
 voltage-gated Na⁺ channels 222

W

water 11, 27, 65, 296
 acids 31
 bacterial cells 202
 degradation 112

emulsions 28
 exclusion of 90
 excretion 322
 hydrophobic interactions 28-9
 nutrients 362
 O₂⁻ 284
 recycling 328-9, 329
 solvents 26-7
 structure 26
 water-soluble vitamins 366-9, 367, 369
 waterbinding polysaccharides 40
 watersplitting enzyme 130
 wavelength, constant 102, 103
 waxes 47
 work 17
 energy 16
 wort 148, 149

X

xanthine 187
 oxidase 186, 187, 420
 xyloglucan 41
 D-xylose 38, 39, 42
 xylulose 5-phosphate 153

Y

yeasts 148, 149

Z